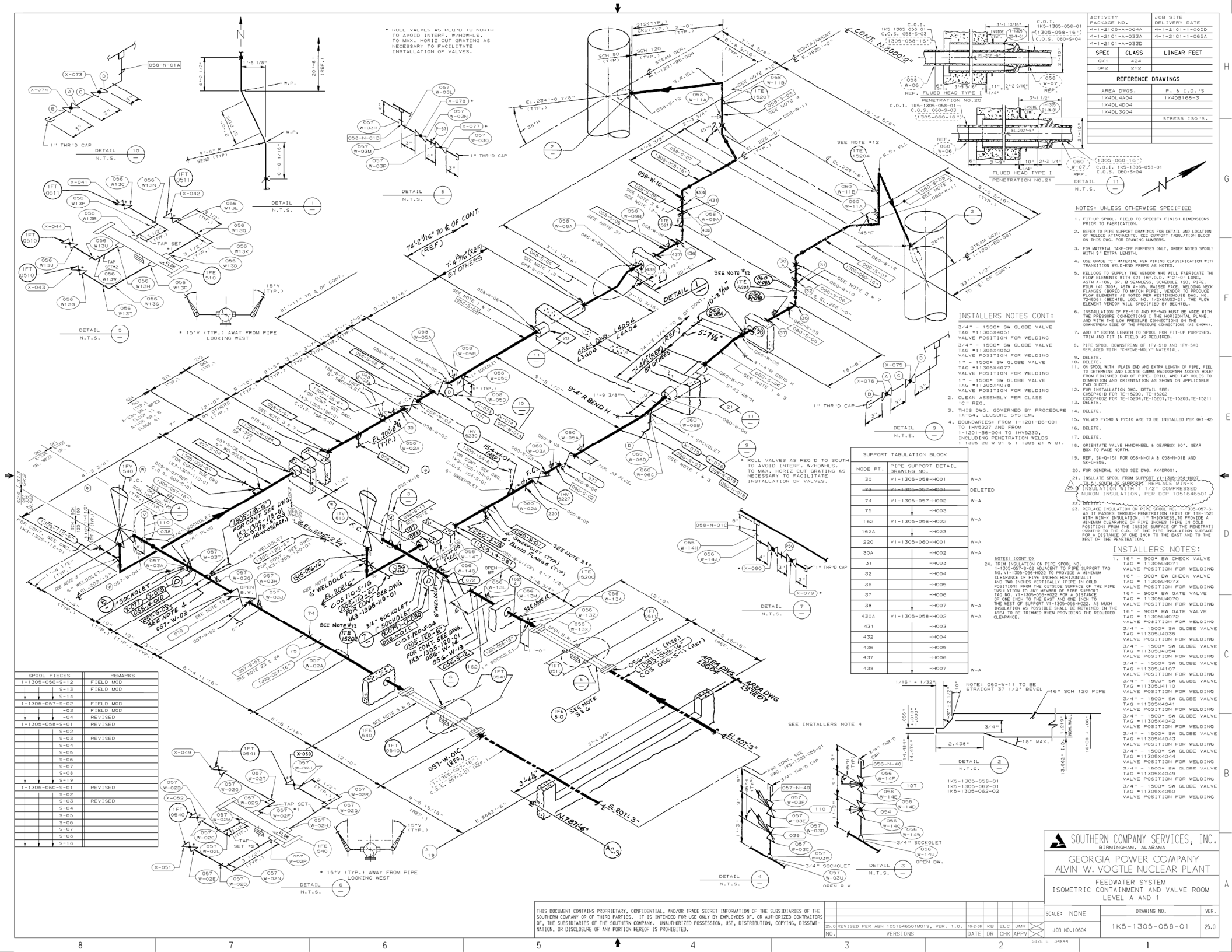


| ACTIVITY PACKAGE NO. | JOB SITE DELIVERY DATE |
|---|---|
| 4-1-2100-A-004A | 4-1-2101-1-005D |
| 4-1-2101-A-033A | 4-1-2101-1-065A |
| 4-1-2101-A-033D | |

| SPEC | CLASS | LINEAR FEET |
|---|---|---|
| GK1 | 424 | |
| GK2 | 212 | |

REFERENCE DRAWINGS

| AREA DWGS. | P. & I.D.'S |
|---|---|
| 1X4DL4A04 | 1X4D9168-3 |
| 1X4DL4D04 | |
| 1X4DL3G04 | |
| | STRESS :SO'S. |

NOTES: UNLESS OTHERWISE SPECIFIED

1. FIT-UP SPOOL. FIELD TO SPECIFY FINISH DIMENSIONS PRIOR TO FABRICATION.
2. REFER TO PIPE SUPPORT DRAWINGS FOR DETAIL AND LOCATION OF WELDED ATTACHMENTS. SEE SUPPORT TABULATION BLOCK ON THIS DWG. FOR DRAWING NUMBERS.
3. FOR MATERIAL TAKE-OFF PURPOSES ONLY, ORDER NOTED SPOOLS WITH 9" EXTRA LENGTH.
4. USE GRADE "C" MATERIAL PER PIPING CLASSIFICATION WITH TRANSITION WELD-END PREPS AS NOTED.
5. KELLOGG TO SUPPLY THE VENDOR WHO WILL FABRICATE THE FLOW ELEMENTS WITH (2) 16"O.D. *12'-0" LONG, ASTM A-106, GR. B SEAMLESS, SCHEDULE 120, PIPE. FOUR (4) 300#, ASTM A-105, RAISED FACE, WELDING NECK FLANGES (BORED TO MATCH PIPE). VENDOR TO PRODUCE FLOW ELEMENTS AS NOTED PER WESTINGHOUSE DWG. NO. 7248D61 (BECHTEL DWG. NO. 1/2X6A03-21. THE FLOW ELEMENT VENDOR WILL SPECIFIED BY BECHTEL.
6. INSTALLATION OF FE-510 AND FE-540 MUST BE MADE WITH THE PRESSURE CONNECTIONS IN THE HORIZONTAL PLANE, AND WITH THE LOW PRESSURE CONNECTIONS ON THE DOWNSTREAM SIDE OF THE PRESSURE CONNECTIONS (AS SHOWN).
7. ADD 9" EXTRA LENGTH TO SPOOL FOR FIT-UP PURPOSES. TRIM AND FIT IN FIELD AS REQUIRED.
8. PIPE SPOOL DOWNSTREAM OF 1FV-510 AND 1FV-540 REPLACED WITH "CHROME-MOLY" MATERIAL.
9. DELETE.
10. DELETE.
11. ON SPOOL WITH PLAIN END AND EXTRA LENGTH OF PIPE, FIEL TO DETERMINE AND LOCATE GAMMA RADIOGRAPH ACCESS HOLE FROM FINISHED END OF PIPE. DRILL AND TAP HOLES TO DIMENSION AND ORIENTATION AS SHOWN ON APPLICABLE PAD SHEET.
12. FOR INSTALLATION DWG. DETAIL SEE: CX50P40!D FOR TE-15200, TE-15202 CX50P40!2 FOR TE-15204,TE-15207,TE-15208,TE-15211
13. DELETE.
14. DELETE.
15. VALVES FY540 & FY510 ARE TO BE INSTALLED PER GK1-42-
16. DELETE.
17. DELETE.
18. ORIENTATE VALVE HANDWHEEL & GEARBOX 90°. GEAR BOX TO FACE NORTH.
19. REF. SK-G-151 FOR 058-N-01A & 058-N-01B AND SK-G-856.
20. FOR GENERAL NOTES SEE DWG. AX4DR001.
21. INSULATE SPOOL FROM SUPPORT V1-1305-058-H007 TO 3" SOUTH OF SUPPORT. REPLACE MIN-K 25.0 INSULATION WITH 1 1/2" COMPRESSED NUKON INSULATION, PER DCP 105164650!
22. DELETE.
23. REPLACE INSULATION ON PIPE SPOOL NO. 1-1305-057-S-AS IT PASSES THROUGH PENETRATION (EAST OF 1TE-1501 WITH MIN-K INSULATION, 1" THICKNESS,TO PROVIDE A MINIMUM CLEARANCE OF FIVE INCHES (FIVE INCHES IN COLD POSITION) FROM THE INSIDE SURFACE OF THE PENETRATION TO THE O.D. OF THE INSULATION SURFACE FOR A DISTANCE OF ONE INCH TO THE EAST AND THE WEST OF THE PENETRATION.

INSTALLERS NOTES CONT:

3/4" - 1500* SW GLOBE VALVE TAG #11305X4051 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4052 VALVE POSITION FOR WELDING
1" - 1500* SW GLOBE VALVE TAG #11305X4077 VALVE POSITION FOR WELDING
1" - 1500* SW GLOBE VALVE TAG #11305X4078 VALVE POSITION FOR WELDING
2. CLEAN ASSEMBLY PER CLASS "C" REQ.
3. THIS DWG. GOVERNED BY PROCEDURE IX-104, CLOSURE SYSTEM.
4. BOUNDARIES: FROM 1-1201-B6-001 TO 1HV5227 AND FROM 1-1201-B6-004 TO 1HV5230, INCLUDING PENETRATION WELDS 1-1305-30-W-01 & 1-1305-21-W-01.

| SUPPORT TABULATION BLOCK | | |
|---|---|---|
| NODE PT. | PIPE SUPPORT DETAIL DRAWING NO. | |
| 30 | V1-1305-058-H001 | W-A |
| ~~73~~ | ~~V1-1305-057-H001~~ | DELETED |
| 74 | V1-1305-057-H002 | W-A |
| 75 | -H003 | |
| 162 | V1-1305-056-H022 | W-A |
| 162A | -H023 | |
| 220 | V1-1305-060-H001 | W-A |
| 30A | -H002 | W-A |
| 31 | -H003 | |
| 32 | -H004 | |
| 36 | -H005 | |
| 37 | -H006 | |
| 38 | -H007 | W-A |
| 430A | V1-1305-058-H002 | W-A |
| 431 | -H003 | |
| 432 | -H004 | |
| 436 | -H005 | |
| 437 | -H006 | |
| 438 | -H007 | W-A |

NOTES: (CONT'D)
24. TRIM INSULATION ON PIPE SPOOL NO. 1-1305-057-S-02 ADJACENT TO PIPE SUPPORT TAG NO. V1-1305-056-H022 TO PROVIDE A MINIMUM CLEARANCE OF FIVE INCHES HORIZONTALLY AND TWO INCHES VERTICALLY (PIPE IN COLD POSITION) FROM THE OUTSIDE SURFACE OF THE PIPE INSULATION TO ANY MEMBER OF PIPE SUPPORT TAG NO. V1-1305-056-H022 FOR A DISTANCE OF ONE INCH TO THE EAST AND ONE INCH TO THE WEST OF SUPPORT V1-1305-056-H022. AS MUCH INSULATION AS POSSIBLE SHALL BE RETAINED IN THE AREA TO BE TRIMMED WHEN PROVIDING THE REQUIRED CLEARANCE.

INSTALLERS NOTES:

1. 16" - 900* BW CHECK VALVE TAG # 11305J4071 VALVE POSITION FOR WELDING
16" - 900* BW CHECK VALVE TAG # 11305J4073 VALVE POSITION FOR WELDING
16" - 900* BW GATE VALVE TAG # 11305J4070 VALVE POSITION FOR WELDING
16" - 900* BW GATE VALVE TAG # 11305J4072 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305J4038 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305J4054 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305J4107 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305J4110 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4041 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4042 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4043 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4044 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4049 VALVE POSITION FOR WELDING
3/4" - 1500* SW GLOBE VALVE TAG #11305X4050 VALVE POSITION FOR WELDING

| SPOOL PIECES | REMARKS |
|---|---|
| 1-1305-056-S-12 | FIELD MOD |
| S-13 | FIELD MOD |
| S-14 | |
| 1-1305-057-S-02 | FIELD MOD |
| -03 | FIELD MOD |
| -04 | REVISED |
| 1-1305-058-S-01 | REVISED |
| S-02 | |
| S-03 | REVISED |
| S-04 | |
| S-05 | |
| S-06 | |
| S-07 | |
| S-08 | |
| S-19 | |
| 1-1305-060-S-01 | REVISED |
| S-02 | |
| S-03 | REVISED |
| S-04 | |
| S-05 | |
| S-06 | |
| S-07 | |
| S-08 | |
| S-18 | |

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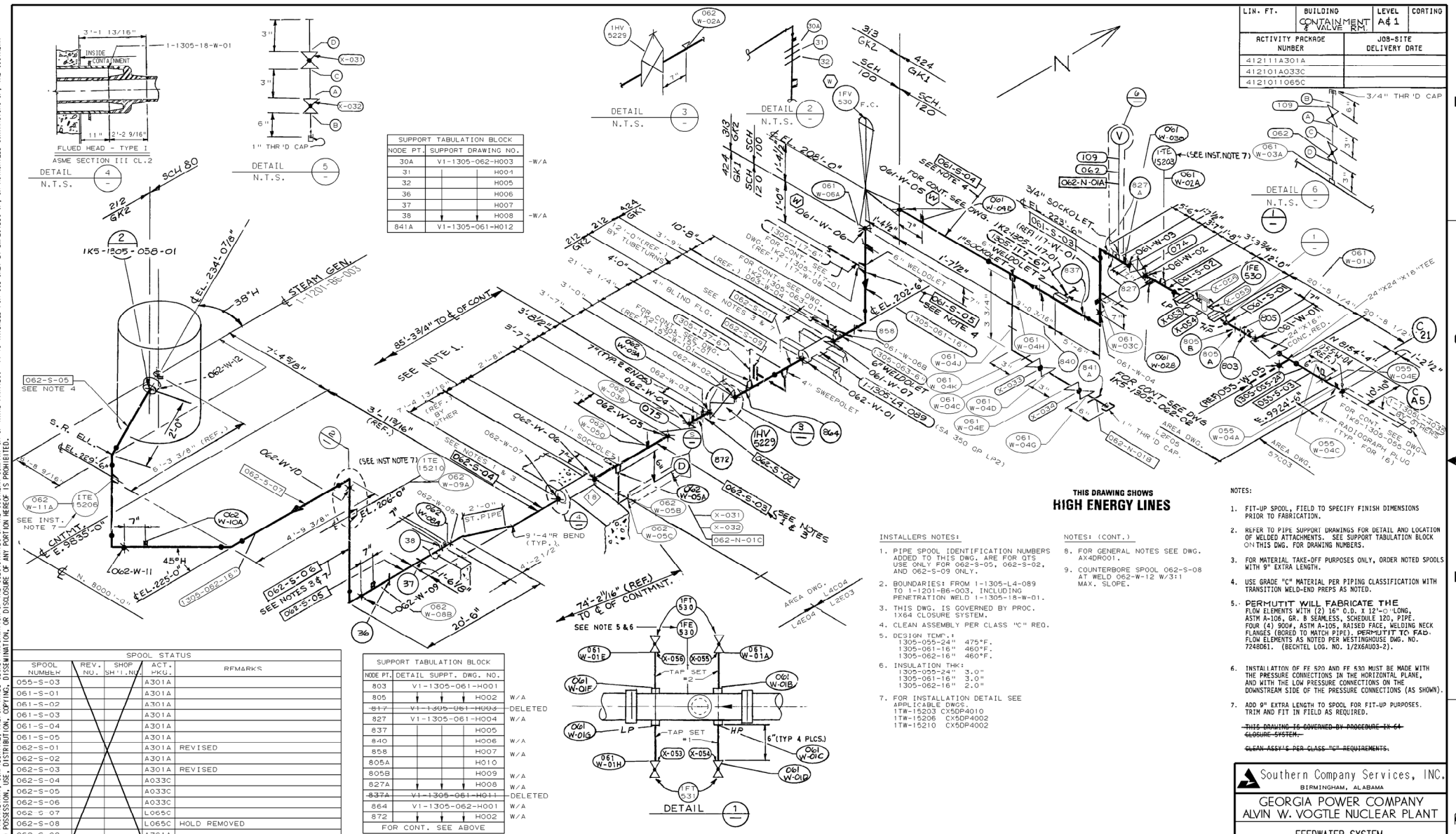
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 25.0 | REVISED PER ABN 105164650!M019, VER. 1.0. | 10-2-08 | KB | ELC | JMR |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FEEDWATER SYSTEM
ISOMETRIC CONTAINMENT AND VALVE ROOM LEVEL A AND 1

SCALE: NONE
JOB NO.10604
DRAWING NO. 1K5-1305-058-01
VER. 25.0
SIZE E 34X44


3'-1 13/16"
1-1305-18-W-01
INSIDE CONTAINMENT
11"
2'-2 9/16"
FLUED HEAD - TYPE I
ASME SECTION III CL.2
DETAIL 4 N.T.S.
3"
3"
6"
X-031
X-032
1" THR'D CAP
DETAIL 5 N.T.S.
1HV 5229
062 W-02A
7"
DETAIL 3 N.T.S.
30A
31
32
W
1FV 530 F.C.
DETAIL 2 N.T.S.
SCH 80
2½ GR2
1K5-1305-058-01
DETAIL 2 N.T.S.
EL. 234'-07⅛"
38"H
STEAM GEN.
1-1201-B6-003
062-W-12
062-S-05 SEE NOTE 4
S.R. ELL.
EL. 229'-6"
2'-0"
6'-3 3/8" (REF.)
062-W-10
062-S-07
CNTMT. EL. 9835'-0"
062 W-11A
ITE 5206
SEE INST. NOTE 7
062 W-10A
7"
4'-9 3/8"
45°H
062-W-11
EL. 225'-0"
1305-062-16
N. 8000'-0"
062-S-06
SEE NOTES 3 & 7
062-S-05
062-W-09
062 W-08B
37
1'-6 1/8"
20'-6"
36
062-W-08
062 W-08A
38
2'-0" ST. PIPE
9'-4"R BEND (TYP.)
4'-2 1/2"
(SEE INST NOTE 7)
ITE 5210
062 W-09A
062-S-04
7"
EL. 206'-0"
062-W-08
18
1
062 W-05A
062 W-05B
062 W-05C
X-031
X-032
062-N-01C
062-S-03
SEE NOTES 1 & 3
062-S-02
1HV 5229
3
864
872
S
D
062-W-01
062-W-02
062-W-03
062-W-04
062-W-05
062-W-06
062-W-07
062 W-03A
062 W-03B
062 W-03C
062 W-05D
062 W-05D
075
1" SOCKOLET
7" (TYP. 2 ENDS)
SEE NOTES 1 & 3
4" BLIND FLG.
SEE NOTES 3 & 7
062-S-01
062-S-09
062-S-08
062-W-13
062-W-14
062-W-15
062-W-16
062-W-17
062-W-18
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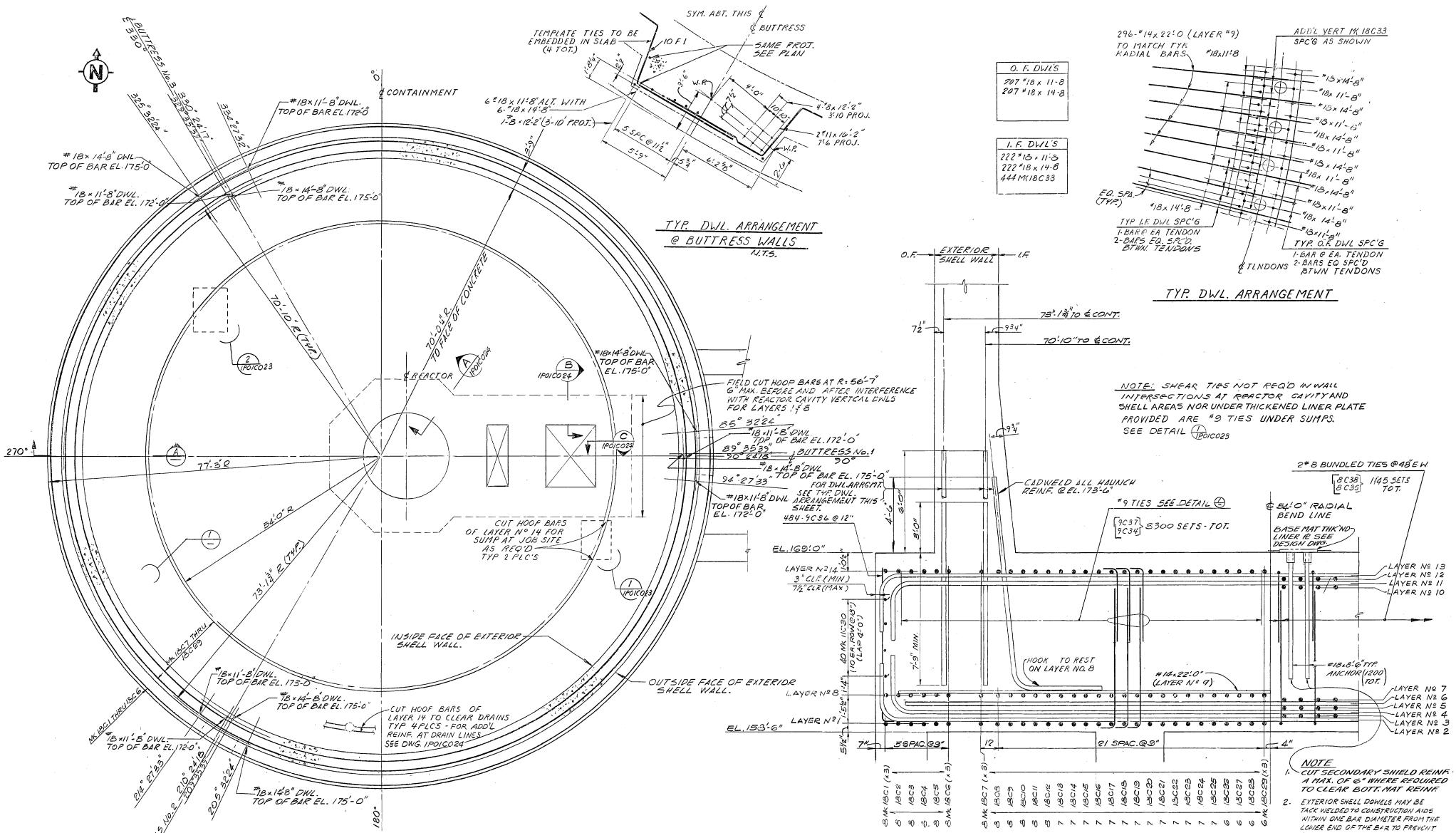
062-W-52




ACT PKG. NO 2-1-2101-C-003A

CADWELDS
211 HORIZ. — LAYER N° 14
211 " " N° 8
211 " " N° 1
633 TOTAL.

| | RELEASE NO. | POUR NO. | BAR LIST NO. |
|---|---|---|---|
| LAYER NO.1 | R-0015 | 1-01C-001 | 1B01C041 & 042 |
| LAYER NO.8 | R-0022 | ↓ | 1B01C095 & 097 |
| LAYER NO. 9 | R-0023 | ↓ | 1B01C098 |
| DOWELS & TIES | R-0024 | 1-01C-001 | 1B01C099 |
| SPACERS | R-0025 | 1-01C-001 | 1B01C100 |
| BUTTRESS DOWELS | R-0024 | 1-01C-001 | 1B01C101 & 102 |
| LAYER NO.14 | R-0040 | 1-01C-002 | 1B01C149 & 150 |
| PLATE ANCHORS | R-0026 | 1-01C-001 | 1B01C102 |

| REF. DESIGN DRWGS |
|---|
| 1X2D01A002 |
| 1X2D01A003 |
| 1X2D01A004 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE UNIT 1
BASE MAT REINF. LAYERS 1, 8, 9, 14 & DOWELS
POUR NOS. 1-01C-001 & 1-01C-002

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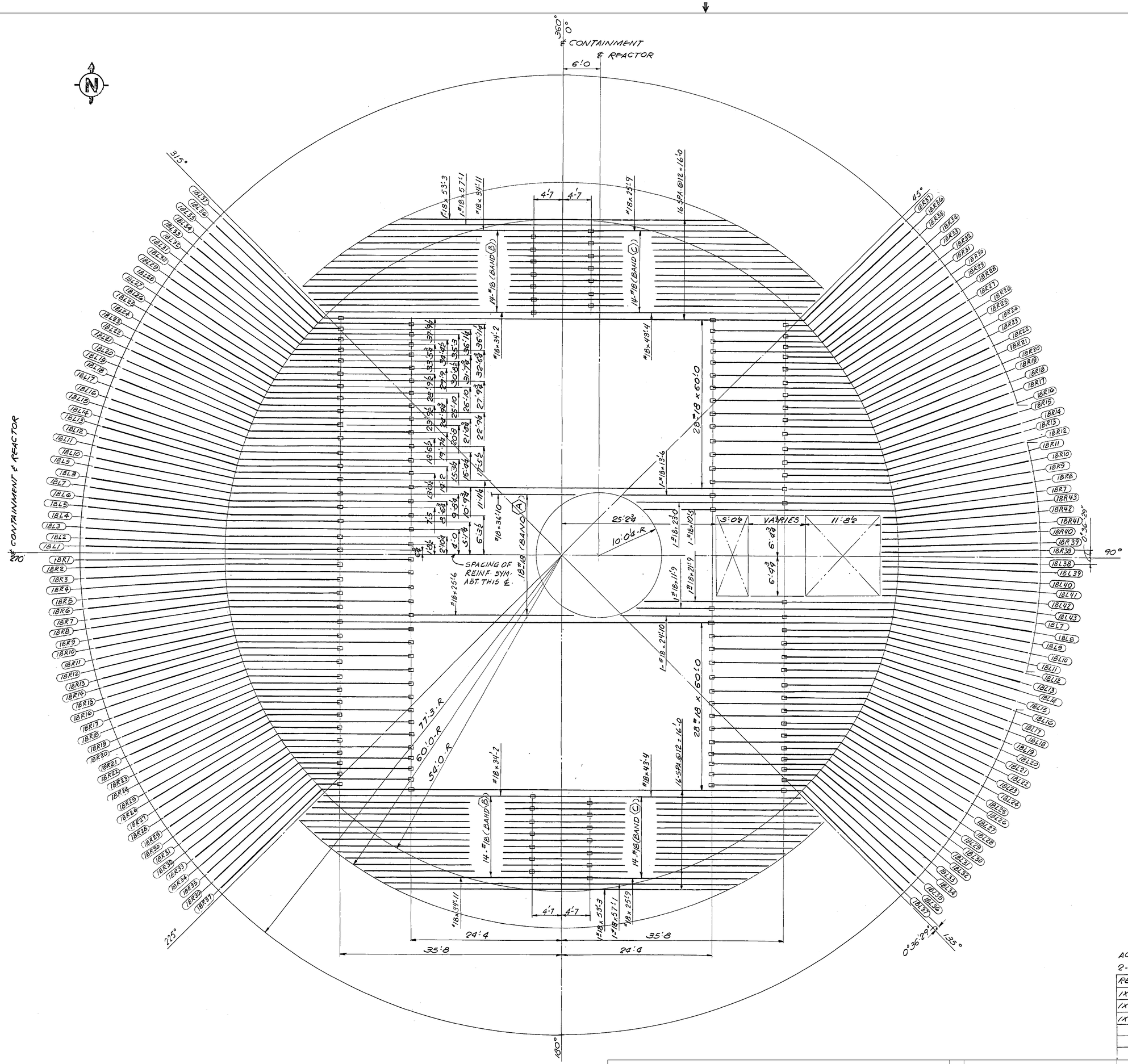
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 9.0 | REVISED TO CLOSE C-FCRB-1939, C-FCRB-746 & C-FCRB-930 (NO CHANGES REQUIRED). | 11-6-07 | DFV | LRP | WLN |

SCALE: 1/8" = 1'-0" 1/3" = 1'-0"
JOB NO. 10604
DRAWING NO. 1P01C016
VER. 9.0
SIZE E 34x44




| BAND (A) |
|---|
| 1-#18 x 25-6 |
| × 35-0 |
| × 22-5 |
| × 32-10 |
| × 20-10 |
| × 31-7 |
| × 19-11 |
| × 31-0 |
| × 19-6 |
| × 30-10 |
| × 19-8 |
| × 31-3 |
| × 20-3 |
| × 32-2 |
| × 21-6 |
| × 33-9 |
| × 23-9 |
| × 36-10 |

| BAND (B) x 2 |
|---|
| 1-#18 x 34-11 |
| × 27-4 |
| × 38-1 |
| × 30-4 |
| × 40-10 |
| × 33-0 |
| × 43-5 |
| × 35-4 |
| × 45-8 |
| × 37-6 |
| × 46-4 |
| × 36-2 |
| × 44-4 |
| × 34-2 |

| BAND (C) x 2 |
|---|
| 1-#18 x 25-9 |
| × 36-7 |
| × 28-11 |
| × 39-6 |
| × 31-9 |
| × 42-2 |
| × 34-2 |
| × 44-6 |
| × 36-5 |
| × 46-8 |
| × 37-2 |
| × 45-4 |
| × 35-2 |
| × 43-4 |

CAD WELDS REQ'D
#18 164 HORIZ

| RELEASE NO | POUR NO. | BAR LIST NO. |
|---|---|---|
| R-0016 | 1-01C-001 | 1801CC43 THRU 1801CC71 |
| | | |
| | | |
| | | |
| | | |
| | | |

ACT. PKG. NO.
2-1-2101-C-003A

| REF. DESIGN DWG. | |
|---|---|
| 1X2D01A0002 | 5 |
| 1X2D01A0003 | 7 |
| 1X2D01A0004 | 4 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE UNIT 1
BASE MAT E-W BARS
LAYER 2 POUR NO. 1-01C-001

SCALE: 3/16" = 1'-0

DRAWING NO. 1P01C017

VER. 2.0

JOB NO. 10604

SIZE E 34X44

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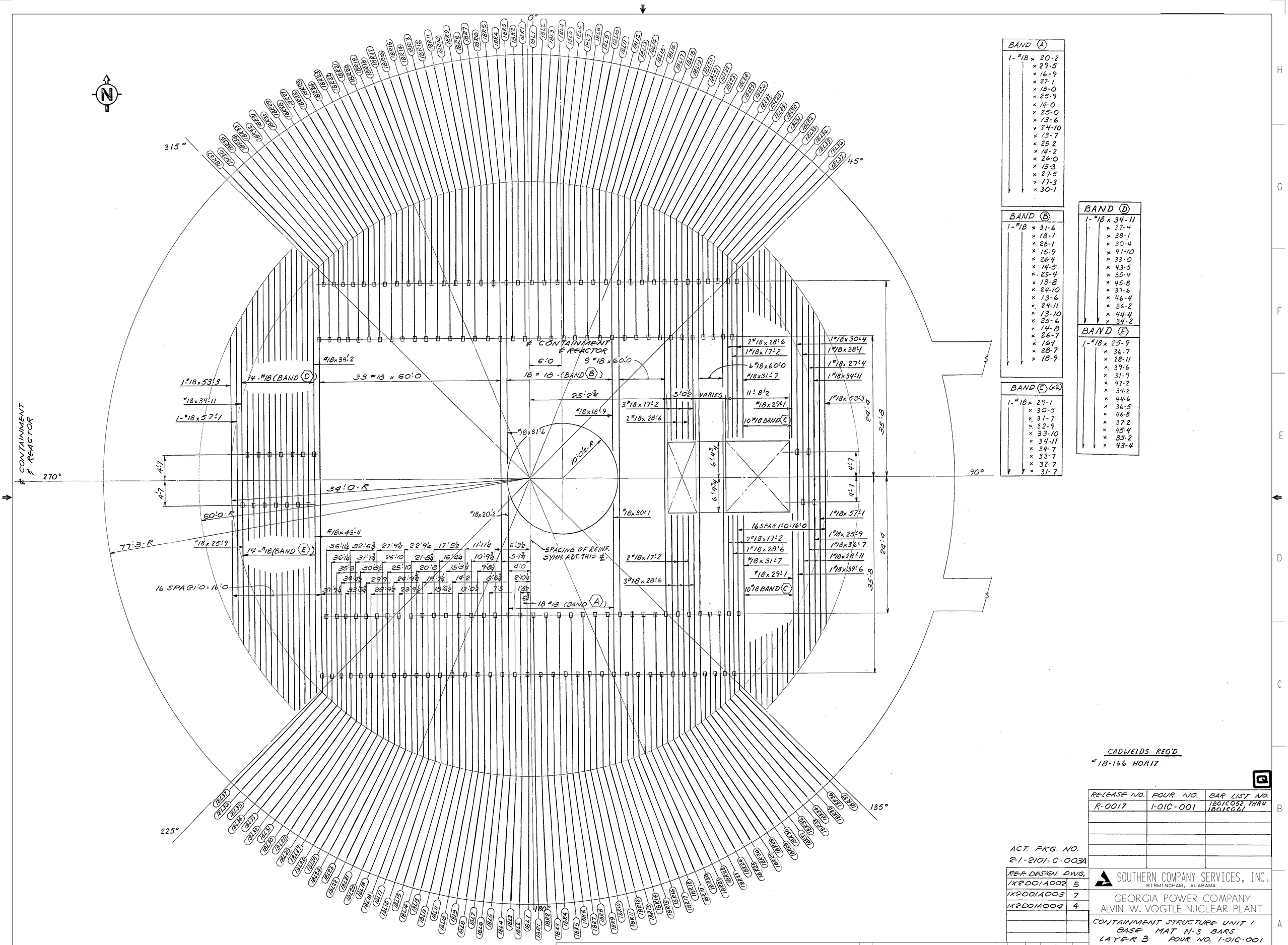
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 2.0 | REACTIVATED PER ABN-V70096, VER 2.0 | 08/01/07 | EL.C | MWD | WLN |




**BAND (A)**

1-#18 x 20-2
× 29-5
× 16-9
× 27-1
× 15-0
× 25-9
× 14-0
× 25-0
× 13-6
× 24-10
× 13-7
× 25-2
× 14-2
× 26-0
× 15-3
× 27-5
× 17-3
× 30-1

**BAND (B)**

1-#18 x 31-6
× 18-1
× 28-1
× 15-9
× 26-4
× 14-5
× 25-4
× 13-8
× 24-10
× 13-6
× 24-11
× 13-10
× 25-6
× 14-8
× 26-7
× 16-1
× 28-7
× 18-9

**BAND (C) (x2)**

1-#18 x 29-1
× 30-5
× 31-7
× 32-9
× 33-10
× 34-11
× 34-7
× 33-7
× 32-7
× 31-7

**BAND (D)**

1-#18 x 34-11
× 27-4
× 38-1
× 30-4
× 41-10
× 33-0
× 43-5
× 35-4
× 45-8
× 37-6
× 46-4
× 36-2
× 44-4
× 34-2

**BAND (E)**

1-#18 x 25-9
× 36-7
× 28-11
× 39-6
× 31-9
× 42-2
× 34-2
× 44-6
× 36-5
× 46-8
× 37-2
× 45-4
× 35-2
× 43-4

CADWELDS REQ'D
#18-166 HORIZ

| RELEASE NO. | POUR NO. | BAR LIST NO. |
|---|---|---|
| R-0017 | 1-01C-001 | 1B01C052 THRU 1B01C061 |
| | | |
| | | |
| | | |
| | | |
| | | |

ACT. PKG. NO.
2-1-2101-C-003A

| REF. DESIGN DWG. | |
|---|---|
| 1X2001A002 | 5 |
| 1X2001A003 | 7 |
| 1X2001A004 | 4 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE UNIT 1
BASE MAT N-S BARS
LAYER 3 POUR NO. 1-01C-001

SCALE: 3/16" = 1'-0

DRAWING NO. 1P01C018

VER. 2.0

JOB NO. 10604

SIZE E 34x44

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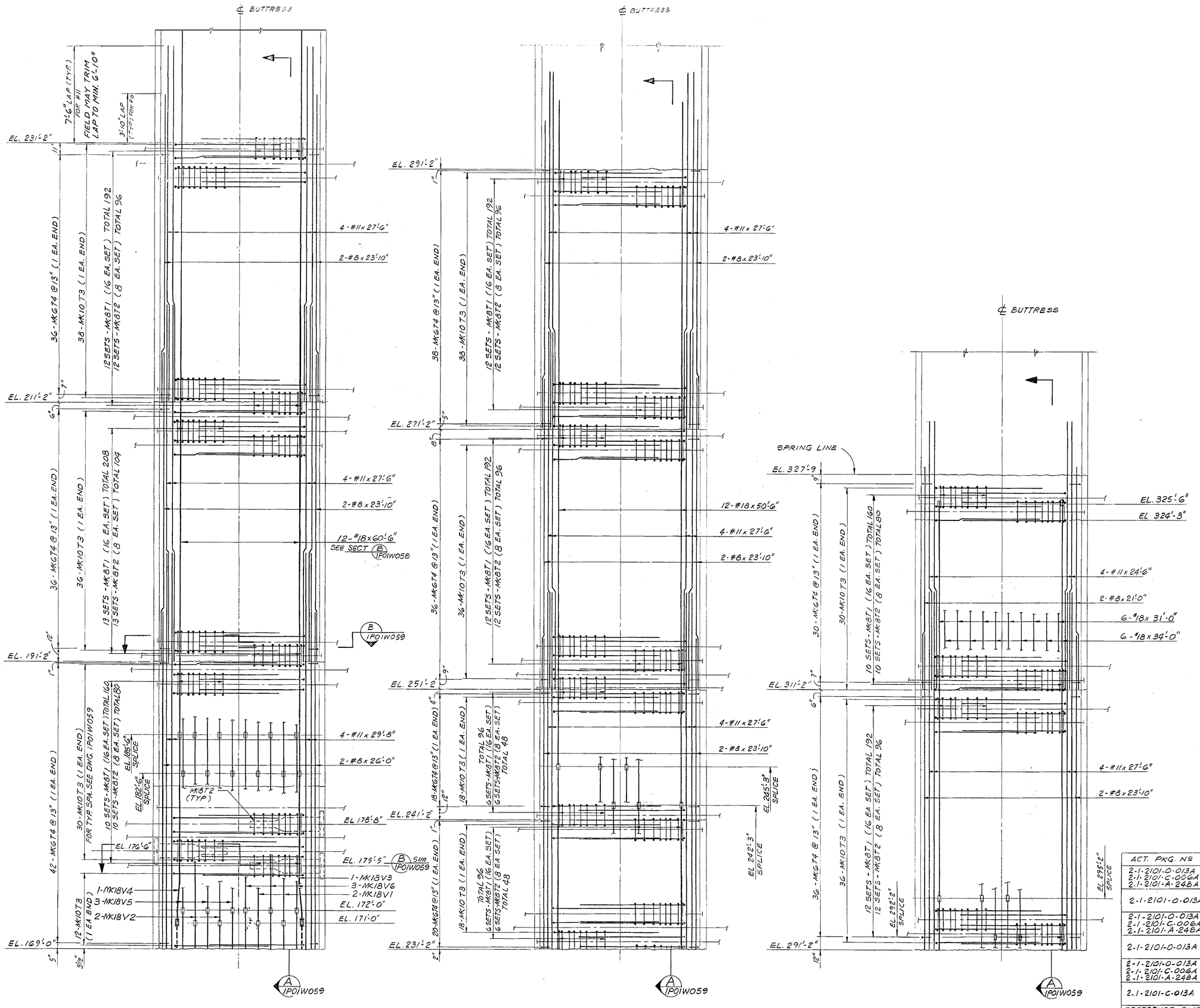
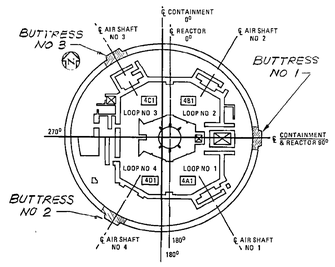
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 2.0 | REACTIVATED PER ABN-V70096, VER 2.0 | 08/01/07 | EL C | MWD | WLN |




KEY PLAN UNIT 1 (EL 171'-9")

## ELEVATION - BUTTRESS REINF.

BUTTRESS Nº1 (SHOWN) BUTTRESS Nº 2 & 3 (SIM.)

### NOTES

1. WORK THIS DWG. WITH DWG. 1P01C01G. #18 BUTTRESS DOWELS SHALL BE CUT AS SHOWN ON THIS DWG.

#18 VERTICAL CADWELDS:

36 - REQ'D. @ EL. 172'-0" & 175'-0"
36 - " " " 242'-3" & 245'-3"
36 - " " " 292'-2" & 295'-2"

| ACT. PKG. Nº | RELEASE Nº | POUR ELEV'S. | REFERENCE BAR LISTS |
|---|---|---|---|
| 2-1-2101-0-013A<br>2-1-2101-C-006A<br>2-1-2101-A-248A | R-0169 | EL. 169'-0" TO EL. 241'-2" | 1B01W265<br>1B01W266<br>1B01W267 |
| 2-1-2101-0-013A | R-0170 | EL. 261'-2" TO EL. 327'-9" | 1B01W268<br>1B01W269 |
| 2-1-2101-0-013A<br>2-1-2101-C-006A<br>2-1-2101-A-248A | R-0171 | EL. 169'-0" TO EL. 241'-2" | 1B01W270<br>1B01W271<br>1B01W272 |
| 2-1-2101-0-013A | R-0172 | EL. 241'-2" TO EL. 327'-9 | 1B01W307<br>1B01W308 |
| 2-1-2101-0-013A<br>2-1-2101-C-006A<br>2-1-2101-A-248A | R-0173 | EL. 169'-0" TO EL. 241'-2" | 1B01W309<br>1B01W310<br>1B01W311 |
| 2-1-2101-C-013A | R-0174 | EL. 241'-2" TO EL. 327'-9" | 1B01W312<br>1B01W313 |

| REFERENCE DWG'S. |
|---|
| DWG. Nº. |
| 1X2D01A002 |
| 1X2D01A004 |
| 1X2D01K008 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE - UNIT 1
BUTTRESS ELEVATION
EL. 169'-0" TO EL. 327'-9"

SCALE: 3/8"=1'-0" U.N.O.
DRAWING NO. 1P01W058
VER. 4.0
JOB NO. 10604
SIZE E 34X44

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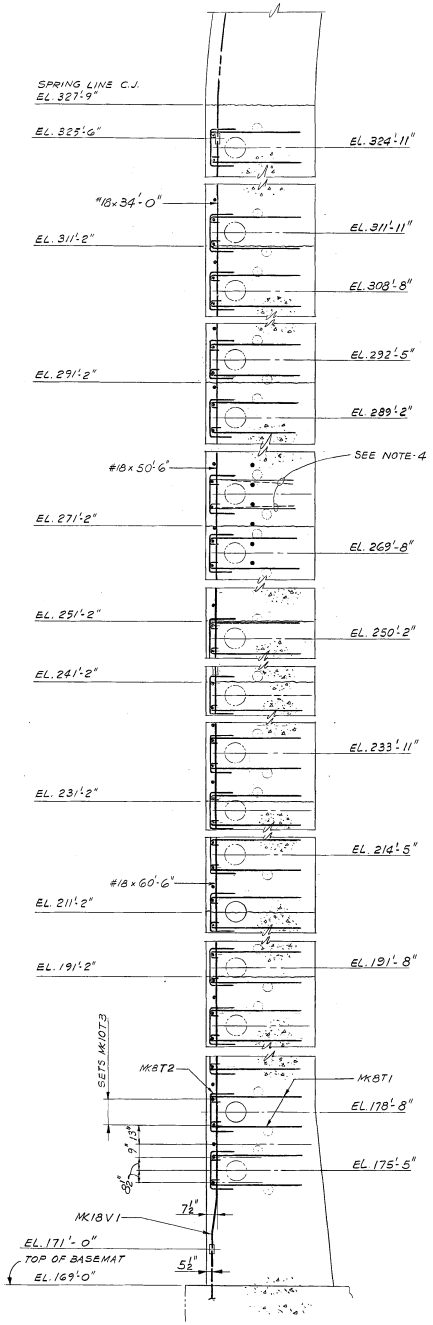
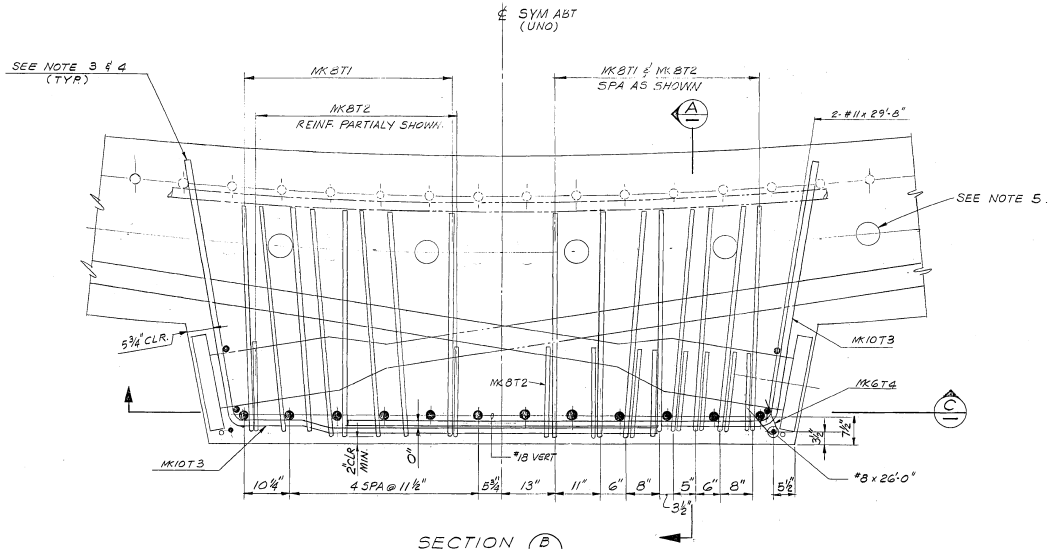
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 4.0 | REACTIVATED PER ABN-V70096, VER 2.0 | 08/01/07 | ELC | MWD | WLN |

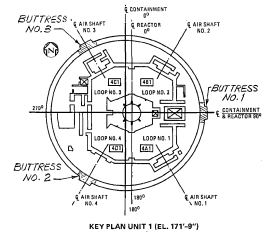



SECTION (A) 1P01W058
3/8"=1'-0"
(REINF. SHOWN PARTIALLY)

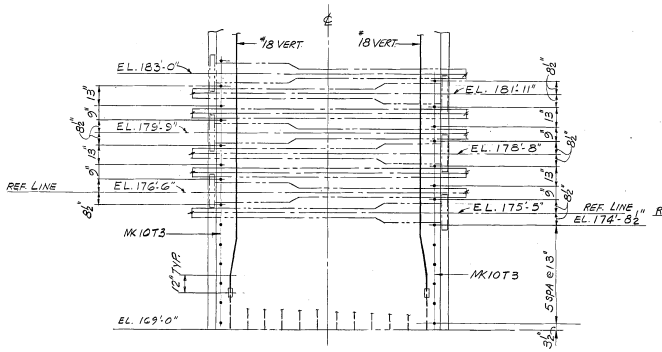



SECTION (B) 1P01W058
1"=1'-0"

BUTTRESS №1 (SHOWN) BUTTRESS №2 & 3 (SIM.)

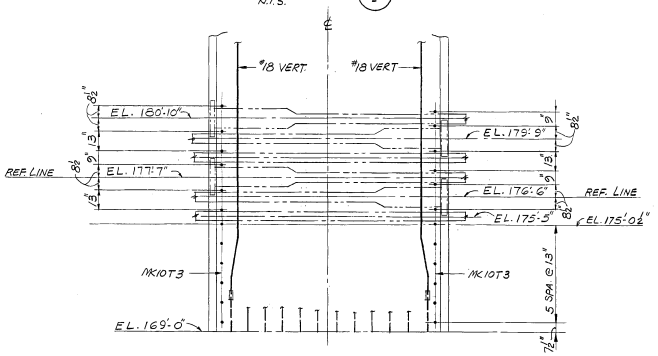



KEY PLAN UNIT 1 (EL. 171'-9")

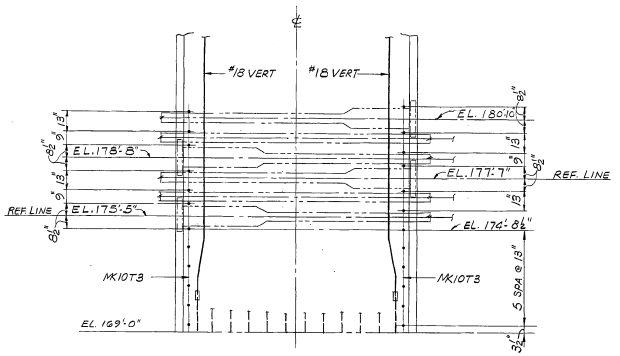



TYP. SPA. OF #10 BARS AT BUTTRESS - #1

SECTION (C) N.T.S.




TYP. SPA. OF #10 BARS AT BUTTRESS - #2
N.T.S.




TYP. SPA. OF #10 BARS AT BUTTRESS - #3
N.T.S.

NOTES:

1. WORK THIS DWG. WITH DWG. 1P01W058.
2. ALL VERT'S TO MATCH DWL'S.
3. FIELD MAY CUT BAR MK10T3 ONLY IF INTERFERENCE WITH LINER PLATE OCCURS. BAR MAY BE CUT NO MORE THAN 1" AWAY FROM STIFFENER INTERFERENCE.
4. THE INTERIOR LEG OF MK8T1 OR MK10T3 MAY BE PLACED ± 20° FROM HORIZONTAL TO AVOID INTERFERENCES.
5. VERTICAL TENDON LOCATIONS VARY FOR EACH BUTTRESS. IF MK8T1 INTERFERES WITH A TENDON, KEEP THE BUTTRESS FACE WORK POINTS AS SHOWN BUT MOVE THE INTERIOR LEG AWAY FROM THE ℄ OF THE BUTTRESS.
6. FIELD MAY REPLACE STANDARD 180° HOOK ON MK8T1 BAR WITH STANDARD 90° HOOK WHERE INTERFERENCE OCCURS WITH TRUMPET.

| REFERENCE DWGS. |
|---|
| DWG. NO. |
| 1X0D1A002 |
| 1X0D1A004 |
| 1X0D1K008 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE - UNIT 1
BUTTRESS, PLAN AND SECTIONS
EL. 169'-0" TO EL. 327'-9"

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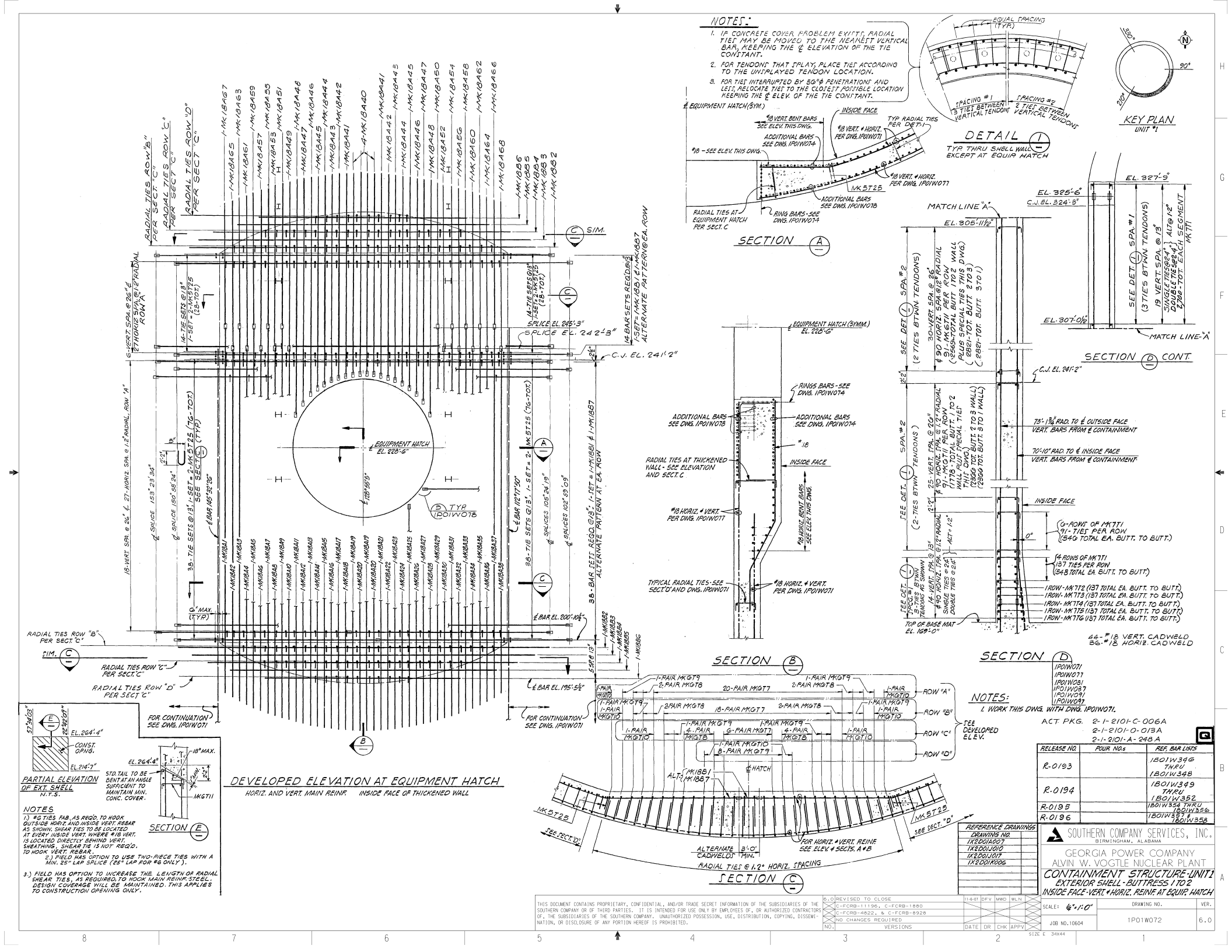
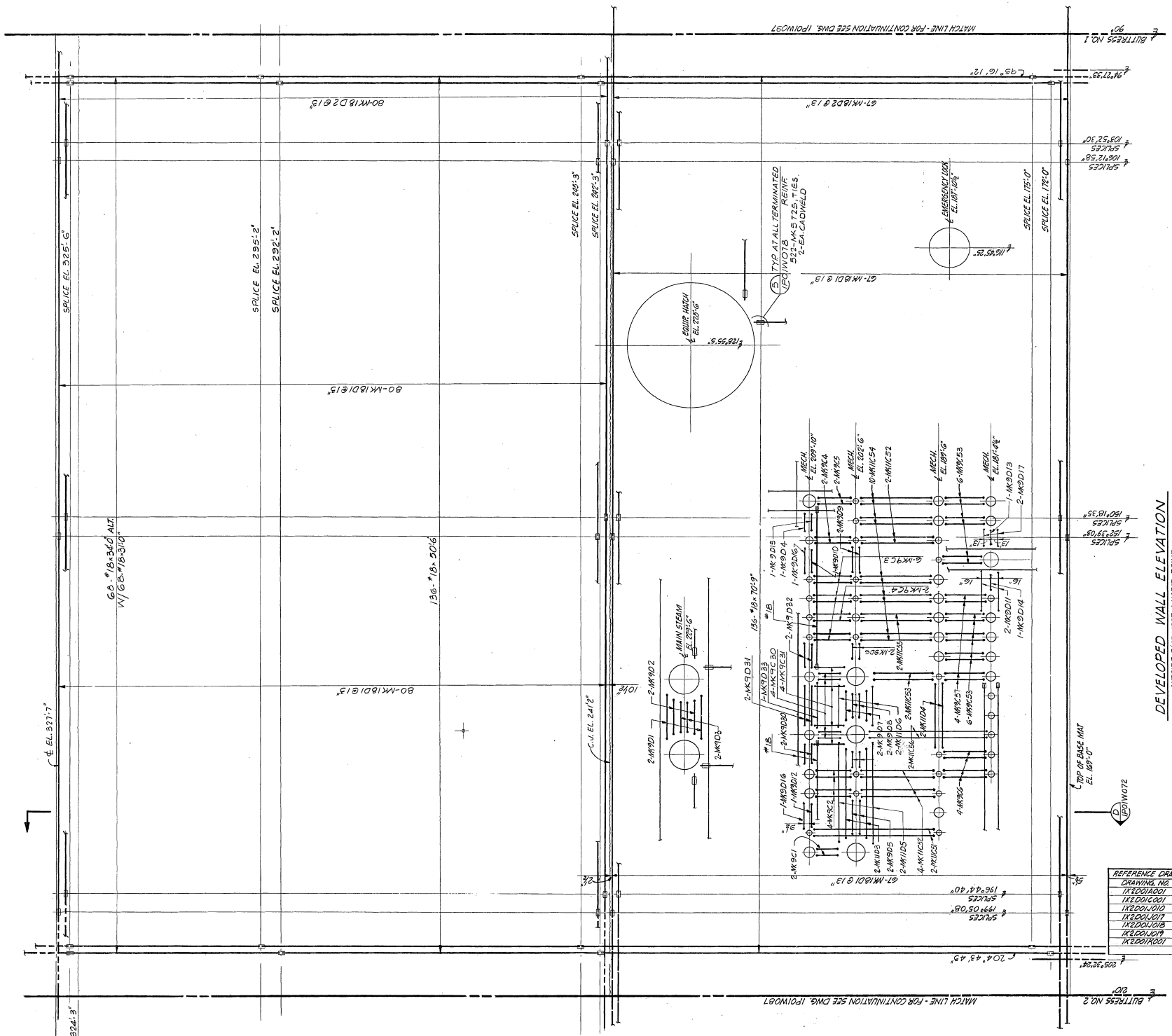
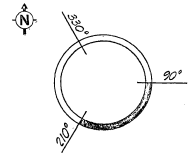
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 5.0 | REVISED TO CLOSE C-FCRB-4012 | 11-6-07 | DFV | LRP | WLN |
| | (NO CHANGES REQUIRED) | | | | |

SCALE: 3/8"=1'-0" U.N.O.
JOB NO. 10604
SIZE E 34X44
DRAWING NO. 1P01W059
VER. 5.0


NOTES:
1. IF CONCRETE COVER PROBLEM EXISTS, RADIAL TIES MAY BE MOVED TO THE NEAREST VERTICAL BAR, KEEPING THE ℄ ELEVATION OF THE TIE CONSTANT.
2. FOR TENDONS THAT SPLAY, PLACE TIES ACCORDING TO THE UNSPLAYED TENDON LOCATION.
3. FOR TIES INTERRUPTED BY SG #8 PENETRATIONS AND LEGS, RELOCATE TIES TO THE CLOSEST POSSIBLE LOCATION KEEPING THE ℄ ELEV. OF THE TIE CONSTANT.
EQUAL SPACING (TYP.)
SPACING #1 3 TIES BETWEEN VERTICAL TENDONS
SPACING #2 2 TIES BETWEEN VERTICAL TENDONS
KEY PLAN UNIT #1
DETAIL 1 TYP THRU SHELL WALL EXCEPT AT EQUIP HATCH
SECTION A
SECTION B
SECTION C
SECTION D
SECTION E CONT.
DEVELOPED ELEVATION AT EQUIPMENT HATCH
HORIZ. AND VERT. MAIN REINF. INSIDE FACE OF THICKENED WALL
PARTIAL ELEVATION OF EXT. SHELL
NOTES
1.) #6 TIES, FAB, AS REQ'D, TO HOOK OUT SIDE HORIZ. AND INSIDE VERT. REBAR AS SHOWN. SHEAR TIES TO BE LOCATED AT EVERY INSIDE VERT. WHERE #18 VERT. IS LOCATED DIRECTLY BEHIND VERT. SHEATHING. SHEAR TIE IS NOT REQ'D. TO HOOK VERT. REBAR.
2.) FIELD HAS OPTION TO USE TWO-PIECE TIES WITH A MIN. 25" LAP SPLICE (25" LAP FOR #6 ONLY).
3.) FIELD HAS OPTION TO INCREASE THE LENGTH OF RADIAL SHEAR TIES, AS REQUIRED, TO HOOK MAIN REINF. STEEL. DESIGN COVERAGE WILL BE MAINTAINED. THIS APPLIES TO CONSTRUCTION OPENING ONLY.
NOTES:
1. WORK THIS DWG. WITH DWG. 1P01W071.
ACT. PKG. 2-1-2101-C-006A
2-1-2101-0-013A
2-1-2101-A-248A
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
CONTAINMENT STRUCTURE-UNIT 1
EXTERIOR SHELL-BUTTRESS 1 TO 2
INSIDE FACE-VERT. & HORIZ. REINF. AT EQUIP. HATCH
DRAWING NO. 1P01W072
VER. 6.0
SCALE: 1/4"=1'-0"
JOB NO. 10604
SIZE E 34X44




KEY PLAN
UNIT #1

NOTES

1. FOR THE TABLE BELOW, FIELD CUT #18'S 6" MAX. BEFORE AND AFTER INTERFERENCE WITH SLEEVES OR WHERE CONCRETE COVER PROBLEMS EXIST AT THE RECESSED PENETRATION FLANGES EXCEPT AT AUX. FEED WATER PENS SEE NOTE 9. REPLACE #18 BETWEEN SLEEVES WITH #9 OR #11 HOOKED BARS AS SHOWN ON ON THIS DWG.
2. WORK VERT. BARS WITH DWG. 1P01C016.
3. HORIZ. CADWELDS MAY BE ADJUSTED TO CLEAR VERT. REINF.
4. IF ANY INTERFERENCES OCCUR DUE TO BLOCKOUTS, PLATES, ETC., REFER TO REINF. NOTE #9 ON DWG. AX209AV018.
5. FOR ADDITIONAL AND TRIM REINF. SEE DWG. 1P01W078.
6. FOR BUTTRESS REINF. SEE DWGS. 1P01W056 AND 1P01W059.
7. FOR RADIAL TIES SEE DWG. 1P01W071.
8. FIELD MAY CUT UP TO 6" OFF THE HOOKS OF THE OUTSIDE FACE #9 OR #11 BARS IF INTERFERENCE WITH LINER PLATE STIFFENERS EXIST.
9. PLACE ALL REINF. AROUND AUX. FEEDWATER PENS PRIOR TO INSTALLATION OF PEN ASSY'S. MAIN REINF. TO BE TERMINATED AT AUX. FEEDWATER PENS, AS SHOWN IN TABLE BELOW, SHALL BE FIELD CUT AT 24" RADIUS FROM ℄ PENS.
10. IF OPTIONAL CONCRETE BLOCKOUT AT AUXILIARY FEEDWATER PENETRATIONS IS PROVIDED AS SHOWN ON DWG. 1X2D01A007, FIELD SHALL ASSURE THAT A CLEAR OPENING OF 19" MIN. RADIUS FROM THE ℄ OF EACH PENETRATION EXISTS IN THE REINF. AND TENDON SHEATHING PRIOR TO CONC. PLACEMENT ADJACENT TO THE BLOCKOUT AREA.
11. WHERE INTERFERENCE OCCURS BETWEEN FIRST TWO #18 VERTICAL OUTER FACE REBAR, ADJACENT TO EACH BUTTRESS, AND THE HORIZ. TENDON SHEATHING, THE VERT. REBAR MAY BE BENT INSIDE THE HORIZ. SHEATHING. THE MAX. CLEAR DIST. BETWEEN BENT BAR AND HORIZ. SHEATHING WILL BE 2". AN ADDITIONAL #11 BAR WILL BE PLACED OUTSIDE THE #18 OUTER FACE HORIZ. REBAR FOR EACH #18 VERT. BENT. THE BENT #18 SHALL EXTEND TO EL. 338'-0" MINIMUM. THE ADDITIONAL #11 REINFORCEMENT WILL CONTINUE TO A VERTICAL ANGLE OF 60° IN THE DOME.
12. ADD'L. #11 BARS PER NOTE N°11 @ INTERFERENCE ADJACENT TO BUTTRESS N°1 MAY BE PLACED INSIDE OF THE #18 HORIZ.

| PENETRATION IDENTIFICATION | | VERTICAL BARS | HORIZONTAL BARS |
|---|---|---|---|
| EMERGENCY LOCK | | 6 | 7 |
| MAIN STEAM | ① | 7 | 6 |
| | ④ | 6 | |
| ALL 34"ϕ PENETRATIONS | | 4 | 6 |
| MECH. PEN. @ ELEV. 209'-10" | 24"ϕ ALL OTHERS EXCEPT AUX. FEED WATER PEN. | 2 | 3 |
| | | | 2 |
| MECH. PEN. AT ELEV. 202'-6" EXCEPT 34"ϕ | | 2 | 2 |
| MECH. PENETRATIONS @ ELEV. 189'-6" | | 2 | 2 |
| MECH. PEN. @ ELEV. 181'-4½" | 12"ϕ | 2 | 1 |
| | 18"ϕ & 22"ϕ | | 3 |
| EQUIPMENT HATCH | | 20 | 20 |
| 18" ϕ AUX. FEEDWATER PEN'S @ EL. 209'-10" | | 4 | 3 |

302 - #18 VERT. CADWELDS REQ'D.
286 - #18 HORIZ. CADWELDS REQ'D (TO EL. 241'-2").
272 - #18 VERT. CADWELDS REQ'D ABOVE EL. 241'-2".
240 - #18 HORIZ. CADWELDS REQ'D ABOVE EL. 241'-2".

ACT. PKG. NO. 2-1-2101-C-006A
2-1-2101-0-013A
2-1-2101-A-248A

| REFERENCE DRAWINGS | RELEASE NO. | POUR NO.s | REF. BAR LISTS |
|---|---|---|---|
| DRAWING NO. | R-0213 | | 1B01W396 & 397 |
| 1X2D01A001 | | | |
| 1X2D01C001 | R-0214 | | 1B01W398 THRU 1B01W400 |
| 1X2D01J010 | | | |
| 1X2D01J017 | | | |
| 1X2D01J018 | R-0215 | | 1B01W404 |
| 1X2D01J019 | R-0216 | | 1B01W405 |
| 1X2D01K001 | R-0219 | | 1B01W406 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT STRUCTURE - UNIT 1
EXTERIOR SHELL - BUTTRESS 1 TO 2
OUTSIDE FACE VERT. & HORIZ. REINF.

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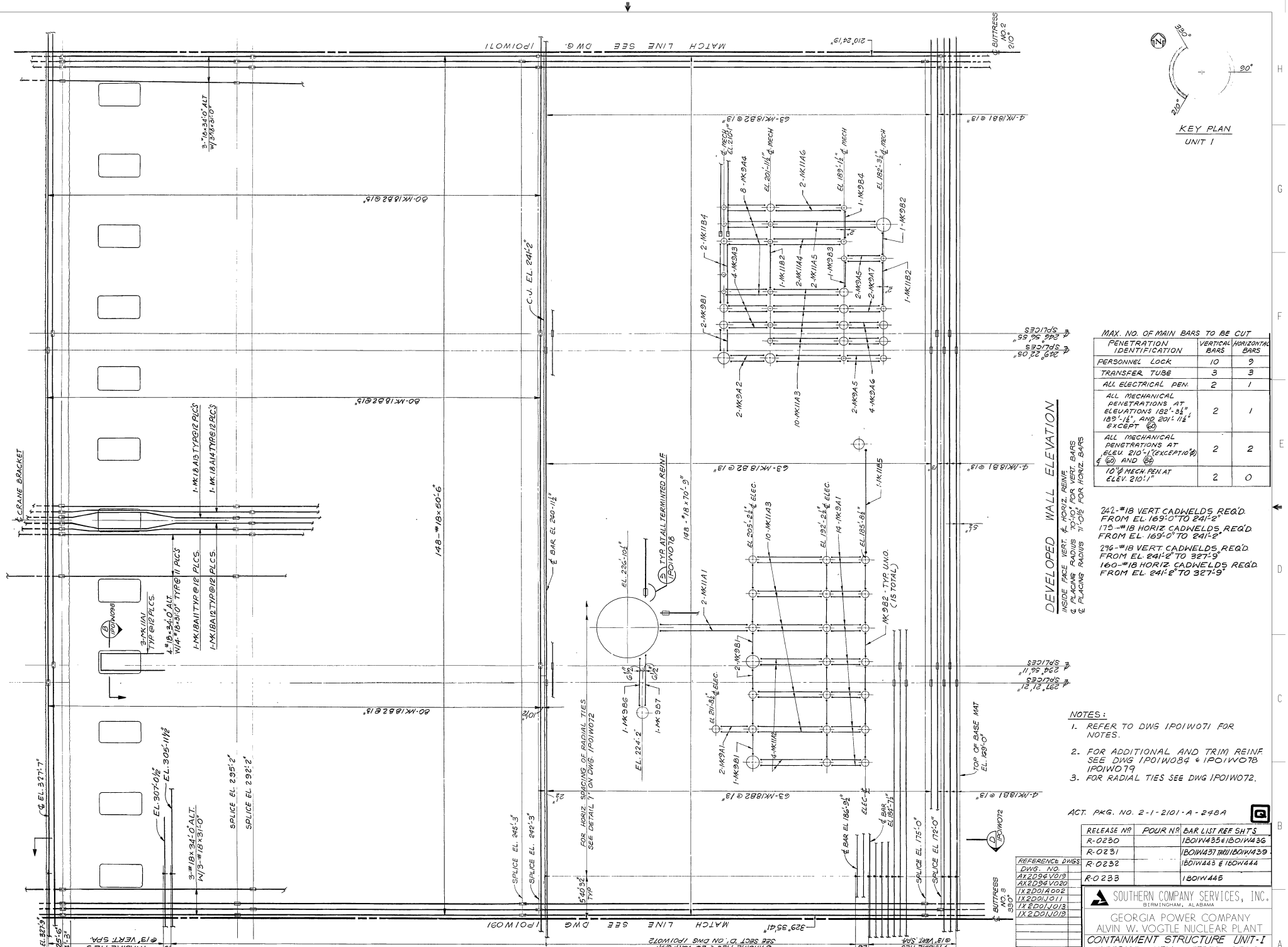
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 6.0 | REVISED TO CLOSE C-FCRB-1923. | 11-6-07 | DFV | LRP | WLN |
| | C-FCRB-2096, C-FCRB-2143, C-FCRB-4169 | | | | |
| | C-FCRB-4467, C-FCRB-4531, C-FCRB-4825 | | | | |
| | C-FCRB-4867, C-FCRB-8065, C-FCRB-8712 | | | | |

SCALE: 3/16"=1'-0" | JOB NO. 10604 | SIZE E 34x44

DRAWING NO. 1P01W077 | VER. 6.0







KEY PLAN
UNIT 1

MAX. NO. OF MAIN BARS TO BE CUT

| PENETRATION IDENTIFICATION | VERTICAL BARS | HORIZONTAL BARS |
|---|---|---|
| PERSONNEL LOCK | 10 | 9 |
| TRANSFER TUBE | 3 | 3 |
| ALL ELECTRICAL PEN. | 2 | 1 |
| ALL MECHANICAL PENETRATIONS AT ELEVATIONS 182'-3½", 189'-1½", AND 201'-11½" EXCEPT (20) | 2 | 1 |
| ALL MECHANICAL PENETRATIONS AT ELEV. 210'-1" (EXCEPTIONS (60) AND (84)) | 2 | 2 |
| 10" ℄ MECH. PEN AT ELEV. 210'-1" | 2 | 0 |

242-#18 VERT. CADWELDS REQ'D. FROM EL. 169'-0" TO 241'-2"
175-#18 HORIZ. CADWELDS REQ'D. FROM EL. 169'-0" TO 241'-2"
296-#18 VERT. CADWELDS REQ'D. FROM EL. 241'-2" TO 327'-9"
160-#18 HORIZ. CADWELDS REQ'D. FROM EL. 241'-2" TO 327'-9"

NOTES:
1. REFER TO DWG 1P01W071 FOR NOTES.
2. FOR ADDITIONAL AND TRIM REINF. SEE DWG 1P01W084 & 1P01W078 1P01W079
3. FOR RADIAL TIES SEE DWG 1P01W072.

ACT. PKG. NO. 2-1-2101-A-248A

| RELEASE NO. | POUR NO. | BAR LIST REF SHTS |
|---|---|---|
| R-0230 | | 1B01W435 & 1B01W436 |
| R-0231 | | 1B01W437 THRU 1B01W439 |
| R-0232 | | 1B01W443 & 1B0W444 |
| R-0233 | | 1B01W445 |

REFERENCE DWGS

| DWG. NO. |
|---|
| AX2D94V019 |
| AX2D94V020 |
| IX2D01A002 |
| IX2D01J011 |
| IX2D01J013 |
| IX2D01J019 |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
CONTAINMENT STRUCTURE UNIT-1
EXTERIOR SHELL-BUTTRESS 2 TO 3
INSIDE FACE-VERT. & HORIZ. REINF.

SCALE: 3/16" = 1'-0" | DRAWING NO. 1P01W081 | VER. 5.0
JOB NO. 10604 | SIZE E 34X44

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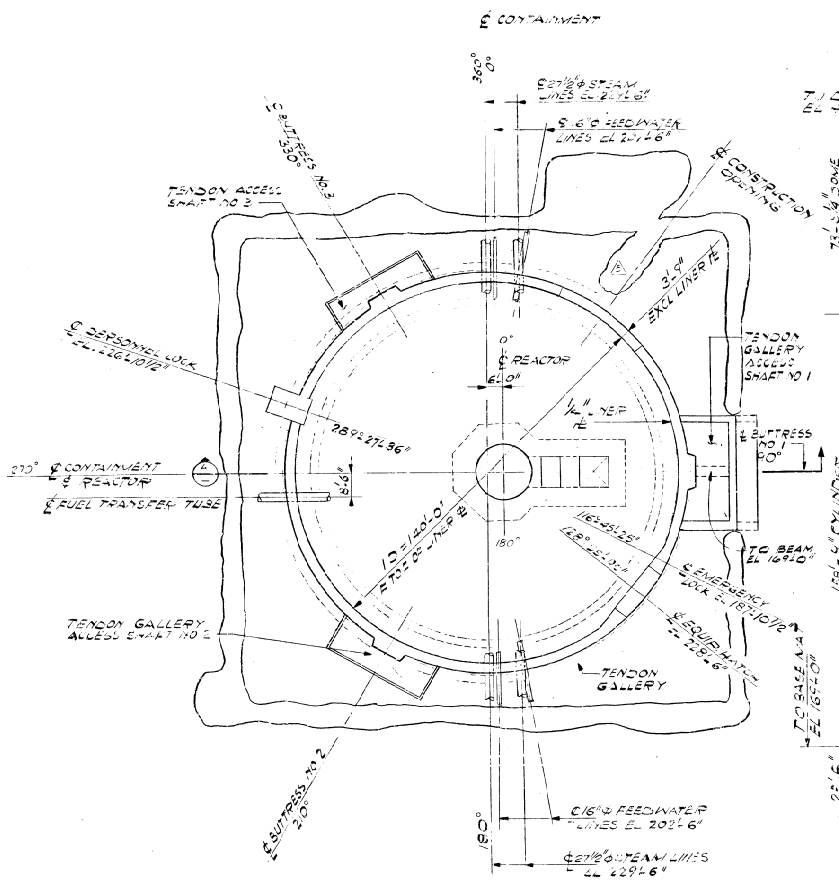
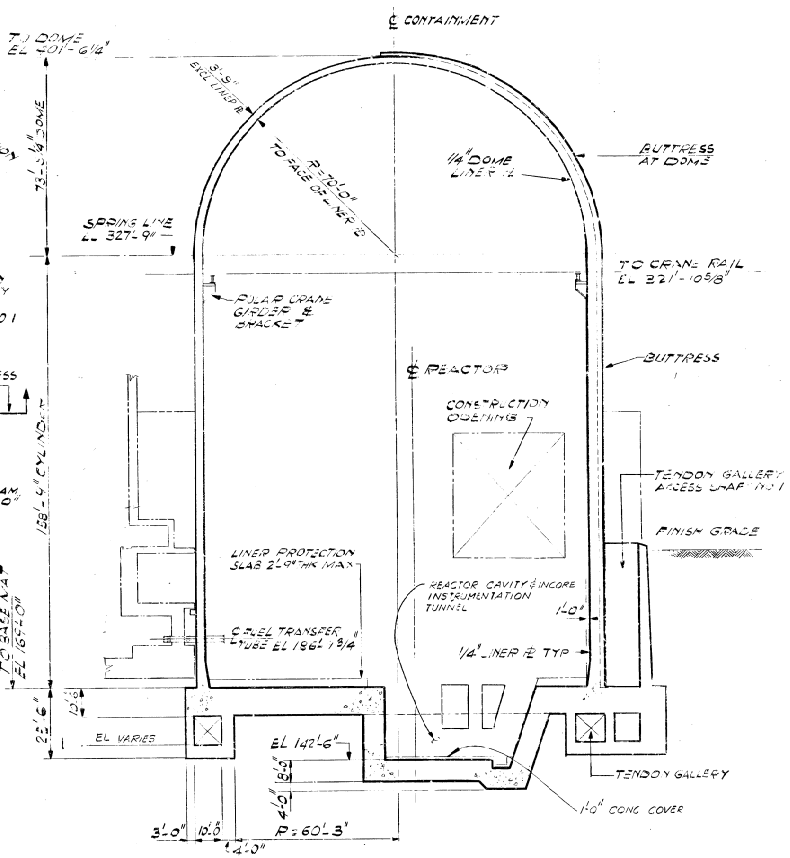
| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 5.0 | REVISED TO CLOSE C-FCRB-4170, C-FCRB-4176, C-FCRB-4468, C-FCRB-4503, C-FCRB-4593, C-FCRB-4824 | 11-6-07 | DFV | LRP | WLN |
| | NO CHANGES REQUIRED | | | | |

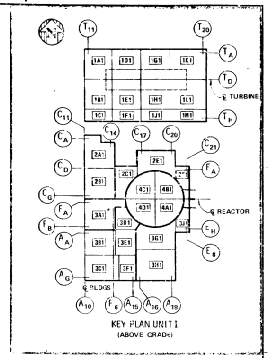



ONLY MAJOR PENETRATIONS SHOWN

PLAN




SECTION




**REFERENCE DRAWINGS:**

| | |
|---|---|
| 1X2001M001 | DRAWING INDEX |
| 1X2001A002 | CONCRETE FORMING |
| 1X2001K001 | PRESTRESSING REQUIREMENTS KEY PLAN |
| 1X2001U001 | LINER PLATE KEY PLAN |
| 1X2001K003 | WALL AND DOME INSERTS |
| 1X2001F002 | POLAR CRANE GIRDER |

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Nov 15, 1977

**BECHTEL**
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
UNIT 1 CONTAINMENT
CONCRETE FORMING
GENERAL ARRANGEMENT

| | |
|---|---|
| SCALE: 1/16"=1'-0" | DRAWING NO. 1X2001A001 |
| JOB NO. 9010 | REV. 5 |

REVISIONS

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | E&L | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | ISSUED FOR CONSTRUCTION | 11/15/77 | TD | | | | | | | |
| 4 | R/Y TITLE, DELETE REVNS, ADD NOTES | 9/28/77 | FM | TD | | | | | | |
| 3 | GENERAL REVD. INSERTS, EL. DIM | 8/8/77 | GL | | | | | | | |
| 2 | INCORP DCN 3 | 4/7/77 | NC | | | | | | | |
| 1 | REV PLAN & SECT & REMOVE HOLD | 3/11/77 | TAJ | | | | | | | |

ENG. MGR. DATE

SIZE E 34 x 44

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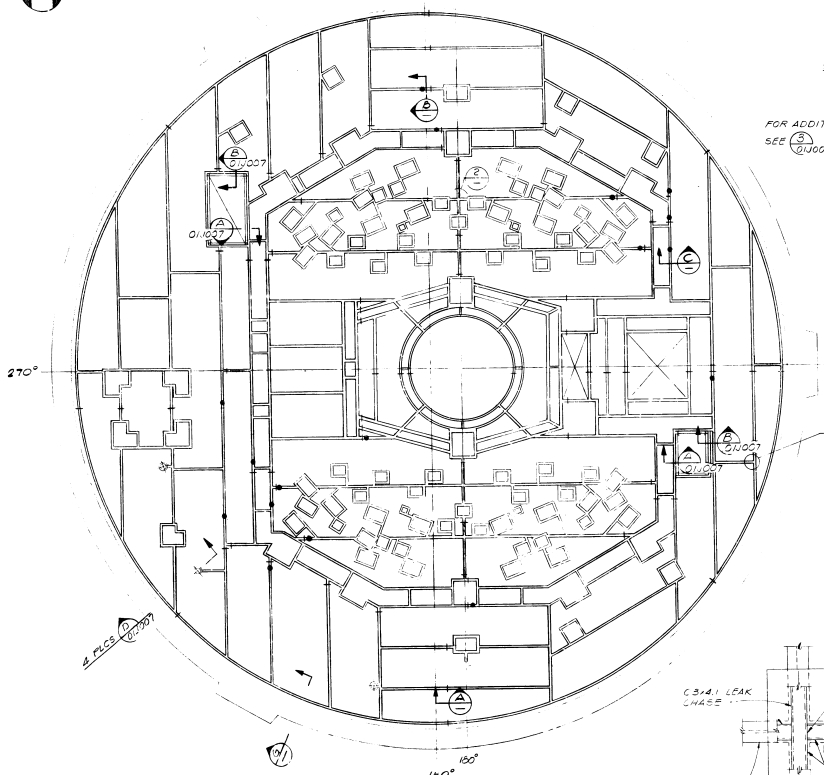
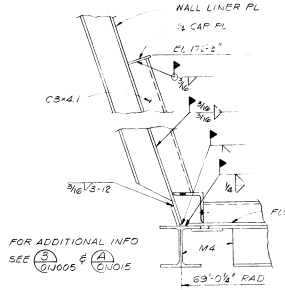
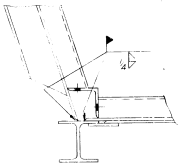
1X2D01J004




PLAN
1/8"=1'-0"

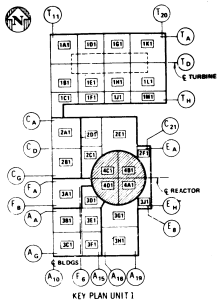



OPTION I

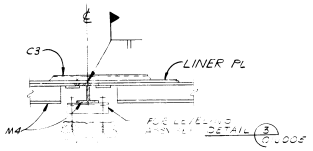



OPTION II

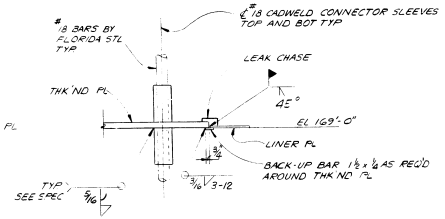
SECTION 3 01J005
3"=1'-0"




KEY PLAN UNIT 1
(BELOW GRADE)

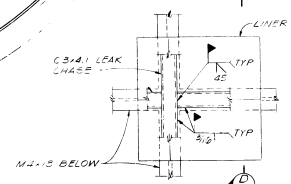



SECTION D
1½"=1'-0"




TYP AT THK'ND PLATE SPLICE TO LINER PLATE

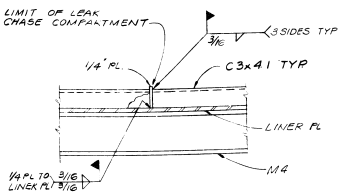
SECTION E 01J006 01J007
1½"=1'-0"



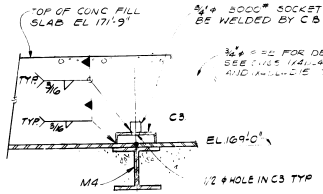

DETAIL 2 01J003
1½"=1'-0"

NOTES:

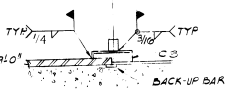
1. FOR GENERAL NOTES AND REFERENCE DWGS. SEE DWG. 1X2D01J001.
2. ALL LEAK CHASE CHANNELS AND ANGLES SHALL BE CUT IN THE FIELD TO THE ACTUAL INSTALLED LINER PLATE JOINT DIMENSIONS.
3. LEAK CHASES SHALL BE INSTALLED AFTER LINER PLATE WELDS HAVE BEEN VACUUM BOX TESTED AND APPROVED BY THE ENGINEER.
4. - | - INDICATES LIMITS OF LEAK CHASE COMPARTMENTS. ALL LEAK CHASES FORMING A COMPARTMENT SHALL BE PROPERLY DETAILED WHERE TWO OR MORE LEAK CHASES ARE IN THE SAME SYSTEM. A HOLE SHALL BE FIELD CUT IN ONE TO ALLOW AIR PASSAGE AFTER EXACT LOCATION OF HOLE HAS BEEN DETERMINED.
5. ⊢●⊣ INDICATES LEAK CHASE **SOCKET PIPE CAP LOCATION**
6. LEAK CHASE PRESSURE TEST RANGE SHALL BE FROM ZERO TO 100 P.S.I.
7. CONTRACTOR MAY INCREASE THE NUMBER OF LEAK CHASE COMPARTMENTS WITH APPROVAL OF THE ENGINEER.
8. AN AS-BUILT LAYOUT SHALL BE MADE SHOWING THE ACTUAL LOCATIONS OF THE LEAK CHASE SYSTEM AND SUBMITTED TO THE ENGINEER FOR INFORMATION ONLY. (7)
9. ALL VACUUM BOX TESTING OF JOINTS SHALL BE COMPLETED AND CERTIFIED PRIOR TO PLACEMENT OF LEAK CHASE SYSTEM.
10. ALL LEAK CHASE TESTING SHALL BE COMPLETED PRIOR TO PLACING OF CONCRETE.
11. LOCATIONS AND DIMENSIONS OF LEAK CHASE SYSTEM AS SHOWN ARE PRELIMINARY. FINAL LOCATIONS & DETAILS TO BE SUBMITTED LATER BY THE CONTRACTOR SUBJECT TO APPROVAL OF THE ENGINEER.
12. FOR R'S LAYOUT, SEE DWG. 1X2D01J003.
13. LEAK CHASE SYSTEM OF REACTOR CAVITY IS SHOWN ON DWG. 1X2D01J005.



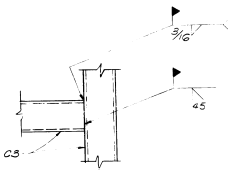

SECTION A
NTS



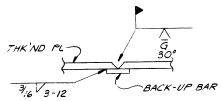

SECTION B
NTS




SECTION C
NTS



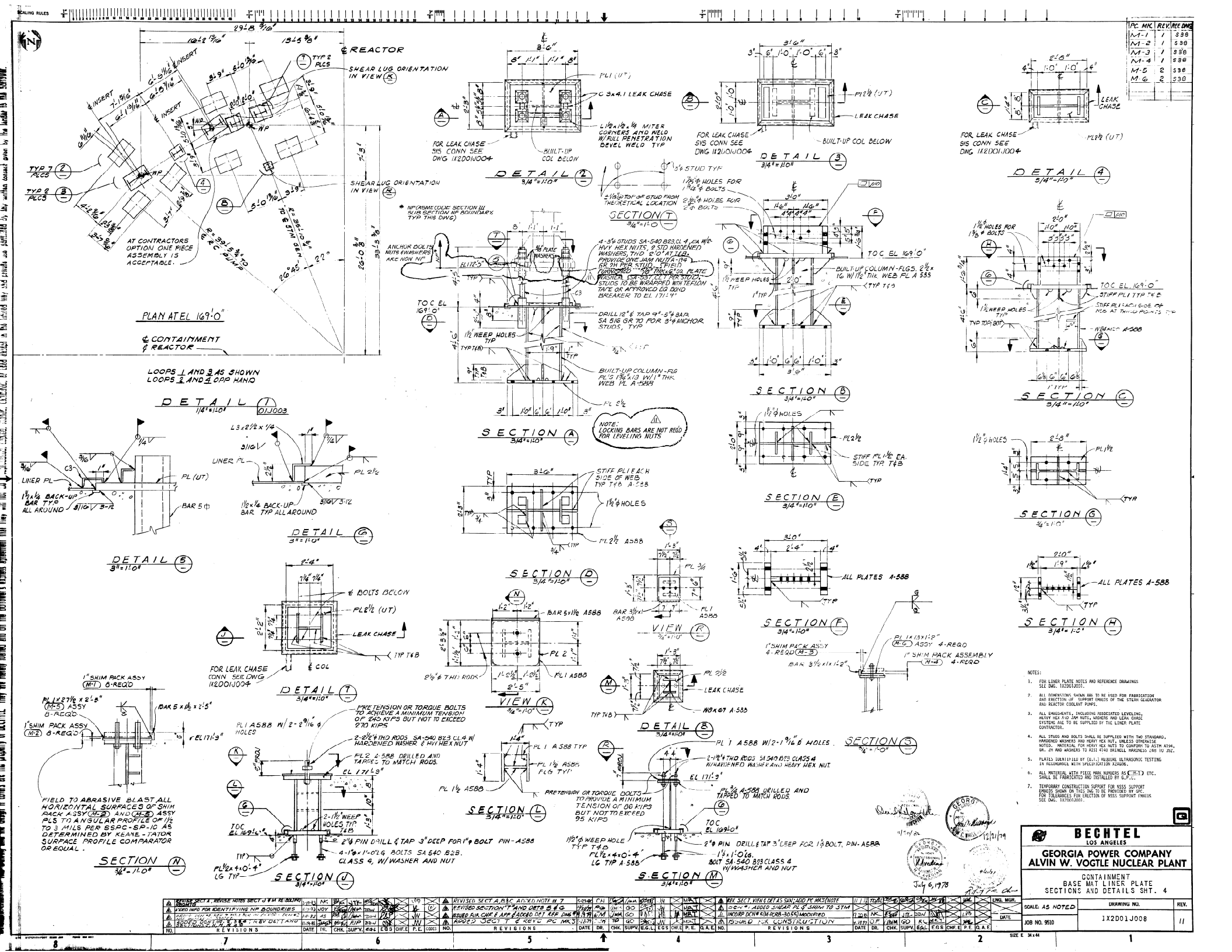

DETAIL 1
NTS




SECTION F 01J003
NTS
TYP AT THK'ND PLATE SPLICES

Nov. 29, 1977

| REVISIONS | | |




| PC. MK. | REV. | REF DWG |
|---|---|---|
| M-1 | 1 | 538 |
| M-2 | 1 | 538 |
| M-3 | 1 | 538 |
| M-4 | 1 | 538 |
| M-5 | 2 | 538 |
| M-6 | 2 | 538 |

NOTES:

1. FOR LINER PLATE NOTES AND REFERENCE DRAWINGS SEE DWG. 1X2D01J001.
2. ALL DIMENTIONS SHOWN ARE TO BE USED FOR FABRICATION AND ERECTION OF SUPPORT CHAIRS OF THE STEAM GENERATOR AND REACTOR COOLANT PUMPS.
3. ALL EMBEDMENTS, INCLUDING ASSOCIATED LEVELING, HEAVY HEX AND JAM NUTS, WASHERS AND LEAK CHASE SYSTEMS ARE TO BE SUPPLIED BY THE LINER PLATE CONTRACTOR.
4. ALL STUDS AND BOLTS SHALL BE SUPPLIED WITH TWO STANDARD, HARDENED WASHERS AND HEAVY HEX NUT, UNLESS OTHERWISE NOTED. MATERIAL FOR HEAVY HEX NUTS TO CONFORM TO ASTM A194, GR. 2H AND WASHERS TO ASTI 4140 DRILNELL HARDNED CAR TO 352.
5. PLATES FURNISHED BY (U.L.) MILBURE ULTRASONIC TESTING IN ACCORDANCE WITH SPECIFICATION X2A00B.
6. ALL MATERIAL WITH PIECE MARK NUMBERS AS (M-1) ETC. SHALL BE FABRICATED AND INSTALLED BY G.P.C.
7. TEMPORARY CONSTRUCTION SUPPORT FOR NSSS SUPPORT EMBEDS SHOWN ON THIS DWG TO BE PROVIDED BY GPC FOR TOLERANCES FOR ERECTION OF NSSS SUPPORT EMBEDS SEE DWG. 1X2D01J001.

July 6, 1978

| REVISIONS | NO. | DATE | DR. | CHK. | SUPV. | E.G.L. | E.G.S. | CH.E. | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| REVISED SECT A, B&C ADDED NOTE # 7 | 5 | 5-24-79 | CAJ | TWA | JMA | GO | JN | MET | | |
| REVISED SECTION "T" AND DETS 5 & 6 | 4 | 3-23-79 | RD | JM | GO | JN | MET | | | |
| ISSUED FOR CHE & APP. ADDED DET REF DWG # # # # | 3 | 1-19-79 | EJM | JMA | GO | JN | MET | | | |
| ADDED SECT T & REVD PC MK'S | 2 | 11-27-78 | TW | JMA | GO | JN | MET | | | |
| REVISED SECT A, B, C, E, F, G, H, J, K, L, M, N & SECTION J | 1 | 10-9-78 | Gough | MG | EJM | DDM | EGS | CHE | PE | |
| REVISE SECT A, REVISE NOTES SECT J & M RE: BOLTING | 0 | 9-11-78 | NC | WA | JTK | DDM | EGS | CHE | PE | |

| REVISIONS | NO. | DATE | DR. | CHK. | SUPV. | E.G.L. | E.G.S. | CH.E. | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| REV. SECT. VIEW & DET AS SUN; ADD PC MK'S (NOTE 1) | 11 | 11/6/79 | JMM | GO | JN | MET | | | | |
| DOOR OPNG ADDED SHEAR PL'S (MIN TO STM) | 10 | 9/28/79 | NC | ESW | JMM | GO | JN | MET | | |
| INCORP DCN # 5M-R08-30-E6 MODIFIED | 9 | 7/20/79 | NC | ESW | JJL | DDM | JN | MET | | |
| ISSUED FOR CONSTRUCTION | 8 | 6/21/79 | JMM | GO | JN | MET | | | | |

| ENG. MGR. | DATE |
|---|---|
| JJS | |

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT
BASE MAT LINER PLATE
SECTIONS AND DETAILS SHT. 4

| SCALE: AS NOTED | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 9510 | 1X2D01J008 | 11 |

SIZE E 34 x 44

SCALING RULES

SPRING LINE EL 327'-9"

SEE DET (7) 01J026

CUT FROM WT 10.5 x 46.5

PL ½

L 3½ x 2½ x ¼

C8 x 11.5

C5 x 6.7

℄ SEAM EL 317'-9"

10'-0"

FIVE PLACE AT 9'-6" AND 2 PLACE AT 10'-0"

℄ SEAM EL 288'-2½"

C8 x 11.5

C5 x 6.7

℄ SEAM EL 277'-9"

10'-5¼"

TYP UNO

C8 x 11.5

C5 x 6.7

¼" LINER PL

L2 x 2 x ¼

ONE PLACE AT 6'-9" AND 9 PLACES AT 10'-0"

(F.P. TYP.

℄ SEAM EL 179'-0"

C8 x 11.5

4'-3"

10'-0"

C5 x 6.7

4'-3"

L2 x 2 x ¼

C8 x 11.5

TOP OF BASE MAT EL 160'-0"

M4 x 13

PL ¼ LINER

1'-0"

(3) 01J005 SIM.

SECTION (A) 01J010 01J011 01J012
1" = 1'-0"

---

NO WELD

L3 x 2

¼" LINER PL

℄ SEAM

L3½ x 2½ x ¼

DETAIL (1)
1½" = 1'-0"

---

℄ VERT SEAM

L3 x 2 VERT STIFF AT 1'-9 1/2" ±OC

BAR 5 x ½

¼" LINER PL

⅝ HORIZ STIFF

SECTION (B) 01J010 01J011 01J012
1" = 1'-0"

---

℄ VERT SEAM

L3x2 VERT STIFF

¼" LINER PL

L3½ x 2½

CA HORIZ STIFF

BAR 5 x ½

NOTE FOR WELD DETS SEE SECTION (C)

SECTION (N) 01J010 01J011 01J012
1" = 1'-0"

---

LINER PL

1" RETURN EA CORNER

EL 171'-9"

PL ½ x 12 x 1'-0" W/ 2" RAD AT CORNER TYP

TO BASE MAT

SECTION (D) 01J010 01J011 01J012
3/4" = 1'-0"

---

TYP 4 SIDES

7/16" ⌀ HOLES

BAR 3 x ⅛ x 0'-3" A36 ON ℄ ½ PL EA WAY

2½" ⌀ BAR ON ℄ ¼ PL EA WAY

℄ EL 180'-0"

SEE DWGS 1X2D01C001 & 1X2D01C002

PL ½ x 14½" x 1'-6½" A516 W/ 2" RAD AT CORNERS

3'-8¾"

SECTION (E) 01J011 01J012

---

℄ EL 178'-9"

4'-3 15/16"

TO BASE MAT

SEE SECT (E) FOR NOTES & DETAILS NOT SHOWN

SECTION (P) 01J010
¾" = 1'-0"

---

P 5000 UNISTRUT OR EQUAL SUPPLIED BY GPC & INSTALLED BY CBI TYP

¼ 2-6 2-6 ℄ TYP

⅛

NOTE
PUNCH MARK LINER PLATE IN SHOP TO FACILITATE FIELD PLACEMENT OF UNISTRUT BEHIND ANGLE STIFFENER

SECTION (F) 01J010 01J011 01J012
1" = 1'-0"

---

3" VARIES

¼" LINER PL

THK'D PL

UNISTRUT

SECTION (G) 01J010 01J011 01J012
NTS

---

PL ¼ LINER

℄ SEAM

UNISTRUT

SECTION (H) 01J010 01J011 01J012
NTS

---

SUPPORT ARM

℄ SUP BOLT

1½" MIN

INTERNAL ELEMENT

FOR CLR REQMTS SEE WALL LINER PL NOTE 10 ON DWG 1X2D01J001

½" MIN

FIELD TRIM AS REQD

LINER PL

UNISTRUT

ELEVATION DETAIL (E)
3" = 1'-0"

---

ADD L3 x 2 x ⅛ x 2'-0" AT UNISTRUTS

STIFF ANGLE TYP

⅜

UNISTRUT

⅜ 2-6 2-6

NOTE
PUNCH MARK LINER PLATE IN SHOP TO FACILITATE FIELD PLACEMENT OF UNISTRUT BEHIND ANGLE STIFFENER.

SECTION (K) 01J005
3" = 1'-0"

---

NOTES:

1. FOR GENERAL NOTES AND REFERENCE DWGS SEE DWG 1X2D01J001.

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. F | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| A | DELETED DETS 2 THRU 9 & REV SECT E | 11-30 | D.P. | VAA | GO | | | | |
| A | REVISED SECTION "B" | 7/1/74 | AG | VAA | GO | | | | |
| A | ISSUED FOR CHF E APPROVAL | 9/11/71 | EM | VAA | GO | | | | |
| A | REVS SECT A & B AS NOTED | 9/27/71 | TD | DEH | | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. F | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| A | ADDED INFORMATION PREVIOUSLY SHOWN ON ADI'S | 01-74 | MJ | | | | | | |
| A | ADDED DET E IN RESPONSE TO AFA # 566 | 1-21-74 | NK | | | | | | |
| A | INCORP DCN 3 (MODIFIED) & CORR. DCN 3 INCAPABLE | 2-24-74 | NK | | | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. F | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| A | IN. CORP DCN #9 | 7/1/74 | | | | | | | |
| A | REVISED DETS A, E & K | 7-15-74 | VAA | | GO | | | | |
| A | ADDED DET F FOR UNIST. WALL LINER & SECT | 7-22-74 | VAA | | GO | | | | |
| A | INCORP DCN 10 & 11 | 8-28-74 | VAA | | GO | | | | |

ENG. MGR. DATE

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT
WALL LINER PLATE
SECTIONS AND DETAILS SHT. 1

SCALE: NOTED

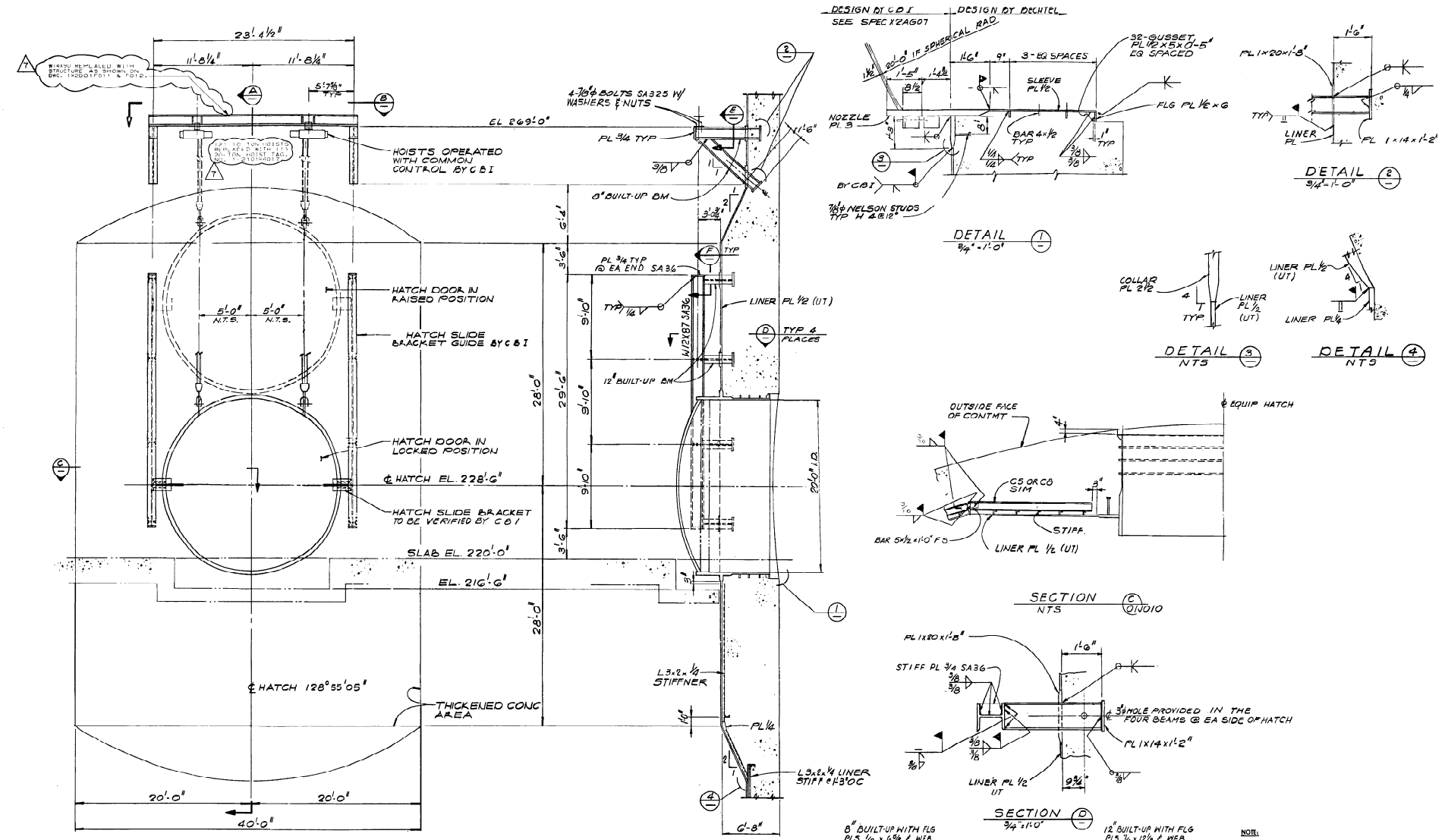
DRAWING NO. 1X2D01J015 REV. 13

JOB NO. 9510

SIZE E 34 x 44

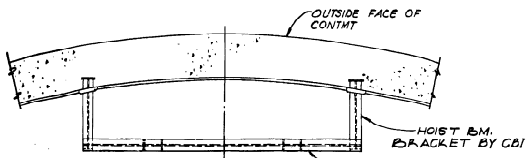
I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS ON THE DATE STATED BELOW AND THAT IT IS

THIS DRAWING AND THE DESIGN IT CONVEYS ARE THE PROPERTY OF SOUTHERN COMPANY SERVICES, INC. THEY ARE MERELY LOANED AND ON THE BORROWER'S EXPRESS AGREEMENT THAT THEY WILL NOT BE REPRODUCED, COPIED, LOANED, EXHIBITED, OR USED EXCEPT IN THE LIMITED WAY AND PRIVATE USE PERMITTED BY ANY WRITTEN CONSENT GIVEN BY THE LENDER TO THE BORROWER.




INTERIOR ELEVATION

EQUIPMENT HATCH ELEVATION
¼" = 1'-0"




SECTION B
¼" = 1'-0"

SECTION A
¼" = 1'-0"

DESIGN BY C B & I
SEE SPEC X2AG07

DESIGN BY BECHTEL




DETAIL 1
¾" = 1'-0"




DETAIL 2
¾" = 1'-0"




DETAIL 3
NTS

DETAIL 4
NTS




SECTION C
NTS
O/N010

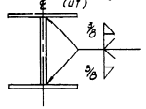



SECTION D
¾" = 1'-0"

8" BUILT-UP WITH FLG PLS ½ x 6⅞ ℄ WEB PL 1/2E x 7 SA 516 GRADE 70 (UT)

SECTION E
1½" = 1'-0"

12" BUILT-UP WITH FLG PLS ⅞ x 12⅛ ℄ WEB PL ½ X 12⅛ SA 516 GRADE 70 (UT)




SECTION F
1½" = 1'-0"

NOTE:
1. FOR GENERAL NOTES AND REFRENCE DRAWINGS SEE DWG. 1X2D01J001.
2. PLATES IDENTIFIED BY (U.T.) REQUIRE ULTRASONIC TESTING IN ACCORDANCE WITH SPECIFICATION X1AG06 - REV 4.
3. ALL DESIGN DETAILS OF EQUIPMENT HATCH, INCLUDING DIMENSIONS, WELD'S AND CONNECTIONS ARE TO BE DETERMINED BY VENDOR IN ACCORDANCE WITH SPECIFICATION X2AG07.

| NO. | REVISIONS | DATE | BY | CHK | SUPV. | ENG. | QUAL. | P.E. | D.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 3 | REVISED PER ADN 06148 | 11-17-95 | MLW | | | | | | |
| 2 | ISSUED FOR CONSTRUCTION | 3/16/95 | DR | CHK | PE | DM | DDPM | DPM | DRN |
| 1 | REVISIONS | DATE | DR | CHK | PE | DM | DDPM | DPM | DRN |

ENG. MGR. DATE

SEE MICROFILM FOR SIGNATURES

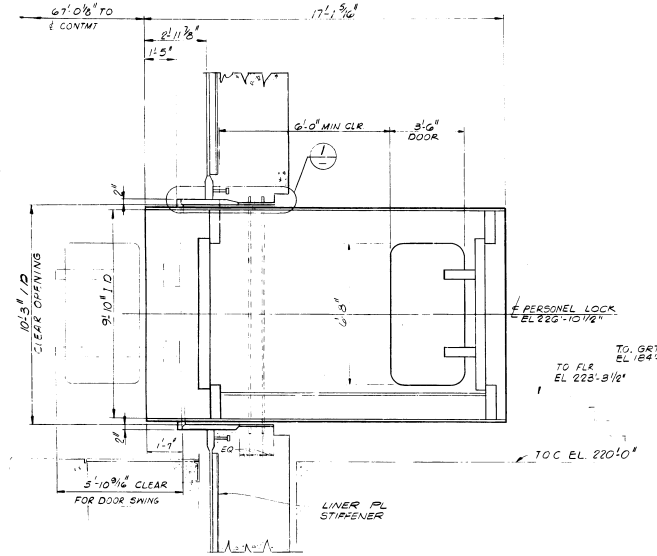
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

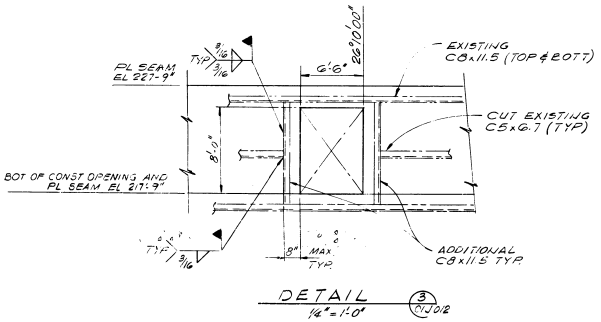
CONTAINMENT
WALL LINER PLATE
HATCH AND LOCK DETAILS SH. 1

| SCALE: AS NOTED | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 10604 | 1X2D01J017 | 7 |

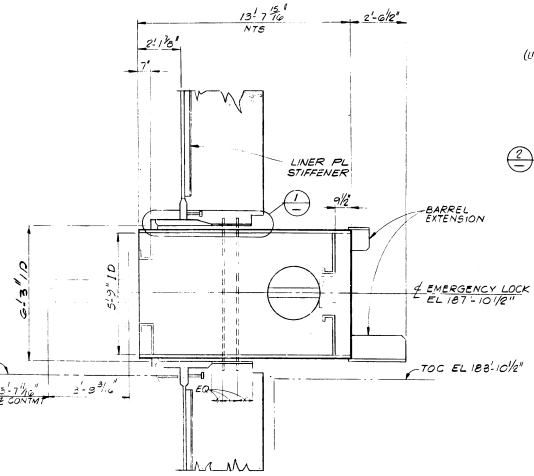
SIZE D 34x44



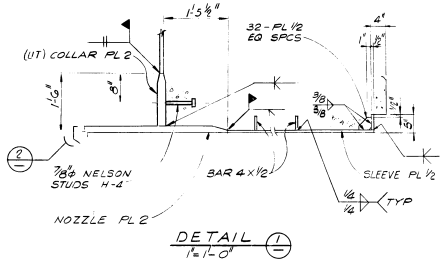

PERSONNEL LOCK - SECTION (A) 01J011
1/2" = 1'-0"



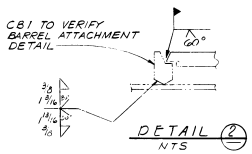

DETAIL (3) 01J012
1/4" = 1'-0"




EMERGENCY LOCK SECTION (B) 01J010
1/2" = 1'-0"




DETAIL (1)
1" = 1'-0"




DETAIL (2)
NTS

NOTES:

1. FOR GENERAL NOTES AND REFERENCE DWGS, SEE DWG. 1X2D01J001.

G/F-R/E

4-4-78

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT
WALL LINER PLATE
HATCH AND LOCK DETAILS SHT. 2

SCALE: AS NOTED
DRAWING NO. 1X2D01J018
REV. 6
JOB NO. 9510
SIZE E 36 x 44

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| A | ISSUE FOR CHIEF ENGINEERS APPROVAL | 1-3-78 | KJ | HK | TK | TK | | | |
| B | REVISED SECTION TO REFLECT LATEST VENDOR INFORMATION | 2-2-78 | NK | HK | TK | DDW | | | |
| C | INCORP DCN # 5 | 3-23-78 | NK | HK | TK | DDW | | | |
| A | INCORP DCN #'S 2, 3 & 4 | 10-6-77 | KJ | HK | TK | TK | | | |
| B | REVISED AS SHOWN | 1-27-78 | FM | TK | GO | EGS | | | |
| C | DCN #1, FIELD WELDS & STUDS CALL OUT | 2-28-78 | TK | TK | GO | EGS | | | |
| D | ISSUED FOR CONSTRUCTION | 4-4-78 | TD | TD | GO | EGS | | | |

ENG. MGR.
DATE

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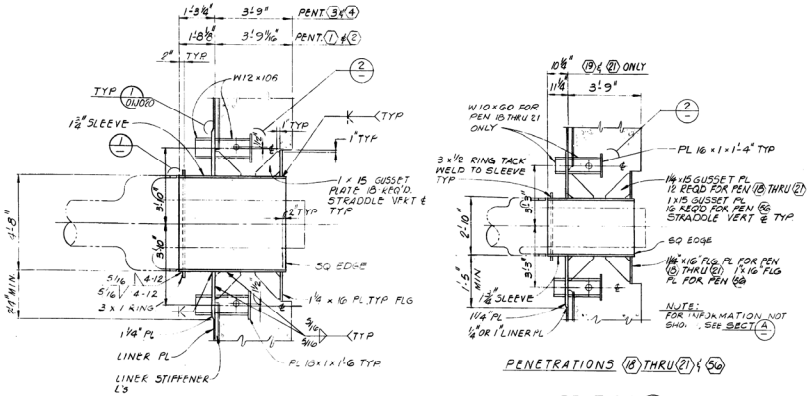
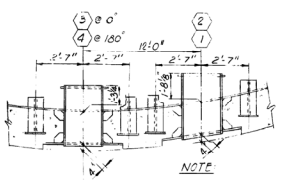
SCALING RULES




PENETRATIONS 1 THRU 4

SECTION A 1/2"=1'-0" D1J010 D1J012 D1J013



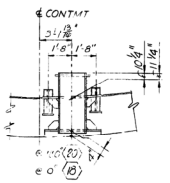

SECTION D NTS D1J010

SECTION E OPP HAND D1J012



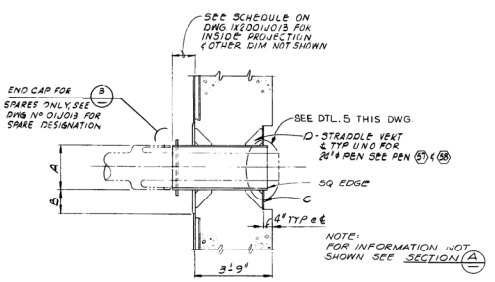

PENETRATIONS 18 THRU 21 & 54

SECTION B 1/2"=1'-0" D1J010 D1J012 D1J013




SECTION F NTS D1J010

SECTION G OPP HAND D1J011

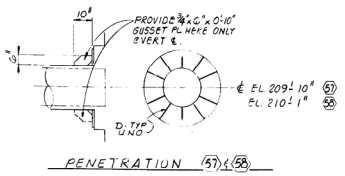



TYPICAL MECHANICAL PENETRATION

SECTION C 1/2"=1'-0" D1J010 D1J011 D1J013 J D1J013

TYP MECHANICAL PEN SCHEDULE

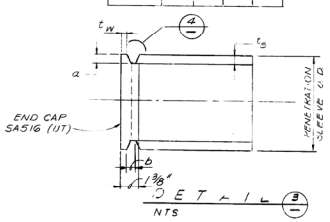
| A | B | C | D | NO. OF GUSSETS |
|---|---|---|---|---|
| 20 & 24 | 13 | 1 x 11 | ¾ x 10 | 10 |
| 20 | 11 | ¾ x 9 | ½ x 8 | 8 |
| 14 | 11 | ¾ x 9 | ¾ x 8 | 6 |
| 12 & 10 | 7 | ½ x 7 | ½ x 6 | 6 |
| 8 | 11 | ¾ x 6 | ½ x 8 | 8 |

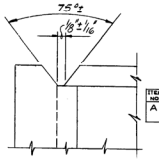



PENETRATION 57 & 58

SECTION H 1/2"=1'-0" D1J010 D1J011

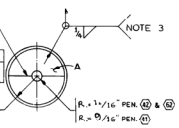
| SLEEVE SIZE | $t_s$ | a (±1/32) | b (±1/32) | $t_w$ (±1/32) |
|---|---|---|---|---|
| 10" ϕ | .365 | .365 | .375 | .5 |
| 12" ϕ | .375 | .375 | .375 | .5 |
| 18" ϕ | .500 | .500 | .50 | .625 |




| ITEM NO | QUANTITY | DESCRIPTION | MAT'L. | SUPPLIER | CODE CLASS |
|---|---|---|---|---|---|
| A | 2 | W" ϕ x 0.5" x O'-10" CUT PER DTL. 5 | SA-36 OR SA-516 GR. 70 | FIELD | MC |

NOTES: (DETAIL 5 ONLY)
1. SA-36 PLATE TO BE IMPACT TESTED PER NE-2300. MIN. TEST TEMP. 15.-15°F.
2. FIELD OPTION TO TACK WELD TO MAINTAIN ALIGNMENT.
3. WELDING ACROSS PLATE SEPARATIONS IS NOT REQUIRED, BUT IS PERMISSIBLE ON PEN. 40 & 41. DO NOT WELD ACROSS ℄ SEPARATIONS ON PEN. 41.

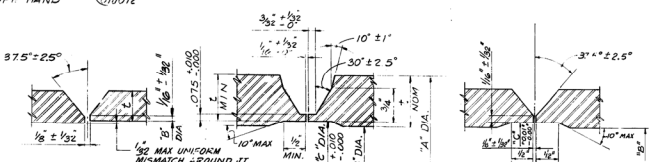



DETAIL 5 N.T.S.

TABLE OF COUNTERBORE "C" DIMENSIONS FOR PENETRATION PIPE SLEEVES TYPES 1B AND 2 WELD END PREPARATIONS

| PENETRATION NUMBER | PENETRATION PIPE SLEEVE OUTSIDE DIA | WALL THICKNESS SCHEDULE | "C" BORE DIMENSION | PIPE SLEEVE MATERIAL |
|---|---|---|---|---|
| 1 THRU 4 | 56" | .125" NOM | 53.336" | SA-516, GR 70 |
| 18 THRU 21/54 | 32" | .125" NOM | 31.336" | |
| 59 AND 60 | 26" | 1" NOM | 24.208" | |
| 58 | 28" | 1" NOM | 26.208" | |
| 85 AND 86 | 24" | 0.65"/XS | 23.064" | SA-333, GR 6 |
| 34 AND 35 | 20" | 0.57"/XS | 19.084" | |
| 7 THRU 9 & 29 40, 43 THRU 46 & 80 & 81 AND 87 | 18" | 0.5/XS | 17.084" | |
| 91 THRU 94 101 THRU 104 | 18" | 0.75"/SCH 60 | 16.646" | |
| 90, 91 AND 93 | 14" | 0.438/SCH 40 | 13.192" | |
| 57 | 24" | 1" NOM | 22.180" | SA-516, GR 70 |

NOTES:
1. FOR GENERAL NOTES AND REFERENCE DWGS, SEE DWG. 1X2D1J001.
2. MATERIAL SHOWN IN PHANTOM BY OTHERS.
3. COMPLETED ASSEMBLIES SHALL BE STRESS RELIEVED PER ASME SECTION III.
4. PLATES IDENTIFIED BY (U.T.) REQUIRE ULTRASONIC TESTING IN ACCORDANCE WITH SPECIFICATION X3AG06 REV 6
5. THE TORQUE ON THE FLANGE CONNECTIONS SHOWN ON SHEET 2 FOR PENETRATIONS 5,55 AND 80 SHOULD INITIALLY BE 150 FT-LBS, AND MAY BE INCREASED IF REQUIRED NOT TO EXCEED A MAXIMUM OF 465 FT. LBS.




TYPE1A
NOMINAL WALL (t) 3/16" THRU 3/8"

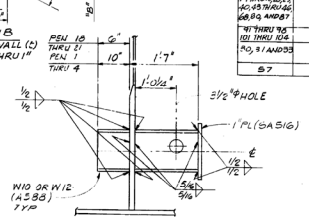
TYPE 2
NOMINAL WALL (t) OVER 1"

NOTE:
WELD-END PREPARATION TYPE 1B & TYPE 2 REQUIRE MACHINING TO COUNTERBORE DIMENSION "C" (SEE TABLE). 10° BORE TOLERANCE APPLICABLE WHILE PIECE IS IN THE SHOP JIG.

WHEN THE I.D. OF ACTUAL PIPE TO BE FABRICATED IS GREATER THAN REQUIRED FOR MINIMUM RING SEATING WHEN MACHINING TO DIMENSION "C" THE METAL REQUIRED FOR MACHINING MAY BE PROVIDED BY DEPOSITING WELD METAL ON THE I.D. OF THE PIPE IN THE AREA TO BE MACHINED.

ABBREVIATION

A NOMINAL PIPE O.D. INCHES.
B NOMINAL PIPE I.D. INCHES.
C MACHINED PIPE I.D., INCHES.
t NOMINAL WALL THK. INCHES




DETAIL 2 1"=1'-0"

WELD PREPARATIONS

DETAIL 1 N.T.S.

A.A.78

| NO. | REVISIONS | DATE | DR. | CHK. | DM |
|---|---|---|---|---|---|
| A | INCORP. ABN 87-VIE0244A001. | 6-3-87 | C.T.B. | [illegible] | [illegible] |
| A | INCORP DCN #1, REVISED TO ADD 1", 2 & 3", ADDED TYPE 1B. | 7-17-91 | NK. | [illegible] | [illegible] |
| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | Q.A.E. |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| A | INCORP. ABN 07573 | 2-3-95 | JH | PB | 2-3-95 | | | | |
| A | INCORP ABN 07268, ABN 03258 & ABN 00488 ADDED SHEET 2 OF 2 | 10-5-95 | MW/A | [illegible] | [illegible] | | | | |
| NO. | REVISIONS | DATE | DR. | CHK. | RE | DM | DM | PDM | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | Bal. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| A | ALL AEL NOM. SHELL ELEMENT ENDED | [illegible] | | | | | | | | |
| A | REVISED MECH. SEAL & DR. 2'-0" [illegible] | [illegible] | | | | | | | | |
| A | [illegible] | [illegible] | | | | | | | | |

ENG. MGR.
DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
WALL LINER PLATE
PENETRATIONS DETAILS SHT. 1

SCALE: AS SHOWN | DRAWING NO. 1X2001J019 | REV. 11
JOB NO. 9510 | SHEET 1 OF 2

SIZE E 34 x 44

8 7 6 5 4 3 2 1

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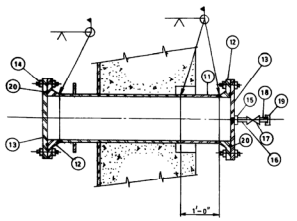


SECTION (J / 01J013)
N.T.S.

HOT! FOR INFORMATION NOT SHOWN SEE SECTION C

PENETRATIONS (5) (55) (90)

SEE MODE 5 AND MODE 6 OPTION (OUTAGE "FIXTURE")

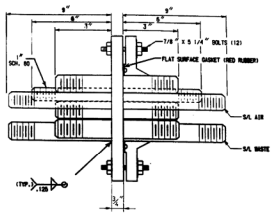
(11) EXTENSION SLEEVE FOR PENETRATION NRS 5, 55 & 90 – 10"ø PIPE, STD. WT, ASME SA-333 GR.6, CLASS 2.
(12) WELDING NECK FLANGE ("R" RING JOINTS) – 10" 150 LBS, ASME SA-105 TO MATCH STD. WT PIPE CLASS 2.
(13) BLIND FLANGE – 10" 150 LBS, WITH "R" RING JOINTS, ASME SA-105, CLASS 2
(14) NUTS & BOLTS – ⅞" ø ASME SA-194 GR. 2H AND ⅞" X 5¼" LG. ASME SA-193 GR. B7.
(15) HALF COUPLING ¾" S.W. 3000 LBS., ASME SA-105, CLASS 2.
(16) ¾"ø X 3" LONG NIPPLE (R.B.E.) – SCH. 80, ASME SA-106 GR. B, CLASS 2.
(17) ¾" GLOBE VALVE – MARK # 378 ASME SA-105, S.W., CLASS 2 (SEE VALVE TABLE FOR VALVE TAG NRS.)
(18) ¾" ø X 3" L. NIPPLE (T.O.E.) – SCH 80 ASME SA-106 GR. B, CLASS 2.
(19) ¾" THREADED CAP – 3000 LBS, ASME SA-105, CLASS 2.
(20) RING JOINT GASKET (R52) – STAINLESS STEEL.

~VALVE TABLE~

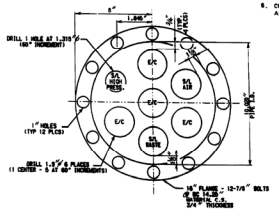
| PENETRATION NR | VALVE TAG NR |
|---|---|
| 5 | 1-1222-X4-017 |
| 55 | 1-1222-X4-018 |
| 90 | 1-1222-X4-019 |

NOTES: (OUTAGE "FIXTURE")

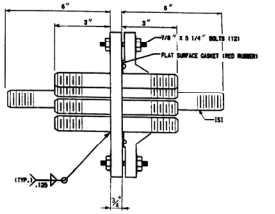
1. WELD ALL PIPE 1/8" FILLET BOTH SIDES OF 10"-150# FLANGE (MACHINE TO 3/4" FLAT FACE) (A-105).
2. TRUE FLANGE SEATING SURFACE AFTER WELDING (IF REQUIRED). MATE TO EXISTING FLANGE.
3. PROCESS IS GTAW ER70S-X OR SMAW E7018.
4. ALL PIPE IS TO BE THREADED ON BOTH ENDS. 1, 1½ TPI – FOR 1.0" ø, 1.5" ø & 2.0" ø / 8 TPI – FOR 3.0" ø.
5. PIPE SLEEVES USED FOR CABLING TO BE SEALED WITH P-90 TO WITHIN ½" OF EACH END.
6. 1.5" ø PIPE OPENINGS NOT USED TO BE CAPPED WITH 1.5 COUPLING WELDED TO 1.5" NO. DRIVE CAP. 3" ø & 2" SLEEVE PONTS TO BE CAPPED WITH STANDARD PIPE CAP WHEN NOT IN USE.
7. 1.5" ø, 2" ø & 3" ø PIPE SCH. 40. 1" ø SCH. 80 A-106.
8. COATING SHALL BE PER SPECIFICATION X1AJ07 AND AS FOLLOWS:
   (a) COAT BLIND FLANG (ITEM 13) WITH FN-3.
   (b) COAT WELD NECK FLANGE INSIDE CONTAINMENT (ITEM 12) WITH FN-36.
   (c) COAT ALL EXPOSED STEEL OUTSIDE THE CONTAINMENT WITH FD-3 OR FD-36.



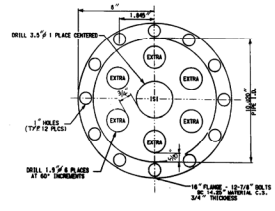

OUTAGE "FIXTURE" PROFILE
(1.5"ø & 1"ø PIPE)
(5) (55) OR (90)



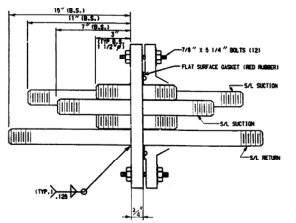

OUTAGE "FIXTURE" ELEVATION
(1.5"ø & 1"ø PIPE)
(5) (55) OR (90)



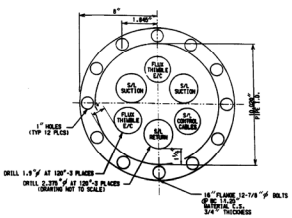

OUTAGE "FIXTURE" PROFILE
(3"ø & 1.5"ø PIPE)
(5) (55) OR (90)




OUTAGE "FIXTURE" ELEVATION
(3"ø & 1.5"ø PIPE)
(5) (55) OR (90)




OUTAGE "FIXTURE" PROFILE
(2"ø & 1.5"ø PIPE)
(5) (55) OR (90)




OUTAGE "FIXTURE" ELEVATION
(2"ø & 1.5"ø PIPE
(5) (55) OR (90)

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

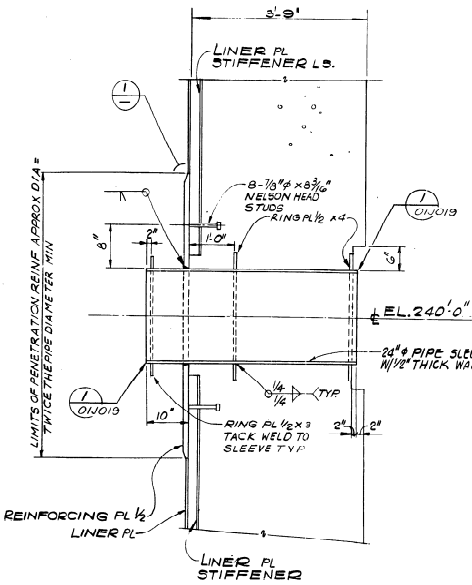
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

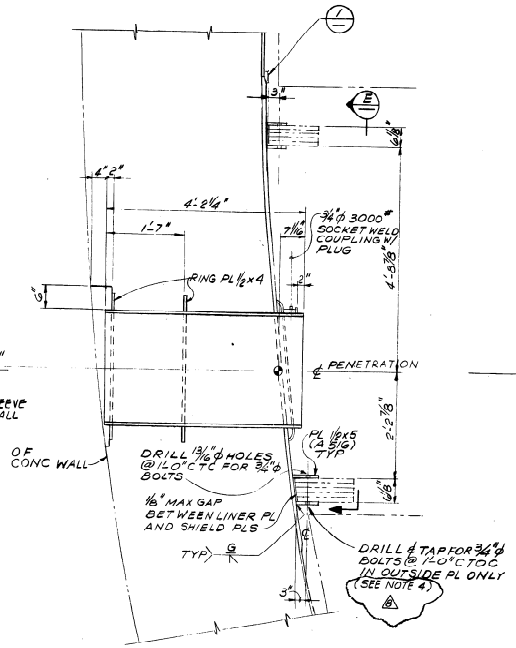
CONTAINMENT
WALL LINER PLATE
PENETRATIONS DETAILS SHT. 1

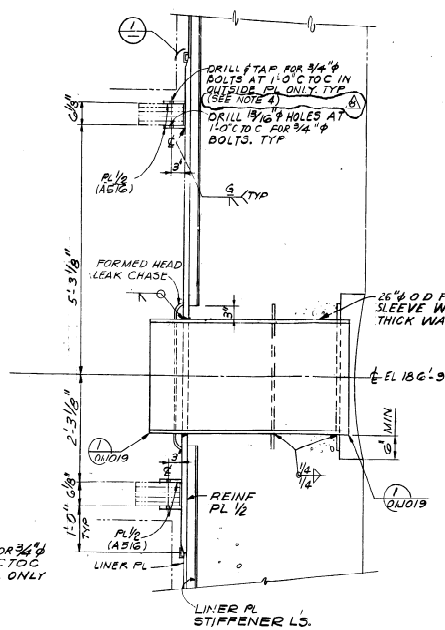
| ENGR. MGR. | | |
|---|---|---|
| SCALE: N.T.S. | DRAWING NO. | REV. |
| DATE | 1X2D01J019 SHEET 2 OF 2 | 11 |
| JOB NO. 9510 | | |

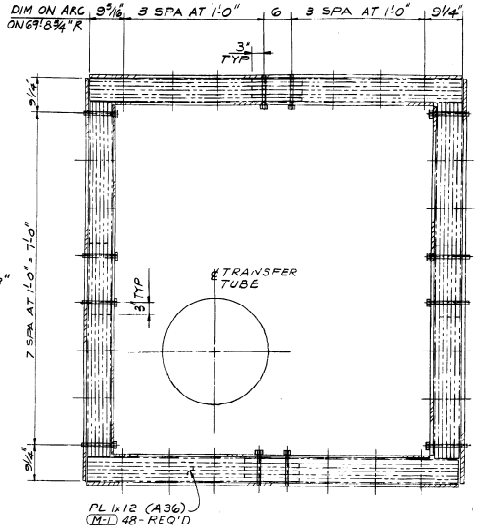
SIZE E 34X44

8 7 6 5 4 3 2 1



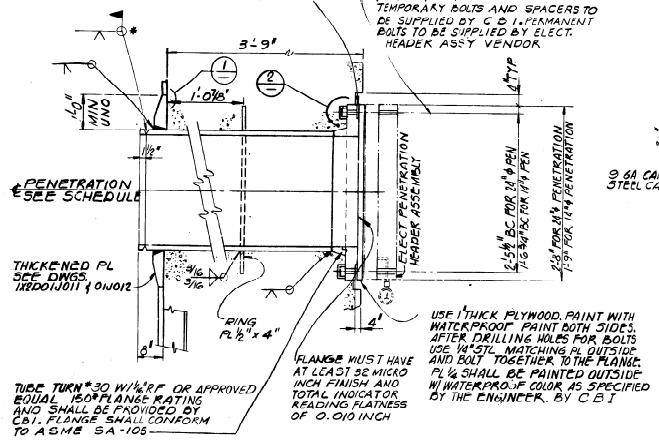




| PC. MK. | REV. | REF. DWG |
|---|---|---|
| M-1 | 4 | 5.80 |

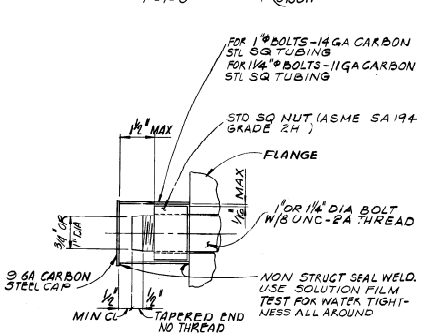
## SECTION (E)
1"=1'-0"

## PURGE LINE PENETRATION (83) & (84) SECTION (B)
1"=1'-0"
0IJ011
0IJ012
0IJ013

* APPLICABLE TO 84"ϕ ELEC PENETRATIONS ONLY (6 TOTAL) WORK TO BE PERFORMED PER REQUIREMENTS OF SPECIFICATION X2AG0G

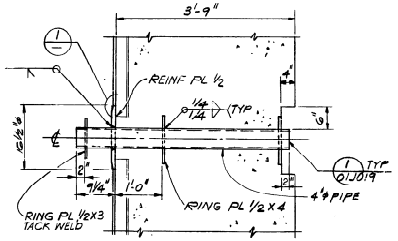



## FUEL TRANSFER TUBE SECTION (D)
1"=1'-0"
0IJ011

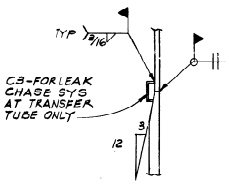



## DETAIL (2)
NTS

## (84) FUEL TRANSFER TUBE SECTION (C)
1"=1'-0"
0IJ011
0IJ013




## POST LOCA PURGE EXHAUST (100) SECTION (F)
0IJ018




## DETAIL (1)
0IJ019

NOTES:
1. FOR GENERAL NOTES AND REFERENCE DRAWING: SEE DWG. 1X2D01J001.
2. WORK THIS DRAWING WITH CONTAINMENT STRUCTURE PENETRATION SCHEDULE, DWG. 1X2D01J013.
3. COMPLETED ASSEMBLIES SHALL BE STRESS RELIEVED PER ASME SECTION III.
4. BOLTS FOR THE PLATES SURROUNDING THE FUEL TRANSFER TUBE MAY BE FABRICATED USING A193 GRADE B7 MATERIAL.

## TYP ELECTRICAL PENETRATION SECTION (A)
NTS
0IJ011
0IJ012
0IJ013

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | INCORP DCN # 3 (C-RCRB-7447) | 3/7/78 | CK | MAC | JTL | MAC | | | LDD | |
| 3 | INCORP DCN # 2 (E-RCRB-3864) (ADDED REF DWG NO.) | 10/6/77 | MAC | JTL | DSH | | | | | |
| 2 | INCORPORATED DCN # 1 | 12/5/77 | DSH | DSH | MAC | DSH | | | JN | |
| 1 | ISSUED FOR CONSTRUCTION | 1/25/77 | MAC | JTL | DSH | | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | REVISED AS SHOWN | 4/3/80 | CM | VMM | GO | MAT | JN | | MAT | |
| 3 | ADDED SECT. E EF, KEY TRANSFER TUBE, LK CHASE | 2/4/79 | CM | VAA | GO | MAT | JN | | MAT | |
| 2 | WELDS, EDGE PREPS & NOTES. | 10/4/78 | FM | TD | GO | EL | JN | | | |
| 1 | CLARIFY FIN SURFACE & ADDED NOTES | 10/4/78 | FM | TD | GO | EL | JN | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | CLARIFY SECTS. & RELEASE HOLD | 5/17/78 | DC | | | | | | | |
| 3 | RELEASE MECH. PEN UP TO EL 210'-0" | 4/4/78 | DC | | | | | | | |
| 2 | ADDED SECT. F, REVISED SECT. C, ADDED NOTE 4 | 2/6/78 | GAR | | | | | | | |
| 1 | REVISED PER CBI LTR BNY-2-2-78 AS NOTED. | 1/27/78 | GAR | | | | | | | |

ENG. MGR. | DATE

I HEREBY CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS, ON THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT PHOTOGRAPHED.

10/5/82

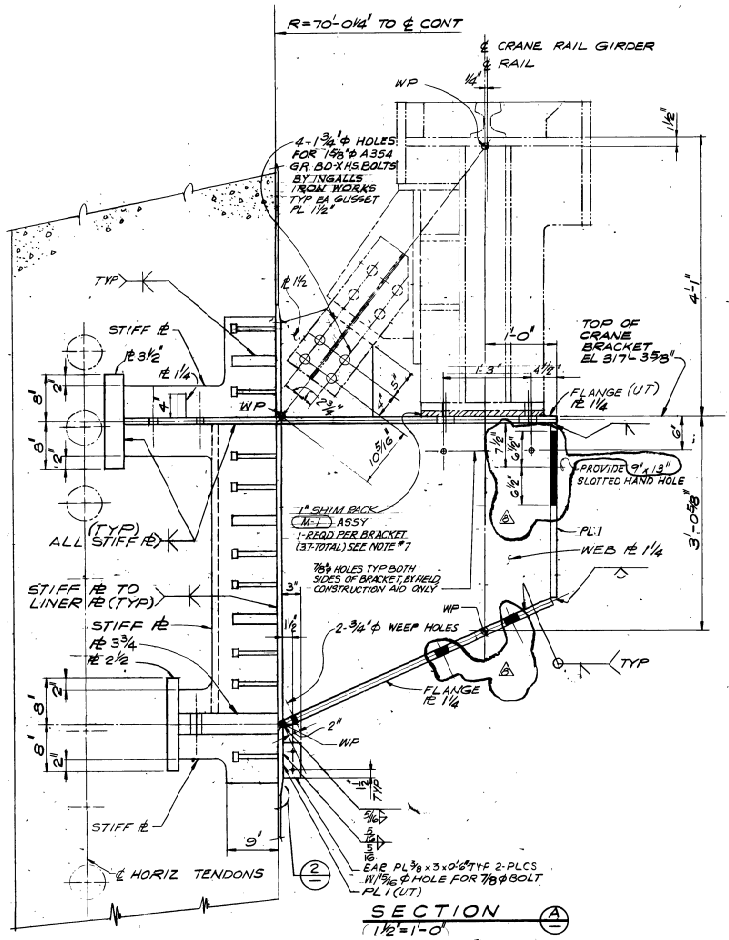
**BECHTEL**
LOS ANGELES

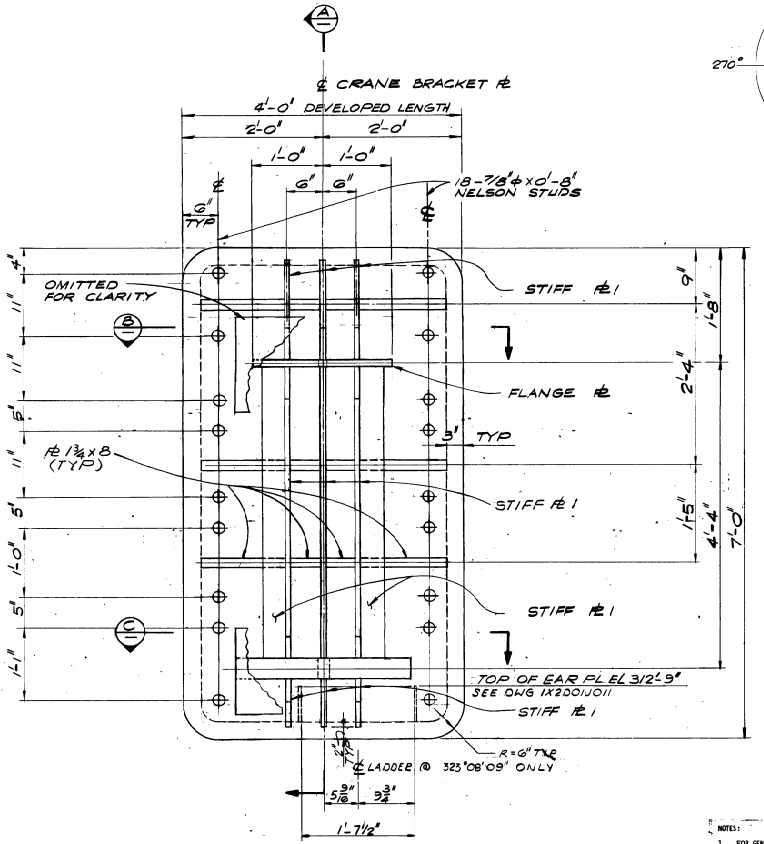
**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

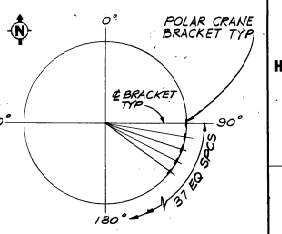
CONTAINMENT
WALL LINER PLATE
PENETRATION DETAILS SHT. 2

SCALE: AS NOTED | DRAWING NO. 1X2D01J020 | REV. 8
JOB NO. 9510

SCALING RULES



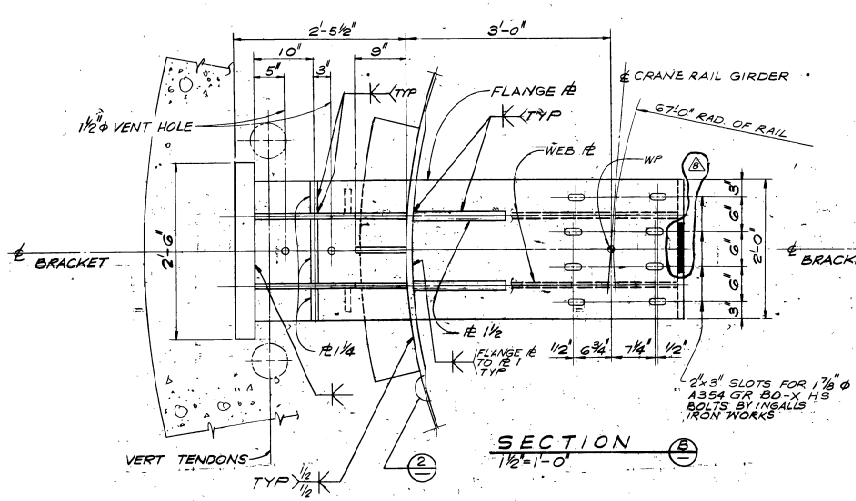



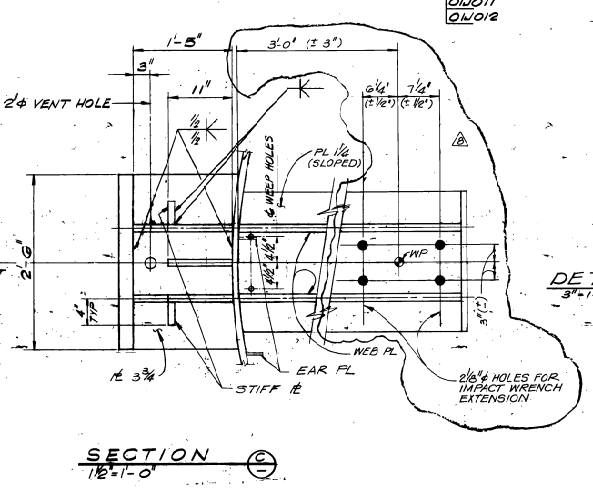



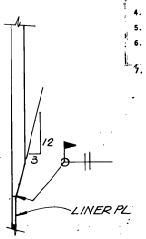

BRACKET LOC PLAN
1" = 20'-0"

| PC. MK. | REV. | REF. DWG |
|---|---|---|
| M-1 | 4 | S-148 |

NOTES:
1. FOR GENERAL JOINTS AND REFERENCE DRAWINGS SEE DRAWING 1X2D01J001.
2. ALL PENETRATION WELDS TO BE FULL PENETRATION
3. PLATES IDENTIFIED BY (U.T.) REQUIRE ULTRASONIC TESTING IN ACCORDANCE WITH SPECIFICATION X2AG06
4. FOR POLAR CRANE GIRDER DETAIL SEE DRAWING 1X2D01J002
5. SUPPORT BRACKET PLATES SHALL CONFORM TO ASME SA-516, GRADE 70.
6. WIDTH SHOWN ON POLAR CRANE BRACKET THICKENED PLATE IS MINIMUM DIMENSION, FABRICATOR MAY INCREASE TO ALIGN WITH 1/4" PL. VERT SEAMS AS REQUIRED.
7. SHIM MATERIAL SHALL BE AS FOLLOWS:
   A) ASTM A-569 FOR PLATE THICKNESSES LESS THAN 1/4"
   B) ASTM A-36 FOR PLATE THICKNESSES EQUAL TO OR GREATER THAN 1/4"




DETAIL 2
3"=1'-0"

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | BSL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| A | INCORP DCN 5 & 6 | 11/7/75 | NC | LB | MWH | DCN | RP | | JW | |
| A | INCORPORATE DCN #364 | 11-19 | RC | | MWH | | RP | | JW | |
| A | INCORPORATE DCN-182 | 2/4/76 | DCN | | MWH | | RP | | JW | |
| A | REVISED AS SHOWN | 4-6-76 | RV | | MWH | JTL | RP | | MBT | |
| A | REMOVED CADWELDS & ADDED SHIM PACK | 11/5/76 | VAA | | MWH | GO | RP | | MBT | |
| A | REV DET 1, SECT A-C & NOTE 6 | 8/16/77 | CJ | | MWH | GO | RP | | JW | |
| A | ADD NOTE 6 AND BOLTS SUPPLIER | 12-8-77 | DB | | MWH | GO | RP | | JW | |
| A | GENERAL REVISION | 3-23-78 | FM | | MWH | GO | RP | | JW | |
| A | ISSUED FOR CONSTRUCTION | 6-1-78 | SL | | DR | GO | RP | | MBT | |

ENG. MGR. DATE

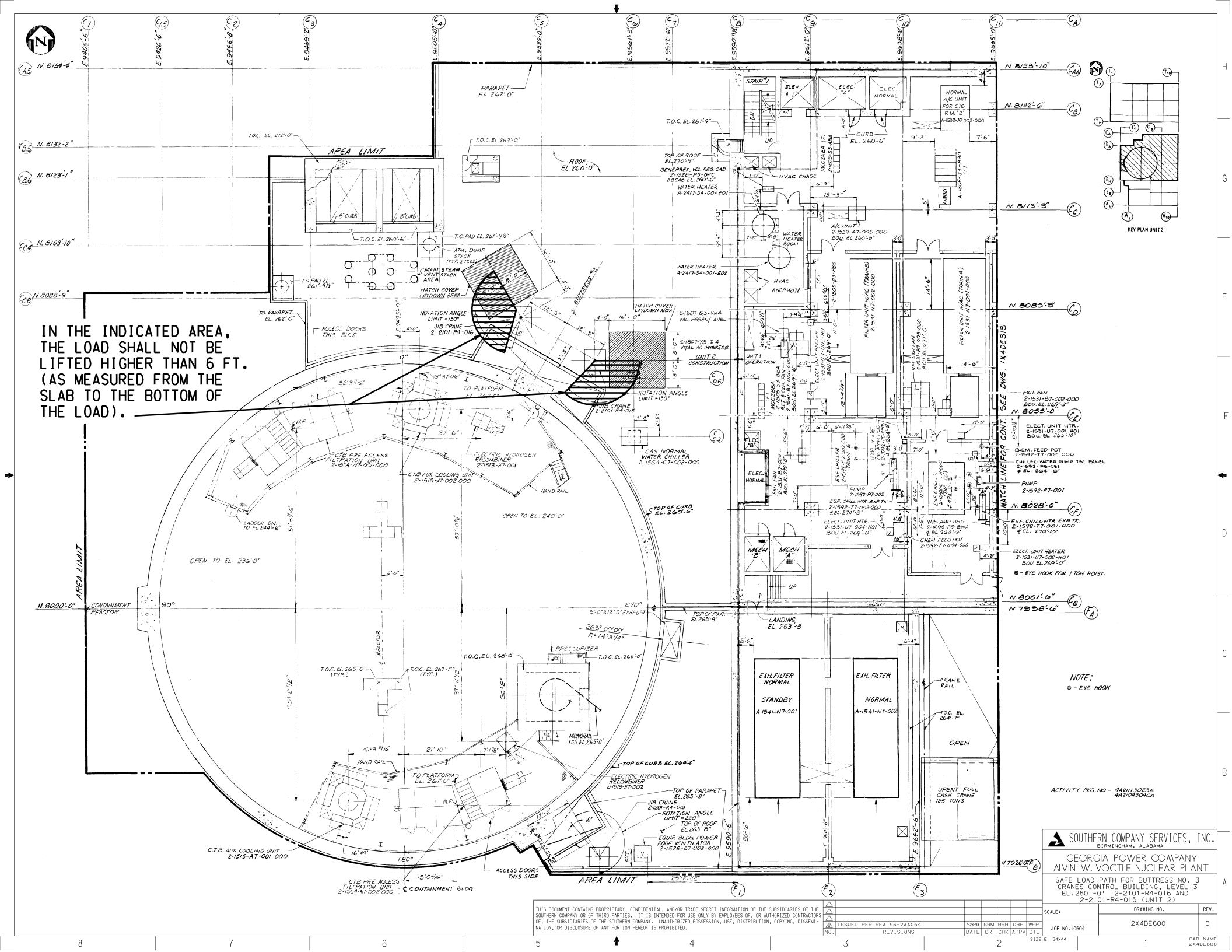
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
WALL LINER PLATE
POLAR CRANE BRACKET

SCALE: AS NOTED
DRAWING NO. 1X2D01J022
REV. A
JOB NO. 9510

SIZE E 34x44

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 1 | ISSUED PER REA 96-VAA054 | 7-28-98 | SRM | RBH | CBH | WFP |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SAFE LOAD PATH FOR BUTTRESS NO. 3 CRANES CONTROL BUILDING, LEVEL 3 EL. 260'-0" 2-2101-R4-016 AND 2-2101-R4-015 (UNIT 2)

SCALE:

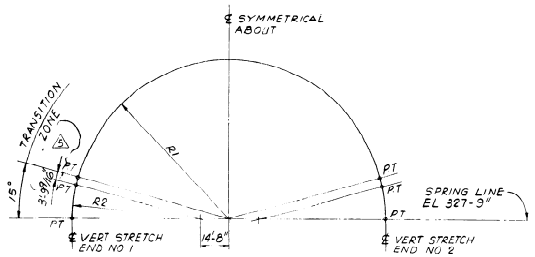
JOB NO. 10604

DRAWING NO. 2X4DE600

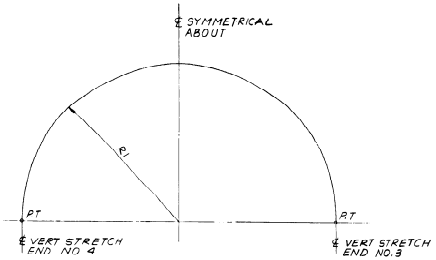
REV. 0

SIZE E 34X44

CAD NAME 2X4DE600



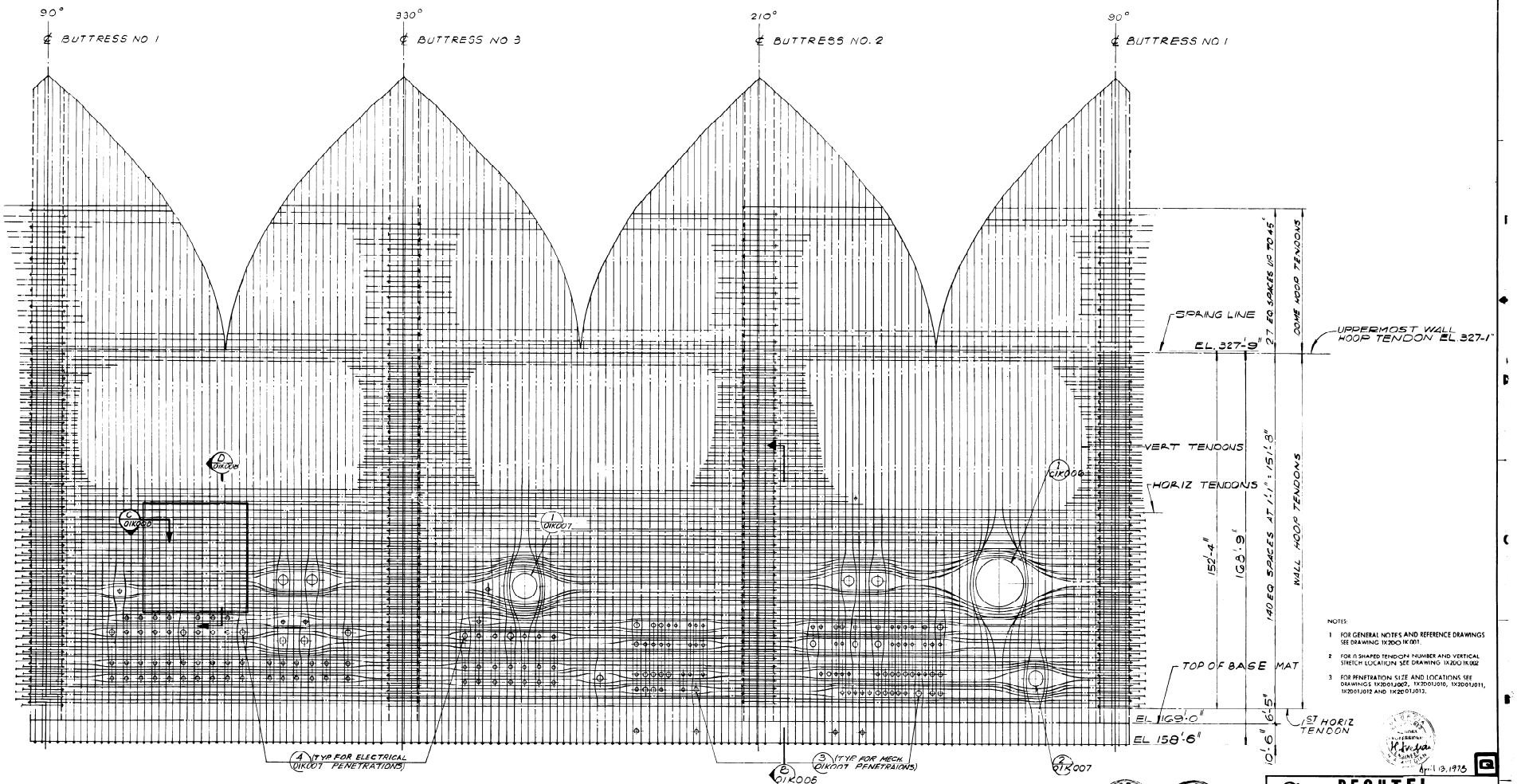

GEOMETRY OF Π SHAPED TENDONS
GROUP NO 1




GEOMETRY OF Π SHAPED TENDONS
GROUP NO 2

Π SHAPED TENDONS RADII AT THE DOME

| TENDON GROUP NO 1 | | | TENDON GROUP NO 2 | | |
|---|---|---|---|---|---|
| TENDON NO | R1 | R2 | TENDON NO | R1 | R2 |
| 1 & 37 | 51'-3 7/16" | 37'-1 7/16" | 38 & 74 | 51'-11 3/16" | |
| 2 & 36 | 53'-4 7/16" | 39'-2 7/16" | 39 & 73 | 54'-0 7/16" | |
| 3 & 35 | 55'-4 1/2" | 41'-2 1/2" | 40 & 72 | 56'-0 1/16" | |
| 4 & 34 | 57'-3 1/8" | 43'-1 3/16" | 41 & 71 | 57'-10 1/16" | |
| 5 & 33 | 59'-0 7/16" | 44'-10 7/16" | 42 & 70 | 59'-7 7/16" | |
| 6 & 32 | 60'-8 3/8" | 46'-6 1/16" | 43 & 69 | 61'-3 3/8" | |
| 7 & 31 | 62'-3 3/8" | 48'-1 7/16" | 44 & 68 | 62'-10 3/16" | |
| 8 & 30 | 63'-8 7/8" | 49'-6 3/4" | 45 & 67 | 64'-3 7/16" | |
| 9 & 29 | 65'-0 7/16" | 50'-10 7/16" | 46 & 66 | 65'-7 1/4" | |
| 10 & 28 | 66'-3 3/8" | 52'-1 1/4" | 47 & 65 | 66'-9 3/8" | |
| 11 & 27 | 67'-4 5/16" | 53'-2 1/4" | 48 & 64 | 67'-10 3/8" | |
| 12 & 26 | 68'-3 7/16" | 54'-1 7/8" | 49 & 63 | 68'-10 1/8" | |
| 13 & 25 | 69'-1 7/8" | 54'-11 1/2" | 50 & 62 | 69'-0 1/4" | |
| 14 & 24 | 69'-10 7/16" | 55'-0 7/16" | 51 & 61 | 70'-4 9/16" | |
| 15 & 23 | 70'-5 7/16" | 56'-3 1/16" | 52 & 60 | 70'-11 9/16" | |
| 16 & 22 | 70'-10 9/16" | 56'-8 5/16" | 53 & 59 | 71'-5" | |
| 17 & 21 | 71'-2 3/16" | 57'-0 1/16" | 54 & 58 | 71'-8 7/8" | |
| 18 & 20 | 71'-5 3/16" | 57'-3 1/4" | 55 & 57 | 71'-11 1/4" | |
| 19 | 71'-6" | 57'-4" | 56 & | 72'-0" | |




NOTES:
1. FOR GENERAL NOTES AND REFERENCE DRAWINGS SEE DRAWING 1X2D01K001.
2. FOR Π SHAPED TENDON NUMBER AND VERTICAL STRETCH LOCATION SEE DRAWING 1X2D01K002
3. FOR PENETRATION SIZE AND LOCATIONS SEE DRAWINGS 1X2D01J002, 1X2D01J010, 1X2D01J011, 1X2D01J012 AND 1X2D01J013.

Apr-13,1976

WALL DEVELOPED ELEVATION (LOOKING FROM OUTSIDE)
1/16" = 1'-0"

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 3 | REVISED AS SHOWN | 11-11-75 | JW | | | | | | |
| 2 | REVISED PER CHG. NO. 1, ADDITION OF #11 C BARS | 7-7-75 | NC | | | | | | |
| 1 | ISSUED FOR CHF E APPROVAL | 11-11-74 | EM | | | | | | |
| 0 | RELEASED FOR CONSTRUCTION | 4-13-76 | MN | | | | | | |

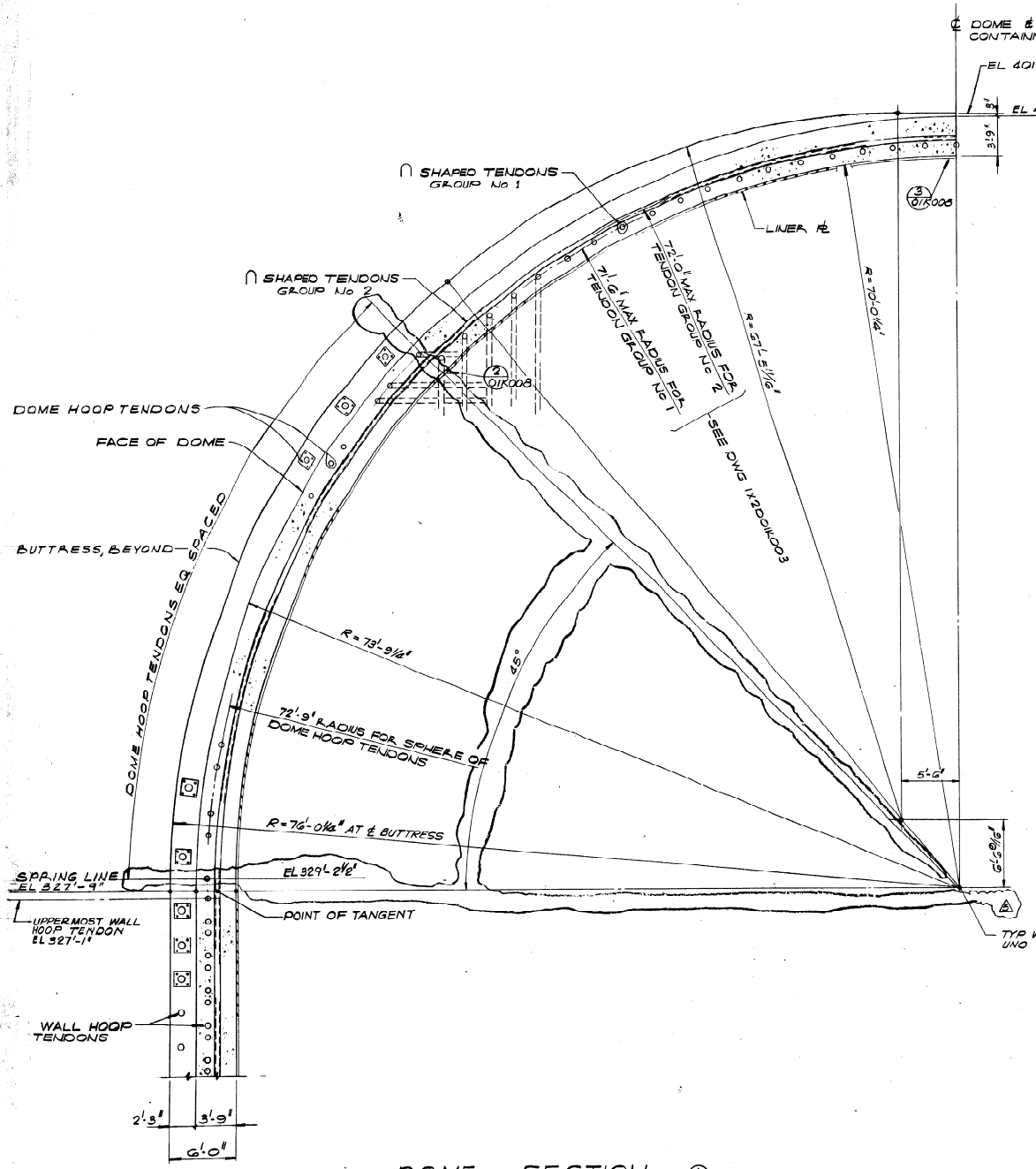
BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

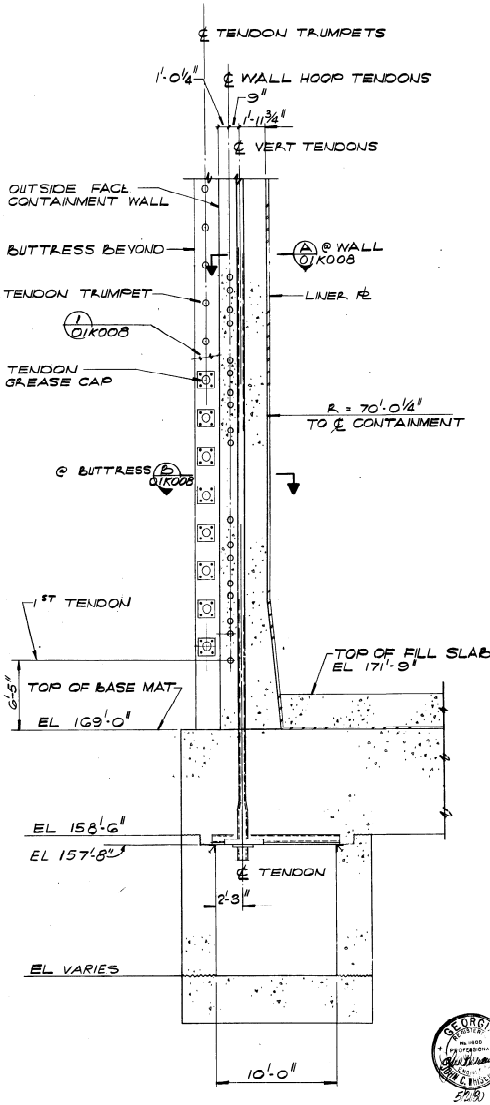
CONTAINMENT
PRESTRESSING REQUIREMENTS
WALL DEVELOPED ELEVATION

SCALE 1/16" = 1'-0" | DRAWING NO. 1X2D01K003 | REV. 5
JOB NO. 9510

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL




DOME SECTION A 01K002




WALL SECTION B 01K003

NOTE:
1. FOR GENERAL NOTES AND REFERENCE DRAWINGS SEE DRAWING 1X2D01K001.

Apr. 1 19, 1978

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT
PRESTRESSING REQUIREMENTS
WALL AND DOME CROSS SECTION

| SCALE: 1/4"-1'-0" | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 9510 | 1X2D01K005 | 5 |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 5 | REVISED DESCRIPTION OF DOME HOOP TENDONS OF 327' EL. INDOOR INSTALLATION REQUIREMENTS | 12-7-8 | NC | | | | | | |
| 4 | REMOVED CONST. JOINT SECT. B | 5-23-8 | CN | | | | | | |
| 3 | ISSUED FOR CHF. E APPROVAL | 9-19-78 | LEM | | | | | | |
| 2 | REV'D DIM'S, RADIUS, EL. TP. TRUMPET & LOCATION OF HOOP TENDONS | 11-3-78 | SL | | | | | | |
| 1 | INDICATED C J 'S & ADDED DIM | 1-4-10 | SL | | | | | | |
| 0 | ISSUED FOR CONSTRUCTION | 2-22-79 | FM | | | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|

ENG. MGR. DATE

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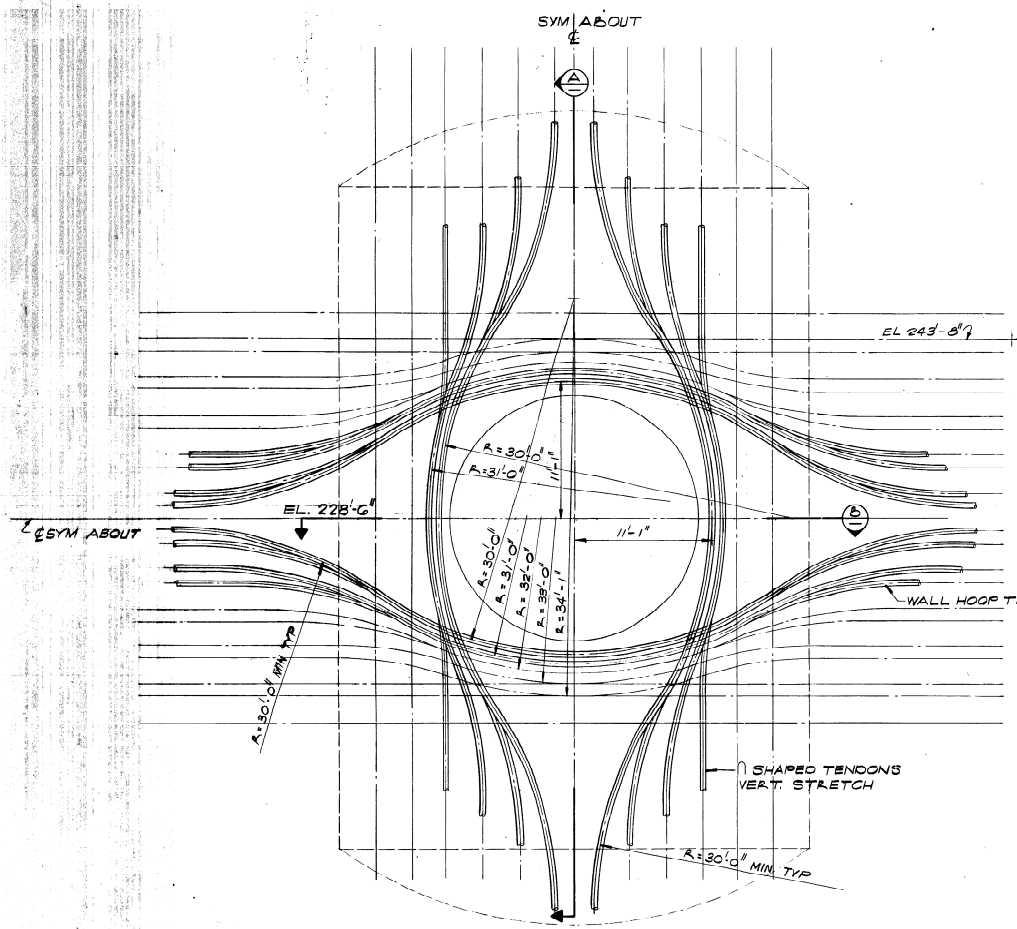
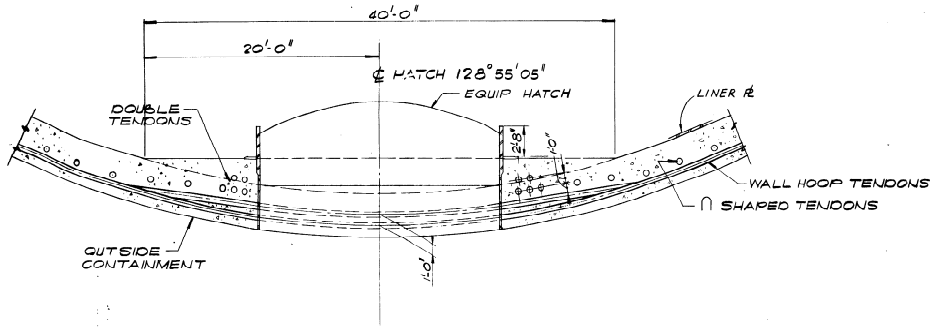
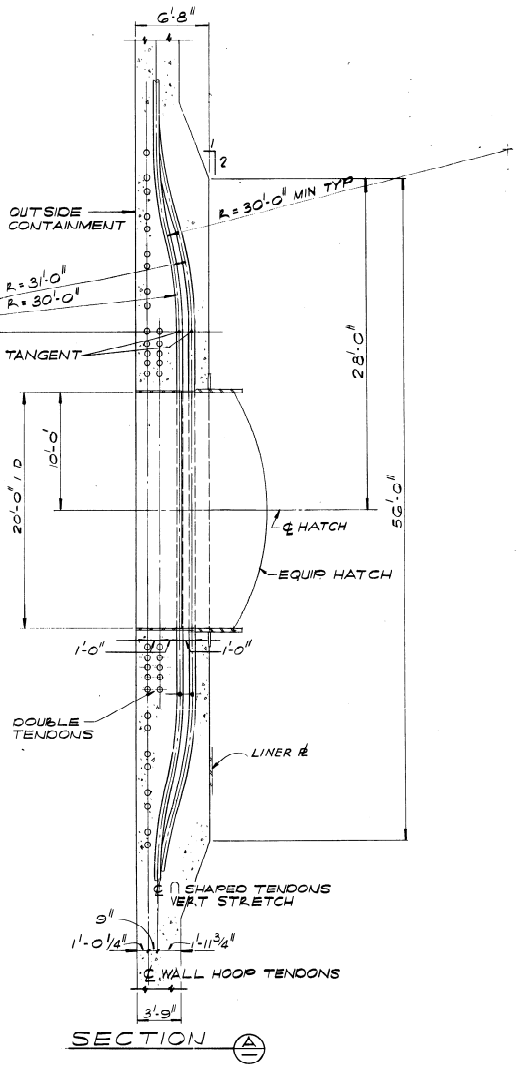



TENDON ELEV AT EQUIPMENT HATCH DETAIL (1) 01K003




SECTION (B)




SECTION (A)

NOTES:
1. FOR GENERAL NOTES AND REFERENCE DRAWINGS SEE DRAWING 1X2D01K001.

Apr. 13, 1978

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
PRESTRESSING REQUIREMENTS
HATCH AND LOCK DETAILS SHT. 1

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 2 | ISSUED FOR CHF E. APPROVAL | 9.19.77 | EM | JMM | GO | LL | XB | MET | |
| 1 | REV. DIM. CLARIFY DWG SECT A-B | 11.11.77 | SSL | JMM | GO | LL | XB | MET | |
| 0 | ISSUED FOR CONSTRUCTION | 6.2.77 | EM | JMM | GO | KC | MET | | |

ENG. MGR. | DATE

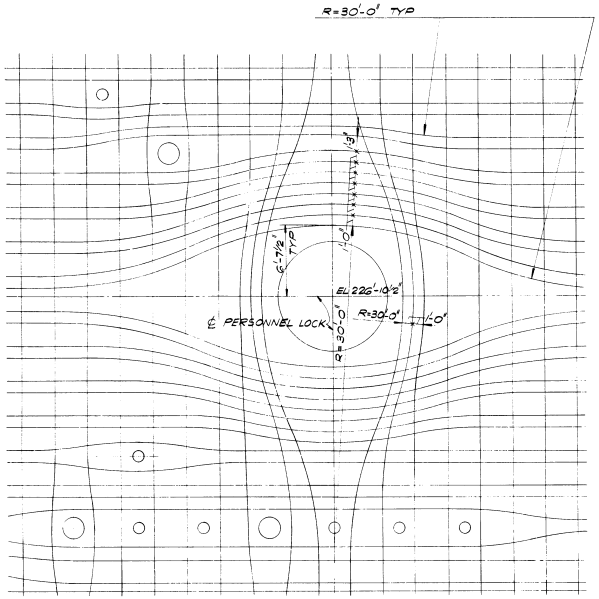
SCALE: 1/2"=1'-0" | DRAWING NO. 1X2D01K006 | REV. 2

JOB NO. 9510

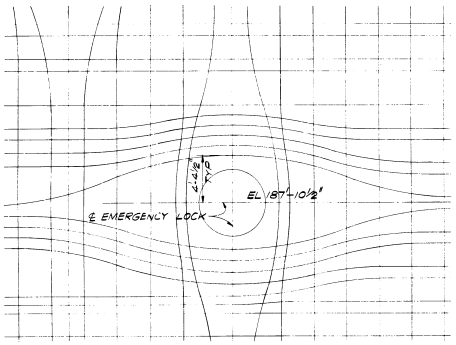
SIZE C 24 x 44

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS ON THE DATE STATED BELOW AND THAT IT

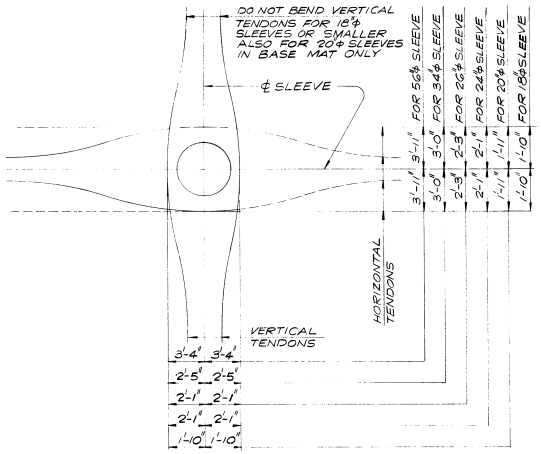
SCALING RULES



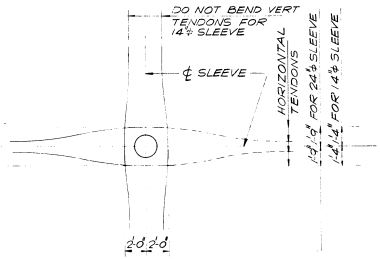

DETAIL PERSONNEL LOCK (1) 01K003




DETAIL EMERGENCY LOCK (2) 01K003




DETAIL MECH PEN (3) 01K007




DETAIL ELEC PEN (4) 01K007

H. N. Wilson
Apr 13 1979

Q

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT
PRESTRESSING REQUIREMENTS
MATCH AND LOCK DETAILS SHT. 2

| ENG. MGR. | | |
|---|---|---|
| DATE | SCALE: 1/4"-0" | DRAWING NO. 1X2D01K007 |
| | JOB NO. 9510 | REV. 2 |

SIZE E 30X44

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 3 | ISSUED FOR CHF. E APPROVAL | 11/27/78 | EM | MM | GG | | YD | | MRT |
| 2 | REDRAWN & REVISED DETAILS | 1/11/79 | SL | | | | | | |

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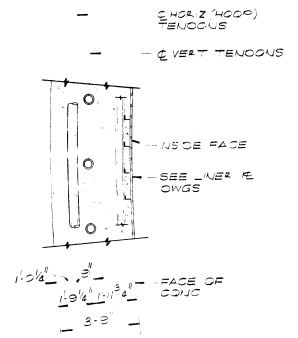
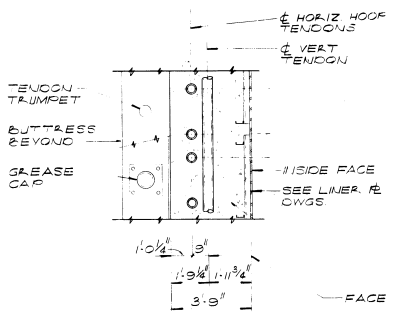
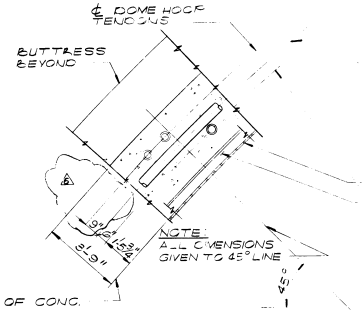


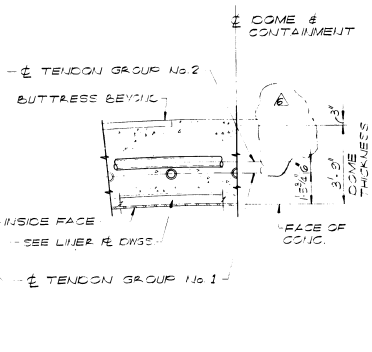

SECTION (A) 01K005
1/2" = 1'-0"



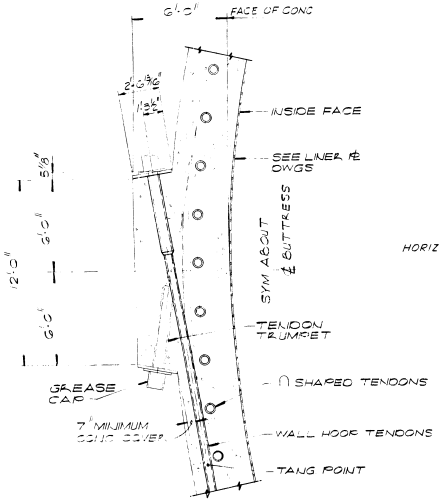

DETAIL (1) 01K005
1/2" = 1'-0"




DETAIL (2) 01K005
1/2" = 1'-0"

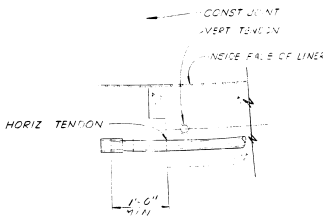



DETAIL (3) 01K005
1/2" = 1'-0"



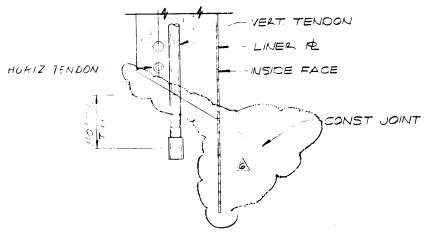

PLAN

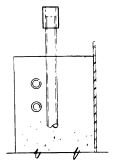
SECTION (B) 01K005
3/8" = 1'-0"




PLAN

SECTION (C) 01K005
NTS
(SEE NOTE 2)



SECTION (D) 01K005
NTS
(SEE NOTE 2)

NOTES:
1. [illegible]
2. [illegible]

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| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 6 | REVISED DET 2 & 3 TO AGREE WITH VENDOR INSTALLATION | | | | | | | | | |
| 5 | REQMTS SHOWN ON 1X2D01K005, REV. SECT D | | | | | | | | | |
| 4 | ISSUED FOR CHF E APPROVAL | 9/17/79 | EM | UNM | | | | | | |
| 3 | REVISED CONST OPENING E | | | | | | | | | |
| 2 | COMPUTER RECAP FOR CLARITY | | | | | | | | | |
| 1 | REV'D DIMS | | | | | | | | | |
| 0 | ISSUED FOR CONSTRUCTION | | | | | | | | | |

ENG. MGR.
DATE

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT
PRESTRESSING REQUIREMENTS
SECTIONS AND DETAILS SHT.1

SCALE: AS NOTED
JOB NO. 9510
DRAWING NO. 1X2D01K008
REV. 6

SIZE E 34 x 44

1X2D01K008

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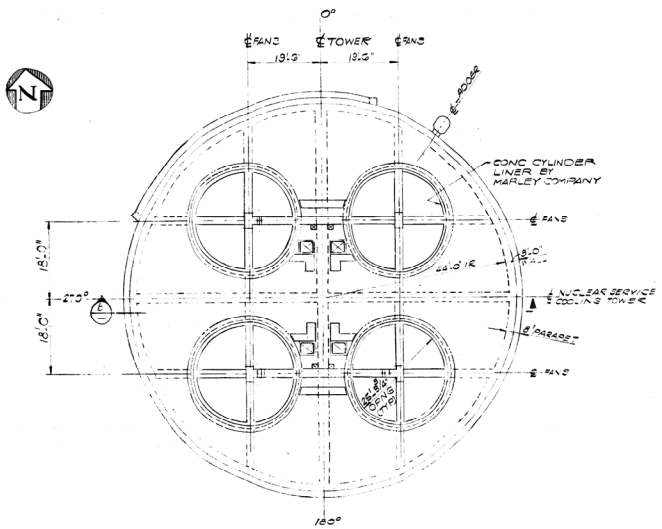
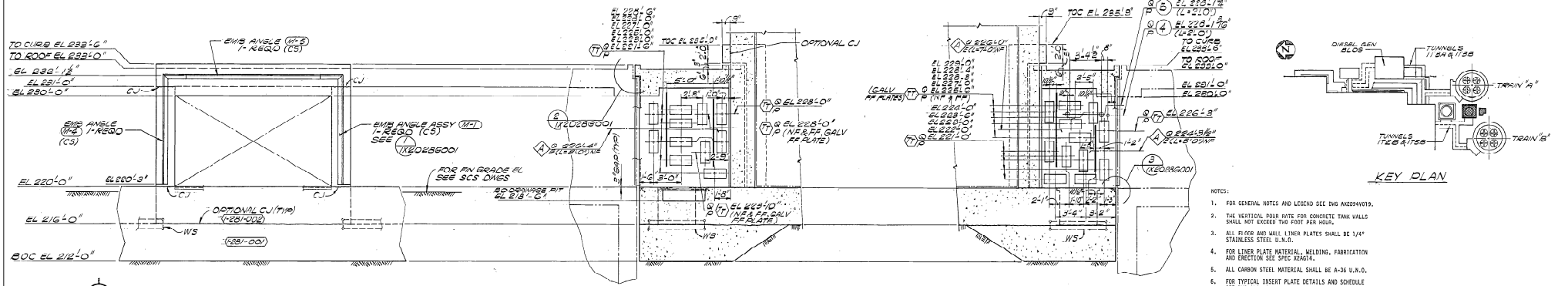








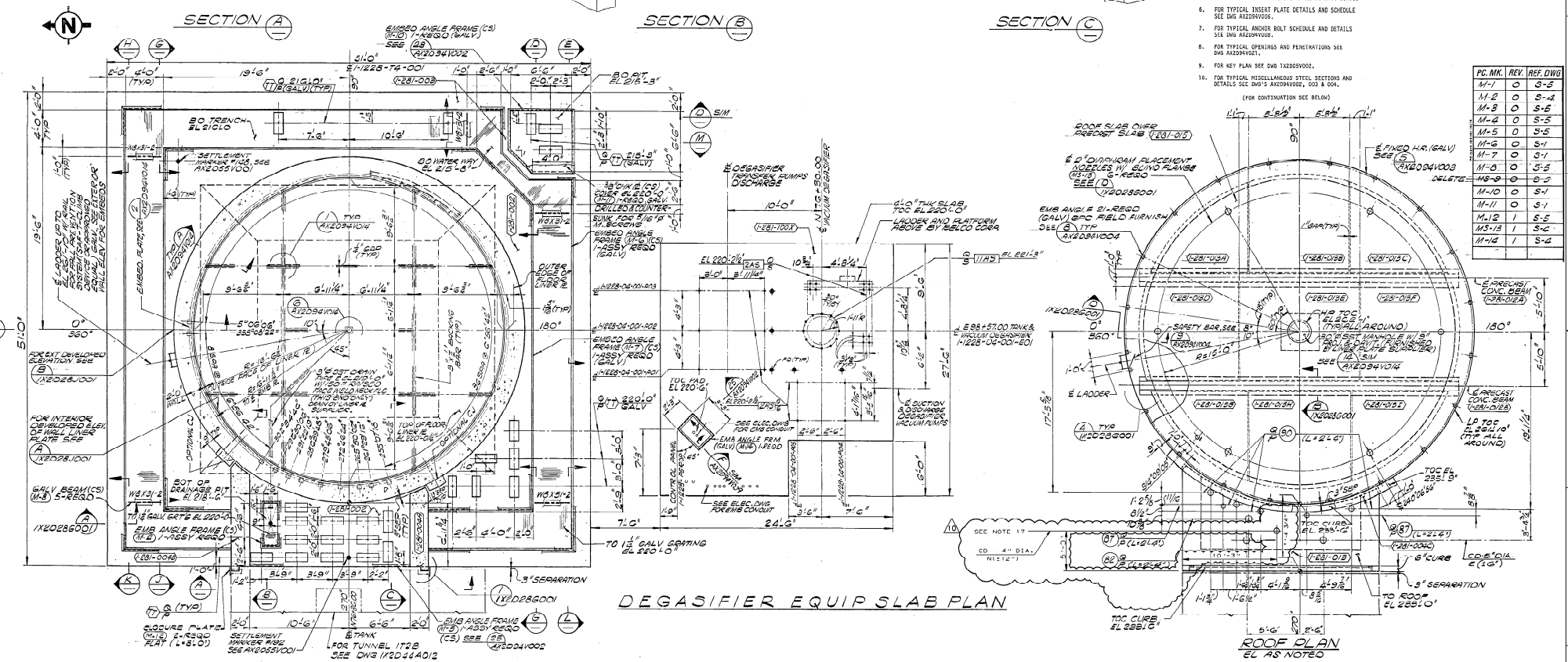
KEY PLAN

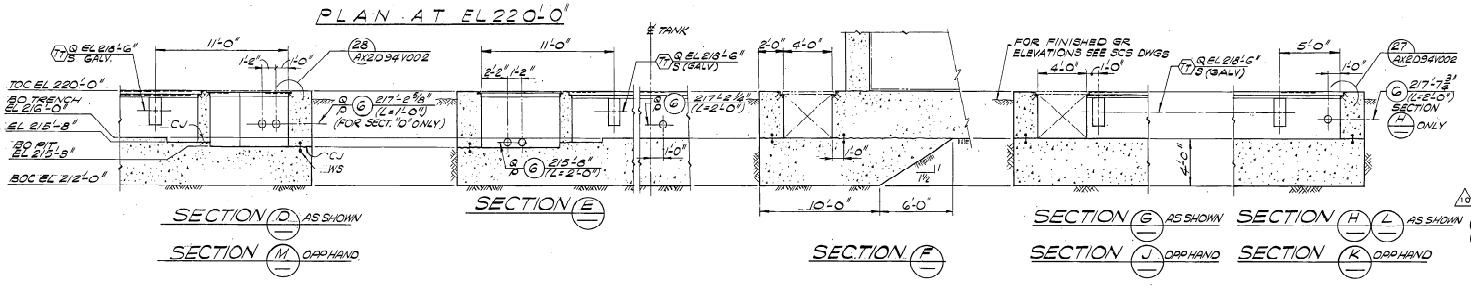
NOTES:

1. FOR GENERAL NOTES AND LEGEND SEE DWG AX2D94V019.
2. THE VERTICAL POUR RATE FOR CONCRETE TANK WALLS SHALL NOT EXCEED TWO FOOT PER HOUR.
3. ALL FLOOR AND WALL LINER PLATES SHALL BE 1/4" STAINLESS STEEL U.N.O.
4. FOR LINER PLATE MATERIAL, WELDING, FABRICATION AND ERECTION SEE SPEC X2AG14.
5. ALL CARBON STEEL MATERIAL SHALL BE A-36 U.N.O.
6. FOR TYPICAL INSERT PLATE DETAILS AND SCHEDULE SEE DWG AX2D94V006.
7. FOR TYPICAL ANCHOR BOLT SCHEDULE AND DETAILS SEE DWG AX2D94V008.
8. FOR TYPICAL OPENINGS AND PENETRATIONS SEE DWG AX2D94V021.
9. FOR KEY PLAN SEE DWG 1X2D05V002.
10. FOR TYPICAL MISCELLANEOUS STEEL SECTIONS AND DETAILS SEE DWG'S AX2D94V002, 003 & 004.

(FOR CONTINUATION SEE BELOW)

| PC. MK. | REV. | REF. DWG |
|---|---|---|
| M-1 | 0 | S-3 |
| M-2 | 0 | S-4 |
| M-3 | 0 | S-5 |
| M-4 | 0 | S-5 |
| M-5 | 0 | S-5 |
| M-6 | 0 | S-1 |
| M-7 | 0 | S-1 |
| M-8 | 0 | S-5 |
| M8-9 | 0 | S-5 |
| M-10 | 0 | S-1 |
| M-11 | 0 | S-1 |
| M-12 | 1 | S-5 |
| M3-13 | 1 | S-4 |
| M-14 | 1 | S-4 |




11. FOR CATEGORY I TANK LINER PLATE TYPICAL SECTIONS AND DETAILS SEE DWG AX2D94V001A.
12. ALL MATERIAL SHALL BE STAINLESS STEEL U.N.O.
13. ELEVATIONS SHOWN ON DEVELOPED WALL ELEVATIONS ARE TO THE ℄ OF NOZZLES. ALL NOZZLES SHALL BE LOCATED RADIALLY U.N.O.
14. ALL EMBEDDED PIPE & ELECTRICAL CONDUIT SHALL BE IN PLACE PRIOR TO PLACING CONCRETE.
15. A COATING OF CARBON ZINC 11 MAY BE USED IN LIEU OF GALVANIZING.
16. THE FOUNDATION CHORDS AND CONCRETE SHALL BE LEVEL AND FLUSH AT THE INTERSECTION WITH THE WALL LINER WITHIN ±1/8 INCH IN ANY 30 FEET OF THE CIRCUMFERENCE, AND WITHIN ±1/4 INCH IN THE TOTAL CIRCUMFERENCE. THE FLOOR EMBEDS MUST BE JOINED TOGETHER WITH A VERTICAL OFFSET NO GREATER THAN ±1/16 INCH TO FACILITATE ATTACHING THE BOTTOM PLATES.
17. ONE REBAR EACH FACE AND DIRECTION MAY BE CUT IF REQUIRED. ONCE CONDUIT IS PLACED, THE VOID SHALL BE FILLED WITH A 4000 PSI (MIN.) GROUT OR PENETRATION SEAL MATERIAL.

THIS DOCUMENT CONTAINS PROPRIETARY, CONFIDENTIAL, AND/OR TRADE SECRET INFORMATION OF THE SUBSIDIARIES OF THE SOUTHERN COMPANY OR OF THIRD PARTIES. IT IS INTENDED FOR USE ONLY BY EMPLOYEES OF, OR AUTHORIZED CONTRACTORS OF, THE SUBSIDIARIES OF THE SOUTHERN COMPANY. UNAUTHORIZED POSSESSION, USE, DISTRIBUTION, COPYING, DISSEMINATION, OR DISCLOSURE OF ANY PORTION HEREOF IS PROHIBITED.

| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 4 | | | | | | |
| 3 | | | | | | |
| 2 | | | | | | |
| 1 | REVISED PER DCP 95-V1N0065 | 2-6-96 | Vap | RBH | DEW | V |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

REACTOR MAKE-UP WATER STORAGE TANK
FORMING PLAN
SECTIONS & DETAILS

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X2D28A001
REV. 10
SIZE E 34x44

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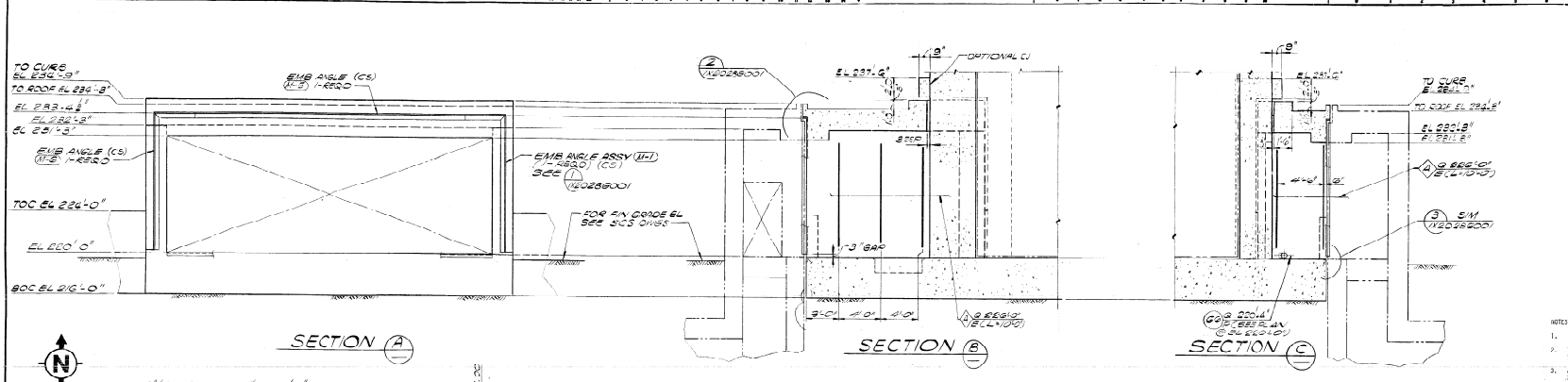
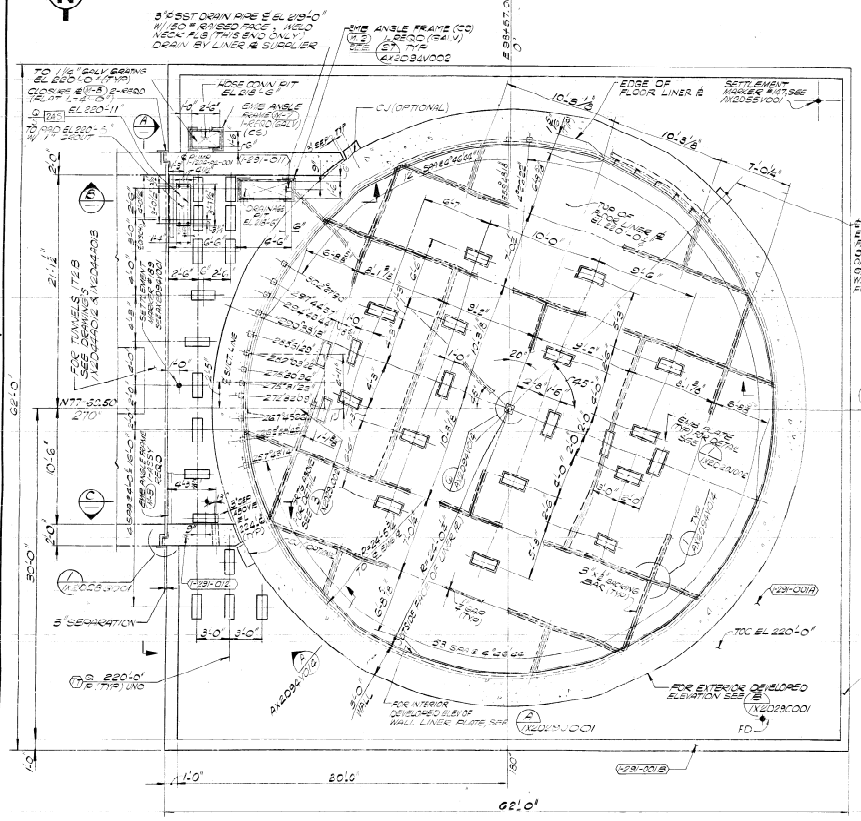
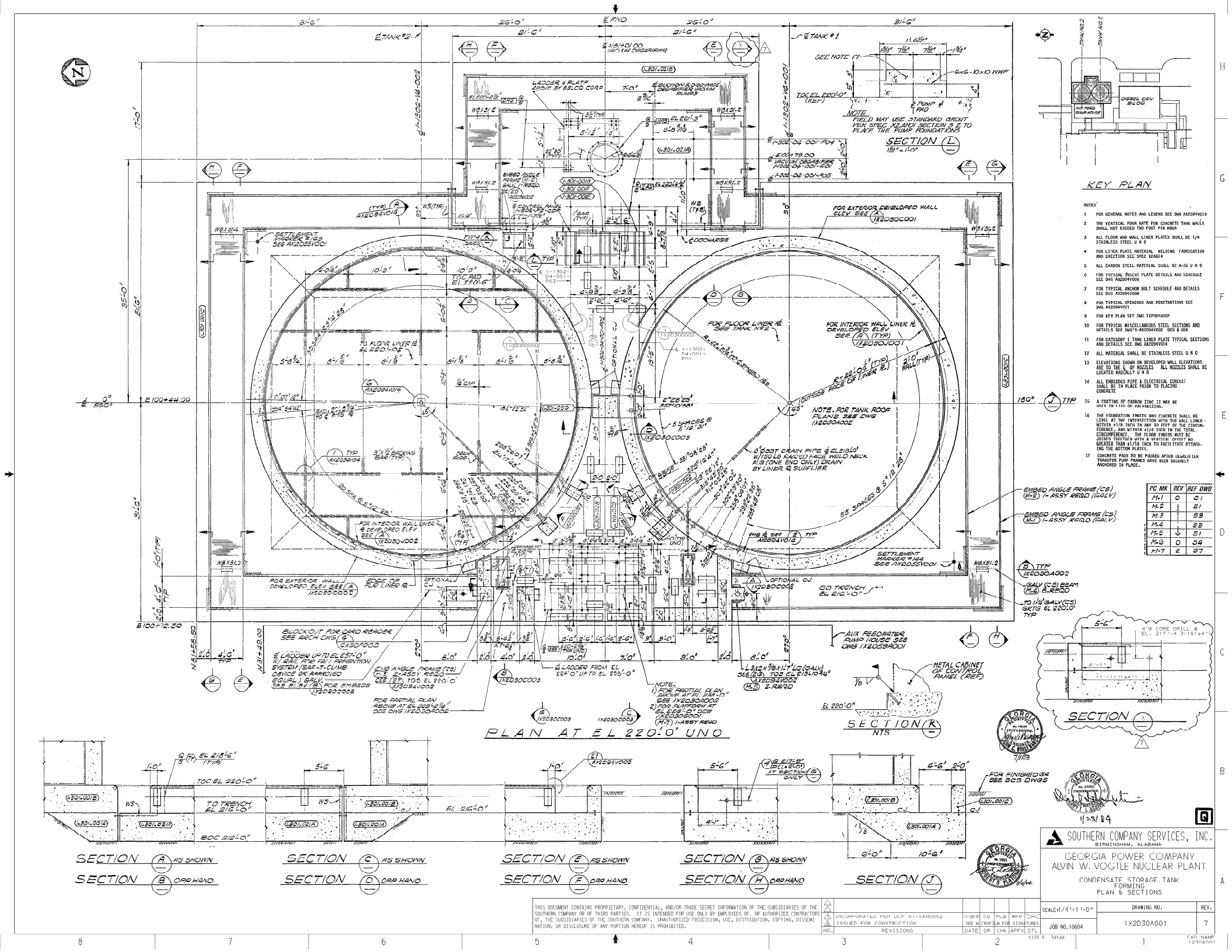
SCALING RULES





3/4" ø DRAIN PIPE @ EL 219'-0" W/ (6) Ø 1/2" HOLES EQUALLY SPACED AT 3" RADIUS FROM PIPE CENTER
EMB ANGLE FRAME (CS) L6X6X1/2 (TYP) A1303AV002
TO TOP OF CURB EL 220'-0" (TYP)
CURB EL 219'-11"
TO TOP OF SLAB EL 220'-11"
HOSE DOWN PIT EL 218'-2"
EMB ANGLE FRAME (CS) L6X6X1/2 (TYP) A1303AV002
CJ (OPTIONAL)
EDGE OF FLOOR LINER @
RETT. ENHNT. RADIUS W/ SEE A1303AV001
TOC EL 220'-0"
TOC EL 219'-0"
TOC EL 221'-0"
TOC EL 224'-0"
TOC EL 227'-0"
TOC EL 230'-0"
TOC EL 233'-0"
TOC EL 236'-0"
TOC EL 239'-0"
TOC EL 242'-0"
TOC EL 245'-0"
TOC EL 248'-0"
TOC EL 251'-0"
TOC EL 254'-0"
TOC EL 257'-0"
TOC EL 260'-0"
TOC EL 263'-0"
TOC EL 266'-0"
TOC EL 269'-0"
TOC EL 272'-0"
TOC EL 274'-0"
EDGE TRAVELS 1728 A1303AV002 & A1304X003
SEE A1303AV001 SCHEDULE
SEE A1303AV001 SCHEDULE
SEE A1303AV001 SCHEDULE


PLAN AT EL 220'-0" UNO
SECTION (L)
SECTION (K)
SECTION (J)
SECTION (A) AS SHOWN
SECTION (B) ORD HAND
SECTION (C) AS SHOWN
SECTION (D) ORD HAND
SECTION (E) AS SHOWN
SECTION (F) ORD HAND
SECTION (G) AS SHOWN
SECTION (H) ORD HAND
KEY PLAN
NOTES:
1 FOR GENERAL NOTES AND LEGEND SEE DWG AX2D94V019
2 THE VERTICAL POUR RATE FOR CONCRETE TANK WALLS SHALL NOT EXCEED TWO FOOT PER HOUR
3 ALL FLOOR AND WALL LINER PLATES SHALL BE 1/4 STAINLESS STEEL U N O
4 FOR LINER PLATE MATERIAL, WELDING FABRICATION AND ERECTION SEE SPEC X2AG14
5 ALL CARBON STEEL MATERIAL SHALL BE A-36 U N O
6 FOR TYPICAL INSERT PLATE DETAILS AND SCHEDULE SEE DWG AX2D94V006
7 FOR TYPICAL ANCHOR BOLT SCHEDULE AND DETAILS SEE DWG AX2D94V008
8 FOR TYPICAL OPENINGS AND PENETRATIONS SEE DWG AX2D94V021
9 FOR KEY PLAN SEE DWG 1X2D05V007
10 FOR TYPICAL MISCELLANEOUS STEEL SECTIONS AND DETAILS SEE DWG'S-AX2D94V002 003 & 004
11 FOR CATEGORY I TANK LINER PLATE TYPICAL SECTIONS AND DETAILS SEE DWG AX2D94V014
12 ALL MATERIAL SHALL BE STAINLESS STEEL U N O
13 ELEVATIONS SHOWN ON DEVELOPED WALL ELEVATIONS ARE TO THE C OF NOZZLES ALL NOZZLES SHALL BE LOCATED RADIALLY U N O
14 ALL EMBEDDED PIPE & ELECTRICAL CONDUIT SHALL BE IN PLACE PRIOR TO PLACING CONCRETE
15 A COATING OF CARBON ZINC II MAY BE USED IN LIEU OF GALVANIZING.
16 THE FOUNDATION EMBEDS AND CONCRETE SHALL BE LEVEL AT THE INTERSECTION WITH THE WALL LINER WITHIN +1/8 INCH IN ANY 30 FEET OF THE CIRCUMFERENCE, AND WITHIN ±1/4 INCH IN THE TOTAL CIRCUMFERENCE. THE FLOOR EMBEDS MUST BE JOINED TOGETHER WITH A VERTICAL OFFSET NO GREATER THAN ±1/16 INCH TO FACILITATE ATTACHING THE BOTTOM PLATES.
17 CONCRETE PADS TO BE POURED AFTER DEGASIFIER TRANSFER PUMP FRAMES HAVE BEEN SECURELY ANCHORED IN PLACE.
PC MK | REV | REF DWG
M-1 | 0 | 91
M-2 | | 91
M-3 | | 58
M-4 | | 58
M-5 | ↓ | 51
M-6 | 0 | 54
M-7 | E | 97
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
CONDENSATE STORAGE TANK FORMING
PLAN & SECTIONS
SCALE 1/4"=1'-0"
DRAWING NO. 1X2D30A001
REV. 7
JOB NO. 10604
SIZE E 36x44
CAD NAME 1230A001
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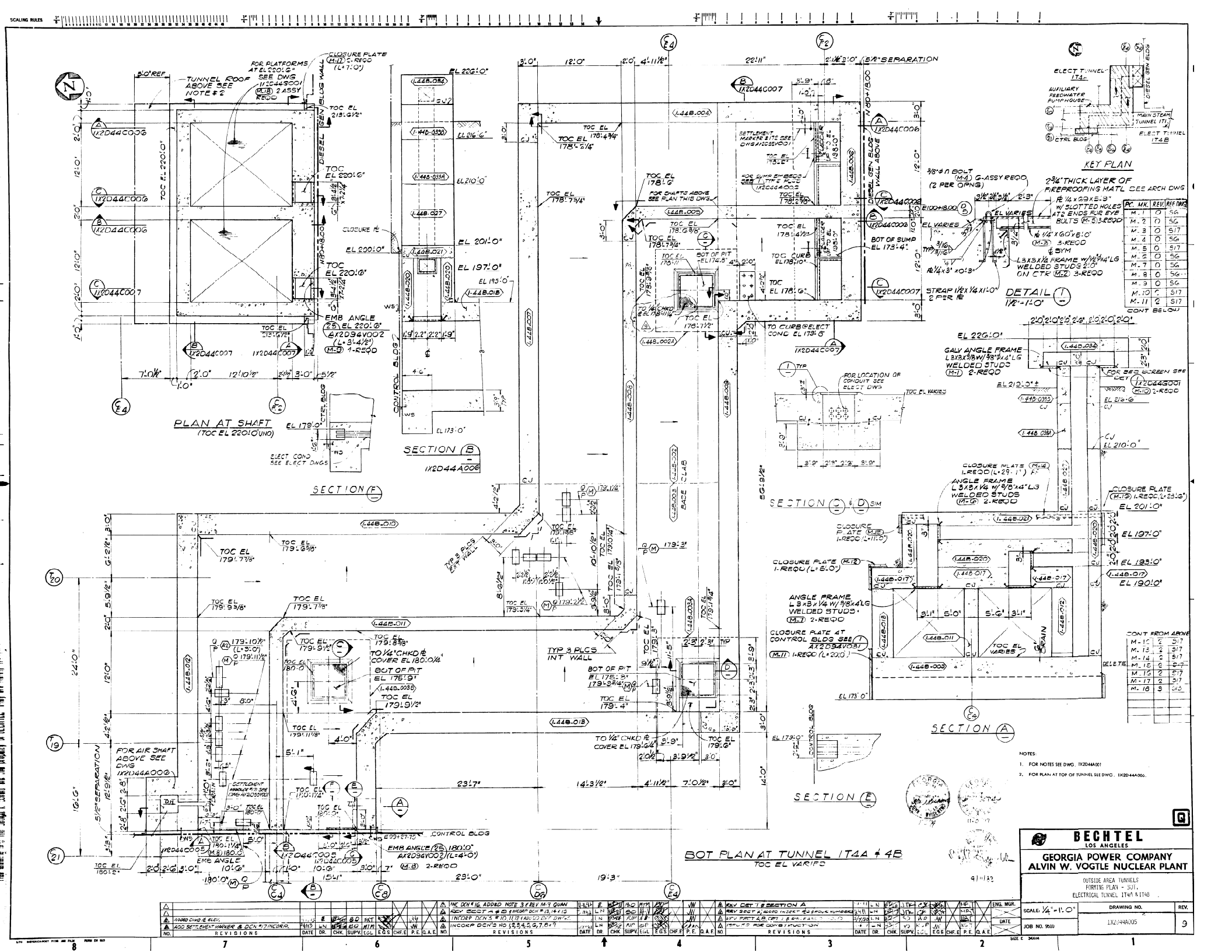
NO. REVISIONS DATE DR CHK APPV DTL
INCORPORATED PER DCP 97-VAN0052 11-26-97 CD PLB WEP
ISSUED FOR CONSTRUCTION SEE MICROFILM FOR SIGNATURES
1/23/84


PLAN AT SHAFT
(TOC EL 220'-0" U.N.O.)
SECTION B
1X2D44A006
SECTION F
SECTION C & D SIM
SECTION A
SECTION E
BOT PLAN AT TUNNEL IT4A & 4B
TOC EL VARIES
KEY PLAN
DETAIL
1/2"=1'-0"
ELECT TUNNEL IT4
AUXILIARY FEEDWATER PUMPHOUSE
DIESEL GEN BLDG
MAIN STEAM TUNNEL IT1
ELECT TUNNEL IT4B
CTRL BLDG
CONTROL BLDG
DIESEL GEN BLDG WALL ABOVE
TOC EL 220'-0"
EL 220'-0"
EL 210'-0"
EL 201'-0"
EL 197'-0"
EL 195'-0"
EL 190'-0"
EL 179'-0"
EL 173'-0"
TOC EL 180'-0"
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TOC EL 178'-3 5/8"
TOC EL 179'-7 5/8"
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SCALING RULES

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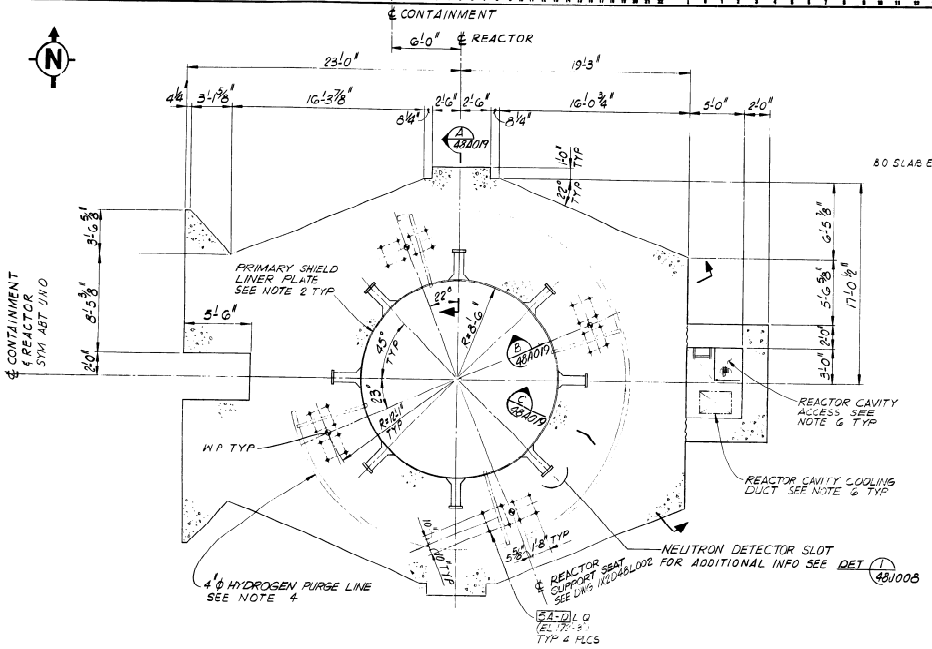
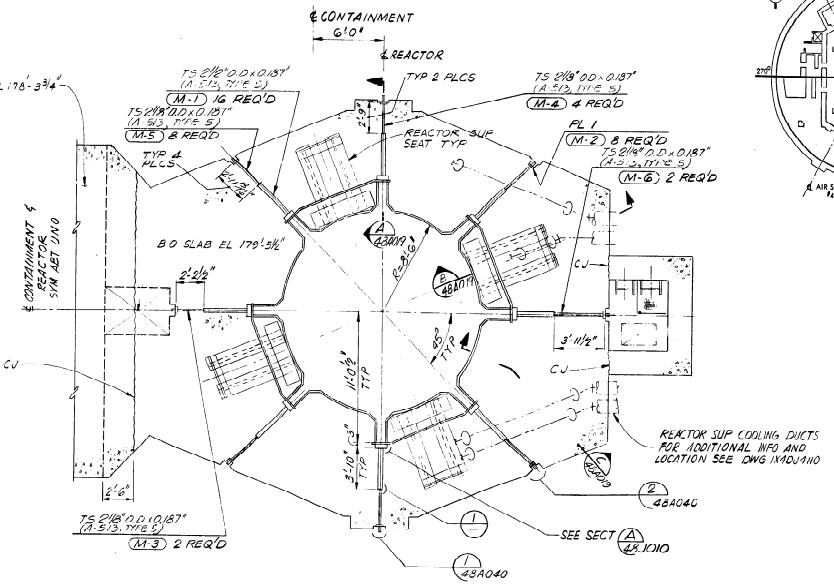
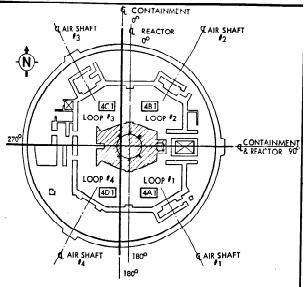



PLAN AT EL 171'-9"




PLAN AT EL 179'-5½"




KEY PLAN UNIT 1

| PC. MK. | REV. | REF. DWG |
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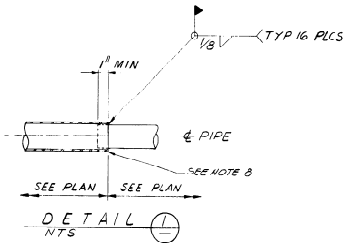
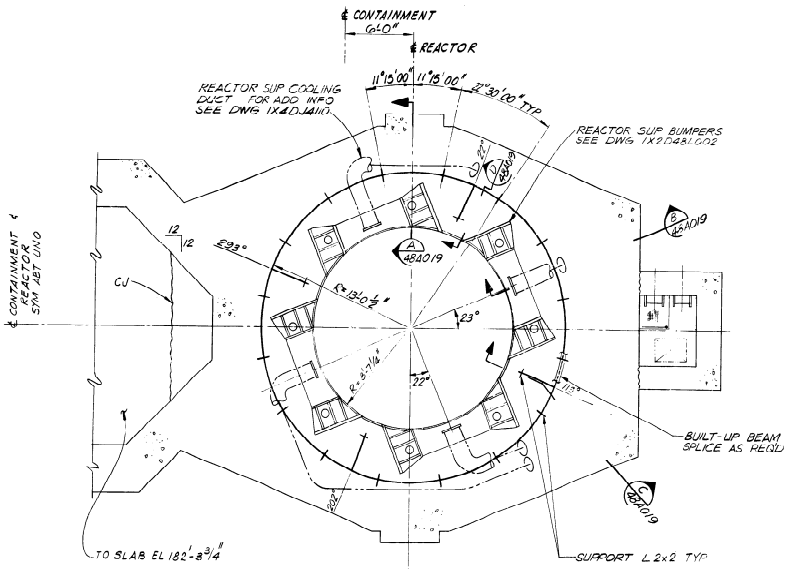
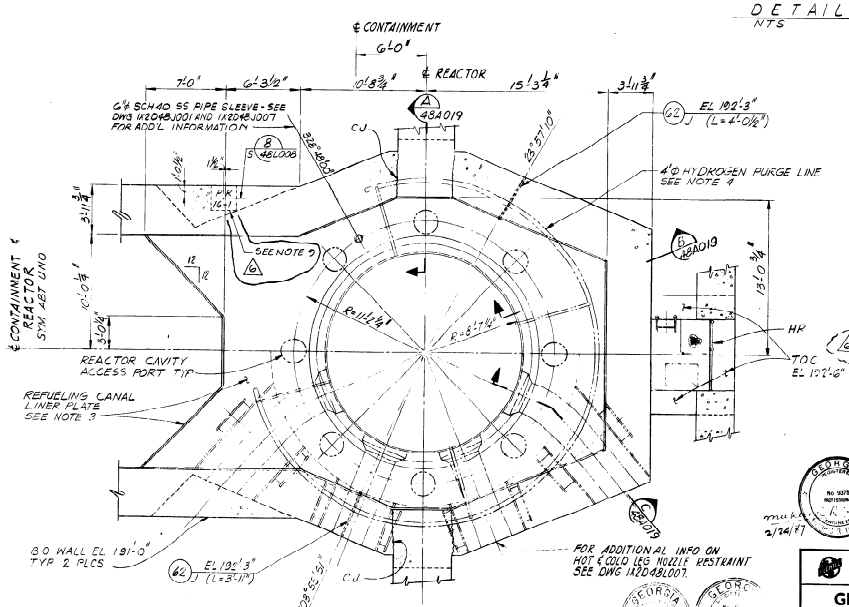
| M-1 | 0 | 576 |
| M-2 | 0 | 576 |
| M-3 | 0 | 576 |
| M-4 | 0 | 576 |
| M-5 | 0 | 576 |
| M-6 | 0 | 576 |




DETAIL 1 NTS




PLAN AT EL 183'-2"




PLAN AT EL 194'-1½"

NOTES:

1. FOR NOTES SEE DWG 1X2D48A004.
2. FOR PRIMARY SHIELD LINER PLATE DETAILS SEE DWGS 1X2D48J008, 1X2D48J010 & 1X27483010.
3. FOR REFUELING CANAL LINER PLATE DETAILS SEE DWGS 1X27D48J001 THRU 1X2D48J005.
4. FOR HYDROGEN PURGE LINE INFORMATION SEE MECH. DWGS 1X2D44A05-A07, 005-807, CGS C07, D05-D07.
5. FOR LOCATION OF N33 EMBEDS SEE 1X2D40L001.
6. FOR REACTOR CAVITY ACCESS & DUCT SUPPORTS SEE DWG 1X2D48C006.
7. FOR LOCATION & DETAILS OF THERMOWELL EMBED SEE DWG C1S0F4013.
8. MACHINE 2½" OF 6"Ø TUBING WALL LESS THAN MEASURED I.D OF 2½" TUBING
9. USAGE OF THIS EMBED SHALL BE IN ACCORDANCE WITH THE PROVISIONS SPECIFIED IN NOTE 13 ON DWG 1X2D40L006.

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 3 | REVISED PER DR CD-6461 | | | | | | | | |
| 2 | INCORP UCN# 5 (C-FCRB-9236) MODIFIED | | | | | | | | |
| 1 | REF DWG NO. ON NOTE 7 SUPERSEDED | | | | | | | | |
| 0 | INCORPORATED DCN#2 | | | | | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 4 | ADDED DCN#1, 2 & 3 - REVISED HOLD | | | | | | | | |
| 3 | REVISED AS SHOWN | | | | | | | | |
| 2 | REMOVED HOLD ALLEY - REVISED CALLOUTS | | | | | | | | |
| 1 | ISSUED FOR CONSTRUCTION | | | | | | | | |

| | | | |
|---|---|---|---|
| ENG. MGR. | | | |
| DATE | | | |

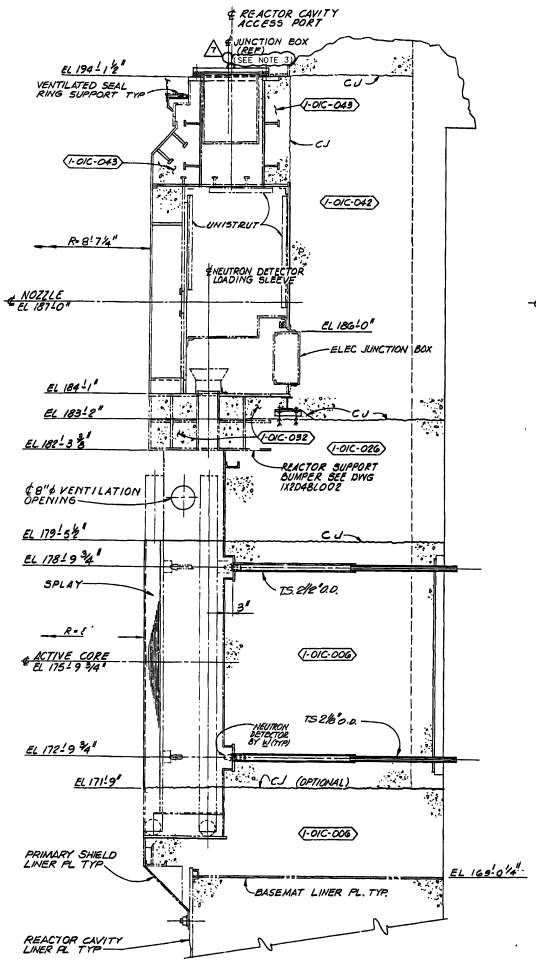
BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

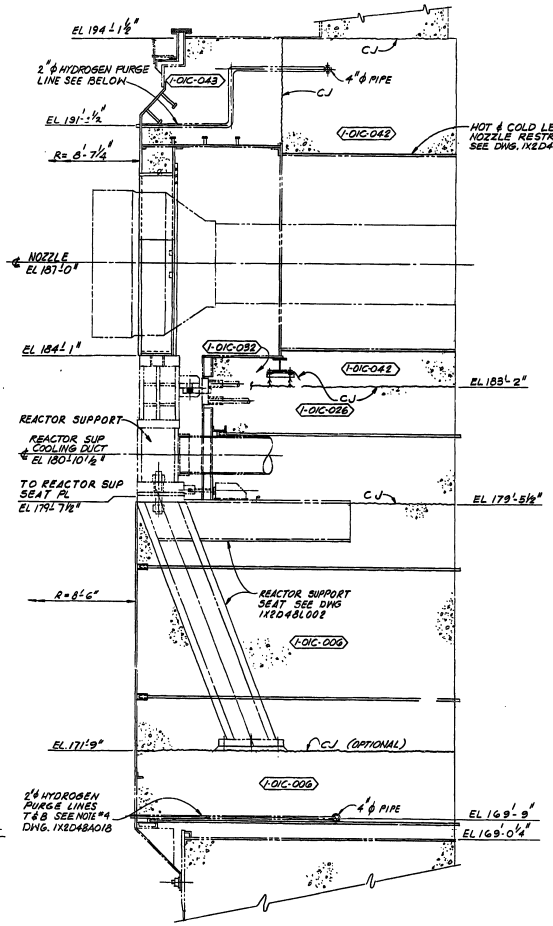
CONTAINMENT INTERNALS
CONCRETE FORMING
PRIMARY SHIELD SHT 1

| SCALE: 1/4" = 1'-0" | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 9510 | 1X2D48A018 | 6 |

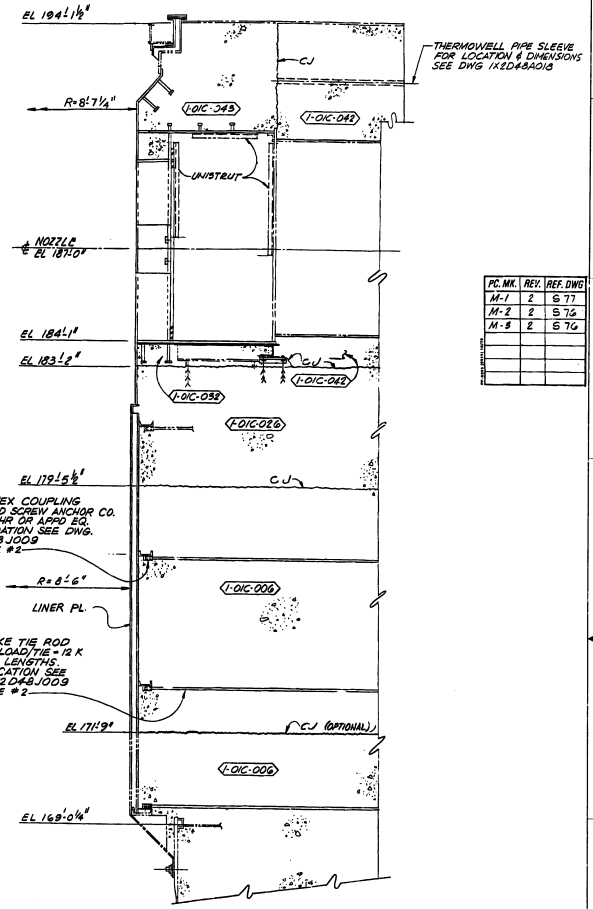
SIZE E 36X44




SECTION (A) 48A018
3/4"=1'-0"

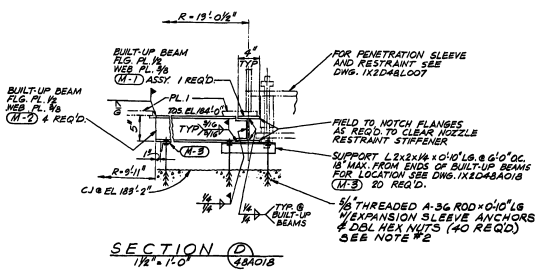



SECTION (B) 48A018
3/4"=1'-0"




SECTION (C) 48A018
3/4"=1'-0"

| PC. MK. | REV. | REF. DWG |
|---|---|---|
| M-1 | 2 | S-77 |
| M-2 | 2 | S-76 |
| M-3 | 2 | S-76 |




SECTION (D) 48A018
1 1/2"=1'-0"

NOTE:
1. FOR NOTES SEE DWG 1X2D48A004 & 1X2D48A018.
2. HEX COUPLINGS, TIE RODS, SUPPORT ANGLES & THREADED RODS ARE SUGGESTED CONSTRUCTION AIDS AND MAY BE FABRICATED FROM MATERIAL STOCKED IN THE FIELD. FIELD SHALL PROVIDE ADEQUATE SUPPORTS TO ENSURE THAT CONSTRUCTION TOLERANCES ARE MAINTAINED.

△7 3. JUNCTION BOX IS OPTIONAL. SEE DETAIL ON DWG 1X2D48J007. (2 48J007)

GEORGIA REGISTERED No. 12826 PROFESSIONAL ENGINEER
4/24/89

GEORGIA REGISTERED No. 12826 PROFESSIONAL ENGINEER
10/2/79

| NO. | REVISIONS | BY | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|---|
| 2 | INCORP. ABN-19808 | JLH | 1/10/83 | SG | | | |
| 1 | ISSUED FOR CONSTRUCTION | | | | | | |
| NO. | REVISIONS | BY | DATE | DR | CHK | SUPV | EGL | EGS | CHFE | PE | QAE |

SEE MICROFILM FOR SIGNATURES

ENG. MGR. DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
CONTAINMENT INTERNALS
CONCRETE FORMING
PRIMARY SHIELD SHT 2

SCALE: AS NOTED | DRAWING NO. 1X2D48A019 | REV. 7
JOB NO. 10604
SIZE E 34x44

THIS DRAWING AND THE DESIGN IT COVERS ARE THE PROPERTY OF SOUTHERN COMPANY SERVICES, INC. THEY ARE MERELY LOANED AND BY THE BORROWER'S EXPRESS AGREEMENT THAT THEY WILL NOT BE REPRODUCED, COPIED, LOANED, EXHIBITED, OR USED EXCEPT IN THE LIMITED WAY AND PRIVATE USE PERMITTED BY ANY WRITTEN CONSENT GIVEN BY THE LENDER TO THE BORROWER

SCALING RULES

| PC MK | REV | REF DWG |
|---|---|---|
| M-1 | 0 | 583 |
| M-2 | | 576 |
| M-3C | | " |
| M-4C | | " |
| M-5C | | " |
| M-GC | 0 | " |
| DELETED | 1 | |
| M-8 | 0 | 583 |
| M-9 | 0 | 583 |
| M-10 | 1 | 584 |
| M-11 | 5 | 584 |
| M-12 | 5 | 584 |
| M-13 | 5 | 584 |
| M-14 | 10 | 584 |
| M-15 | 10 | 584 |
| M-16 | 10 | 584 |
| M-17 | 10 | 584 |
| M-18 | 10 | 584 |
| M-19 | 10 | 584 |

ACCUMULATOR TANK FOR LOCATION SEE DWG 1X2D48E005

TANK

SB ER SCALE

5/8"ø x 14" (SA 193 GR B7) THD ROD (M-1) C-112 REQ'D

HVY HEX NUT TYP

STD WASHER

1½" GROUT

EL 172'-6⅛"

FIN FLOOR

PL 5/8 x 8" SQ

FOR ADDITIONAL INFO SEE TYPE B DWG AX2D94V008

PUMP & PAD

WP FOR LOCATION SEE FLOOR PLANS

5/8"ø x 2'-0" (A36) THD ROD (M-2) 10-REQ'D

1⅛" GROUT

EL 172'-0⅞"

FOR ADDITIONAL INFO SEE SECT A

TACKWELD

REACTOR COOLANT LUBE OIL DRAIN TANK

385 Q (172'-5")

TO 1" GROUT EL 172'-3"

PAD

DET 12

DET 13

DETAIL 1 — 3/8" = 1'-0" — 4BA004 THRU 4BA007

SECTION A — 1" = 1'-0"

DETAIL 2 — 3/4" = 1'-0" — 4BA004 4BA005

SECTION B — 1½" = 1'-0"

DETAIL 3 — 3/4" = 1'-0" — 4BA004 THRU 4BA007

DETAIL 12 — 1½" = 1'-0" — 4BA005

DETAIL 13 — 1½" = 1'-0" — 4BA006

PL ½ 2"ø GROUT HOLES

FOR ADDITIONAL INFO SEE DET G

1¾"ø x 4'-0" (SA193 GR B7) THD ROD (M-3) C-28 REQ'D

FOR ADDITIONAL INFO SEE SECTION A

1⅞"ø x 3½" (SA36) THD ROD (M-4) C-4 ASSY REQ'D

PL 1 x 5" x 3⅛"

FOR ADDITIONAL INFO SEE SECTION A

1"ø x 4½" (SA 36) THD ROD (M-5) C-4 REQ'D

3"ø VIO SLEEVE

PL 1 x 4½" SQ

PL ¾ x 3½" SQ

DETAIL 4 — 3/8" = 1'-0" — 4BA007

DETAIL 5 — 3/8" = 1'-0" — 4BA007

SECTION C — 1" = 1'-0"

DETAIL 6 — 1" = 1'-0" — 4BA003

SECTION D — 1½" = 1'-0"

DETAIL 7 — 1½" = 1'-0" — 4BA005

DETAIL 8 — 1½" = 1'-0" — 4BA003

SECTION E — 1½" = 1'-0"

SECTION F — 1½" = 1'-0"

PL ½ x 20" x 8'-1" W/ 8" ø HOLES (M-11) 1 REQ'D

PL ½ W/ 2-1⅛" ø HOLES (M-12) 1 REQ'D

PL ½ W/ 2-7/8" ø HOLES (M-13) 1 REQ'D

FOR ADDITIONAL INFO SEE SECTION C

FOR ADDITIONAL INFO SEE SECTION F

FOR ADDITIONAL INFO SEE SECTION E

PS (L=1'-8") SEE PLANS FOR LOC

FACE OF WALL

T.O.S. EL 171'-0"

EL 171'-0"

EL 171'-7¾"

LB x 2'-8" x 5/16 LLV SEE PLANS FOR LENGTH (M-5) ASSY.

EL VARIES

NOTE: FOR ALT DETAIL SEE SECT G (1X2D48A005)

SECTION G — 1½" = 1'-0" TYP @ WALL — 4BA006 4BA007

SYM ABT ℄

A3 x 2'-0" x 94V002 SEE PLANS FOR LENGTH (M-6) ASSY

T.O.S. EL 171'-5¾"

EL 171'-5¾"

PS (L=1'-0") SEE PLANS FOR LOC

EL VARIES

SECTION H — 1½" = 1'-0" TYP @ SLAB — 4BA006 4BA007

8" SQ ø

1½" TYP

℄ LIFTING LUG

1⅝" ø HOLE

DETAIL 11 — NTS — 4BA052

REPLACE REMOVABLE SHIELD W/ 1½" ø CADWELD PLATED SLOTTED (AS STD) AT TOP LAYER W/ RECESS WITH SEALANT

LUG

4-½" ø x 5½" LG ANCHORS W/ 4" HOOK

1½" ø SUPERIOR TYPE S PICK-UP INSERT W/ 1½" ø COIL BOLTS OR APPROVED EQUAL (12,500 LBS CAPACITY)

TACK WELD

SECTION L — NTS

EL 198'-10"

TYP 3/16

TYP 3/16

PL ¼ A36

L 3x3x¼ (TYP)

EL 192'-10"

4"(TYP) ℄ HATCH

3'-1"

4'-1"

5'-1"

SUGGESTED CONSTRUCTION SEQUENCE:

1. GLUE SHIELDING FOAM TO CONTAINMENT SHELL WALL LINER PL.
2. WELD PL ¼ TO PERIMETER L 6"x4"x ⅜ FRAME AND VERTICAL L 3"x3"x¼ (TYP @ PLCS)
3. ATTACH ASSEMBLY DESCRIBED ABOVE IN NOTE 2 TO BLOCKOUT FORMWORK AND PLACE INTO FINAL VERTICAL POSITION.

SECTION R — ½" = 1'-0"

SECTION M — NTS

SECTION N — NTS

SECTION P — NTS

NOTCH ANGLE AS REQ'D TO ACCOMMODATE EMBED PL

R=70'-0" TO ℄ CONTAINMENT

3" THICK SHIELDING FOAM REF.

2'-1 1/2" **

270°

ANGLE FRAME ASSY (A)

9" EA END TYP

SEATING NOTCH SHOWN FOR CLARITY

CONTAINMENT SHELL WALL

STEEL HATCH COVER

LINER PLATE

¼ THK

4 ¾" THREADED LIFTING EYEBOLTS FURNISH & FIELD INSTALL (TYP) SEE DET 14

L 3x3x¼ w/ ⅜"ø x 4" LG WELDED STUDS @ 1'-4"±OC MAX AND 2" FROM EA END (MIN 2 STUDS/ANGLE) 6 REQ'D

DETAIL 10 — 3/4" = 1'-0" — 4BA011

NOTE **: THESE CLEAR OPENING DIMENSIONS ARE AT ℄ 142'-10"

SYM ABT ℄

3/8" GAP-FILL GAP W/ 5000 PSI, NON-SHRINK GROUT

REMOVABLE GETS 11'-8" x 11'-8"

CONTAINMENT SHELL WALL

SHIELDING FOAM (REF)

CLOSER (TYP) 1'-1/4"

DRILL & TAP 8"ø x 1½" DEEP FOR ¾"ø THREADED LIFTING EYEBOLTS & PLCS PER HATCH COVER DET (TYP)

PL ½ x

BAR ⅝ x 1'-9" x 2'-9" (BY FIELD)

CONTMT WALL LINER PLATE

NOTE 1: DIMENSIONS OF HATCH CLOSURE TO BE VERIFIED TO THE AS BUILT DIM'S FOR THE GAPS INDICATED ON DWG. FOR AS BUILT DIMS SEE CHERR-4475. ALL ASSEMBLIES SHALL BE ASS EXCEPT LIFTING EYEBOLTS.

L 6 x 4 x ⅜ W/ ⅝"ø x 4" LG WELDED STUDS @ 1'-0" OC L.H. (TYP)

EL 198'-10"

EL 192'-10"

5-1" PLE BOLTED W/ 2-1⅛"ø x 0'-4¾" CSK HD BOLTS (M-17) 1-ASSY

SECTION J — 3/4" = 1'-0"

SECTION K — 3/4" = 1'-0"

SEE NOTE 5

EL 198'-10"

1½"ø x 0'-6¼" CSK HD BOLTS (TYP)

EACH OF THE SEVEN STEEL HATCH COVERS CONSISTS OF 3-1" PLE BOLTED W/ 6-⅜"ø x 0'-2¾" CSK HD BOLTS (TYP) (M-15) 1-ASSY

4-1" PLS BOLTED W/ 8-⅜"ø x 0'-2¾" CSK HD BOLTS (2-411BEE) (M-16) 1-ASSY

3 SIDES

GETS

L2 x 2 x ¼ (M-14) 1-ASSY

2-1" PLE BOLTED W/ 2-1⅛"ø x 0'-1¾" CSK HD BOLTS (TYP) (M-18) 1-ASSY

DETAIL 14 — 1½" = 1'-0"

DETAIL 15 — 1½" = 1'-0"

NOTES:

1. FOR NOTES SEE DWG 1X2D48A004.
2. FOR ADDITIONAL INFORMATION SEE DWG AX2D94V008.
3. ANCHOR BOLT FIELD PLACEMENT SHALL BE BASED ON THE FOLLOWING:
   a. NOTE 12 ON DWG AX2D94V008.
   b. THE LOCATION OF ANCHOR BOLTS WITHOUT SLEEVES ± 1/8".
4. EYEBOLTS TO BE COMMERCIAL GRADE NON-Q ITEM.
5. FIELD MAY SUBSTITUTE THICKER PLATES TO PROVIDE EQUIVALENT THICKNESS OF LAYERED PLUG SHIELDING PLATES.

GEORGIA REGISTERED PROFESSIONAL ENGINEER

E.S.D.

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | REV. PER DCN'S (INCRP-GEAS) FROM DWG 1X2D48A011 | 5-19-83 | MK | WC | JVW | DDW | | | | |
| 3 | REV DET 10, 14 & 15; SECT J & K; AND ADDED PC MK'S (NEW) (INCRP-5942) | 3-8-83 | MRC | JVW | DDW | | | | | |
| 2 | REV SECT K & DET 10 FOR CLARIFICATION | 1-31-83 | JCY | JVW | DDW | | | | | |
| 1 | REVISED AS SHOWN (INCRP-5942) REVISE SECT A | 1-14-83 | MK | JVW | DDW | | | | | |
| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
| 4 | ADDED DCN # 1 & 2 | 8-14-82 | MRC | JVW | DDW | | | | | |
| 3 | ADDED DET'S 12 & 13, SECTS M, N & P REV DET 4 | 8-14-82 | MRC | JVW | DDW | | | | | |
| 2 | REVISED DET 10 & SECT J & ADDED SECT R | 7-13-82 | MRC | JVW | DDW | | | | | |
| 1 | ADDED DET'S 11 & 14 & SECTS K & L | 6-30-82 | MRC | JVW | DDW | | | | | |
| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |
| 4 | INC. DCN'S 1 & 2, MOVED DCN 2 & ENLGD COVER | 4-13-83 | MK | WC | JVW | DDW | | | | |
| 3 | REVISED AS SHOWN | 1-19-83 | RM | JTL | DDW | | | | | |
| 2 | INC. DCN'S (C-RKB-15,140) ON DWG 1X2D48A051 | 10-6-82 | RJ | WC | JVW | DDW | | | | |
| 1 | ISSUED FOR CONSTRUCTION | 5-13-80 | DR | WC | JVW | DDW | | | | |
| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGL | EGS | CHF. E | P.E. | Q.A.E. |

ENG. MGR. — DATE

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT INTERNALS CONCRETE FORMING SECTIONS & DETAILS SH. 1

SCALE: AS NOTED — DRAWING NO. — REV.

JOB NO. 9510 — 1X2D48A050 — 12

SIZE E 36X44

I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS ON THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT SUPPLIED TO MICROGRAPHICS.

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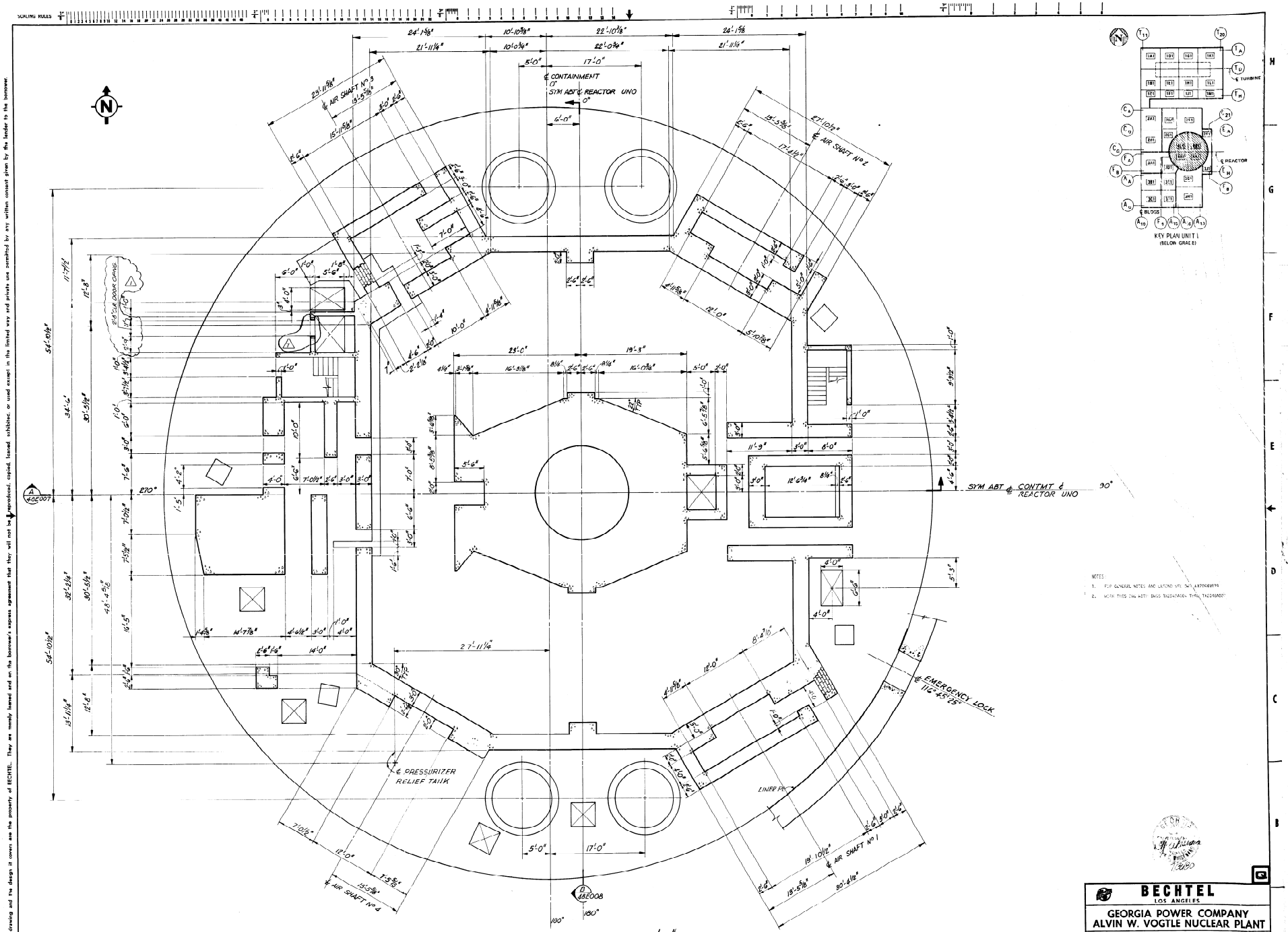
7X2D48A050







NOTES:
1. FOR GENERAL NOTES AND LAYOUT SEE DWG 1X2D16W019
2. WORK THIS DWG WITH ENGR TAG/CATALOG TYPE 1X2D16M007

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| NO | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 4 | ISSUED FOR CHF E APPROVAL INCORP DCN# 1 | 9-1-2 | TK | SS-JH | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 3 | REVISED AS SHOWN | 1-3-3 | LSW | [illegible] | JTK | [illegible] | [illegible] | [illegible] | [illegible] |
| 2 | REVISED ADDED/DELETED DIMENSIONS | 2-13 | RV | [illegible] | JTT | [illegible] | [illegible] | [illegible] | [illegible] |
| 1 | REDRAWN & GENERAL REVISION | 4-8-10 | RM | [illegible] | GO | [illegible] | [illegible] | [illegible] | [illegible] |

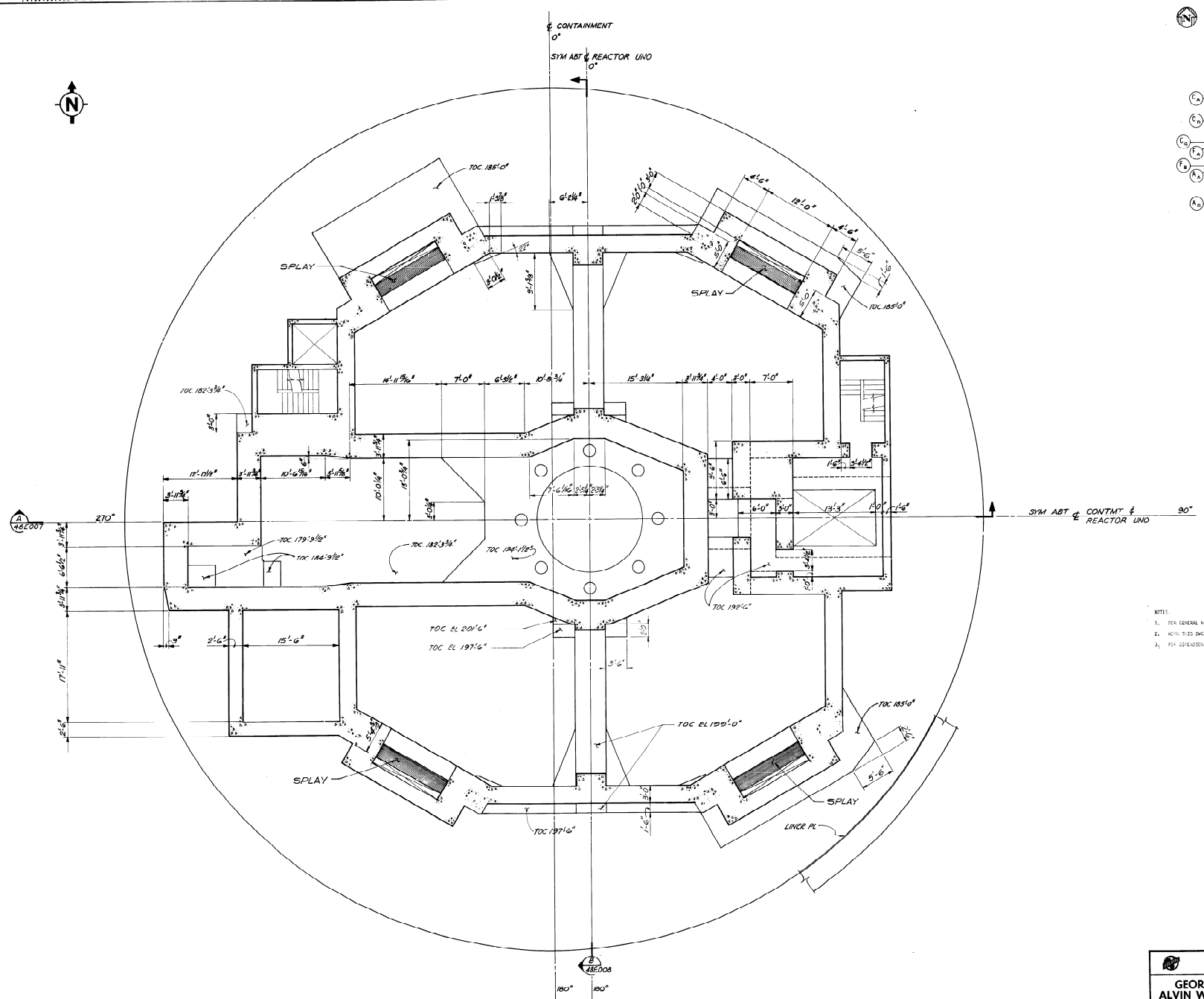
ENG. MGR. | DATE

**BECHTEL**
LOS ANGELES

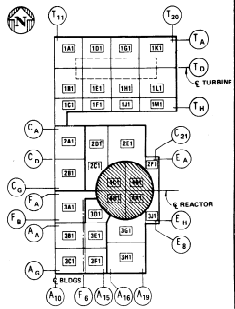
**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT INTERNALS
GENERAL ARRANGEMENT
PLAN AT EL 171'-9"

SCALE: 3/16"=1'-0" | DRAWING NO. 1X2D48E003 | REV. 7
JOB NO. 9510
SIZE E 36044




PLAN @ ELEVATION 195'-0"




NOTES:
1. FOR GENERAL NOTES AND LEGEND SEE DWG XXDDAAYYYY
2. WORK THIS DWG WITH DWGS XXDDAAGGGG THRU XXDDAAHHHH
3. FOR DIMENSIONS NOT SHOWN SEE DWGS XXDDAABBDD

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 1 | INCORPORATED COMMENTS | 4/12/84 | J.C. | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 2 | ERECT. DIS SHOWN | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 3 | REVISED, ADDED & DELETED DIMENSIONS | 5/1/84 | M.V. | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| 4 | REDRAWN & GENERAL REVISION | 9/6/84 | RM | [illegible] | GD | [illegible] | [illegible] | [illegible] | [illegible] |
| 5 | ISSUED FOR CHF E APPROVAL | 11/16/83 | JDY | [illegible] | [illegible] | DDH | JBT | JN | JW |

ENG. MGR. [illegible] DATE

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT INTERNALS
GENERAL ARRANGEMENT
PLAN AT EL. 195'-0"

SCALE: 3/16"=1'-0" | DRAWING NO. 1X1DAAE004 | REV. 8
JOB NO. 9510
SIZE E 36044

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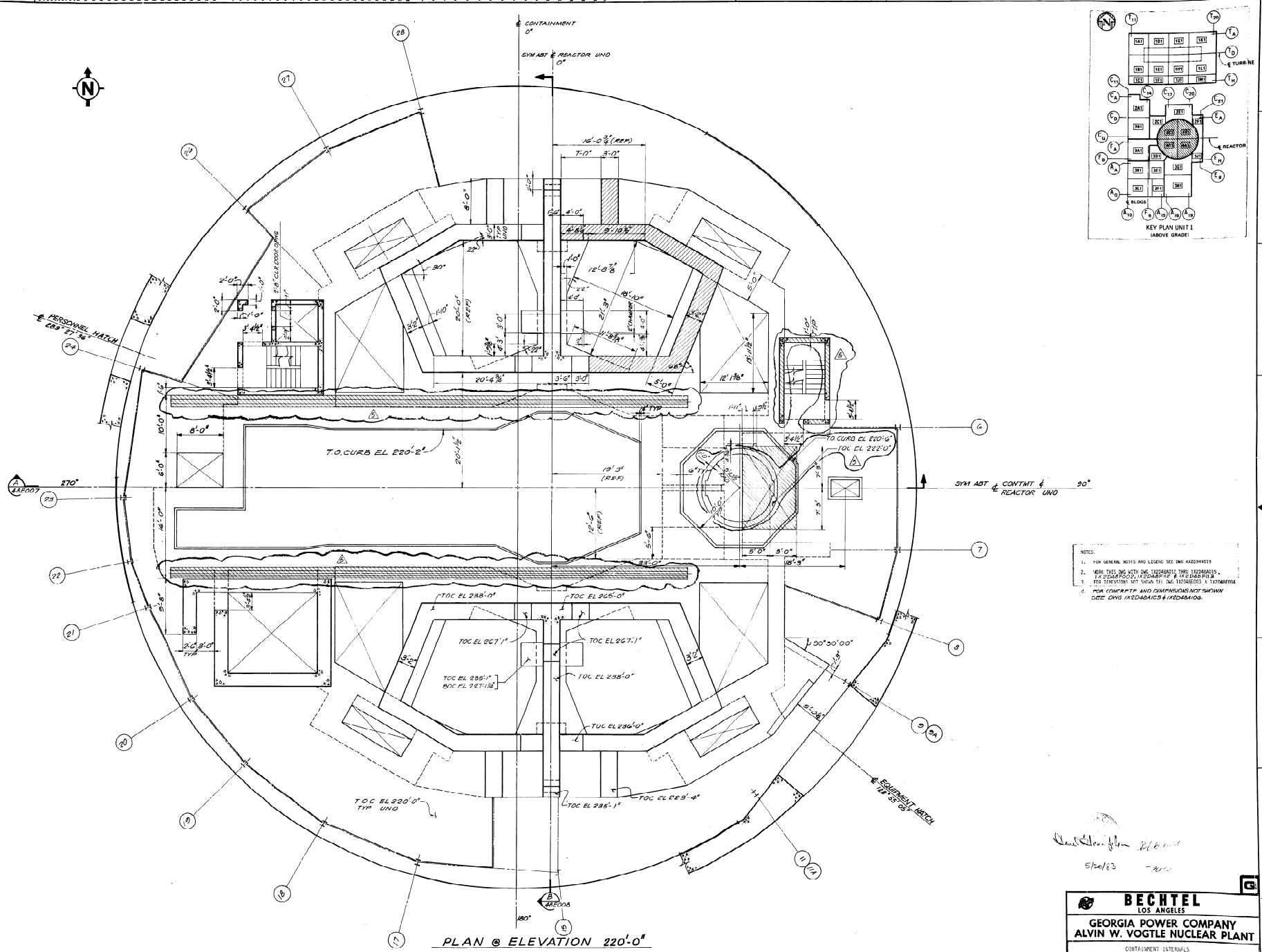


NOTES:
1. FOR GENERAL NOTES AND LEGEND SEE DWG AX2D94V019
2. WORK THIS DWG WITH DWG 1X2D48A012 THRU 1X2D48A015, 1X2D48F002, 1X2D48F112 & 1X2D48F113
3. FOR DIMENSIONS NOT SHOWN SEE DWG 1X2D48E003 & 1X2D48E004
4. FOR CONCRETE AND DIMENSIONS NOT SHOWN SEE DWG 1X2D48A1C5 & 1X2D48A106.

5/20/83

**BECHTEL**
LOS ANGELES

**GEORGIA POWER COMPANY**
**ALVIN W. VOGTLE NUCLEAR PLANT**

CONTAINMENT INTERNALS
GENERAL ARRANGEMENT
PLAN AT EL. 220'-0"

SCALE: 3/16"=1'-0" | DRAWING NO. 1X2D48E005 | REV. 5
JOB NO. 9510

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 5 | REVISED CIVIL CONFIGURATION & ADDED DIM'S | | | | | | | | |
| 4 | GENERAL REVISION | | | | | | | | |
| 3 | REV AS SHOWN | | | | | | | | |
| 2 | ISSUED FOR CONSTRUCTION | | | | | | | | |
| 1 | INCORP DCNS # 566 | 2-21-83 | MC | JOY | | | | | |
| 0 | ISSUED FOR CHF E APPROVAL, INCORP DCNS 941, ADDED TEMP BLOCKOUT OVER SEAL TABLE | 1-14-83 | MC | JOY | | | | | |

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1X2D48E007

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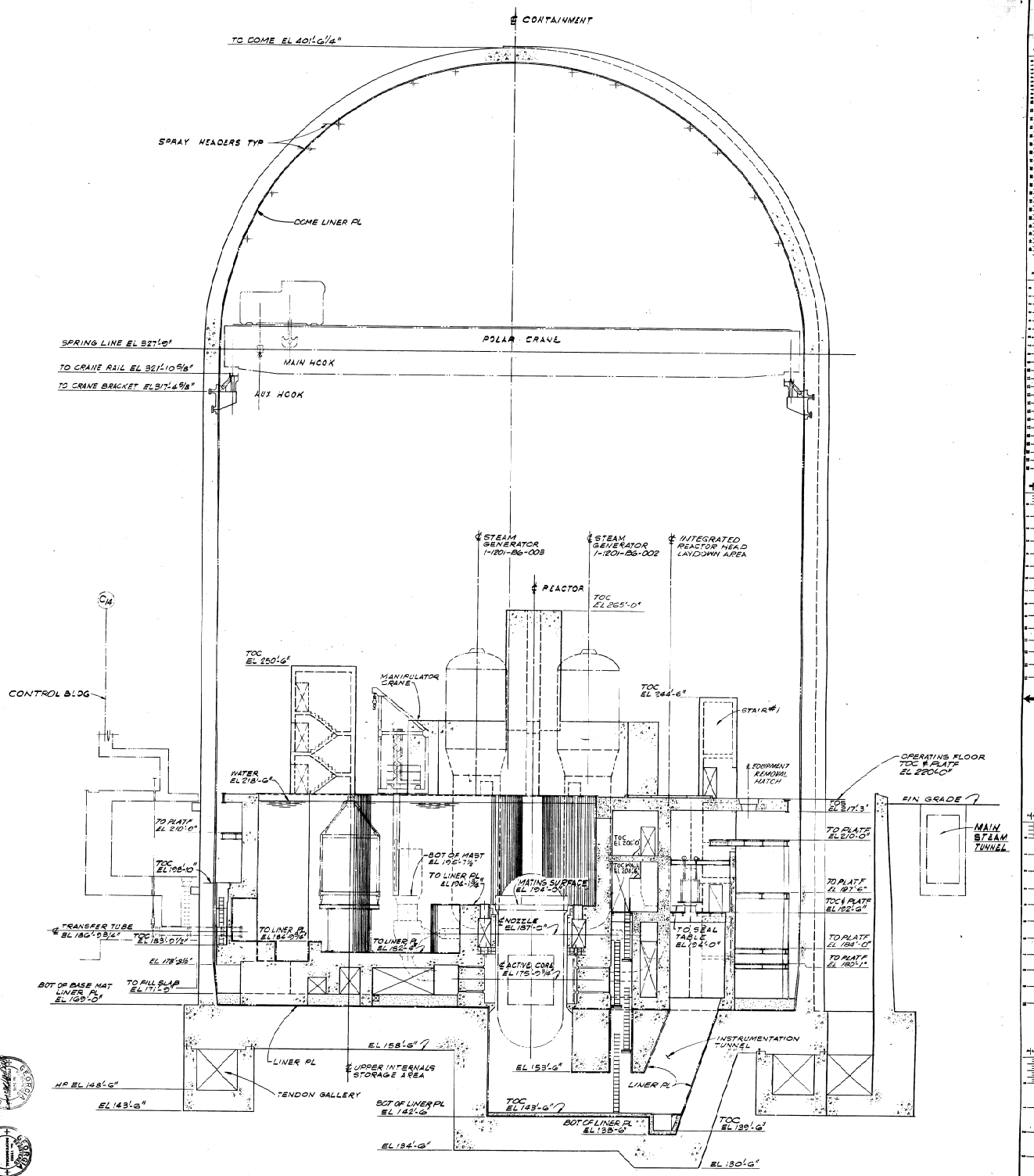


SECTION A

46E008
46E004
46E005
46E006

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGE | CHE | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| | GENERAL REVISIONS | | | | | | | | |
| 1 | ISSUED FOR CONSTRUCTION | 5-27-77 | CWH | | | | | | |
| 2 | REVISED PER CCAS 1 THROUGH 5 | 8-19-77 | CWH | | | | | | |
| 3 | REVISED PER CCAS 6 THROUGH 9 | 9-2-77 | CWH | | | | | | |
| 4 | REVISED PER CCAS 10 THROUGH 13 | 9-7-77 | CWH | | | | | | |
| 5 | REVISED PER CCAS 14 THROUGH 17 | 9-12-77 | CWH | | | | | | |
| 6 | REVISED PER CCAS 18 THROUGH 21 | 9-19-77 | CWH | | | | | | |
| 7 | REVISED PER CCAS 22 THROUGH 25 | 9-26-77 | CWH | | | | | | |
| 8 | REVISED PER CCAS 26 THROUGH 29 | 10-3-77 | CWH | | | | | | |
| 9 | REVISED PER CCAS 30 THROUGH 33 | 10-10-77 | CWH | | | | | | |
| 10 | REVISED PER CCAS 34 THROUGH 37 | 10-17-77 | CWH | | | | | | |
| 11 | REVISED PER CCAS 38 THROUGH 41 | 10-24-77 | CWH | | | | | | |
| 12 | REVISED PER CCAS 42 THROUGH 45 | 10-31-77 | CWH | | | | | | |
| 13 | REVISED PER CCAS 46 THROUGH 49 | 11-7-77 | CWH | | | | | | |
| 14 | REVISED PER CCAS 50 THROUGH 53 | 11-14-77 | CWH | | | | | | |
| 15 | REVISED PER CCAS 54 THROUGH 57 | 11-21-77 | CWH | | | | | | |
| 16 | REVISED PER CCAS 58 THROUGH 61 | 11-28-77 | CWH | | | | | | |
| 17 | REVISED PER CCAS 62 THROUGH 65 | 12-5-77 | CWH | | | | | | |
| 18 | REVISED PER CCAS 66 THROUGH 69 | 12-12-77 | CWH | | | | | | |
| 19 | REVISED PER CCAS 70 THROUGH 73 | 12-19-77 | CWH | | | | | | |
| 20 | REVISED PER CCAS 74 THROUGH 77 | 12-26-77 | CWH | | | | | | |
| 21 | REVISED PER CCAS 78 THROUGH 81 | 1-2-78 | CWH | | | | | | |
| 22 | REVISED PER CCAS 82 THROUGH 85 | 1-9-78 | CWH | | | | | | |
| 23 | REVISED PER CCAS 86 THROUGH 89 | 1-16-78 | CWH | | | | | | |
| 24 | REVISED PER CCAS 90 THROUGH 93 | 1-23-78 | CWH | | | | | | |
| 25 | REVISED PER CCAS 94 THROUGH 97 | 1-30-78 | CWH | | | | | | |
| 26 | REVISED PER CCAS 98 THROUGH 101 | 2-6-78 | CWH | | | | | | |
| 27 | REVISED PER CCAS 102 THROUGH 105 | 2-13-78 | CWH | | | | | | |
| 28 | REVISED PER CCAS 106 THROUGH 109 | 2-20-78 | CWH | | | | | | |
| 29 | REVISED PER CCAS 110 THROUGH 113 | 2-27-78 | CWH | | | | | | |
| 30 | REVISED PER CCAS 114 THROUGH 117 | 3-6-78 | CWH | | | | | | |
| 31 | REVISED PER CCAS 118 THROUGH 121 | 3-13-78 | CWH | | | | | | |
| 32 | REVISED PER CCAS 122 THROUGH 125 | 3-20-78 | CWH | | | | | | |
| 33 | REVISED PER CCAS 126 THROUGH 129 | 3-27-78 | CWH | | | | | | |
| 34 | REVISED PER CCAS 130 THROUGH 133 | 4-3-78 | CWH | | | | | | |
| 35 | REVISED PER CCAS 134 THROUGH 137 | 4-10-78 | CWH | | | | | | |
| 36 | REVISED PER CCAS 138 THROUGH 141 | 4-17-78 | CWH | | | | | | |
| 37 | REVISED PER CCAS 142 THROUGH 145 | 4-24-78 | CWH | | | | | | |
| 38 | REVISED PER CCAS 146 THROUGH 149 | 5-1-78 | CWH | | | | | | |
| 39 | REVISED PER CCAS 150 THROUGH 153 | 5-8-78 | CWH | | | | | | |
| 40 | REVISED PER CCAS 154 THROUGH 157 | 5-15-78 | CWH | | | | | | |
| 41 | REVISED PER CCAS 158 THROUGH 161 | 5-22-78 | CWH | | | | | | |
| 42 | REVISED PER CCAS 162 THROUGH 165 | 5-29-78 | CWH | | | | | | |
| 43 | REVISED PER CCAS 166 THROUGH 169 | 6-5-78 | CWH | | | | | | |
| 44 | REVISED PER CCAS 170 THROUGH 173 | 6-12-78 | CWH | | | | | | |
| 45 | REVISED PER CCAS 174 THROUGH 177 | 6-19-78 | CWH | | | | | | |
| 46 | REVISED PER CCAS 178 THROUGH 181 | 6-26-78 | CWH | | | | | | |
| 47 | REVISED PER CCAS 182 THROUGH 185 | 7-3-78 | CWH | | | | | | |
| 48 | REVISED PER CCAS 186 THROUGH 189 | 7-10-78 | CWH | | | | | | |
| 49 | REVISED PER CCAS 190 THROUGH 193 | 7-17-78 | CWH | | | | | | |
| 50 | REVISED PER CCAS 194 THROUGH 197 | 7-24-78 | CWH | | | | | | |
| 51 | REVISED PER CCAS 198 THROUGH 201 | 7-31-78 | CWH | | | | | | |
| 52 | REVISED PER CCAS 202 THROUGH 205 | 8-7-78 | CWH | | | | | | |
| 53 | REVISED PER CCAS 206 THROUGH 209 | 8-14-78 | CWH | | | | | | |
| 54 | REVISED PER CCAS 210 THROUGH 213 | 8-21-78 | CWH | | | | | | |
| 55 | REVISED PER CCAS 214 THROUGH 217 | 8-28-78 | CWH | | | | | | |
| 56 | REVISED PER CCAS 218 THROUGH 221 | 9-4-78 | CWH | | | | | | |
| 57 | REVISED PER CCAS 222 THROUGH 225 | 9-11-78 | CWH | | | | | | |
| 58 | REVISED PER CCAS 226 THROUGH 229 | 9-18-78 | CWH | | | | | | |
| 59 | REVISED PER CCAS 230 THROUGH 233 | 9-25-78 | CWH | | | | | | |
| 60 | REVISED PER CCAS 234 THROUGH 237 | 10-2-78 | CWH | | | | | | |
| 61 | REVISED PER CCAS 238 THROUGH 241 | 10-9-78 | CWH | | | | | | |
| 62 | REVISED PER CCAS 242 THROUGH 245 | 10-16-78 | CWH | | | | | | |
| 63 | REVISED PER CCAS 246 THROUGH 249 | 10-23-78 | CWH | | | | | | |
| 64 | REVISED PER CCAS 250 THROUGH 253 | 10-30-78 | CWH | | | | | | |
| 65 | REVISED PER CCAS 254 THROUGH 257 | 11-6-78 | CWH | | | | | | |
| 66 | REVISED PER CCAS 258 THROUGH 261 | 11-13-78 | CWH | | | | | | |
| 67 | REVISED PER CCAS 262 THROUGH 265 | 11-20-78 | CWH | | | | | | |
| 68 | REVISED PER CCAS 266 THROUGH 269 | 11-27-78 | CWH | | | | | | |
| 69 | REVISED PER CCAS 270 THROUGH 273 | 12-4-78 | CWH | | | | | | |
| 70 | REVISED PER CCAS 274 THROUGH 277 | 12-11-78 | CWH | | | | | | |
| 71 | REVISED PER CCAS 278 THROUGH 281 | 12-18-78 | CWH | | | | | | |
| 72 | REVISED PER CCAS 282 THROUGH 285 | 12-25-78 | CWH | | | | | | |
| 73 | REVISED PER CCAS 286 THROUGH 289 | 1-1-79 | CWH | | | | | | |
| 74 | REVISED PER CCAS 290 THROUGH 293 | 1-8-79 | CWH | | | | | | |
| 75 | REVISED PER CCAS 294 THROUGH 297 | 1-15-79 | CWH | | | | | | |
| 76 | REVISED PER CCAS 298 THROUGH 301 | 1-22-79 | CWH | | | | | | |
| 77 | REVISED PER CCAS 302 THROUGH 305 | 1-29-79 | CWH | | | | | | |
| 78 | REVISED PER CCAS 306 THROUGH 309 | 2-5-79 | CWH | | | | | | |
| 79 | REVISED PER CCAS 310 THROUGH 313 | 2-12-79 | CWH | | | | | | |
| 80 | REVISED PER CCAS 314 THROUGH 317 | 2-19-79 | CWH | | | | | | |
| 81 | REVISED PER CCAS 318 THROUGH 321 | 2-26-79 | CWH | | | | | | |
| 82 | REVISED PER CCAS 322 THROUGH 325 | 3-5-79 | CWH | | | | | | |
| 83 | REVISED PER CCAS 326 THROUGH 329 | 3-12-79 | CWH | | | | | | |
| 84 | REVISED PER CCAS 330 THROUGH 333 | 3-19-79 | CWH | | | | | | |
| 85 | REVISED PER CCAS 334 THROUGH 337 | 3-26-79 | CWH | | | | | | |
| 86 | REVISED PER CCAS 338 THROUGH 341 | 4-2-79 | CWH | | | | | | |
| 87 | REVISED PER CCAS 342 THROUGH 345 | 4-9-79 | CWH | | | | | | |
| 88 | REVISED PER CCAS 346 THROUGH 349 | 4-16-79 | CWH | | | | | | |
| 89 | REVISED PER CCAS 350 THROUGH 353 | 4-23-79 | CWH | | | | | | |
| 90 | REVISED PER CCAS 354 THROUGH 357 | 4-30-79 | CWH | | | | | | |
| 91 | REVISED PER CCAS 358 THROUGH 361 | 5-7-79 | CWH | | | | | | |
| 92 | REVISED PER CCAS 362 THROUGH 365 | 5-14-79 | CWH | | | | | | |
| 93 | REVISED PER CCAS 366 THROUGH 369 | 5-21-79 | CWH | | | | | | |
| 94 | REVISED PER CCAS 370 THROUGH 373 | 5-28-79 | CWH | | | | | | |
| 95 | REVISED PER CCAS 374 THROUGH 377 | 6-4-79 | CWH | | | | | | |
| 96 | REVISED PER CCAS 378 THROUGH 381 | 6-11-79 | CWH | | | | | | |
| 97 | REVISED PER CCAS 382 THROUGH 385 | 6-18-79 | CWH | | | | | | |
| 98 | REVISED PER CCAS 386 THROUGH 389 | 6-25-79 | CWH | | | | | | |
| 99 | REVISED PER CCAS 390 THROUGH 393 | 7-2-79 | CWH | | | | | | |
| 100 | REVISED PER CCAS 394 THROUGH 397 | 7-9-79 | CWH | | | | | | |
| 101 | REVISED PER CCAS 398 THROUGH 401 | 7-16-79 | CWH | | | | | | |
| 102 | REVISED PER CCAS 402 THROUGH 405 | 7-23-79 | CWH | | | | | | |
| 103 | REVISED PER CCAS 406 THROUGH 409 | 7-30-79 | CWH | | | | | | |
| 104 | REVISED PER CCAS 410 THROUGH 413 | 8-6-79 | CWH | | | | | | |
| 105 | REVISED PER CCAS 414 THROUGH 417 | 8-13-79 | CWH | | | | | | |
| 106 | REVISED PER CCAS 418 THROUGH 421 | 8-20-79 | CWH | | | | | | |
| 107 | REVISED PER CCAS 422 THROUGH 425 | 8-27-79 | CWH | | | | | | |
| 108 | REVISED PER CCAS 426 THROUGH 429 | 9-3-79 | CWH | | | | | | |
| 109 | REVISED PER CCAS 430 THROUGH 433 | 9-10-79 | CWH | | | | | | |
| 110 | REVISED PER CCAS 434 THROUGH 437 | 9-17-79 | CWH | | | | | | |
| 111 | REVISED PER CCAS 438 THROUGH 441 | 9-24-79 | CWH | | | | | | |
| 112 | REVISED PER CCAS 442 THROUGH 445 | 10-1-79 | CWH | | | | | | |
| 113 | REVISED PER CCAS 446 THROUGH 449 | 10-8-79 | CWH | | | | | | |
| 114 | REVISED PER CCAS 450 THROUGH 453 | 10-15-79 | CWH | | | | | | |
| 115 | REVISED PER CCAS 454 THROUGH 457 | 10-22-79 | CWH | | | | | | |
| 116 | REVISED PER CCAS 458 THROUGH 461 | 10-29-79 | CWH | | | | | | |
| 117 | REVISED PER CCAS 462 THROUGH 465 | 11-5-79 | CWH | | | | | | |
| 118 | REVISED PER CCAS 466 THROUGH 469 | 11-12-79 | CWH | | | | | | |
| 119 | REVISED PER CCAS 470 THROUGH 473 | 11-19-79 | CWH | | | | | | |
| 120 | REVISED PER CCAS 474 THROUGH 477 | 11-26-79 | CWH | | | | | | |
| 121 | REVISED PER CCAS 478 THROUGH 481 | 12-3-79 | CWH | | | | | | |
| 122 | REVISED PER CCAS 482 THROUGH 485 | 12-10-79 | CWH | | | | | | |
| 123 | REVISED PER CCAS 486 THROUGH 489 | 12-17-79 | CWH | | | | | | |
| 124 | REVISED PER CCAS 490 THROUGH 493 | 12-24-79 | CWH | | | | | | |
| 125 | REVISED PER CCAS 494 THROUGH 497 | 12-31-79 | CWH | | | | | | |
| 126 | REVISED PER CCAS 498 THROUGH 501 | 1-7-80 | CWH | | | | | | |
| 127 | REVISED PER CCAS 502 THROUGH 505 | 1-14-80 | CWH | | | | | | |
| 128 | REVISED PER CCAS 506 THROUGH 509 | 1-21-80 | CWH | | | | | | |
| 129 | REVISED PER CCAS 510 THROUGH 513 | 1-28-80 | CWH | | | | | | |
| 130 | REVISED PER CCAS 514 THROUGH 517 | 2-4-80 | CWH | | | | | | |
| 131 | REVISED PER CCAS 518 THROUGH 521 | 2-11-80 | CWH | | | | | | |
| 132 | REVISED PER CCAS 522 THROUGH 525 | 2-18-80 | CWH | | | | | | |
| 133 | REVISED PER CCAS 526 THROUGH 529 | 2-25-80 | CWH | | | | | | |
| 134 | REVISED PER CCAS 530 THROUGH 533 | 3-3-80 | CWH | | | | | | |
| 135 | REVISED PER CCAS 534 THROUGH 537 | 3-10-80 | CWH | | | | | | |
| 136 | REVISED PER CCAS 538 THROUGH 541 | 3-17-80 | CWH | | | | | | |
| 137 | REVISED PER CCAS 542 THROUGH 545 | 3-24-80 | CWH | | | | | | |
| 138 | REVISED PER CCAS 546 THROUGH 549 | 3-31-80 | CWH | | | | | | |
| 139 | REVISED PER CCAS 550 THROUGH 553 | 4-7-80 | CWH | | | | | | |
| 140 | REVISED PER CCAS 554 THROUGH 557 | 4-14-80 | CWH | | | | | | |
| 141 | REVISED PER CCAS 558 THROUGH 561 | 4-21-80 | CWH | | | | | | |
| 142 | REVISED PER CCAS 562 THROUGH 565 | 4-28-80 | CWH | | | | | | |
| 143 | REVISED PER CCAS 566 THROUGH 569 | 5-5-80 | CWH | | | | | | |
| 144 | REVISED PER CCAS 570 THROUGH 573 | 5-12-80 | CWH | | | | | | |
| 145 | REVISED PER CCAS 574 THROUGH 577 | 5-19-80 | CWH | | | | | | |
| 146 | REVISED PER CCAS 578 THROUGH 581 | 5-26-80 | CWH | | | | | | |
| 147 | REVISED PER CCAS 582 THROUGH 585 | 6-2-80 | CWH | | | | | | |
| 148 | REVISED PER CCAS 586 THROUGH 589 | 6-9-80 | CWH | | | | | | |
| 149 | REVISED PER CCAS 590 THROUGH 593 | 6-16-80 | CWH | | | | | | |
| 150 | REVISED PER CCAS 594 THROUGH 597 | 6-23-80 | CWH | | | | | | |
| 151 | REVISED PER CCAS 598 THROUGH 601 | 6-30-80 | CWH | | | | | | |
| 152 | REVISED PER CCAS 602 THROUGH 605 | 7-7-80 | CWH | | | | | | |
| 153 | REVISED PER CCAS 606 THROUGH 609 | 7-14-80 | CWH | | | | | | |
| 154 | REVISED PER CCAS 610 THROUGH 613 | 7-21-80 | CWH | | | | | | |
| 155 | REVISED PER CCAS 614 THROUGH 617 | 7-28-80 | CWH | | | | | | |
| 156 | REVISED PER CCAS 618 THROUGH 621 | 8-4-80 | CWH | | | | | | |
| 157 | REVISED PER CCAS 622 THROUGH 625 | 8-11-80 | CWH | | | | | | |
| 158 | REVISED PER CCAS 626 THROUGH 629 | 8-18-80 | CWH | | | | | | |
| 159 | REVISED PER CCAS 630 THROUGH 633 | 8-25-80 | CWH | | | | | | |
| 160 | REVISED PER CCAS 634 THROUGH 637 | 9-1-80 | CWH | | | | | | |
| 161 | REVISED PER CCAS 638 THROUGH 641 | 9-8-80 | CWH | | | | | | |
| 162 | REVISED PER CCAS 642 THROUGH 645 | 9-15-80 | CWH | | | | | | |
| 163 | REVISED PER CCAS 646 THROUGH 649 | 9-22-80 | CWH | | | | | | |
| 164 | REVISED PER CCAS 650 THROUGH 653 | 9-29-80 | CWH | | | | | | |
| 165 | REVISED PER CCAS 654 THROUGH 657 | 10-6-80 | CWH | | | | | | |
| 166 | REVISED PER CCAS 658 THROUGH 661 | 10-13-80 | CWH | | | | | | |
| 167 | REVISED PER CCAS 662 THROUGH 665 | 10-20-80 | CWH | | | | | | |
| 168 | REVISED PER CCAS 666 THROUGH 669 | 10-27-80 | CWH | | | | | | |
| 169 | REVISED PER CCAS 670 THROUGH 673 | 11-3-80 | CWH | | | | | | |
| 170 | REVISED PER CCAS 674 THROUGH 677 | 11-10-80 | CWH | | | | | | |
| 171 | REVISED PER CCAS 678 THROUGH 681 | 11-17-80 | CWH | | | | | | |
| 172 | REVISED PER CCAS 682 THROUGH 685 | 11-24-80 | CWH | | | | | | |
| 173 | REVISED PER CCAS 686 THROUGH 689 | 12-1-80 | CWH | | | | | | |
| 174 | REVISED PER CCAS 690 THROUGH 693 | 12-8-80 | CWH | | | | | | |
| 175 | REVISED PER CCAS 694 THROUGH 697 | 12-15-80 | CWH | | | | | | |
| 176 | REVISED PER CCAS 698 THROUGH 701 | 12-22-80 | CWH | | | | | | |
| 177 | REVISED PER CCAS 702 THROUGH 705 | 12-29-80 | CWH | | | | | | |
| 178 | REVISED PER CCAS 706 THROUGH 709 | 1-5-81 | CWH | | | | | | |
| 179 | REVISED PER CCAS 710 THROUGH 713 | 1-12-81 | CWH | | | | | | |
| 180 | REVISED PER CCAS 714 THROUGH 717 | 1-19-81 | CWH | | | | | | |
| 181 | REVISED PER CCAS 718 THROUGH 721 | 1-26-81 | CWH | | | | | | |
| 182 | REVISED PER CCAS 722 THROUGH 725 | 2-2-81 | CWH | | | | | | |
| 183 | REVISED PER CCAS 726 THROUGH 729 | 2-9-81 | CWH | | | | | | |
| 184 | REVISED PER CCAS 730 THROUGH 733 | 2-16-81 | CWH | | | | | | |
| 185 | REVISED PER CCAS 734 THROUGH 737 | 2-23-81 | CWH | | | | | | |
| 186 | REVISED PER CCAS 738 THROUGH 741 | 3-2-81 | CWH | | | | | | |
| 187 | REVISED PER CCAS 742 THROUGH 745 | 3-9-81 | CWH | | | | | | |
| 188 | REVISED PER CCAS 746 THROUGH 749 | 3-16-81 | CWH | | | | | | |
| 189 | REVISED PER CCAS 750 THROUGH 753 | 3-23-81 | CWH | | | | | | |
| 190 | REVISED PER CCAS 754 THROUGH 757 | 3-30-81 | CWH | | | | | | |
| 191 | REVISED PER CCAS 758 THROUGH 761 | 4-6-81 | CWH | | | | | | |
| 192 | REVISED PER CCAS 762 THROUGH 765 | 4-13-81 | CWH | | | | | | |
| 193 | REVISED PER CCAS 766 THROUGH 769 | 4-20-81 | CWH | | | | | | |
| 194 | REVISED PER CCAS 770 THROUGH 773 | 4-27-81 | CWH | | | | | | |
| 195 | REVISED PER CCAS 774 THROUGH 777 | 5-4-81 | CWH | | | | | | |
| 196 | REVISED PER CCAS 778 THROUGH 781 | 5-11-81 | CWH | | | | | | |
| 197 | REVISED PER CCAS 782 THROUGH 785 | 5-18-81 | CWH | | | | | | |
| 198 | REVISED PER CCAS 786 THROUGH 789 | 5-25-81 | CWH | | | | | | |
| 199 | REVISED PER CCAS 790 THROUGH 793 | 6-1-81 | CWH | | | | | | |
| 200 | REVISED PER CCAS 794 THROUGH 797 | 6-8-81 | CWH | | | | | | |
| 201 | REVISED PER CCAS 798 THROUGH 801 | 6-15-81 | CWH | | | | | | |
| 202 | REVISED PER CCAS 802 THROUGH 805 | 6-22-81 | CWH | | | | | | |
| 203 | REVISED PER CCAS 806 THROUGH 809 | 6-29-81 | CWH | | | | | | |
| 204 | REVISED PER CCAS 810 THROUGH 813 | 7-6-81 | CWH | | | | | | |
| 205 | REVISED PER CCAS 814 THROUGH 817 | 7-13-81 | CWH | | | | | | |
| 206 | REVISED PER CCAS 818 THROUGH 821 | 7-20-81 | CWH | | | | | | |
| 207 | REVISED PER CCAS 822 THROUGH 825 | 7-27-81 | CWH | | | | | | |
| 208 | REVISED PER CCAS 826 THROUGH 829 | 8-3-81 | CWH | | | | | | |
| 209 | REVISED PER CCAS 830 THROUGH 833 | 8-10-81 | CWH | | | | | | |
| 210 | REVISED PER CCAS 834 THROUGH 837 | 8-17-81 | CWH | | | | | | |
| 211 | REVISED PER CCAS 838 THROUGH 841 | 8-24-81 | CWH | | | | | | |
| 212 | REVISED PER CCAS 842 THROUGH 845 | 8-31-81 | CWH | | | | | | |
| 213 | REVISED PER CCAS 846 THROUGH 849 | 9-7-81 | CWH | | | | | | |
| 214 | REVISED PER CCAS 850 THROUGH 853 | 9-14-81 | CWH | | | | | | |
| 215 | REVISED PER CCAS 854 THROUGH 857 | 9-21-81 | CWH | | | | | | |
| 216 | REVISED PER CCAS 858 THROUGH 861 | 9-28-81 | CWH | | | | | | |
| 217 | REVISED PER CCAS 862 THROUGH 865 | 10-5-81 | CWH | | | | | | |
| 218 | REVISED PER CCAS 866 THROUGH 869 | 10-12-81 | CWH | | | | | | |
| 219 | REVISED PER CCAS 870 THROUGH 873 | 10-19-81 | CWH | | | | | | |
| 220 | REVISED PER CCAS 874 THROUGH 877 | 10-26-81 | CWH | | | | | | |
| 221 | REVISED PER CCAS 878 THROUGH 881 | 11-2-81 | CWH | | | | | | |
| 222 | REVISED PER CCAS 882 THROUGH 885 | 11-9-81 | CWH | | | | | | |
| 223 | REVISED PER CCAS 886 THROUGH 889 | 11-16-81 | CWH | | | | | | |
| 224 | REVISED PER CCAS 890 THROUGH 893 | 11-23-81 | CWH | | | | | | |
| 225 | REVISED PER CCAS 894 THROUGH 897 | 11-30-81 | CWH | | | | | | |
| 226 | REVISED PER CCAS 898 THROUGH 901 | 12-7-81 | CWH | | | | | | |
| 227 | REVISED PER CCAS 902 THROUGH 905 | 12-14-81 | CWH | | | | | | |
| 228 | REVISED PER CCAS 906 THROUGH 909 | 12-21-81 | CWH | | | | | | |
| 229 | REVISED PER CCAS 910 THROUGH 913 | 12-28-81 | CWH | | | | | | |
| 230 | REVISED PER CCAS 914 THROUGH 917 | 1-4-82 | CWH | | | | | | |
| 231 | REVISED PER CCAS 918 THROUGH 921 | 1-11-82 | CWH | | | | | | |
| 232 | REVISED PER CCAS 922 THROUGH 925 | 1-18-82 | CWH | | | | | | |
| 233 | REVISED PER CCAS 926 THROUGH 929 | 1-25-82 | CWH | | | | | | |
| 234 | REVISED PER CCAS 930 THROUGH 933 | 2-1-82 | CWH | | | | | | |
| 235 | REVISED PER CCAS 934 THROUGH 937 | 2-8-82 | CWH | | | | | | |
| 236 | REVISED PER CCAS 938 THROUGH 941 | 2-15-82 | CWH | | | | | | |
| 237 | REVISED PER CCAS 942 THROUGH 945 | 2-22-82 | CWH | | | | | | |
| 238 | REVISED PER CCAS 946 THROUGH 949 | 3-1-82 | CWH | | | | | | |
| 239 | REVISED PER CCAS 950 THROUGH 953 | 3-8-82 | CWH | | | | | | |
| 240 | REVISED PER CCAS 954 THROUGH 957 | 3-15-82 | CWH | | | | | | |
| 241 | REVISED PER CCAS 958 THROUGH 961 | 3-22-82 | CWH | | | | | | |
| 242 | REVISED PER CCAS 962 THROUGH 965 | 3-29-82 | CWH | | | | | | |
| 243 | REVISED PER CCAS 966 THROUGH 969 | 4-5-82 | CWH | | | | | | |
| 244 | REVISED PER CCAS 970 THROUGH 973 | 4-12-82 | CWH | | | | | | |
| 245 | REVISED PER CCAS 974 THROUGH 977 | 4-19-82 | CWH | | | | | | |
| 246 | REVISED PER CCAS 978 THROUGH 981 | 4-26-82 | CWH | | | | | | |
| 247 | REVISED PER CCAS 982 THROUGH 985 | 5-3-82 | CWH | | | | | | |
| 248 | REVISED PER CCAS 986 THROUGH 989 | 5-10-82 | CWH | | | | | | |
| 249 | REVISED PER CCAS 990 THROUGH 993 | 5-17-82 | CWH | | | | | | |
| 250 | REVISED PER CCAS 994 THROUGH 997 | 5-24-82 | CWH | | | | | | |
| 251 | REVISED PER CCAS 998 THROUGH 1001 | 5-31-82 | CWH | | | | | | |
| 252 | REVISED PER CCAS 1002 THROUGH 1005 | 6-7-82 | CWH | | | | | | |
| 253 | REVISED PER CCAS 1006 THROUGH 1009 | 6-14-82 | CWH | | | | | | |
| 254 | REVISED PER CCAS 1010 THROUGH 1013 | 6-21-82 | CWH | | | | | | |
| 255 | REVISED PER CCAS 1014 THROUGH 1017 | 6-28-82 | CWH | | | | | | |
| 256 | REVISED PER CCAS 1018 THROUGH 1021 | 7-5-82 | CWH | | | | | | |
| 257 | REVISED PER CCAS 1022 THROUGH 1025 | 7-12-82 | CWH | | | | | | |
| 258 | REVISED PER CCAS 1026 THROUGH 1029 | 7-19-82 | CWH | | | | | | |
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| 260 | REVISED PER CCAS 1034 THROUGH 1037 | 8-2-82 | CWH | | | | | | |
| 261 | REVISED PER CCAS 1038 THROUGH 1041 | 8-9-82 | CWH | | | | | | |
| 262 | REVISED PER CCAS 1042 THROUGH 1045 | 8-16-82 | CWH | | | | | | |
| 263 | REVISED PER CCAS 1046 THROUGH 1049 | 8-23-82 | CWH | | | | | | |
| 264 | REVISED PER CCAS 1050 THROUGH 1053 | 8-30-82 | CWH | | | | | | |
| 265 | REVISED PER CCAS 1054 THROUGH 1057 | 9-6-82 | CWH | | | | | | |
| 266 | REVISED PER CCAS 1058 THROUGH 1061 | 9-13-82 | CWH | | | | | | |
| 267 | REVISED PER CCAS 1062 THROUGH 1065 | 9-20-82 | CWH | | | | | | |
| 268 | REVISED PER CCAS 1066 THROUGH 1069 | 9-27-82 | CWH | | | | | | |
| 269 | REVISED PER CCAS 1070 THROUGH 1073 | 10-4-82 | CWH | | | | | | |
| 270 | REVISED PER CCAS 1074 THROUGH 1077 | 10-11-82 | CWH | | | | | | |
| 271 | REVISED PER CCAS 1078 THROUGH 1081 | 10-18-82 | CWH | | | | | | |
| 272 | REVISED PER CCAS 1082 THROUGH 1085 | 10-25-82 | CWH | | | | | | |
| 273 | REVISED PER CCAS 1086 THROUGH 1089 | 11-1-82 | CWH | | | | | | |
| 274 | REVISED PER CCAS 1090 THROUGH 1093 | 11-8-82 | CWH | | | | | | |
| 275 | REVISED PER CCAS 1094 THROUGH 1097 | 11-15-82 | CWH | | | | | | |
| 276 | REVISED PER CCAS 1098 THROUGH 1101 | 11-22-82 | CWH | | | | | | |
| 277 | REVISED PER CCAS 1102 THROUGH 1105 | 11-29-82 | CWH | | | | | | |
| 278 | REVISED PER CCAS 1106 THROUGH 1109 | 12-6-82 | CWH | | | | | | |
| 279 | REVISED PER CCAS 1110 THROUGH 1113 | 12-13-82 | CWH | | | | | | |
| 280 | REVISED PER CCAS 1114 THROUGH 1117 | 12-20-82 | CWH | | | | | | |
| 281 | REVISED PER CCAS 1118 THROUGH 1121 | 12-27-82 | CWH | | | | | | |
| 282 | REVISED PER CCAS 1122 THROUGH 1125 | 1-3-83 | CWH | | | | | | |
| 283 | REVISED PER CCAS 1126 THROUGH 1129 | 1-10-83 | CWH | | | | | | |
| 284 | REVISED PER CCAS 1130 THROUGH 1133 | 1-17-83 | CWH | | | | | | |
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| 288 | REVISED PER CCAS 1146 THROUGH 1149 | 2-14-83 | CWH | | | | | | |
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| 297 | REVISED PER CCAS 1182 THROUGH 1185 | 4-18-83 | CWH | | | | | | |
| 298 | REVISED PER CCAS 1186 THROUGH 1189 | 4-25-83 | CWH | | | | | | |
| 299 | REVISED PER CCAS 1190 THROUGH 1193 | 5-2-83 | CWH | | | | | | |
| 300 | REVISED PER CCAS 1194 THROUGH 1197 | 5-9-83 | CWH | | | | | | |
| 301 | REVISED PER CCAS 1198 THROUGH 1201 | 5-16-83 | CWH | | | | | | |
| 302 | REVISED PER CCAS 1202 THROUGH 1205 | 5-23-83 | CWH | | | | | | |
| 303 | REVISED PER CCAS 1206 THROUGH 1209 | 5-30-83 | CWH | | | | | | |
| 304 | REVISED PER CCAS 1210 THROUGH 1213 | 6-6-83 | CWH | | | | | | |
| 305 | REVISED PER CCAS 1214 THROUGH 1217 | 6-13-83 | CWH | | | | | | |
| 306 | REVISED PER CCAS 1218 THROUGH 1221 | 6-20-83 | CWH | | | | | | |
| 307 | REVISED PER CCAS 1222 THROUGH 1225 | 6-27-83 | CWH | | | | | | |
| 308 | REVISED PER CCAS 1226 THROUGH 1229 | 7-4-83 | CWH | | | | | | |
| 309 | REVISED PER CCAS 1230 THROUGH 1233 | 7-11-83 | CWH | | | | | | |
| 310 | REVISED PER CCAS 1234 THROUGH 1237 | 7-18-83 | CWH | | | | | | |
| 311 | REVISED PER CCAS 1238 THROUGH 1241 | 7-25-83 | CWH | | | | | | |
| 312 | REVISED PER CCAS 1242 THROUGH 1245 | 8-1-83 | CWH | | | | | | |
| 313 | REVISED PER CCAS 1246 THROUGH 1249 | 8-8-83 | CWH | | | | | | |
| 314 | REVISED PER CCAS 1250 THROUGH 1253 | 8-15-83 | CWH | | | | | | |
| 315 | REVISED PER CCAS 1254 THROUGH 1257 | 8-22-83 | CWH | | | | | | |
| 316 | REVISED PER CCAS 1258 THROUGH 1261 | 8-29-83 | CWH | | | | | | |
| 317 | REVISED PER CCAS 1262 THROUGH 1265 | 9-5-83 | CWH | | | | | | |
| 318 | REVISED PER CCAS 1266 THROUGH 1269 | 9-12-83 | CWH | | | | | | |

1X2D48E008

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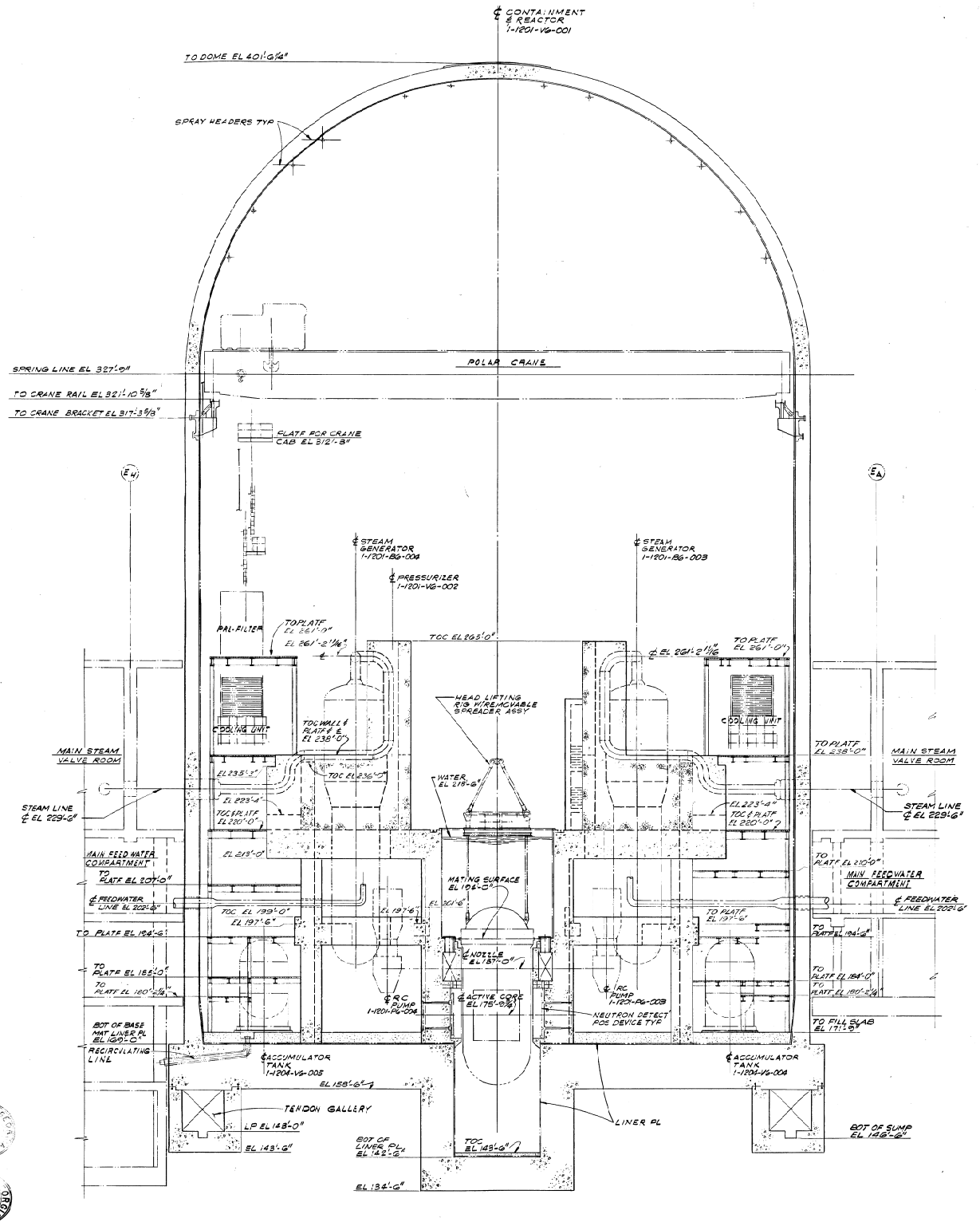


SECTION B

48E005
48E004
48E005
48E006

GENERAL REVISIONS

| NO. | DESCRIPTION | DATE | BY | CHK | SUPV | DISC | CIVIL | PIPING | ELEC | I&C | HVAC | NUCL | QA |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | RELEASED, HOLD & REISSUED AS SHOWN | 10/8/79 | KEW | VMM | MCL | | | | | | | | |
| 5 | ISSUED FOR CONSTRUCTION | 8/17/79 | KEW | VMM | MCL | | | | | | | | |
| 4 | ISSUED FOR CONSTRUCTION | 8/10/79 | KEW | VMM | MCL | | | | | | | | |
| 3 | ISSUED FOR CONSTRUCTION | 7/27/79 | KEW | VMM | MCL | | | | | | | | |
| 2 | ISSUED FOR CONSTRUCTION | 7/13/79 | KEW | VMM | MCL | | | | | | | | |
| 1 | ISSUED FOR CONSTRUCTION | 6/29/79 | KEW | VMM | MCL | | | | | | | | |
| 0 | ISSUED FOR CONSTRUCTION | 6/15/79 | KEW | VMM | MCL | | | | | | | | |

SEE E 94-H

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT INTERNALS
GENERAL ARRANGEMENT
SECTION LOOKING WEST

| SCALE | JOB NO. | DRAWING NO. | REV. |
|---|---|---|---|
| NONE | 5513 | 1X2D48E008 | 4 |

GPC PLANT VOGTLE CONSTR MICROGRAPHICS
I CERTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS TAKEN IN THE NORMAL AND REGULAR COURSE OF BUSINESS, ON THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT SUBMITTED TO MICROGRAPHICS.

1/15/87
DATE

CAMERA OPERATOR

SECTION SUPERVISOR

30X

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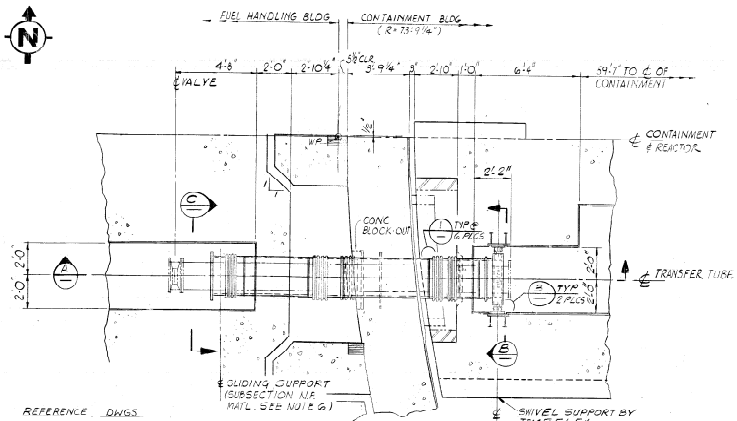
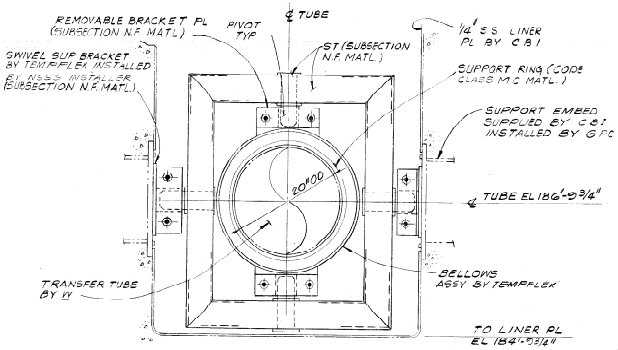
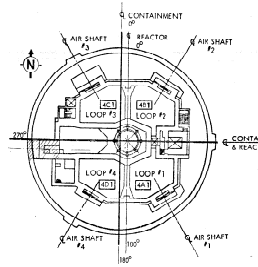



REFERENCE DWGS.
1X2D48JJ004 REFUELING CANAL LINER PL
1X2D01J020 WALL LINER PL REN.
AX2D09J013 LINER PL SECT

## PLAN AT FUEL TRANSFER TUBE
3/8"=1'-0"

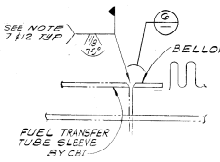


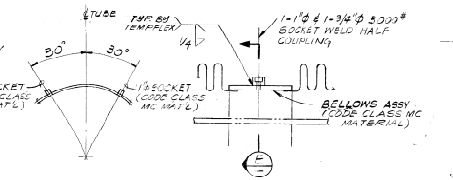

NOTE:
SWIVEL SUPPORT SHALL BE DESIGNED FOR BOTH HORIZ & VERT MOVEMENT PER SPEC NO. XZAH08



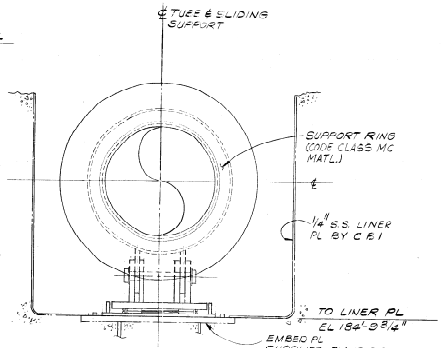

KEY PLAN UNIT 1 (EL 195'-0")

## SECTION
1/2"=1'-0"

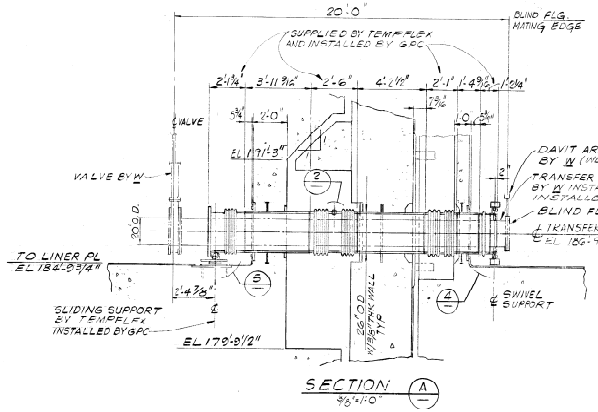


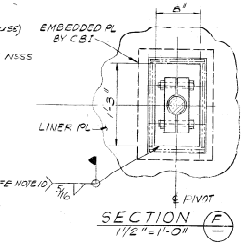

## SECTION
1/2"=1'-0"



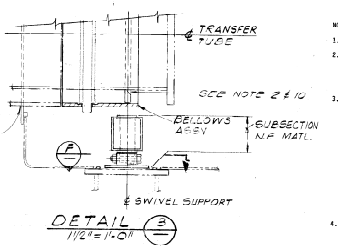

## SECTION
1/2"=1'-0"




## SECTION
3/8"=1'-0"



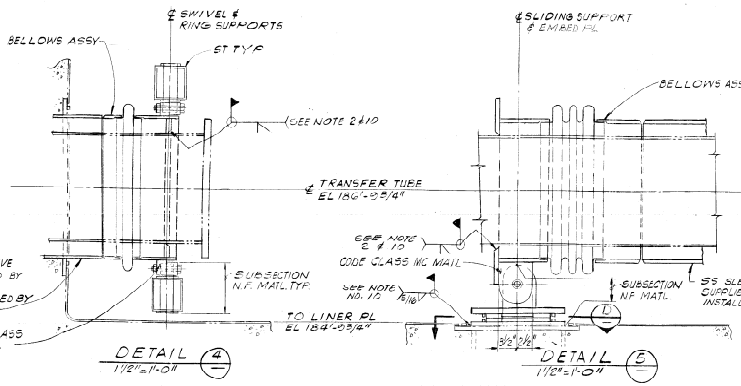

## SECTION
1/2"=1'-0"



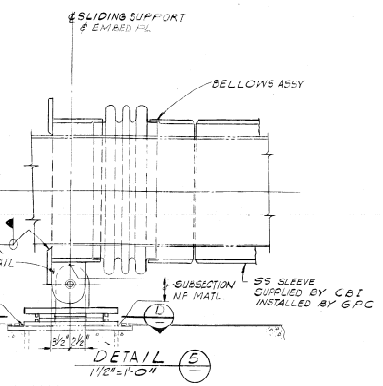

## DETAIL
1/2"=1'-0"

NOTE:

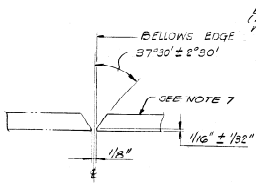
1. ALL MATERIAL SHALL CONFORM TO SPEC. NO. XZAH08.
2. ALL CIRCUMFERENTIAL FIELD WELDS TO THE TRANSFER TUBE (BELLOWS, SUPPORTS, ETC.) SHOULD BE MADE IN INTERMITTENT AND OPPOSING SEQUENCE TO CONTROL TWISTING DISTORTION. DO NOT WELD TO THE TUBE CIRCUMFERENCE IN ONE CONTINUOUS PASS.
3. THE FOLLOWING PHYSICAL DATA AND LOADS FOR THE FUEL TRANSFER TUBE ASSEMBLY (SUPPLIED BY OTHERS) IS AS FOLLOWS:
   A. TRANSFER TUBE ONLY (EMPTY) _____ 100 LB/FT
   B. TRANSFER TUBE ONLY (FILLED WITH WATER) 220 LB/FT
   C. VALVE _____ 600 LB
   D. SLIP ON FLANGE (FUEL HANDLING BLDG. SIDE) 2008 LB
   E. BOLTING FLANGE ASSEMBLY (REACTOR SIDE) 500 LB
   F. FUEL ASSEMBLY, CAN AND FUEL CONTAINED (EVENLY DISTRIBUTED OVER A 7'-0" LENGTH) 2900 LB
4. THE FUEL TRANSFER TUBE ASSEMBLY WILL BE SUBJECTED TO A MAXIMUM SLIDING LOAD (ALONG THE TUBE AXIS) OF APPROXIMATELY 2000 POUNDS BASED ON TWO DRIVE CHAINS WITH THE FUEL CANISTER IN A FULLY LOADED CONDITION.
5. THE SUPPORT SCHEME MUST PROVIDE ADEQUATE SUPPORT OF THE FUEL TRANSFER TUBE NEAR BOTH ENDS DURING REFUELING PROCESS. THE SUPPORTS MUST BE ABLE TO SUPPORT THE FUEL TRANSFER TUBE FOR ALL LOADS, AND ALL RESULTING MOMENTS AND FORCES RESULTING FROM THEIR APPLICATION AT ANY POINT.
6. THE DESIGNATION BETWEEN ASME CODE SECTION III, CLASS MC PRESSURE BOUNDARY MATERIAL AND SUBSECTION NF SUPPORT MATERIAL IS IDENTIFIED ON THE DRAWING. MATERIAL PROVIDED BY TEMPFLEX IS ASME CODE, SECTION III, CLASS MC AND.
7. THE FOLLOWING WELDS SHALL BE JOINED, BY AN ORGANIZATION WHICH HAS A CERTIFICATE OF AUTHORIZATION FOR ASME SECTION III, CODE CLASS MC AND WELDS SHALL BE DONE IN ACCORDANCE WITH THE REQUIREMENTS OF ASME SECTION III, CLASS MC UNDER THE PROVISIONS OF CODE CLASS 2.
8. THE WESTINGHOUSE FUEL TRANSFER TUBE ASSEMBLY, EXCEPT THE VALVE IS DESIGNED AND FABRICATED AS AN ASME CODE SECTION III, CLASS MC COMPONENT. THE VALVE IS NONE CODE.
9. THE SLIDING SUPPORT AND SWIVEL SUPPORT SHALL BE DESIGNED AS LINEAR COMPONENT SUPPORTS AS DEFINED BY THE ASME CODE, SUBSECTION NF.
10. THE NUCLEAR WELDS SHALL BE DONE BY THE NSSS INSTALLER IN ACCORDANCE WITH THE REQUIREMENTS OF ASME SECTION III, CLASS MC, UNDER THE PROVISIONS OF CODE CLASS 2. THE NOZZLE-TO-SHELL WELDS SHALL BE FULL PENETRATION AND FINAL PASS.
11. FILLER MATERIAL TO BE COMPATIBLE WITH BUT NEED NOT BE SPECIFICALLY "QA" MATERIALS. WELDING WHERE APPLICABLE SHALL BE IN ACCORDANCE WITH SUBSECTION "QW".
12. ASME SECTION III CODE WELDS AND FIELD INSTALLATION ACTIVITIES SHALL BE PERFORMED BY ASME SECTION IX QUALIFIED WELDERS.




## DETAIL
1/2"=1'-0"

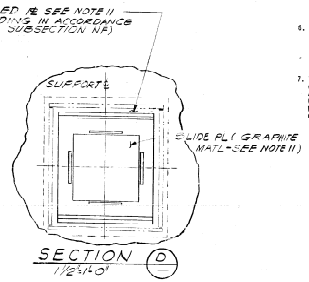



## DETAIL
1/2"=1'-0"




END PREPARATION

## DETAIL
NTS




## SECTION
1/2"=1'-0"

Oct 2, 1978

3-2-79

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 5 | INCORPORATE (AM-P08-IC-004) REVISIONS AS SHOWN | 3/7/78 | KS | JM | ST | | | | |
| 4 | REVISED AS SHOWN | 4-6-78 | ON | JM | ST | | | | |
| 3 | GEN REVISION | 9-77 | RC | GO | SG | N | | MET | |
| 2 | ISSUED FOR CHF E APPROVAL | 12-77 | EM | GO | SG | N | | MET | |
| 1 | REVISED SECT B & C AND "HUB" | 2-78 | RCM | GO | SG | N | | MET | |
| 0 | ISSUED FOR CONSTRUCTION | 10-78 | RM | GO | SG | N | | MET | |

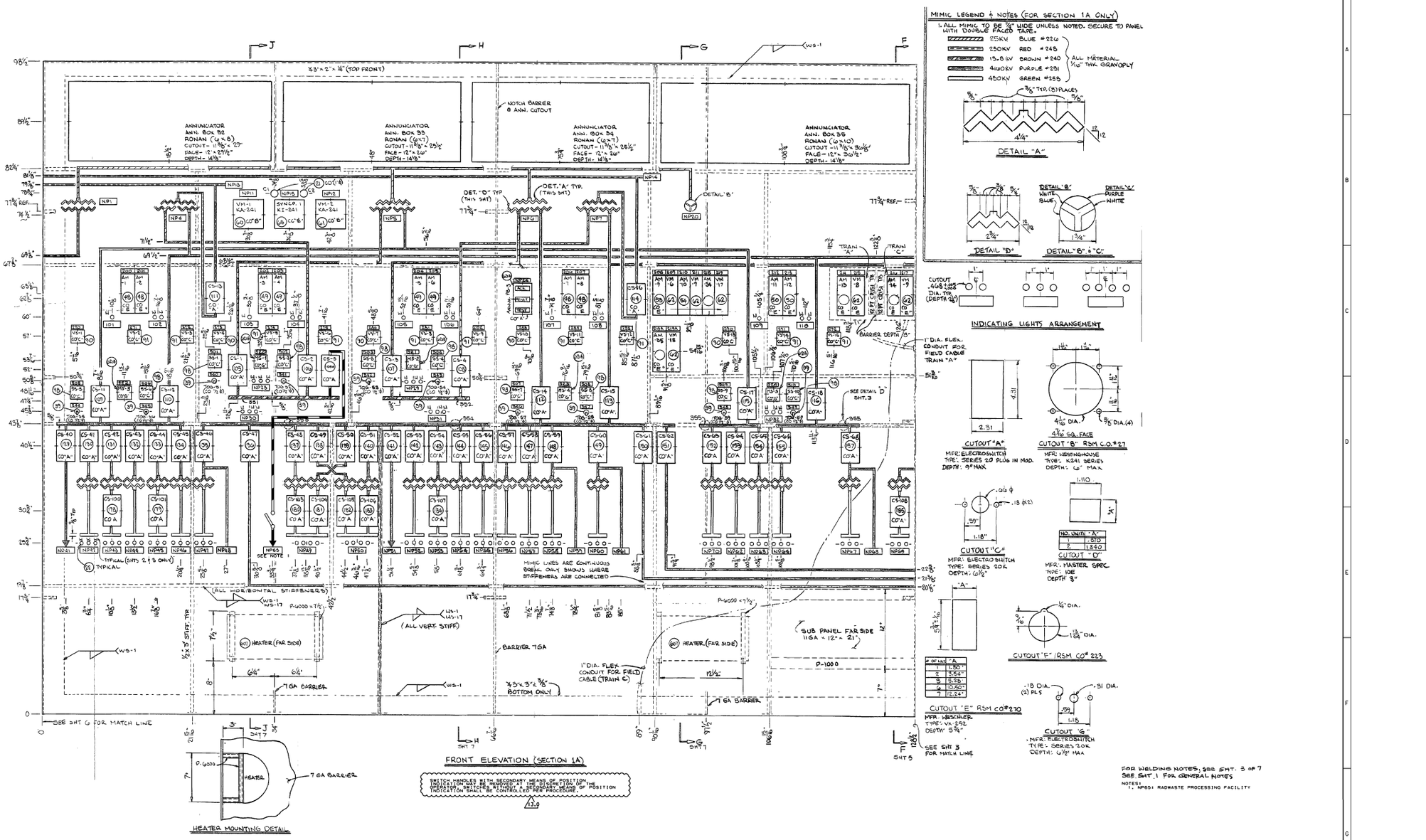
PECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

CONTAINMENT INTERNALS
FUEL TRANSFER HOUSING & SUPPORT
PLAN, SECTIONS & DETAILS

ENG. MGR.
DATE
SCALE: AS NOTED
DRAWING NO. 1X2D48JJ006
REV. G
IOR NO. 9510

I IDENTIFY THAT THE IMAGE CONTAINED ON THIS FRAME WAS MADE IN THE NORMAL AND REGULAR COURSE OF BUSINESS IN THE DATE STATED BELOW AND THAT IT IS AN ACCURATE REPRODUCTION OF THE DOCUMENT SUBMITTED TO MICROGRAPHICS.




NUCLEAR SAFETY RELATED

| REV | BY | DESCRIPTION |
|---|---|---|
| 0 | | ECO NO. 13599 |
| A | | ECO NO. 1317 |
| 7 | | ECO NO. 1515 |
| 8 | | ECO NO. 1812 |
| 9 | | ECO NO. 1915 |
| A | | ECO NO. 1961 |
| B | | ECO NO. 2015 |
| C | | ECO NO. 2067 |
| 1 | | ECO NO. 7266 |
| 2 | | FOR APPROVAL |

| | |
|---|---|
| TITLE | MECH DWG., ELECTRICAL AUXILIARY BOARD SECTION 1A ( 1B UNIT NO.1 (1B1816U3007) |
| PROJECT MANAGER | BILL CLEVELAND |
| DRAWING | 3784-402-301 |
| CUSTOMER FIELD REP. APPROVED | GA. POWER CO./VOGTLE NUCLEAR PLANT |
| DRAWING | C. Bolton |
| DRAWN BY | E. SIMS |
| | RELIANCE ELECTRIC COMPANY / CUSTOM CONTROLS STONE MOUNTAIN, GEORGIA 30083 |

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT. N/A
PAGE N/A
FIGURE N/A

| VERSION | DATE |
|---|---|
| 13.0 | 02/08/10 |

REVISED BY SNC PER ABN C092064901E005, VER.1.0
SEE WORKFLOW FOR PREVIOUS VER. SIGNATURES.

| BY | CHK'D | APPR.1 | APPR.2 |
|---|---|---|---|
| TSL | ELC | JMR | X |

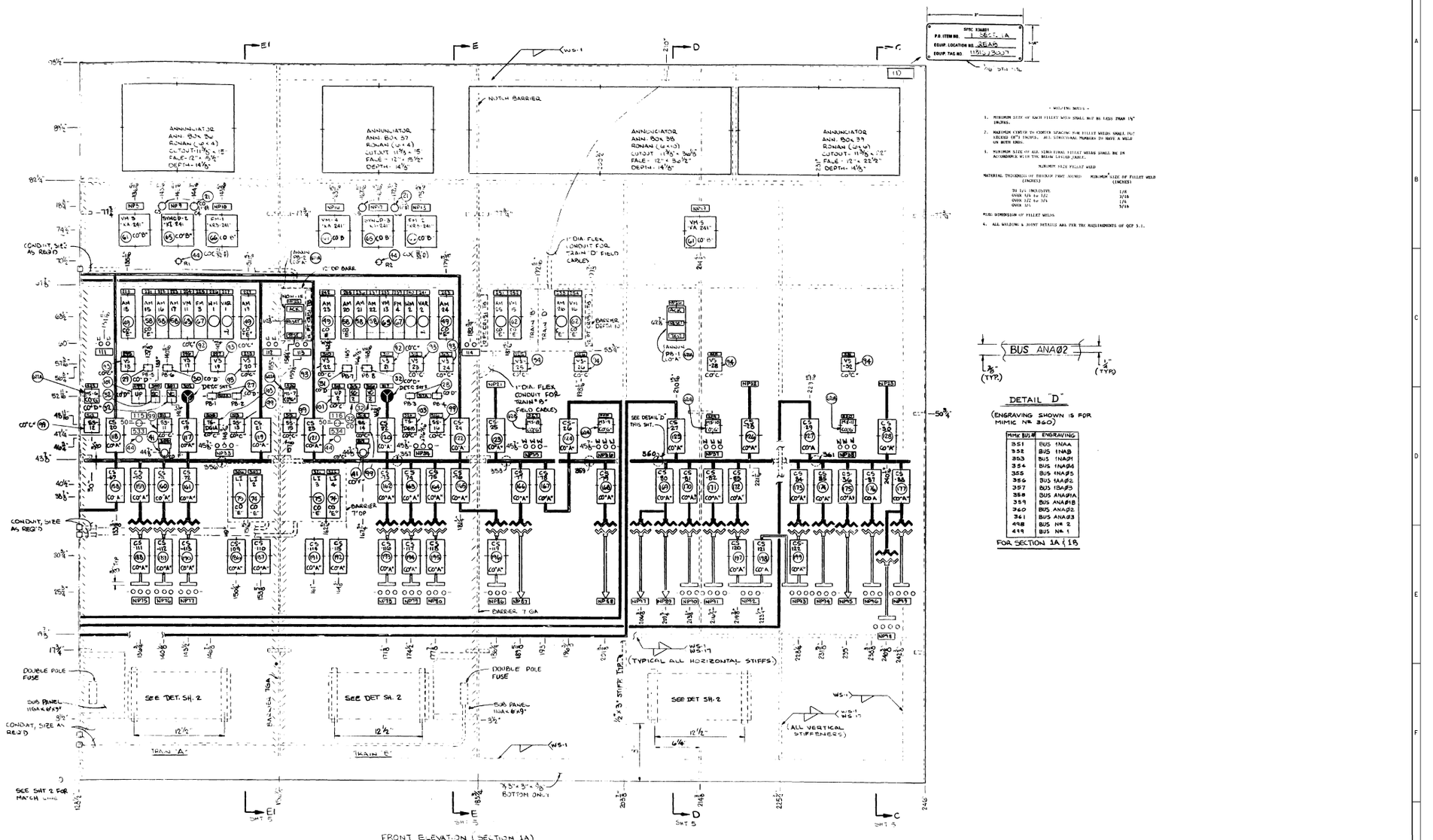
CAD MICROSTATION TSL/01 1X3AE01-00020.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

TITLE: MECHANICAL DRAWING ELECTRICAL AUXILIARY BOARD 1-1816-U3-007 SECTION 1A & 1B

VENDOR: RELIANCE ELECTRIC P.O.#: PAV-426

DRAWING NO. 1X3AE01-00020




FRONT ELEVATION (SECTION 1A)




DETAIL "D"

(ENGRAVING SHOWN IS FOR MIMIC Nº 360)

| MIMIC BUS # | ENGRAVING |
|---|---|
| 351 | BUS 1NAA |
| 352 | BUS 1NAB |
| 353 | BUS 1NA01 |
| 354 | BUS 1NA04 |
| 355 | BUS 1NA05 |
| 356 | BUS 1AA02 |
| 357 | BUS 1BA03 |
| 358 | BUS ANA01A |
| 359 | BUS ANA01B |
| 360 | BUS ANA02 |
| 361 | BUS ANA03 |
| 498 | BUS Nº 2 |
| 499 | BUS Nº 1 |

FOR SECTION 1A & 1B

SWITCH HANDLES WITH SECONDARY MEANS OF POSITION INDICATION MAY BE REMOVED AT THE DISCRETION OF THE OPERATOR. SWITCHES WITHOUT A SECONDARY MEANS OF POSITION INDICATION SHALL BE CONTROLLED PER PROCEDURE.

12.0

- WELDING NOTES -

1. MINIMUM SIZE OF EACH FILLET WELD SHALL NOT BE LESS THAN 1/8" INCHES.
2. MAXIMUM CENTER TO CENTER SPACING FOR FILLET WELDS SHALL NOT EXCEED 24" INCHES. ALL STRUCTURAL MEMBERS TO HAVE A WELD ON BOTH ENDS.
3. MINIMUM SIZE OF ALL SEMI-FINAL FILLET WELDS SHALL BE IN ACCORDANCE WITH THE BELOW LISTED TABLE.

MINIMUM SIZE FILLET WELD

| MATERIAL THICKNESS OF THICKER PART JOINED (INCHES) | MINIMUM SIZE OF FILLET WELD (INCHES) |
|---|---|
| TO 1/4 INCLUSIVE | 1/8 |
| OVER 1/4 TO 1/2 | 3/16 |
| OVER 1/2 TO 3/4 | 1/4 |
| OVER 3/4 | 5/16 |

4. ALL WELDING & JOINT DETAILS ARE PER THE REQUIREMENTS OF QCP 3.1.

NUCLEAR SAFETY RELATED

| IN FIELD | AS BUILT | NO. | DESCRIPTION | BY | DATE |
|---|---|---|---|---|---|
| | | 10 | ECO NO. 13399 | | |
| | | 9 | ECO NO. 1377 | | |
| | | 8 | ECO NO. 1376 | | |
| | | 7 | ECO Nº 1372 | | |
| | | 6 | ECO NO. 1214 | | |
| | | 5 | ECO NO. 1612 | | |
| | | 4 | ECO NO. 1613 | | |
| | | 3 | ECO NO. 1519 | | |
| | | 2 | ECO NO. 1047 | | |
| | | 1 | ECO NO. 725 | | |
| | | 0 | FOR ANNUNCIATOR | | |

SEE SHT 1 FOR GENERAL NOTES

| | |
|---|---|
| TITLE | MECH. DWG. ELECTRICAL AUXILIARY BOARD SECTION 1A & 1B, UNIT NO. 1 (1X3AE01S001) |
| PROJECT MANAGER | BILL CLEVELAND |
| DESIGNED BY | TSW |
| DRAWN BY | TSW |
| CHECKED BY | JMR |
| APPROVED BY | C. Eatmon |
| DRAWING NO. | 1X3AE01-00001 |
| SCALE | 1/2" = 1'-0" |
| SHEET NO. | 1 OF 7 |

RELIANCE CUSTOM CONTROLS ELECTRIC COMPANY STONE MOUNTAIN, GEORGIA 30083

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT.: N/A
PAGE: N/A
FIGURE: N/A

VERSION 12.0 | DATE 02/09/10

REVISED BY SNC PER ABN C092064901E004, VER. 1.0

SEE MICROFILM FOR PREVIOUS VER. SIGNATURES.

| BY | CHK'D | APPR.1 | APPR.2 |
|---|---|---|---|
| TSL | ELC | JMR | X |

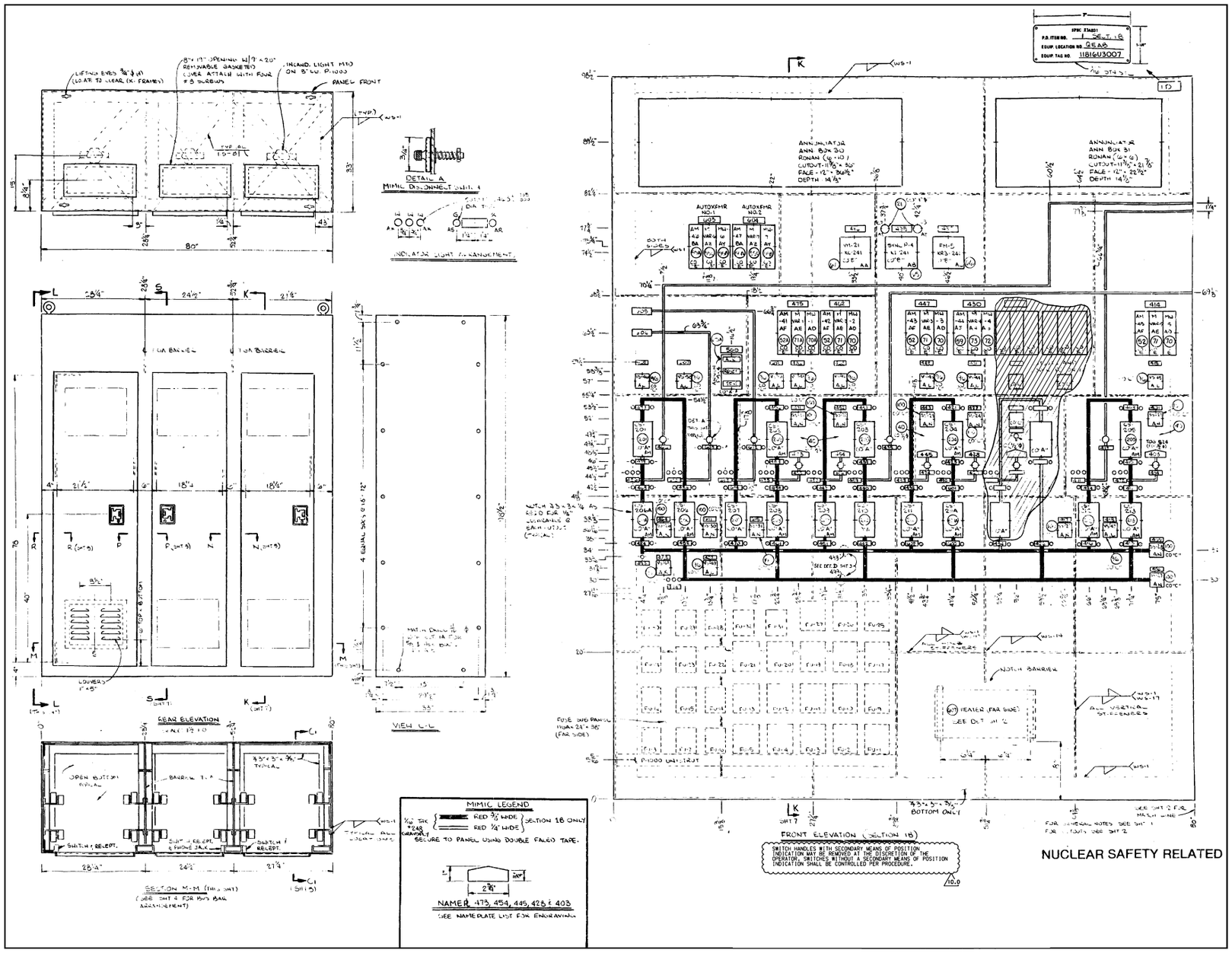
CAD MICROSTATION TSL/01 1X3AE01-00021.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1

TITLE: MECHANICAL DRAWING ELECTRICAL AUXILIARY BOARD 1-1816-U3-007 SECTION 1A & 1B

VENDOR: RELIANCE ELECTRIC | P.O.#: PAV-426

DRAWING NO. 1X3AE01-00021




| | | | | | |
|---|---|---|---|---|---|
| | | | 1 | ECO NO. 377 | |
| | | | 0 | ECO NO. 1512 | |
| | | | B | ECO NT 1872 | |
| | | | A | ECO NT 1928 | |
| | | | 3 | ECO NO. 1648 | |
| | | | 2 | ECO NO. 2647 | |
| | | | 1 | ECO NO. 1744 | |
| | | | 0 | INITIAL RELEASE | |
| DATE | BY | CHK'D | REV. NO. | REVISION | |

As Built

MECH. DWG. ELECTRICAL AUXILIARY BOARD SECTION 1A & 1B UNIT NO. 1 (1816U3007)

PROJECT MANAGER: Bill Cleveland — PROJ. ENGINEER: Frank Dyer — CUSTOMER ORDER NO.: 310 Equipt Nos. 1 & 2 — CUSTOMER ORDER NO.: Day #42

CUSTOMER: GA Power Co (Vogtle Nuclear Plant) — MECH. DWG. NO.: E-2413 SHT 1 — SCALE: 1/2 IN. = 1'-0" — SIZE: E

RELIANCE ELECTRIC COMPANY — CUSTOM CONTROLS — STONE MOUNTAIN, GEORGIA 30083

THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A
TAB/SECT.: N/A
PAGE: N/A
FIGURE: N/A

VERSION 10.0 | DATE 02/09/10

REVISED BY SNC PER ABN C092064901E002. VER.1.0
SEE WORKFLOW FOR PREVIOUS VER. SIGNATURES.

| BY | CHK'D | APPR.1 | APPR.2 |
|---|---|---|---|
| TSL | ELC | JMR | |

CAD: MICROSTATION | TSL/02 | 1X3AE01-00024.DGN

Southern Nuclear Operating Company, Inc.
FOR
VOGTLE ELECTRIC GENERATING PLANT
UNIT NO. 1

TITLE:
MECHANICAL DRAWING
ELECTRICAL AUXILIARY BOARD
1-1816-U3-007
SECTION 1A & 1B

VENDOR: RELIANCE ELECTRIC | P.O.#: PAV-426

DRAWING NO. 1X3AE01-00024

| SYMBOLS | DESCRIPTION |
|---|---|
| | GENERATING UNIT |
| 1500 | MOTOR, NUMERALS INDICATE HP. |
| 480-120V | POTENTIAL TRANSFORMER NUMERALS INDICATE RATIO FOR MCC CONTROL TRANSFORMER CONNECT TO 460V TAP TO CORRECT FOR VOLTAGE DROP (AS REQUIRED) |
| | AUTO TRANSFORMER |
| 1505 | CURRENT TRANSFORMER NUMERALS INDICATE RATIO |
| 3500/5 | BUSHING TYPE CURRENT TRANSFORMER |
| A B C B B B J L Δ C B A, C, C A, C A, C A | PHASOR INDICATORS |
| | TEST SWITCH (CURRENT) |
| | LIGHTNING ARRESTER |
| | METAL OXIDE ARRESTER |
| | CIRCUIT BREAKER, DRAW OUT TYPE |
| | CIRCUIT BREAKER, WITH MAGNETIC OVERLOAD TRIP COIL |
| | CIRCUIT BREAKER |
| | DISCONNECT SWITCH |
| | THERMAL OVERLOAD ELEMENT |
| | FUSE |
| | RESISTOR |
| | INCOMING CABLE |
| | OUTGOING CABLE |
| | OUTSIDE CONTAINMENT REACTOR CABLE PENETRATION INSIDE CONTAINMENT |
| | COMPUTER INPUT |
| T | FEEDER ENTRY T-DENOTES TOP: B, BOTTOM |
| 27 | RELAY OR DEVICE, NUMERALS INDICATE FUNCTION NUMBERS SEE DEVICE TABLE |
| | GROUND |
| | NON REVERSING STARTERS |
| | REVERSING STARTERS |
| | DISCONNECT LINK |
| | POWER RECEPTACLES |
| LOCAL STARTER SIZE1 | INDICATES LOCAL STARTER INSCRIBED NUMERAL INDICATES STARTER SIZE |
| | VOLTAGE REGULATOR |
| | BATTERY |
| | SHUNT |
| | TRANSFER SWITCH |

| SYMBOLS | DESCRIPTION |
|---|---|
| 70 | MOTOR ACTUATED RHEOSTAT SEE DEVICE TABLE |
| R | REDUCED VOLTAGE STARTER |
| | DIODE |
| | CAPACITOR |
| SN | SOLID NEUTRAL |
| M.L. | MAIN LUGS |
| RTD | RESISTIVE TEMPERATURE DETECTOR |
| | DISCONNECT SAFETY SWITCH MARK NO. EBT-30A, EBU-60A ETC; SEE MASTER BILL OF MATERIAL. |
| | BUS BAR |

DEVICE FUNCTION NUMBERS

| | |
|---|---|
| 2 | TIME-DELAY STARTING, OR CLOSING RELAY |
| 8 | CONTROL POWER DISCONNECTING DEVICE |
| 12 | OVER-SPEED DEVICE |
| 13 | SYNCHRONOUS-SPEED DEVICE |
| 14 | UNDER-SPEED DEVICE |
| 15 | SPEED OR FREQUENCY, MATCHING DEVICE |
| 21 | DISTANCE RELAY |
| 25 | SYNCHRONIZING OR SYNCHRONISM - CHECK DEVICE |
| 27 | UNDERVOLTAGE RELAY |
| 30 | ANNUNCIATOR RELAY |
| 32 | DIRECTIONAL POWER RELAY |
| 33 | POSITION SWITCH |
| 37 | UNDERCURRENT OR UNDERPOWER RELAY |
| 38 | BEARING PROTECTIVE DEVICE |
| 39 | MECHANICAL CONDITION MONITOR |
| 40 | FIELD RELAY |
| 41 | FIELD CIRCUIT BREAKER |
| 42 | RUNNING CIRCUIT BREAKER |
| 46 | REVERSE-PHASE, OR PHASE-BALANCE, CURRENT RELAY |
| 47 | PHASE-SEQUENCE VOLTAGE RELAY |
| 49 | MACHINE, OR TRANSFORMER, THERMAL RELAY |
| 50 | INSTANTANEOUS OVERCURRENT, OR RATE-OF-RISE RELAY |
| 51 | AC TIME OVERCURRENT RELAY |
| 52 | AC CIRCUIT BREAKER |
| 53 | EXCITER OR DC GENERATOR RELAY |
| 59 | OVERVOLTAGE RELAY |
| 60 | VOLTAGE OR CURRENT BALANCE RELAY |
| 62 | TIME-DELAY STOPPING OR OPENING RELAY |
| 64 | GROUND PROTECTIVE RELAY |
| 67 | AC DIRECTIONAL OVERCURRENT RELAY |
| 72 | DC CIRCUIT BREAKER |
| 74 | ALARM RELAY |
| 76 | DC OVERCURRENT RELAY |
| 81 | FREQUENCY RELAY |
| 86 | LOCKING-OUT RELAY |
| 87 | DIFFERENTIAL PROTECTIVE RELAY |
| 90 | REGULATING DEVICE |
| 98 | BYPASS STATUS RELAY |

SYSTEM VOLTAGE PREFIXES FOR DEVICE FUNCTION NUMBERS

NONE - 480 VOLT OR BELOW
1 - 4160 VOLT
2 - 13800 VOLT
3 - 25 KV
4 - 230 KV
5 - 500 KV

EXAMPLE: 127- 4160 V SYSTEM UNDERVOLTAGE RELAY

GENERAL NOTES:

2. NOTES RELATED TO A SPECIFIC SUBJECT ON A DWG ARE SHOWN ON THAT DWG.
3. RELAYS AND DEVICES IN WHICH THERE IS MORE THAN ONE WITHIN ONE CUBICLE/COMPARTMENT/SECTION THE NUMERALS 1, 2, 3 ETC., WILL BE ADDED I.E., 27, 27-1, 27-2, 27-3.
4. NUMERAL AT LOWER RIGHT SIDE OF AN INSTRUMENT/DEVICE DENOTES QUANTITY, WHERE NONE IS SHOWN INSTRUMENT QUANTITY IS ONE.
5. THIS DRAWING PRESENTS SPECIAL SYMBOLS, STANDARD OR SELF EVIDENT SYMBOLS ARE NOT INCLUDED.

Deleted E, NO SIMILS
Removed
12-2-96

BECHTEL
LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

GENERAL NOTES & SYMBOLS
DEVICE FUNCTION NUMBERS & SYMBOL LIST
FOR ONE LINE DIAGRAMS

SCALE: —
DRAWING NO. 1X3D-AA-A00E
REV. 5
JOB NO. 9320

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E |
|---|---|---|---|---|---|---|---|---|---|
| 1 | INCORP. ABN - 00400 | 4-95 | MER | | | | | | |
| 2 | INCORP. RCN 1 | 9/96 | | | | | | | |
| 3 | REVISED AS SHOWN | 4-23-91 | | | | | | | |
| 4 | INSULED INFORMATION COPY TO SCSI | 8-8-95 | | | | | | | |
| 5 | INFORMATION COPY SENT TO SCSI | 2-29-96 | | | | | | | |
| 6 | ISSUED FOR CONSTRUCTION | 4-29-96 | | | | | | | |

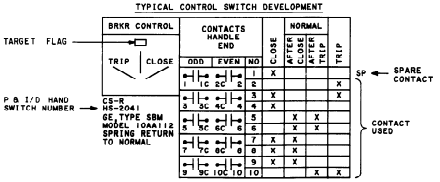
ENG. MGR.
DATE

SCALING RULES 1/1 3/16"=1'-0"

| SYMBOL | DESCRIPTION |
|---|---|
| 72 | DC CIRCUIT BREAKER |
| 252 | AC CIRCUIT BREAKER DRAW-OUT TYPE |
| | LOW VOLTAGE CIRCUIT BREAKER DRAW-OUT TYPE AC(52) OR DC(72) AS DENOTED BY FUNCTION NUMBER |
| 49 | THERMAL DEVICE |
| 49 | THERMAL DEVICE OPERATED CONTACT |
| 42 | CONTACTOR OPERATING COIL FOR FULL VOLTAGE NON-REVERSING STARTER |
| 42 O / 42 C | CONTACTOR OPERATING COILS FOR FULL VOLTAGE REVERSING STARTER |
| 79 152 a 80 | CIRCUIT BREAKER DRAW-OUT AUXILIARY SWITCH (CLOSES WHEN THE BREAKER IS CLOSED) |
| 69 152 b 70 | CIRCUIT BREAKER FIXED AUXILIARY SWITCH (OPENS ONLY WHEN THE BREAKER IS CLOSED AND IN OPERATING POSITION) |
| 83 33 a 84 | CIRCUIT BREAKER POSITION CELL SWITCH (CLOSES WHEN THE BREAKER IS NOT IN OPERATING POSITION) |
| 83 b | CIRCUIT BREAKER POSITION CELL SWITCH (OPENS WHEN BREAKER IS NOT IN OPERATING POSITION) ⚠ |
| 42-0 a | AUXILIARY CONTACT |
| | SEPARABLE CONNECTORS (ENGAGED) |
| 3A | FUSE |
| 350Ω 20W | FIXED RESISTOR |
| 2000Ω 20W SET @ 1800Ω | ADJUSTABLE RESISTOR |
| 5 50 6 | DRAW-OUT PROTECTIVE RELAY WITH CURRENT SHORTING DEVICE |
| | LOAD TEST JACK |
| | TEST SWITCH MAKE BEFORE BREAK |
| | TEST SWITCH THREE-POLE GANGED (POTENTIAL) |
| | TEST SWITCH (CURRENT) |
| | TEST SWITCH WITH TEST JACK (CURRENT) |
| | INSTRUMENT OR RELAY SHUNT (INSTRUMENT OR RELAY CONNECTIONS ARE THE TERMINALS INSIDE THE RECTANGLE) |
| | THERMOCOUPLE |

| SYMBOLS | DESCRIPTION |
|---|---|
| 50/5 | CURRENT TRANSFORMER NUMBERS - DENOTE DEVICE RATIO |
| 480-120V | POTENTIAL TRANSFORMER NUMBERS - DENOTE DEVICE RATIO (NOTE 2) ⚠ |
| A 0-15A | INDICATING INSTRUMENT<br>A -AMMETER<br>F -FREQUENCY METER<br>V -VOLTMETER<br>VAR -VARMETER<br>W -WATTMETER<br>NUMBERS - DENOTE DEVICE RANGE |
| A | TRANSDUCER<br>A -CURRENT<br>F -FREQUENCY<br>V -VOLTAGE<br>VAR -VARS<br>W -WATTS |
| G WITH XFMR, G WITH RES, G | INDICATING LIGHTS<br>CODE COLOR FUNCTIONS<br>R - RED EQUIP OPERATING, BREAKER CLOSED, ON<br>G - GREEN EQUIP NOT OPERATING, BREAKER OPEN, OFF<br>A - AMBER TRIP OVERLOAD<br>Y - YELLOW SPECIAL USE<br>W - WHITE ELECT POTENTIAL AVAILABLE, OR SYS STATUS, OR MONITORING<br>B - BLUE SPECIAL USE<br>C - CLEAR SPECIAL USE<br>N - NEON BULB FOR LOCKOUT RELAY COIL SUPERVISION |
| N R | |
| 1 W 2 3 | PUSH-TO-TEST LIGHTS (FOR COLOR AND FUNCTIONS SEE ABOVE) |
| | RECTIFIER, DIODE |
| 10μF | CAPACITOR |
| | OUTSIDE CONTAINMENT<br>ELECTRICAL PENETRATION ASSEMBLY<br>INSIDE CONTAINMENT |
| M | MOTOR |
| | MOTOR SERIES WOUND |
| | MOTOR SHUNT WOUND |
| | CHASSIS CONNECTION |
| | GROUND |
| | CONNECTION |
| | NO CONNECTION |

TYPICAL CONTROL SWITCH DEVELOPMENT




TYPE OF SWITCH (1)
NUMERICAL SUFFIX IF REQUIRED, TO DIFFERENTIATE TWO OR MORE SWITCHES OF THE SAME TYPE AT THE SAME LOCATION
LOCATION OF SWITCH (2)
TYPE OF CONTACT(3)
TYPE OF SWITCH(1)
LOCATION OF SW(2)
TYPE OF CONTACT(4)

1. TYPE OF SWITCH

CS -CONTROL SWITCH
SLS -SELECTOR SWITCH
SS -SYNCHRONIZING SWITCH
TS -TEST SWITCH
TRS -TRANSFER SWITCH
PB -PUSHBUTTON

2. LOCATION OF SWITCH OR PUSHBUTTON

R -CONTROL ROOM
L -LOCAL(NEAR CONTROLLED EQUIPMENT)
LR -LOCAL-REMOTE(OTHER THAN LOCAL OR CONTROL ROOM)

3. TYPE OF CONTACT FOR CONTROL SWITCH (CS) ONLY

CLOSE, START, TRIP, STOP — CONTACT CLOSES WHEN THE SWITCH IS IN POSITION INDICATED
CIN. AUTO — CLOSED IN NORMAL OR IN AUTO, CONTACT CLOSES IN NORMAL OR IN AUTO POSITION ONLY (CONTACT SHOWN NORMALLY CLOSED)
CAC — CLOSED AFTER CLOSED, DEFINES CONTACT WITH MEMORY ACTION WHICH CLOSES IN THE CLOSE/START AND REMAINS CLOSED UPON SWITCH RETURN TO NORMAL POSITION.
CAT — CLOSED AFTER TRIP, DEFINES CONTACT WITH MEMORY ACTION WHICH CLOSES IN THE TRIP/STOP AND REMAINS CLOSED UPON SWITCH RETURN TO NORMAL POSITION.
OPL — OPEN WHEN IN PULL TO LOCK POSITION (CONTACT OPENS ONLY WHEN HANDLE IS PUSHED IN)
CPL — CLOSED WHEN IN PULL TO LOCK POSITION (CONTACT CLOSES ONLY WHEN HANDLE IS PUSHED IN)

4. TYPE OF CONTACT

CLOSE, START, TRIP, STOP, OFF, REMOTE, AUTO, MANUAL — POSITION OF THE SWITCH IN WHICH THE CONTACT IS CLOSED

62 TDE 2.5-50 SEC — DELAY ON ENERGIZING TIMING RELAY.

THE INSTANTANEOUS CONTACT (INST) CHANGES STATUS IMMEDIATELY WHEN THE COIL (TDE) IS ENERGIZED, TDC (TIME DELAY CLOSING) CONTACT AND TDO (TIME DELAY OPENING) CONTACT CHANGES STATUS 10 SECONDS AFTER MAINTAINED ENERGIZATION OF THE COIL. ALL CONTACTS RETURN TO AS SHOWN POSITIONS WITH NO DELAY WHEN THE COIL IS DE-ENERGIZED.

62 TDD 2.5-50 SEC — DELAY ON DE-ENERGIZING TIMING RELAY.

WHEN THE COIL (TDD) IS ENERGIZED ALL CONTACTS CHANGE STATUS IMMEDIATELY AND REMAIN IN THAT POSITION. WHEN THE COIL IS DE-ENERGIZED THE INSTANTANEOUS CONTACT RETURNS TO AS SHOWN POSITION WITH NO DELAY, TDC (TIME DELAY CLOSING) CONTACT AND TDO (TIME DELAY OPENING) CONTACT RETURN TO AS SHOWN POSITION AFTER 5 SECONDS.

BX — MECHANICALLY LATCHING AUXILIARY RELAY WITH (O) -OPERATING AND (R) RESET COILS
(RESET CUT-OFF CONTACT SHOWN)

AX — AUXILIARY RELAY
(CONTACT STATUS SHOWN WHEN RELAY IS DE-ENERGIZED)

250 2409 VLV CLOSED — SOLENOID OPERATED LIMIT SWITCH
(CONTACT CLOSES WHEN VALVE IS IN POSITION INDICATED)

PSH 2035 CLOSES ON HIGH PRESS
LSL 1838 OPENS ON LOW LEVEL — PROCESS SWITCH
CONTACTS ARE SHOWN IN STANDARD REFERENCE POSITIONS; LOW PRESSURE, LOW LEVEL, LOW FLOW, ETC. (FUNCTIONAL DESCRIPTIONS SUCH AS "CLOSES ON" OR "OPENS ON" SHALL BE USED TO DESCRIBE WHEN THE CONTACT CHANGE STATUS TO PERFORM ITS FUNCTION)

VALVE NOT CLOSED
VALVE NOT OPEN — MOTOR OPERATED VALVE LIMIT SWITCH
(CONTACT CLOSES WHEN VALVE IS IN POSITION INDICATED)

DEVICE FUNCTION NUMBERS

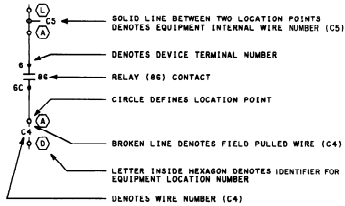
| | |
|---|---|
| 2 | TIME-DELAY STARTING OR CLOSING RELAY |
| 8 | CONTROL POWER DISCONNECTING DEVICE |
| 12 | OVER-SPEED DEVICE |
| 13 | SYNCHRONOUS-SPEED DEVICE |
| 14 | UNDER-SPEED DEVICE |
| 15 | SPEED OR FREQUENCY, MATCHING DEVICE |
| 21 | DISTANCE RELAY |
| 25 | SYNCHRONIZING OR SYNCHRONISM - CHECK DEVICE |
| 27 | UNDERVOLTAGE RELAY |
| 30 | ANNUNCIATOR RELAY |
| 32 | DIRECTIONAL POWER RELAY |
| 33 | POSITION SWITCH |
| 37 | UNDERCURRENT OR UNDERPOWER RELAY |
| 38 | BEARING PROTECTIVE DEVICE |
| 39 | MECHANICAL CONDITION MONITOR |
| 40 | FIELD RELAY |
| 41 | FIELD CIRCUIT BREAKER |
| 42 | RUNNING CIRCUIT BREAKER |
| 46 | REVERSE-PHASE, OR PHASE-BALANCE, CURRENT RELAY |
| 47 | PHASE-SEQUENCE VOLTAGE RELAY |
| 49 | MACHINE, OR TRANSFORMER, THERMAL RELAY |
| 50 | INSTANTANEOUS OVERCURRENT, OR RATE-OF RISE RELAY |
| 51 | AC TIME OVERCURRENT RELAY |
| 52 | AC CIRCUIT BREAKER |
| 53 | EXCITER OR DC GENERATOR RELAY |
| 59 | OVERVOLTAGE RELAY |
| 60 | VOLTAGE BALANCE RELAY |
| 62 | TIME-DELAY STOPPING OR OPENING RELAY |
| 64 | GROUND PROTECTIVE RELAY |
| 67 | AC DIRECTIONAL OVERCURRENT RELAY |
| 71 | LIQUID OR GAS LEVEL RELAY |
| 72 | DC CIRCUIT BREAKER |
| 74 | ALARM RELAY |
| 76 | DC OVERCURRENT RELAY |
| 81 | FREQUENCY RELAY |
| 86 | LOCKING-OUT RELAY |
| 87 | DIFFERENTIAL PROTECTIVE RELAY |
| 90 | REGULATING DEVICE |
| 98 | BYPASS STATUS RELAY |
| 63 | LIQUID OR GAS PRESSURE OR VACCUUM RELAY |
| 80 | LIQUID OR GAS FLOW RELAY |
| 96 | SMOKE DETECTOR RELAY |

SYSTEM VOLTAGE PREFIXES FOR DEVICE FUNCTION NUMBERS

| | |
|---|---|
| NONE | -480 VOLT OR BELOW |
| 1 | -4160 VOLT |
| 2 | -13800 VOLT |
| 3 | -25KV |
| 4 | -230KV |
| 5 | -500KV |

EXAMPLE: 127-4160V SYSTEM UNDERVOLTAGE RELAY

TYPICAL WIRE, TERMINAL, AND LOCATION NUMBERS




NOTES:

1. UNIT DESIGNATION FOR HAND SWITCHES, FIELD PROCESS DEVICES ETC. IS THE SAME UNIT DESIGNATOR OF THE EQUIPMENT TAG NUMBER. UNLESS OTHERWISE NOTED UNIT DESIGNATOR FOR THE ABOVE DEVICES IS OMITTED.
2. ⚠ ALTHOUGH MCC E/D'S REFLECT OVERALL VOLTAGE RATIOS OF 480-120V FOR CONTROL TRANSFORMERS, EACH TRANSFORMER SHALL HAVE A DUAL RATING OF 480/460-120V WITH THE PRIMARY SET AT THE 460V TAP.
3. ALL CT LEADS AT 13.8KV AND 4.16KV SWGR SHALL BE TERMINATED BY USING SHORTING TYPE TERMINAL BLOCK. A SHORTING TYPE TB IS NOT REQUIRED AT CONTROL BOARDS & PANELS.

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF.E | P.E. | Q.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 3 | ADDED NOTE 2 AND NOTE 3 PER GB-2788 | 8-4-11 | | | | | | | |
| 2 | ISSUED FOR CONSTRUCTION & VENDOR INFO. | 7-26-11 | | | | | | | |

ENG. MGR. DATE

BECHTEL LOS ANGELES

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

GENERAL NOTES & SYMBOLS
DEVICE FUNCTION NUMBERS & SYMBOL LIST
FOR ELEMENTARY DIAGRAMS

SCALE: NONE
JOB NO. 9510
DRAWING NO. 1X3D-AA-A00M
REV. 1
SIZE E 24 x 36

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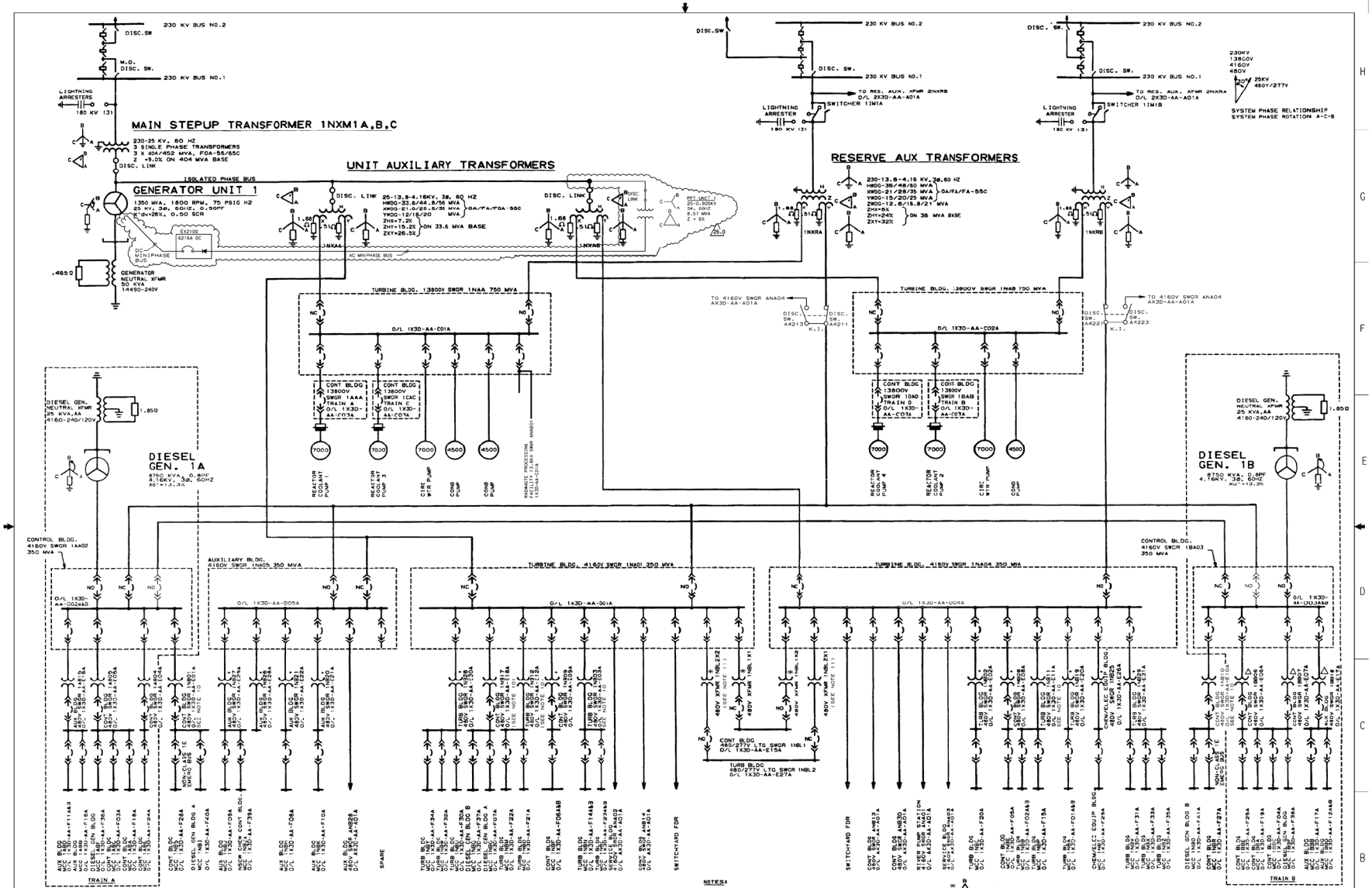
MAIN STEPUP TRANSFORMER 1NXM1A,B,C
230-25 KV, 60 HZ
3 SINGLE PHASE TRANSFORMERS
3 X 404/452 MVA, FOA-55/65C
Z =9.0% ON 404 MVA BASE
UNIT AUXILIARY TRANSFORMERS
RESERVE AUX TRANSFORMERS
GENERATOR UNIT 1
1350 MVA, 1800 RPM, 75 PSIG H2
25 KV, 3Ø, 60HZ, 0.9PF
X'd=28%, 0.50 SCR
GENERATOR NEUTRAL XFMR
50 KVA
14490-240V
ISOLATED PHASE BUS
AC MINIPHASE BUS
DC MINIPHASE BUS
230 KV BUS NO.2
230 KV BUS NO.1
DISC.SW
M.O. DISC. SW.
DISC. SW.
DISC. LINK
LIGHTNING ARRESTERS
180 KV (3)
SWITCHER 1IM1A
SWITCHER 1IM1B
TO RES. AUX. XFMR 2NXRB O/L 2X3D-AA-A01A
TO RES. AUX. XFMR 2NXRA O/L 2X3D-AA-A01A
230-13.8-4.16 KV, 30, 60 HZ
HMDG-36/48/60 MVA
XMDG-21/28/35 MVA
YMDG-15/20/25 MVA
ZMDG-12.6/16.8/21 MVA
ZHX=8%
ZHY=24%
ZXY=32%
ON 36 MVA BASE
1NXRA
1NXRB
1NXAA
1NXAB
EX2100
421A DC
SYSTEM PHASE RELATIONSHIP
SYSTEM PHASE ROTATION A-C-B
TURBINE BLDG. 13800V SWGR 1NAA 750 MVA
TURBINE BLDG. 13800V SWGR 1NAB 750 MVA
O/L 1X3D-AA-C01A
O/L 1X3D-AA-C02A
CONT BLDG 13800V SWGR 1AAA TRAIN A O/L 1X3D-AA-C07A
CONT BLDG 13800V SWGR 1CAC TRAIN C O/L 1X3D-AA-C07A
CONT BLDG 13800V SWGR 1DAD TRAIN D O/L 1X3D-AA-C07A
CONT BLDG 13800V SWGR 1BAB TRAIN B O/L 1X3D-AA-C07A
REACTOR COOLANT PUMP 1
REACTOR COOLANT PUMP 3
CIRC WTR PUMP
COND PUMP
REACTOR COOLANT PUMP 4
REACTOR COOLANT PUMP 2
CIRC WTR PUMP
COND PUMP
BARMETRIC PROCESSING FACILITY 13.8kV SWGR ANB01 1X3D-AA-C01A
TO 4160V SWGR ANA04 AX3D-AA-A01A
TO 4160V SWGR ANA04 AX3D-AA-A01A
DISC. SW. A4213
DISC. SW. A4211
DISC. SW. A4221
DISC. SW. A4223
K.I.
DIESEL GEN. 1A
8750 KVA, 0.8PF
4.16KV, 3Ø, 60HZ
DIESEL GEN. 1B
DIESEL GEN. NEUTRAL XFMR 25 KVA,AA 4160-240/120V
CONTROL BLDG. 4160V SWGR 1AA02 350 MVA
CONTROL BLDG. 4160V SWGR 1BA03 350 MVA
AUXILIARY BLDG. 4160V SWGR 1NA05 350 MVA
TURBINE BLDG. 4160V SWGR 1NA01 350 MVA
TURBINE BLDG. 4160V SWGR 1NA04 350 MVA
O/L 1X3D-AA-D02AB
O/L 1X3D-AA-D05A
O/L 1X3D-AA-D01A
O/L 1X3D-AA-D04A
O/L 1X3D-AA-D03AM
480V XFMR 1NBL2X2
480V XFMR 1NBL1X1
480V XFMR 1NBL2X1
480V XFMR 1NBL1X2
CONT BLDG 480/277V LTG SWGR 1NBL1 O/L 1X3D-AA-E15A
TURB BLDG 480/277V LTG SWGR 1NBL2 O/L 1X3D-AA-E27A
TRAIN A
TRAIN B
SPARE
SWITCHYARD FOR
NOTES:
1. FOR DEVICE TABLE, GENERAL NOTES, REFERENCE DWG & LEGEND SEE DWG 1X3D-AA-A00B.
2. CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000 KVA AA 80°C RISE, Z=5.75%
3. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000/1333 KVA AA/FA 115°C RISE, Z=5.75%
4. LIGHTING LOAD CENTER TRANSFORMERS 4160-480Y/277V 750/1000 KVA AA/FA 115°C RISE, Z=5.75%
5. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 300 KVA AA 115°C RISE, Z=2.7%
6. DELETED
7. FOR SWYD DETAILS, SEE DWG AX3D-AA-L50A.
8. FOR SWITCHING/LOAD DISPATCH EQUIPMENT NUMBERS, SEE DWG 13XD-AA-A01B
9. FOR LOAD CENTER TRANSFORMER PRIMARY CONNECTIONS REPRESENTED IN NOTES 2-5, PHASES A,B AND C ARE CONNECTED TO TRANSFORMER TERMINALS H1,H2 AND H3, RESPECTIVELY.
10. NON-CLASS 1E LOAD CENTER TRANSFORMERS 4160-480V 1000/1333 KVA AA/FA 80°C RISE, Z=5.75%
11. LIGHTING LOAD CENTER TRANSFORMER 4160-480Y/277V 750/1000 KVA AA/FA 80°C RISE. Z=5.75%
12. K.I. = KEY INTERLOCK (THE KEY INTERLOCK PREVENTS BOTH SWITCHES FROM BEING CLOSED AT THE SAME TIME.)
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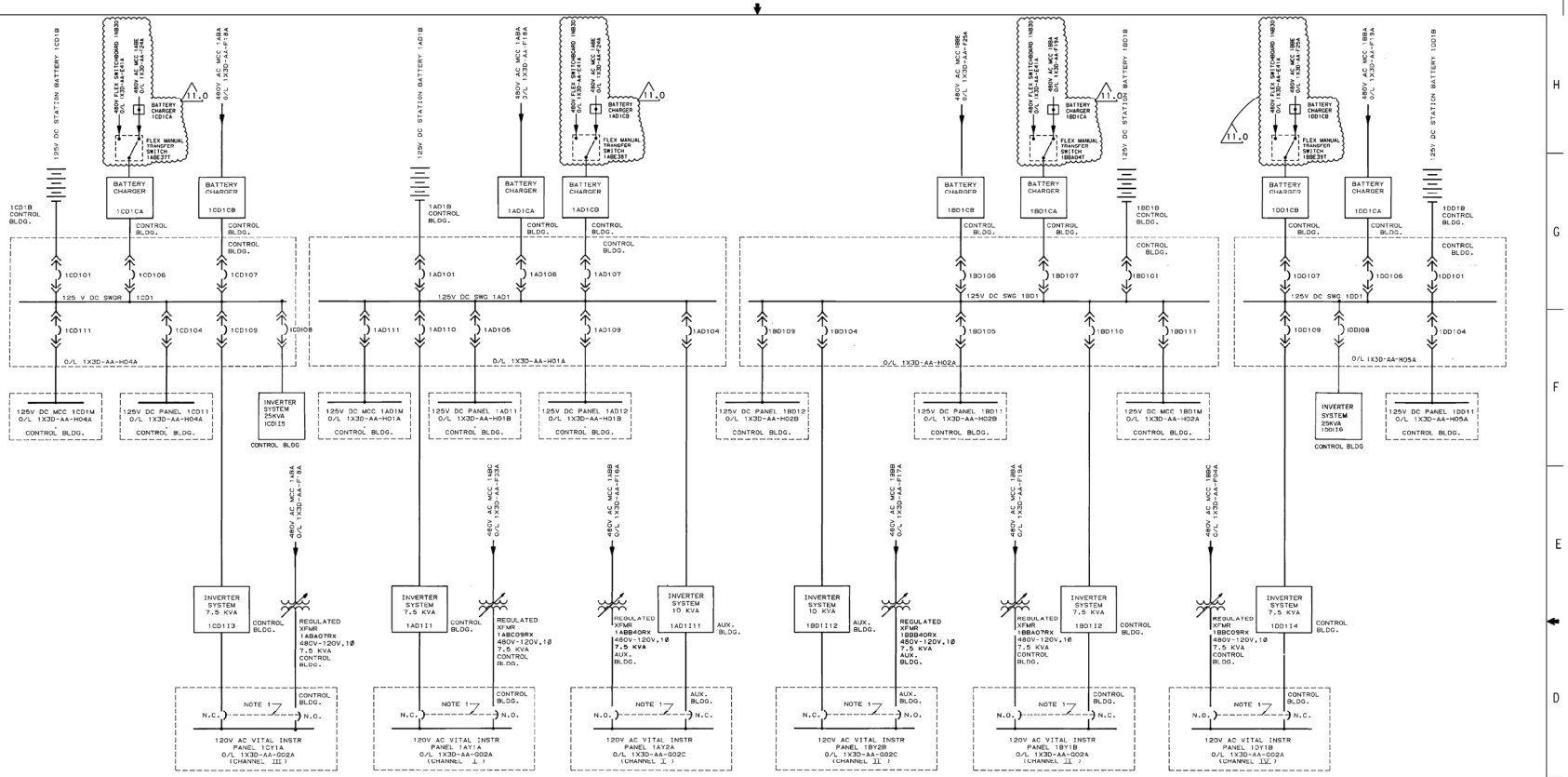
26.0 REVISED PER ABN 1049001501ED15, VER. 3.0 5-02-08 TSL RBH JMR
NO. VERSIONS DATE DR CHK APPV
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
MAIN ONE LINE
UNIT 1
SCALE: NONE
DRAWING NO. 1X3D-AA-A01A
VER. 26.0
JOB NO. 10604
SIZE E 34x44
DRAWING CATEGORY: CRITICAL




NOTE:
1. MECHANICAL INTERLOCKS ARE PROVIDED SUCH THAT ONLY ONE BREAKER OF THE TWO INTERLOCKED IS CLOSED.

NUCLEAR SAFETY RELATED

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

MAIN ONE LINE
CLASS 1E 125V DC AND
120V VITAL AC SYSTEMS

SCALE: NONE

DRAWING NO. 1X3D-AA-G01A

VER. 11.0

JOB NO.10604

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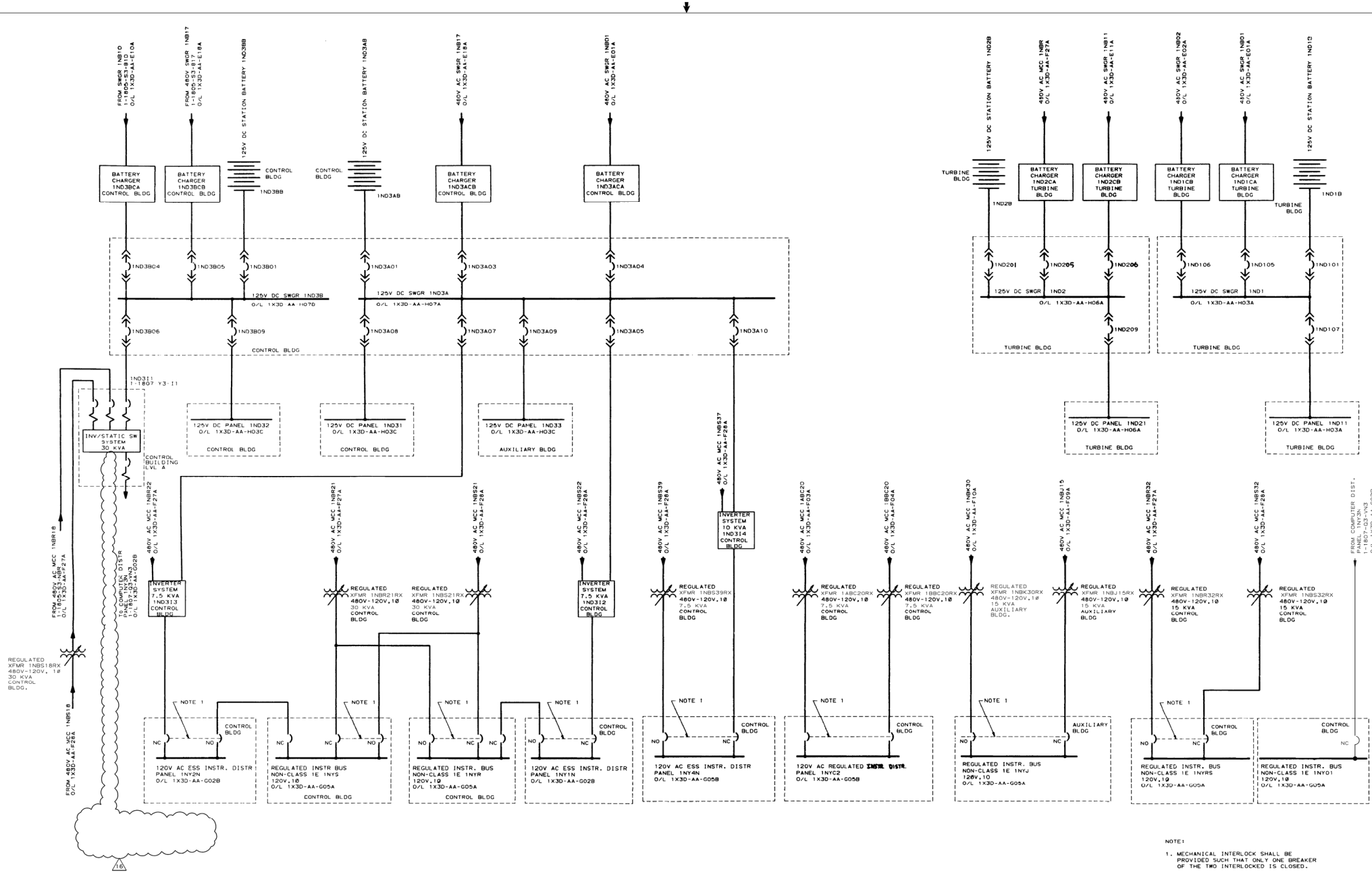
|---|---|---|---|---|
| 11.0 REVISED PER ABN SNC473105E010, VER. 1.0 | 09/18/15 | TDA | ALM | WLN |

SIZE E 34x44

DRAWING CATEGORY: CRITICAL




NOTE:

1. MECHANICAL INTERLOCK SHALL BE PROVIDED SUCH THAT ONLY ONE BREAKER OF THE TWO INTERLOCKED IS CLOSED.

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 1 | INCORP. PER 01-V1N0009 | 12-05-01 | CD | EGG | MWD | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

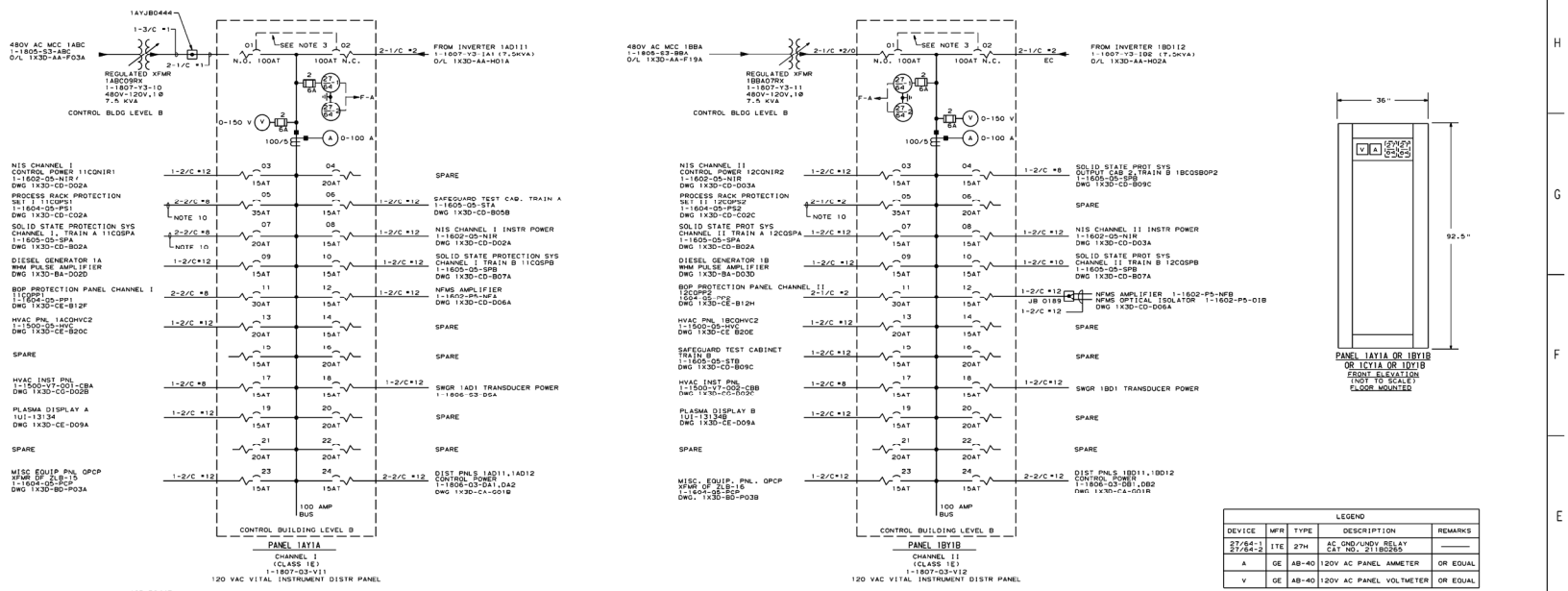
MAIN ONE LINE
NON-CLASS 1E 125V DC AND
120V ESSENTIAL AC SYSTEM

SCALE: NONE

JOB NO. 10604

| DRAWING NO. | REV. |
|---|---|
| 1X3D-AA-G01B | 16 |

SIZE E 34X44




| LEGEND | | | | |
|---|---|---|---|---|
| DEVICE | MFR | TYPE | DESCRIPTION | REMARKS |
| 27/64-1<br>27/64-2 | ITE | 27H | AC GND/UNDV RELAY<br>CAT NO. 211B0265 | ——— |
| A | GE | AB-40 | 120V AC PANEL AMMETER | OR EQUAL |
| V | GE | AB-40 | 120V AC PANEL VOLTMETER | OR EQUAL |

| FUNCTION | DESCRIPTION |
|---|---|
| F | |
| A | ANNUNCIATOR |

NOTES:

1. ALL BREAKERS SHOWN WITH #2 AWG OR LARGER FIELD WIRING CABLE SHALL BE PROVIDED WITH A 6" PIGTAIL OF #2 AWG PER POLE. (#4 AWG FOR BRANCH BREAKERS).
2. BREAKER INTERRUPTING RATINGS ARE 14,000 AMPS AT 120 VAC. (25,000 FOR INPUT BREAKERS).
3. ONLY ONE OF TWO INTERLOCKED BREAKERS FOR EACH PANEL CAN BE CLOSED AT ONE TIME.
4. THE SYSTEM IS UNGROUNDED.
5. ALL BREAKERS, EXCEPT AS NOTED, ARE 15 AT.
6. ALL BREAKERS ARE 2 POLE, 100 AMP FRAME.
7. EACH BREAKER SHALL HAVE AN AUXILIARY "B" BREAKER ALARM CONTACT WHICH SHALL BE WIRED TO AN INTERPOSING RELAY.THE OUTPUT OF THIS RELAY SHALL BE WIRED TO STATES CO. TYPE ZWM TERMINAL BLOCKS INTERNAL IN THE PANEL.
8. UNLESS OTHERWISE NOTED, ALL CABLES ARE ROUTED IN TRAY.
9. ALL #4 AWG & SMALLER CABLE SHALL BE TERMINATED AT THE DISTR. PNLS. USING LUGS MK NOS. PRM,PRQ, OR PRX.
10. CONNECT IN PARALLEL TO ONE #6 PIGTAIL OF #10 AWG.
11. FOR GENERAL NOTES AND SYMBOLS REFER TO DWG. 1X3D-AA-A00A.
12. INDIVIDUAL ELECTRICAL LOADS FOR PANELS 1AY1A, 1BY1B, 1CY1A & 1DY1B ARE LISTED AND TOTALED IN CALCULATION X3CN01.

NUCLEAR SAFETY RELATED [A] [B] [C] [D]

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| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 30.0 | REVISED PER ABN SNC473105E007, VER. 1.0 | 10/01/15 | TOA | ALM | WLN |

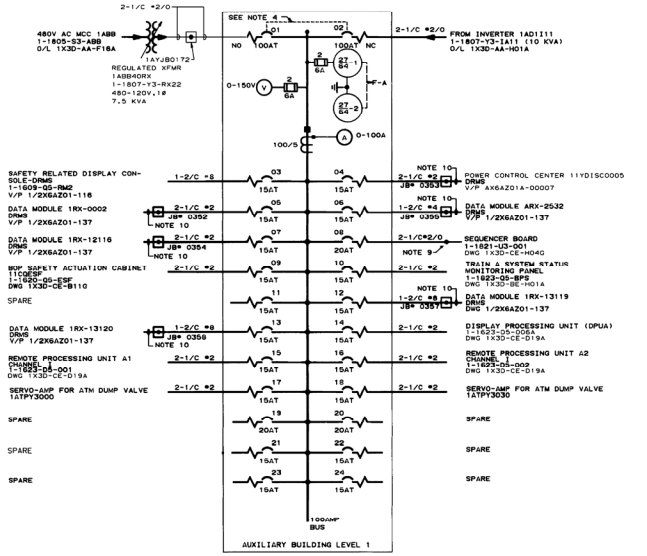
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

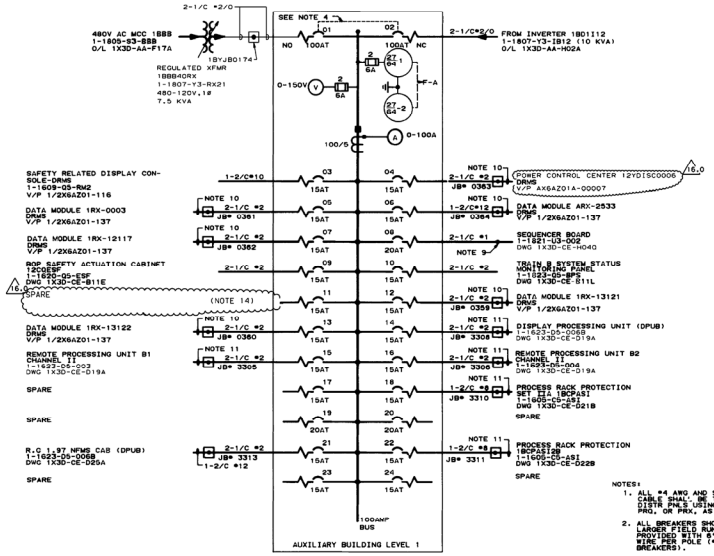
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

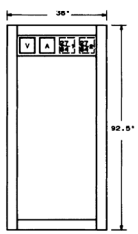
ONE LINE DIAGRAM
120V AC CLASS 1E
VITAL INSTRUMENT DIST PNLS
1AY1A, 1BY1B, 1CY1A & 1DY1B

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X3D-AA-G02A
VER. 30.0

SIZE E 34X44
DRAWING CATEGORY: CRITICAL




PANEL 1AY2A & 1BY2B
FRONT ELEVATION
(NOT TO SCALE)
FLOOR MOUNTED

| LEGEND | | | | |
|---|---|---|---|---|
| DEVICE | MFR | TYPE | DESCRIPTION | REMARKS |
| 27/64-1 27/64-2 | 1TE | 27H | AC GND/UNDV RELAY CAT# 2119D26G | — |
| V | GE | AB-40 | 120V AC PNL VOLTMETER | OR EQUAL |
| A | GE | AB-40 | 120V AC PNL AMMETER | OR EQUAL |

| FUNCTION F | DESCRIPTION |
|---|---|
| A | ANNUNCIATOR |

NOTES:
1. ALL #4 AWG AND SMALLER FIELD RUN CABLE SHALL BE TERMINATED AT THE DISTR PNLS USING LUGS MK NOS. PRM, PRQ, OR PRX, AS APPROPRIATE.
2. ALL BREAKERS SHOWN WITH A #2 AWG OR LARGER FIELD RUN CABLE SHALL BE PROVIDED WITH 6" PIGTAIL OF #2 AWG WIRE PER POLE (#4AWG FOR BRANCH BREAKERS).
3. BREAKER INTERRUPTING RATING ARE 18,000 AMPS AT 120V AC.(25,000 AMPS FOR INPUT BREAKER).
4. ONLY ONE OF TWO INTERLOCKED BREAKERS FOR EACH PANEL CAN BE CLOSED AT ONE TIME.
5. THE SYSTEMS ARE UNGROUNDED.
6. ALL BREAKERS ARE 2 POLE, 100 AMP FRAME.
7. EACH BREAKER SHALL HAVE AN AUXILIARY OR BREAKER ALARM CONTACT WHICH SHALL BE WIRED TO AN INTERPOSING RELAY. THE OUTPUT OF THIS RELAY SHALL BE WIRED TO STATES CO. TYPE XW TERMINAL BLOCKS INTERNAL IN THE PANEL.
8. UNLESS OTHERWISE NOTED, ALL CABLES ARE ROUTED IN TRAY.
9. A BUTT SPLICE IS TO BE MADE TO 1-2/C #10 CABLE AT THE LOAD.
10. CABLE 1-2/C #12 SHIELD (A2SS) FROM JUNCTION BOX TO LOAD. JUNCTION BOX TO BE LOCATED WITHIN 10 FT OF LOAD.
11. CABLE 1-2/C #12 (A2SS) FROM JUNCTION BOX TO LOAD.
12. FOR GENERAL NOTES AND SYMBOLS REFER TO DWG.1X3D-AA-A00A.
13. INDIVIDUAL ELECTRICAL LOADS FOR PANELS 1AY2A & 1BY2B ARE LISTED AND TOTALED IN CALCULATION X3CH01.
14. FIELD SIDE OF CABLE 1BY2B11SA IS SPARED. CABLE IS LIFTED AND TAPED. CARE WILL NEED TO BE TAKEN PRIOR TO CLOSING THE BREAKER.

NUCLEAR SAFETY RELATED A & B

Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

120V AC CLASS 1E
VITAL INSTR DISTR PNLS
1-1807-03-V15 & V16

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| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 16.0 | REVISED PER ABN SNC121508E002, VER. 2.0 | 12/03/15 | SMD | SO | WLN |

SCALE: NONE
JOB NO.N/A
SIZE E 34x44

| DRAWING NO. | VER. |
|---|---|
| 1X3D-AA-G02C | 16.0 |

DWG CATEGORY: CRITICAL

| EQUIPMENT TAG NUMBER | EQUIPMENT DESCRIPTION | PROJECT CLASSIFICATION (ELECTRICAL SYSTEM) | PROCUREMENT SPECIFICATION | BUS NUMBER | BUS VOLTAGE | NAMEPLATE RATING HP (OR KW) | REFERENCE DWG. 1X3D- | LOSS OF COOLANT ACCIDENT SI PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL, PROCESS, ACCIT, OR SEQUENCER START SEE TABLE 1 | LOSS OF COOLANT ACCIDENT SI PREFERRED OFFSITE SOURCE AVAILABLE: TIME (SECONDS) TO START (NOTE 7) | LOSS OF COOLANT ACCIDENT SI PREFERRED OFFSITE SOURCE AVAILABLE: RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL, PROCESS, ACCIT, OR SEQUENCER START SEE TABLE 1 | LOSS OF OFFSITE POWER & SUBSEQUENT SI: TIME (SEC) TO START AFTER D/G BRKR IS CLOSED (NOTE 3.b) | LOSS OF OFFSITE POWER & SUBSEQUENT SI: RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER: MANUAL, PROCESS, ACCIT, OR SEQUENCER START SEE TABLE 1 | LOSS OF OFFSITE POWER: TIME (SEC) TO START AFTER D/G BRKR IS CLOSED (NOTE 3.c) | LOSS OF OFFSITE POWER: RUNNING TIME AFTER START (NOTE 4) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11202P4001M01 | NUC SERV CLG WTR PP | 11E | X4 | 1AA02 | 4160V | 700 | BD-K04A | S | 25.5 | C | S | 25.5 | C | S | 25.5 | C |
| 11202P4003M01 | NUC SERV CLG WTR PP | 11E | X4 | 1AA02 | 4160V | 700 | BD-K04C | S | 25.5 | C | S | 25.5 | C | S | 25.5 | C |
| 11202P4005M01 | NUC SERV CLG WTR PP (SPARE) | 11E | X4 | 1AA02 | 4160V | 700 | BD-K04E | NOTE. 5 | 30.5 | — | NOTE 5 | 30.5 | — | NOTE 5 | 30.5 | — |
| 11203P4001M01 | COMP CLG WTR PP | 11E | X4 | 1AA02 | 4160V | 300 | BD-L01A | S | 20.5 | C | S | 20.5 | C | S | 20.5 | C |
| 11203P4003M01 | COMP CLG WTR PP | 11E | X4 | 1AA02 | 4160V | 300 | BD-L01C | S | 20.5 | C | S | 20.5 | C | S | 20.5 | C |
| 11203P4005M01 | COMP CLG WTR PP (SPARE) | 11E | X4 | 1AA02 | 4160V | 300 | BD-L01E | NOTE 5 | 25.5 | — | NOTE 5 | 25.5 | — | NOTE 5 | 25.5 | — |
| 11204P6003M01 | SAFETY INJ PP | 11E | X6 | 1AA02 | 4160V | 450 | BD-D01C | S | 5.5 | C | S | 5.5 | C | M | >30.5 | C |
| 11205P6001M01 | RESID HT REMVL PP | 11E | X6 | 1AA02 | 4160V | 400 | BD-E01A | S | 10.5 | C | S | 10.5 | C | M | >30.5 | C |
| 11206P6001M01 | CONTAINMENT SPRAY PP | 11E | X6 | 1AA02 | 4160V | 400 | BD-J01A | NOTE 6 | 15.5 | C | NOTE 6 | 15.5 | C | M | >30.5 | C |
| 11208P6002M01 | CENTRIFUGAL CHARG PP | 11E | X6 | 1AA02 | 4160V | 600 ** | BD-C01A | S | 0.5 | C | S | 0.5 | C | S | 0.5 | C |
| 11217P4001M01 | AUX COMP CLG WTR PP | 11E | X4 | 1AA02 | 4160V | 600 | BD-L03A | M | >30.5 | C | M | >30.5 | C | S | 15.5 | C |
| 11302P4003M01 | AUX FDWTR PP-MOTOR DRIVEN | 11E | X4 | 1AA02 | 4160V | 900 | BC-F04A | S | 20.5 | C | S | 20.5 | C | 1HV-1975 * | AUX CCW RETRN ISO | 11E | X5 | 1ABB | 480V | .66 | BD-L03G | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC |
| 11592C7001M01 | ESF CHILLER (COMPRESSOR) | 11E | X4 | 1AA02 | 4160V | 400 | BG-G02A | P | >30.5 | C | P | >30.5 | C | 1HV-1979 * | AUX CCW SUPPLY ISO | 11E | X5 | 1ABB | 480V | .66 | BD-L03J | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC |
| | | | | | | | | | | | | | | DISTR PANEL 1AYB1 | DISTRIBUTION PANEL | 11E | X3 | 1ABB | 480V | 15KVA | AA-F16A | N | 0 | C | N | 0.5 | C |
| 11805S3B04 | LOAD CENTER TRANS. 1AB04 | 11E | X3 | 1AB04 | 4160 -480V | 1000KVA | BA-D02F | — | — | C | — | 0 | C | — | 0 | C | 1PV-3000 | ATMOSPHERIC DUMP VALVE | 11E | X5 | 1ABB | 480V | 1.5 | BC-Q03G | K | 0 | 20 SEC | K | 0.5 | 20 SEC |
| | SECONDARY BREAKER 1AB04 | 11E | X3 | 1AB04 | 480V | — | BA-E02A | — | — | C | S | 0.5 | C | S | 0.5 | C | 1PV-3030 | ATMOSPHERIC DUMP VALVE | 11E | X5 | 1ABB | 480V | 1.5 | BC-Q03G | K | 0 | 20 SEC | K | 0.5 | 20 SEC |
| 11501A7001M01 | CTMT CLG UNIT HIGH SPEED | 11E | X4 | 1AB04 | 480V | 125 | BG-B01A | NOTE 10 | NA | NA | NOTE 10 | NA | NA | 5 NOTE 15 | 50.5 | C | 1HV-11600 | NSCW PUMP P4001 OUT | 11E | X5 | 1AB | 480V | 2.0 | BD-K04Z | I | 70.5 | 15 SEC | I | 70.5 | 15 SEC |
| | CTMT CLG UNIT LOW SPEED | 11E | X4 | 1AB04 | 480V | 62.5 | BG-B03F | S | 30.5 | C | S | 30.5 | C | NOTE 10 | NA | NA | 1HV-5137 | AUX FD PP P4003 DISCH MOV | 11E | X5 | 1ABB | 480V | 1.3 | BC-F08C | L | 0 | 10 SEC | L | 0.5 | 10 SEC |
| 11501A7002M01 | CTMT CLG UNIT HIGH SPEED | 11E | X4 | 1AB04 | 480V | 125 | BG-B01B | NOTE 10 | NA | NA | NOTE 10 | NA | NA | 5 NOTE 15 | 50.5 | C | 1HV-5139 | AUX FD PP P4003 DISCH MOV | 11E | X5 | 1ABB | 480V | 1.3 | BC-F08D | L | 0 | 10 SEC | L | 0.5 | 10 SEC |
| | CTMT CLG UNIT LOW SPEED | 11E | X4 | 1AB04 | 480V | 62.5 | BG-B03G | S | 30.5 | C | S | 30.5 | C | NOTE 10 | NA | NA | 11540B7005M01 * | TB AB ELEC TUNNEL VENT FAN | 11E | X4 | 1ABB | 480V | 10 | BG-K01E | MM | 0 | C | MM | 0.5 | C |
| 11501A7005M01 | CTMT CLG UNIT HIGH SPEED | 11E | X4 | 1AB04 | 480V | 125 | BG-B01E | NOTE 10 | NA | NA | NOTE 10 | NA | NA | S | 30.5 | C | PNL 1AYC1 | 120/240 DISTRIB PANEL | 11E | X3 | 1ABC | 480V | 10KVA | AA-F03A | N | 0 | C | N | 0.5 | C |
| | CTMT CLG UNIT LOW SPEED | 11E | X4 | 1AB04 | 480V | 62.5 | BG-B03K | S | 30.5 | C | S | 30.5 | C | NOTE 10 | NA | NA | 11532B7001M01 | CBSF BATT RM EXH FAN | 11E | X4 | 1ABC | 480V | 1.5 | BG-C04N | L | 0 | C | L | 0.5 | C |
| 11501A7006M01 | CTMT CLG UNIT HIGH SPEED | 11E | X4 | 1AB04 | 480V | 125 | BG-B01F | NOTE 10 | NA | NA | NOTE 10 | NA | NA | S | 30.5 | C | 11532B7003M01 | CBSF BATT RM EXH FAN (STBY) | 11E | X4 | 1ABC | 480V | 1.5 | BG-C04Q | L | 0 | C | L | 0.5 | C |
| | CTMT CLG UNIT LOW SPEED | 11E | X4 | 1AB04 | 480V | 62.5 | BG-B03L | S | 30.5 | C | S | 30.5 | C | NOTE 10 | NA | NA | 11532A7001M01 | CBSF ELEC EQUIP RM A/C | 11E | X4 | 1ABC | 480V | 40 | BG-C04A | L | 0 | C | L | 0.5 | C |
| | | | | | | | | | | | | | | | | | 11539A7001M01 NOTE 19 | CB AUX RELAY RM A C | 11E | X4 | 1ABC | 480V | 2 | BG-C07M | L | 0 | C | L | 0.5 | C |
| 11805S3B05 | LOAD CENTER TRANS. 1AB05 | 11E | X3 | 1AB05 | 4.16 -480V | 1000KVA | BA-D02L | — | — | C | — | 0 | C | — | 0 | C | 11807Y31Q | REG XFMR 1ABC09RX | 11E | X3 | 1ABC | 480V | 7.5KVA | AA-G01A | N | 0 | C | N | 0.5 | C |
| | SECONDARY BREAKER 1AB05 | 11E | X3 | 1AB05 | 480V | — | BA-E02P | — | — | C | S | 0.5 | C | S | 0.5 | C | | | | | | | | | | | | | | |
| 11566B7001M01 | DGB VENT FAN | 11E | X4 | 1AB05 | 480V | 50 | BG-F01B | I | >30.5 | C | I | >30.5 | C | I | >30.5 | C | | | | | | | | | | | | | | |
| 11566B7003M01 | DGB VENT FAN | 11E | X4 | 1AB05 | 480V | 50 | BG-F01C | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | | | | | | | | | | | | | | |
| 11531N7001H01 | CB CONTROL ROOM HTR | 11E | X4 | 1AB05 | 480V | 118KW | BG-C05R | N NOTE 11 | >30.5 | C | N NOTE 11 | >30.5 | C | N NOTE 11 | >30.5 | C | 11808T3116 | LTG ISOLATION XFMR 1ABC23RX | 11E | X3 | 1ABC | 480V | 15KVA | AA-F03A | N | 0 | C | N | 0.5 | C |
| 11531N7001M01 | CB CONTROL ROOM FIL UNIT | 11E | X4 | 1AB05 | 480V | 125 | BG-C01E | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | | | | | | | | | | | | | | |
| 11513H7001000 | CTB HYDROGEN RECOMBINER | 11E | X6 | 1AB05 | 480V | 75KW | BG-B02U | M | >30.5 | C | M | >30.5 | C | M | >30.5 | C | | | | | | | | | | | | | | |
| 11805S3B15 | LOAD CENTER TRANS. 1AB15 | 11E | X3 | 1AB15 | 4.16 -480V | 1000KVA | BA-D02K | — | — | C | — | 0 | C | — | 0 | C | 11807Y3RX25 | RGD POS IND CAB ISO XFMR 1ABC20RX | 11E | X3 | 1ABC | 480V | 7.5KVA | AA-G03B | N | 0 | C | N | 0.5 | C |
| | SECONDARY BREAKER 1AB15 | 11E | X3 | 1AB15 | 480V | — | BA-E02R | — | — | C | S | 0.5 | C | S | 0.5 | C | 1HV-2584A | CTB COOLING UNIT (DAMPER) | 11E | X4 | 1ABC | 480V | .13 | BG-B03X | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC |
| 11202W4001M01 | NUC SERV CLG TWR FAN | 11E | X4 | 1AB15 | 480V | 100 | BD-K03A | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | 1HV-2584B | CTB COOLING UNIT (DAMPER) | 11E | X4 | 1ABC | 480V | .13 | BG-B03X | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC |
| 11202W4001M02 | NUC SERV CLG TWR FAN | 11E | X4 | 1AB15 | 480V | 100 | BD-K03B | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | 1HV-12742 | CBSF BATTERY RM EXH | 11E | X4 | 1ABC | 480V | .2 | BG-C04S | I | 0 | 10 SEC | I | 0.5 | 10 SEC |
| 11202W4001M03 | NUC SERV CLG TWR FAN | 11E | X4 | 1AB15 | 480V | 100 | BD-K03C | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | 1HV-2628A | CTB NORM PURGE EXH ISO | 11E | X5 | 1ABC | 480V | 1.33 | BG-B04G | L | 0 | 10 SEC | L | 0.5 | 10 SEC |
| 11202W4001M04 | NUC SERV CLG TWR FAN | 11E | X4 | 1AB15 | 480V | 100 | BD-K03D | P | >30.5 | C | P | >30.5 | C | P | >30.5 | C | 1HV-12748 | CBSF BATTERY RM FAN (STBY) | 11E | X4 | 1ABC | 480V | .2 | BG-C04U | I | 0 | 10 SEC | I | 0.5 | 10 SEC |
| 11213P6002M01 | SPENT FUEL PIT PUMP | 11E | X6 | 1AB15 | 480V | 100 | BD-H01B | M | >30.5 | C | M | >30.5 | C | M | >30.5 | C | 1HV-8808C | ACCUM LOOP NO. 3 | 11E | X6 | 1ABC | 480V | 10.3 | BD-D02T | L | 0 | 10 SEC | L | 0.5 | 10 SEC |
| 11561N7001M01 | PIPE PEN RM FILTER/EXH FAN | 11E | X4 | 1AB15 | 480V | 75 | BG-D01D | S | 15.5 | C | S NOTE 17 | 15.5 | C | 1HV-19055 * | THERM BARR CW RCP 003 | 11E | X4 | 1ABC | 480V | .7 | BD-L03M | K | 0 | 10 SEC | K | 0.5 | 10 SEC |
| 11561N7001H01 | PIPE PEN FLTR/EXH UNIT HTR | 11E | X4 | 1AB15 | 480V | 80KW | BG-D01C | 12K NOTE 17 | 15.5 | C | 12K NOTE 17 | 15.5 | C | 1HV-19057 * | THERM BARR CW RCP 004 | 11E | X4 | 1ABC | 480V | .7 | BD-L03N | K | 0 | 10 SEC | K | 0.5 | 10 SEC |
| | | | | | | | | | | | | | | 1HV-8146 | CHARG TO REAC COOL SYS ISO | 11E | X6 | 1ABC | 480V | .7 | BD-C03T | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC |
| 11805S3B01 | LOAD CENTER TRANS. 1NB01 | 62E | X3 | 1NB01 | 4.16 -480V | 1000KVA | BA-D02G | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | —— | 0 | C | | | | | | | | | | | | | | |
| | SECONDARY BREAKER | 62E | X3 | 1NB01 | 480V | —— | BA-E01A | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | S | 10.5 | C | | | | | | | | | | | | | | |
| 11511B7001M01 | CTB CAV CLG FAN (NON 1E) | 62E | X4 | 1NB01 | 480V | 75 | BG-B02D | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | S | 10.5 | C | | | | | | | | | | | | | | |
| 11805G3PB1 | FDR TO HTR PNL 1NBPB1 | 62E | X3 | 1NB01 | 480V | 483KW | BD-B01K | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | K | 10.5 | C | | | | | | | | | | | | | | |
| 11806B3CN5 | BATTERY CHARGER 1ND3ACA | 62E | X3 | 1NB01 | 480V | 150KVA | BA-E03P | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | MM | 10.5 | C | | | | | | | | | | | | | | |
| 11806B3CN1 | BATTERY CHARGER 1ND1CA | 62E | X3 | 1NB01 | 480V | 100KVA | BA-E03F | SEE NOTE 3.a | >30.5 | C | SEE NOTE 3.a | >30.5 | C | MM | 10.5 | C | | | | | | | | | | | | | | |
| 11806B3CAA | BATTERY CHARGER 1AD1CA | 11E | X3 | 1ABA | 480V | 100KVA | AA-G01A | N | 0 | C | N | 0.5 | C | N | 0.5 | C | | | | | | | | | | | | | | |
| 11806B3CCB | BATTERY CHARGER 1CD1CB (NOTE 19) | 11E | X3 | 1ABA | 480V | 75KVA | AA-G01A | N | 0 | C | N | 0.5 | C | N | 0.5 | C | | | | | | | | | | | | | | |
| 11807Y312 | REG XFMR 1ABA07RX | 11E | X3 | 1ABA | 480V | 7.5KVA | AA-G02A | N | 0 | C | N | 0.5 | C | N | 0.5 | C | | | | | | | | | | | | | | |
| 11531B7002M01 | CBCR CHLR RM VENT FAN | 11E | X4 | 1ABA | 480V | 5.0 | BG-C01B | K | 0 | C | K | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| A1542N7001M01 | FHB PA FLTR EXH FAN | 11E | X4 | 1ABA | 480V | 40 | BG-E01A | K | 0 | C | K | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| A1542N7001H01 | FHB PA EXH UNIT HTR | 11E | X4 | 1ABA | 480V | 20KW | AA-F18A | 1,K | 0 | C | 1,K | 0.5 | C | 1,K | 0.5 | C | | | | | | | | | | | | | | |
| 11592P7001M01 | CB ESF CHILLED WATER PUMP | 11E | X4 | 1ABA | 480V | 30 | BG-G02C | L NOTE 16A | 120 | C | L NOTE 16A | 120 | C | M | >30.5 | C | | | | | | | | | | | | | | |
| 11592C7001M02 &M03 | CB ESF CHILLER PURGE COMP. B | 11E | X4 | 1ABA | 480V | 0.75 | AA-F18A | 1,K NOTE 16B,C | >30.5 | C | 1,K NOTE 16B,C | >30.5 | C | 1,K NOTE 16B,C | >30.5 | C | | | | | | | | | | | | | | |
| 11539A7005M01 | CB NORMAL AC RM ESF AC FAN MOT (NOTE 19) | 11E | X4 | 1ABA | 480V | 5 | BG-C07R | L | 0 | C | L | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| DISTR PNL 1AYA1 | 120/240 DISTRIBUTION PANEL | 11E | X3 | 1ABA | 480V | 15KVA | AA-F18A | N | 0 | C | N | 0.5 | C | N | 0.5 | C | | | | | | | | | | | | | | |
| 1HV-12114 | MCR AIR INTAKE ISO | 11E | X4 | 1ABA | 480V | 1.9 | BG-C01Q | K | 0 | 10 SEC | K | 0.5 | 10SEC | K | 0.5 | 10SEC | | | | | | | | | | | | | | |
| 1HV-12130 | CBCR RTRN AIR FAN IN | 11E | X4 | 1ABA | 480V | .2 | BG-C01X | I NOTE 11 | >30.5 | 10SEC | I NOTE 11 | >30.5 | 10 SEC | I NOTE 11 | >30.5 | 10SEC | | | | | | | | | | | | | | |
| 1HV-12118 | CBCR FLTR UNIT IN | 11E | X4 | 1ABA | 480V | .2 | BG-C01V | M | >30.5 | 10 SEC | M | >30.5 | 10 SEC | M | >30.5 | 10SEC | | | | | | | | | | | | | | |
| 1HV-12128 | CBCR FLTR UNIT OUT | 11E | X4 | 1ABA | 480V | .2 | BG-C01X | I NOTE 11 | >30.5 | 10 SEC | I NOTE 11 | >30.5 | 10 SEC | I NOTE 11 | >30.5 | 10SEC | | | | | | | | | | | | | | |
| 1HV-11605 | NSCW PUMP P4005 OUT | 11E | X5 | 1ABB | 480V | 1.9 | BD-K04Z | I | 70.5 | 15SEC | I | 70.5 | 15SEC | I | 70.5 | 15SEC | | | | | | | | | | | | | | |
| 1HV-8508A | AH CHG PUMP MINIFLOW | 11E | X6 | 1ABB | 480V | .7 | BD-C01T | L | 0 | 10SEC | L | 0.5 | 10SEC | I | 0.5 | 10SEC | | | | | | | | | | | | | | |
| 1HV-8509A | AH CHG PUMP MINIFLOW | 11E | X6 | 1ABB | 480V | .7 | BD-C01V | M | >30.5 | 10SEC | M | >30.5 | 10 SEC | M | >30.5 | 10SEC | | | | | | | | | | | | | | |
| 11807Y31A11 | INVERTER 1AD1111 | 11E | X3 | 1ABB | 480V | 10KVA | AA-G01A | SEE NOTE 14 | 0 | C | SEE NOTE 14 | 0.5 | C | SEE NOTE 14 | 0.5 | C | | | | | | | | | | | | | | |
| 11807Y3RX22 | REG XFMR 1ABB40RX | 11E | X3 | 1ABB | 480V | 7.5KVA | AA-G01A | N | 0 | C | N | 0.5 | C | N | 0.5 | C | | | | | | | | | | | | | | |
| 11202P4008M01 | NS CW XFR PUMP | 11E | X4 | 1ABB | 480V | 30 | BD-K01B | M | >30.5 | 4DAYS | M | >30.5 | 4DAYS | M | >30.5 | C | | | | | | | | | | | | | | |
| 11555A7005M01 | ELEC MCC ROOM COOLER | 11E | X4 | 1ABB | 480V | 2.0 | BD-D05E | L | 0 | C | L | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| 11555A7011M01 | CCWP ROOM COOLER FAN | 11E | X4 | 1ABB | 480V | 5.0 | BD-D05E | L | 0 | C | L | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| 11555A7017M01 | SFP PUMP ROOM CLR | 11E | X4 | 1ABB | 480V | 5.0 | BD-D05U | L | 0 | C | L | 0.5 | C | K | 0.5 | C | | | | | | | | | | | | | | |
| 1HV-11606 | NSCW PUMP P4003 OUT | 11E | X5 | 1ABB | 480V | 1.9 | BD-K04Z | I | 70.5 | 15SEC | I | 70.5 | 15SEC | I | 70.5 | 15SEC | | | | | | | | | | | | | | |

** SEE NOTE 21

NOTES:

1.) LOSS OF OFFSITE POWER (LOP) IS DEFINED AS A LOSS OF VOLTAGE ON THE 4.16KV BUS.

2.) TIME (SECONDS IS DEFINED AS THE TIME THE DIESEL GENERATOR BREAKER IS CLOSED FOR ALL LOP CONDITIONS AND THE TIME THE SI SIGNAL IS RECEIVED IN THE SAFETY FEATURES (SF) SEQUENCER WITH PREFERRED OFFSITE SOURCE AVAILABLE TO THE 4.16KV BUS.

b. DURING LOP CONDITIONS THE MAXIMUM TIME AFTER THE D.G. RECEIVES THE START SIGNAL TO THE FIRST LOAD STEP IS 10 SECONDS. THIS 10 SECOND INTERVAL INCLUDES A MAXIMUM OF 9.5 SECONDS FOR THE DIESEL GENERATOR TO START AND COME UP TO RATED VOLTAGE AND FREQUENCY PLUS 0.5 SECONDS, AFTER THE DIESEL GENERATOR BREAKER CLOSES, TO ENERGIZE THE UNLOADED 480V LOADCENTER TRANSFORMERS.

3.) a. THE 4.16KV BREAKERS SUPPLYING 1NB01 AND 1NB10 (15) WILL BE TRIPPED DURING SI CONDITIONS BY (SDPS) POWER CAN BE RESTORED MANUALLY TO BUS 1NB01 AND 1NB10 UNDER ADMINISTRATIVE PROCEDURE AFTER THE SI SIGNAL HAS BEEN MANUALLY BYPASSED.

b. DURING LOP CONDITIONS AND SUBSEQUENT SI ALL BUSSES (EXCEPT 1NB01,1NB10,1NBS,1NBR,1NBO AND 1NB1) WILL HAVE POWER AVAILABLE FOR LOADING 0.5 SECONDS AFTER THE DIESEL GENERATOR BREAKER CLOSES.

c. DURING LOP CONDITIONS ALL BUSSES WILL HAVE POWER AVAILABLE FOR LOADING 0.5 SECONDS AFTER THE DIESEL GENERATOR BREAKER CLOSES.

4.) a. C MEANS CONTINOUS S MEANS STANDBY

b. UNLESS OTHERWISE INDICATED MOV(M) ARE ASSUMED TO OPERATE FOR 10 SECONDS ONLY.

5.) THIS PUMP WILL BE SEQUENCED ON ONLY IF ONE OR BOTH OF THE PRIMARY PUMPS FAIL TO START ON THE PREVIOUSLY SEQUENCED STEP

6.) THE SF SEQUENCER WILL ONLY PERMIT THIS LOAD TO START FOR 1 SECOND PERIODS OF TIME AT THE BEGINNING OF THE INDICATED SEQUENCE STEP, ALL SUBSEQUENT STEPS AND AT ANYTIME AFTER 30.5 SECONDS

7.) THE FIRST LOAD STEP WILL OCCUR (WITH PREFERRED OFFSITE SOURCE AVAILABLE) 0.5 SECONDS AFTER THE SI SIGNAL IS RECEIVED AT THE SF SEQUENCER

10.) UNDER SI CONDITIONS, THESE LOADS ARE SEQUENCED ON AT LOW SPEED ONLY FOR LOP CONDITIONS THEY ARE SEQUENCED ON AT HIGH SPEED ONLY

11.) THESE LOADS ARE INTERLOCKED WITH OTHER LOADS WHICH ARE BLOCKED BY SEQUENCER UNITL 30.5 SECONDS

12.) TRAIN B SIMILAR EXCEPT WHERE NOTED BY ASTERISK (*) OR IN TRAIN B TABLE (SEE NOTES 18,19 &20)

13.) SI MEANS SAFETY INJECTION SYSTEM ACTUATED.

14.) THIS LOAD FED FROM BREAKER CLOSED ONLY DURING STARTUP OF THE LOAD

15.) SEQUENCER PROVIDES MOTOR AUTO-START CIRCUIT WITH START SIGNAL AT 30.5 SECOND STEP. HOWEVER MOTOR WILL START AT 50.5 SECONDS DUE TO AN ADDITIONAL TIME DELAY OF 20 SECONDS BY AGASTAT TIME DELAY RELAY IN THE AUTO-START CIRCUIT.

NOTES CONT

16 A) SI PROVIDES CB ESF CHILLER WATER PUMP AUTO-START CIRCUIT WITH START SIGNAL, HOWEVER PUMP MOTOR START WILL BE DELAYED APPROXIMATELY 120 SEC. DUE TO ADDITION OF AGASTAT TIME DELAY RELAY IN THE AUTO-START CIRCUIT.

B) CB ESF CHILLER PURGE COMP MTR IS INTERLOCKED WITH THE ESF CHILLER COMPRESSOR WHICH IS BLOCKED BY THE SEQUENCER UNTIL 30.5 SEC

C) CB ESF CHILLER OIL PUMP MOTOR IS INTERLOCKED WITH ESF CHILLER CHILLED WATER FLOW SWITCH. THIS SWITCH OPERATES DUE TO FLOW FROM THE CB ESF CHILLED WATER PUMP APPROXIMATELY 120 SECONDS AFTER THE SI SIGNAL IS INITIATED.

17. FAN 1-1561-N7-001 WILL LOAD SHED BUT WILL NOT AUTO-START UNLESS A CVI SIGNAL IS PRESENT. HEATER 1-1561-N7-001-H01 IS INTERLOCKED WITH FAN 1-1561-N7-001 AND IS ALSO PROCESS CONTROLLED.

TABLE 1

| INITIATING LETTER DESIGNATOR | TIME SEC TO OPERATE AFTER DIESEL GENERATOR BREAKER CLOSES | STARTING INITIATED BY |
|---|---|---|
| K | 0.5 | PROCESS DEVICES (THERMOSTATS PRESSURE SWITCHES ETC) |
| L | 0.5 | ACCIDENT SIGNAL (SI) OR LOP |
| M | >30.5 | MANUAL (ASSUME NO MANUAL ACTIONS WILL OCCUR UNTIL SEQUENCER TIMING CYCLE IS COMPLETED) |
| N | 0.5 | PANELS & OTHER LOADS FED FROM BRKRS ONLY-WHICH REMAIN CLOSED |
| P | >30.5 (BLOCKED BY SEQUENCER) | PROCESS DEVICE |
| S | <30.5 | AUTOMATICALLY BY SEQUENCER CAN BE STRIED MANUALLY ONLY AFTER 30.5 SEC |
| MM | 0.5 | MNL STARTED WITH MEMORY (SLIP OR MNTN START CONT) LOAD WILL RE-START UPON RESTORATION OF POWER |
| I | VARIOUS | INTERLOCKED WITH OTHER LOADS |

NOTES CONT.

18. THOSE LOADS MARKED BY AN ASTERISK (*) ARE UNIQUE TO THEIR PARTICULAR TRAIN AND DO NOT HAVE AN EQUIVALENT OPPOSITE TRAIN LOAD.

19. THESE LOADS DIFFER FROM THEIR CORRESPONDING OPPOSITE TRAIN LOAD. BOTH LOADS ARE LISTED UNDER THEIR RESPECTIVE TRAIN.

20. TRAIN A AND B LOADS WHICH DIFFER BY .2 HP OR LESS ARE LISTED ONLY UNDER TRAIN A. ALL OTHER LISTED DATA IS SIMILIAR.

21. THE UNIT 1, TRAIN A CENTRIFUGAL CHARGING PUMP MOTOR (1-1208P6002M01) WILL PRODUCE 715 HORSE-POWER DURING LOCA CONDITIONS.

**NUCLEAR SAFETY RELATED** [A] [B]

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| A | INCORP. PER ABN-36788 | 11-28-01 | CD | EOG | MWD | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GENERATOR
1A AND TRAIN "A" AC BUSSES
LOADING TABLE - PAGE 1 OF 2

SCALE: NONE | DRAWING NO. 1X3D-AA-K02A | REV. 15

JOB NO.10604

SIZE E 34x44

| EQUIPMENT TAG NUMBER | EQUIPMENT DESCRIPTION | PROJECT CLASSIFICATION (ELECTRICAL SYSTEM) | PROCUREMENT SPECIFICATION | BUS NUMBER | BUS VOLTAGE | NAMEPLATE RATING HP (OR KW) | REFERENCE DWG. 1X3D- | LOSS OF COOLANT ACCIDENT,(SI), PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SECONDS) TO START (NOTE 7) | RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SEC) TO START BRKR IS CLOSED (NOTE 3b) | RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SEC) TO START BRKR IS CLOSED (NOTE 3c) | RUNNING TIME AFTER START (NOTE 4) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PNL 1AYD1 | 120/240 DISTRIBUTION PANEL | 11E | X3 | 1ABD | 480V | 15.0KVA | AA-F11A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 11555A7013M01 | CHARGING PUMP ROOM COOLER | 11E | X4 | 1ABD | 480V | 5.0 | BG-D05N | I,K | O | C | I,K | 0.5 | C | I,K | 0.5 | C |
| 11555A7015M01 | SIS PUMP ROOM COOLER | 11E | X4 | 1ABD | 480V | 3.0 | BG-D05R | I,K | O | C | I,K | 0.5 | C | I,K | 0.5 | C |
| 11555A7001M01 | MCC ROOM COOLER | 11E | X4 | 1ABD | 480V | 5.0 | BG-D05A | L | O | C | L | 0.5 | C | K | 0.5 | C |
| 11555A7003M01 | SWGR & MCC ROOM COOLER | 11E | X4 | 1ABD | 480V | 2.0 | BG-D05C | L | O | C | L | 0.5 | C | K | 0.5 | C |
| 11208P6006M01 | BORIC ACID TRANSFER PUMP | 11E | X6 | 1ABD | 480V | 20.8 | BD-C01E | I | O | C | I | 0.5 | C | I | 0.5 | C |
| 11555A7009M01 | CTMT SPRAY PUMP COOLER | 11E | X4 | 1ABD | 480V | 3.0 | BG-D05J | I,K | O | C | I,K | 0.5 | C | I,K | 0.5 | C |
| 11555A7007M01 | RHR PUMP ROOM COOLER | 11E | X4 | 1ABD | 480V | 5.0 | BG-D05G | I,K | O | C | I,K | 0.5 | C | I,K | 0.5 | C |
| 11513P5HMAF01 | CTMT H2 MONITORING PANEL | 11E | X5 | 1ABD | 480V | 1.7 KVA | AA-F11A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 1HV-2138 | RC COOLING COIL OUT ISO | 11E | X5 | 1ABD | 480V | .13 | BD-K04M | L | O | 40SEC | L | 0.5 | 40SEC | M | >30.5 | 40SEC |
| 1HV-1830 | CTMT AIR COOLER CW OUT | 11E | X5 | 1ABD | 480V | .13 | BD-K04V | L | O | 30SEC | L | 0.5 | 30SEC | M | >30.5 | 30SEC |
| 1LV-0112D | REFUEL W TK TO CHG PUMP | 11E | X6 | 1ABD | 480V | 1.6 | BD-C02H | L | O | 15SEC | L | 0.5 | 15SEC | K | 0.5 | 15SEC |
| 1HV-8812A | RHR PUMP INLET | 11E | X6 | 1ABD | 480V | 7.8 | BD-E02E | M | >30.5 | 20SEC | M | >30.5 | 20SEC | M | >30.5 | 20SEC |
| 1HV-8110 * | CHG PUMP TO MINIFLO ISO | 11E | X6 | 1ABD | 480V | .7 | BD-C03J | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8920 | SI INLET B MINIFLO ISO | 11E | X6 | 1ABD | 480V | 0.66 | BD-D02B | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8821A | RCS HT LEG LP 1/4HDR BYP | 11E | X6 | 1ABD | 480V | 1.0 | BD-D03D | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-8802A | RCS HT LEG 1/4HDR ISO | 11E | X6 | 1ABD | 480V | 1.0 | BD-D02G | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8923A | SI PUMP INLET | 11E | X6 | 1ABD | 480V | 1 | BD-D03F | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-8716A | RHR NO. 1 COLD LEG ISO | 11E | X6 | 1ABD | 480V | 1.6 | BD-E02N | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-8807A | SI PUMP SUCTION HDR | 11E | X6 | 1ABD | 480V | .7 | BD-D02F | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-1806 | CTMT AIR COOLER CW INLET | 11E | X5 | 1ABD | 480V | .13 | BD-K04N | L | O | 30SEC | L | 0.5 | 30SEC | M | >30.5 | 30SEC |
| 1HV-8811A | CTMT SUMP ISO | 11E | X6 | 1ABD | 480V | 7.8 | BD-E03F | K,L | O | 20SEC | K,L | 0.5 | 20SEC | M | >30.5 | 20SEC |
| 1HV-9003A | CTMT SPRAY PUMP SUCTION | 11E | X6 | 1ABD | 480V | 2.0 | BD-J02J | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-8814 | SI PUMP MINIFLO ISO | 11E | X6 | 1ABD | 480V | .7 | BD-D03C | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8801A | BORON INJ TK DISC ISO | 11E | X6 | 1ABD | 480V | 3.2 (13) | BD-D02E | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10SEC |
| 1FV-0610 | RHR PUMP 1 MINIFLO | 11E | X6 | 1ABD | 480V | .7 | BD-E02C | K | O | 10SEC | K | 0.5 | 10SEC | K | 0.5 | 10SEC |
| 1HV-9017A | CSP RFL WTR STOR TK IN | 11E | X6 | 1ABD | 480V | 2.0 | BD-J02G | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-8924 * | CHG PUMP HDR TO/FR SIS PA | 11E | X6 | 1ABD | 480V | 1 | BD-C04V | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-9002A | CSP SUCTION | 11E | X6 | 1ABD | 480V | 2.0 | BD-J02E | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-1808 | CTMT AIR COOL CW INLET | 11E | X5 | 1ABD | 480V | .13 | BD-K04U | L | O | 30SEC | L | 0.5 | 30SEC | M | >30.5 | 30SEC |
| 1HV-9001A | CTMT SPRAY HDR ISO | 11E | X6 | 1ABD | 480V | 1.6 | BD-J02C | L | O | 15SEC | L | 0.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-1822 | CTMT AIR COOL CW OUTLET | 11E | X5 | 1ABD | 480V | .13 | BD-K04P | L | O | 30SEC | L | 0.5 | 30SEC | M | >30.5 | 30SEC |
| 1HV-2134 | REACTOR CAVITY CC INLET ISO | 11E | X5 | 1ABD | 480V | .13 | BD-K04L | L | O | 40SEC | L | 0.5 | 40SEC | M | >30.5 | 40SEC |
| 1HV-8485A | CHARGING PUMP A DISCH | 11E | X6 | 1ABD | 480V | 1.0 | BD-C05A | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8471A | CHARGING PUMP A SUCTION | 11E | X6 | 1ABD | 480V | .7 | BD-C05C | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| PNL 1AYE1 | 120/240V DISTRIBUTION PANEL | 11E | X3 | 1ABE | 480V | 15 KVA | AA-F24A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 11516B7001M01 | CTB POST LOCA CAV PURGE UNIT | 11E | X4 | 1ABE | 480V | 5 | BG-B02X | L | O | C | L | 0.5 | C | MM | 0.5 | C |
| 11806B3CAB | BATTERY CHARGER 1AD1CB | 11E | X3 | 1ABE | 480V | 100KVA | AA-H02A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 11806B3CCA | BATTERY CHARGER 1CD1CA (NOTE 19) | 11E | X3 | 1ABE | 480V | 75KVA | AA-H04A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 1HV-2624A | CTB POST LOCA PURGE EXH ISO | 11E | X5 | 1ABE | 480V | 0.13 | BG-B04A | L | O | 20SEC | L | 0.5 | 20SEC | M | >30.5 | 20SEC |
| 1HV-8701A | RHR LOOP 1 INLET ISO | 11E | X6 | 1ABE | 480V | 2.6 | BD-E02G | K | O | 120SEC | K | 0.5 | 120SEC | K | 0.5 | 120SEC |
| 1HV-8112 | RCP SEAL WATER ISO | 11E | X6 | 1ABE | 480V | 0.70 | BD-C03L | M | >30.5 | 15SEC | M | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-9380A | CTMT ATMOS UNIT SERV AIR | 11E | X4 | 1ABE | 480V | 0.33 | BH-R01D | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8808A | ACCUM LOOP 1 ISO | 11E | X6 | 1ABE | 480V | 10.3 | BD-D02R | L | O | 10SEC | L | 0.5 | 10SEC | K | 0.5 | 10SEC |
| 1HV-2626A | CTB NORM PURGE SPLY ISO | 11E | X5 | 1ABE | 480V | 1.33 | BG-B04E | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10 SEC |
| 1HV-2582A | CTB CLR A7001 UNIT (DAMPER) | 11E | X4 | 1ABE | 480V | .13 | BG-B03V | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10 SEC |
| 1HV-2582B | CTB CLR A7002 UNIT (DAMPER) | 11E | X4 | 1ABE | 480V | .13 | BG-B03V | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10 SEC |
| 1HV-19051 * | THERM BARR CW RCP 001 | 11E | X4 | 1ABE | 480V | 0.70 | BD-L03K | K | O | 10SEC | K | 0.5 | 10SEC | K | 0.5 | 10 SEC |
| 1HV-19053 * | THERM BARR CW RCP 002 | 11E | X4 | 1ABE | 480V | 0.70 | BD-L03L | K | O | 10SEC | K | 0.5 | 10SEC | K | 0.5 | 10 SEC |
| 1HV-8000A | PRESSURIZER POWER RELIEF ISO | 11E | X6 | 1ABE | 480V | 2.0 | BD-B02A | K | O | 10SEC | K | 0.5 | 10SEC | K | 0.5 | 10 SEC |
| PNL 1AYF1 | 120/240V DISTRIBUTION PANEL | 11E | X3 | 1ABF | 480V | 15KVA | AA-F36A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 11593B7001M01 | AUX FWP HSE FAN | 11E | X4 | 1ABF | 480V | 5.0 | BG-G03A | K | O | C | K | 0.5 | C | MM | 0.5 | C |
| 12403P4001M01 | DSL OIL STORAGE TK PUMP | 11E | X4 | 1ABF | 480V | 1.5 | BH-G01X | K | O | C | K | 0.5 | C | K | 0.5 | C |
| 12403P4002M01 | DSL OIL STORAGE TK PUMP | 11E | X4 | 1ABF | 480V | 1.5 | BH-G01Z | K | O | C | K | 0.5 | C | K | 0.5 | C |
| 11540B7001M01 | DSL TUNNEL VENT FAN | 11E | X4 | 1ABF | 480V | 3.0 | BG-K01C | K | O | C | K | 0.5 | C | K | 0.5 | C |
| 1HV-12051 | DGB FAN B7003 DAMPER | 11E | X4 | 1ABF | 480V | .2 | BG-F01M | NOTE 11 | >30.5 | 10SEC | NOTE 11 | >30.5 | 10SEC | NOTE 11 | >30.5 | 10SEC |
| 1180873114 | LTG ISO XFMR 1ABF13RX | 11E | X3 | 1ABF | 480V | 15KVA | AA-F36A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 1HV-5119 | COND STOR TK TO PUMP | 11E | X5 | 1ABF | 480V | .25 | BC-F02D | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-12006 | AUX FWP RM INLET | 11E | X4 | 1ABF | 480V | .2 | BG-G01M | I | O | 10SEC | I | 0.5 | 10SEC | I | 0.5 | 10SEC |
| 1HV-12050 | DGB FAN B7001 DAMPER | 11E | X4 | 1ABF | 480V | .2 | BG-F01M | NOTE 11 | >30.5 | 10SEC | NOTE 11 | >30.5 | 10SEC | NOTE 11 | >30.5 | 10SEC |

| EQUIPMENT TAG NUMBER | EQUIPMENT DESCRIPTION | PROJECT CLASSIFICATION (ELECTRICAL SYSTEM) | PROCUREMENT SPECIFICATION | BUS NUMBER | BUS VOLTAGE | NAMEPLATE RATING HP (OR KW) | REFERENCE DWG. 1X3D- | LOSS OF COOLANT ACCIDENT,(SI), PREFERRED OFFSITE SOURCE AVAILABLE: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SECONDS) TO START (NOTE 7) | RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER & SUBSEQUENT SI: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SEC) TO START BRKR IS CLOSED (NOTE 3b) | RUNNING TIME AFTER START (NOTE 4) | LOSS OF OFFSITE POWER: MANUAL, PROCESS, ACCIDENT OR SEQUENCER START SEE TABLE 1 | TIME (SEC) TO START BRKR IS CLOSED (NOTE 3c) | RUNNING TIME AFTER START (NOTE 4) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11808B3CDA | BATTERY CHARGER 1DD1CA (NOTE 19) | 11E | X3 | 1BBA | 480V | 50KVA | AA-G01A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 11539A7002M01 NOTE 19 | CB AUX RELAY RM AC UNIT | 11E | X4 | 1BBA | 480V | 2.0 | BG-C07N | L | O | C | L | 0.5 | C | MM | 0.5 | C |
| 11539A7006M01 NOTE 19 | CB CABLE SPREADING RM ESF AC UNIT | 11E | X4 | 1BBA | 480V | 2.0 | BG-C07S | L,K | O | C | L,K | 0.5 | C | MM,K | 0.5 | C |
| 1HV-8111B * | CHG PMP MINIFLOW ISO. | 11E | X6 | 1BBB | 480V | .7 | BD-C05L | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8840 * | RHR HOT LEG INJECT. CROSSOVER | 11E | X6 | 1BBB | 480V | 10.3 | BD-D03J | M | >30.5 | 20SEC | M | >30.5 | 20SEC | M | >30.5 | 20SEC |
| 1HV-8111A * | CHG PMP MINIFLOW ISO. | 11E | X6 | 1BBD | 480V | .7 | BD-C03K | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8103A * | RCP 1 SEAL WATER INLET VALVE | 11E | X6 | 1BBD | 480V | .7 | BD-C04L | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8103B * | RCP 2 SEAL WATER INLET VALVE | 11E | X6 | 1BBD | 480V | .7 | BD-C04H | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8438 * | CHARGING PUMP DISCHARGE | 11E | X6 | 1BBD | 480V | 1.0 | BD-C05P | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-3548 * | RCS HOT LEG SMPL | 11E | X5 | 1BBE | 480V | .13 | BD-N01N | L | O | 15SEC | L | 0.5 | 15SEC | M | >30.5 | 15SEC |
| 1HV-2627A * | CNTNMT BLDG. NORMAL PURGE SPLY | 11E | X5 | 1BBE | 480V | 1.3 | BG-B04F | L | O | 10SEC | L | 0.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8103C * | RCP 3 SEAL WATER INLET VALVE | 11E | X6 | 1BBE | 480V | .7 | BD-C04J | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-8103D * | RCP 4 SEAL WATER INLET VALVE | 11E | X6 | 1BBE | 480V | .7 | BD-C04U | M | >30.5 | 10SEC | M | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 1HV-2624B * | CTMT BLDG POST LOCA PURGE EXH ISO VALVE | 11E | X5 | 1BBE | 480V | .13 | BG-B04B | L | O | 20SEC | L | 0.5 | 20SEC | M | >30.5 | 20SEC |
| 1HV-2041 * | THERMAL BARRIOR CW RETURN VALVE | 11E | X4 | 1BBE | 480V | 1.0 | BD-L03P | K | O | 10SEC | K | 0.5 | 10SEC | K | 0.5 | 10SEC |
| 11806B3CDB | BATTERY CHARGER 1DD1CB (NOTE 19) | 11E | X3 | 1BBE | 480V | 50KVA | AA-H05A | N | O | C | N | 0.5 | C | N | 0.5 | C |
| 1HV-3500 * | RCS HOT LEG SAMPLE | 62E | X5 | 1NBR | 480V | .13 | BD-N01A | SEE NOTE 3a | >30.5 | 15SEC | SEE NOTE 3a | >30.5 | 15SEC | M | >30.5 | 15SEC |
| 11808T3003 * | AUX BLDG LVL A LTG TRANSFORMER 1NBR29X | 62E | X3 | 1NBR | 480V | 30KVA | AA-F27A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11806B3CN3 * | BATTERY CHARGER 1ND2CA | 62E | X3 | 1NBR | 480V | 50KVA | AA-H06A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y311 | INVERTER 1ND3II (NOTE 19) | 62E | X3 | 1NBR | 480V | 30KVA | AA-H07B | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11301K4001M09 * | MAIN TURB GEAR DRIVE MTR | 62E | X4 | 1NB1O | 480V | 40 | BC-Q52C | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | M | >30.5 | C |
| 11307T4501M03 * | MAIN TURB LO RES TG OP | 62E | X4 | 1NB1O | 480V | 50 | BC-L02A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | M | >30.5 | C |
| 11301K4001M09 * | MAIN TURB GEAR DRIVE MTR | 62E | X4 | 1NB1O | 480V | 40 | BC-Q52C | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | M | >30.5 | C |
| 11307T4501M03 * | MAIN TURB LO RES TG OP | 62E | X4 | 1NB1O | 480V | 50 | BC-L02A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | M | >30.5 | C |
| 11807Y3RX24 | REG XFMR 1NBS32RX | 62E | X3 | 1NBS | 480V | 15KVA | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| PNL 1NYS1 | 120/240V DISTRIBUTION PANEL | 62E | X3 | 1NBS | 480V | 10.0KVA | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y3RX14 * | INVERTER 1ND314 | 62E | X3 | 1NBS | 480V | 10 KVA | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y3RX27 * | REG TRANSFORMER 1NBS39RX | 62E | X3 | 1NBS | 480V | 7.5KVA | AA-F28A | SEE NOTE 3a | >30.5 | S | SEE NOTE 3a | >30.5 | S | N | 10.5 | S |
| 11808T3020 | LTG XFMR 1NBS47X | 62E | X3 | 1NBS | 480V | 15KVA | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y3RX16 * | REG TRANSFORMER 1NBS18RX | 62E | X3 | 1NBS | 480V | 30KVA | AA-G01B | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y3RX18 | REG TRANSFORMER 1NBS21RX | 62E | X3 | 1NBS | 480V | 30KVA | AA-G01B | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11807Y312 | INVERTER 1ND312 | 62E | X6 | 1NBS | 480V | 7.5KVA | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 11228P4001M01 | REAC MAKE-UP WTR PP 005 | 62E | X4 | 1NBS | 480V | 40.0 | BD-L05A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 11817U3007B | CONT $H_2$ MON HEAT TRACE PNL(ST BY) | 62E | X3 | 1NBS | 480V | 15kW | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 1HV-12646 | CTB REACTOR SUPPLY COOLING UNIT | 62E | X4 | 1NBS | 480V | .13 | BG-B05U | SEE NOTE 3a | >30.5 | 10SEC | SEE NOTE 3a | >30.5 | 10SEC | I | 10.5 | 10SEC |
| 1HV-12644 | CTB REACTOR SUPPLY COOLING UNIT | 62E | X4 | 1NBS | 480V | .13 | BG-B05S | SEE NOTE 3a | >30.5 | 10SEC | SEE NOTE 3a | >30.5 | 10SEC | I | 10.5 | 10SEC |
| 1HV-8098 * | EXCESS LTDWN HT EXCH INLET | 62E | X6 | 1NBS | 480V | .7 | BD-C05R | SEE NOTE 3a | >30.5 | 10SEC | SEE NOTE 3a | >30.5 | 10SEC | M | >30.5 | 10SEC |
| 11817U3001A | HEAT TRACE SI SYSTEM | 62E | X3 | 1NBS | 480V | 15KW | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 11817U3003A | HEAT TRACE (NORMAL) CVCS | 62E | X3 | 1NBS | 480V | 15KW | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 11817U3005A | HEAT TRACE REACTOR MAKE-UP WATER | 62E | X3 | 1NBS | 480V | 15KW | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 11817U3002A | AUX HEAT TRACE PNL (LVL A) | 62E | X3 | 1NBS | 480V | 15KW | AA-F28A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001H01 | JACKET WTR HTR | 62E | X4 | 1NBI | 480V | 75KW | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001H02 | LUBE OIL HTR | 62E | X4 | 1NBI | 480V | 45KW | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001E01 | AIR COMPRESSOR AFTERCOOLER | 62E | X4 | 1NBI | 480V | .75 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001E02 | AIR COMPRESSOR AFTERCOOLER | 62E | X4 | 1NBI | 480V | .75 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001M07 | LUBE OIL CIRC PUMP | 62E | X4 | 1NBI | 480V | 15 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001M04 | JACKET WTR CIRC PUMP | 62E | X4 | 1NBI | 480V | 3.0 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001M01 | AIR START COMPRESSOR | 62E | X4 | 1NBI | 480V | 30.0 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| 12403G4001M02 | AIR START COMPRESSOR | 62E | X4 | 1NBI | 480V | 30.0 | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |
| PNL 1NYI1 | 120/240 V DIST. PNL. | 62E | X3 | 1NBI | 480V | 10KVA | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | N | 10.5 | C |
| 12403G4001G01 | GEN. SP. HTR. | 62E | X4 | 1NBI | 480V | 7.2KW | AA-F40A | SEE NOTE 3a | >30.5 | C | SEE NOTE 3a | >30.5 | C | K | 10.5 | C |

TRAIN B LOADS
(SEE NOTES 18 & 20)

FOR NOTES & TABLES SEE SHEET 1 DRAWING NO 1X3D-AA-K02A.

**NUCLEAR SAFETY RELATED** A B Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

DIESEL GENERATOR
1A AND TRAIN "A" AC BUSSES
LOADING TABLE - PAGE 2 OF 2

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 3 | INCORPORATED PER DCP 97-V1N0071 | 10-27-00 | CD | EGG | MWD | |
| 2 | ISSUED FOR CONSTRUCTION | SEE MICROFILM FOR SIGNATURES | | | | |

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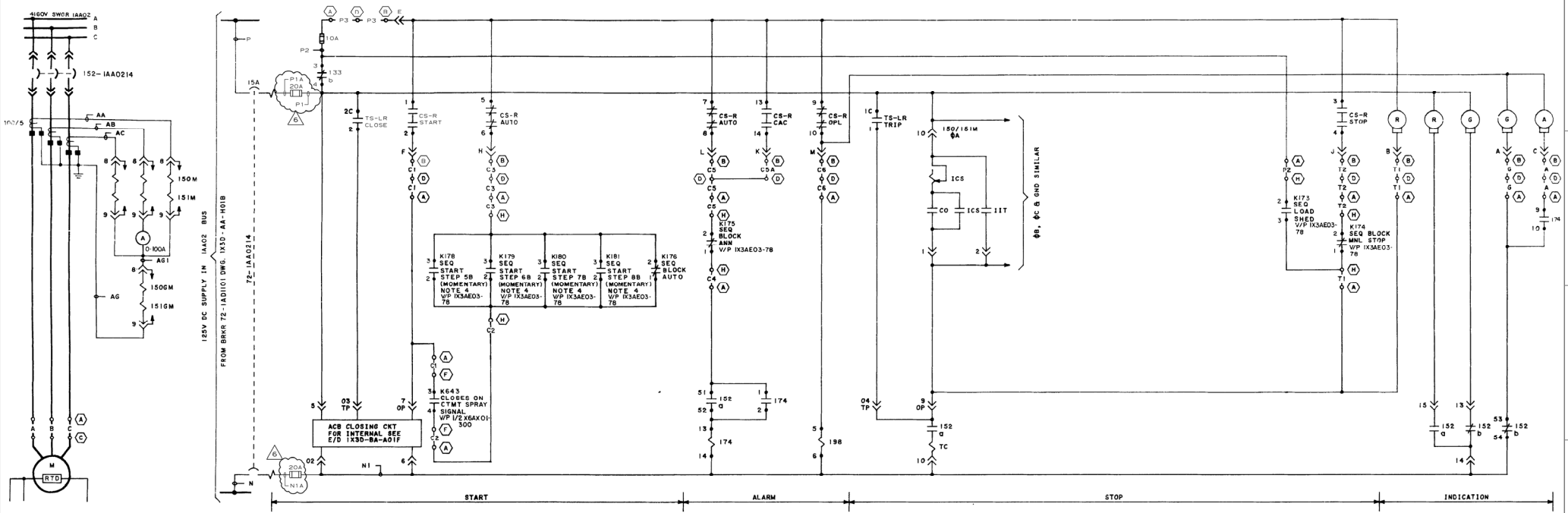
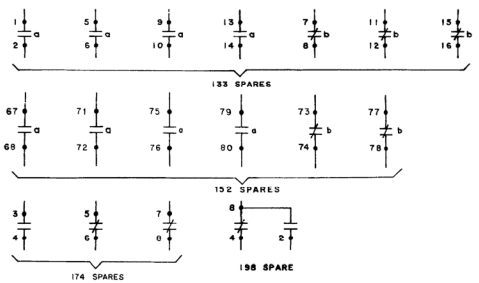
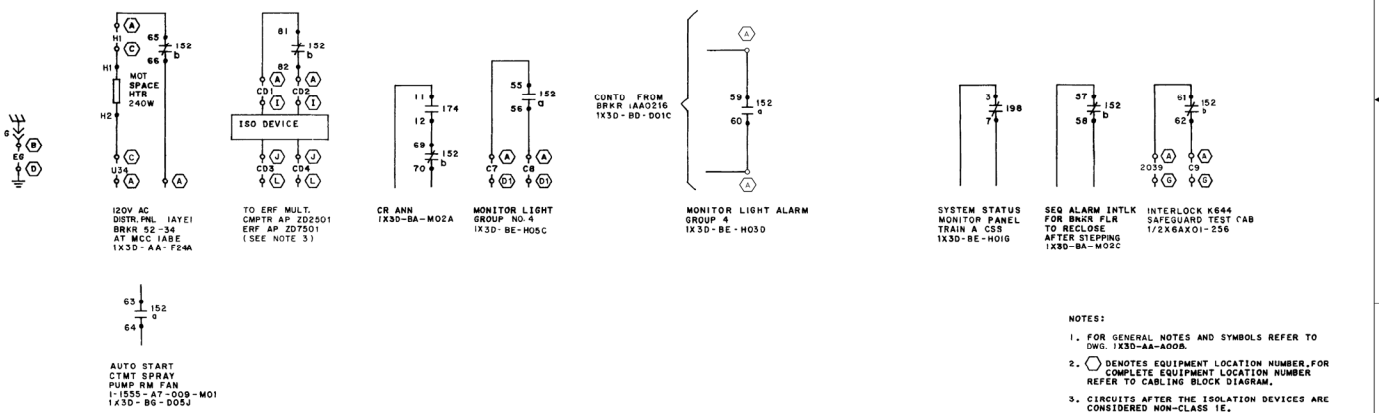
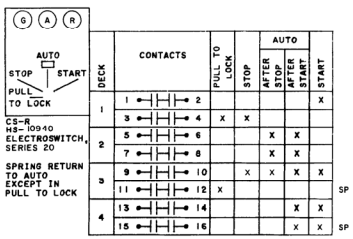
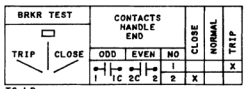
JOB NO. 10604
DRAWING NO. 1X3D-AA-K02B
REV. 13
SIZE E 34x44

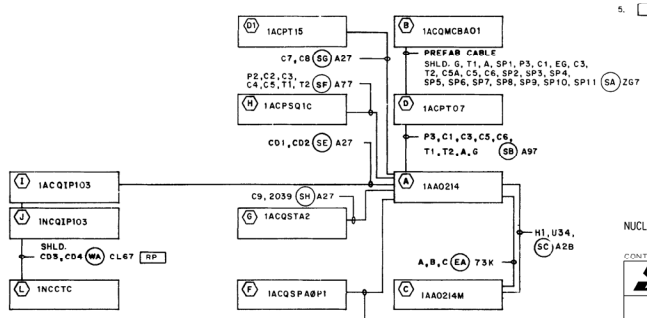



NOTES:

1. FOR GENERAL NOTES AND SYMBOLS REFER TO DWG. 1X3D-AA-A008.
2. ⬡ DENOTES EQUIPMENT LOCATION NUMBER. FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRAM.
3. CIRCUITS AFTER THE ISOLATION DEVICES ARE CONSIDERED NON-CLASS 1E.
4. SEQUENCER START CONTACT CLOSES DURING SAFETY INJECTION (SI).
5. ▭ THE TWO LETTERS INSIDE THE BOX INDICATE THE STARTUP NUMBER OF THE CORRESPONDING CABLE

NUCLEAR SAFETY RELATED [A]

CONTAINMENT SPRAY PUMP MOTOR [Q]

| DEVICE | MFR | TYPE | MODEL NO | REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|---|---|---|---|
| 150/151M | W | CO-5 | ST* 1456C05A04 | 1X4DB131 | P & I DIAGRAM | — |
| 150/151GM | W | CO-9 | ST* 1456C05A05 | 1X3D-AA-D02B | ONE LINE DIAGRAM | — |
| 174 | GE | HFA | 12HFA51A42H | N/A | LOGIC DIAGRAM | — |
| 198 | GE | HMA | 12HMA11B6 | 1/2X6AD02-3 | BECHTEL V/P | — |

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 3 | INCORPORATED PER DCP 96-VAN0031 | 9-23-98 | JGM | SM | MWD | |
| 2 | ISSUED FOR CONSTRUCTION | SEE MICROFILM FOR SIGNATURES | | | | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

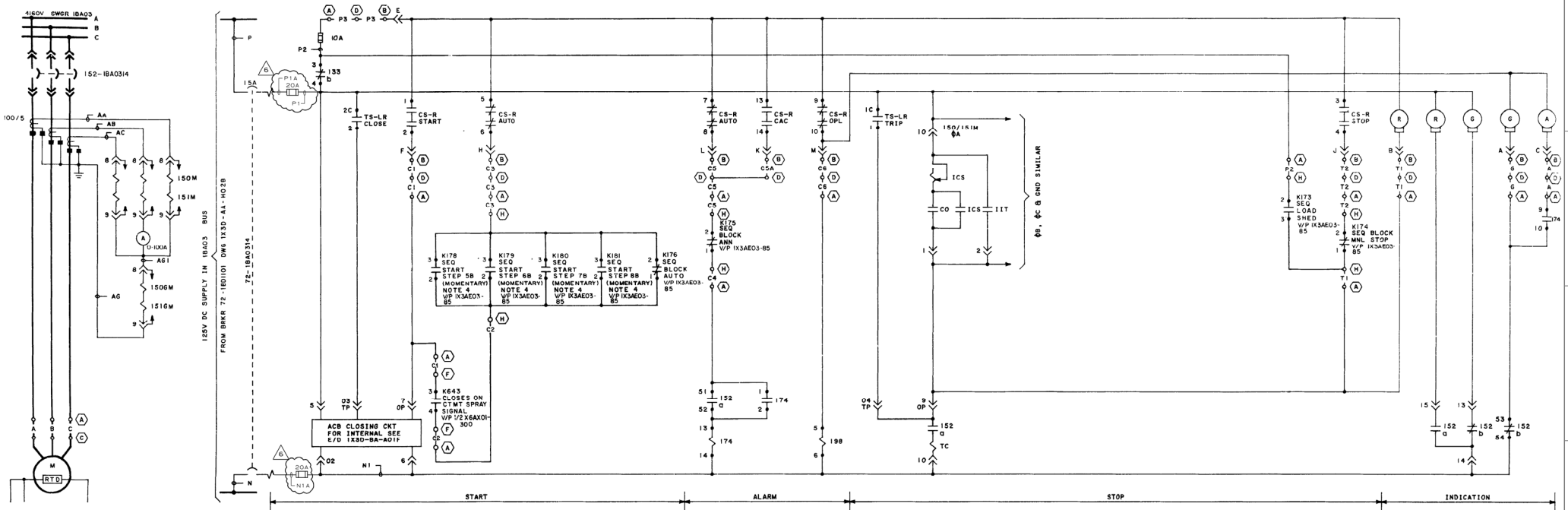
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

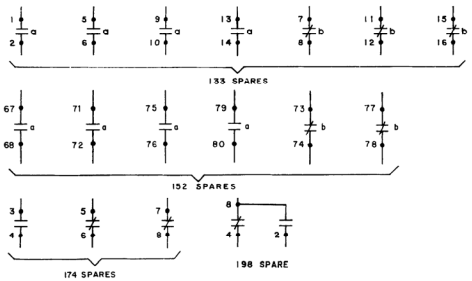
ELEMENTARY DIAGRAM
CONTAINMENT SPRAY SYSTEM
1-1206-P6-001-M01

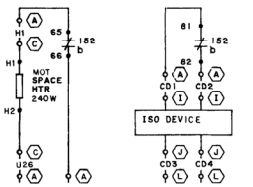
SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X3D-BD-J01A
REV. 6

SIZE E 34X44
CAD NAME 13BDJ01A

8 7 6 5 4 3 2 1




| AUTO STOP | PULL TO LOCK | START |
|---|---|---|

CS-R
NS-10941
ELECTROSWITCH, SERIES 20

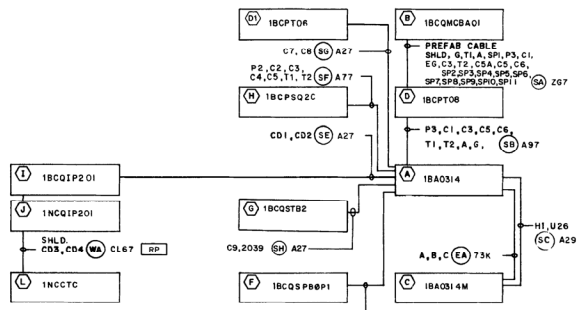
SPRING RETURN TO AUTO EXCEPT IN PULL TO LOCK

| DECK | CONTACTS | PULL TO LOCK | STOP | AUTO AFTER STOP | AUTO AFTER START | AUTO START | |
|---|---|---|---|---|---|---|---|
| 1 | 1-2 | | | | | X | |
| | 3-4 | X | X | | | | |
| 2 | 5-6 | | | X | X | | |
| | 7-8 | | | X | X | | |
| 3 | 9-10 | | X | X | X | X | |
| | 11-12 | X | | | | | SP |
| 4 | 13-14 | | | | X | X | |
| | 15-16 | | | | X | X | SP |

| BRKR TEST | CONTACTS HANDLE END | | | CLOSE | NORMAL | TRIP |
|---|---|---|---|---|---|---|
| TRIP / CLOSE | ODD | EVEN | NO | | | |
| | 1C | 2C | 1 | | | X |
| | | | 2 | X | | |

TS-LR

GE TYPE SB-1
MODEL 16SB1B11
SPRING RETURN TO NORMAL




SCHEME 1NCBDJ01B
SEE NOTE 3

SCHEME 1BA0314
1I206P6002M01
S.U. NO 8K

NOTES:

1. FOR GENERAL NOTES AND SYMBOLS REFER TO DWGS 1X3D-AA-A008.
2. ⬡ DENOTES EQUIPMENT LOCATION NUMBER. FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRAM.
3. CIRCUITS AFTER THE ISOLATION DEVICES ARE CONSIDERED NON-CLASS 1E.
4. SEQUENCER START CONTACT CLOSES DURING SAFETY INJECTION (S1).
5. ▭ THE TWO LETTERS INSIDE THE BOX INDICATE THE STARTUP NUMBER OF THE CORRESPONDING CABLE

| DEVICE | MFR | TYPE | MODEL NO | REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|---|---|---|---|
| 150/151M | W | CO-5 | ST* 1456C05A04 | 1X4DB131 | P & I DIAGRAM | — |
| 150/151GM | W | CO-9 | ST* 1456C05A05 | 1X3D-AA-D03B | ONE LINE DIAGRAM | — |
| 174 | GE | HFA | 12HFA51A42H | N/A | LOGIC DIAGRAM | — |
| 198 | GE | HMA | 12HMA11B6 | 1/2X6AD02-4 | BECHTEL V/P | — |

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| 4 | INCORPORATED PER DCP 96-VAN0031 | 9-10-98 | CD | EGG | MWD | |
| 3 | ISSUED FOR CONSTRUCTION | SEE MICROFILM FOR SIGNATURES | | | | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

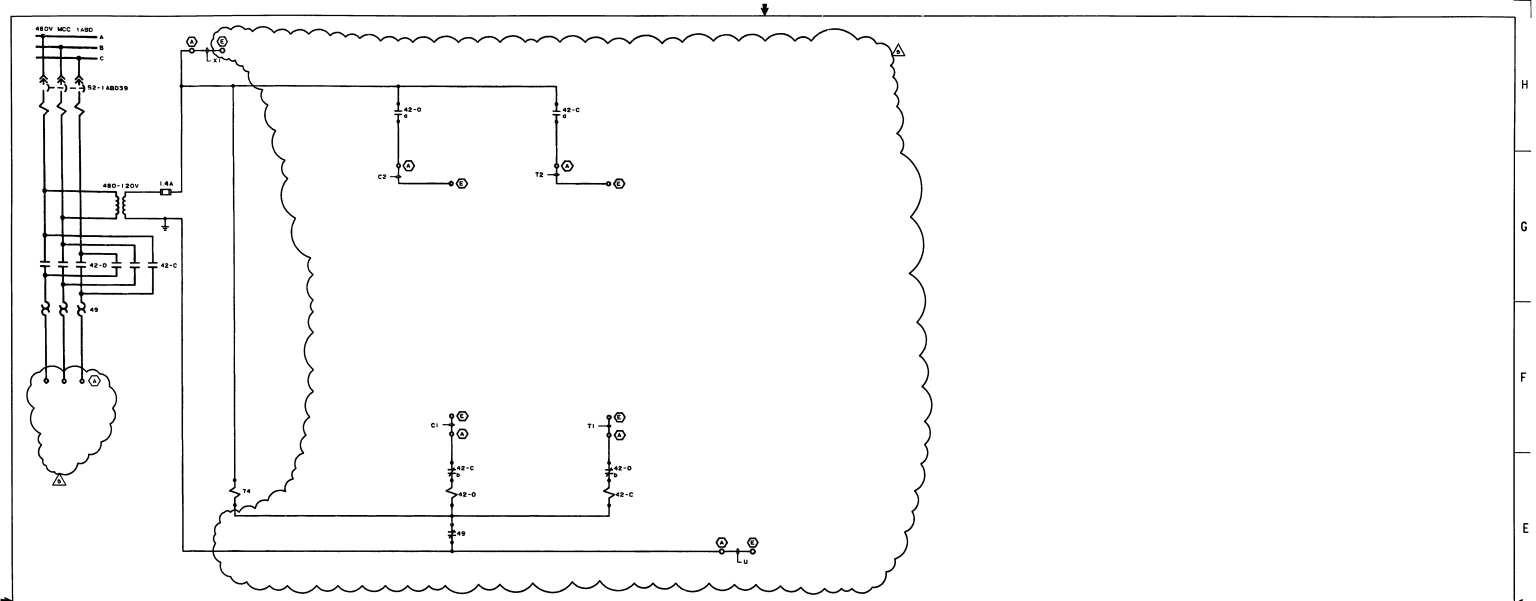
ELEMENTARY DIAGRAM
CONTAINMENT SPRAY SYSTEM
1-1206-P6-002-M01

SCALE: NONE
JOB NO. 10604

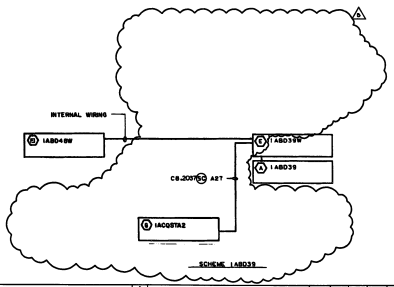
DRAWING NO. 1X3D-BD-J01B
REV. 6

SIZE E 34X44
CAD NAME 13BDJ01B

8 7 6 5 4 3 2 1
H G F E D C B A




NOTES:
1. FOR GENERAL NOTES AND SYMBOLS REFER TO D/L 1X3D-AA-A00B
2. ○ DENOTES EQUIPMENT LOCATION NUMBER. FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRAM.
3. SPARE CONTACT WITH A "Δ" IS WIRED TO UNIT TERMINAL BLOCKS.




NUCLEAR SAFETY RELATED A

| DEVICE | MFR | TYPE | MODEL NO | REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|---|---|---|---|
| 74 | CH | D26M | D26AR13A | 1X4D8131 | F & I DIAGRAM | — |
| | | | | 1X3D-AA-F11A | ONE LINE DIAGRAM | — |
| | | | | N/A | LOGIC DIAGRAM | — |

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| | INCORP. ABN-42515 | 5/96 | GS | GAB | GAB | |
| | ISSUED FOR CONSTRUCTION | SEE MICROFILM FOR SIGNATURES | | | | |

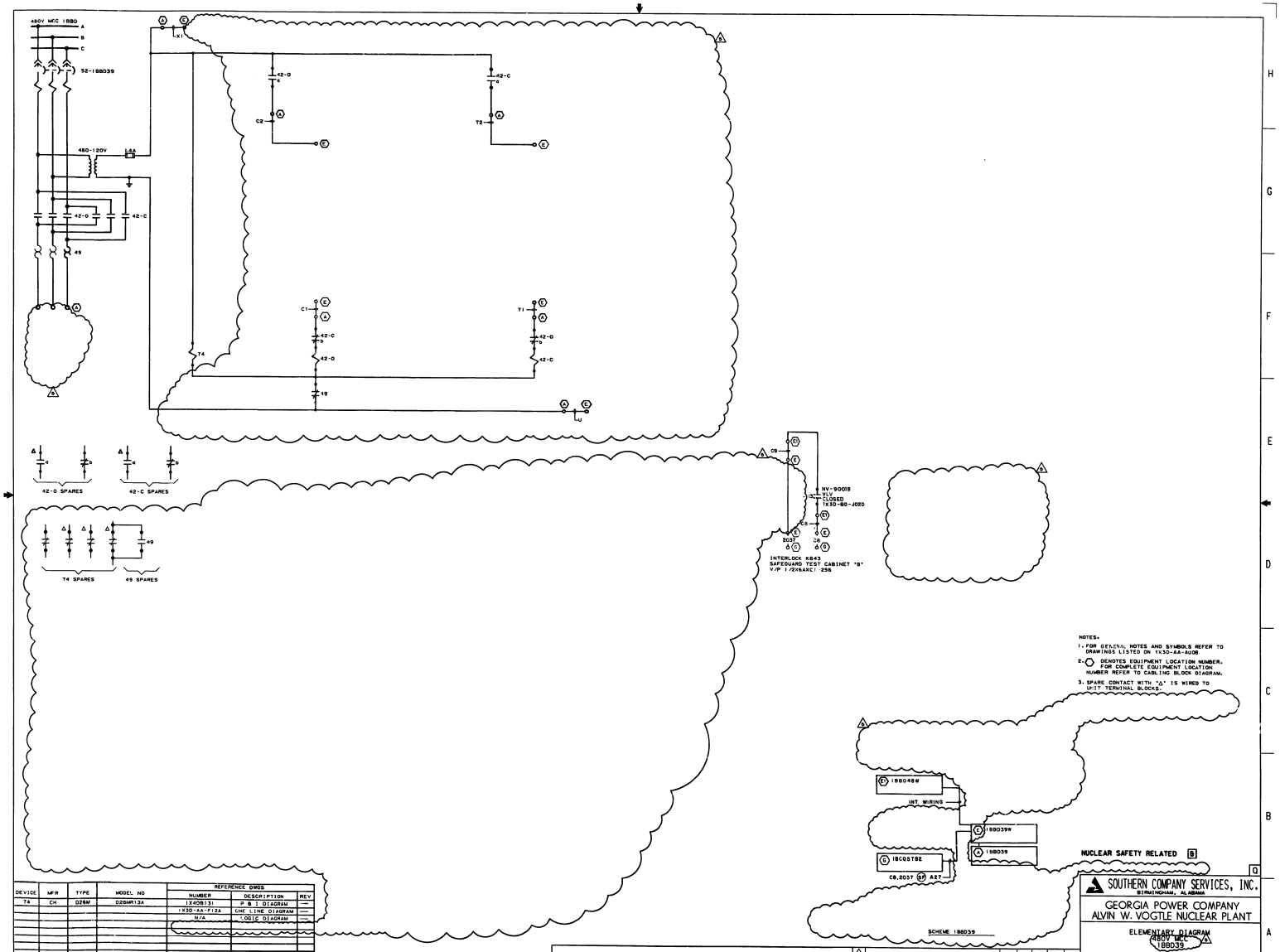
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

ELEMENTARY DIAGRAM
480V MCC 1ABD39

| SCALE: NONE | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 10604 | 1X3D-BD-J02A | 5 |

SIZE E 34X44




NOTES:
1. FOR GENERAL NOTES AND SYMBOLS REFER TO DRAWINGS LISTED ON 1X3D-AA-A00B.
2. ○ DENOTES EQUIPMENT LOCATION NUMBER. FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRAM.
3. SPARE CONTACT WITH "A" IS WIRED TO UNIT TERMINAL BLOCKS.

NUCLEAR SAFETY RELATED B

| DEVICE | MFR | TYPE | MODEL NO |
|---|---|---|---|
| 74 | CH | D26MR13A | D26MR13A |

| REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|
| 1X4D81.31 | P & I DIAGRAM | — |
| 1X3D-AA-F12A | ONE LINE DIAGRAM | — |
| N/A | LOGIC DIAGRAM | — |

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 2 | INCORP. ABN-42506 | 7/11/86 | CS | RCB | RCB | |
| 1 | ISSUED FOR CONSTRUCTION | SEE WORKFLOW FOR SIGNATURES | | | | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

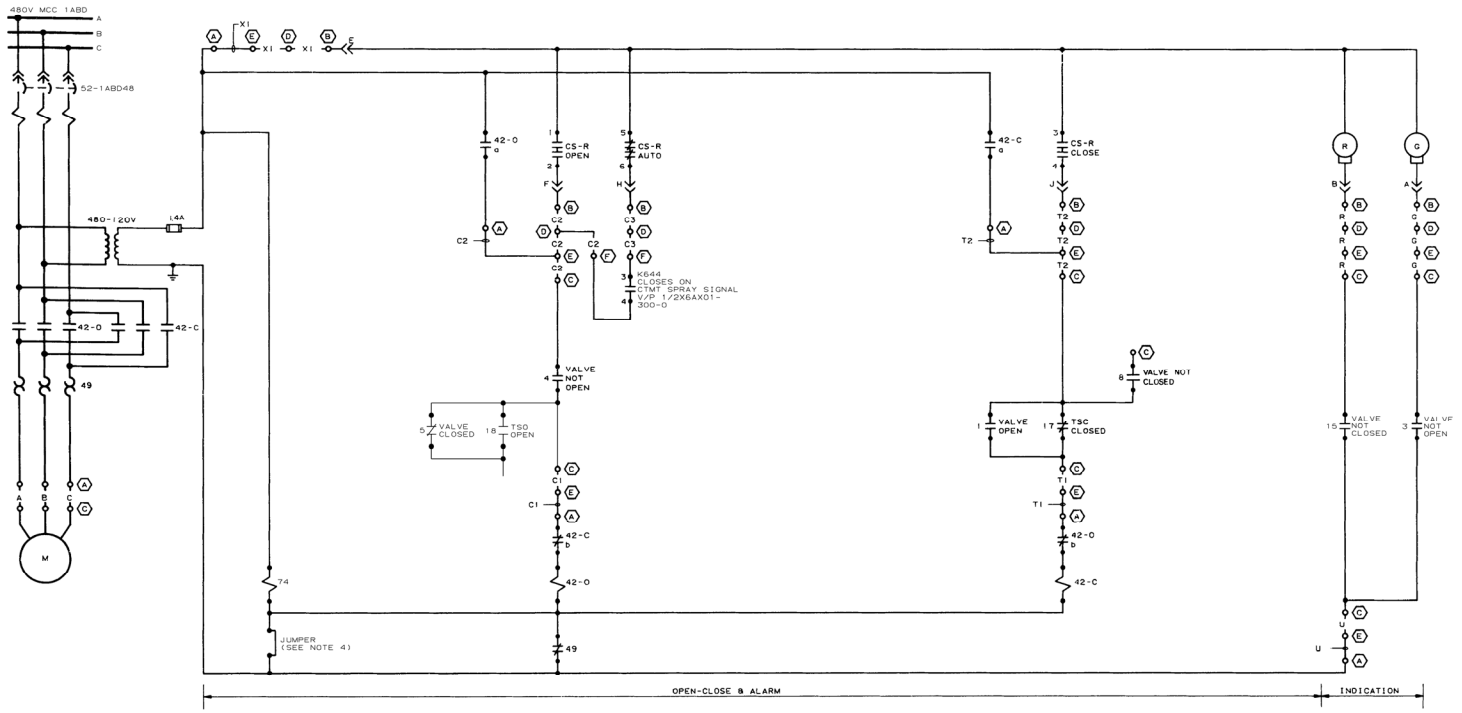
ELEMENTARY DIAGRAM
480V MCC
1BB039

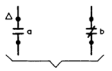
| SCALE: NONE | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 10604 | 1X3D-BD-J02B | 5 |

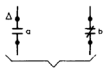
SIZE E 34X44

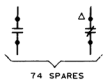
8 7 6 5 4 3 2 1

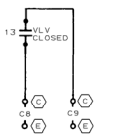
H G F E D C B A

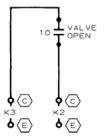


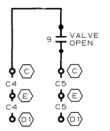



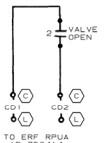



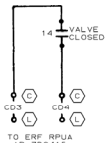



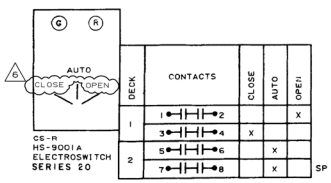


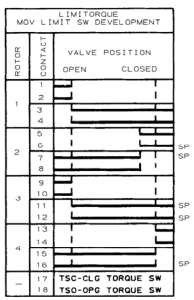


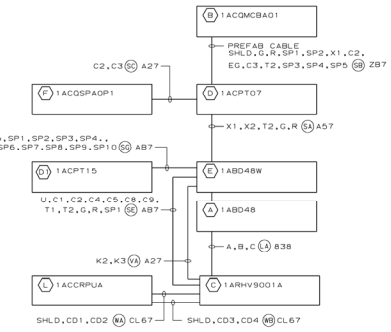




NOTES:
1. FOR GENERAL NOTES AND SYMBOLS REFER TO O/L 1X3D-AA-A008.
2. ⬡ DENOTES EQUIPMENT LOCATION NUMBER. FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRM.
3. SPARE CONTACT WITH "A" IS WIRED TO UNIT TERMINAL BLOCKS.
4. FIELD TO REMOVE JUMPER FROM THERMAL OVERLOAD CONTACT DURING INITIAL START-UP & PERIODIC TESTING.




CS-R
HS-9001A
ELECTROSWITCH
SERIES 20




NUCLEAR SAFETY RELATED A
CONTAINMENT SPRAY HEADER ISO VALVE Q

| DEVICE | MFR | TYPE | MODEL NO | REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|---|---|---|---|
| 74 | C-H | D26M | D26MR13A | 1X4DB131 | P & I DIAGRAM | |
| | | | | 1X3D-AA-F11B | ONE LINE DIAGRAM | |
| | | | | NA | LOGIC DIAGRAM | |
| | | | | 1/2X6AD01-4 | BECHTEL V/P | |

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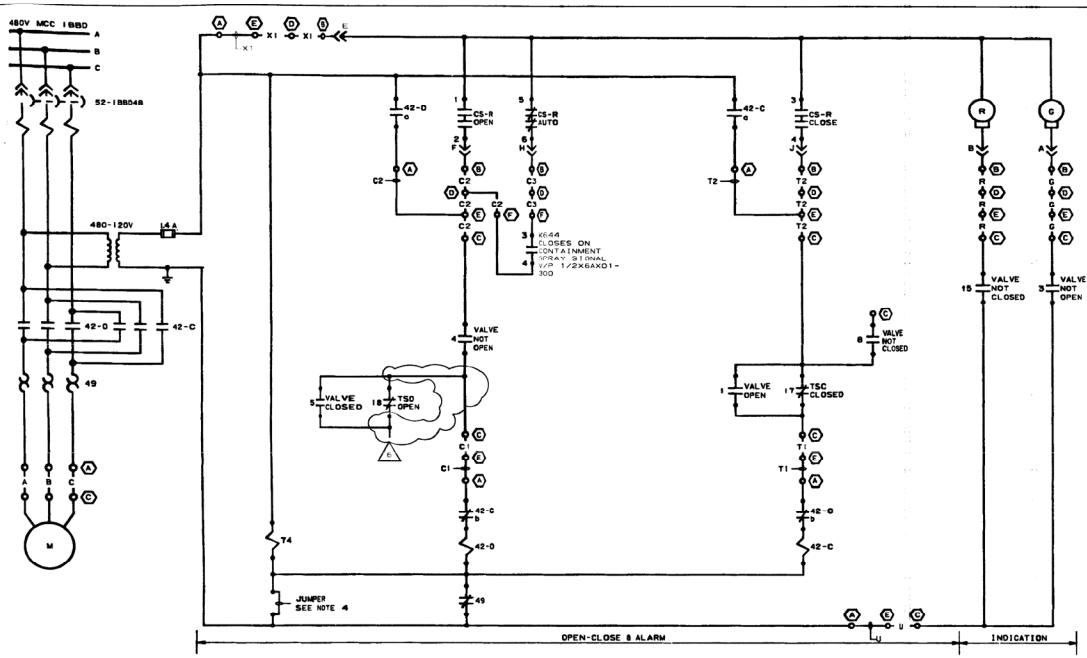
| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| | INCORP. PER ABN-36241 | 04-26-01 | CD | SM | TSL | |

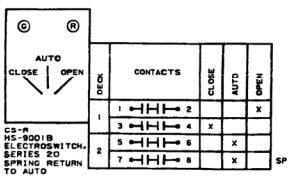
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

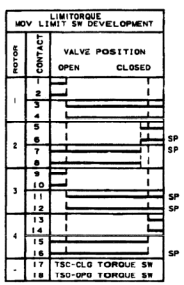
ELEMENTARY DIAGRAM
CONTAINMENT SPRAY SYSTEM
1HV-9001A

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X3D-BD-J02C
REV. 6
SIZE E 34X44

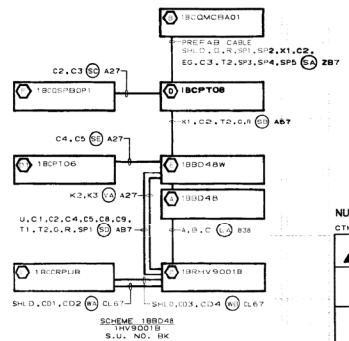



CS-R
HS-9001B
ELECTROSWITCH,
SERIES 20
SPRING RETURN
TO AUTO




SOLID LINE DENOTES CONTACT CLOSED




NOTES:
1. FOR GENERAL NOTES AND SYMBOLS REFER TO DRAWINGS LISTED ON 1X3D-AA-A00B.
2. ◇ DENOTES EQUIPMENT LOCATION NUMBER FOR COMPLETE EQUIPMENT LOCATION NUMBER REFER TO CABLING BLOCK DIAGRAM.
3. SPARE CONTACT WITH "△" IS WIRED TO UNIT TERMINAL BLOCKS.
4. FIELD TO REMOVE JUMPER FROM THERMAL OVERLOAD CONTACT DURING INITIAL START-UP & PERIODIC TESTING.

NUCLEAR SAFETY RELATED B
CTMT SPRAY PUMP 2 TO SPRAY HDR VLV Q

| DEVICE | MFR | TYPE | MODEL NO | REFERENCE DWGS NUMBER | DESCRIPTION | REV |
|---|---|---|---|---|---|---|
| 74 | CH | D26M | D26MR13A | 1X4DB131 | P & I DIAGRAM | |
| | | | | 1X3D-AA-F12B | ONE LINE DIAGRAM | |
| | | | | N/A | LOGIC DIAGRAM | |
| | | | | 1/2X6AD01-5 | BECHTEL V/P | |

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| NO. | REVISIONS | DATE | DR | CHK | RE | DM | DM | DM | PDM |
|---|---|---|---|---|---|---|---|---|---|
| A | INCORP. ABN-11343. | 5-4-95 | GB | DMM | CGB | MAP | | | |
| B | ISSUED FOR CONSTRUCTION & VENDOR ENGRG. | 7/5/80 | SEE MICROFILM FOR SIGNATURES | | | | | | |

| NO. | REVISIONS | DATE | DR | CHK | RGE | EGL | EGS | CHF | EPE | QAE |
|---|---|---|---|---|---|---|---|---|---|---|

ENG. MGR. DATE

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

ELEMENTARY DIAGRAM
CONTAINMENT SPRAY SYSTEM
1HV-9001B

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X3D-BD-J02D
REV. 6
SIZE E 34X44

TURBINE BLDG

LEVEL 3

LEVEL 2

LEVEL 1

LEVEL A

ELECT. TUNNEL

NOTES:

1. THE EQUIPMENT NUMBERS ARE AS SHOWN ON RISER DIAGRAM:
   A. THE UNIT
   B. FLOOR LEVEL AND
   C. DEVICE (SPEAKER, HANDSET, OR SPEAKER AMPLIFIER)
   S, H, AND A RESPECTIVELY.

   THUS, SPEAKER NO. 18 IN FLOOR LEVEL 2 IS:

   12S18
   - SPEAKER NUMBER
   - SPEAKER
   - FLOOR LEVEL 2
   - UNIT 1

2. (TG T14) THIS SYMBOL REPRESENTS THE COLUMNS THE DEVICE IS NEAREST.

REFERENCES:

1. CX3D-CE-G89A-CONNECTION DIAGRAM
2. CX3D-CE-G89B-CONNECTION DIAGRAM
3. CX3D-CE-G89C-CONNECTION DIAGRAM
4. CX3D-CE-G89D-CONNECTION DIAGRAM

REVISION NOTES:

1. A. REVISED ROUTING TO BECHTEL INTERFACE FOR 1NNJB6097
   B. ADDED EQUIP IN ELECT. TUNNEL
2. REVISED SPEAKER 11S38 FROM 1 TO 3.

SCHEME NO. 1NNPAGE1
START UP NO. QF

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| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 8.0 | REVISED PER ABN-V02337, VER. 1.0 | 11-1-10 | Vep | TSL | JMR |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TELEPHONE/PAGE SYSTEM
RISER DIAGRAM TURBINE BUILDING
UNIT NO.1 & MISC OUTDOOR STR

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X3DG001 | 8.0 |

SIZE E 34X44

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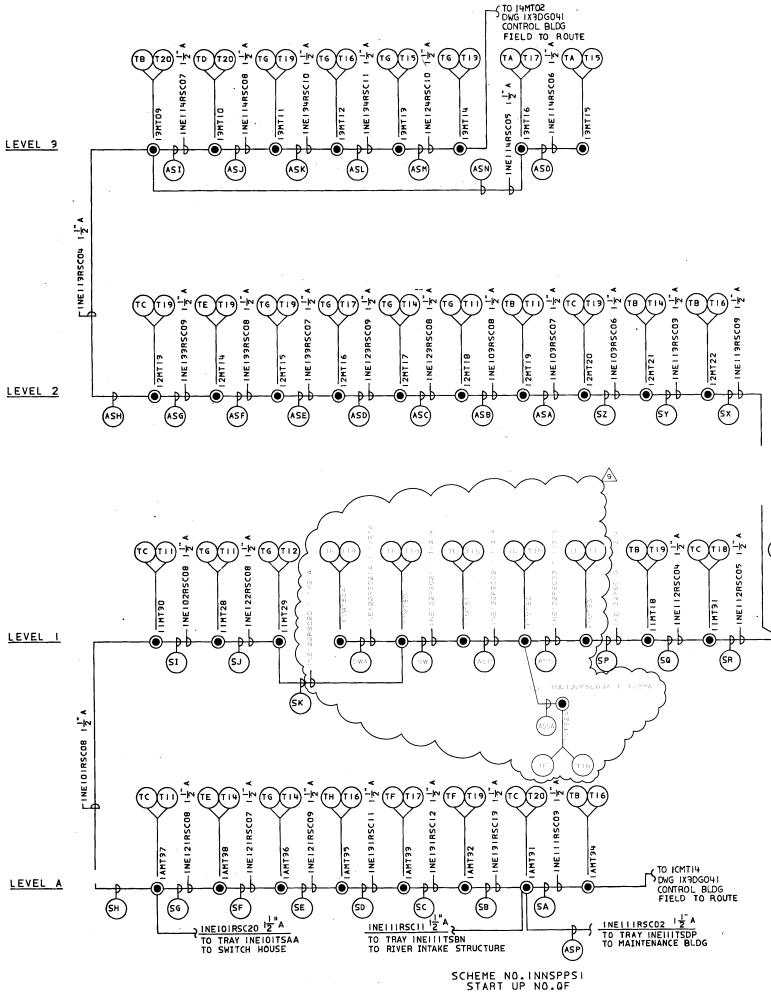
SCALING RULES

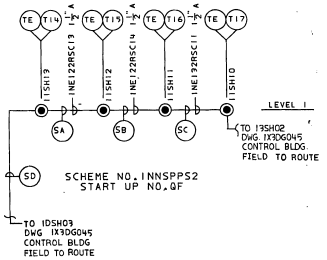
TURBINE BUILDING

REVISION NOTES:
1. DELETED CONNECTION DIAGRAMS; ADDED NOTE 3.
2. INCORPORATE DCN 1, ADDED NOTE 4.

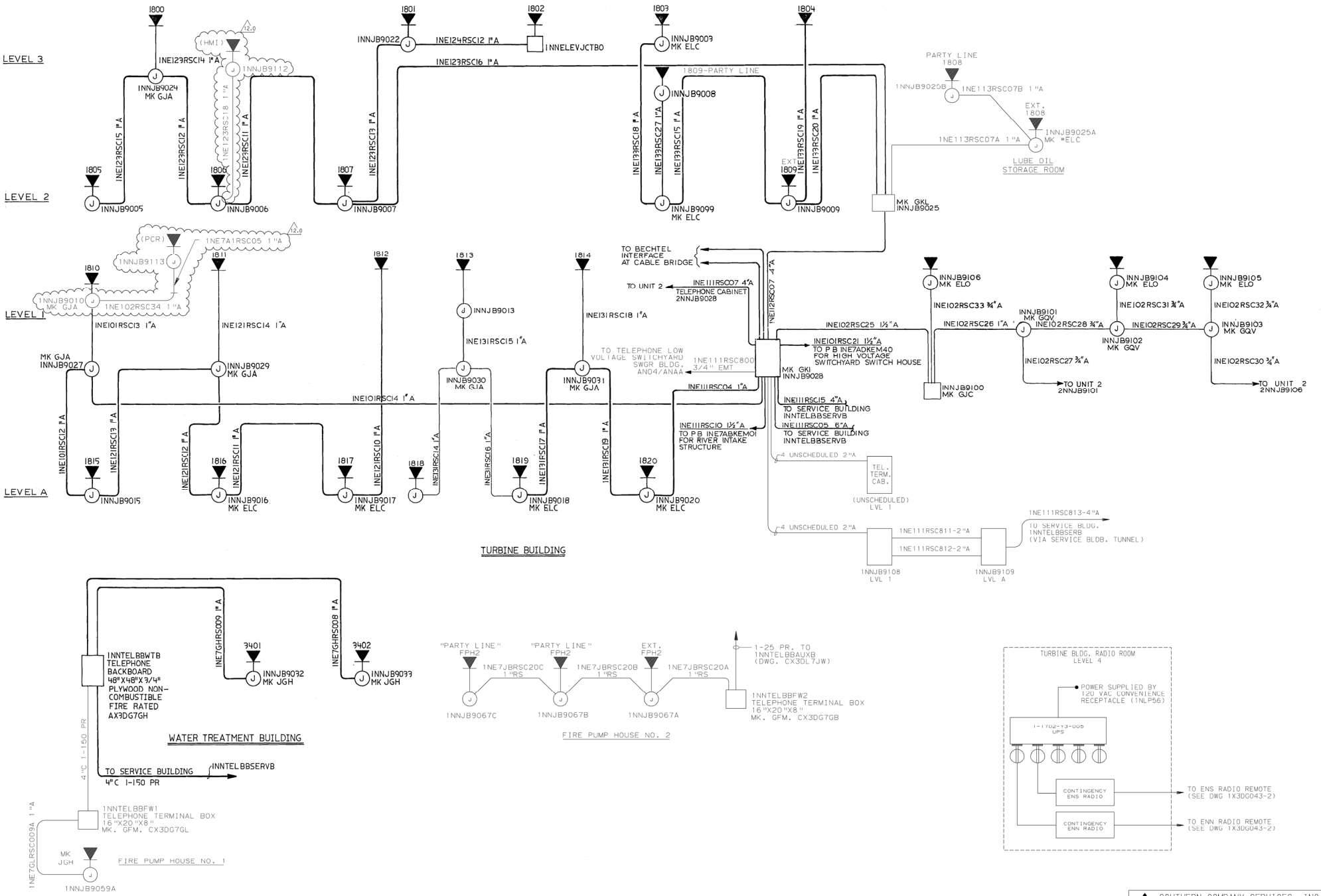
NOTES:
1. ALL CABLE TO BE 14/C. CABLE CODE CR67
2. ALL (SA) NUMBERS TO BE PRECEEDED BY SCHEME NO. SHOWN BELOW RISER DIAGRAM
3. REFER TO DWG AX3DG090 FOR CABLE TERMINATION INFORMATION
4. JACK BOX EE-980 LOCATION NO'S SHALL BE CONSIDERED TO REFER TO THE ACTUAL JACK BOX AND THE ASSOCIATED TERMINAL BOX (IF APPLICABLE).




TURBINE BUILDING
MAIN STEAM ATMOSPHERIC DUMP STATION




| REVISIONS | | |

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| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 12.0 | REVISED PER ABN 10490015O1E141, VER. 2.0 | 8-27-09 | vep | MWD | JMR |

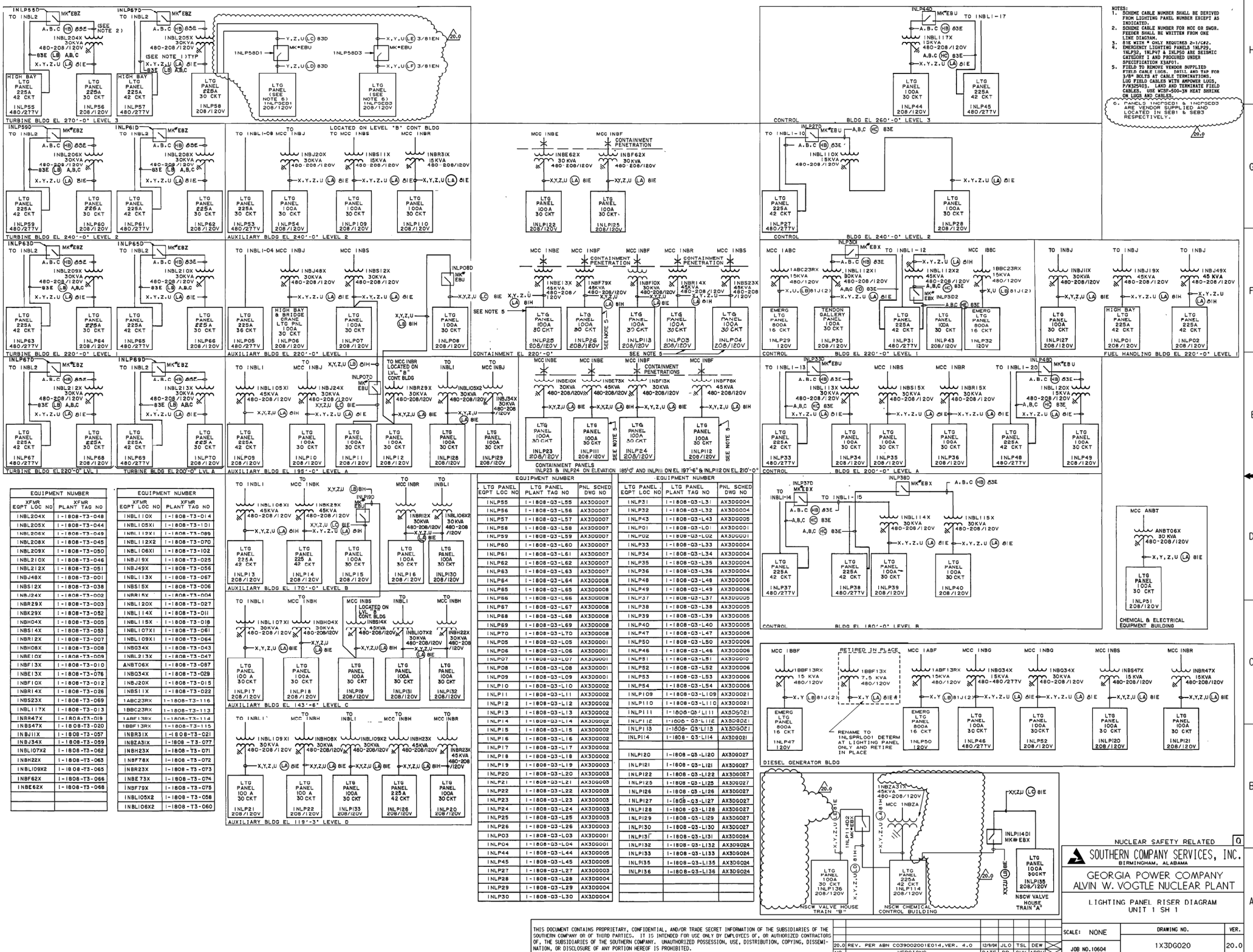
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SOUTHERN BELL SYSTEM
BLOCK DIAGRAM - TURBINE BUILDING
UNIT NO.1 & MISC. OUTDOOR STR

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X3DG004 | 12.0 |

SIZE E 34x44




NOTES:
1. SCHEME CABLE NUMBER SHALL BE DERIVED FROM LIGHTING PANEL NUMBER EXCEPT AS INDICATED.
2. SCHEME CABLE NUMBER FOR MCC OR SWGR. FEEDER SHALL BE WRITTEN FROM ONE LINE DIAGRAM.
3. 81E WITH * ONLY REQUIRES 2-1/C#2.
4. EMERGENCY LIGHTING PANELS INLP32, INLP37 & INLP50 ARE SEISMIC CATEGORY I AND PROCURED UNDER SPECIFICATION X3AF01.
5. FIELD TO REMOVE VENDOR SUPPLIED FIELD CABLE LUGS. DRILL AND TAP FOR 3/8" BOLTS AT CABLE TERMINATIONS. LUG FIELD CABLES WITH AMPOWER LUGS, P/N325403. LAND AND TERMINATE FIELD CABLES. USE WCSP-600-3N HEAT SHRINK ON LUGS AND CABLES.
6. PANELS INCP5CD1 & INCP5CD3 ARE VENDOR SUPPLIED AND LOCATED IN SEB1 & SEB3 RESPECTIVELY.

| EQUIPMENT NUMBER | |
|---|---|
| XFMR EQPT LOC NO | XFMR PLANT TAG NO |
| INBL204X | 1-1808-T3-048 |
| INBL205X | 1-1808-T3-044 |
| INBL206X | 1-1808-T3-049 |
| INBL208X | 1-1808-T3-045 |
| INBL209X | 1-1808-T3-050 |
| INBL210X | 1-1808-T3-046 |
| INBL212X | 1-1808-T3-051 |
| INBJ48X | 1-1808-T3-001 |
| INBS12X | 1-1808-T3-038 |
| INBJ24X | 1-1808-T3-002 |
| INBR29X | 1-1808-T3-003 |
| INBK29X | 1-1808-T3-052 |
| INBH04X | 1-1808-T3-005 |
| INBS14X | 1-1808-T3-053 |
| INBR12X | 1-1808-T3-007 |
| INBH08X | 1-1808-T3-008 |
| INBE10X | 1-1808-T3-009 |
| INBF13X | 1-1808-T3-010 |
| INBE13X | 1-1808-T3-076 |
| INBF10X | 1-1808-T3-012 |
| INBR14X | 1-1808-T3-026 |
| INBS23X | 1-1808-T3-069 |
| INBLI17X | 1-1808-T3-013 |
| INBR47X | 1-1808-T3-019 |
| INBS47X | 1-1808-T3-020 |
| INBJIIX | 1-1808-T3-057 |
| INBJ34X | 1-1808-T3-059 |
| INBL107X2 | 1-1808-T3-062 |
| INBH22X | 1-1808-T3-063 |
| INBL109X2 | 1-1808-T3-065 |
| INBF62X | 1-1808-T3-066 |
| INBE62X | 1-1808-T3-068 |

| EQUIPMENT NUMBER | |
|---|---|
| XFMR EQPT LOC NO | XFMR PLANT TAG NO |
| INBLI10X | 1-1808-T3-014 |
| INBL105X1 | 1-1808-T3-101 |
| INBL112X1 | 1-1808-T3-089 |
| INBL112X2 | 1-1808-T3-070 |
| INBL106X1 | 1-1808-T3-102 |
| INBJ19X | 1-1808-T3-025 |
| INBJ49X | 1-1808-T3-056 |
| INBL113X | 1-1808-T3-067 |
| INBS15X | 1-1808-T3-006 |
| INRR15X | 1-1808-T3-004 |
| INBL120X | 1-1808-T3-027 |
| INBL114X | 1-1808-T3-011 |
| INBL115X | 1-1808-T3-018 |
| INBL107X1 | 1-1808-T3-061 |
| INBL109X1 | 1-1808-T3-064 |
| INBG34X | 1-1808-T3-043 |
| INBL213X | 1-1808-T3-047 |
| ANBT06X | 1-1808-T3-087 |
| INBG34X | 1-1808-T3-029 |
| INBJ20X | 1-1808-T3-015 |
| INBS11X | 1-1808-T3-022 |
| 1ABC23RX | 1-1808-T3-116 |
| 1BBC23RX | 1-1808-T3-113 |
| 1ABF13RX | 1-1808-T3-114 |
| 1BBF13RX | 1-1808-T3-115 |
| INBR31X | 1-1808-T3-021 |
| INBZA31X | 1-1808-T3-077 |
| INBH23X | 1-1808-T3-071 |
| INBF78X | 1-1808-T3-072 |
| INBR23X | 1-1808-T3-073 |
| INBE73X | 1-1808-T3-074 |
| INBF79X | 1-1808-T3-075 |
| INBL105X2 | 1-1808-T3-058 |
| INBL106X2 | 1-1808-T3-060 |

| EQUIPMENT NUMBER | | |
|---|---|---|
| LTG PANEL EQPT LOC NO | LTG PANEL PLANT TAG NO | PNL SCHED DWG NO |
| INLP55 | 1-1808-Q3-L55 | AX3DG007 |
| INLP56 | 1-1808-Q3-L56 | AX3DG007 |
| INLP57 | 1-1808-Q3-L57 | AX3DG007 |
| INLP58 | 1-1808-Q3-L58 | AX3DG007 |
| INLP59 | 1-1808-Q3-L59 | AX3DG007 |
| INLP60 | 1-1808-Q3-L60 | AX3DG007 |
| INLP61 | 1-1808-Q3-L61 | AX3DG007 |
| INLP62 | 1-1808-Q3-L62 | AX3DG007 |
| INLP63 | 1-1808-Q3-L63 | AX3DG007 |
| INLP64 | 1-1808-Q3-L64 | AX3DG008 |
| INLP65 | 1-1808-Q3-L65 | AX3DG008 |
| INLP66 | 1-1808-Q3-L66 | AX3DG008 |
| INLP67 | 1-1808-Q3-L67 | AX3DG008 |
| INLP68 | 1-1808-Q3-L68 | AX3DG008 |
| INLP69 | 1-1808-Q3-L69 | AX3DG008 |
| INLP70 | 1-1808-Q3-L70 | AX3DG008 |
| INLP05 | 1-1808-Q3-L05 | AX3DG001 |
| INLP06 | 1-1808-Q3-L06 | AX3DG001 |
| INLP07 | 1-1808-Q3-L07 | AX3DG001 |
| INLP08 | 1-1808-Q3-L08 | AX3DG001 |
| INLP09 | 1-1808-Q3-L09 | AX3DG001 |
| INLP10 | 1-1808-Q3-L10 | AX3DG002 |
| INLP11 | 1-1808-Q3-L11 | AX3DG002 |
| INLP12 | 1-1808-Q3-L12 | AX3DG002 |
| INLP13 | 1-1808-Q3-L13 | AX3DG002 |
| INLP14 | 1-1808-Q3-L14 | AX3DG002 |
| INLP15 | 1-1808-Q3-L15 | AX3DG002 |
| INLP16 | 1-1808-Q3-L16 | AX3DG002 |
| INLP17 | 1-1808-Q3-L17 | AX3DG002 |
| INLP18 | 1-1808-Q3-L18 | AX3DG002 |
| INLP19 | 1-1808-Q3-L19 | AX3DG003 |
| INLP20 | 1-1808-Q3-L20 | AX3DG003 |
| INLP21 | 1-1808-Q3-L21 | AX3DG003 |
| INLP22 | 1-1808-Q3-L22 | AX3DG003 |
| INLP23 | 1-1808-Q3-L23 | AX3DG003 |
| INLP24 | 1-1808-Q3-L24 | AX3DG003 |
| INLP25 | 1-1808-Q3-L25 | AX3DG003 |
| INLP26 | 1-1808-Q3-L26 | AX3DG003 |
| INLP03 | 1-1808-Q3-L03 | AX3DG001 |
| INLP04 | 1-1808-Q3-L04 | AX3DG001 |
| INLP44 | 1-1808-Q3-L44 | AX3DG005 |
| INLP45 | 1-1808-Q3-L45 | AX3DG005 |
| INLP27 | 1-1808-Q3-L27 | AX3DG003 |
| INLP28 | 1-1808-Q3-L28 | AX3DG004 |
| INLP29 | 1-1808-Q3-L29 | AX3DG004 |
| INLP30 | 1-1808-Q3-L30 | AX3DG004 |

| EQUIPMENT NUMBER | | |
|---|---|---|
| LTG PANEL EQPT LOC NO | LTG PANEL PLANT TAG NO | PNL SCHED DWG NO |
| INLP31 | 1-1808-Q3-L31 | AX3DG004 |
| INLP32 | 1-1808-Q3-L32 | AX3DG004 |
| INLP43 | 1-1808-Q3-L43 | AX3DG005 |
| INLP01 | 1-1808-Q3-L01 | AX3DG001 |
| INLP02 | 1-1808-Q3-L02 | AX3DG001 |
| INLP33 | 1-1808-Q3-L33 | AX3DG004 |
| INLP34 | 1-1808-Q3-L34 | AX3DG004 |
| INLP35 | 1-1808-Q3-L35 | AX3DG004 |
| INLP36 | 1-1808-Q3-L36 | AX3DG004 |
| INLP48 | 1-1808-Q3-L48 | AX3DG006 |
| INLP49 | 1-1808-Q3-L49 | AX3DG006 |
| INLP37 | 1-1808-Q3-L37 | AX3DG005 |
| INLP38 | 1-1808-Q3-L38 | AX3DG005 |
| INLP39 | 1-1808-Q3-L39 | AX3DG005 |
| INLP40 | 1-1808-Q3-L40 | AX3DG005 |
| INLP47 | 1-1808-Q3-L47 | AX3DG006 |
| INLP50 | 1-1808-Q3-L50 | AX3DG006 |
| INLP46 | 1-1808-Q3-L46 | AX3DG006 |
| INLP51 | 1-1808-Q3-L51 | AX3DG010 |
| INLP52 | 1-1808-Q3-L52 | AX3DG006 |
| INLP53 | 1-1808-Q3-L53 | AX3DG006 |
| INLP54 | 1-1808-Q3-L54 | AX3DG006 |
| INLP109 | 1-1808-Q3-L109 | AX3DG021 |
| INLP110 | 1-1808-Q3-L110 | AX3DG021 |
| INLP111 | 1-1808-Q3-L111 | AX3DG021 |
| INLP112 | 1-1808-Q3-L112 | AX3DG021 |
| INLP113 | 1-1808-Q3-L113 | AX3DG021 |
| INLP114 | 1-1808-Q3-L114 | AX3DG021 |
| | | |
| INLP120 | 1-1808-Q3-L120 | AX3DG027 |
| INLP121 | 1-1808-Q3-L121 | AX3DG027 |
| INLP122 | 1-1808-Q3-L122 | AX3DG027 |
| INLP125 | 1-1808-Q3-L125 | AX3DG027 |
| INLP126 | 1-1808-Q3-L126 | AX3DG027 |
| INLP127 | 1-1808-Q3-L127 | AX3DG027 |
| INLP128 | 1-1808-Q3-L128 | AX3DG027 |
| INLP129 | 1-1808-Q3-L129 | AX3DG027 |
| INLP130 | 1-1808-Q3-L130 | AX3DG027 |
| INLP131 | 1-1808-Q3-L131 | AX3DG024 |
| INLP132 | 1-1808-Q3-L132 | AX3DG024 |
| INLP133 | 1-1808-Q3-L133 | AX3DG024 |
| INLP135 | 1-1808-Q3-L135 | AX3DG024 |
| INLP136 | 1-1808-Q3-L136 | AX3DG024 |

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| NO. | VERSIONS | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 20.0 | REV. PER ABN CO3900200IE014,VER. 4.0 | 12/9/04 | JLO | TSL | DEW |

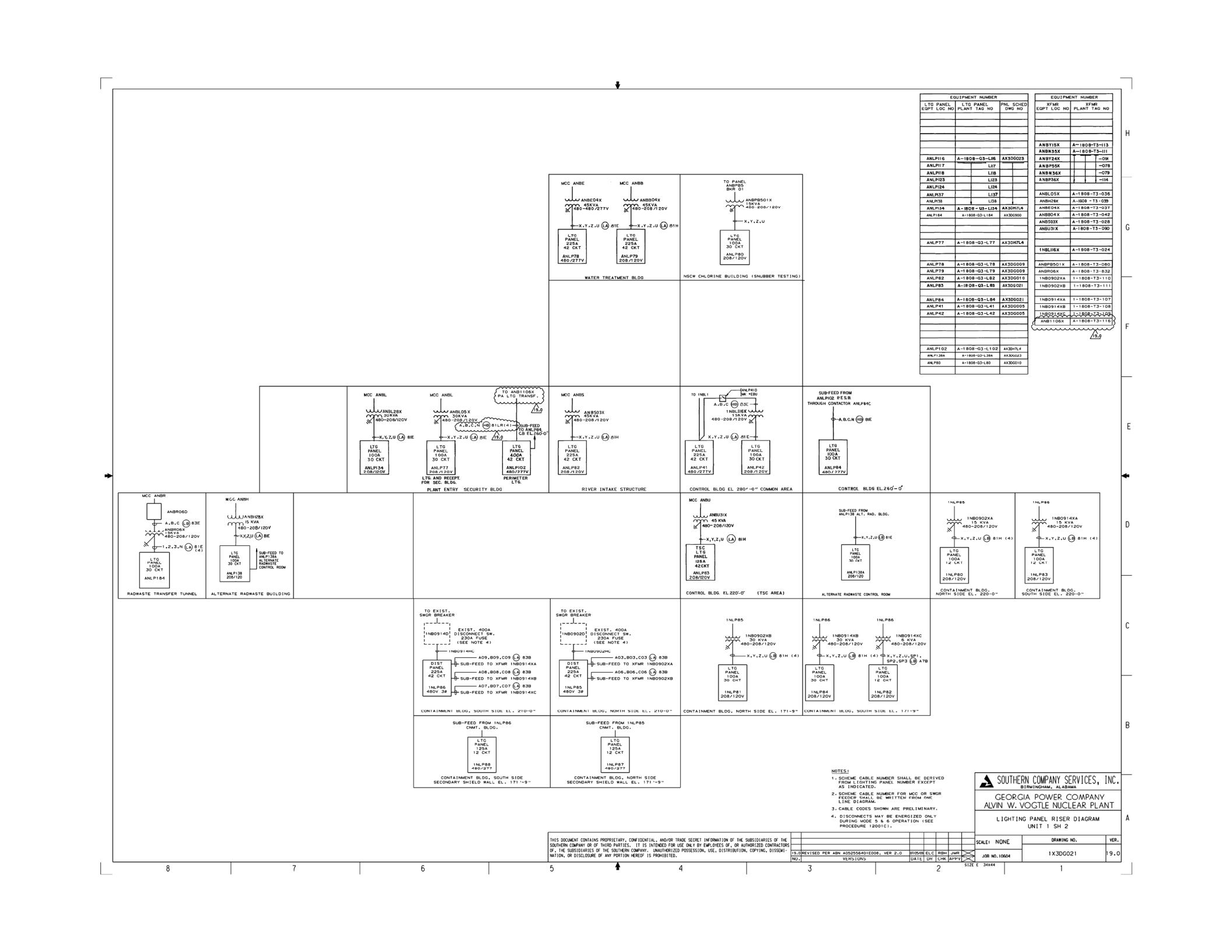
NUCLEAR SAFETY RELATED Q

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

LIGHTING PANEL RISER DIAGRAM
UNIT 1 SH 1

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X3DG020
VER. 20.0
SIZE E 34X44


EQUIPMENT NUMBER
LTG PANEL EQPT LOC NO
LTG PANEL PLANT TAG NO
PNL SCHED DWG NO
ANLP116 A-1808-Q3-L116 AX3DG023
ANLP117 L117
ANLP118 L118
ANLP123 L123
ANLP124 L124
ANLP137 L137
ANLP138 L138
ANLP134 A-1808-Q3-L134 AX3DH7L4
ANLP184 A-1808-Q3-L184 AX3DG900
ANLP77 A-1808-Q3-L77 AX3DH7L4
ANLP78 A-1808-Q3-L78 AX3DG009
ANLP79 A-1808-Q3-L79 AX3DG009
ANLP82 A-1808-Q3-L82 AX3DG010
ANLP83 A-1808-Q3-L83 AX3DG021
ANLP84 A-1808-Q3-L84 AX3DG021
ANLP41 A-1808-Q3-L41 AX3DG005
ANLP42 A-1808-Q3-L42 AX3DG005
ANLP102 A-1808-Q3-L102 AX3DH7L4
ANLP138A A-1808-Q3-L38A AX3DG023
ANLP80 A-1808-Q3-L80 AX3DG010
EQUIPMENT NUMBER
XFMR EQPT LOC NO
XFMR PLANT TAG NO
ANBY15X A-1808-T3-113
ANBN35X A-1808-T3-111
ANBY24X -091
ANBP55X -078
ANBN36X -079
ANBP36X -114
ANBL05X A-1808-T3-036
ANBH28X A-1808 - T3 - 039
ANBE04X A-1808-T3-037
ANBB04X A-1808-T3-042
ANB503X A-1808-T3-028
ANBU31X A-1808-T3-090
1NBL116X A-1808-T3-024
ANBPB501X A-1808-T3-080
ANBR06X A-1808-T3-832
1NB0902XA 1-1808-T3-110
1NB0902XB 1-1808-T3-111
1NB0914XA 1-1808-T3-107
1NB0914XB 1-1808-T3-108
1NB0914XC 1-1808-T3-109
ANB1106X A-1808-T3-116
19.0
MCC ANBE
ANBE04X 45KVA 480-480/277V
X,Y,Z,U LA 81E
LTG PANEL 225A 42 CKT
ANLP78 480/277V
MCC ANBB
ANBB04X 45KVA 480-208/120V
X,Y,Z,U LA 81H
LTG PANEL 225A 42 CKT
ANLP79 208/120V
WATER TREATMENT BLDG
TO PANEL ANBPB5 BKR 01
ANBPB501X 15KVA 480-208/120V
X,Y,Z,U
LTG PANEL 100A 30 CKT
ANLP80 208/120V
NSCW CHLORINE BUILDING (SNUBBER TESTING)
MCC ANBL
ANBL28X 30KVA 480-208/120V
X,Y,Z,U LA 81E
LTG PANEL 100A 30 CKT
ANLP134 208/120V
MCC ANBL
ANBL05X 30KVA 480-208/120V
A,B,C,N HB 81LR(4)
X,Y,Z,U LA 81E
TO ANB1106X PA LTG TRANSF.
19.0
SUB-FEED TO ANLP84 CB EL.260'-0"
LTG PANEL 225A 30 CKT
ANLP77 208/120V
LTG. AND RECEPT. FOR SEC. BLDG.
LTG PANEL 400A 42 CKT
ANLP102 480/277V
PERIMETER LTG.
PLANT ENTRY SECURITY BLDG
MCC ANBS
ANBS03X 45KVA 480-208/120V
X,Y,Z,U LA 81H
LTG PANEL 225A 42 CKT
ANLP82 208/120V
RIVER INTAKE STRUCTURE
TO 1NBL1
ANLP41D MK "EBU"
A,B,C HB 83C
1NBL116X 15KVA 480-208/120V
X,Y,Z,U LA 81E
LTG PANEL 225A 42 CKT
ANLP41 480/277V
LTG PANEL 100A 30 CKT
ANLP42 208/120V
CONTROL BLDG EL 280'-0" COMMON AREA
SUB-FEED FROM ANLP102 PESS THROUGH CONTACTOR ANLP84C
A,B,C,N HB 81E
LTG PANEL 100A 30 CKT
ANLP84 480/277V
CONTROL BLDG EL.260'-0"
MCC ANBR
ANBR06D
A,B,C LB 83E
ANBR06X 15KVA 480-208/120V
1,2,3,N LA 81E (4)
LTG PANEL 100A 30 CKT
ANLP184
RADWASTE TRANSFER TUNNEL
MCC ANBH
ANBH28X 15 KVA 480-208/120V
X,Y,Z,U LA 81E
LTG PANEL 100A 30 CKT
ANLP138 208/120
SUB-FEED TO ANLP138A ALTERNATE RADWASTE CONTROL ROOM
ALTERNATE RADWASTE BUILDING
MCC ANBU
ANBU31X 45 KVA 480-208/120V
X,Y,Z,U LA 81H
TSC LTG PANEL 125A 42CKT
ANLP85 208/120V
CONTROL BLDG. EL.220'-0" (TSC AREA)
SUB-FEED FROM ANLP138 ALT. RAD. BLDG.
X,Y,Z,U LB 81E
LTG PANEL 100A 30 CKT
ANLP138A 208/120
ALTERNATE RADWASTE CONTROL ROOM
1NLP85
1NB0902XA 15 KVA 480-208/120V
X,Y,Z,U LB 81H (4)
LTG PANEL 100A 12 CKT
1NLP80 208/120V
CONTAINMENT BLDG. NORTH SIDE EL. 220'-0"
1NLP86
1NB0914XA 15 KVA 480-208/120V
X,Y,Z,U LB 81H (4)
LTG PANEL 100A 12 CKT
1NLP83 208/120V
CONTAINMENT BLDG. SOUTH SIDE EL. 220'-0"
TO EXIST. SWGR BREAKER
1NB0914D
EXIST. 400A DISCONNECT SW. 230A FUSE (SEE NOTE 4)
1NB0914HC
DIST PANEL 225A 42 CKT
1NLP86 480V 3Ø
A09,B09,C09 LA 83B
SUB-FEED TO XFMR 1NB0914XA
A08,B08,C08 LA 83B
SUB-FEED TO XFMR 1NB0914XB
A07,B07,C07 LA 83B
SUB-FEED TO XFMR 1NB0914XC
CONTAINMENT BLDG. SOUTH SIDE EL. 210'-0"
TO EXIST. SWGR BREAKER
1NB0902D
EXIST. 400A DISCONNECT SW. 230A FUSE (SEE NOTE 4)
1NB0902HC
DIST PANEL 225A 42 CKT
1NLP85 480V 3Ø
A03,B03,C03 LA 83B
SUB-FEED TO XFMR 1NB0902XA
A06,B06,C06 LA 83B
SUB-FEED TO XFMR 1NB0902XB
CONTAINMENT BLDG. NORTH SIDE EL. 210'-0"
1NLP85
1NB0902XB 30 KVA 480-208/120V
X,Y,Z,U LB 81H (4)
LTG PANEL 100A 30 CKT
1NLP81 208/120V
CONTAINMENT BLDG. NORTH SIDE EL. 171'-9"
1NLP86
1NB0914XB 30 KVA 480-208/120V
X,Y,Z,U LB 81H (4)
LTG PANEL 100A 30 CKT
1NLP84 208/120V
1NLP86
1NB0914XC 6 KVA 480-208/120V
X,Y,Z,U,SP1, SP2,SP3 LB A7B
LTG PANEL 100A 12 CKT
1NLP82 208/120V
CONTAINMENT BLDG. SOUTH SIDE EL. 171'-9"
SUB-FEED FROM 1NLP86 CNMT. BLDG.
LTG PANEL 125A 12 CKT
1NLP88 480/277
CONTAINMENT BLDG. SOUTH SIDE SECONDARY SHIELD WALL EL. 171'-9"
SUB-FEED FROM 1NLP85 CNMT. BLDG.
LTG PANEL 125A 12 CKT
1NLP87 480/277
CONTAINMENT BLDG. NORTH SIDE SECONDARY SHIELD WALL EL. 171'-9"
NOTES:
1. SCHEME CABLE NUMBER SHALL BE DERIVED FROM LIGHTING PANEL NUMBER EXCEPT AS INDICATED.
2. SCHEME CABLE NUMBER FOR MCC OR SWGR FEEDER SHALL BE WRITTEN FROM ONE LINE DIAGRAM.
3. CABLE CODES SHOWN ARE PRELIMINARY.
4. DISCONNECTS MAY BE ENERGIZED ONLY DURING MODE 5 & 6 OPERATION (SEE PROCEDURE 12001C).
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NO. VERSIONS DATE DR CHK APPV
19.0 REVISED PER ABN A052556401E008, VER 2.0 07/25/09 ELC RBH JMR
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
LIGHTING PANEL RISER DIAGRAM
UNIT 1 SH 2
SCALE: NONE
DRAWING NO. 1X3DG021
VER. 19.0
JOB NO. 10604
SIZE E 34x44
8 7 6 5 4 3 2 1
H G F E D C B A

# CONTAINMENT BUILDING

## LEVEL 4

E04D TO 12J246 SEE DWG. 1X3DG033
AB2 TO 1BJ201 SEE DWG. 1X3DG033
ES4D TO 13J251 SEE DWG. 1X3DG033

| TB1 | | | TB19 | | | |
|---|---|---|---|---|---|---|
| GREEN | L1 | 1 | BLACK | H | 221 | LEG A |
| WHITE | L2 | 2 | WHITE | N | 222 | A |
| RED | L1 | 3 | WHITE | GND | 223 | GND |
| WHITE | L2 | 4 | BLACK | H | 224 | LEG B |
| WHITE/BLACK | L1 | 5 | WHITE | N | 225 | B |
| WHITE | L2 | 6 | | H | 226 | |
| RED/BLACK | L1 | 7 | | N | 227 | |
| WHITE | L2 | 8 | | GND | 228 | |
| WHITE | L1 | 9 | | | | |
| ORANGE | L2 | 10 | | | | |
| WHITE | L1 | 11 | | | | |
| BLUE | L2 | 12 | | | | |
| SPARE | | 13 | | | | |

## LEVEL 3

GREEN/BLACK — SPARE
WHITE — SPARE

NOTE: 1. WIRING DIAGRAM USING ES4D CABLE AT COMMUNICATIONS TEST & DISTRIBUTION CABINET.

## LEVEL 2

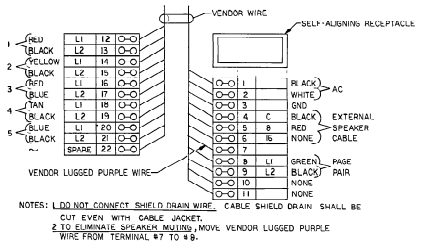
E08 TO 12J246 SEE DWG. 1X3DG033
A2B TO 1BJ201 SEE DWG. 1X3DG033
E08 TO 13J251 SEE DWG. 1X3DG033

| TB1 | | | TB19 | | | |
|---|---|---|---|---|---|---|
| GREEN | L1 | 1 | BLACK | H | 221 | LEG A |
| BLACK | L2 | 2 | WHITE | N | 222 | A |
| RED | L1 | 3 | WHITE | GND | 223 | |
| BLACK | L2 | 4 | BLACK | H | 224 | LEG B |
| YELLOW | L1 | 5 | WHITE | N | 225 | B |
| BLACK | L2 | 6 | | H | 226 | |
| RED | L1 | 7 | | N | 227 | |
| BLUE | L2 | 8 | | GND | 228 | |
| TAN | L1 | 9 | | | | |
| BLACK | L2 | 10 | | | | |
| BLUE | L1 | 11 | | | | |
| BLACK | L2 | 12 | | | | |
| | SPARE | 13 | | | | |

NOTE: 1. WIRING DIAGRAM USING E08 CABLE AT COMMUNICATIONS TEST & DISTRISTIBUTION CARNET.

## LEVEL 1

WIRING DIAGRAM FOR GAI-TRONICS COMMUNICATIONS WIRING DIAGRAM TELEPHONE/PAGE HANDSETS & AMPLIFIERS USING "E08" CABLE

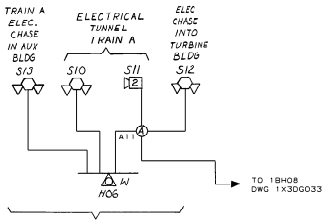



## LEVEL A

NOTES: 1. DO NOT CONNECT SHIELD DRAIN WIRE. CABLE SHIELD DRAIN SHALL BE CUT EVEN WITH CABLE JACKET.
2. TO ELIMINATE SPEAKER MUTING, MOVE VENDOR LUGGED PURPLE WIRE FROM TERMINAL #7 TO #8.

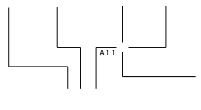
## LEVEL B

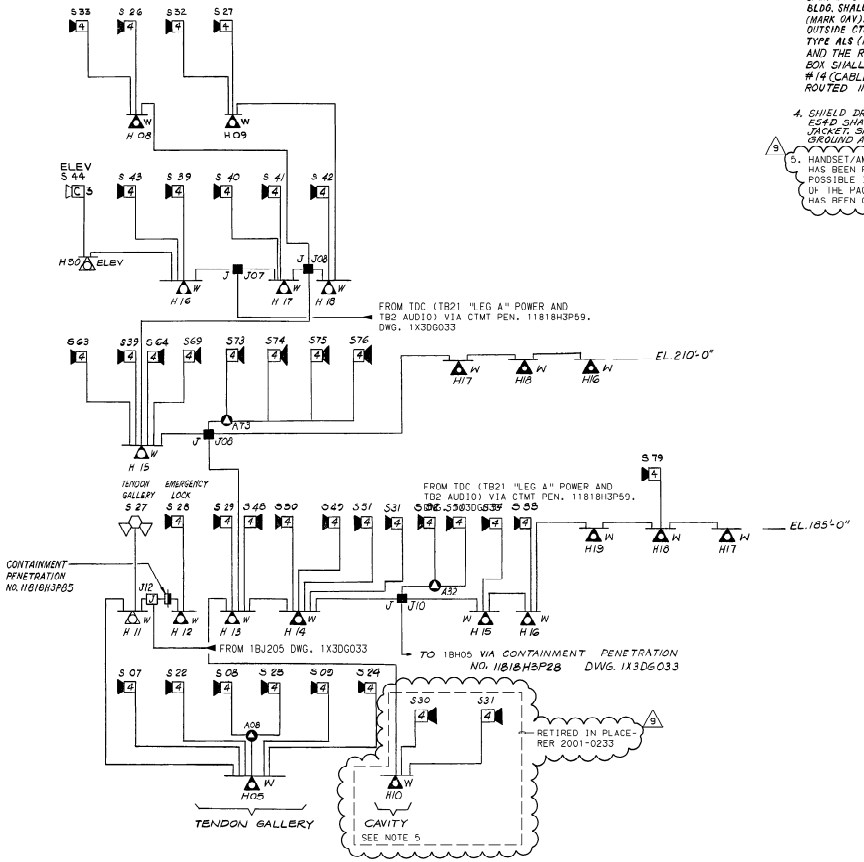
## LEVEL C




ELECTRICAL TUNNEL AND CHASES BETWEEN AUXILIARY AND TURBINE BLDGS

## LEVEL D




NOTES:

1. TO OBTAIN THE FULL EQUIPMENT NUMBER, PREFIX ALL NUMBERS SHOWN ON RISER DIAGRAMS WITH:
   a. THE UNIT
   b. FLOOR LEVEL AND
   c. DEVICE(SPEAKER, HANDSET OR SPEAKER AMPLIFIER) G, H AND A RESPECTIVELY.
   THUS, SPEAKER NO. 18 IN FLOOR LEVEL 2 IN:

2. FOR KEY TO SYMBOLS SEE DWG. 1X3DG030.
3. ALL INTERCONNECTING CABLE BETWEEN HANDSET STATIONS, SPEAKER AMPLIFIERS, JUNCTION BOXES AND THE TEST AND DISTRIBUTION CABINET ZONE CONNECTIONS SHALL BE 16/C CABLE (1 PR #14 & 7 PR #18, CABLE CODE ES4D) AND IS TO BE ROUTED IN 1½ CONDUIT MINIMUM. SPEAKER CABLE FOR SPEAKERS INSIDE CTMT. BLDG. SHALL BE 2/C #16, TYPE CUS (MARK OAV). SPEAKER CABLE FOR SPEAKERS OUTSIDE CTMT. BLDG. SHALL BE 2/C #16, TYPE ALS (MARK OAU). THE ZONE MERGE UNITS AND THE REMOTE PLANT ALARM SWITCH BOX SHALL BE CONNECTED WITH 12/C #14 (CABLE CODE AB1) AND SHALL BE ROUTED IN 1½ CONDUIT MINIMUM.
4. SHIELD DRAIN WIRE PROVIDED IN CABLE ES4D SHALL BE CUT EVEN WITH CABLE JACKET. SHIELD TO BE ISOLATED FROM GROUND AT ALL LOCATIONS.
5. HANDSET/AMPLIFIER BENEATH THE VESSEL HAS BEEN REMOVED AS TO ELIMINATE POSSIBLE INTERFERENCE WITH THE REST OF THE PAGE SYSTEM. AMPLIFIER BOX HAS BEEN COVERED BY A BLANK PLATE. (9)

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| 9 | INCORP. PER ABN-39354 | 12-19-03 | Vap | MDN | DEW | X |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TELEPHONE/PAGE SYSTEM
RISER DIAGRAM
UNIT 1 SHEET 1

| SCALE: NONE | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 10604 | 1X3DG031 | 9 |

SIZE E 34x44

# AUXILIARY BUILDING

LEVEL 4

LEVEL 3

LEVEL 2

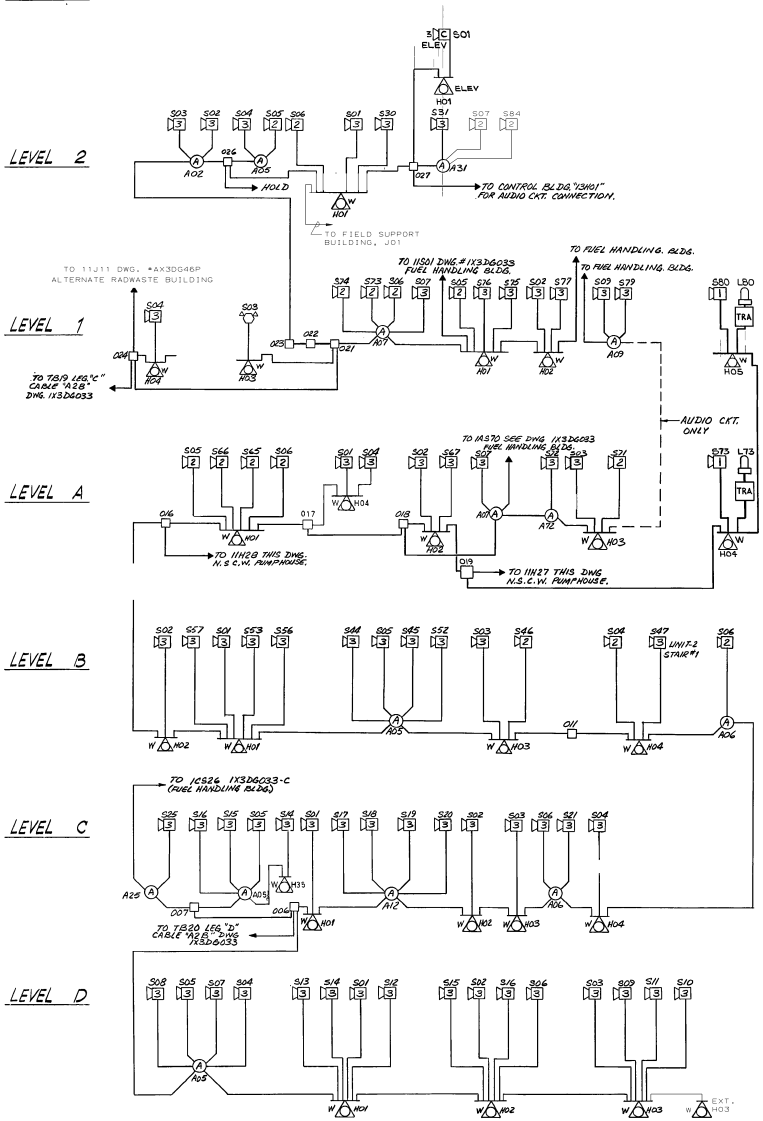
LEVEL 1

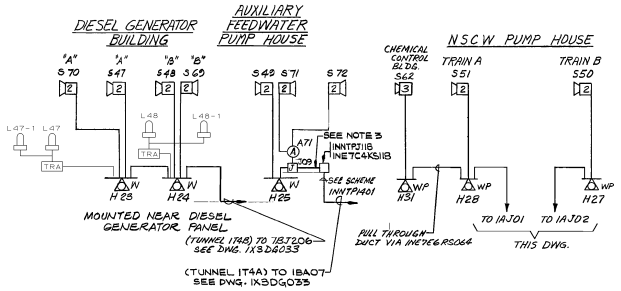
LEVEL A

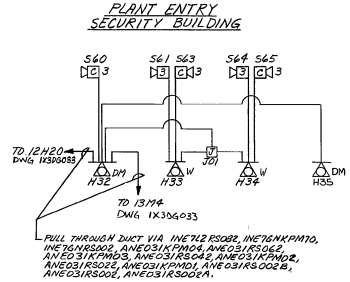
LEVEL B

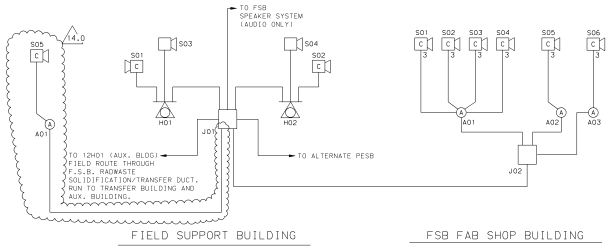
LEVEL C

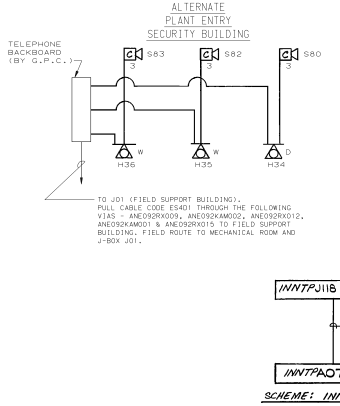
LEVEL D

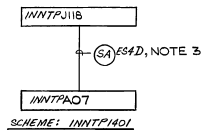



SCHEME: INNTPI401

NOTES:

1. FOR GENERAL NOTES SEE DWG. 1X3DG031, SHEET 1.
2. TO BE ROUTED IN MECHANICAL TUNNELS.
3. FIELD TO CONTINUE. ROUTE OF CABLE IN UNSCHEDULED RACEWAYS FOR COMPLETION OF WIRING REQUIREMENTS PER RISER DIAGRAM.

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| 14.0 | REVISED PER ABN-81373, VERSION 1.0 | 12/29/05 | ELC | RBH | DEW |

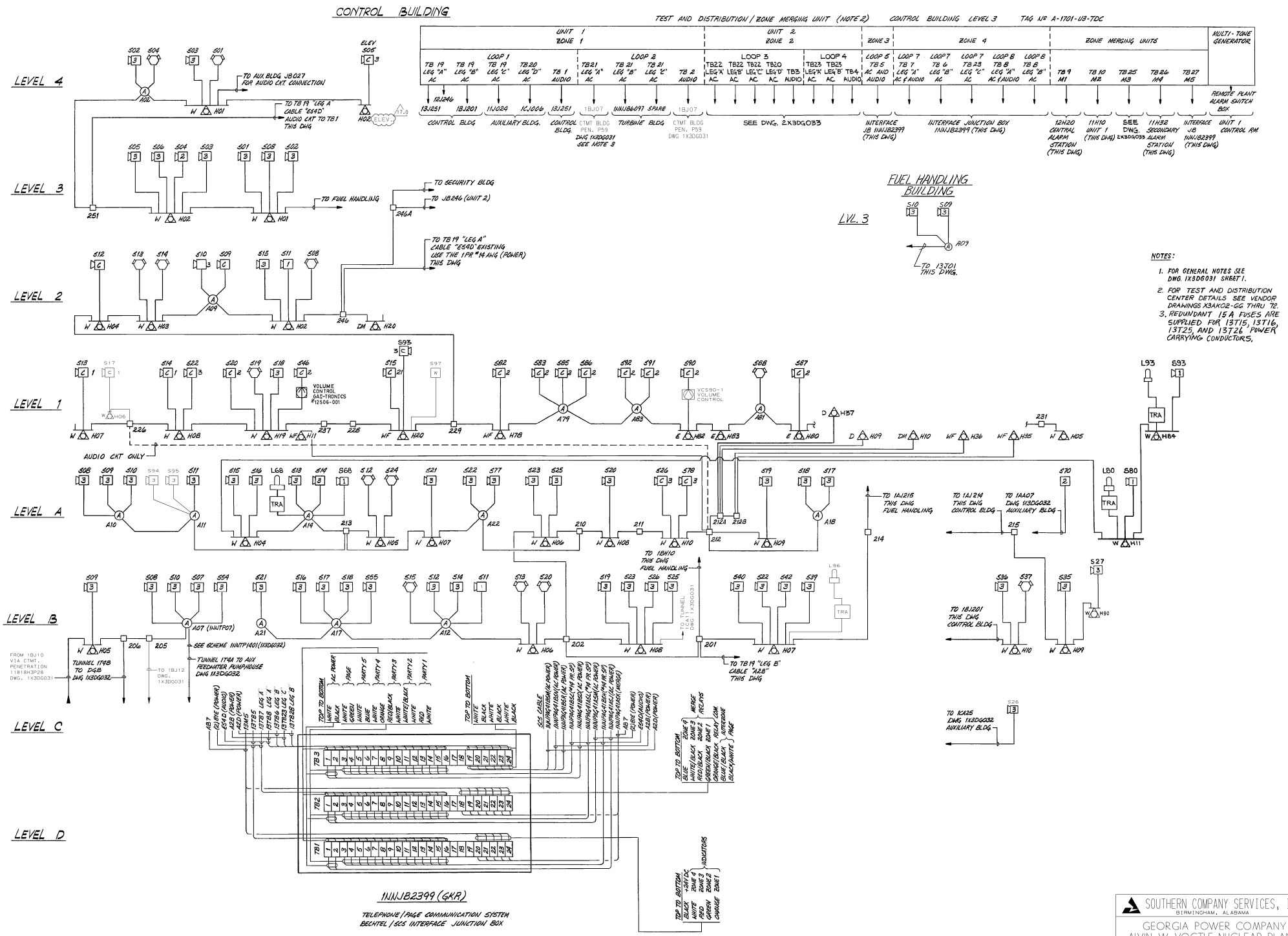
SIZE E 34X44

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

TELEPHONE/PAGE SYSTEM
RISER DIAGRAM
UNIT 1 SHEET 2

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X3DG032 | 14.0 |

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SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

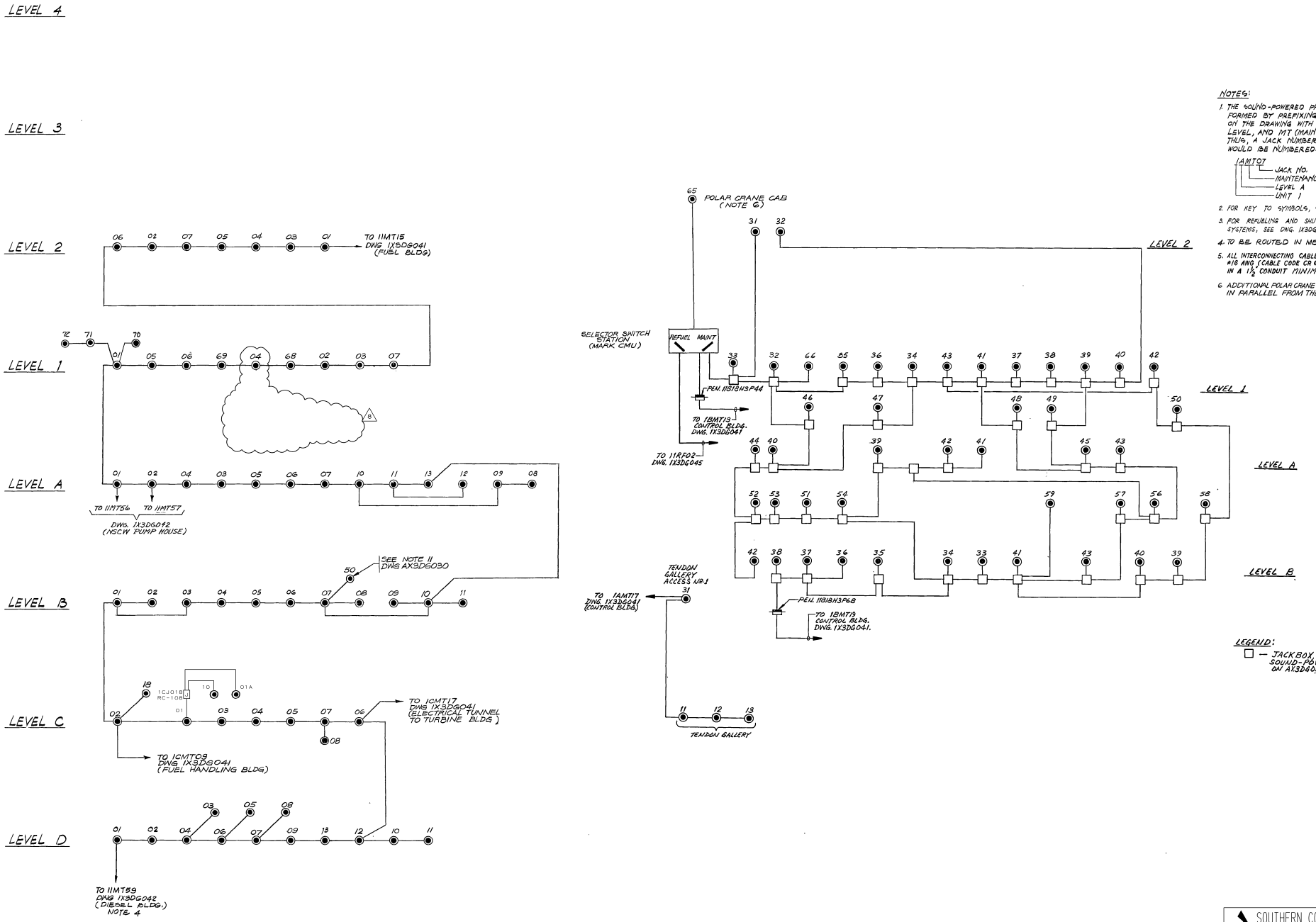
TELEPHONE/PAGE SYSTEM
RISER DIAGRAM
UNIT 1 SHEET 3

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO. 10604 | 1X3DG033 | 17.0 |

SIZE E 34X44

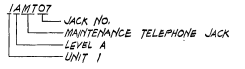
AUXILIARY BUILDING

CONTAINMENT BUILDING




NOTES:

1. THE SOUND-POWERED PHONE JACK NUMBER IS FORMED BY PREFIXING THE NUMBERS AS SHOWN ON THE DRAWING WITH THE UNIT NUMBER, FLOOR LEVEL, AND MT (MAINTENANCE TELEPHONE). THUS, A JACK NUMBERED 07 ON LEVEL A WOULD BE NUMBERED:




2. FOR KEY TO SYMBOLS, SEE DWG. AX3DG030.
3. FOR REFUELING AND SHUTDOWN SOUND-POWERED PHONE SYSTEMS, SEE DWG. 1X3DG045, SHEET 1.
4. TO BE ROUTED IN MECHANICAL TUNNELS
5. ALL INTERCONNECTING CABLE IS TO BE 7 PAIR (1 SPARE) #16 AWG (CABLE CODE CR 87) AND IS TO BE ROUTED IN A 1½" CONDUIT MINIMUM..
6. ADDITIONAL POLAR CRANE JACKS COULD BE WIRED IN PARALLEL FROM THE CAB JACK.

LEGEND:

□ — JACK BOX, SEE MAINTENANCE SOUND-POWER SYSTEM DETAILS ON AX3D6030 FOR MARK #.

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|---|---|---|---|---|---|---|
| 8 | INCORP. PER DCP 98-VAN0055 | 01/07/02 | ELC | EGG | TSL | X |

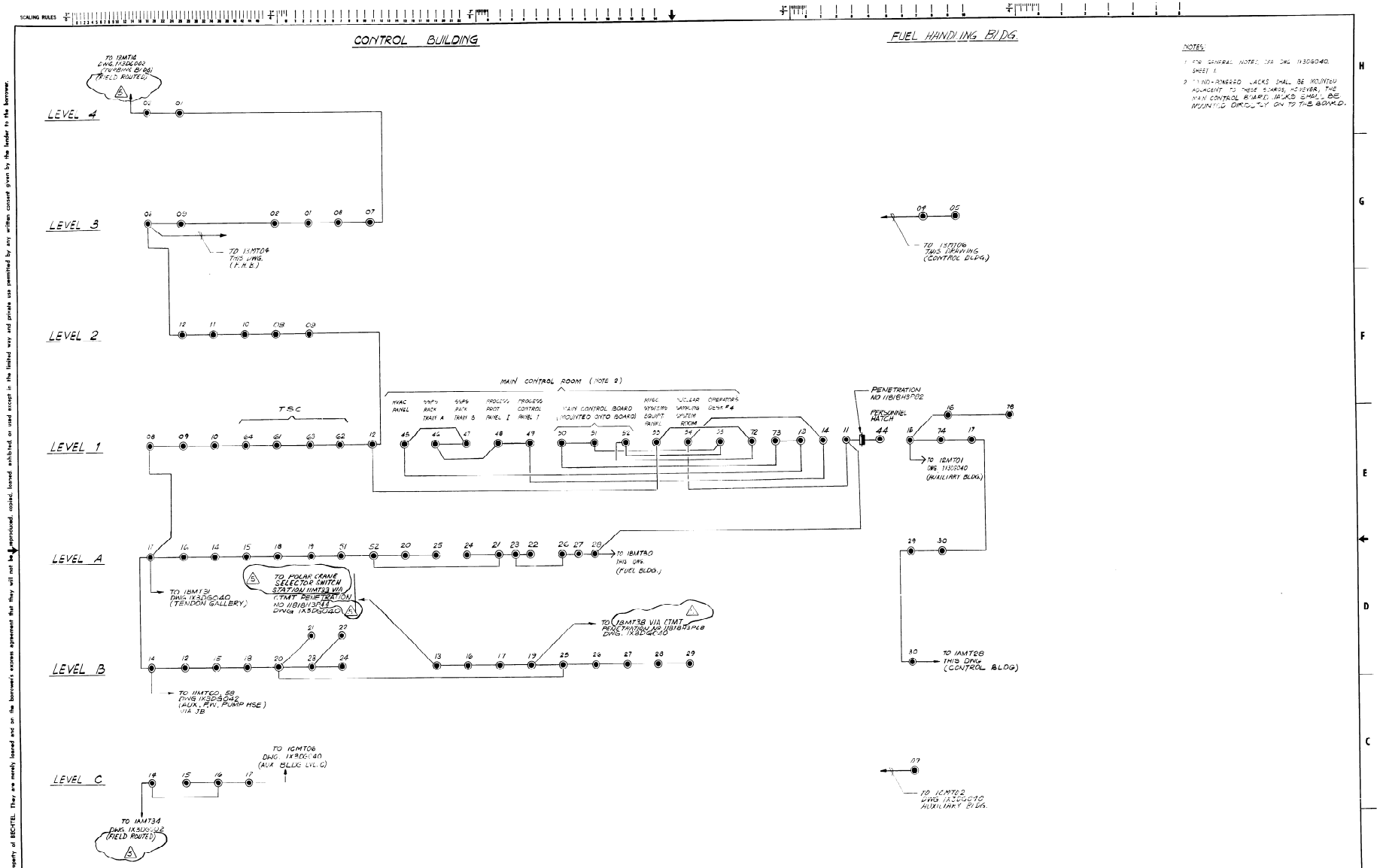
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SOUND-POWERED PHONE SYSTEM
RISER DIAGRAM-MAINTENANCE SYSTEM
UNIT 1 SHEET 1

| SCALE: NONE | DRAWING NO. | REV. |
|---|---|---|
| JOB NO. 10604 | 1X3DG040 | 8 |

SIZE E 34X44




NOTES:

1. FOR GENERAL NOTES SEE DWG 1X3D0040, SHEET 1.
2. WIND-POWERED JACKS SHALL BE MOUNTED ADJACENT TO THESE BOARDS, HOWEVER, THE MAIN CONTROL BOARD JACKS SHALL BE MOUNTED DIRECTLY ON TO THE BOARD.

| REVISIONS | | |

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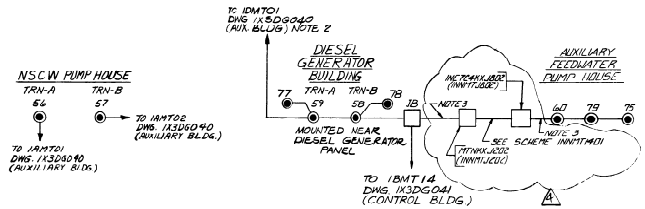
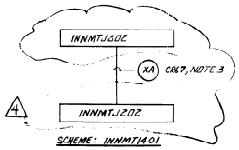
NOTES:

1. FOR GENERAL NOTES, SEE DWG. IX3DG040, SHEET 1.
2. TO BE ROUTED IN MECHANICAL TUNNELS.
3. FIELD TO CONTINUE ROUTE OF CABLE IN UNSCHEDULED RACEWAYS FOR CONNECTION OF WIRING REQUIREMENTS PER RISER DIAGRAM.
4. -



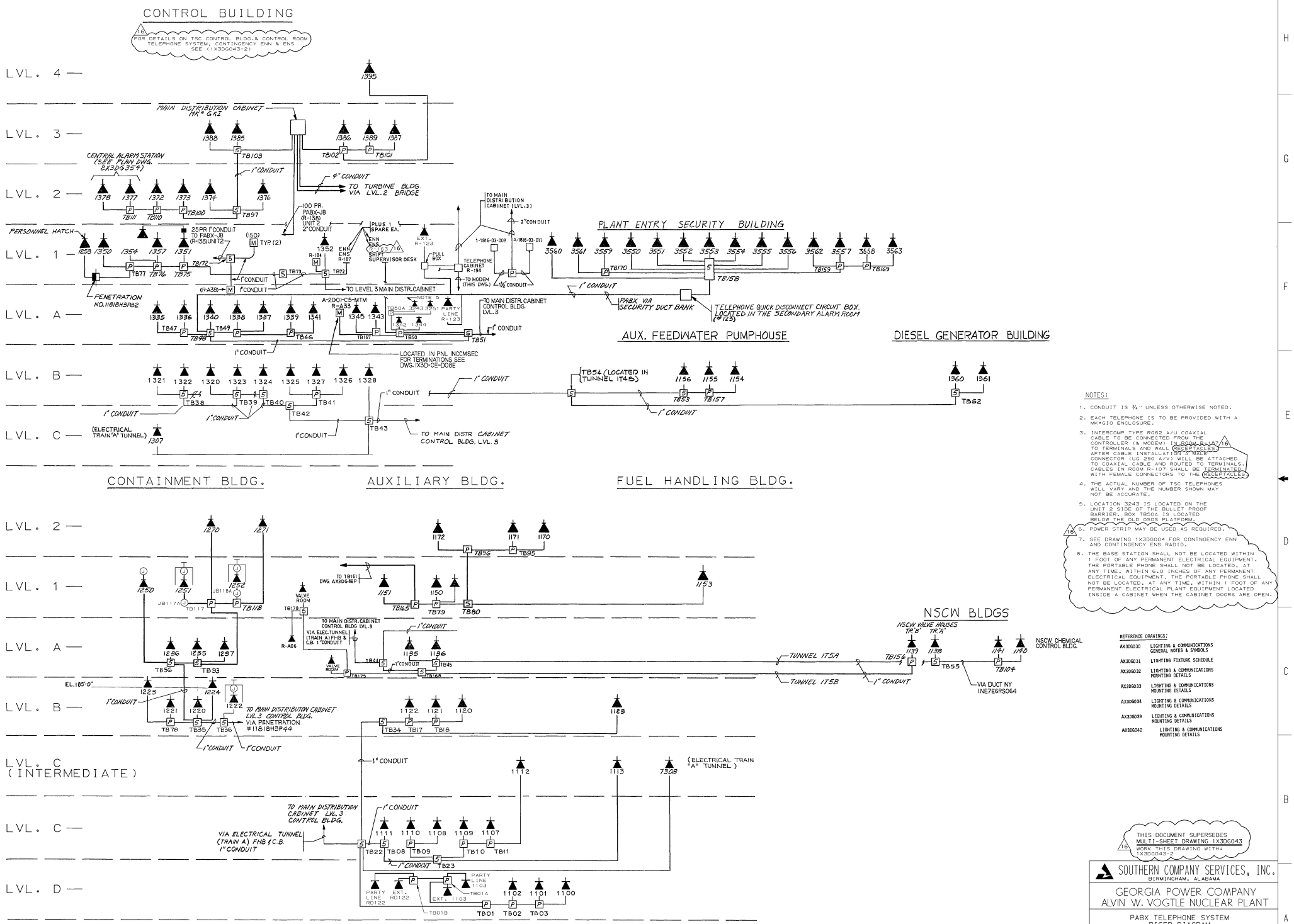




| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | G.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 4 | INCORPORATED DCN #1 (E-PCRF-13390) FFE | 4-13-81 | DEL. | JM | JJ | | | | |
| 3 | REVISED PER DCN #2 | | | | | | | | |
| 2 | REVISED NUMBERS IN D.G.B. | | | | | | | | |
| 1 | ISSUED FOR CONSTRUCTION | | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 4 | INCORPORATED DCN #1 (E-PCRF-13390) FFE | 4-13-81 | DEL. | JM | JJ | | | | |
| 3 | REVISED PER DCN #2 | 10-16-80 | DEL. | JM | JJ | | | | |
| 2 | REVISED NUMBERS IN D.G.B. | 9-10-80 | DEL. | JM | JJ | | | | |
| 1 | ISSUED FOR CONSTRUCTION | 6-10-80 | DEL. | JM | JJ | | | | |

| NO. | REVISIONS | DATE | DR. | CHK. | SUPV. | EGS | CHF. E | P.E. | G.A.E. |
|---|---|---|---|---|---|---|---|---|---|
| 4 | INCORPORATED DCN #1 (E-PCRF-13390) FFE | 4-13-81 | DEL. | JM | JJ | | | | |
| 3 | REVISED PER DCN #2 | 10-16-80 | DEL. | JM | JJ | | | | |
| 2 | REVISED NUMBERS IN D.G.B. | 9-10-80 | DEL. | JM | JJ | | | | |
| 1 | ISSUED FOR CONSTRUCTION | 6-10-80 | DEL. | JM | JJ | | | | |

| | | | | | | | | |




NOTES:

1. CONDUIT IS ¾" UNLESS OTHERWISE NOTED.
2. EACH TELEPHONE IS TO BE PROVIDED WITH A MK*GIO ENCLOSURE.
3. INTERCOMP TYPE RG62 A/U COAXIAL CABLE TO BE CONNECTED FROM THE CONTROLLER (& MODEM) IN ROOM R-147 TO TERMINALS AND WALL RECEPTACLES AFTER CABLE INSTALLATION A MALE CONNECTOR (UG 290 A/V) WILL BE ATTACHED TO COAXIAL CABLE AND ROUTED TO TERMINALS. CABLES IN ROOM R-107 SHALL BE TERMINATED WITH FEMALE CONNECTORS TO THE RECEPTACLES
4. THE ACTUAL NUMBER OF TSC TELEPHONES WILL VARY AND THE NUMBER SHOWN MAY NOT BE ACCURATE.
5. LOCATION 3243 IS LOCATED ON THE UNIT 2 SIDE OF THE BULLET PROOF BARRIER. BOX TB50A IS LOCATED BELOW THE OLD OSOS PLATFORM.
6. POWER STRIP MAY BE USED AS REQUIRED.
7. SEE DRAWING 1X3DG004 FOR CONTNGENCY ENN AND CONTINGENCY ENS RADIO.
8. THE BASE STATION SHALL NOT BE LOCATED WITHIN 1 FOOT OF ANY PERMANENT ELECTRICAL EQUIPMENT. THE PORTABLE PHONE SHALL NOT BE LOCATED, AT ANY TIME, WITHIN 6.0 INCHES OF ANY PERMANENT ELECTRICAL EQUIPMENT. THE PORTABLE PHONE SHALL NOT BE LOCATED, AT ANY TIME, WITHIN 1 FOOT OF ANY PERMANENT ELECTRICAL PLANT EQUIPMENT LOCATED INSIDE A CABINET WHEN THE CABINET DOORS ARE OPEN.

REFERENCE DRAWINGS:

| | |
|---|---|
| AX3DG030 | LIGHTING & COMMUNICATIONS GENERAL NOTES & SYMBOLS |
| AX3DG031 | LIGHTING FIXTURE SCHEDULE |
| AX3DG032 | LIGHTING & COMMUNICATIONS MOUNTING DETAILS |
| AX3DG033 | LIGHTING & COMMUNICATIONS MOUNTING DETAILS |
| AX3DG034 | LIGHTING & COMMUNICATIONS MOUNTING DETAILS |
| AX3DG039 | LIGHTING & COMMUNICATIONS MOUNTING DETAILS |
| AX3DG040 | LIGHTING & COMMUNICATIONS MOUNTING DETAILS |

16 THIS DOCUMENT SUPERSEDES MULTI-SHEET DRAWING 1X3DG043 WORK THIS DRAWING WITH: 1X3DG043-2

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

PABX TELEPHONE SYSTEM
RISER DIAGRAM
UNIT 1

SCALE: NONE

DRAWING NO. 16 1X3DG043-1

REV. 16

JOB NO.10604

SIZE E 34X44

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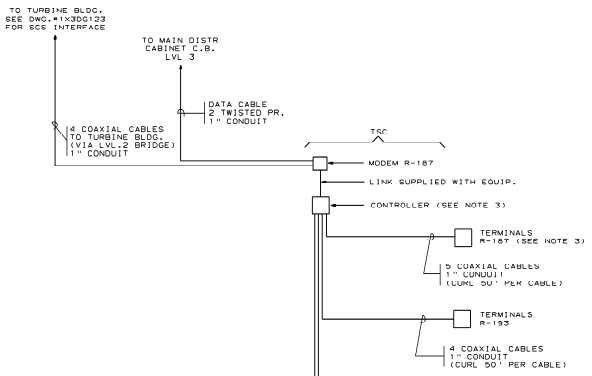
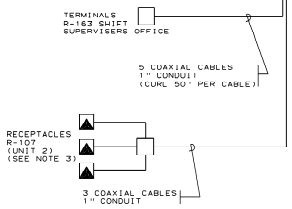
| 16 | INCORP. PER ABN-36907 & DRAWING CLARIFICATION | 02-22-02 | PM | EL.C | MWD | |

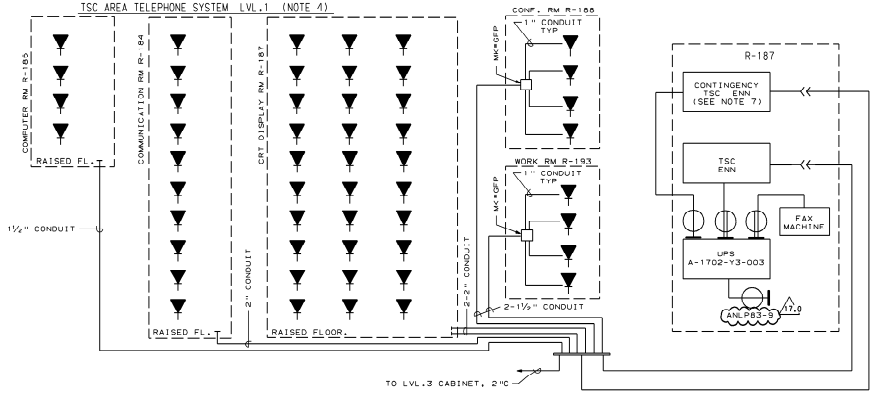
## CONTROL BUILING

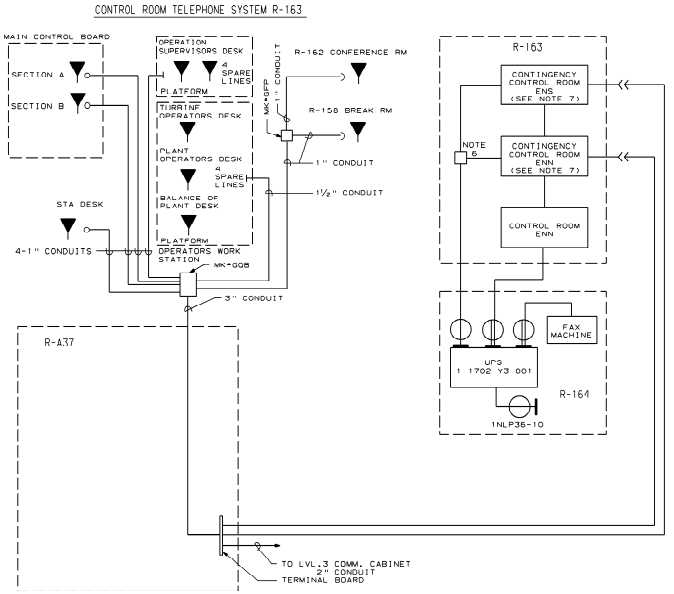



TSC —

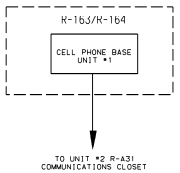
LVL. 1 —




NOTES:
FOR NOTES SEE DRAWING 1X3DG043-1




THIS DOCUMENT SUPERSEDES
MULTI-SHEET DRAWING 1X3DG043
WORK THIS DRAWING WITH:
1X3DG043-1

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SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

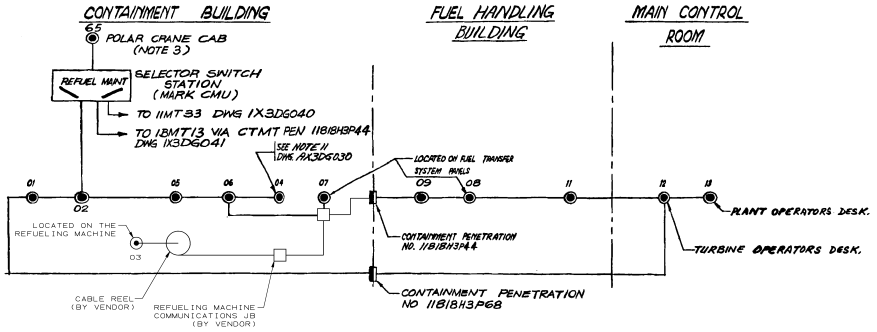
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

PABX TELEPHONE SYSTEM
RISER DIAGRAM
UNIT 1

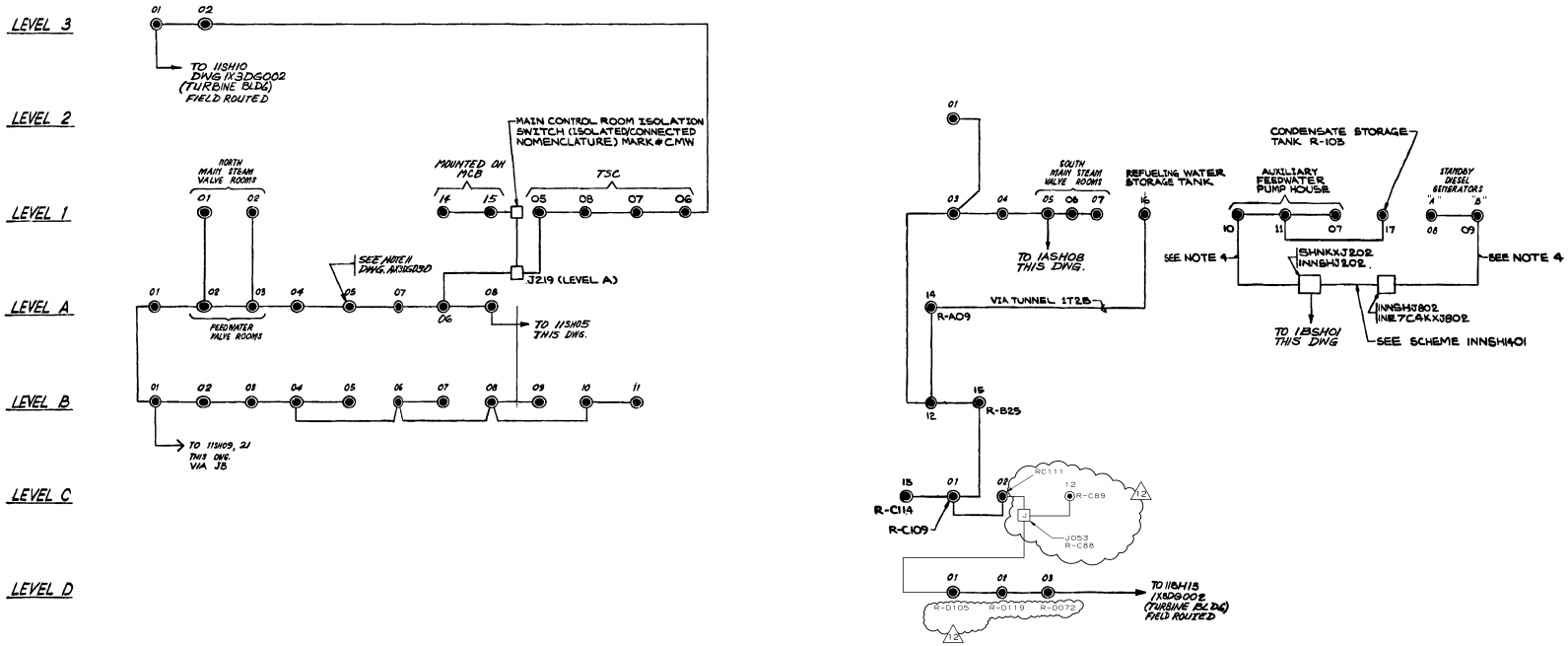
| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X3DG043-2 | 17.0 |

SIZE E 34x44

## REFUEL SYSTEM

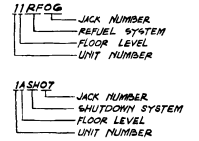



## SHUTDOWN SYSTEM




NOTES:

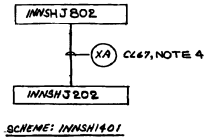
1. THE SOUND-POWERED PHONE JACK NUMBER IS FORMED BY PREFIXING THE NUMBER AS SHOWN ON THE DRAWING WITH THE UNIT NUMBER, FLOOR LEVEL, AND RF, (REFUEL SYSTEM) OR SH, (SHUTDOWN SYSTEM). I. JACK NUMBERED 06 IN THE REFUEL SYSTEM WOULD BE:




2. ALL INTERCONNECTING CABLE FOR THE SHUTDOWN AND REFUEL SYSTEM SHALL BE A 1 PAIR - 16 AWG CABLE (CABLE CODE CL67) AND IS TO BE ROUTED IN ¾" CONDUIT MINIMUM.

3. ADDITIONAL POLAR CRANE JACKS COULD BE WIRED IN PARALLEL FROM THE CAB JACK.

4. FIELD TO CONTINUE ROUTE OF CABLE IN UNSCHEDULED RACEWAYS FOR COMPLETION OF WIRING REQUIREMENTS PER RISER DIAGRAM.




SCHEME: INNSH1401

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 2 | INCORPORATED PER DCP 92-VAN0029 | 1-21-99 | JGM | SM | GLB | |
| 1 | ISSUED FOR CONSTRUCTION | SEE MICROFILM FOR SIGNATURES | | | | |

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

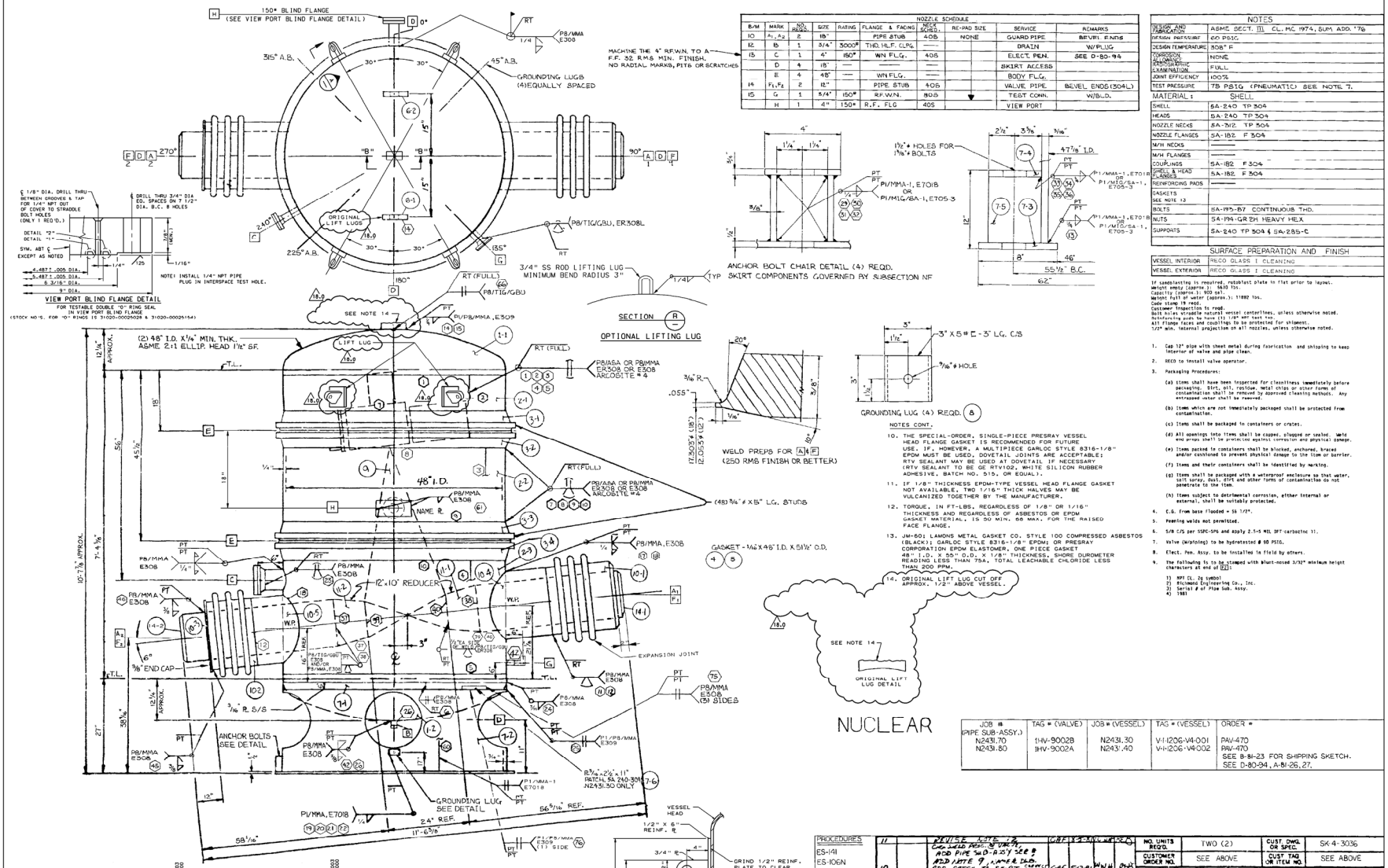
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

SOUND POWERED PHONE SYSTEM
RISER DIAGRAM-REFUEL & SHUTDOWN SYSTEMS
UNIT 1 SHEET 1

| SCALE: NONE | DRAWING NO. | REV. |
|---|---|---|
| JOB NO.10604 | 1X3DG045 | 12 |

SIZE E 34X44

CAD NAME 1X3DG045




| NOZZLE SCHEDULE | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| B/M | MARK | NO. REQD. | SIZE | RATING | FLANGE & FACING | NECK SCHED. | RE-PAD SIZE | SERVICE | REMARKS |
| 10 | $A_1, A_2$ | 2 | 18" | | PIPE STUB | 40S | NONE | GUARD PIPE | BEVEL ENDS |
| 12 | B | 1 | 3/4" | 3000# | THD. HLF. CPLG. | —— | | DRAIN | W/PLUG |
| 13 | C | 1 | 4" | 150# | WN FLG., | 40S | | ELECT. PEN. | SEE D-80-94 |
| | D | 4 | 18" | —— | —— | —— | | SKIRT ACCESS | |
| | E | 4 | 48" | —— | WN FLG. | —— | | BODY FLG. | |
| 14 | $F_1, F_2$ | 2 | 12" | | PIPE STUB | 40S | | VALVE PIPE | BEVEL ENDS (304L) |
| 15 | G | 1 | 3/4" | 150# | RF.WN. | 80S | | TEST CONN. | W/BLD. |
| | H | 1 | 4" | 150# | R.F. FLG | 40S | | VIEW PORT | |

MACHINE THE 4" RF.W.N. TO A F.F. 32 RMS MIN. FINISH. NO RADIAL MARKS, PITS OR SCRATCHES

| NOTES | |
|---|---|
| DESIGN AND FABRICATION | ASME SECT. III CL. MC 1974 SUM ADD. '76 |
| DESIGN PRESSURE | 60 PSIG |
| DESIGN TEMPERATURE | 308° F |
| CORROSION ALLOWANCE | NONE |
| RADIOGRAPHIC EXAMINATION | FULL |
| JOINT EFFICIENCY | 100% |
| TEST PRESSURE | 75 PSIG (PNEUMATIC) SEE NOTE 7. |
| MATERIAL : | SHELL |
| SHELL | SA-240 TP 304 |
| HEADS | SA-240 TP 304 |
| NOZZLE NECKS | SA-312 TP 304 |
| NOZZLE FLANGES | SA-182 F 304 |
| W/H NECKS | —— |
| W/H FLANGES | —— |
| COUPLINGS | SA-182 F 304 |
| SHELL & HEAD FLANGES | SA-182 F 304 |
| REINFORCING PADS | —— |
| GASKETS SEE NOTE 13 | |
| BOLTS | SA-193-B7 CONTINUOUS THD. |
| NUTS | SA-194-GR 2H HEAVY HEX |
| SUPPORTS | SA-240 TP 304 & SA-285-C |
| | |
| SURFACE PREPARATION AND FINISH | |
| VESSEL INTERIOR | RECO GLASS I CLEANING |
| VESSEL EXTERIOR | RECO GLASS I CLEANING |

If sandblasting is required, nitrobast plate in flat prior to layout.
Weight empty (approx.): 5610 lbs.
Capacity (approx.): 900 cu.
Weight full of water (approx.): 11882 lbs.
Code stamp 19 req'd.
Customer inspection to req'd.
Bolt holes straddle natural vessel centerlines, unless otherwise noted.
Reinforcing pads to have (1) 1/8" NPT test tap.
All flange faces and couplings to be protected for shipment.
1/2" min. internal projection on all nozzles, unless otherwise noted.

1. Cap 12" pipe with sheet metal during fabrication and shipping to keep interior of valve and pipe clean.
2. RECO to install valve operator.
3. Packaging Procedures:
   (a) Items shall have been inspected for cleanliness immediately before packaging. Dirt, oil, residue, metal chips or other forms of contamination shall be removed by approved cleaning methods. Any entrapped water shall be removed.
   (b) Items which are not immediately packaged shall be protected from contamination.
   (c) Items shall be packaged in containers or crates.
   (d) All openings into items shall be capped, plugged or sealed. Weld end preps shall be protected against corrosion and physical damage.
   (e) Items packed in containers shall be blocked, anchored, braced and/or cushioned to prevent physical damage to the item or barrier.
   (f) Items and their containers shall be identified by marking.
   (g) Items shall be packaged with a waterproof enclosure so that water, salt spray, dust, dirt and other forms of contamination do not penetrate to the item.
   (h) Items subject to detrimental corrosion, either internal or external, shall be suitably protected.
4. C.G. from base flooded = 56 1/2".
5. Peening welds not permitted.
6. S/B C/S per SSPC-SP6 and apply 2.5-5 MIL DFT carbozinc 11.
7. Valve (W/piping) to be hydratested @ 90 PSIG.
8. Elect. Pen. Assy. to be installed in field by others.
9. The following is to be stamped with blunt-nosed 3/32" minimum height characters at end of [12]:
   1) NPT (L. 2g symbol)
   2) Richmond Engineering Co., Inc.
   3) Serial # of Pipe Sub. Assy.
   4) 1981
10. THE SPECIAL-ORDER, SINGLE-PIECE PRESRAY VESSEL HEAD FLANGE GASKET IS RECOMMENDED FOR FUTURE USE. IF, HOWEVER, A MULTIPIECE GARLOK STYLE 8316-1/8" EPDM MUST BE USED, DOVETAIL JOINTS ARE ACCEPTABLE; RTV SEALANT MAY BE USED AT DOVETAIL IF NECESSARY (RTV SEALANT TO BE GE RTV102, WHITE SILICON RUBBER ADHESIVE, BATCH NO. 513, OR EQUAL).
11. IF 1/8" THICKNESS EPDM-TYPE VESSEL HEAD FLANGE GASKET NOT AVAILABLE, TWO 1/16" THICK HALVES MAY BE VULCANIZED TOGETHER BY THE MANUFACTURER.
12. TORQUE, IN FT-LBS, REGARDLESS OF 1/8" OR 1/16" THICKNESS AND REGARDLESS OF ASBESTOS OR EPDM GASKET MATERIAL, IS 50 MIN. 60 MAX. FOR THE RAISED FACE FLANGE.
13. JM-601 LAMONS METAL GASKET CO. STYLE 100 COMPRESSED ASBESTOS (BLACK); GARLOC STYLE 8316-1/8" EPDM; OR PRESRAY CORPORATION EPDM ELASTOMER, ONE PIECE GASKET 48" I.D. X 55" O.D. X 1/8" THICKNESS, SHORE DUROMETER READING LESS THAN 75A, TOTAL LEACHABLE CHLORIDE LESS THAN 200 PPM.
14. ORIGINAL LIFT LUG CUT OFF APPROX. 1/2" ABOVE VESSEL.

| JOB # (PIPE SUB-ASSY.) | TAG # (VALVE) | JOB # (VESSEL) | TAG # (VESSEL) | ORDER # |
|---|---|---|---|---|
| N2431.70<br>N2431.80 | 1HV-9002B<br>1HV-9002A | N2431.30<br>N2431.40 | V-1-1206-V4-001<br>V-1-1206-V4-002 | PAV-470<br>PAV-470<br>SEE B-81-23 FOR SHIPPING SKETCH.<br>SEE D-80-94, A-81-26, 27. |

| PROCEDURES |
|---|
| ES-141 |
| ES-106N |
| ES-123-N2431 |
| ES-134 |
| ES-140 |
| ES-118NC |
| ES-107NC |
| PB/MMA |
| PI/PB/MMA |
| P8/ASA |
| PT/MMA-1 |
| PB/TIG/GBU |
| PI/MIG/SA-1 |
| ES-100 |

| REV. NO. | CHANGE NOTE NO. | DESCRIPTION | BY | DATE | APPROVAL ENG. | APPROVAL QA |
|---|---|---|---|---|---|---|
| 11 | | REVISE NOTE 12 | CAF | 1-23-80 | WNH | KJH |
| 10 | | CHG WELD PREP @ 9:00 HR ADD PIPE SUB-N2431 SEE 9 | CAF | 5-29-79 | WNH | KJH |
| 9 | | ADD NOTE 9, NAME & DATE STAMP ADD PATCH 19.30 FT² (APPROX) | CAF | 2-2-79 | WNH | KJH |
| 8 | SCRN | REVISE SHELL LENGTHS | CAF | 1-22-79 | WNH | KJH |
| 7 | | ADD RT CHARTS, CHG NOZZ. ORIENT | CAF | 11-9-78 | WNH | KJH |
| 6 | | ADD CUST. REV. DATA | CAF | 7-28-78 | WNH | KJH |
| 5 | | ADD CHGS | CAF | 5-5-78 | WNH | KJH |
| 4 | | ADD 304 S | ESB | 4-24-78 | WNH | KJH |
| 3 | | REV. PER SKETCH | BSB | 3-17-78 | WNH | KJH |
| 2 | | REV PER CUST. 2-28-78 | CAF | 3-14-78 | WNH | KJH |
| 1 | | PER PRE CUSTOMER COMMENTS | BSB | 10-7-77 | WNH | KJH |

| | |
|---|---|
| NO. UNITS REQ'D. | TWO (2) |
| CUST. DWG. OR SPEC. | SK-4-3036 |
| CUSTOMER ORDER NO. | SEE ABOVE |
| CUST. TAG OR ITEM NO. | SEE ABOVE |

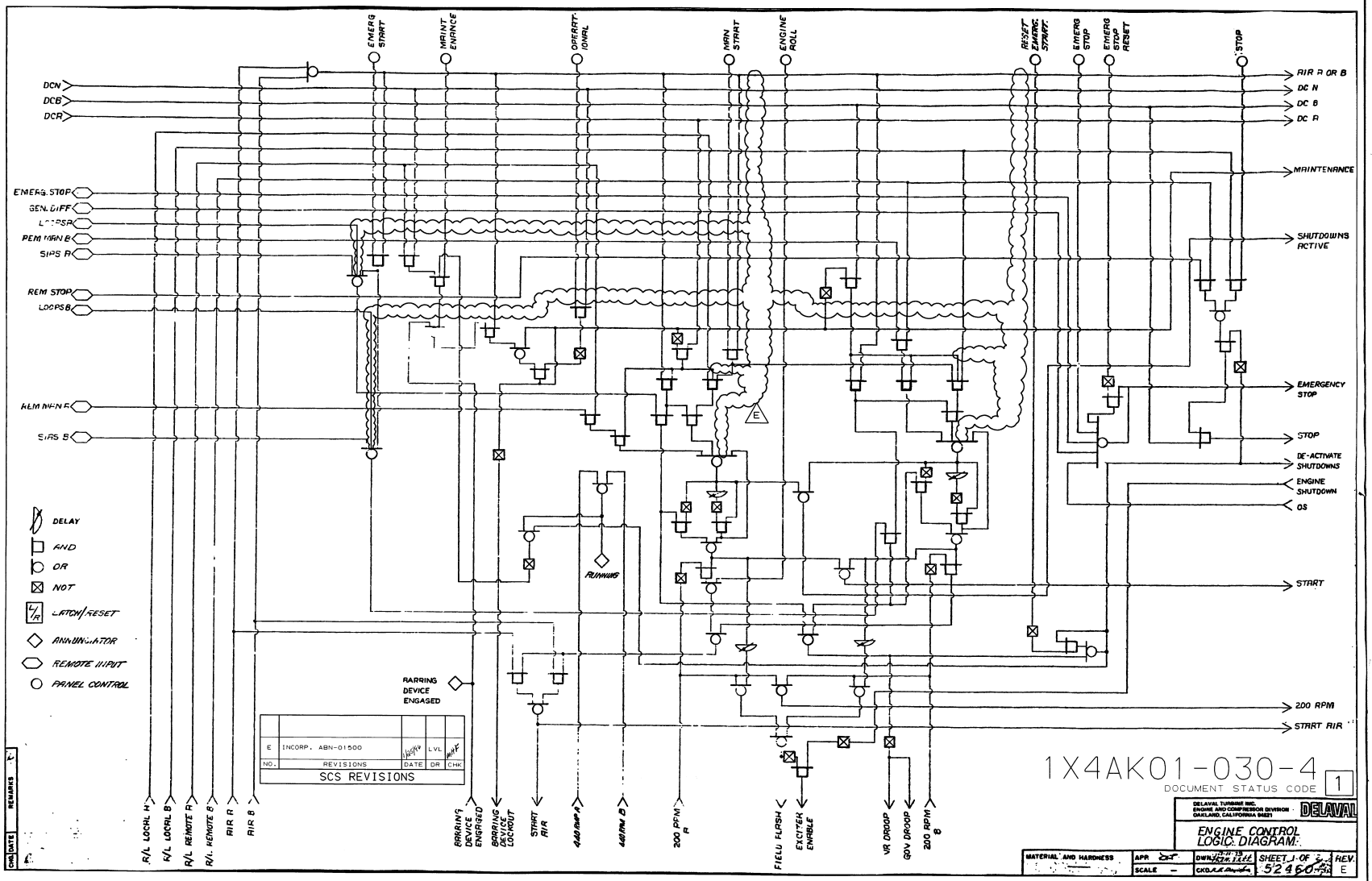
**RICHMOND ENGINEERING CO., INC.**
RICHMOND VIRGINIA

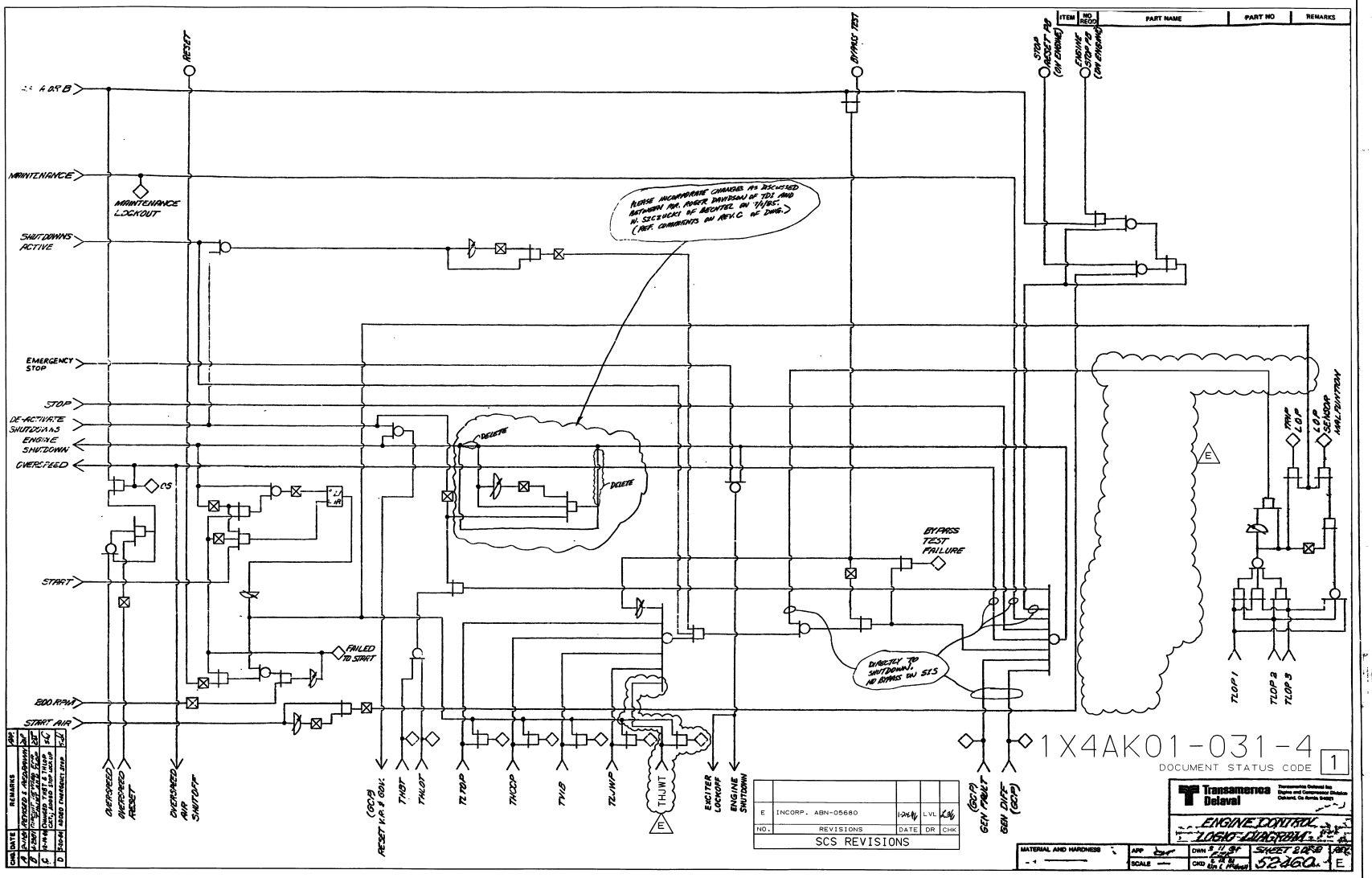
CONTAINMENT SPRAY ENCAPSULATION VESSEL
GEORGIA POWER CO., ALVIN W. VOGTLE NUCLEAR PLANT
BY: BECHTEL

| JOB NO. | DATE | DRAWN BY | DRAWING NO. | REV. |
|---|---|---|---|---|
| SEE ABOVE | 3/23/78 | BLB | D-78-128 | |
| | CHECKED BY | APPROVED | | |
| | BT | | | |

DOCUMENT STATUS CODE 1

| | | | | |
|---|---|---|---|---|
| THIS DWG. IS REFERENCED IN VENDOR MANUAL: N/A | CAD MICROSTATION JL0/01 | 1X4AH04-00025.DGN | | |
| TAB/SECT. N/A PAGE N/A FIGURE N/A | Southern Nuclear Operating Company, Inc. FOR VOGTLE ELECTRIC GENERATING PLANT UNIT NO. 1 | | | |
| VERSION 18.0 DATE 3-21-07 | TITLE: CONTAINMENT SPRAY ENCAPSULATION VESSEL | | | |
| REVISED BY SNC PER ABN C9990001010004, VER. 1.0 SEE MICROFILM FOR PREVIOUS VER. SIGNATURES. | | | | |
| BY: JL0 | CHK'D: RBH | APPR.1: JMR | APPR.2: X | VENDOR: RICHMOND ENG. P.O.#: |
| SIZE E | DRAWING NO. 1X4AH04-00025 | | | |


EMERG START
MAINT ENANCE
OPERAT- IONAL
MAN START
ENGINE ROLL
RESET ELECTRIC START
EMERG STOP
EMERG STOP RESET
STOP
DCN
DCB
DCA
EMERG STOP
GEN. DIFF
LOOPSA
PEM MAN B
SIPS A
REM STOP
LOOPS B
REM MAN A
SIPS B
AIR A OR B
DC N
DC B
DC A
MAINTENANCE
SHUTDOWNS ACTIVE
EMERGENCY STOP
STOP
DE-ACTIVATE SHUTDOWNS
ENGINE SHUTDOWN
OS
START
200 RPM
START AIR
DELAY
AND
OR
NOT
LATCH/RESET
ANNUNCIATOR
REMOTE INPUT
PANEL CONTROL
RUNNING
BARRING DEVICE ENGAGED
R/L LOCAL A
R/L LOCAL B
R/L REMOTE A
R/L REMOTE B
AIR A
AIR B
BARRING DEVICE ENGAGED
BARRING DEVICE LOCKOUT
START AIR
440 RPM A
440 RPM B
200 RPM A
FIELD FLASH
EXCITER ENABLE
VR DROOP
GOV DROOP
200 RPM B
E INCORP. ABN-01500
NO. REVISIONS DATE DR CHK
SCS REVISIONS
1X4AK01-030-4
DOCUMENT STATUS CODE 1
DELAVAL TURBINE INC. ENGINE AND COMPRESSOR DIVISION OAKLAND, CALIFORNIA 94621
DELAVAL
ENGINE CONTROL LOGIC DIAGRAM:
MATERIAL AND HARDNESS
APR
SCALE –
DWN
CKD
SHEET 1 OF 2
REV. E


RESET
BYPASS TEST
STOP (ON ENGINE)
RESET PB (ON ENGINE)
ENGINE STOP PB (ON ENGINE)
2 of 4 OR B
MAINTENANCE
MAINTENANCE LOCKOUT
SHUTDOWNS ACTIVE
EMERGENCY STOP
STOP
DE-ACTIVATE SHUTDOWNS
ENGINE SHUTDOWN
OVERSPEED
CS
START
800 RPM
START AIR
DELETE
DELETE
PLEASE INCORPORATE CHANGES AS DISCUSSED BETWEEN MR. ROGER DAVIDSON OF TDI AND M. SZCZUCKI OF BOOSTEC ON 7/7/95. (REF. COMMENTS ON REV. C OF DWG.)
BYPASS TEST FAILURE
DIRECTLY TO SHUTDOWN, NO BYPASS ON SIS
FAILED TO START
TLOP 1
TLOP 2
TLOP 3
OVERSPEED LATCHED RESET
OVERSPEED AIR SHUTOFF
(GCP) RESET V.A.R.'S 60%
THST
THLOP
TE TDP
THCDP
TVLS
TLVHP
THJWT
EXCITER LOCKOFF
ENGINE SHUTDOWN
(GCP) GEN PMNT
GEN DIFF (GCP)
ITEM
NO RECD
PART NAME
PART NO
REMARKS
1X4AK01-031-4
DOCUMENT STATUS CODE 1
Transamerica Delaval
ENGINE CONTROL
LOGIC DIAGRAM
SHEET 8 OF 8
52460
E
INCORP. ABN-05680
REVISIONS
DATE
DR
CHK
SCS REVISIONS
MATERIAL AND HARDNESS
APP
SCALE
DWN
CHD

## LEGEND AND SYMBOLS FOR P & ID'S

| DRAWING NO | SYSTEM NO | B/S | DESCRIPTION |
|---|---|---|---|
| 1X4DB100 | | B | P & ID AND FLOW DIAGRAM LEGEND |
| 1X4DB101 | | B | INSTRUMENTATION IDENTIFICATION & SYMBOLS |
| 1X4DB102 | | B | INSTRUMENTATION IDENTIFICATION & SYMBOLS |

## P & ID'S–NSSS SYSTEMS

| | | | |
|---|---|---|---|
| AX4DB104–1 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (WASTE FD TK) |
| AX4DB104–2 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (DECANT & CMT DRUM.) |
| AX4DB104–3 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (CAUST & EVP.C.F.TK) |
| AX4DB104–4 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (WASTE FD & SCBR) |
| AX4DB104–5 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (OFFGAS TREATMT.) |
| AX4DB104–6 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (FLUID BED DRYER) |
| AX4DB104–7 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (DRYWASTE PROC.) |
| AX4DB104–8 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (TRASH HANDLING) |
| AX4DB104–9 | 1418 | B | PLANT DEMIN. WATER SYS (DECONTAMINATION WATER) |
| AX4DB105–1 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (SOB SPT RES TRNFR) |
| AX4DB105–2 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (WPSL SPT RES TRNFR) |
| AX4DB105–3 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS(POLYMER DRUMMING STATION) |
| AX4DB105–4 | 1215 | B | AUXILIARY BLDG. & MISC. DRAINS (TRANSFER BLDG) |
| AX4DB105–5 | 1903 | B | RDWST SOLIDIF. & VOL. REDUCT. SYS (BKFSH FLTR CRUD TRFR) |
| AX4DB105–6 | 1903 | B | RDWST SOLIDIF. & VOL. REDUCT. SYS (CMT/PLMR STO & FIL SBSYS) |
| AX4DB105–7 | 1225 | B | SOLIDIFICATION BUILDING DRAINS |
| AX4DB105–9 | 2401 | B | INSTRUMENT & SERVICE AIR SYSTEMS. |
| AX4DB106–1 | 1225 | B | SOLIDIFICATION BUILDING DRAINS |
| AX4DB106–2 | 1225 | B | SOLIDIFICATION BUILDING DRAINS |
| AX4DB106–3 | 1225 | B | SOLIDIFICATION BUILDING DRAINS |
| AX4DB106–4 | 1422 | B | RADWASTE SOLIDIF.BLDG COOLING WATER SYS |
| AX4DB106–5 | 1609 | | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (SELECTIVE CUBICLE MONITORING SYSTEM) |
| AX4DB106–6 | 1423 | B | RADWASTE PUMP SEAL WATER SYSTEM |
| AX4DB106–7 | 1903 | | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (DRY ASH STORAGE AND TRANSFER SYSTEM) |
| AX4DB106–8 | 1903 | B | RADWASTE SOLIDIF. & VOL. REDUCT. SYS (SAMPLE/REAGENT ADDITION) |
| 1X4DB110 | 2702 | B | POST ACCIDENT SAMPLING SYSTEM |
| 1X4DB111 | 1201 | B | REACTOR COOLANT SYSTEM |
| 1X4DB112 | 1201 | B | REACTOR COOLANT SYSTEM |
| 1X4DB113 | 1201 | B | RTD BYPASS REACTOR COOLANT SYSTEM |
| 1X4DB114 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB115 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB116–1 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB116–2 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB117 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB118 | 1208 | B | CHEMICAL & VOLUME CONTROL SYSTEM |
| 1X4DB119 | 1204 | B | SAFETY INJECTION SYSTEM |
| 1X4DB120 | 1204 | B | SAFETY INJECTION SYSTEM |
| 1X4DB121 | 1204 | B | SAFETY INJECTION SYSTEM |
| 1X4DB122 | 1205 | B | RESIDUAL HEAT REMOVAL SYSTEM |
| AX4DB123–1 | 1210 | B | BORON RECYCLE SYSTEM |
| AX4DB123–2 | 1210 | B | BORON RECYCLE SYSTEM |
| 1X4DB124 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| AX4DB124–1 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| AX4DB124–2 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| AX4DB124–3 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| AX4DB124–4 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| AX4DB124–5 | 1901 | S | WASTE PROCESSING SYSTEM–LIQUID |
| 1X4DB125 | 1901 | B | WASTE PROCESSING SYSTEM–LIQUID |
| 1X4DB126 | 1901 | B | WASTE PROCESSING SYSTEM–LIQUID |
| 1X4DB127 | 1901 | B | WASTE PROCESSING SYSTEM–LIQUID |
| 1X4DB128 | 1902 | B | WASTE PROCESSING SYSTEM–GAS |
| 1X4DB129 | 1902 | B | WASTE PROCESSING SYSTEM–GAS |
| 1X4DB130 | 1213 | B | SPENT FUEL COOLING & PURIFICATION SYSTEM |
| 1X4DB131 | 1206 | B | CONTAINMENT SPRAY SYSTEM |
| 1X4DB132 | 1222 | B | MISCELLANEOUS LEAK DETECTION SYSTEM |
| 1X4DB133–1 | 1202 | B | NUCLEAR SERVICE CLG WTR SYS |
| 1X4DB133–2 | 1202 | B | NUCLEAR SERVICE CLG WTR SYS |
| 1X4DB134 | 1202 | B | NUCLEAR SERVICE CLG WTR SYS |
| 1X4DB135–1 | 1202 | B | NUCLEAR SERVICE CLG WTR SYS |
| 1X4DB135–2 | 1202 | B | NUCLEAR SERVICE CLG WTR SYS |
| 1X4DB136 | 1203 | B | COMPONENT COOLING WATER SYS |
| 1X4DB137 | 1203 | B | COMPONENT COOLING WATER SYS |
| 1X4DB138–1 | 1217 | B | AUX COMPONENT COOLING WATER SYS |
| 1X4DB138–2 | 1217 | B | AUX COMPONENT COOLING WATER SYS |
| 1X4DB139 | 1217 | B | AUX COMPONENT COOLING WATER SYS |
| 1X4DB140 | 1212 | B | NUCLEAR SAMPLING SYS–LIQUID |
| 1X4DB141 | 1211 | B | NUCLEAR SAMPLING SYS–GASEOUS |
| 1X4DB142–1 | 1225 | B | CONTROL BLDG DRAINS SYS |
| 1X4DB142–2 | 1225 | B | CONTROL BLDG DRAINS SYS |
| 1X4DB143 | 1214 | B | CONTAINMENT & AUX BLDG DRNS–RADIOACTIVE |
| 1X4DB144–1 | 1214 | B | CONTAINMENT & AUX BLDG DRNS–RADIOACTIVE |
| 1X4DB144–2 | 1214 | B | CONTAINMENT & AUX BLDG DRNS–RADIOACTIVE |
| 1X4DB145–1 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–2 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–3 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–4 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–5 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–6 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB145–7 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB146–1 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB146–2 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB146–3 | 1215 | B | AUX BLDG & MISC DRAINS SYSTEM |
| 1X4DB147–1 | 1218 | B | AUX BLDG FLOOD RETAINING RMS ALARMS & DRAIN SYS |
| 1X4DB147–2 | 1218 | B | AUX BLDG FLOOD RETAINING RMS ALARMS & DRAIN SYS |
| 1X4DB148 | 1224 | B | BACKFLUSHABLE FILTERS SYS |

## FLOW DIAGRAMS–NSSS SYSTEMS

| | | | |
|---|---|---|---|
| 1X4DB149–1 | 1202,1203 1217 | B | COOLING WATER SYS FLOW DIAGRAM |
| 1X4DB149–2 | 1202,1203 1217 | B | COOLING WATER SYS FLOW DIAGRAM |
| 1X4DB149–3 | 1202,1203 1217 | B | COOLING WATER SYS FLOW DIAGRAM |
| 1X4DB149–4 | 1202,1203 1217 | B | COOLING WATER SYS FLOW DIAGRAM |

## P & ID'S–BULK GAS SYSTEMS

| | | | |
|---|---|---|---|
| AX4DB010 | 263C | B | BULK GASES SUPPLY SYSTEM (FUEL GAS) |
| AX4DB011 | 263C | B | BULK GASES SUPPLY SYSTEM (FUEL GAS) |
| AX4DB012 | 263C | B | BULK GASES SUPPLY SYSTEM (FUEL GAS) |
| AX4DB013 | 263C | B | BULK GASES SUPPLY SYSTEM (FUEL GAS) |
| AX4DB015 | 263C | S | BULK GASES SUPPLY SYSTEM (FUEL GAS) |
| AX4DB020 | 2632 | B | BULK GASES SUPPLY SYSTEM (ARGON) |
| AX4DB021 | 2632 | B | BULK GASES SUPPLY SYSTEM (ARGON) |
| AX4DB022 | 2632 | B | BULK GASES SUPPLY SYSTEM (ARGON) |
| AX4DB023 | 2632 | B | BULK GASES SUPPLY SYSTEM (ARGON) |
| AX4DB025 | 2632 | S | BULK GASES SUPPLY SYSTEM (ARGON) |
| AX4DB030 | 2634 | B | BULK GASES SUPPLY SYSTEM (OXYGEN) |
| AX4DB031 | 2634 | B | BULK GASES SUPPLY SYSTEM (OXYGEN) |
| AX4DB032 | 2634 | B | BULK GASES SUPPLY SYSTEM (OXYGEN) |
| AX4DB033 | 2634 | B | BULK GASES SUPPLY SYSTEM (OXYGEN) |
| AX4DB035 | 2634 | S | BULK GASES SUPPLY SYSTEM (OXYGEN) |
| AX4DB040 | 2636 | B | BULK GASES SUPPLY SYSTEM (COMPRESSED AIR) |
| AX4DB041 | 2636 | B | BULK GASES SUPPLY SYSTEM (COMPRESSED AIR) |
| AX4DB042 | 2636 | B | BULK GASES SUPPLY SYSTEM (COMPRESSED AIR) |
| AX4DB043 | 2636 | B | BULK GASES SUPPLY SYSTEM (COMPRESSED AIR) |
| AX4DB045 | 2636 | S | BULK GASES SUPPLY SYSTEM (COMPRESSED AIR) |

## TURBINE PLANT HEAT BALANCE DIAGRAMS

| | | | |
|---|---|---|---|
| 1X4DC101 | | S | TURBINE VWO MAXIMUM CALC.– NOT SUARANTEE |
| 1X4DC103 | | S | REACTOR WARRANTY |
| 1X4DC104 | 22.0 | S | 75% LOAD CASE |
| 1X4DC105 | | S | 50% LOAD CASE |
| 1X4DC106 | | S | 25% LOAD CASE |
| 1X4DC107 | | S | 15% LOAD CASE |
| 1X4DC109 | | S | 98.3% UFM OUT OF SERVICE |

## P & ID'S–BALANCE OF PLANT SYSTEMS

| DRAWING NO | SYSTEM NO | B/S | DESCRIPTION |
|---|---|---|---|
| 1X4DB050 | VARIOUS | S | CHEM. CLNG. 2 FDWTR. MAIN STEAM AUX. |
| 1X4DB051 | VARIOUS | S | CHEM. PHASE 4 CONDENSATE PPG. |
| 1X4DB052 | VARIOUS | S | CHEM. CLNG. 2 FDWTR. MAIN STEAM AUX. |
| 1X4DB053 | VARIOUS | S | CLNG. PHASE 3 HTR. SHELL EXTR. DRAIN SYS. |
| 1X4DB054 | VARIOUS | S | CLNG. PHASE 3 HTR. SHELL EXTR. DRAIN SYS. |
| 1X4DB055 | VARIOUS | S | CHEM. CLNG. PHASE 1&2 COMMON. |
| 1X4DB150 | 1401 | S | CIRCULATING WATER SYSTEM |
| 1X4DB150–1 | 1401 | S | CIRCULATING WATER SYSTEM |
| 1X4DB150–2 | 2424 | S | CONDENSER WATERBOX CATHODIC PROTECTION SYS. |
| 1X4DB151–1 | 1405 | S | TURBINE PLANT CLG WATER SYS |
| 1X4DB151–2 | 1405 | B | TURBINE PLANT CLG WATER SYS |
| 1X4DB151–3 | 1405 | S | TURBINE PLANT CLG WATER SYS |
| AX4DB152–1 | 1402 | S | RIVER INTAKE STRUCTURE SYS |
| AX4DB152–2 | 1420 | S | WASTE WATER EFFLUENT SYSTEM |
| AX4DB152–3 | 1420 | S | WASTE WATER EFFLUENT SYSTEM |
| AX4DB152–4 | 1402 | S | RIVER WATER INTAKE STRUCTURE SYSTEM–CHLORINE |
| AX4DB152–5 | 1420 | S | WASTE WATER EFFLUENT DECHLORINATION |
| AX4DB152–6 | 1420 | S | WASTE WATER EFFLUENT START UP PONDS |
| AX4DB152–7 | 1420 | S | WASTE WATER EFFLUENT SYS MISC.SUMPS |
| CX4DB153 | 1408,2417 | S | PLANT MAKEUP WTR WELL & POTABLE WATER SYS |
| 1X4DB154–1 | 1404 | S | TURBINE PLANT CLOSED CCOLING WATER SYSTEM |
| 1X4DB154–2 | 1404 | S | TURBINE PLANT CLOSED CCOLING WATER SYSTEM |
| 1X4DB155 | 2623,2419 | S | CONSTRUCTION & UTILITY WATER SYSTEM |
| 1X4DB156 | 1310 | S | FEEDWATER HEATER VENT SYSTEM |
| 1X4DB157 | 1411 | S | CONDENSATE CHEMICAL INJECTION SYS |
| AX4DB157–1 | 1411 | S | CONDENSATE CHEMICAL INJECTION SYS |
| AX4DB158–1 | 1410 | S | CIRC WATER CHEMICAL INJECTION SYS |
| AX4DB158–2 | 1410 | S | CIRC WATER CHEM INJECTION SYS (ACID & DISPERSANT) |
| AX4DB158–3 | 1410 | S | CIRC WATER CHEMICAL INJECTION SYS CHLORINE |
| 1X4DB159–1 | 1301 | B | MAIN STEAM SYSTEM |
| 1X4DB159–2 | 1301 | B | MAIN STEAM SYSTEM |
| 1X4DB159–3 | 1301 | B | MAIN STEAM SYSTEM |
| 1X4DB160–1 | 1301 | S | MAIN STEAM SYSTEM |
| 1X4DB160–2 | 1301 | S | MAIN STEAM SYSTEM |
| 1X4DB160–3 | 1301 | S | TURBINE STEAM SEAL SYS |
| 1X4DB161–1 | 1302 | B | AUXILIARY FEEDWATER SYSTEM– CONDENSATE STORAGE & DEGASIFIER SYSTEM |
| 1X4DB161–2 | 1302 | B | AUXILIARY FEEDWATER SYSTEM |
| 1X4DB161–3 | 1302 | B | AUXILIARY FEEDWATER SYSTEM– AUX. FEEDWATER PUMP TURBINE DRIVER |
| 1X4DB162–1 | 1303 | S | EXTRACTION STEAM SYSTEM |
| 1X4DB162–2 | 1303 | S | EXTRACTION STEAM SYSTEM |
| 1X4DB162–3 | 1303 | S | CONDENSER MANIFOLDS FOR EXTRACTION & MISC. DRAINS |
| 1X4DB163–1 | 1304 | S | FEEDWATER HEATER DRAIN SYS MSR.NO.6A HTR DRAINS |
| 1X4DB163–2 | 1304 | S | FEEDWATER HEATER DRAIN SYS MSR.NO.6B HTR DRAINS |
| 1X4DB163–3 | 1304 | S | FEEDWATER HEATER DRAIN SYS NO. 4A, NO. 5A, HTR DRN PUMP A DRNS |
| 1X4DB163–4 | 1304 | S | FEEDWATER HEATER DRAIN SYS NO. 4B, NO. 5B, HTR DRN PUMP B DRNS |
| 1X4DB163–5 | 1304 | S | FEEDWATER HEATER DRAIN SYS NO. 3A, 2A, 1A, HTR DRNS |
| 1X4DB163–6 | 1304 | S | FEEDWATER HEATER DRAIN SYS NO. 3B, 2B, 1B, HTR DRNS |
| 1X4DB163–7 | 1304 | S | FEEDWATER HEATER DRAIN SYS NO. 3C, 2C, 1C, HTR DRNS |
| AX4DB164–1 | 2419 | S | UTILITY WATER SYSTEM |
| 1X4DB164–2 | 2419 | B | UTILITY WATER SYSTEM |
| 1X4DB165 | 1413 | S | NSCW CHEMICAL INJECTION SYS–CHLORINE |
| 1X4DB166 | 1306 | S | TURBINE DRIVE STEAM SYSTEM |
| 1X4DB167–1 | 1307 | S | TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM |
| 1X4DB167–2 | 1307 | S | TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM |
| 1X4DB167–3 | 1307 | S | TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM |
| 1X4DB167–4 | 1307 | S | TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM |
| 1X4DB167–5 | 1307 | S | TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM |
| 1X4DB168–1 | 1305 | S | CONDENSATE & FEEDWATER SYS |
| 1X4DB168–2 | 1305 | S | CONDENSATE & FEEDWATER SYS |
| 1X4DB168–3 | 1305 | B | CONDENSATE & FEEDWATER SYS |
| 1X4DB170–1 | 2403 | B | DIESEL GEN SYSTEM (TRAIN–A) |
| 1X4DB170–2 | 2403 | B | DIESEL GEN SYSTEM (TRAIN–B) |
| 1X4DB171–1 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–2 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–3 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–4 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–5 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–6 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–7 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–8 | 1311 | B | TURBINE PLANT SAMPLING SYS |
| 1X4DB171–9 | 1311 | S | TURBINE PLANT SAMPLING SYS |
| AX4DB172 | 1311 | S | AUXILIARY BOILER SAMPLING SYSTEM |
| 1X4DB173 | 2301 | S | FIRE PROTECTION PIPING– TURBINE BLDG UNIT NO. 1 |
| CX4DB173–1 | 2301 | S | FIRE PROTECTION PUMP HOUSE 1 & 2 |
| CX4DB173–2 | 2301 | S | FIRE PROTECTION–YARD PPG SYS |
| CX4DB173–3 | 2301 | S | FIRE PROTECTION–TURBINE BLDG UNIT 1 SYS |
| CX4DB173–4 | 2301 | S | FIRE PROTECTION–TURBINE BLDG UNIT 2 SYS |
| CX4DB173–5 | 2301 | S | FIRE PROT. PPG.–HV SWYD. VLV. HOUSE.–UNIT 1 |
| AX4DB173–10 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. UNIT 1 |
| AX4DB173–11 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. 1ST FL |
| AX4DB173–12 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. 2ND FL |
| AX4DB173–13 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. 3RD FL |
| AX4DB173–14 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. HOSE RISER |
| AX4DB173–15 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. HOSE STA DATA |
| AX4DB173–16 | 2301 | S | FIRE PROT. PPG. SERV. BLDG. HAL & DET SYS |
| AX4DB173–17 | 2301 | S | FIRE PROT. PPG. PRODUCTION WHSE SYS. |
| AX4DB173–18 | 2301 | S | FIRE PROT. PPG. PRODUCTION WHSE SYS. |
| AX4DB173–19 | 2301 | S | FIRE PROT. PPG. PRODUCTION WHSE SYS. |
| AX4DB173–20 | 2301 | S | FIRE PROT. PPG. PRODUCTION WHSE SYS. |
| AX4DB173–21 | 2301 | S | FIRE PROT. PPG. MAINT. BLDG. |
| AX4DB173–22 | 2301 | S | FIRE PROT. PPG. MAINT. BLDG. |
| AX4DB173–23 | 2301 | S | FIRE PROT. PPG. MAINT. BLDG. |
| AX4DB173–24 | 2301 | S | FIRE PROT. PPG. MAINT. BLDG. |
| AX4DB173–25 | 2301 | S | MAINT. BLDG. PROD. WHSE. ADMIN BLDG. F.P. FIRE HOSE |
| AX4DB173–26 | 2301 | S | F.P. PPG. PROD. WHSE HOSE STANDPIPE. |
| AX4DB173–27 | 2301 | S | P & ID FIRE PROTECTION SUPPRESSION AND DETECTION SYSTEM COMPUTER CONTROL AND ALARM |
| 1X4DB173–500 | 2301 | S | FIRE PROT. TURB. BLDG. UNIT |
| 1X4DB173–501 | 2301 | S | FIRE PROT. TURB. BLDG. UNIT 1 |
| 1X4DB173–502 | 2301 | S | FIRE PROT. TURB. BLDG. UNIT 1 |
| 1X4DB173–503 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–504 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–505 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–507 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–508 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–510 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–511 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–512 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–513 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–514 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–515 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–517 | 2301 | S | FIRE PROT. PPG. TURB. BLDG. UNIT 1 |
| 1X4DB173–518 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–520 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–521 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–522 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–523 | 2301 | S | FIRE PROT. PPG. LOW VOLT. SWYD. UNIT 1 |
| 1X4DB173–524 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–525 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–526 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–527 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–528 | 2301 | S | FIRE PROTECTION PPG– LOW VOLTAGE SWITCHYARD UNIT 1 |
| 1X4DB173–531 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–532 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–533 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–534 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–535 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–536 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–537 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–538 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–539 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–540 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| 1X4DB173–541 | 2301 | S | FIRE PROTECTION PPG– TURBINE BLDG. UNIT 1 |
| CX4DB173–543 | 2301 | S | FIRE PROT. PPG– HV SWYD UNITS 1 & 2–VLV. HSE. NO. 1 |
| CX4DB173–544 | 2301 | S | FIRE PROT. PPG– HV SWYD UNITS 1 & 2–VLV. HSE. NO. 1 |
| CX4DB173–545 | 2301 | S | FIRE PROT. PPG– HV SWYD UNITS 1 & 2–VLV. HSE. NO. 1 |
| CX4DB173–546 | 2301 | S | FIRE PROT. PPG– HV SWYD UNITS 1 & 2–VLV. HSE. NO. 1 |
| CX4DB173–547 | 2301 | S | FIRE PROT. PPG– HV SWYD UNITS 1 & 2–VLV. HSE. NO. 1 |
| CX4DB173–548 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 2 |
| CX4DB173–549 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 2 |
| CX4DB173–550 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 2 |
| CX4DB173–551 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 3 |

## P & ID'S–BALANCE OF PLANT SYSTEMS (CONT.)

| DRAWING NO | SYSTEM NO | B/S | DESCRIPTION |
|---|---|---|---|
| CX4DB173–552 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 3 |
| CX4DB173–553 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 3 |
| CX4DB173–557 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 4 |
| CX4DB173–558 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 4 |
| CX4DB173–559 | 2301 | S | FIRE PROT. PPG– HIGH VOLT. SWITCHYARD UNITS 1 & 2 VALVEHOUSE NO. 4 |
| 1X4DB173–6 | 2301 | S | FIRE PROTECTION PIPING–LOW VOLTAGE SWITCHYARD VALVE HOUSE– UNIT NO. 1 |
| CX4DB173–6 | 2301 | S | FIRE PROTECTION HIGH VOLTAGE SWITCHYARD–UNITS 1 & 2 |
| 1X4DB173–7 | 2301 | S | FIRE PROTECTION SYSTEM |
| AX4DB173–8 | 2301 | S | FIRE PROTECTION–ADMINISTRATIVE BLDG. |
| AX4DB173–9 | 2301 | S | FIRE PROTECTION PIPING– WATER TREATMENT BUILDING |
| 1X4DB174–1 | 2302,2304 | B | FIRE PROTECTION HALON SYS |
| 1X4DB174–2 | 2301 | B | FIRE PROTECTION WATER SYSTEM |
| 1X4DB174–3 | 2301 | B | FIRE PROTECTION WATER SYSTEM |
| 1X4DB174–4 | 2301 | B | FIRE PROTECTION WATER SYSTEM |
| 1X4DB174–5 | 2301 | B | FIRE PROTECTION WATER SYSTEM |
| 1X4DB174–6 | 2303 | B | FIRE PROTECTION WATER SYSTEM |
| 1X4DB175–1 | 2401,2420 | S | INSTRUMENT & SERVICE AIR SYSTEM |
| 1X4DB175–2 | 2401,2420 | S | INSTRUMENT & SERVICE AIR SYSTEM |
| 1X4DB175–3 | 2401 | S | INSTRUMENT & SERVICE AIR SYSTEM |
| 1X4DB175–4 | 2401 | S | INSTRUMENT & SERVICE AIR SYSTEM |
| AX4DB175–5 | 2420 | S | INSTRUMENT AIR SYSTEM–OUTSIDE AREAS |
| 1X4DB175–5 | 2420 | S | INSTRUMENT AIR SYSTEM (TURBINE BLDG.) |
| 1X4DB175–6 | 2420 | S | INSTRUMENT AIR SYSTEM (TURBINE BLDG.) |
| 1X4DB175–7 | 2420 | S | INSTRUMENT AIR SYSTEM (TURBINE BLDG.) |
| AX4DB176–1 | 2402 | S | AUXILIARY GAS SYSTEM |
| AX4DB176–2 | 2406,2418 | S | AUXILIARY GAS SYSTEM |
| 1X4DB176–3 | 2402,2406 | B | AUXILIARY GAS SYSTEM |
| AX4DB177 | 1409 | S | PLANT MAKEUP WTR TREATMENT SYS |
| AX4DB177–1 | 2420 | S | INSTRUMENT AIR SYSTEM–WATER TREATMENT BLDG. |
| AX4DB178 | 1409 | S | PLANT MAKEUP WTR TREATMENT SYS |
| 1X4DB179–1 | 1407 | B | STEAM GENERATOR BLOWDOWN SYS |
| 1X4DB179–2 | 1407 | B | STEAM GENERATOR BLOWDOWN SYS |
| 1X4DB180–1 | 2412 | S | TURBINE BLDG DRAIN SYSTEM |
| 1X4DB180–2 | 2412 | B | TURBINE BLDG DRAIN SYSTEM |
| 1X4DB180–3 | 2412 | S | TURBINE BLDG ROOF DRAINS SUSPENDED DRAINS |
| 1X4DB180–4 | 2412 | S | TURBINE BLDG DRAIN SYSTEM |
| AX4DB181 | 1314 | B | WASTE EVAPORATOR STEAM SUPPLY SYS |
| AX4DB181–1 | 1314 | B | WASTE EVAPORATOR STEAM SUPPLY SYS |
| 1X4DB182 | 1309 | S | CONDENSER AIR EJECTION SYSTEM |
| 1X4DB183 | 1226,1227 | B | FUEL HANDLING BLDG & ELECT CHASE TUNNEL DRNS |
| 1X4DB184 | 1228 | B | REACTOR MAKE–UP WATER STORAGE TANK & DEGASIFIER. |
| 1X4DB185–1 | 1414 | S | CONDENSATE FILTER DEMINERALIZER SYS |
| 1X4DB185–2 | 1414 | S | CONDENSATE FILTER DEMINERALIZER SYS |
| 1X4DB185–3 | 1414 | S | CONDENSATE FILTER DEMINERALIZER SYS |
| 1X4DB185–5 | 1414 | B | CONDENSATE FILTER DEMIN. BACKWASH SEPARATION SYS |
| 1X4DB185–6 | 2420 | B | INSTRUMENT AIR SYSTEM CONTROL BUILDING |
| 1X4DB186–1 | 2401 | B | SERVICE AIR SYSTEM |
| 1X4DB186–2 | 2420 | B | INSTRUMENT AIR SYSTEM |
| AX4DB186–3 | 2401 | B | SERVICE AIR SYSTEM |
| 1X4DB186–4 | 2420 | B | INST. AIR SYS. CONTAIN. BLDG. |
| 1X4DB186–5 | 2420 | B | INST. AIR AUX BLDG. |
| 1X4DB186–6 | 2420 | B | INST. AIR AUX. BLDG. |
| 1X4DB186–7 | 2420 | B | INST. AIR AUX. BLDG. |
| 1X4DB186–8 | 2420 | B | INST. AIR CONT. BLDG. |
| 1X4DB186–9 | 2420 | B | PPG. & INSTR. DIAG. INSTR. AIR SYS. |
| 1X4DB187 | 1419 | S | CONDENSER TUBE CLEANING SYS |
| 1X4DB188 | 2401 | S | SERVICE AIR SYSTEM (TURBINE BLDG.) |
| AX4DB189 | 1417 | S | PLANT MAKEUP WATER TREATMENT WASTE NEUTRALIZATION SYSTEM |
| AX4DB190–1 | 1418 | S | PLANT DEMINERALIZED WATER SYSTEM |
| AX4DB190–2 | 1418 | B | PLANT DEMINERALIZED WATER SYSTEM |
| 1X4DB191 | 1324 | B | TURBINE GENERATOR HYDROGEN SEAL OIL SYS |
| 1X4DB192 | 1413 | B | NUCLEAR SERVICE COOLING WATER CHEMICAL INJECTION SYS & CHLORINE ANALYSIS SYS |
| 1X4DB193 | 1326 | S | TURBINE GENERATOR STATOR COOLING SYS |
| 1X4DB194 | 1615 | S | TURBINE GENERATOR ELECTROHYD.CONTROL SYS |
| AX4DB195–1 | 1322 | S | AUXILIARY STEAM SYSTEM BOILER |
| AX4DB195–2 | 1322 | S | AUXILIARY STEAM SYSTEM DEAERATOR CHEM. FEED |
| AX4DB195–3 | 1322 | S | AUXILIARY STEAM SYSTEM STEAM DISTRIBUTION – UNIT 1 |
| AX4DB195–4 | 1322 | S | AUXILIARY STEAM SYSTEM STEAM DISTRIBUTION – UNIT 2 |
| AX4DB195–5 | 1322 | S | AUXILIARY STEAM SYSTEM FUEL OIL |
| 1X4DB196 | 1325 | S | TURBINE GENERATOR GAS SYSTEM |
| AX4DB196–1 | 1325 | S | TURBINE GENERATOR GAS SYS. |
| 1X4DB197 | 1316 | S | TURBINE D

## EQUIPMENT

- DOUBLE FLOW TURBINE
- SINGLE FLOW TURBINE
- TANK WITH DIAPHRAGM
- CONE ROOF TANK
- PRESSURE VESSEL (SHOWN ON HORIZONTAL TANKS, RECEIVERS, DRYERS, SEPARATORS, ETC.)
- PRESSURIZED GAS BOTTLE
- AFTERCOOLER
- MIXER
- VACUUM PUMP
- VERTICAL SUMP PUMP
- VERTICAL PUMP (WET PIT TYPE)
- VERTICAL PUMP (CANNED TYPE)
- POSITIVE DISPLACEMENT PUMP
- ROTARY PUMP
- CENTRIFUGAL PUMP
- HEAT EXCHANGER (SHELL AND U-TUBE TYPE)
- HEAT EXCHANGER (SHELL AND STRAIGHT TUBE TYPE)
- HEAT EXCHANGER (OTHER)
- SAMPLE COOLER

## VALVE SYMBOLS

- GATE NORMALLY OPEN
- GATE DOUBLE DISK
- STEM LEAKOFF
- GATE NORMALLY CLOSED
- LOCKED OPEN (LO)
- LOCKED CLOSED (LC)
- GLOBE NORMALLY OPEN
- GLOBE NORMALLY CLOSED
- NEEDLE
- STOP CHECK
- CHECK
- BUTTERFLY
- BALL NORMALLY OPEN
- BALL NORMALLY CLOSED
- PLUG
- DIAPHRAGM
- BACKFLOW PREVENTER
- ANGLE
- STOP CHECK ANGLE
- THREE WAY
- FOUR WAY
- VALVE EXTENSION STEM
- VALVE WITH INDICATOR POST
- AIR WRENCH OPERATED VALVE (AWO)
- TEST BOUNDARY (TB)
- PINCH VALVE NORMALLY OPEN
- PINCH VALVE NORMALLY CLOSED
- KNIFE GATE VALVE

NOTE: ALL PROCESS VALVES AND CONTROL SOLENOID VALVES ARE SHOWN ON P & I DIAGRAMS IN THEIR POSITION FOR NORMAL PLANT OPERATION.

## LINES

- MAIN PROCESS
- SECONDARY PROCESS
- PHANTOM SECONDARY REPETITION OF LINES AND OR EQUIPMENT
- STEAM TRACED LINE
- BY OTHER VENDORS
- ELECTRICAL HEAT TRACING
- ELECTRICAL REDUNDANT HEAT TRACING
- TUNNEL 1T1 NOTE 1 — PIPING LOCATED WITHIN TUNNEL
- BURIED LINE — BURIED PIPING

## FIRE PROTECTION SYMBOLS

- DELUGE TYPE DIRECTIONAL FOG SYSTEM
- PRE-ACTION SPRINKLER SYSTEM HYDRAULICALLY BALANCED
- STANDARD WET PIPE SPRINKLER SYSTEM OF ORDINARY HAZARD PIPE SCHEDULE
- FIRE HYDRANT
- FHS — FIRE HOSE STATION
- ALARM CHECK WITH WATER GONG
- R-A5 — ARCHITECTURAL ROOM NUMBER (TYP)
- 057 — PRE-ACTION SPRINKLER SYSTEM DELUGE VALVE NUMBER. (TYP)
- HOSE ADAPTOR WITH PRESSURE REDUCER

## MISCELLANEOUS

- AV — ACTIVE VALVE SEE NOTE 5
- AV-F — VALVES LISTED IN FSAR ACTIVE VALVE TABLES 3.9.B.3-9 AND 3.9.N.3-2
- 1 — CONTAINMENT PENETRATION NUMBER
- FURNISHED WITH ASSOCIATED EQUIPMENT
- W — FURNISHED BY WESTINGHOUSE
- NOT FURNISHED BY WESTINGHOUSE
- I — DESIGNATION FOR NOZZLE ON CONDENSER
- 1 — CONTAINMENT ISOLATION VALVE
- EA — MOTOR CONNECTS TO EMERGENCY POWER SOURCE "A" ON FAILURE OF NORMAL POWER
- EB — MOTOR CONNECTS TO EMERGENCY POWER SOURCE "B" ON FAILURE OF NORMAL POWER
- EJECTOR OR EDUCTOR
- WELDED CAP
- SCREWED CAP
- HOSE CONNECTION
- BLIND FLANGE
- FLEXIBLE CONNECTION
- EXHAUST HEAD
- ATMOSTPHERIC VENT
- EXPANSION JOINT
- LCV — FLOAT TYPE DRAIN TRAP
- XCV — ALL OTHER TYPE DRAIN TRAPS
- RT — RESIN TRAP
- PP — PRESSURE TEST POINT
- AP — SAMPLE POINT
- CONTAINMENT BUILDING PENETRATION
- A, B — AUXILIARY BUILDING BOUNDARY
- SPRAY NOZZLE
- LOOP SEAL
- QUICK DISCONNECT COUPLING
- REDUCER-INCREASER
- FILTER-WOUND
- F — BASKET STRAINER SIMPLEX
- F — BASKET STRAINER DUPLEX
- S — STRAINER
- Y — Y-STRAINER
- F — FILTER
- EQUIPMENT OR OPEN DRAIN
- FLOOR DRAIN
- LEAK DETECTING DRAIN
- BARRIER
- PIPE WITH THERMAL SLEEVE
- TEMPORARY STARTUP STRAINER IN REMOVABLE SPOOL
- REMOVABLE SPOOL
- WATER HAMMER ARRESTOR
- SPECTACLE FLANGE

## ABBREVIATIONS

ARB – ALTERNATE RADWASTE BUILDING
ATSI – ADVANCED TURBINE SUPERVISORY INSTRUMENTATION
BD – THERMAL REGENERATION SIGNAL
CS – CONTAINMENT SUMP
CWS – CLEAN WATER SUMP
CCDT – COMPONENT COOLING WATER DRAIN TANK
D – LOCAL DRAIN
DH – DRAIN HEADER
DW – DEMINERALIZED WATER
EHC – ELECTROHYDRAULIC CONTROL
ERFCS – EMERGENCY RESPONSE FACILITIES COMPUTER SYSTEM
FC – FAIL CLOSE
FO – FAIL OPEN
FDT – FLOOR DRAIN TANK
H – HYDROGEN FROM HYDROGEN MANIFOLD
ISS – INSIDE SECONDARY SHIELDING
IRC – INSIDE REACTOR CONTAINMENT
N – NITROGEN FROM NITROGEN MANIFOLD
OSS – OUTSIDE SECONDARY SHIELDING
ORC – OUTSIDE REACTOR CONTAINMENT
RCDT – REACTOR COOLANT DRAIN TANK
RCPS – REACTOR CONTROL AND PROTECTION SYSTEM
RCP – REACTOR COOLANT PUMP
RHT – RECYCLE HOLDUP TANK
RMS – RADIATION MONITOR SYSTEM
RMWST – REACTOR MAKE-UP WATER STORAGE TANK
RTD – RESISTANCE TEMPERATURE DETECTOR
RWST – REFUELING WATER STORAGE TANK
SC – SPEED CONTROL
WHT – WASTE HOLD-UP TANK
TC – TEST CONNECTION
TV – TEST VENT
PV – PLANT VENT
V – VENT

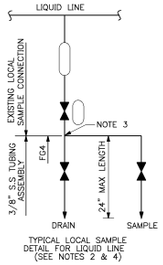
CI-A-A (A), CI-B-A (B) — CONTAINMENT ISOLATION (CI)– PHASE (A)–TRAIN (A) (B)
SI – A, SI – B — SAFETY INJECTION SIGNAL TRAIN (A) (B)
CVI – A, CVI – B — CONTAINMENT VENTILATION TRAIN ISOLATION (A) (B)
CRI – A, CRI – B — CONTROL ROOM ISOLATION TRAIN (A) (B)
SLI – A, SLI – B — STEAM LINE ISOLATION TRAIN (A) (B)
TT — TURBINE TRIP SIGNAL
FI – A, FI – B — FEEDWATER ISOLATION TRAIN A, TRAIN B
CS – A, CS – B — CONTAINMENT SPRAY TRAIN A, TRAIN B
FHBI – A, FHBI – B — FUEL HANDLING BUILDING ISOLATION SIGNAL TRAIN (A) (B)

## NUCLEAR STEAM SUPPLY SYSTEM ABBREVIATIONS

RCS – REACTOR COOLANT SYSTEM
CVCS – CHEMICAL & VOLUME CONTROL SYSTEM
SIS – SAFETY INJECTION SYSTEM
RHRS – RESIDUAL HEAT REMOVAL SYSTEM
BRS – BORON RECYCLE SYSTEM
WPSL – WASTE PROCESSING SYSTEM LIQUID
WPSG – WASTE PROCESSING SYSTEM GAS
SFPCS – SPENT FUEL COOLING & PURIFICATION SYSTEM
CSS – CONTAINMENT SPRAY SYSTEM
NSCWS – NUCLEAR SERVICE COOLING WATER SYSTEM
CCWS – COMPONENT COOLING WATER SYSTEM
ACCWS – AUX COMPONENT COOLING WATER SYSTEM
SS – SAMPLING SYSTEM
SGBS – STEAM GENERATOR BLOWDOWN SYSTEM
SGFP – STEAM GENERATOR FEED PUMP

## EQUIPMENT LETTER DESIGNATOR

A H.V. AND A.C. EQUIPMENT, AIR HANDLING (FANS, BLOWERS)
B STEAM BOILER
C COMPRESSOR
D DEMINERALIZER, RECOMBINER EVAPORATOR PKG.
E HEAT EXCHANGER, COOLER, CHILLER, DEMISTER, CONDENSER, STEAM GENERATOR.
F FILTER, STRAINER, SILENCER, FLAME ARRESTOR, EXHAUST MUFFLER, SCREENS, BREATHER, FLOW TEST POINT DEVICE.
G DIESEL ENGINE
H HANGERS, SUPPORTS
J EXHAUSTER, EDUCTOR, EJECTOR.
K STEAM TURBINE, MOISTURE SEPARATOR.
L LINES (PIPE, FITTINGS, FLANGES, PIPE SPOOLS)
M MOTORS
N FORGED FLUED HEAD.
P PUMPS (CENTRIFUGAL OR POSITIVE DISPLACEMENT)
Q AGITATOR, BLENDER, MIXER
R
S
T TANK (ATMOSPHERE)
U VALVES
V VESSEL (PRESSURE-INCLUDING REACTOR)
W COOLING TOWER
Y FLEXIBLE AND EXPANSION JOINTS
a METAL (CLASS 1)
b METAL (NON NUCLEAR)
c RUBBER
Z TRAP



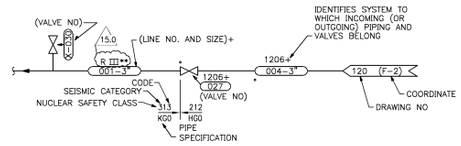
TYPICAL LOCAL SAMPLE DETAIL FOR LIQUID LINE (SEE NOTES 2 & 4)

## PLANT NUMBERING SYSTEM

ALL IDENTIFYING NUMBERS ON THE P & ID DRAWINGS ARE PRECEDED BY THE UNIT AND SYSTEM NUMBER, SUBJECT AND DISCIPLINE.

NOTE: SEE DWG. 1X4DB102 FOR DESCRIPTION OF NUMBERING SYSTEM FOR INSTRUMENTS AND VALVES NUMBERED BY THE CONTROLS GROUP.

| | PLANT | * UNIT | * SYSTEM | # SUBJECT | DISCIPLINE |
|---|---|---|---|---|---|
| LINE : | | 1,2 A OR C | 1206 | L . O | 4 , 2 |
| VALVE : | | 1,2 A OR C | 1206 | U/X . M | 4 , 2 |
| EQUIPMENT : | | 1,2 A OR C | 1206 | (SEE EQUIPMENT LETTER DESIGNATOR THIS DWG) | 4 (EXCEPT AS NOTED) |
| * AS APPLICABLE | | | | | |




X INDICATES EITHER ROOT, NONPROCESS DRAIN OR VENT VALVE, ALL ARE 1" EXCEPT: 1) AS NOTED, 2) FOR LINES LESS THAN 1" ARE LINE SIZE, 3) FOR FLOW MEASUREMENTS SEE CX5DP0002

* LINE NUMBER WILL CHANGE WHEN PIPE SPECIFICATION AND/OR PROJECT CLASSIFICATION CHANGES.

• SIZE OF VALVE IS SAME AS LINE SIZE, UNLESS OTHERWISE NOTED.

# SEE DC-1000-G FOR DESIGNATION.

** RADIATION ZONE DESIGNATOR

| | |
|---|---|
| NR | NO RADIATION |
| R I | RAD LEVEL < .25 MR/HR |
| R II | RAD LEVEL 0.25-2.5 MR/HR |
| R III | RAD LEVEL 2.5-15 MR/HR |
| R IV | RAD LEVEL 15-100 MR/HR |
| R V | RAD LEVEL > 100 MR/HR |

15.0 + VALVES AND LINE NUMBERS ARE LISTED WITH SYSTEM NUMBER SHOWN ON TITLE BLOCK OF THE DRAWING ON WHICH THEY APPEAR, UNLESS OTHERWISE NOTED.

ALL DRAWING NUMBERS IN COORDINATE ARROWS ARE PRECEDED BY "X4DB"

## HEATING, VENTILATION & AIR CONDITIONING SYMBOLS

- BACK DRAFT DAMPER
- SUPPLY AIR
- EXHAUST AIR
- RETURN AIR
- EXFILTRATION AIR
- CF — ACTIVATED CHARCOAL FILTER
- HEF — HIGH EFFICIENCY FILTER
- HEPA — HIGH EFFICIENCY PARTICULATE AIR FILTER
- PF — PREFILTER
- ME — MOISTURE ELIMINATOR
- C/C — COOLING COIL
- H C — ELECTRIC HEATING COIL
- M AD — AIR COOLED CONDENSING UNIT (LIQUID CHILLER)
- H C — ELECTRIC UNIT HEATER
- RELIEF DAMPER
- DAMPER
- MODULATING DAMPERS
- FIRE DAMPER
- MANUAL BALANCING DAMPER
- OPPOSED BLADE DAMPER
- PARALLEL BLADE DAMPER
- TORNADO PROOF DAMPER
- EXT WALL LOUVER
- PROPELLER FAN
- CENTRIFUGAL BLOWER
- CENTRIFUGAL BLOWER WITH INLET VANES
- VANE AXIAL FAN
- P — PNEUMATIC OPERATOR
- M — MOTOR OPERATOR
- MANUAL OPERATOR
- PL — PLENUM
- MB — MIXING BOX
- DUCT SILENCER
- SUPPLY REGISTER
- RETURN REGISTER

NOTE
1. TUNNEL IDENTIFICATION NUMBER

| | |
|---|---|
| 1T1 | MAIN STEAM AND FEEDWATER TUNNEL |
| 1T2A | NSCW TUNNEL TRAIN "A" |
| 1T2B | NSCW TUNNEL TRAIN "B" |
| 1T3A | NSCW TUNNEL TRAIN "A" |
| 1T3B | NSCW TUNNEL TRAIN "B" |
| 1T4A | ELECTRICAL TUNNEL TRAIN "A" |
| 1T4B | ELECTRICAL TUNNEL TRAIN "B" |
| 1T5A | ELECTRICAL TUNNEL TRAIN "A" |
| 1T5B | ELECTRICAL TUNNEL TRAIN "B" |
| 1T6A | AUX FEEDWATER TUNNEL TRAIN "A" |
| 1T6B | AUX FEEDWATER TUNNEL TRAIN "B" |
| 1T7 | MAIN STEAM TUNNEL |
| 1T8 | ELECTRICAL TUNNEL |

2. THE 3/8"ø STAINLESS STEEL TUBING ASSEMBLY MAY BE ADDED TO LOCAL LIQUID SAMPLE POINTS, WHEREVER REQUIRED IN THE PLANT. THE SAMPLE TUBING ASSEMBLY SHALL NOT BE CONNECTED TO SEISMIC CATEGORY I PIPING/COMPONENTS. THE TUBING ASSEMBLY LENGTH SHALL BE A MAXIMUM OF 24". TYGON HOSE MAY BE ROUTED FROM THE DRAIN CONNECTION TO THE NEAREST COMPATIBLE DRAIN. TUBING ASSEMBLY SHALL BE LESS THAN 10 POUNDS.
3. REMOVE CAP AND ADD TUBING ASSEMBLY. ALL CONNECTIONS OF THE TUBING ASSEMBLY ARE TO BE THREADED AND/OR SWAGELOCK TYPE FITTINGS.
4. CONSTRUCTION SPECS X4AZ01, DIV, P1, P2, P4 AND P5 SHALL BE FOLLOWED WHILE MAKING THE CHANGE.
5. REFER TO DC-1000-M.

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
INSTRUMENTATION IDENTIFICATION & SYMBOLS

CAD 1X4DB101
ACAD2002 NJ – 01

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| NO. | | DATE | DR | CHK | APPV. | DTL |
|---|---|---|---|---|---|---|
| 15.0 | REVISED PER ABN-52219, VER. 1.0 | 10-11-05 | NJ | LCF | LKB | X |

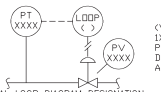
SIZE E 34X44

SCALE: NONE
DRAWING NO. 1X4DB101
VER. 15.0
JOB NO.10604

DRAWING CATEGORY: CRITICAL

| FIRST LETTER | SECOND AND SUCCEEDING LETTERS | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | DISPLAY DEVICES | | | | | | CONTROLLING DEVICES | | | | | | SENSING DEVICES | | | | | | |
| MEASURED VARIABLE | SYMBOL FOR MEASURED VARIABLES | INDICATING | RECORDING (SEE NOTE 17) | INTEGRATING INDICATOR (SEE NOTE 5) | SCAN (SEE NOTE 7) | COMPUTER ISOLATION CARD | ALARM | INDICATOR | RECORDING | BLIND | CONTROL VALVE | SELF ACTUATED VALVE | FINAL CONTROL ELEMENT (SEE NOTE 11) | SWITCH (SEE NOTE 6) | PRIMARY ELEMENT | BLIND TRANSMITTER | INDICATING TRANSMITTER | LOCAL OBSERVATION GLASS | TEST CONNECTION | RELAY OR CONVERTER (BLIND) |
| TYPICAL SYMBOL | ( ) | ( ) I | ( ) R | ( ) QI | ( ) J ( ) | (W SCOPE) | | ( ) IC | ( ) RC | ( ) G | ( ) V | ( ) CV | ( ) Z | ( ) S ( ) | ( ) E | ( ) T | ( ) IT | ( ) G | ( ) P | ( ) Y |
| ANALYSIS (SEE NOTE 1) | A | AI | AR | | AJ ( ) | | AA | AIC | ARC | AG | AV | | AZ | AS( ) | AE | AT | AIT | | AP | AY |
| BURNER FLAME | B | BI | BR | | BJ ( ) | | | | | BG | BV | | | BS( ) | BE | BT | | BG | BP | BY |
| CONDUCTIVITY (SEE NOTE 1) | C | CI | CR | | CJ ( ) | | | CIC | CRC | | CV | | CZ | CS( ) | CE | CT | CIT | | CP | CY |
| DENSITY | D | DI | DR | | DJ ( ) | | | DIC | DRC | | DV | | DZ | DS( ) | DE | DT | DIT | | DP | DY |
| VOLTAGE (EMF) | E | EI | ER | | EJ ( ) | | | EIC | ERC | EG | | | EZ | ES( ) | EE | ET | EIT | | | EY |
| FLOW (SEE NOTE 10) | F | FI | FR | FQI | FJ ( ) | FD | | FIC | FRC | FG | FV | FCV | FZ | FS( ) | FE | FT | FIT | FG | FP | FY |
| FLOW RATIO | FF | FFI | FFR | | FFJ ( ) | | | FFIC | FFRC | FFG | FFV | | FFZ | | | | | | | |
| GAGING (DIMENTIONAL) | G | GI | GR | | GJ ( ) | | | GIC | GRC | GG | GV | | GZ | GS( ) | GE | GT | GIT | | | |
| HAND | H | | | | | | | HIC | | HG | HV | HCV | HZ | HS( ) | | | | | | |
| CURRENT | I | II | IR | IQI | IJ ( ) | | | IIC | IRC | IG | IC | | IZ | IS( ) | IE | IT | IIT | | | IY |
| POWER | J | JI | JR | JQI | JJ ( ) | | | JIC | JRC | JG | JC | | JZ | JS( ) | JE | JT | JIT | | | JY |
| TIME | K | KI | KR | KQI | KJ ( ) | | | KIC | KRC | KG | KC | | KZ | KS( ) | | KT | KIT | | | KY |
| LEVEL | L | LI | LR | | LJ ( ) | LD | | LIC | LRC | LG | LC | LV | LCV | LZ | LS( ) | LE | LT | LIT | LG | LP | LY |
| MOISTURE | M | MI | MR | | MJ ( ) | | | MIC | MRC | MG | MC | MV | | MZ | MS( ) | ME | MT | MIT | | MP | MY |
| USER'S CHOICE (SEE NOTE 2) | N | | | | | | | | | | | | | | | | | | | |
| TORQUE | O | OI | OR | | OJ ( ) | | | OIC | ORC | OG | OC | OV | | OZ | OS( ) | OE | OT | | | | OY |
| PRESSURE | P | PI | PR | | PJ ( ) | PD | | PIC | PRC | PG | PC | PV | PCV | PZ | PS( ) | PE | PT | PIT | | PP | PY |
| PRESSURE DIFFERENTIAL | PD | PDI | PDR | | | | | PDIC | PDRC | PDG | PDC | PDV | PDCV | PDZ | PDS( ) | | PDT | PDIT | | | |
| QUANTITY OR EVENT (SEE NOTE 5) | Q | QI | QR | QQI | QJ ( ) | | | QIC | QRC | QG | QC | QV | | QZ | QS( ) | | QT | QIT | | | QY |
| RADIATION (SEE NOTE 11) | R | RI | RR | RQI | RJ ( ) | | RA | RIC | RRC | | RC | | | RZ | RS( ) | RE | RT | RIT | | RP | RY |
| SPEED OR FREQUENCY | S | SI | SR | SQI | SJ ( ) | | | SIC | SRC | SG | SC | | | SZ | SS( ) | | ST | SIT | | | SY |
| TEMPERATURE | T | TI | TR | | TJ ( ) | TD | | TIC | TRC | TG | TC | TV | TCV | TZ | TS( ) | TE | TT | TIT | | TW | TY |
| TEMPERATURE DIFFERENTIAL | TD | TDI | TDR | | | | | TDIC | TDRC | TDG | TDC | TDV | TDCV | TDZ | TDS( ) | (SEE NOTE 8) | | | | (SEE NOTE 8) | |
| MULTI-VARIABLE | U | UI | UR | | UJ ( ) | | | UIC | URC | UG | UC | UV | | UZ | US( ) | | | | | | UY |
| VISCOSITY | V | VI | VR | | VJ ( ) | | | VIC | VRC | VG | VC | VV | | VZ | VS( ) | VE | VT | VIT | | | VY |
| WEIGHT | W | WI | WR | | | | | | | | | | | | | | | | | | |
| UNCLASSIFIED (SEE NOTE 4) | X | XI | XR | | XJ ( ) | | | XIC | XRC | XG | XC | XV | | XZ | XS( ) | XE | XT | XIT | | | XY |
| USERS CHOICE (SEE NOTE 2) | Y | | | | | | | | | | | | | | | | | | | | YY |
| POSITION | Z | ZI | ZR | | ZJ ( ) | | | ZIC | ZRC | ZG | ZC | | | ZZ | ZS( ) | ZE | ZT | ZIT | | | ZY |

NOTES:
(1) THE LETTER "A" IS USED FOR ALL ALALYSIS VARIABLES EXCEPT CONDUCTIVITY. THE FOLLOWING TERMS MAY BE PLACED OUTSIDE THE INSTRUMENT CIRCLE OF A LOOP TO DENOTE THE SPECIFIC VARIABLES.
CC –CATION CONDUCTIVITY
CL –CHLORINE
CO –CARBON MONOXIDE
COMB –COMBUSTIBLE
H –DISSOLVED HYDROGEN
$H_2$ –GASEOUS HYDROGEN
NG –SODIUM
$N_2H_4$ –HYDRAZINE
NOX –NITROGEN OXIDES
O –DISSOLVED OXYGEN
$O_2$ –GASEOUS OXYGEN
PH –PH
SC –SPECIFIC CONDUCTIVITY
$Si\ O_2$ –SILICA
SMOKE –SMOKE DENSITY
SO –SULPHUR DIOXIDE
TRB –TURBIDITY
(2) A USERS CHOICE LETTER IS INTENDED TO COVER A MEANING THAT WOULD BE USED REPETITIVELY IN A PARTICULAR PROJECT. WHEN USED, THE LETTER MAY HAVE ONE MEANING AS A FIRST LETTER AND ANOTHER MEANING AS A SUCCEEDING LETTER. THE MEANINGS NEED BE DEFINED ONLY ONCE IN A LEGEND, OR OTHERWISE FOR THAT PROJECT. FOR EXAMPLE, THE LETTER "N" MAY BE DEFINED AS TURBIDITY AS A FIRST LETTER AND TELEVISION MONITOR AS A SECOND LETTER.
(3) THE EQUATION OF DESCRIPTION DENOTING THE FUNCTION OF THE RELAY "Y" SHOULD BE SHOWN ON THE P & ID, FOR EXAMPLE, A–B+C+K, LP SELECTOR, VOLUME BOOSTER.
(4) "X" IS USED TO REPRESENT ANY "SPECIAL" VARIABLES AND MAY BE DEFINED AS REQUIRED. FOR EXAMPLE, MASS FLOW RECORDERS WHICH RECEIVE A SIGNAL FROM A MULTIPLYING RELAY WHICH COMBINES THE PRODUCT OF DENSITY AND FLOW. THIS ITEM IS NOT TO BE CONFUSED WITH "U" MULTI–VARIABLE SYMBOL.
(5) WHEN "Q" IS USED AS A SECOND SUCCEEDING LETTER IT MAY DENOTE AN INTEGRATING MODIFIER FOR EXAMPLE, "FQI" IS AN INDICATING FLOW INTEGRATOR (OR TOTALIZER). NOTE THAT THE INTEGRATING FUNCTION SHALL BE SHOWN WITH SEPARATE IDENTIFICATION. FOR EXAMPLE, FQI/FRS OR FR/FQIS. "Q" MAY ALSO INDICATE ELECTRIC POWER SUPPLY, E.G. UQ.
(6) STARTUP AND SHUTDOWN DEVICES ARE USUALLY BLIND, BUT MAY BE INDICATING OR RECORDING. IF SO, ADD "I" OR "R" AFTER MEASURED VARIABLE. FOR EXAMPLE, FIS, TRS SWITCH FUNCTIONS MAY BE FURTHER MODIFIED BY "L" FOR LOW AND "H" FOR HIGH.
(7) THE DESIGNATION "AJ( )" MAY DENOTE A SCANNING ANALYZER INDICATOR, RECORDER, TRANSMITTER, ETC. BY USING THE DESIGNATION AJI, AJR, AJT, ETC., RESPECTIVELY.
(8) "TW" DENOTES AN EMPTY THERMOWELL, "TE" DENOTES A THERMOWELL WITH THERMOCOUPLE OR RTD AND HEAD SUITABLE FOR USE WITH A SECONDARY INSTRUMENT. "TI" DENOTES DIAL THERMOMETER EITHER BI–METALIC OR FILLED SYSTEM.
(9) PRESSURE RELIEF VALVES AND RUPTURE DISKS SHALL BE IDENTIFIED AS "PSV" AND "PSE" RESPECTIVELY.
(10) "FO" IS USED TO DESIGNATE A RESTRICTION ORIFICE. "FT" IS USED TO DESIGNATE A FILTER.
(11) "RA" IS RADIATION ALARM UNIT. "RN" IS RADIATION SAMPLING NOZZLES. "RX" IS RADIATION SYSTEM DATA PROCESSING MODULE.
(12) DESCRIPTOR "LOOP" REFERENCES LOOP DIAGRAM.



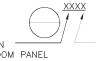
TYPICAL LOOP DIAGRAM DESIGNATION

(V ) REFERS TO LOOP DWG. 1X5DV IF (XXXX) IS NOT PREFIXED WITH "V", SEE LOOP DWG. INDEX 1X5DV001 FOR APPROPRIATE DWG. REFERENCE.

(13) LOCATIONS FOR NOTATIONS SURROUNDING INSTRUMENT SYMBOLS.

1. EXPLANATION OF INSTR. FUNCTION SEE MISCELLANEOUS, ZONE (D–3)
2. DESIGNATOR FOR THE PANEL ON WHICH INSTRUMENT IS LOCATED, E.G. QXXX REPRESENTS CONTROL ROOM PANEL (ALSO SEE NOTE 16).
3. SAFETY CHANNEL (1, 2, 3 OR 4), LOAD GROUP (A, B, C, OR D) OR POST ACCIDENT MONITOR (P, P1 OR F2).
4. SUPPLIER DESIGNATION, (SEE MISCELLANEOUS SECTION ON 1X4DB101).
5. UNIT DESIGNATOR PREFIXES INSTRUMENT DESCRIPTOR SHOWN IN BUBBLE ONLY WHEN IT IS DIFFERENT THAN THE UNIT DESCRIPTOR OF THE DRAWING NUMBER.
6. SEE "INSTRUMENTS", ZONE (C–8).

(14) SOLENOID VALVE–BODY PORTS THAT ARE CLOSED IN NORMAL OPERATION ARE BLACKENED. THE OPERATING CONDITION SHOWN FOR ALL VALVE BODIES CORRESPONDS TO FULL–LOAD OF NORMAL OPERATION REGARDLESS OF THE TYPE OF ACTUATOR OR WHETHER THE ACTUATOR IS ENERGIZED OR DE–ENERGIZED DURING THIS MODE OF OPERATION.
(15) LEVEL INSTRUMENT. THE UPPER INSTRUMENT LEAD LINE IS OMITTED IF LEVEL IS MEASURED BY PRESSURE INSTEAD OF DIFFERENTIAL PRESSURE.
(16) THE SPECIFIC BOARD AND LOCATION AT WHICH AN INSTRUMENT IS MOUNTED IS REFERENCED OUTSIDE THE INSTRUMENT CIRCLE. E.G.



PANEL LOCATION
Q–CONTROL ROOM PANEL
P–LOCAL PANEL
C–COMPUTER
V–HVAC SYSTEM LOCAL PANEL

PANEL IDENTIFICATION
MCB–MAIN CONTROL BOARD
PCB–PROCESS CONTROL PANEL (SEE EQUIPMENT INDEX GROUP 5 FOR COMPLETE LIST OF PANELS)
CHARACTERS SHOWN ARE LAST TWO OR THREE OF EQUIPMENT TAG NO.

(17) MULTIPLE POINT RECORDERS ARE NOT SHOWN ON THE P & ID. SEE REFERENCED LOOP DIAGRAMS.
(18) EACH OF THE LOCAL PANEL ANNUNCIATORS RETRANSMIT A COMMON TROUBLE ALARM TO THE CONTROL ROOM EXCEPT FOR THE DIESEL GENERATOR ANNUNCIATOR WHICH RETRANSMITS ALL ALARMS INDIVIDUALLY TO THE CONTROL ROOM.
(19) ALL ELECTRIC MOTOR OPERATED VALVES FAILURE MODE IS "FAIL LOCKED" (FL).
(20) CONSULT PSMS COMPUTER LIST 1/2X6AZ02–10, OD6, FOR INPUTS UTILIZED FOR PSMS COMPUTER, CHECK IPC DATABASE FOR POINTS DATA LINKED TO/FROM PSMS.

GENERAL NOTES:
(A) ALL INSTRUMENT IDENTIFICATIONS ARE BASED UPON ISA STANDARD "S5. 1.1–1967". FOR FURTHER DETAILS REFER TO ISA STANDARD.
(B) WHERE SPECIAL DESIGNATION IS REQUIRED, PILOT LIGHTS SHALL BE IDENTIFIED WITH THE PARTICULAR VARIABLE LETTER, FOLLOWED BY SECOND LETTER "L".

## LINES

- CONNECTION TO PROCESS, MECHANICAL LINK OR INSTRUMENT INPUT
- PNEUMATIC SIGNAL
- ELECTRICAL SIGNAL
- CAPILLARY TUBING (FILLED SYSTEM)
- HYDRAULIC SIGNAL
- HEAT TRACING
- A.S. INSTRUMENT AIR SUPPLY
- WIRELESS SIGNAL

## INSTRUMENTS

- LOCAL INSTRUMENT INCLUDING TRANSMITTER FOR SINGLE MEASURED VARIABLE
- LOCAL INSTRUMENT FOR TWO MEASURED VARIABLES OR MORE THAN ONE FUNCTION
- PANEL MOUNTED INSTRUMENT FOR SINGLE MEASURED VARIABLE
- INSTRUMENT SENSES AMBIENT CONDITION
- INSTRUMENT MOUNTED BEHIND PANEL
- INSTRUMENT IS PRINTED CIRCUIT CARD AND NOT INDIVIDUALLY LISTED IN INSTRUMENT INDEX

## COMPUTERS

- PE, E → IPC
- PES, ES → IPC AND PLANT SAFETY MONITORING SYSTEM COMPUTER SEE NOTE 20

## SENSORS–ANALYSIS

(NOTE 7)

- OFF–LINE TYPE
- IN–LINE TYPE

## TYPICAL SENSORS

PRESSURE

- DIAPHRAGM SEAL–LINE OR EQUIPMENT MOUNTED
- DIAPHRAGM SEAL CONNECTION, PIPED

FLOW

- ORIFICE PLATE OR RESTRICTION ORIFICE (NOTE 10)
- VENTURI TUBE OR FLOW NOZZLE
- PITOT OR PITOT VENTURI TUBE
- TURBINE OR PROPELLER–TYPE PRIMARY ELEMENT
- ROTAMETER INDICATOR (FI) OR TRANSMITTER (FT)
- IN–LINE INSTRUMENT E.G. MAGNETIC FLOWMETER, DISPLACEMENT METER, MASS FLOWMETER FLOW SIGHT GLASS
- FLOW STRAIGHTENING VANES
- CLAMP ON – SONIC BEAM MEASUREMENT

THE TYPE OF FLOW INSTRUMENT MAY BE NAMED OUTSIDE THE INSTRUMENT CIRCLE, E.G. MAGNETIC FLOW METER DISPLACEMENT METER, MASS FLOW, ETC.

LEVEL

- GAGE GLASS, FLOAT OR DISPLACEMENT–TYPE LEVEL INSTRUMENT OR TRANSMITTER
- (NOTE 15) DIFFERENTIAL–PRESSURE TYPE LEVEL TRANSMITTER
- (NOTE 15) FLANGE–MOUNTED DIFFERENTIAL–PRESSURE TYPE LEVEL TRANSMITTER
- INTERNAL BALL–FLOAT–TYPE LEVEL INSTRUMENT OR TRANSMITTER
- GAGE–BOARD–TYPE LEVEL INSTRUMENT

TEMPERATURE (NOTE 8)

- DUAL OR DUPLEX THERMOCOUPLE IN ONE WELL
- TE, TC, RTD: SIGNAL THERMOCOUPLE OR RTD NORMALLY RECORDED, OPTIONALLY INDICATED (IF APPLICABLE)
- DIAL THERMOMETER BI–METALLIC
- DIAL THERMOMETER FILLED SYSTEM

RADIATION

- OFF–LINE RADIATION MONITOR
- IN–LINE RADIATION MONITOR

THE LETTER N IS USED ONLY WHEN NEUTRON FLUX IS THE MEASURED VARIABLE, THE LETTER R IS USED ONLY WHEN RADIOACTIVITY IS THE MEASURED VARIABLE.

## CONTROL VALVE BODIES

- GLOBE, FO: FO INDICATES FAIL OPEN; FC INDICATES FAIL CLOSED; FL INDICATES FAIL LOCKED (POSITION DOES NOT CHANGE); FI INDICATES FAIL INDETERMINATE
- GATE, FO
- ANGLE
- BUTTERFLY (NO NORMALLY OPEN, NC NORMALLY CLOSED)
- BALL
- THREE WAY (NOTE 14), F–FAIL POSITION
- FOUR–WAY
- CHECK
- DIAPHRAGM
- PLUG
- PINCH
- UNCLASSIFIED (TYPE OF BODY IS WRITTEN IN OR ADJACENT TO SYMBOL) NOTE: LINE DRAWN THROUGH VALVE BODY OPTIONAL

## ACTUATORS

- AO (IF REQUIRED) PNEUMATIC ACTUATOR; AO INDICATES AIR OPENS; AC INDICATES AIR CLOSES
- DIAPHRAGM, PRESSURE–BALANCED
- (ELECTRIC) (NOTE 19) ROTARY MOTOR (SHOWN TYPICALLY WITH ELECTRIC SIGNAL)
- CYLINDER, SINGLE–ACTING, ALSO DOUBLE–ACTING CYLINDER THAT IS ASSEMBLED WITH PILOT, SO THAT ACTUATOR ASSEMBLY IS ACTUATED BY ONE CONTROLLED INPUT
- AS: CYLINDER, DOUBLE–ACTING WITH ACTUATING PILOT VALVE
- CYLINDER, DOUBLE–ACTING ASSEMBLED WITHOUT PILOT
- HAND ACTUATOR (MOUNTED AT TOP SIDE, OR BOTTOM OF ACTUATED DEVICE AS APPLICABLE.)
- ELECTRO–HYDRAULIC
- UNCLASSIFIED (TYPE OF ACTUATOR TO BE WRITTEN ADJACENT TO THE SYMBOL)
- S: SOLENOID (MANUAL OPTIONAL)

## SELF–ACTUATED DEVICES

FLOW

- (UPSTREAM REG.) FCV (DOWNSTREAM REG.) FLOW REGULATORS, SELF–CONTAINED

LEVEL

- LCV: LEVEL REGULATOR WITH MECHANICAL LINKAGE

PRESSURE

- PCV: PRESSURE–REDUCING REGULATOR, SELF–CONTAINED
- PCV: PRESSURE–REDUCING REGULATOR, WITH EXTERNAL PRESSURE TAP
- PDCV: DIFFERENTIAL–PRESSURE–REDUCING REGULATOR WITH INTERNAL & EXTERNAL PRESSURE TAPS
- PCV: BACK–PRESSURE REGULATOR SELF–CONTAINED
- PCV: BACK–PRESSURE REGULATOR WITH EXTERNAL PRESSURE TAP
- PSV: PRESSURE RELIEF OR SAFETY VALVE, ANGLE PATTERN, SPRING OR WEIGHT–LOADED, OR WITH INTEGRAL PILOT. (NOTE 9)
- PSV: PRESSURE RELIEF OR SAFETY VALVE, STRAIGHT–THROUGH PATTERN, SPRING– OR WEIGHT–LOADED, OR WITH INTEGRAL PILOT.
- PSV: VACUUM RELIEF, ANGLE PATTERN, SPRING– OR WEIGHT–LOADED, OR WITH INTEGRAL PILOT.
- PS, PSV: PRESSURE RELIEF OR SAFETY VALVE, ANGLE PATTERN, TRIPPED BY INTEGRAL SOLENOID
- PSE: RUPTURE DISK OR SAFETY HEAD FOR PRESSURE RELIEF (NOTE 9).
- PSE: RUPTURE DISK OR SAFETY HEAD FOR VACUUM RELIEF
- PI, PI, PCV: 2–STAGE PRESSURE–REDUCING REGULATOR WITH DUAL PRESSURE INDICATOR, SELF–CONTAINED.

TEMPERATURE

- TCV: TEMPERATURE REGULATOR FILLED SYSTEM TYPE

## HAND–ACTUATED DEVICES

- HS: HAND–ACTUATED ELECTRIC SWITCH
- HCV: HAND CONTROL VALVE IN PROCESS LINE REQUISITIONED BY INSTR. GROUP.
- HS: HAND–ACTUATED SWITCHING VALVE IN PNEUMATIC SIGNAL LINE TWO WAY SHOWN–THREE OR FOUR WAY ADD LINES AS REQUIRED.
- HO: MANUALLY ADJUSTABLE RESTRICTION ORIFICE IN SIGNAL LINE

## MISCELLANEOUS

- 1:1 RELAY (VOLUME BOOSTER) (NOTE 3)
- HIGH MEASURED VARIABLE SELECTOR IF HIGH SIGNAL SELECTOR ADD (SIGNAL)
  - HIGH SELECTOR
  - LOW SELECTOR
- SQUARE ROOT EXTRACTOR
- I/P: CURRENT TO PNEUMATIC CONVERTER
- INTERLOCK (1/4)
- PURGE CONNECTION
- ALARM (ANNUNCIATOR) H–HIGH L–LOW P–PANEL LOCATION * O–OPEN C–CLOSED HH–HIGH HIGH LL–LOW LOW
  * ALARMS WITH NO PANEL LOCATION SHOWN ARE LOCATED IN THE CONTROL ROOM ON PANEL QMCB, QPCP QHVC OR QEAB (SEE NOTE 18).
- ZS: POSITION SWITCH
- L: PILOT LIGHT (GENERAL NOTE B)
- DISTRIBUTED CONTROL INTERCONNECTING LOGIC CONTROLLER WITH BINARY OR SEQUENTIAL LOGIC FUNCTIONS
- DFW: THIS SYMBOL INDICATES AN INPUT OR OUTPUT TO DIGITAL FEEDWATER DCS (13.0)

NOTE:
THIS DRAWING IS BASED ON THE INSTRUMENT SOCIETY OF AMERICA STANDARD NO. S5. 1–1967

| CAD | 1X4DB102 |
|---|---|
| ACAD2007 | JLO – 03 |

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 13.0 | REVISED PER ABN 103900250IJ172, VER. 1.0 | 10/10/12 | JLO | ELC | JMR | X |

SOUTHERN COMPANY

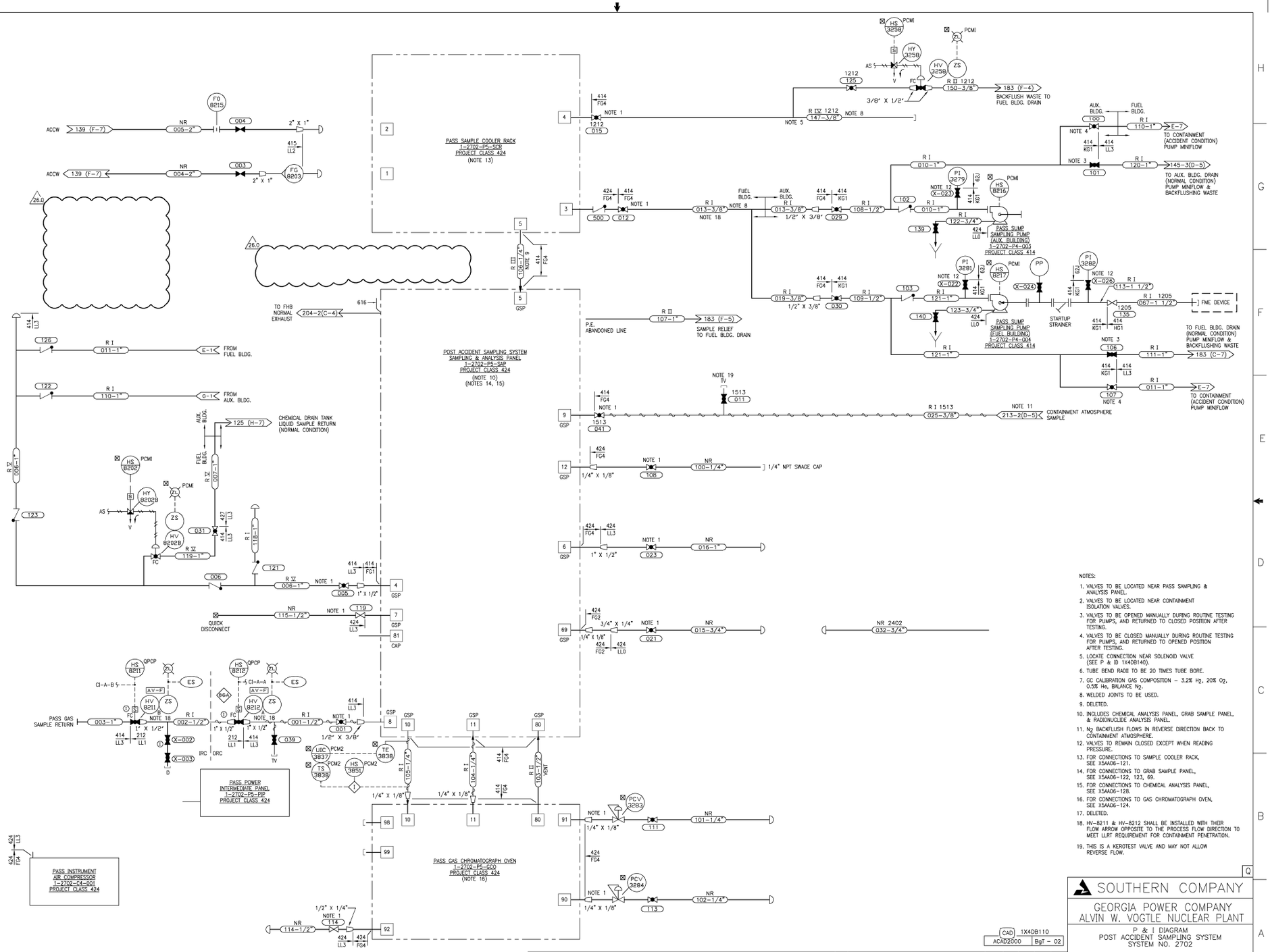
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENTATION
IDENTIFICATION & SYMBOLS

| SCALE: | NONE | DRAWING NO. | VER. |
|---|---|---|---|
| JOB NO. | 10604 | 1X4DB102 | 13.0 |

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. VALVES TO BE LOCATED NEAR PASS SAMPLING & ANALYSIS PANEL.
2. VALVES TO BE LOCATED NEAR CONTAINMENT ISOLATION VALVES.
3. VALVES TO BE OPENED MANUALLY DURING ROUTINE TESTING FOR PUMPS, AND RETURNED TO CLOSED POSITION AFTER TESTING.
4. VALVES TO BE CLOSED MANUALLY DURING ROUTINE TESTING FOR PUMPS, AND RETURNED TO OPENED POSITION AFTER TESTING.
5. LOCATE CONNECTION NEAR SOLENOID VALVE (SEE P & ID 1X4DB140).
6. TUBE BEND RADII TO BE 20 TIMES TUBE BORE.
7. GC CALIBRATION GAS COMPOSITION – 3.2% $H_2$, 20% $O_2$, 0.5% He, BALANCE $N_2$.
8. WELDED JOINTS TO BE USED.
9. DELETED.
10. INCLUDES CHEMICAL ANALYSIS PANEL, GRAB SAMPLE PANEL, & RADIONUCLIDE ANALYSIS PANEL.
11. $N_2$ BACKFLUSH FLOWS IN REVERSE DIRECTION BACK TO CONTAINMENT ATMOSPHERE.
12. VALVES TO REMAIN CLOSED EXCEPT WHEN READING PRESSURE.
13. FOR CONNECTIONS TO SAMPLE COOLER RACK, SEE X5AA06-121.
14. FOR CONNECTIONS TO GRAB SAMPLE PANEL, SEE X5AA06-122, 123, 69.
15. FOR CONNECTIONS TO CHEMICAL ANALYSIS PANEL, SEE X5AA06-128.
16. FOR CONNECTIONS TO GAS CHROMATOGRAPH OVEN, SEE X5AA06-124.
17. DELETED.
18. HV-8211 & HV-8212 SHALL BE INSTALLED WITH THEIR FLOW ARROW OPPOSITE TO THE PROCESS FLOW DIRECTION TO MEET LLRT REQUIREMENT FOR CONTAINMENT PENETRATION.
19. THIS IS A KEROTEST VALVE AND MAY NOT ALLOW REVERSE FLOW.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
POST ACCIDENT SAMPLING SYSTEM
SYSTEM NO. 2702

CAD 1X4DB110
ACADZ000 BgT – 02

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB110
VER. 26.0

| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 26.0 | REVISED PER ABN-V03117, VER.1.0 | 10-9-12 | BgT | CJK | AAN | |

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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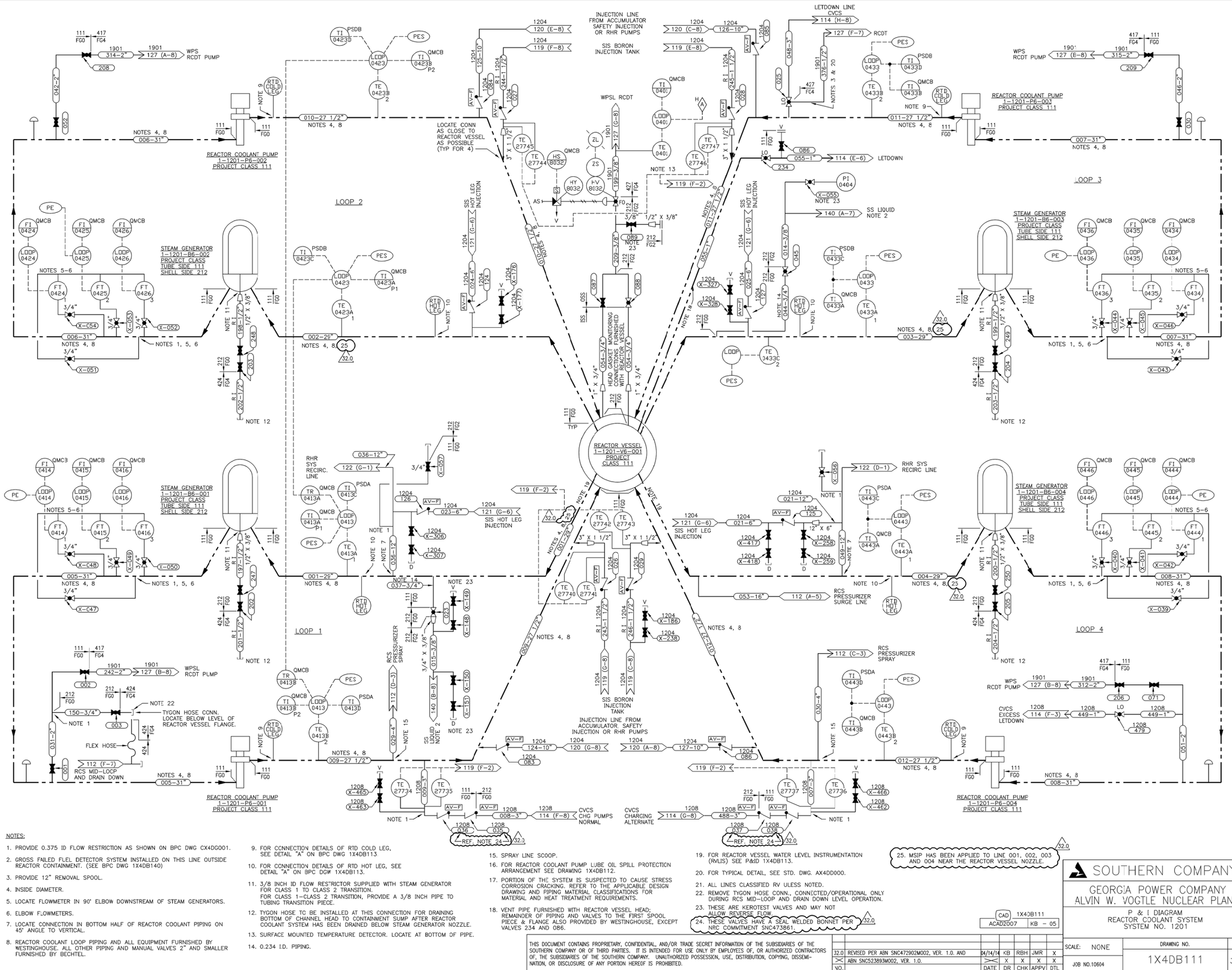


NOTES:

1. PROVIDE 0.375 ID FLOW RESTRICTION AS SHOWN ON BPC DWG CX4DG001.
2. GROSS FAILED FUEL DETECTOR SYSTEM INSTALLED ON THIS LINE OUTSIDE REACTOR CONTAINMENT. (SEE BPC DWG 1X4DB140)
3. PROVIDE 12" REMOVAL SPOOL.
4. INSIDE DIAMETER.
5. LOCATE FLOWMETER IN 90° ELBOW DOWNSTREAM OF STEAM GENERATORS.
6. ELBOW FLOWMETERS.
7. LOCATE CONNECTION IN BOTTOM HALF OF REACTOR COOLANT PIPING ON 45° ANGLE TO VERTICAL.
8. REACTOR COOLANT LOOP PIPING AND ALL EQUIPMENT FURNISHED BY WESTINGHOUSE. ALL OTHER PIPING AND MANUAL VALVES 2" AND SMALLER FURNISHED BY BECHTEL.
9. FOR CONNECTION DETAILS OF RTD COLD LEG, SEE DETAIL "A" ON BPC DWG 1X4DB113
10. FOR CONNECTION DETAILS OF RTD HOT LEG, SEE DETAIL "A" ON BPC DGW 1X4DB113.
11. 3/8 INCH ID FLOW RESTRICTOR SUPPLIED WITH STEAM GENERATOR FOR CLASS 1 TO CLASS 2 TRANSITION. FOR CLASS 1–CLASS 2 TRANSITION, PROVIDE A 3/8 INCH PIPE TO TUBING TRANSITION PIECE.
12. TYGON HOSE TO BE INSTALLED AT THIS CONNECTION FOR DRAINING BOTTOM OF CHANNEL HEAD TO CONTAINMENT SUMP AFTER REACTOR COOLANT SYSTEM HAS BEEN DRAINED BELOW STEAM GENERATOR NOZZLE.
13. SURFACE MOUNTED TEMPERATURE DETECTOR. LOCATE AT BOTTOM OF PIPE.
14. 0.234 I.D. PIPING.
15. SPRAY LINE SCOOP.
16. FOR REACTOR COOLANT PUMP LUBE OIL SPILL PROTECTION ARRANGEMENT SEE DRAWING 1X4DB112.
17. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
18. VENT PIPE FURNISHED WITH REACTOR VESSEL HEAD; REMAINDER OF PIPING AND VALVES TO THE FIRST SPOOL PIECE & FLANGE ALSO PROVIDED BY WESTINGHOUSE, EXCEPT VALVES 234 AND 086.
19. FOR REACTOR VESSEL WATER LEVEL INSTRUMENTATION (RVLIS) SEE P&ID 1X4DB113.
20. FOR TYPICAL DETAIL, SEE STD. DWG. AX4DD000.
21. ALL LINES CLASSIFIED RV ULESS NOTED.
22. REMOVE TYGON HOSE CONN., CONNECTED/OPERATIONAL ONLY DURING RCS MID-LOOP AND DRAIN DOWN LEVEL OPERATION.
23. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
24. THESE VALVES HAVE A SEAL WELDED BONNET PER NRC COMMITMENT SNC473861. 32.0
25. MSIP HAS BEEN APPLIED TO LINE 001, 002, 003 AND 004 NEAR THE REACTOR VESSEL NOZZLE. 32.0

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|---|---|---|---|---|---|---|
| 32.0 | REVISED PER ABN SNC472902M002, VER. 1.0. AND | 04/14/16 | KB | RBH | JMR | X |
| | ABN SNC523893M002, VER. 1.0. | >< | X | X | X | X |

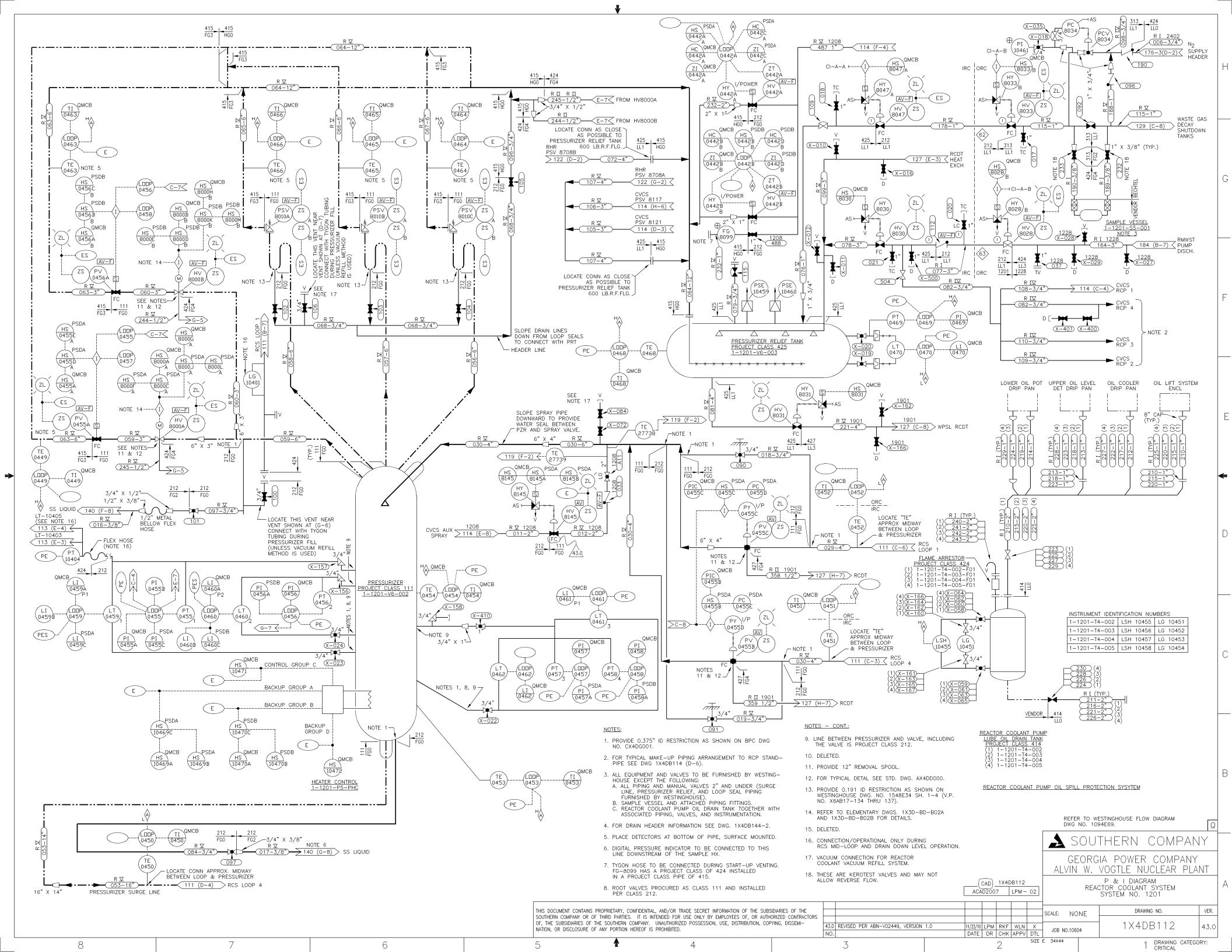
CAD 1X4DB111
ACAD2007 KB – 05

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
REACTOR COOLANT SYSTEM
SYSTEM NO. 1201

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB111 | 32.0 |

SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL




| INSTRUMENT IDENTIFICATION NUMBERS | | |
|---|---|---|
| 1-1201-T4-002 | LSH 10455 | LG 10451 |
| 1-1201-T4-003 | LSH 10456 | LG 10452 |
| 1-1201-T4-004 | LSH 10457 | LG 10453 |
| 1-1201-T4-005 | LSH 10458 | LG 10454 |

NOTES:

1. PROVIDE 0.375" ID RESTRICTION AS SHOWN ON BPC DWG NO. CX4DG001.
2. FOR TYPICAL MAKE-UP PIPING ARRANGEMENT TO RCP STAND-PIPE SEE DWG 1X4DB114 (D-6).
3. ALL EQUIPMENT AND VALVES TO BE FURNISHED BY WESTING-HOUSE EXCEPT THE FOLLOWING:
   A. ALL PIPING AND MANUAL VALVES 2" AND UNDER (SURGE LINE, PRESSURIZER RELIEF, AND LOOP SEAL PIPING FURNISHED BY WESTINGHOUSE).
   B. SAMPLE VESSEL AND ATTACHED PIPING FITTINGS.
   C. REACTOR COOLANT PUMP OIL DRAIN TANK TOGETHER WITH ASSOCIATED PIPING, VALVES, AND INSTRUMENTATION.
4. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB144-2.
5. PLACE DETECTORS AT BOTTOM OF PIPE, SURFACE MOUNTED.
6. DIGITAL PRESSURE INDICATOR TO BE CONNECTED TO THIS LINE DOWNSTREAM OF THE SAMPLE HX.
7. TYGON HOSE TO BE CONNECTED DURING START-UP VENTING. FG-8099 HAS A PROJECT CLASS OF 424 INSTALLED IN A PROJECT CLASS PIPE OF 415.
8. ROOT VALVES PROCURED AS CLASS 111 AND INSTALLED PER CLASS 212.

NOTES - CONT.:

9. LINE BETWEEN PRESSURIZER AND VALVE, INCLUDING THE VALVE IS PROJECT CLASS 212.
10. DELETED.
11. PROVIDE 12" REMOVAL SPOOL.
12. FOR TYPICAL DETAL SEE STD. DWG. AX4DD000.
13. PROVIDE 0.191 ID RESTRICTION AS SHOWN ON WESTINGHOUSE DWG. NO. 1548E34 SH. 1-4 (V.P. NO. X6AB17-134 THRU 137).
14. REFER TO ELEMENTARY DWGS. 1X3D-BD-B02A AND 1X3D-BD-B02B FOR DETAILS.
15. DELETED.
16. CONNECTION/OPERATIONAL ONLY DURING RCS MID-LOOP AND DRAIN DOWN LEVEL OPERATION.
17. VACUUM CONNECTION FOR REACTOR COOLANT VACUUM REFILL SYSTEM.
18. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.

REFER TO WESTINGHOUSE FLOW DIAGRAM DWG NO. 1094E89.

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|---|---|---|---|---|---|---|
| 43.0 | REVISED PER ABN-V02449, VERSION 1.0 | 11/23/10 | LPM | RKF | WLN | X |

CAD 1X4DB112
ACAD2007 | LPM- 02

SOUTHERN COMPANY

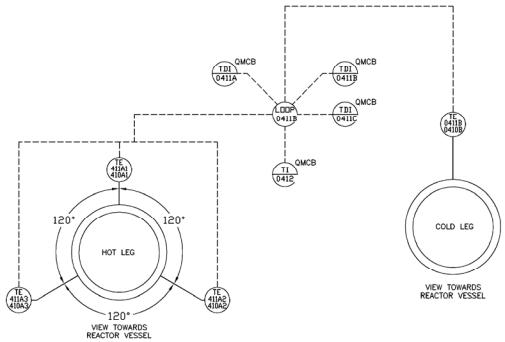
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P. & I DIAGRAM
REACTOR COOLANT SYSTEM
SYSTEM NO. 1201

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB112 | 43.0 |

SIZE E 34x44

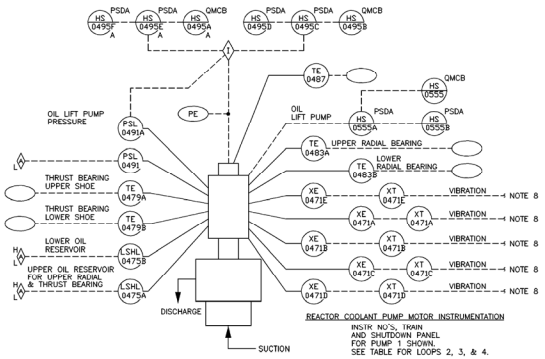
DRAWING CATEGORY: CRITICAL



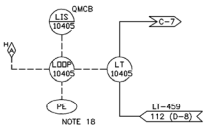

DETAIL "A"
NOTE: INSTRUMENT NOS. AND SAFETY CHANNEL FOR LOOP NO. 1 SHOWN. SEE TABLE FOR LOOPS 2, 3 AND 4.

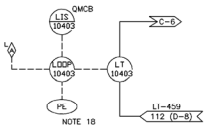
RTD INSTRUMENTATION

| INSTRUMENT | LOOP 1 | LOOP 2 | LOOP 3 | LOOP 4 |
|---|---|---|---|---|
| SAFETY CHANNEL | 1 | 2 | 3 | 4 |
| TRAIN | A | B | C | D |
| TEMPERATURE ELEMENT (HOT LEG – ACTIVE) (HOT LEG – SPARE) | TE 411A1 TE 410A1 | TE 421A1 TE 420A1 | TE 431A1 TE 430A1 | TE 441A1 TE 440A1 |
| TEMPERATURE ELEMENT (HOT LEG – ACTIVE) (HOT LEG – SPARE) | TE 411A2 TE 410A2 | TE 421A2 TE 420A2 | TE 431A2 TE 430A2 | TE 441A2 TE 440A2 |
| TEMPERATURE ELEMENT (HOT LEG – ACTIVE) (HOT LEG – SPARE) | TE 411A3 TE 410A3 | TE 421A3 TE 420A3 | TE 431A3 TE 430A3 | TE 441A3 TE 440A3 |
| TEMPERATURE ELEMENT (COLD LEG – ACTIVE) (COLD LEG – SPARE) | TE 0411B TE 0410B | TE 0421B TE 0420B | TE 0431B TE 0430B | TE 0441B TE 0440B |
| DIFFERENTIAL TEMPERATURE INDICATOR | TDI 0411A | TDI 0421A | TDI 0431A | TDI 0441A |
| OVERPOWER DIFFERENTIAL TEMPERATURE INDICATOR | TDI 0411B | TDI 0421B | TDI 0431B | TDI 0441B |
| OVERTEMPERATURE DIFFERENTIAL TEMPERATURE INDICATOR | TDI 0411C | TDI 0421C | TDI 0431C | TDI 0441C |
| AVERAGE TEMPERATURE INDICATOR | TI 0412 | TI 0422 | TI 0432 | TI 0442 |



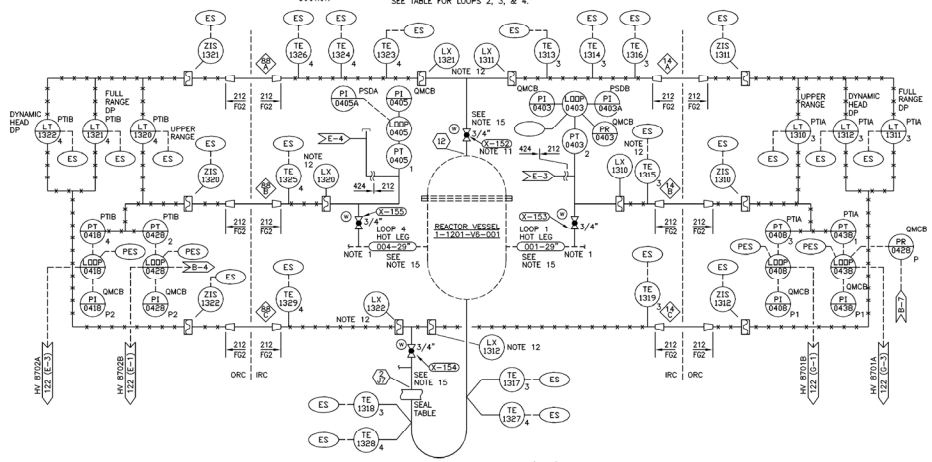

REACTOR COOLANT PUMP MOTOR INSTRUMENTATION
INSTR NO'S, TRAIN AND SHUTDOWN PANEL FOR PUMP 1 SHOWN. SEE TABLE FOR LOOPS 2, 3, & 4.




REACTOR COOLANT PUMP INSTRUMENTATION

| INSTRUMENT | LOOP 1 | LOOP 2 | LOOP 3 | LOOP 4 |
|---|---|---|---|---|
| SAFETY CHANNEL | 1 | 2 | 3 | 4 |
| TRAIN | A | B | C | D |
| SHUTDOWN PANEL | PSDA | PSDB | PSDB | PSDA |
| REACTOR COOLANT PUMP SWITCHES | HS 0495A THRU HS 0495F | HS 0496A THRU HS 0496F | HS 0497A THRU HS 0497F | HS 0498A THRU HS 0498F |
| OIL LIFT PUMP CONTROL SWITCHES | HS 0555 | HS 0556 | HS 0557 | HS 0558 |
| | HS 0555A | HS 0556A | HS 0557A | HS 0558A |
| | HS 0555B | HS 0556B | HS 0557B | HS 0558B |
| OIL LIFT PUMP PRESSURE | PSL 0491 | PSL 0492 | PSL 0493 | PSL 0494 |
| | PSL 0491A | PSL 0492A | PSL 0493A | PSL 0494A |
| THRUST BEARING UPPER SHOE | TE 0479A | TE 0480A | TE 0481A | TE 0482A |
| THRUST BEARING LOWER SHOE | TE 0479B | TE 0480B | TE 0481B | TE 0482B |
| LOWER OIL RESERVOIR | LSHL 0475B | LSHL 0476B | LSHL 0477B | LSHL 0478B |
| UPPER OIL RESERVOIR FOR UPPER RADIAL & THRUST BEARING | LSHL 0475A | LSHL 0476A | LSHL 0477A | LSHL 0478A |
| STATOR WINDING | TE 0487 | TE 0488 | TE 0489 | TE 0490 |
| UPPER RADIAL BEARING | TE 0483A | TE 0484A | TE 0485A | TE 0486A |
| LOWER RADIAL BEARING | TE 0483B | TE 0484B | TE 0485B | TE 0486B |
| VIBRATION PROXIMITY PROBE (KO) | XE/XT-0471E | XE/XT-0472E | XE/XT-0473E | XE/XT-0474E |
| VIBRATION PROXIMITY PROBE (P2) | XE/XT-0471D | XE/XT-0472D | XE/XT-0473D | XE/XT-0474D |
| VIBRATION PROXIMITY PROBE (P1) | XE/XT-0471C | XE/XT-0472C | XE/XT-0473C | XE/XT-0474C |
| VIBRATION VELOCITY SEISMOPROBE (SP2) | XE/XT-0471B | XE/XT-0472B | XE/XT-0473B | XE/XT-0474B |
| VIBRATION VELOCITY SEISMOPROBE (SP1) | XE/XT-0471A | XE/XT-0472A | XE/XT-0473A | XE/XT-0474A |




REACTOR VESSEL WATER LEVEL INSTRUMENTATION (RVLIS)
REFER TO WESTINGHOUSE RVLIS CAPILLARY SCHEMATIC DWG. NO. 1686E48

NOTES:

1. PROVIDE 0.375 I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG CX4DG001.
2. DELETED.
3. DELETED.
4. DELETED.
5. DELETED.
6. ALL EQUIPMENT BY (W) UNLESS OTHERWISE NOTED.
7. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
8. FOR VIBRATION TRANSMITTER DETAILS SEE DWG 1/2 X6AB09-88.
9. REFERENCE WESTINGHOUSE SCHEMATIC 1465F78 (X6AB02-270) AND 2343D84 (X6AB02-268).
10. ALL RVLIS COMPONENTS PROVIDED BY WESTINGHOUSE.
11. PROVIDE .375 I.D. FLOW RESTRICTION AS SHOWN ON WESTINGHOUSE DWG. 2654C41 RECHTEL LOG. X6AB02-217.
12. THE DESCRIPTOR "LX" PERTAINS TO A SEPARABLE COMPONENT OF THE ISOLATOR (ZIS).
13. DELETED.
14. ALL LINES ARE CLASSIFIED R (R) UNLESS OTHERWISE NOTED.
15. "PROJECT CLASS IS 212 FROM TAP THROUGH ROOT VALVE".
16. DELETED.
17. DELETED.
18. LT-10403 & LT-10405, CONNECTED/OPERATIONAL ONLY DURING RCS MID-LOOP AND DRAIN DOWN LEVEL OPERATION. SENSING LINE TO LT-0459 CONNECTED/OPERATIONAL ONLY DURING RCS MID-LOOP AND DRAIN DOWN LEVEL OPERATION.
19. REFER TO X6AB02-00223 FOR CLASS 1 TO CLASS 2 BREAK AT THE SEAL TABLE.

27.0

REFER TO WESTINGHOUSE FLOW DIAGRAM NO. 1094E75.

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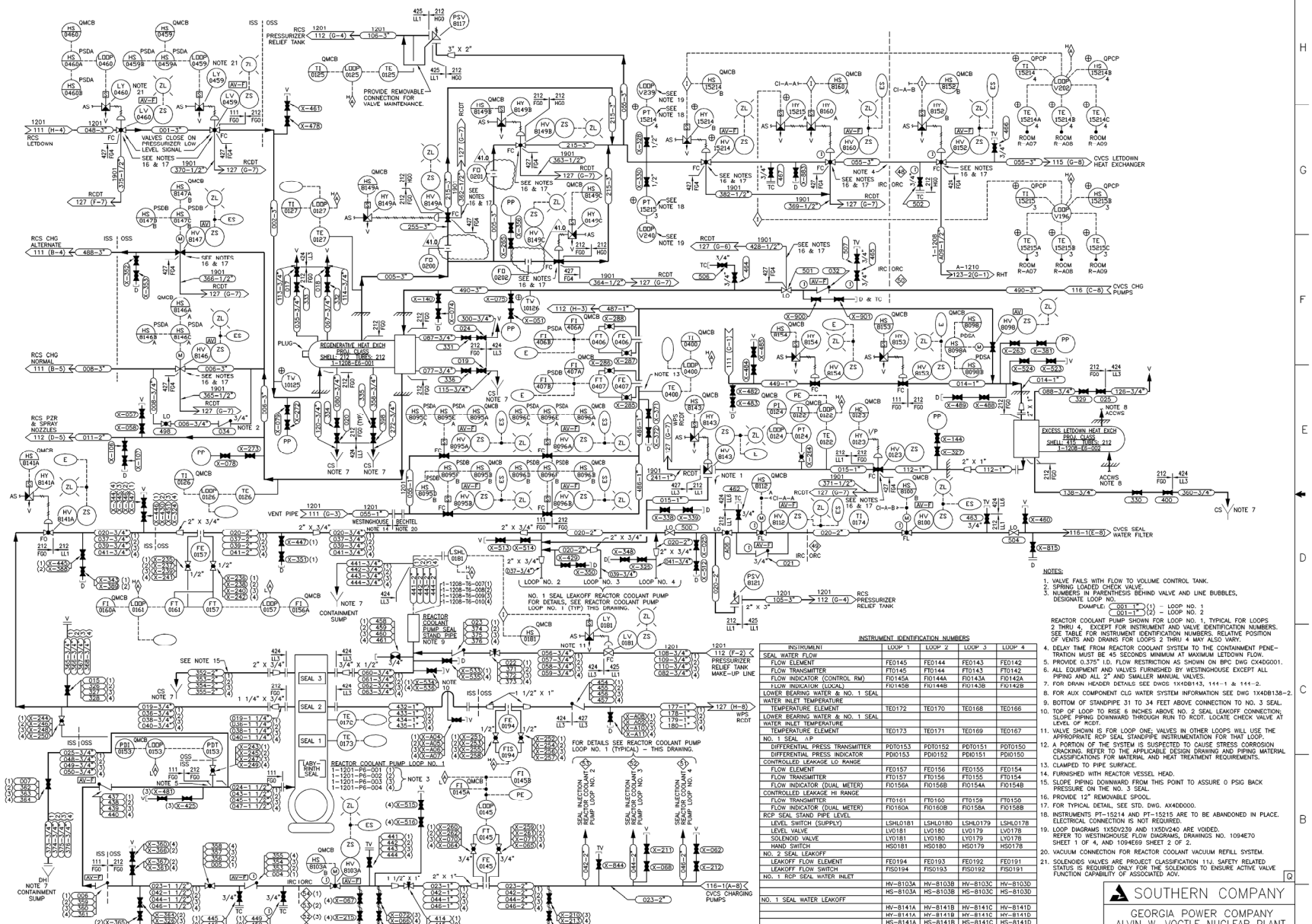
| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 27.0 | REVISED PER ABN-V70469 VER 1.0 | 3-7-08 | NJ | MWD | PML | X |

CAD 1X4DB113
ACAD2007 NJ – 02

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
RTD BY-PASS REACTOR COOLANT
SYSTEM NO. 1201

SCALE: NONE
DRAWING NO. 1X4DB113
VER. 27.0
JOB NO.10604
SIZE E 34044
DRAWING CATEGORY: CRITICAL



NOTES:

1. VALVE FAILS WITH FLOW TO VOLUME CONTROL TANK.
2. SPRING LOADED CHECK VALVE.
3. NUMBERS IN PARENTHESIS BEHIND VALVE AND LINE BUBBLES, DESIGNATE LOOP NO.
   EXAMPLE: 001-1" (1) – LOOP NO. 1
   001-1" (2) – LOOP NO. 2
   REACTOR COOLANT PUMP SHOWN FOR LOOP NO. 1, TYPICAL FOR LOOPS 2 THRU 4, EXCEPT FOR INSTRUMENT AND VALVE IDENTIFICATION NUMBERS. SEE TABLE FOR INSTRUMENT IDENTIFICATION NUMBERS. RELATIVE POSITION OF VENT AND DRAINS FOR LOOPS 2 THRU 4 MAY ALSO VARY.
4. DELAY TIME FROM REACTOR COOLANT SYSTEM TO THE CONTAINMENT PENETRATION MUST BE 45 SECONDS MINIMUM AT MAXIMUM LETDOWN FLOW.
5. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG CX4DQ001.
6. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL 2" AND SMALLER MANUAL VALVES.
7. FOR DRAIN HEADER DETAILS SEE DWGS 1X4DB143, 144-1 & 144-2.
8. FOR AUX COMPONENT CLG WATER SYSTEM INFORMATION SEE DWG 1X4DB138-2.
9. BOTTOM OF STANDPIPE 31 TO 34 FEET ABOVE CONNECTION TO NO. 3 SEAL.
10. TOP OF LOOP TO RISE 6 INCHES ABOVE NO. 2 SEAL LEAKOFF CONNECTION; SLOPE PIPING DOWNWARD THROUGH RUN TO RCDT. LOCATE CHECK VALVE AT LEVEL OF RCDT.
11. VALVE SHOWN IS FOR LOOP ONE; VALVES IN OTHER LOOPS WILL USE THE APPROPRIATE RCP SEAL STANDPIPE INSTRUMENTATION FOR THAT LOOP.
12. A PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
13. CLAMPED TO PIPE SURFACE.
14. FURNISHED WITH REACTOR VESSEL HEAD.
15. SLOPE PIPING DOWNWARD FROM THIS POINT TO ASSURE 0 PSIG BACK PRESSURE ON THE NO. 3 SEAL.
16. PROVIDE 12" REMOVABLE SPOOL.
17. FOR TYPICAL DETAIL, SEE STD. DWG. AX4DD000.
18. INSTRUMENTS PT-15214 AND PT-15215 ARE TO BE ABANDONED IN PLACE. ELECTRICAL CONNECTION IS NOT REQUIRED.
19. LOOP DIAGRAMS 1X5DV239 AND 1X5DV240 ARE VOIDED. REFER TO WESTINGHOUSE FLOW DIAGRAMS, DRAWINGS NO. 1094E70 SHEET 1 OF 4, AND 1094E69 SHEET 2 OF 2.
20. VACUUM CONNECTION FOR REACTOR COOLANT VACUUM REFILL SYSTEM.
21. SOLENOIDS VALVES ARE PROJECT CLASSIFICATION 11J. SAFETY RELATED STATUS IS REQUIRED ONLY FOR THE SOLENOIDS TO ENSURE ACTIVE VALVE FUNCTION CAPABILITY OF ASSOCIATED AOV.

INSTRUMENT IDENTIFICATION NUMBERS

| INSTRUMENT | LOOP 1 | LOOP 2 | LOOP 3 | LOOP 4 |
|---|---|---|---|---|
| SEAL WATER FLOW | | | | |
| FLOW ELEMENT | FE0145 | FE0144 | FE0143 | FE0142 |
| FLOW TRANSMITTER | FT0145 | FT0144 | FT0143 | FT0142 |
| FLOW INDICATOR (CONTROL RM) | FI0145A | FI0144A | FI0143A | FI0142A |
| FLOW INDICATOR (LOCAL) | FI0145B | FI0144B | FI0143B | FI0142B |
| LOWER BEARING WATER & NO. 1 SEAL WATER INLET TEMPERATURE | | | | |
| TEMPERATURE ELEMENT | TE0172 | TE0170 | TE0168 | TE0166 |
| LOWER BEARING WATER & NO. 1 SEAL WATER INLET TEMPERATURE | | | | |
| TEMPERATURE ELEMENT | TE0173 | TE0171 | TE0169 | TE0167 |
| NO. 1 SEAL ΔP | | | | |
| DIFFERENTIAL PRESS TRANSMITTER | PDT0153 | PDT0152 | PDT0151 | PDT0150 |
| DIFFERENTIAL PRESS INDICATOR | PDI0153 | PDI0152 | PDI0151 | PDI0150 |
| CONTROLLED LEAKAGE LO RANGE | | | | |
| FLOW ELEMENT | FE0157 | FE0156 | FE0155 | FE0154 |
| FLOW TRANSMITTER | FT0157 | FT0156 | FT0155 | FT0154 |
| FLOW INDICATOR (DUAL METER) | FI0156A | FI0156B | FI0154A | FI0154B |
| CONTROLLED LEAKAGE HI RANGE | | | | |
| FLOW TRANSMITTER | FT0161 | FT0160 | FT0159 | FT0158 |
| FLOW INDICATOR (DUAL METER) | FI0160A | FI0160B | FI0158A | FI0158B |
| RCP SEAL STAND PIPE LEVEL | | | | |
| LEVEL SWITCH (SUPPLY) | LSHL0181 | LSHL0180 | LSHL0179 | LSHL0178 |
| LEVEL VALVE | LV0181 | LV0180 | LV0179 | LV0178 |
| SOLENOID VALVE | LY0181 | LY0180 | LY0179 | LY0178 |
| HAND SWITCH | HS0181 | HS0180 | HS0179 | HS0178 |
| NO. 2 SEAL LEAKOFF | | | | |
| LEAKOFF FLOW ELEMENT | FE0194 | FE0193 | FE0192 | FE0191 |
| LEAKOFF FLOW SWITCH | FIS0194 | FIS0193 | FIS0192 | FIS0191 |
| NO. 1 RCP SEAL WATER INLET | | | | |
| | HV-8103A | HV-8103B | HV-8103C | HV-8103D |
| | HS-8103A | HS-8103B | HS-8103C | HS-8103D |
| NO. 1 SEAL WATER LEAKOFF | | | | |
| | HV-8141A | HV-8141B | HV-8141C | HV-8141D |
| | HY-8141A | HY-8141B | HY-8141C | HY-8141D |
| | HS-8141A | HS-8141B | HS-8141C | HS-8141D |

CAD 1X4DB114
ACAD2002 | TSL – 02

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 41.0 | REVISED PER ABN 1090006501M002, VER. 1.0 | 10/07/09 | TSL | MWD | JMR | X |

SIZE E 34X44

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

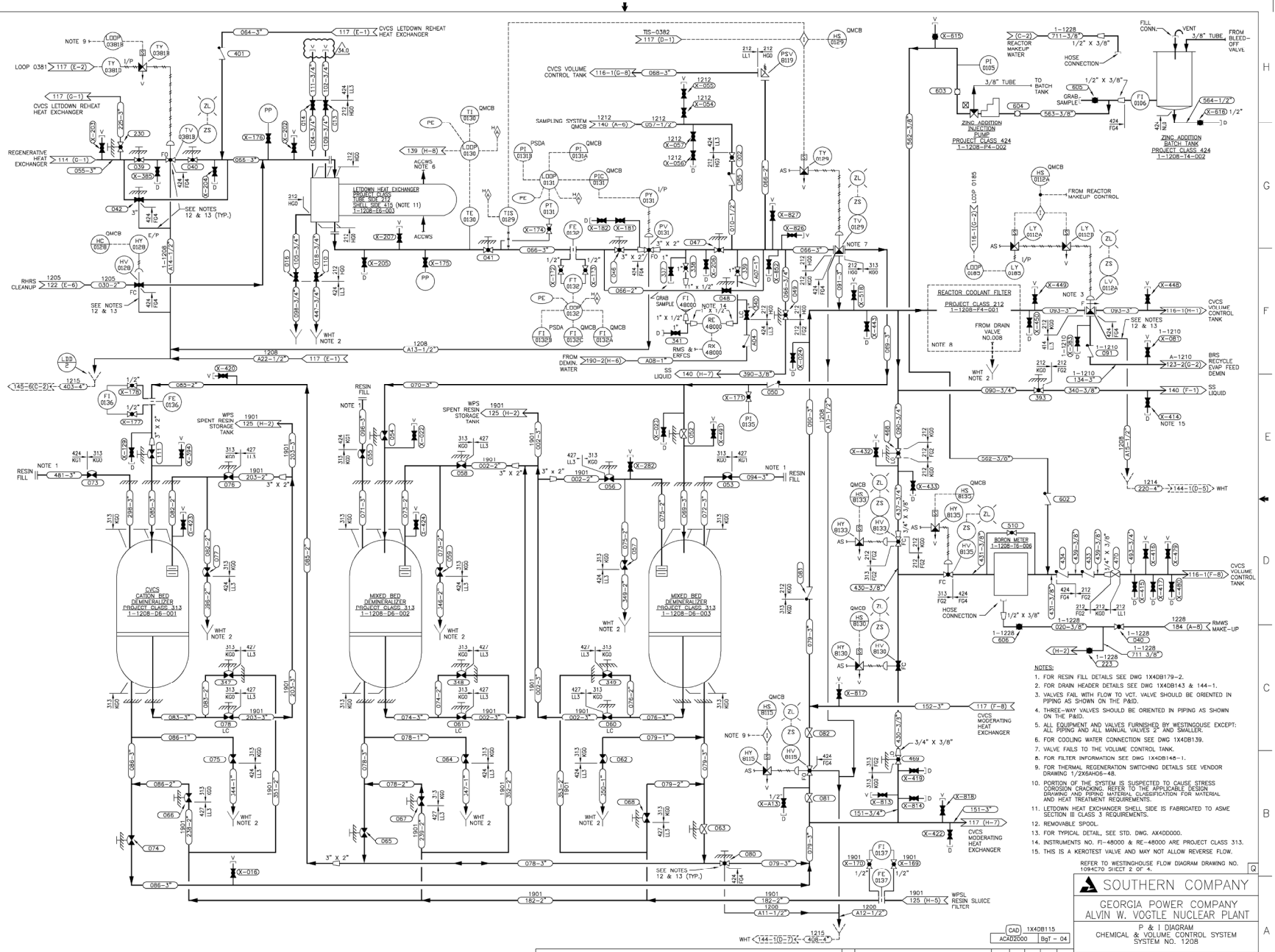
SCALE: NONE

DRAWING NO. 1X4DB114

VER. 41.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR RESIN FILL DETAILS SEE DWG 1X4DB179-2.
2. FOR DRAIN HEADER DETAILS SEE DWG 1X4DB143 & 144-1.
3. VALVES FAIL WITH FLOW TO VCT. VALVE SHOULD BE ORIENTED IN PIPING AS SHOWN ON THE P&ID.
4. THREE-WAY VALVES SHOULD BE ORIENTED IN PIPING AS SHOWN ON THE P&ID.
5. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGOUSE EXCEPT: ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
6. FOR COOLING WATER CONNECTION SEE DWG 1X4DB139.
7. VALVE FAILS TO THE VOLUME CONTROL TANK.
8. FOR FILTER INFORMATION SEE DWG 1X4DB148-1.
9. FOR THERMAL REGENERATION SWITCHING DETAILS SEE VENDOR DRAWING 1/2X6AH06-48.
10. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATION FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
11. LETDOWN HEAT EXCHANGER SHELL SIDE IS FABRICATED TO ASME SECTION III CLASS 3 REQUIREMENTS.
12. REMOVABLE SPOOL.
13. FOR TYPICAL DETAIL, SEE STD. DWG. AX4DD000.
14. INSTRUMENTS NO. FI-48000 & RE-48000 ARE PROJECT CLASS 313.
15. THIS IS A KEROTEST VALVE AND MAY NOT ALLOW REVERSE FLOW.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094C70 SHEET 2 OF 4.

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| NO. | VERSIONS | DATE | DN | CHK | APPV |
|---|---|---|---|---|---|
| 34.0 | REVISED PER ABN-W01936, VER. 1.0 | 1-11-12 | BgT | JWL | AAN |

CAD 1X4DB115 | ACAD2000 | BgT – 04

SOUTHERN COMPANY

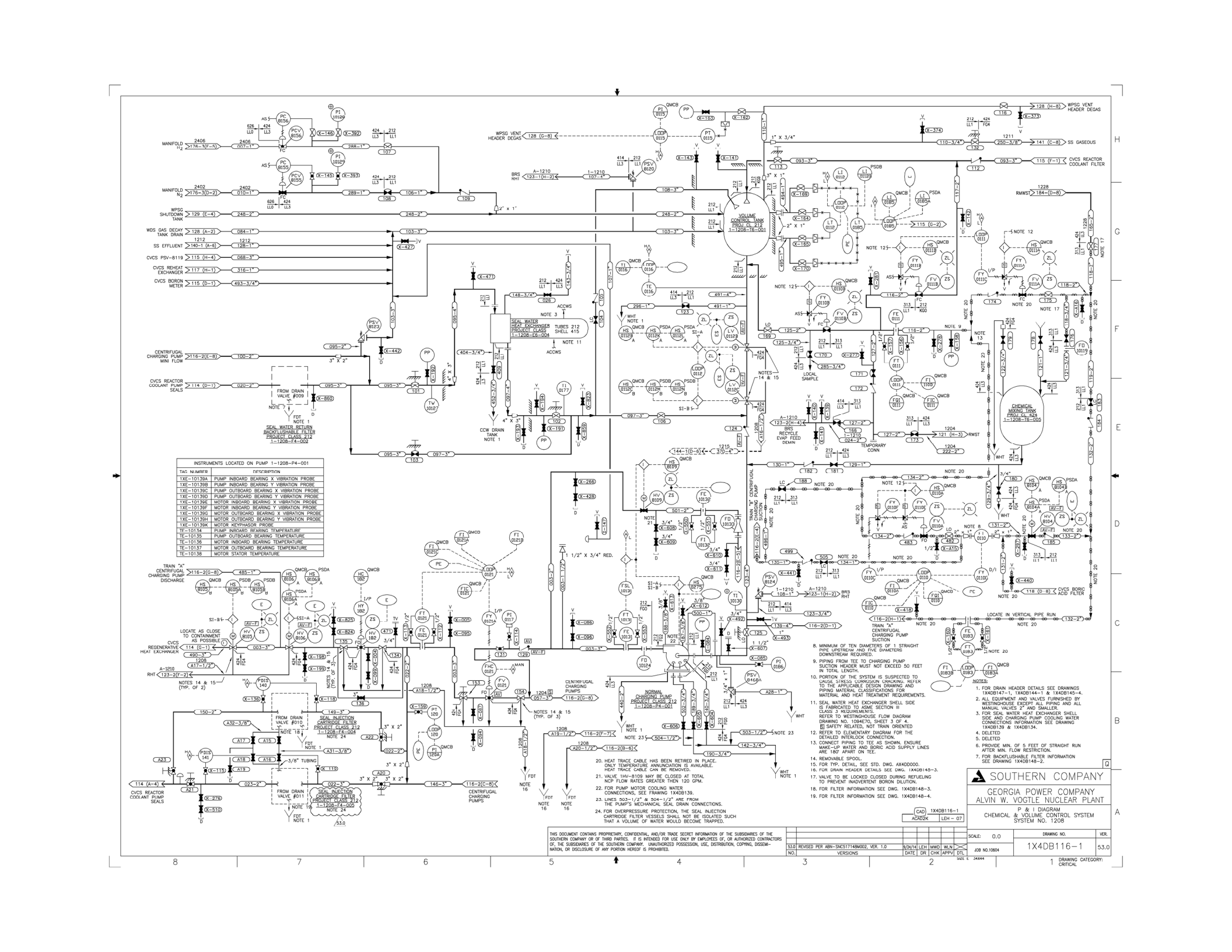
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

SCALE: NONE | DRAWING NO. 1X4DB115 | VER. 34.0 | JOB NO. 10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



| INSTRUMENTS LOCATED ON PUMP 1-1208-P4-001 | |
|---|---|
| TAG NUMBER | DESCRIPTION |
| 1XE-10139A | PUMP INBOARD BEARING X VIBRATION PROBE |
| 1XE-10139B | PUMP INBOARD BEARING Y VIBRATION PROBE |
| 1XE-10139C | PUMP OUTBOARD BEARING X VIBRATION PROBE |
| 1XE-10139D | PUMP OUTBOARD BEARING Y VIBRATION PROBE |
| 1XE-10139E | MOTOR INBOARD BEARING X VIBRATION PROBE |
| 1XE-10139F | MOTOR INBOARD BEARING Y VIBRATION PROBE |
| 1XE-10139G | MOTOR OUTBOARD BEARING X VIBRATION PROBE |
| 1XE-10139H | MOTOR OUTBOARD BEARING Y VIBRATION PROBE |
| 1XE-10139K | MOTOR KEYPHASOR PROBE |
| TE-10134 | PUMP INBOARD BEARING TEMPERATURE |
| TE-10135 | PUMP OUTBOARD BEARING TEMPERATURE |
| TE-10136 | MOTOR INBOARD BEARING TEMPERATURE |
| TE-10137 | MOTOR OUTBOARD BEARING TEMPERATURE |
| TE-10138 | MOTOR STATOR TEMPERATURE |

8. MINIMUM OF TEN DIAMETERS OF 1 STRAIGHT PIPE UPSTREAM AND FIVE DIAMETERS DOWNSTREAM REQUIRED.
9. PIPING FROM TEE TO CHARGING PUMP SUCTION HEADER MUST NOT EXCEED 50 FEET IN TOTAL LENGTH.
10. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
11. SEAL WATER HEAT EXCHANGER SHELL SIDE IS FABRICATED TO ASME SECTION III CLASS 3 REQUIREMENTS. REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E70, SHEET 3 OF 4. (S) SAFETY RELATED, NOT TRAIN ORIENTED
12. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK CONNECTION.
13. CONNECT PIPING TO TEE AS SHOWN. ENSURE MAKE-UP WATER AND BORIC ACID SUPPLY LINES ARE 180° APART ON TEE.
14. REMOVABLE SPOOL.
15. FOR TYP. DETAIL, SEE STD. DWG. AX4D0000.
16. FOR DRAIN HEADER DETAILS SEE DWG. 1X4DB145-3.
17. VALVE TO BE LOCKED CLOSED DURING REFUELING TO PREVENT INADVERTENT BORON DILUTION.
18. FOR FILTER INFORMATION SEE DWG. 1X4DB148-3.
19. FOR FILTER INFORMATION SEE DWG. 1X4DB148-4.
20. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.
21. VALVE 1HV-8109 MAY BE CLOSED AT TOTAL NCP FLOW RATES GREATER THEN 120 GPM.
22. FOR PUMP MOTOR COOLING WATER CONNECTIONS, SEE FRAWING 1X4DB139.
23. LINES 503-1/2" & 504-1/2" ARE FROM THE PUMP'S MECHANICAL SEAL DRAIN CONNECTIONS.
24. FOR OVERPRESSURE PROTECTION, THE SEAL INJECTION CARTRIDGE FILTER VESSELS SHALL NOT BE ISOLATED SUCH THAT A VOLUME OF WATER WOULD BECOME TRAPPED.

NOTES:
1. FOR DRAIN HEADER DETAILS SEE DRAWINGS 1X4DB147-1, 1X4DB144-1 & 1X4DB145-4.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
3. FOR SEAL WATER HEAT EXCHANGER SHELL SIDE AND CHARGING PUMP COOLING WATER CONNECTIONS INFORMATION SEE DRAWING 1X4DB139 & 1X4DB134.
4. DELETED
5. DELETED
6. PROVIDE MIN. OF 5 FEET OF STRAIGHT RUN AFTER MIN. FLOW RESTRICTION.
7. FOR BACKFLUSHABLE FILTER INFORMATION SEE DRAWING 1X4DB148-2.

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

CAD 1X4DB116-1
ACAD2K LEH – 07

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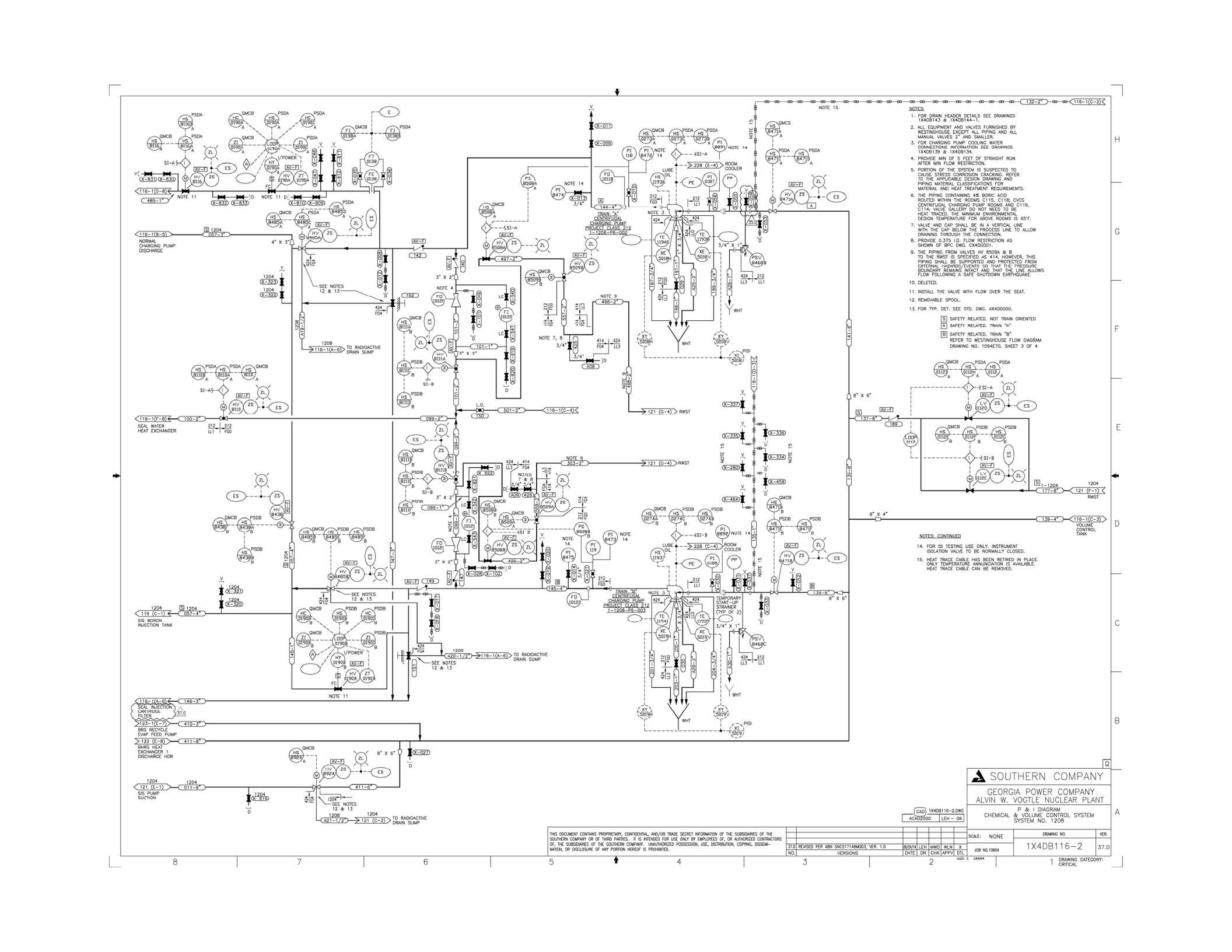
| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 53.0 | REVISED PER ABN-SNC511148M002, VER. 1.0 | 9/24/14 | LEH | MWD | WLN | |

SCALE: 0.0
JOB NO.10604
DRAWING NO. 1X4DB116-1
VER. 53.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL


NOTES:
1. FOR DRAIN HEADER DETAILS SEE DRAWINGS 1X4DB143 & 1X4DB144-1.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
3. FOR CHARGING PUMP COOLING WATER CONNECTIONS INFORMATION SEE DRAWINGS 1X4DB139 & 1X4DB134.
4. PROVIDE MIN OF 5 FEET OF STRAIGHT RUN AFTER MIN FLOW RESTRICTION.
5. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
6. THE PIPING CONTAINING 4% BORIC ACID ROUTED WITHIN THE ROOMS C115, C118; CVCS CENTRIFUGAL CHARGING PUMP ROOMS AND C119, C114; VALVE GALLERY DO NOT NEED TO BE HEAT TRACED. THE MINIMUM ENVIRONMENTAL DESIGN TEMPERATURE FOR ABOVE ROOMS IS 65°F.
7. VALVE AND CAP SHALL BE IN A VERTICAL LINE WITH THE CAP BELOW THE PROCESS LINE TO ALLOW DRAINING THROUGH THE CONNECTION.
8. PROVIDE 0.375 I.D. FLOW RESTRICTION AS SHOWN OF BPC DWG. CX4DG001.
9. THE PIPING FROM VALVES HV 8509A & B TO THE RWST IS SPECIFIED AS 414. HOWEVER, THIS PIPING SHALL BE SUPPORTED AND PROTECTED FROM EXTERNAL HAZARDS/EVENTS SO THAT THE PRESSURE BOUNDARY REMAINS INTACT AND THAT THE LINE ALLOWS FLOW FOLLOWING A SAFE SHUTDOWN EARTHQUAKE.
10. DELETED.
11. INSTALL THE VALVE WITH FLOW OVER THE SEAT.
12. REMOVABLE SPOOL.
13. FOR TYP. DET. SEE STD. DWG. AX4DD000.
S SAFETY RELATED. NOT TRAIN ORIENTED
A SAFETY RELATED. TRAIN "A"
B SAFETY RELATED. TRAIN "B"
REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E70. SHEET 3 OF 4
NOTES: CONTINUED
14. FOR ISI TESTING USE ONLY. INSTRUMENT ISOLATION VALVE TO BE NORMALLY CLOSED.
15. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.
NORMAL CHARGING PUMP DISCHARGE
SEAL WATER HEAT EXCHANGER
SIS BORON INJECTION TANK
SEAL INJECTION CAM/HUGGL FILTER
BRS RECYCLE EVAP FEED PUMP
RHRS HEAT EXCHANGER 1 DISCHARGE HDR
SIS PUMP SUCTION
TRAIN "A" CENTRIFUGAL CHARGING PUMP PROJECT CLASS 212 1-1208-P6-002
TRAIN "B" CENTRIFUGAL CHARGING PUMP PROJECT CLASS 212 1-1208-P6-003
TEMPORARY START-UP STRAINER (TYP OF 2)
ROOM COOLER
LUBE OIL
TO RADIOACTIVE DRAIN SUMP
VOLUME CONTROL TANK
RWST
SEE NOTES 12 & 13
NOTE 11
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208
CAD 1X4DB116-2.DWG
ACAD2000 LEH – 06
SCALE: NONE
DRAWING NO. 1X4DB116-2
VER. 37.0
JOB NO.10604
37.0 REVISED PER ABN SNC517148M003, VER. 1.0 08/24/11 LEH MWD WLN X
NO. VERSIONS DATE DR CHK APPV DTL
SIZE E 34344
DRAWING CATEGORY: CRITICAL
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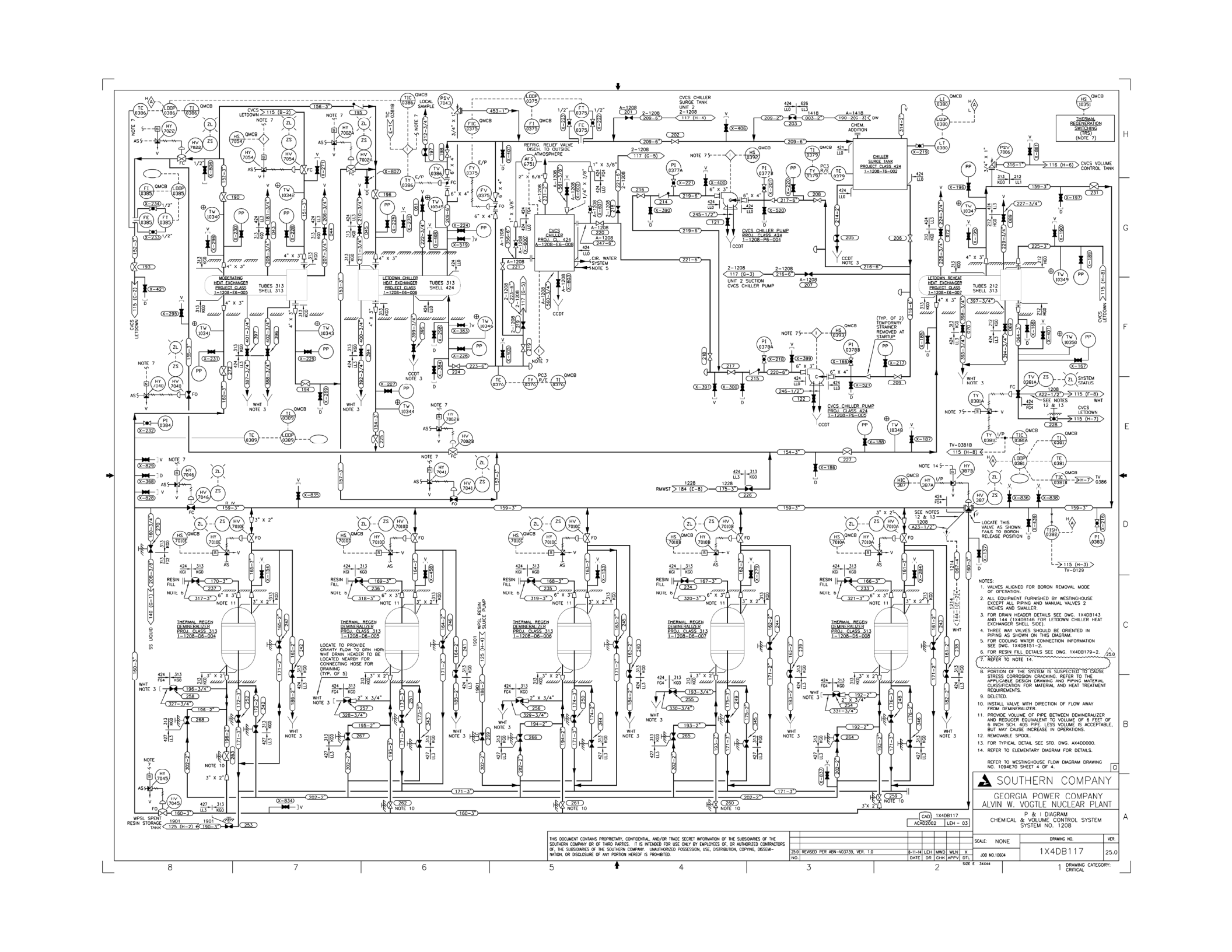


NOTES:

1. VALVES ALIGNED FOR BORON REMOVAL MODE OF OPERATION.
2. ALL EQUIPMENT FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND MANUAL VALVES 2 INCHES AND SMALLER.
3. FOR DRAIN HEADER DETAILS SEE DWG. 1X4DB143 AND 144 (1X4DB146 FOR LETDOWN CHILLER HEAT EXCHANGER SHELL SIDE).
4. THREE WAY VALVES SHOULD BE ORIENTED IN PIPING AS SHOWN ON THIS DIAGRAM.
5. FOR COOLING WATER CONNECTION INFORMATION SEE DWG. 1X4DB151-2.
6. FOR RESIN FILL DETAILS SEE DWG. 1X4DB179-2. 25.0
7. REFER TO NOTE 14.
8. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATION FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
9. DELETED.
10. INSTALL VALVE WITH DIRECTION OF FLOW AWAY FROM DEMINERALIZER.
11. PROVIDE VOLUME OF PIPE BETWEEN DEMINERALIZER AND REDUCER EQUIVALENT TO VOLUME OF 6 FEET OF 6 INCH SCH. 40S PIPE. LESS VOLUME IS ACCEPTABLE, BUT MAY CAUSE INCREASE IN OPERATIONS.
12. REMOVABLE SPOOL.
13. FOR TYPICAL DETAIL SEE STD. DWG. AX4DD000.
14. REFER TO ELEMENTARY DIAGRAM FOR DETAILS.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E70 SHEET 4 OF 4.

CAD 1X4DB117
ACAD2002 LEH - 03

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL SYSTEM
SYSTEM NO. 1208

SCALE: NONE
DRAWING NO. 1X4DB117
VER. 25.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

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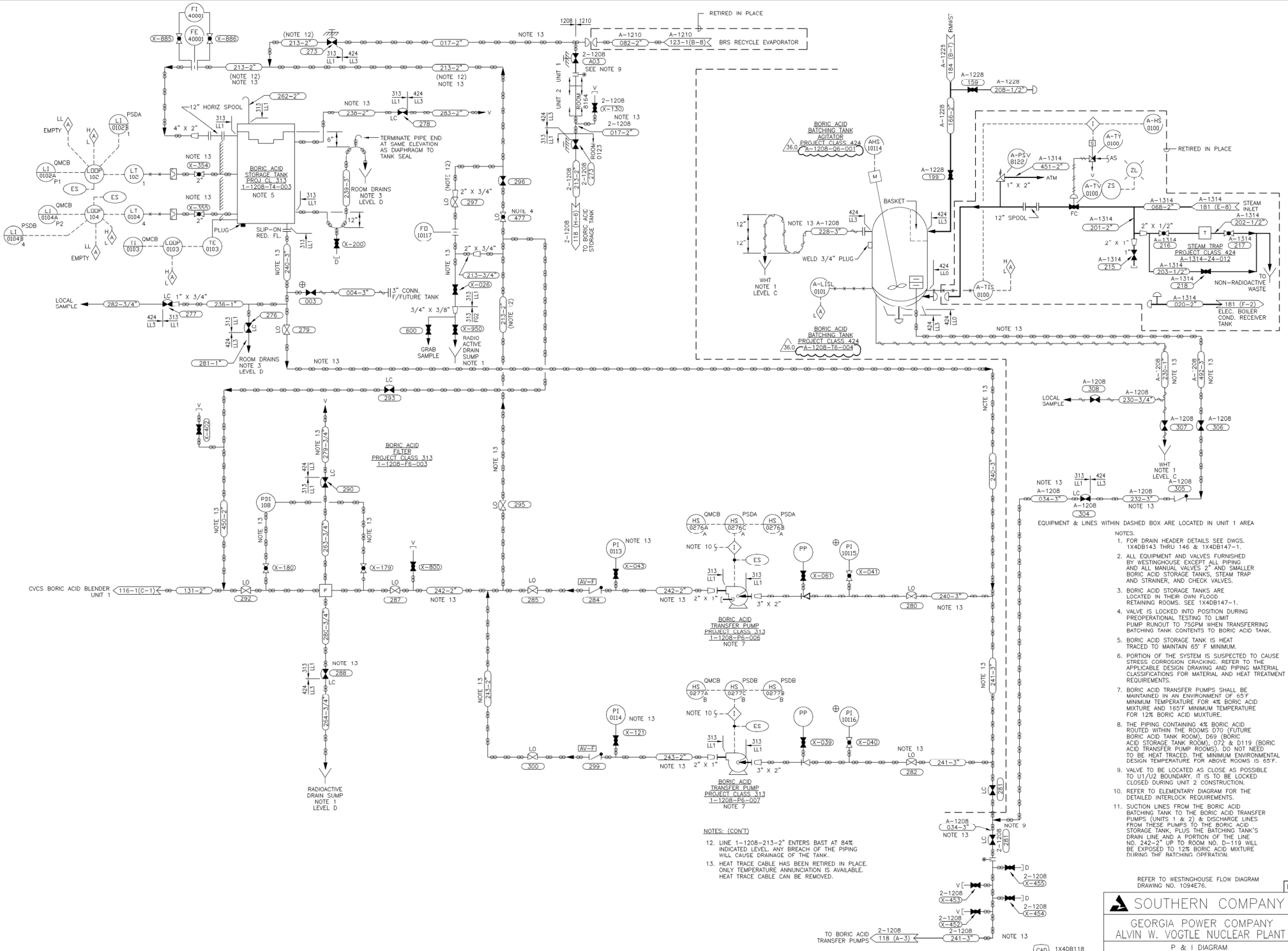
| NO. | REVISED PER ABN-V03739, VER. 1.0 | DATE | DR | CHK | APPV |
|---|---|---|---|---|---|
| 25.0 | | 8-11-14 | LEH | MWD | WLN |

DTL X

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| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 36.0 | REVISED PER ABN- V70034, VER. 1.0 | 5-3-06 | CYN | LCF | LKB | X |

CAD 1X4DB118
ACAD2002 CYN- 02

SOUTHERN COMPANY

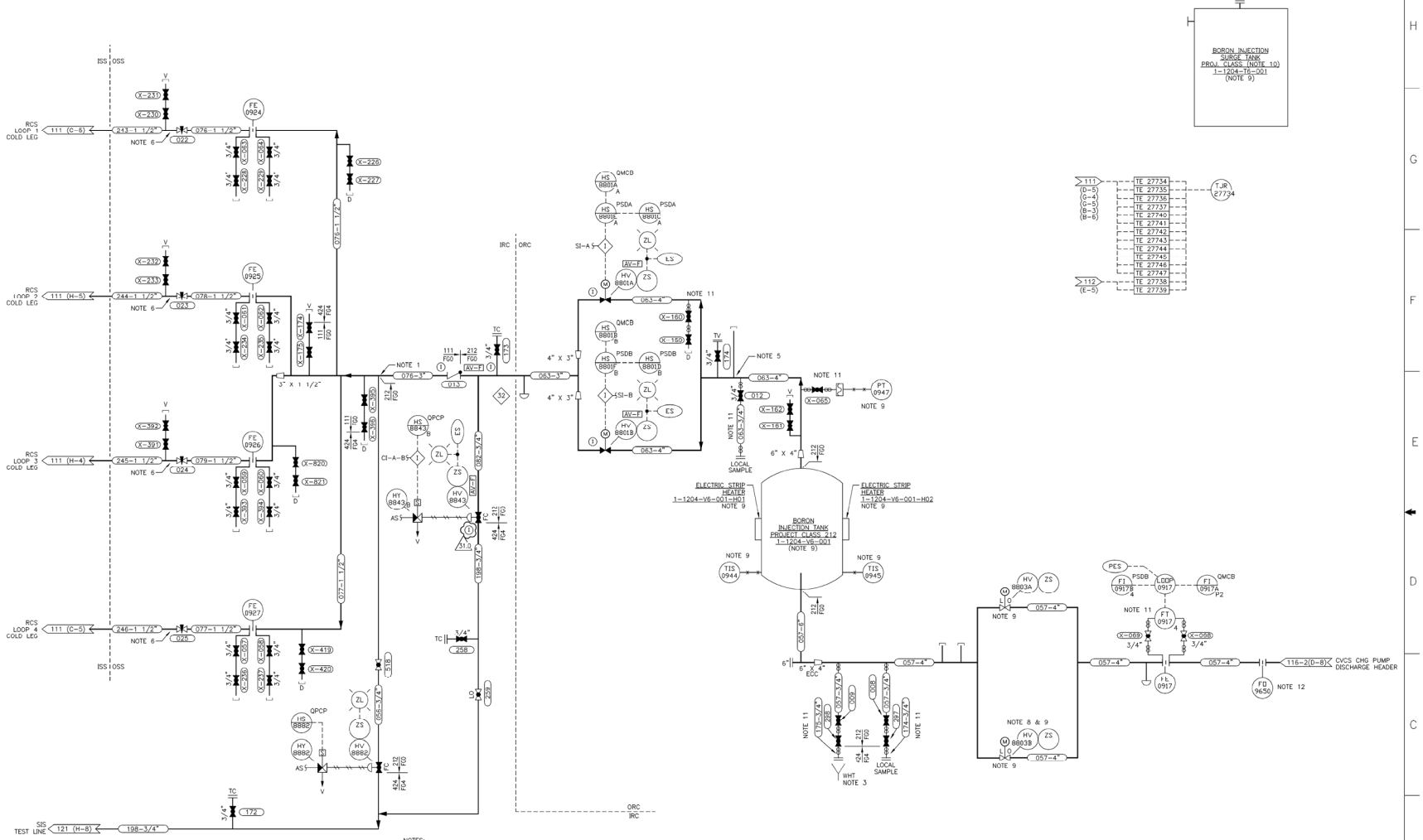
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CHEMICAL & VOLUME CONTROL
SYSTEM NO. 1208

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB118 | 36.0 |

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN IN BPC DWG CX4DG001.
2. ALL EQUIPMENT AND VALVES PROVIDED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2 INCH AND SMALLER, AND EQUIPMENT MARKED ⊕.
3. FOR DRAIN HEADER DETAILS SEE DWG 1X4DB144.
4. LOOP TO EXTEND 12" BELOW AND ABOVE NOZZLE. TOTAL LINE LENGTH INCLUDING LOOP TO BE AS SHORT AS POSSIBLE.
5. LOCATE CONNECTION CLOSE TO VALVE HVB801A AND B.
6. ADJUST VALVES IN FIELD TO LIMIT CENTRIFUGAL CHARGING PUMP RUNOUT, THEN LOCK IN POSITION. REFER TO SIS START-UP TEST PROCEDURE GAE/GBE-SU-2.3.3. BECHTEL LOG X6AA13-B.
7. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
8. AIR AND ELECTRIC POWER SUPPLY ARE DISCONNECTED. VALVES/INSTRUMENTS ARE INSTALLED BUT NOT REQUIRED FOR SYSTEM OPERATION.
9. ELECTRIC POWER SUPPLY IS DISCONNECTED. INSTRUMENTS/EQUIPMENT ARE INSTALLED BUT NOT REQUIRED FOR SYSTEM OPERATION.
10. ASME OR ANSI CODES ARE NOT APPLICABLE TO EQUIPMENT AND PIPING WITHIN ABANDONED SYSTEM BOUNDARY.
11. HEAT TRACE CABLE HAS BEEN RETIRED IN PLACE. ONLY TEMPERATURE ANNUNCIATION IS AVAILABLE. HEAT TRACE CABLE CAN BE REMOVED.
12. ORIFICE ASSEMBLY; SEE VENDOR DRAWING 1X6AG01-10001 & 10002.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E71, SHEET 1 OF 3.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204

CAD 1X4DB119
ACAD2000 | BgT - 04

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 31.0 | REVISED PER ABN-V02157, VER.1.0 | 5-12-10 | BgT | LRP | PML | |

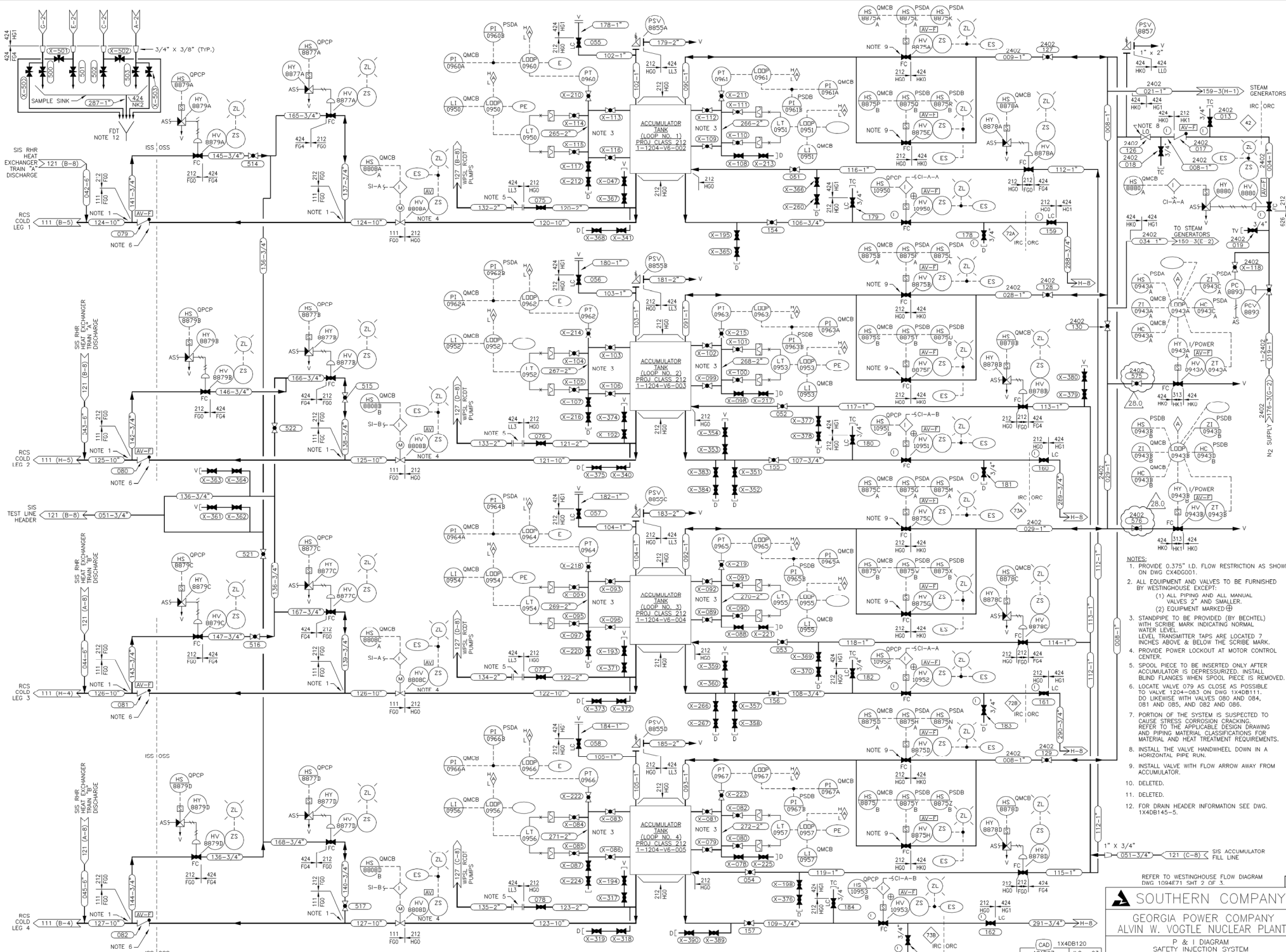
SIZE E 34X44

SCALE: NONE

DRAWING NO. 1X4DB119 VER. 31.0

JOB NO. 10604

DRAWING CATEGORY: CRITICAL




NOTES:

1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON DWG CX4DG001.
2. ALL EQUIPMENT AND VALVES TO BE FURNISHED BY WESTINGHOUSE EXCEPT:
   (1) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
   (2) EQUIPMENT MARKED ⊕
3. STANDPIPE TO BE PROVIDED (BY BECHTEL) WITH SCRIBE MARK INDICATING NORMAL WATER LEVEL.
   LEVEL TRANSMITTER TAPS ARE LOCATED 7 INCHES ABOVE & BELOW THE SCRIBE MARK.
4. PROVIDE POWER LOCKOUT AT MOTOR CONTROL CENTER.
5. SPOOL PIECE TO BE INSERTED ONLY AFTER ACCUMULATOR IS DEPRESSURIZED. INSTALL BLIND FLANGES WHEN SPOOL PIECE IS REMOVED.
6. LOCATE VALVE 079 AS CLOSE AS POSSIBLE TO VALVE 1204-083 ON DWG 1X4DB111. DO LIKEWISE WITH VALVES 080 AND 084, 081 AND 085, AND 082 AND 086.
7. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
8. INSTALL THE VALVE HANDWHEEL DOWN IN A HORIZONTAL PIPE RUN.
9. INSTALL VALVE WITH FLOW ARROW AWAY FROM ACCUMULATOR.
10. DELETED.
11. DELETED.
12. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB145-5.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204

CAD 1X4DB120
ACAD07 JLO - 03

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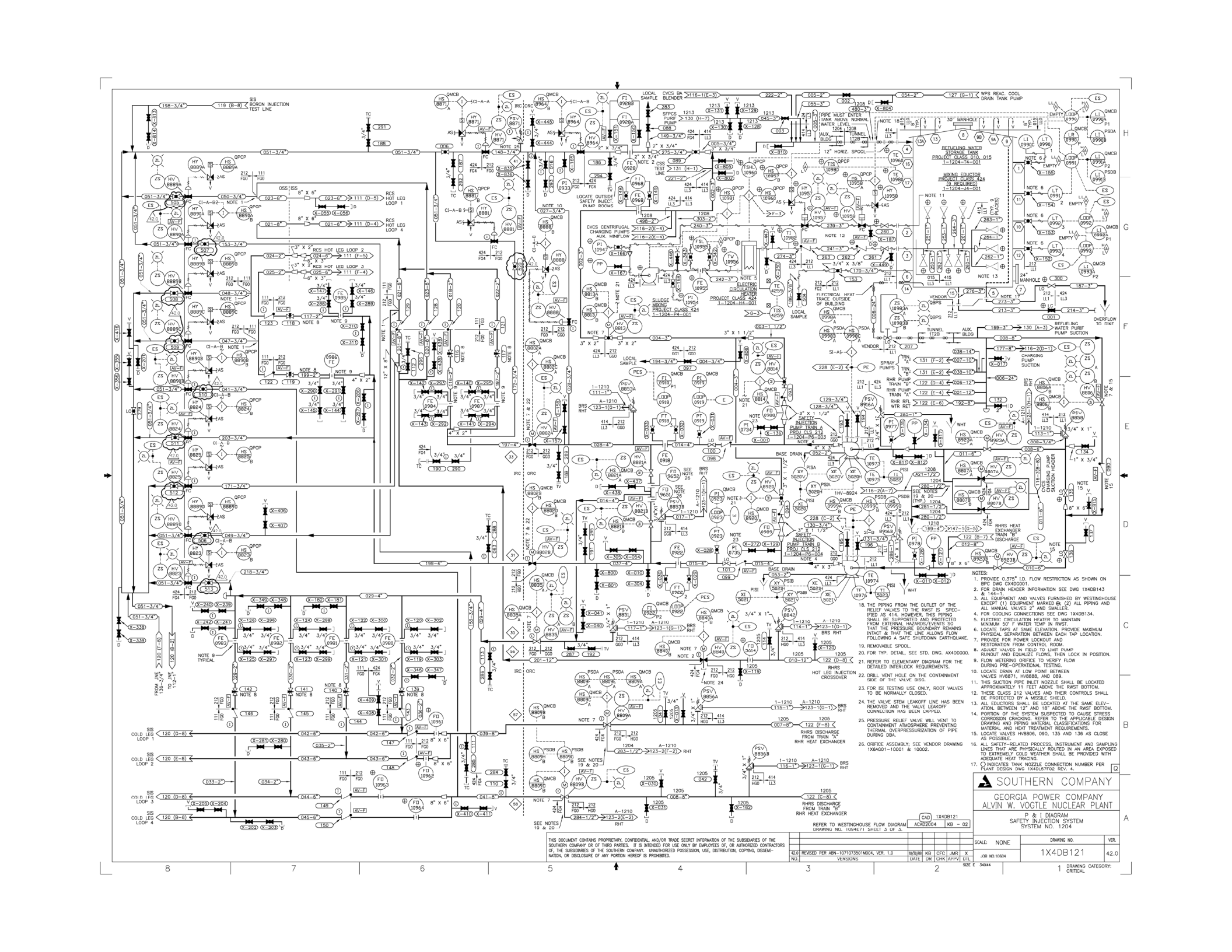
| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 28.0 | REVISED PER ABN 1039001201M016, VER. 1.0 | 10/8/12 | JLO | WLM | JMR | X |

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB120
VER. 28.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




NOTES:

1. PROVIDE 0.375" I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG CX4DG001.
2. FOR DRAIN HEADER INFORMATION SEE DWG 1X4DB143 & 144-1.
3. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT (1) EQUIPMENT MARKED @; (2) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
4. FOR COOLING CONNECTIONS SEE DWG 1X4DB134.
5. ELECTRIC CIRCULATION HEATER TO MAINTAIN MINIMUM 50° F WATER TEMP IN RWST.
6. LOCATE TAPS AT SAME ELEVATION. PROVIDE MAXIMUM PHYSICAL SEPARATION BETWEEN EACH TAP LOCATION.
7. PROVIDE FOR POWER LOCKOUT AND RESTORATION FROM CONTROL ROOM.
8. ADJUST VALVES IN FIELD TO LIMIT PUMP RUNOUT AND EQUALIZE FLOWS, THEN LOCK IN POSITION.
9. FLOW METERING ORIFICE TO VERIFY FLOW DURING PRE-OPERATIONAL TESTING.
10. LOCATE DRAIN AT LOW POINT BETWEEN VALVES HV8871, HV8888, AND 089.
11. THIS SUCTION PIPE INLET NOZZLE SHALL BE LOCATED APPROXIMATELY 11 FEET ABOVE THE RWST BOTTOM.
12. THESE CLASS 212 VALVES AND THEIR CONTROLS SHALL BE PROTECTED BY A MISSILE SHIELD.
13. ALL EDUCTORS SHALL BE LOCATED AT THE SAME ELEVATION, BETWEEN 12" AND 18" ABOVE THE RWST BOTTOM.
14. PORTION OF THE SYSTEM SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
15. LOCATE VALVES HV8806, 090, 135 AND 136 AS CLOSE AS POSSIBLE.
16. ALL SAFETY-RELATED PROCESS, INSTRUMENT AND SAMPLING LINES THAT ARE PHYSICALLY ROUTED IN AN AREA EXPOSED TO EXTREMELY COLD WEATHER SHALL BE PROVIDED WITH ADEQUATE HEAT TRACING.
17. ○ INDICATES TANK NOZZLE CONNECTION NUMBER PER PLANT DESIGN DWG 1X4DL57F02 REV. 4. Q
18. THE PIPING FROM THE OUTLET OF THE RELIEF VALVES TO THE RWST IS SPECIFIED AS 414. HOWEVER, THIS PIPING SHALL BE SUPPORTED AND PROTECTED FROM EXTERNAL HAZARDS/EVENTS SO THAT THE PRESSURE BOUNDARY REMAINS INTACT & THAT THE LINE ALLOWS FLOW FOLLOWING A SAFE SHUTDOWN EARTHQUAKE.
19. REMOVABLE SPOOL.
20. FOR TYP. DETAIL, SEE STD. DWG. AX4DD000.
21. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK REQUIREMENTS.
22. DRILL VENT HOLE ON THE CONTAINMENT SIDE OF THE VALVE DISC.
23. FOR ISI TESTING USE ONLY, ROOT VALVES TO BE NORMALLY CLOSED.
24. THE VALVE STEM LEAKOFF LINE HAS BEEN REMOVED AND THE VALVE LEAKOFF CONNECTION HAS BEEN CAPPED.
25. PRESSURE RELIEF VALVE WILL VENT TO CONTAINMENT ATMOSPHERE PREVENTING THERMAL OVERPRESSURIZATION OF PIPE DURING DBA.
26. ORIFICE ASSEMBLY; SEE VENDOR DRAWING 1X8AG01-10001 & 10002.

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| NO. | VERSIONS | DATE | DN | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 42.0 | REVISED PER ABN-10710735501M004, VER. 1.0 | 10/29/09 | KB | CFC | JMR | X |

CAD 1X4DB121 ACAD2004 KB – 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SAFETY INJECTION SYSTEM
SYSTEM NO. 1204

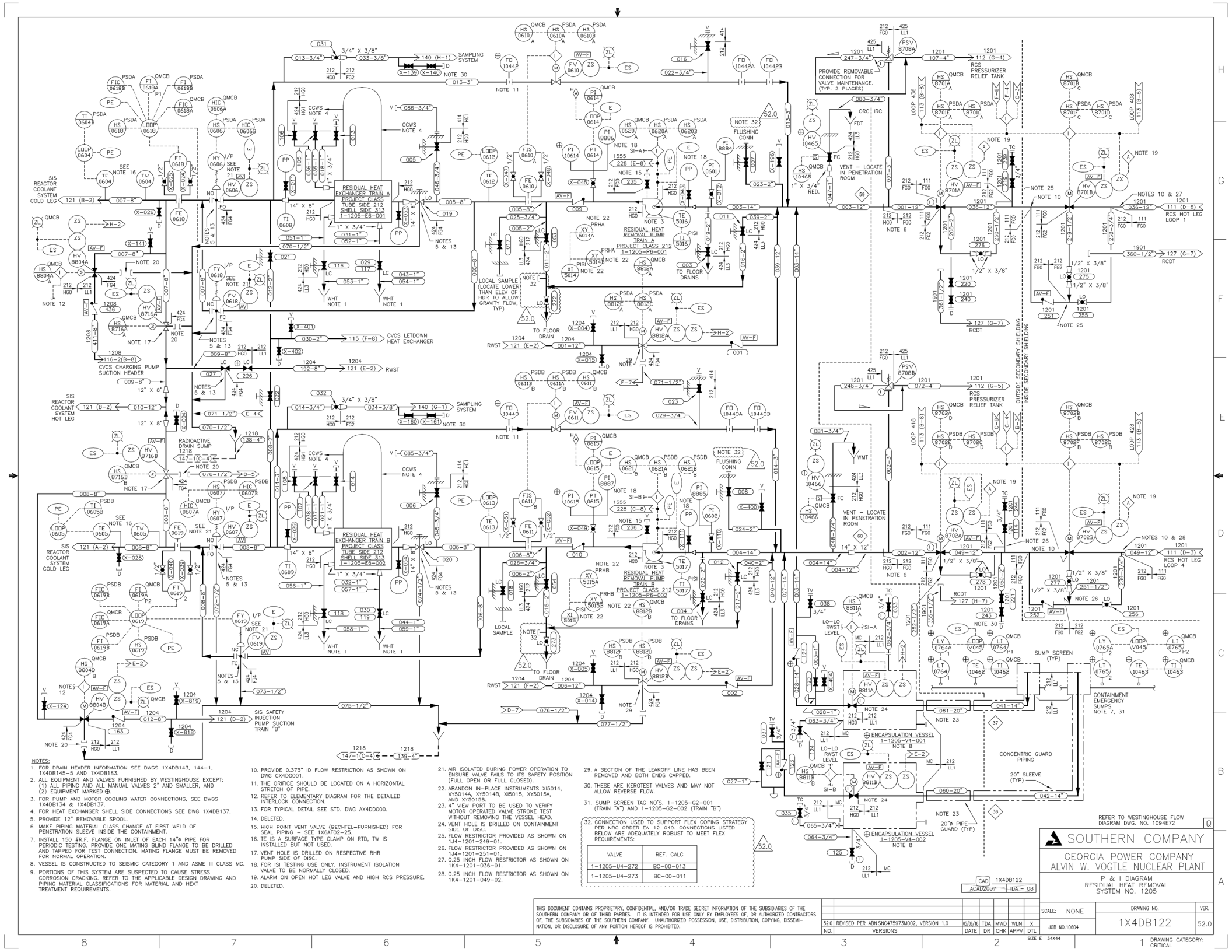
SCALE: NONE

DRAWING NO. 1X4DB121 VER. 42.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWGS 1X4DB143, 144-1, 1X4DB145-5 AND 1X4DB183.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT: (1) ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER, AND (2) EQUIPMENT MARKED ⊕.
3. FOR PUMP AND MOTOR COOLING WATER CONNECTIONS, SEE DWGS 1X4DB134 & 1X4DB137.
4. FOR HEAT EXCHANGER SHELL SIDE CONNECTIONS SEE DWG 1X4DB137.
5. PROVIDE 12" REMOVABLE SPOOL.
6. MAKE PIPING MATERIAL CLASS CHANGE AT FIRST WELD OF PENETRATION SLEEVE INSIDE THE CONTAINMENT.
7. INSTALL 150 #R.F. FLANGE ON INLET OF EACH 14"ø PIPE FOR PERIODIC TESTING. PROVIDE ONE MATING BLIND FLANGE TO BE DRILLED AND TAPPED FOR TEST CONNECTION. MATING FLANGE MUST BE REMOVED FOR NORMAL OPERATION.
8. VESSEL IS CONSTRUCTED TO SEISMIC CATEGORY 1 AND ASME III CLASS MC.
9. PORTIONS OF THIS SYSTEM ARE SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
10. PROVIDE 0.375" ID FLOW RESTRICTION AS SHOWN ON DWG CX4DG001.
11. THE ORIFICE SHOULD BE LOCATED ON A HORIZONTAL STRETCH OF PIPE.
12. REFER TO ELEMENTARY DIAGRAM FOR THE DETAILED INTERLOCK CONNECTION.
13. FOR TYPICAL DETAIL SEE STD. DWG AX4DD000.
14. DELETED.
15. HIGH POINT VENT VALVE (BECHTEL–FURNISHED) FOR SEAL PIPING – SEE 1X6AF02–25.
16. TE IS A SURFACE TYPE CLAMP ON RTD, TW IS INSTALLED BUT NOT USED.
17. VENT HOLE IS DRILLED ON RESPECTIVE RHR PUMP SIDE OF DISC.
18. FOR ISI TESTING USE ONLY. INSTRUMENT ISOLATION VALVE TO BE NORMALLY CLOSED.
19. ALARM ON OPEN HOT LEG VALVE AND HIGH RCS PRESSURE.
20. DELETED.
21. AIR ISOLATED DURING POWER OPERATION TO ENSURE VALVE FAILS TO ITS SAFETY POSITION (FULL OPEN OR FULL CLOSED).
22. ABANDON IN–PLACE INSTRUMENTS XI5014, XY5014A, XY5014B, XI5015, XY5015A, AND XY5015B.
23. 4" VIEW PORT TO BE USED TO VERIFY MOTOR OPERATED VALVE STROKE TEST WITHOUT REMOVING THE VESSEL HEAD.
24. VENT HOLE IS DRILLED ON CONTAINMENT SIDE OF DISC.
25. FLOW RESTRICTOR PROVIDED AS SHOWN ON 1J4–1201–249–01.
26. FLOW RESTRICTOR PROVIDED AS SHOWN ON 1J4–1201–251–01.
27. 0.25 INCH FLOW RESTRICTOR AS SHOWN ON 1K4–1201–036–01.
28. 0.25 INCH FLOW RESTRICTOR AS SHOWN ON 1K4–1201–049–02.
29. A SECTION OF THE LEAKOFF LINE HAS BEEN REMOVED AND BOTH ENDS CAPPED.
30. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
31. SUMP SCREEN TAG NO'S. 1–1205–G2–001 (TRAIN "A") AND 1–1205–G2–002 (TRAIN "B")
32. CONNECTION USED TO SUPPORT FLEX COPING STRATEGY PER NRC ORDER EA–12–049. CONNECTIONS LISTED BELOW ARE ADEQUATELY ROBUST TO MEET FLEX REQUIREMENTS:

| VALVE | REF. CALC |
|---|---|
| 1–1205–U4–272 | BC–00–013 |
| 1–1205–U4–273 | BC–00–011 |

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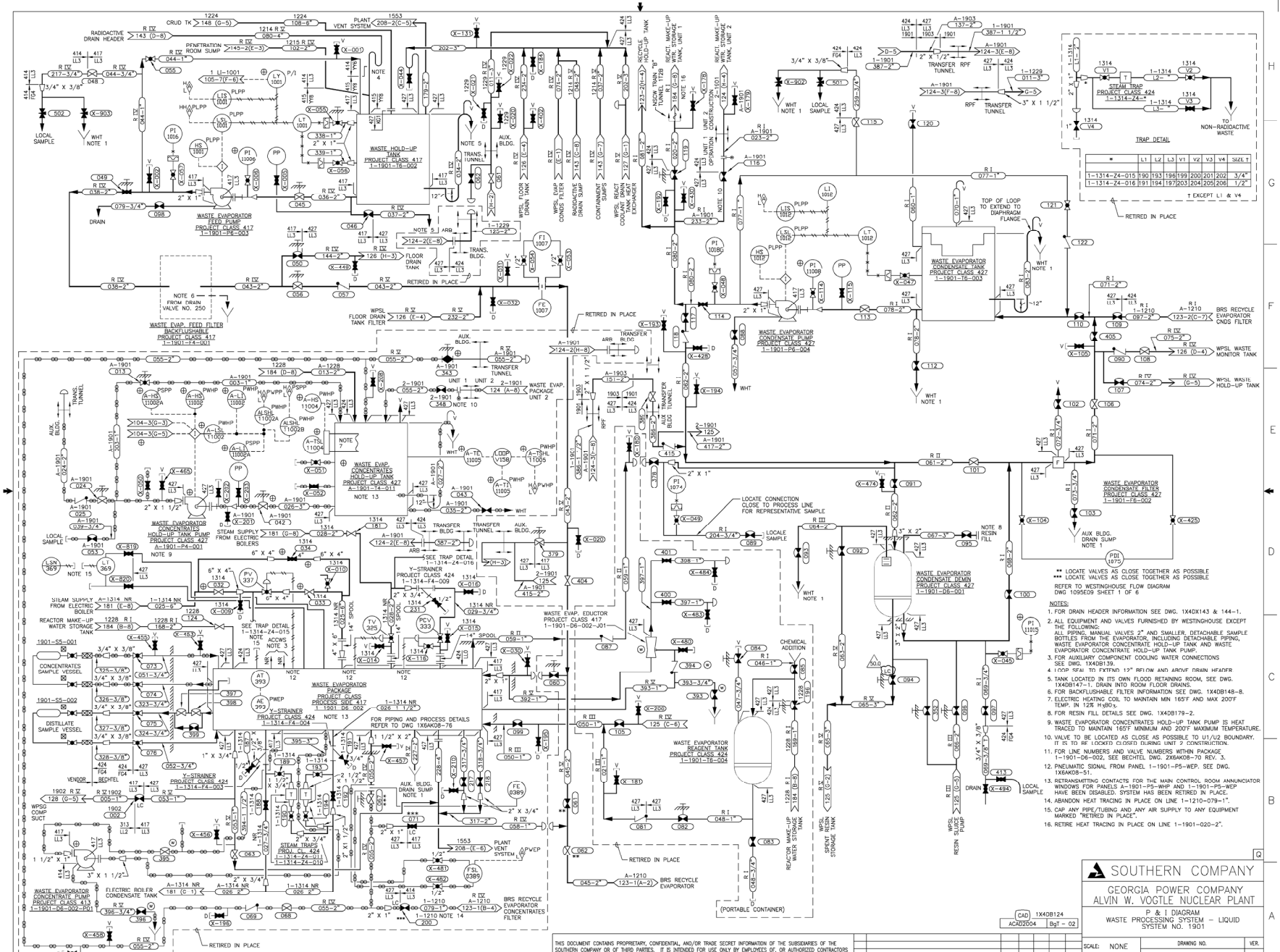
| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 52.0 | REVISED PER ABN SNC475973M002, VERSION 1.0 | 05/06/16 | TDA | MWD | WLN | X |

CAD 1X4DB122
ACAD2007 IDA – 08

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
RESIDUAL HEAT REMOVAL
SYSTEM NO. 1205

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB122
VER. 52.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




| * | L1 | L2 | L3 | V1 | V2 | V3 | V4 | SIZE T |
|---|---|---|---|---|---|---|---|---|
| 1-1314-Z4-015 | 190 | 193 | 196 | 199 | 200 | 201 | 202 | 3/4" |
| 1-1314-Z4-016 | 191 | 194 | 197 | 203 | 204 | 205 | 206 | 1/2" |

T EXCEPT L1 & V4

NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DX143 & 144-1.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT THE FOLLOWING: ALL PIPING, MANUAL VALVES 2" AND SMALLER, DETACHABLE SAMPLE BOTTLES FROM THE EVAPORATOR, INCLUDING DETACHABLE PIPING, WASTE EVAPORATOR CONCENTRATE HOLD-UP TANK AND WASTE EVAPORATOR CONCENTRATE HOLD-UP TANK PUMP.
3. FOR AUXILIARY COMPONENT COOLING WATER CONNECTIONS SEE DWG. 1X4DB139.
4. LOOP SEAL TO EXTEND 12" BELOW AND ABOVE DRAIN HEADER
5. TANK LOCATED IN ITS OWN FLOOD RETAINING ROOM, SEE DWG. 1X4DB147-1. DRAIN INTO ROOM FLOOR DRAINS.
6. FOR BACKFLUSHABLE FILTER INFORMATION SEE DWG. 1X4DB148-8.
7. ELECTRIC HEATING COIL TO MAINTAIN MIN 160°F AND MAX 200°F TEMP. IN 12% $H_2BO_3$.
8. FOR RESIN FILL DETAILS SEE DWG. 1X4DB179-2.
9. WASTE EVAPORATOR CONCENTRATES HOLD-UP TANK PUMP IS HEAT TRACED TO MAINTAIN 165°F MINIMUM AND 200°F MAXIMUM TEMPERATURE.
10. VALVE TO BE LOCATED AS CLOSE AS POSSIBLE TO U1/U2 BOUNDARY. IT IS TO BE LOCKED CLOSED DURING UNIT 2 CONSTRUCTION.
11. FOR LINE NUMBERS AND VALVE NUMBERS WITHIN PACKAGE 1-1901-D6-002, SEE BECHTEL DWG. 2X6AK08-70 REV. 3.
12. PNEUMATIC SIGNAL FROM PANEL 1-1901-P5-WEP. SEE DWG. 1X6AK08-51.
13. RETRANSMITTING CONTACTS FOR THE MAIN CONTROL ROOM ANNUNCIATOR WINDOWS FOR PANELS A-1901-P5-WHP AND 1-1901-P5-WEP HAVE BEEN DISABLED. SYSTEM HAS BEEN RETIRED IN PLACE.
14. ABANDON HEAT TRACING IN PLACE ON LINE 1-1210-079-1".
15. CAP ANY PIPE/TUBING AND ANY AIR SUPPLY TO ANY EQUIPMENT MARKED "RETIRED IN PLACE".
16. RETIRE HEAT TRACING IN PLACE ON LINE 1-1901-020-2".

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| NO. | VERSIONS | DATE | DN | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 50.0 | REVISED PER ABN-V01041, VER.1.0 | 10-31-07 | BgT | MWD | WLN | X |

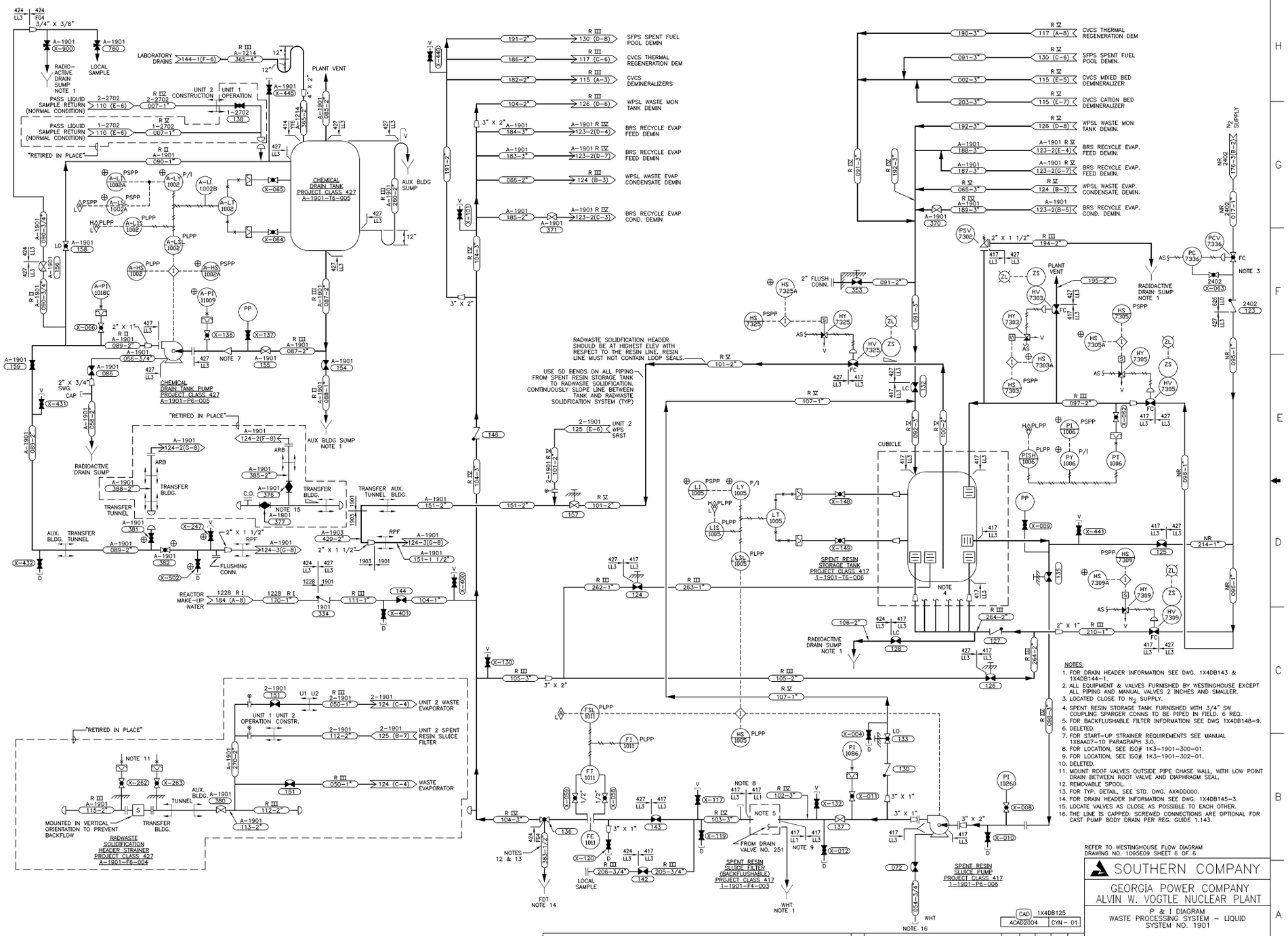
CAD 1X4DB124
ACAD2004 BgT - 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE PROCESSING SYSTEM – LIQUID
SYSTEM NO. 1901

SCALE: NONE
DRAWING NO. 1X4DB124
VER. 50.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 & 1X4DB144-1.
2. ALL EQUIPMENT & VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND MANUAL VALVES 2 INCHES AND SMALLER.
3. LOCATED CLOSE TO $N_2$ SUPPLY.
4. SPENT RESIN STORAGE TANK FURNISHED WITH 3/4" SW COUPLING SPARGER COMNS TO BE PIPED IN FIELD. 6 REQ.
5. FOR BACKFLUSHABLE FILTER INFORMATION SEE DWG 1X4DB148-9.
6. DELETED.
7. FOR START-UP STRAINER REQUIREMENTS SEE MANUAL 1X6AA07-10 PARAGRAPH 3.0.
8. FOR LOCATION, SEE ISO# 1K3-1901-300-01.
9. FOR LOCATION, SEE ISO# 1K3-1901-302-01.
10. DELETED.
11. MOUNT ROOT VALVES OUTSIDE PIPE CHASE WALL, WITH LOW POINT DRAIN BETWEEN ROOT VALVE AND DIAPHRAGM SEAL.
12. REMOVABLE SPOOL.
13. FOR TYP. DETAIL, SEE STD. DWG. AX4DD000.
14. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB145-3.
15. LOCATE VALVES AS CLOSE AS POSSIBLE TO EACH OTHER.
16. THE LINE IS CAPPED. SCREWED CONNECTIONS ARE OPTIONAL FOR CAST PUMP BODY DRAIN PER REG. GUIDE 1.143.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1095E09 SHEET 6 OF 6

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 35.0 | REDRAWN PER ABN-V70125, VER. 1.0 | 09/13/06 | CYN | LCF | LKB | X |

CAD 1X4DB125
ACAD2004 CYN- 01

SOUTHERN COMPANY

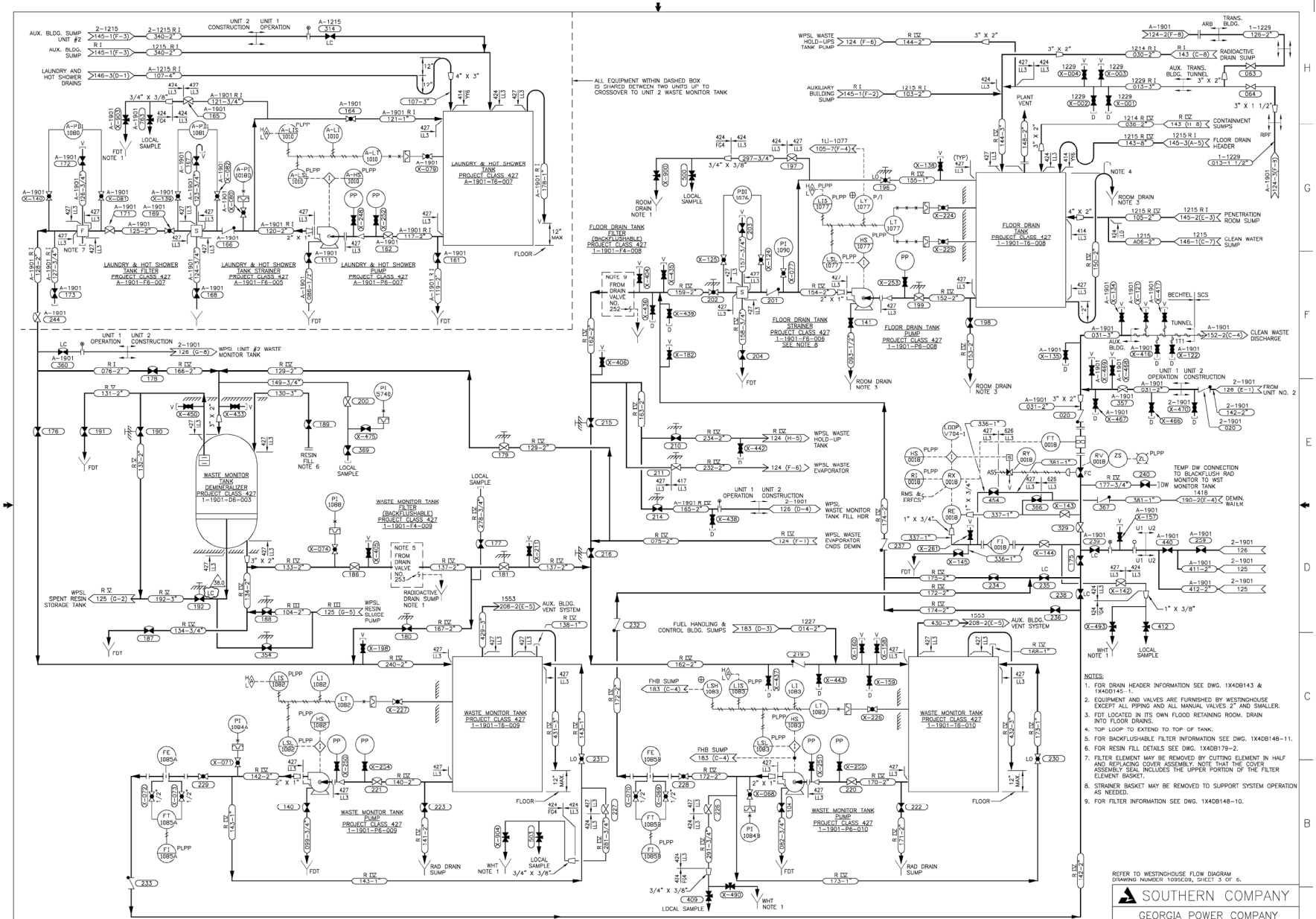
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE PROCESSING SYSTEM – LIQUID
SYSTEM NO. 1901

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB125 | 35.0 |

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 & 1X4DB145-1.
2. EQUIPMENT AND VALVES ARE FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2" AND SMALLER.
3. FDT LOCATED IN ITS OWN FLOOD RETAINING ROOM. DRAIN INTO FLOOR DRAINS.
4. TOP LOOP TO EXTEND TO TOP OF TANK.
5. FOR BACKFLUSHABLE FILTER INFORMATION SEE DWG. 1X4DB148-11.
6. FOR RESIN FILL DETAILS SEE DWG. 1X4DB179-2.
7. FILTER ELEMENT MAY BE REMOVED BY CUTTING ELEMENT IN HALF AND REPLACING COVER ASSEMBLY. NOTE THAT THE COVER ASSEMBLY SEAL INCLUDES THE UPPER PORTION OF THE FILTER ELEMENT BASKET.
8. STRAINER BASKET MAY BE REMOVED TO SUPPORT SYSTEM OPERATION AS NEEDED.
9. FOR FILTER INFORMATION SEE DWG. 1X4DB148-10.

REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NUMBER 1095C09, SHEET 3 OF 6.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE PROCESSING SYSTEM – LIQUID
SYSTEM NO. 1901

CAD 1X4DB126
ACAD2004 BgT – 02

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| NO. | VERSION | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 38.0 | REVISED PER ABN-V01041, VER.1.0 | 10-31-07 | BgT | MWD | WLN | X |

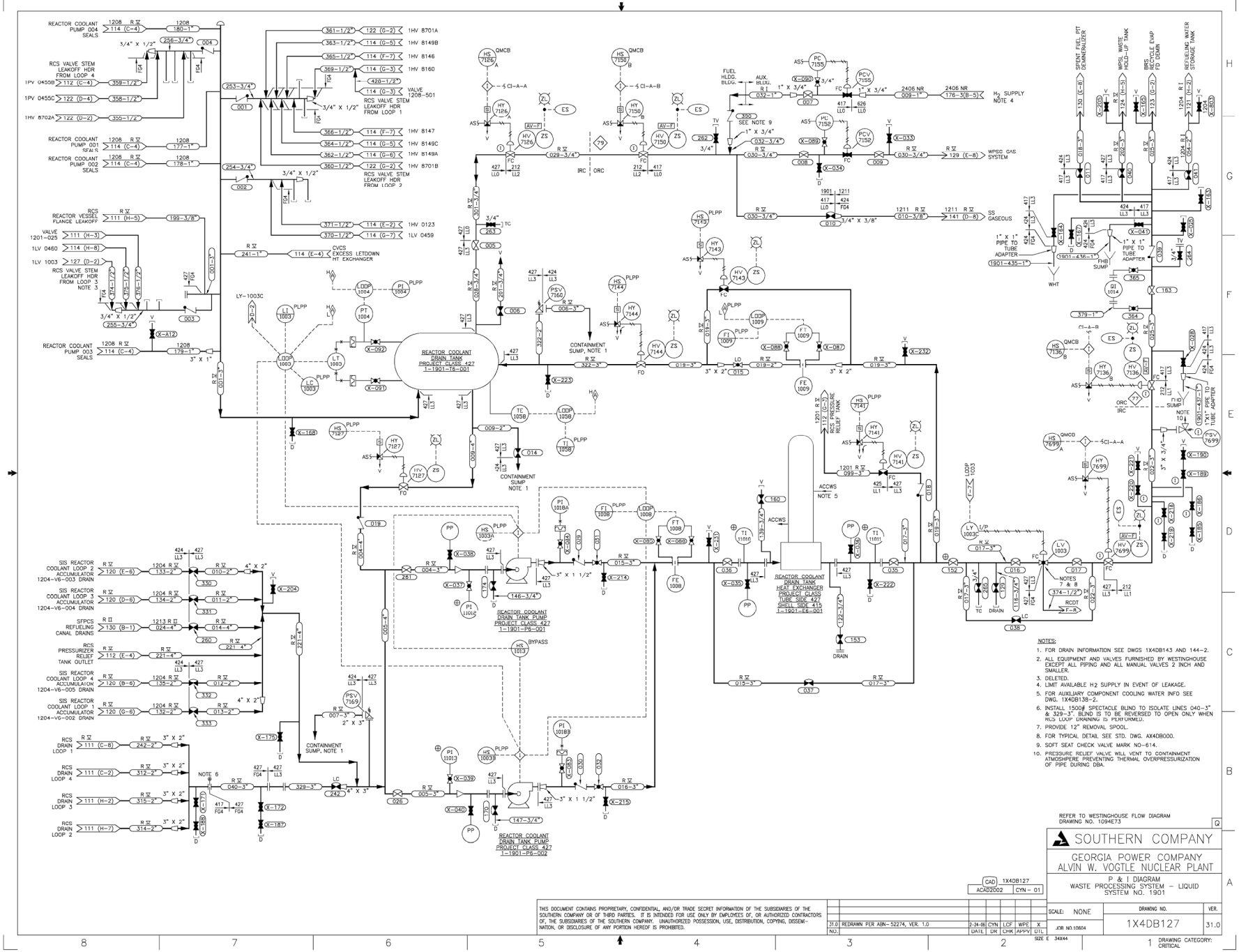
SIZE E 34X44

SCALE: NONE

DRAWING NO. 1X4DB126 VER. 38.0

JOB NO.10604

1 DRAWING CATEGORY: CRITICAL


REACTOR COOLANT PUMP 004 SEALS
REACTOR COOLANT PUMP 001 SEALS
REACTOR COOLANT PUMP 002 SEALS
REACTOR COOLANT PUMP 003 SEALS
RCS VALVE STEM LEAKOFF HDR FROM LOOP 4
RCS VALVE STEM LEAKOFF HDR FROM LOOP 1
RCS VALVE STEM LEAKOFF HDR FROM LOOP 2
RCS VALVE STEM LEAKOFF HDR FROM LOOP 3 NOTE 3
RCS REACTOR VESSEL FLANGE LEAKOFF
CVCS EXCESS LETDOWN HT EXCHANGER
REACTOR COOLANT DRAIN TANK PRO-REF CLASS 422 1-1901-P6-001
CONTAINMENT SUMP, NOTE 1
REACTOR COOLANT DRAIN TANK PUMP PROJECT CLASS 427 1-1901-P6-001
REACTOR COOLANT DRAIN TANK HEAT EXCHANGER PROJECT CLASS TUBE SIDE 427 SHELL SIDE 415 1-1901-E6-001
REACTOR COOLANT DRAIN TANK PUMP PROJECT CLASS 427 1-1901-P6-002
RCS PRESSURE RELIEF TANK
ACCWS
ACCWS NOTE 5
SPENT FUEL PIT DEMINERALIZER
WPSL WASTE HOLD-UP TANK
BRS RECYCLE EVAP FD DEMIN
REFUELING WATER STORAGE TANK
SS GASEOUS
WPSC GAS SYSTEM
H2 SUPPLY NOTE 4
FHB SUMP
T110 SUMP
RCDT
SIS REACTOR COOLANT LOOP 2 ACCUMULATOR 1204-V6-003 DRAIN
SIS REACTOR COOLANT LOOP 3 ACCUMULATOR 1204-V6-004 DRAIN
SFPCS REFUELING CANAL DRAINS
RCS PRESSURIZER RELIEF TANK OUTLET
SIS REACTOR COOLANT LOOP 4 ACCUMULATOR 1204-V6-005 DRAIN
SIS REACTOR COOLANT LOOP 1 ACCUMULATOR 1204-V6-002 DRAIN
RCS DRAIN LOOP 1
RCS DRAIN LOOP 4
RCS DRAIN LOOP 3
RCS DRAIN LOOP 2
BYPASS
DRAIN
NOTES:
1. FOR DRAIN INFORMATION SEE DWGS 1X4DB143 AND 144-2.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2 INCH AND SMALLER.
3. DELETED.
4. LIMIT AVAILABLE H2 SUPPLY IN EVENT OF LEAKAGE.
5. FOR AUXILIARY COMPONENT COOLING WATER INFO SEE DWG. 1X4DB138-2.
6. INSTALL 1500# SPECTACLE BLIND TO ISOLATE LINES 040-3" & 329-3". BLIND IS TO BE REVERSED TO OPEN ONLY WHEN RCS LOOP DRAINING IS PERFORMED.
7. PROVIDE 12" REMOVAL SPOOL.
8. FOR TYPICAL DETAIL SEE STD. DWG. AX4DB000.
9. SOFT SEAT CHECK VALVE MARK NO-614.
10. PRESSURE RELIEF VALVE WILL VENT TO CONTAINMENT ATMOSPHERE PREVENTING THERMAL OVERPRESSURIZATION OF PIPE DURING DBA.
REFER TO WESTINGHOUSE FLOW DIAGRAM DRAWING NO. 1094E73
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
WASTE PROCESSING SYSTEM – LIQUID
SYSTEM NO. 1901
CAD 1X4DB127
ACAD2002 CYN– 01
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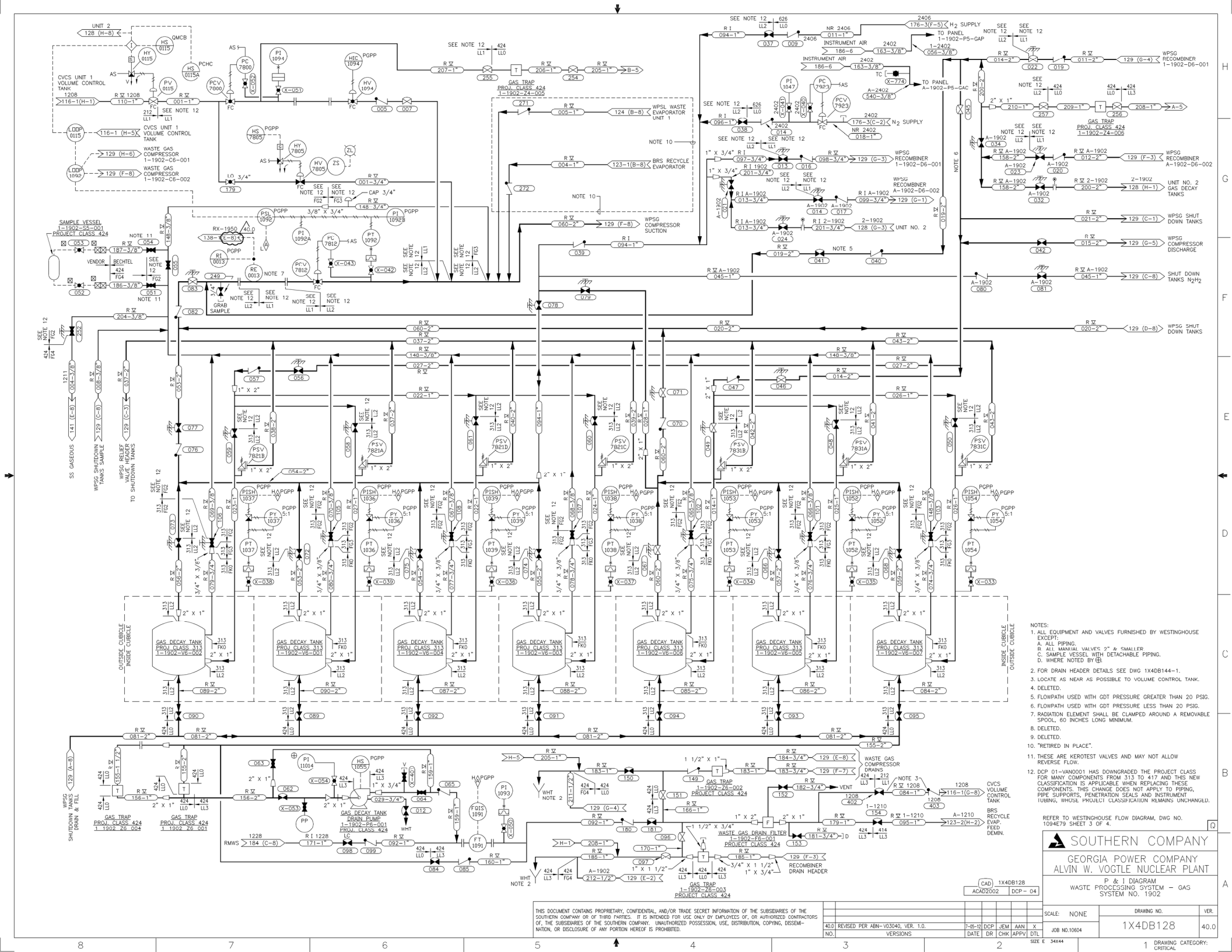
31.0 REDRAWN PER ABN– 52274, VER. 1.0 2-24-06 CYN LCF WPE X
SCALE: NONE
DRAWING NO. 1X4DB127
VER. 31.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




NOTES:
1. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT:
   A. ALL PIPING.
   B. ALL MANUAL VALVES 2" & SMALLER.
   C. SAMPLE VESSEL WITH DETACHABLE PIPING.
   D. WHERE NOTED BY Ⓓ.
2. FOR DRAIN HEADER DETAILS SEE DWG 1X4DB144-1.
3. LOCATE AS NEAR AS POSSIBLE TO VOLUME CONTROL TANK.
4. DELETED.
5. FLOWPATH USED WITH GDT PRESSURE GREATER THAN 20 PSIG.
6. FLOWPATH USED WITH GDT PRESSURE LESS THAN 20 PSIG.
7. RADIATION ELEMENT SHALL BE CLAMPED AROUND A REMOVABLE SPOOL, 60 INCHES LONG MINIMUM.
8. DELETED.
9. DELETED.
10. "RETIRED IN PLACE".
11. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.
12. DCP 01-VAN0001 HAS DOWNGRADED THE PROJECT CLASS FOR MANY COMPONENTS FROM 313 TO 417 AND THIS NEW CLASSIFICATION IS APPLICABLE WHEN REPLACING THESE COMPONENTS. THIS CHANGE DOES NOT APPLY TO PIPING, PIPE SUPPORTS, PENETRATION SEALS AND INSTRUMENT TUBING, WHOSE PROJECT CLASSIFICATION REMAINS UNCHANGED.

REFER TO WESTINGHOUSE FLOW DIAGRAM, DWG NO. 1094E79 SHEET 3 OF 4.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE PROCESSING SYSTEM – GAS
SYSTEM NO. 1902

| SCALE: | NONE | DRAWING NO. | VER. |
|---|---|---|---|
| JOB NO. | 10604 | 1X4DB128 | 40.0 |

CAD 1X4DB128
ACAD2002 | DCP- 04

| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 40.0 | REVISED PER ABN-V03040, VER. 1.0. | 7-05-12 | DCP | JEM | AAN | X |

SIZE E 34X44

DRAWING CATEGORY: 1 CRITICAL

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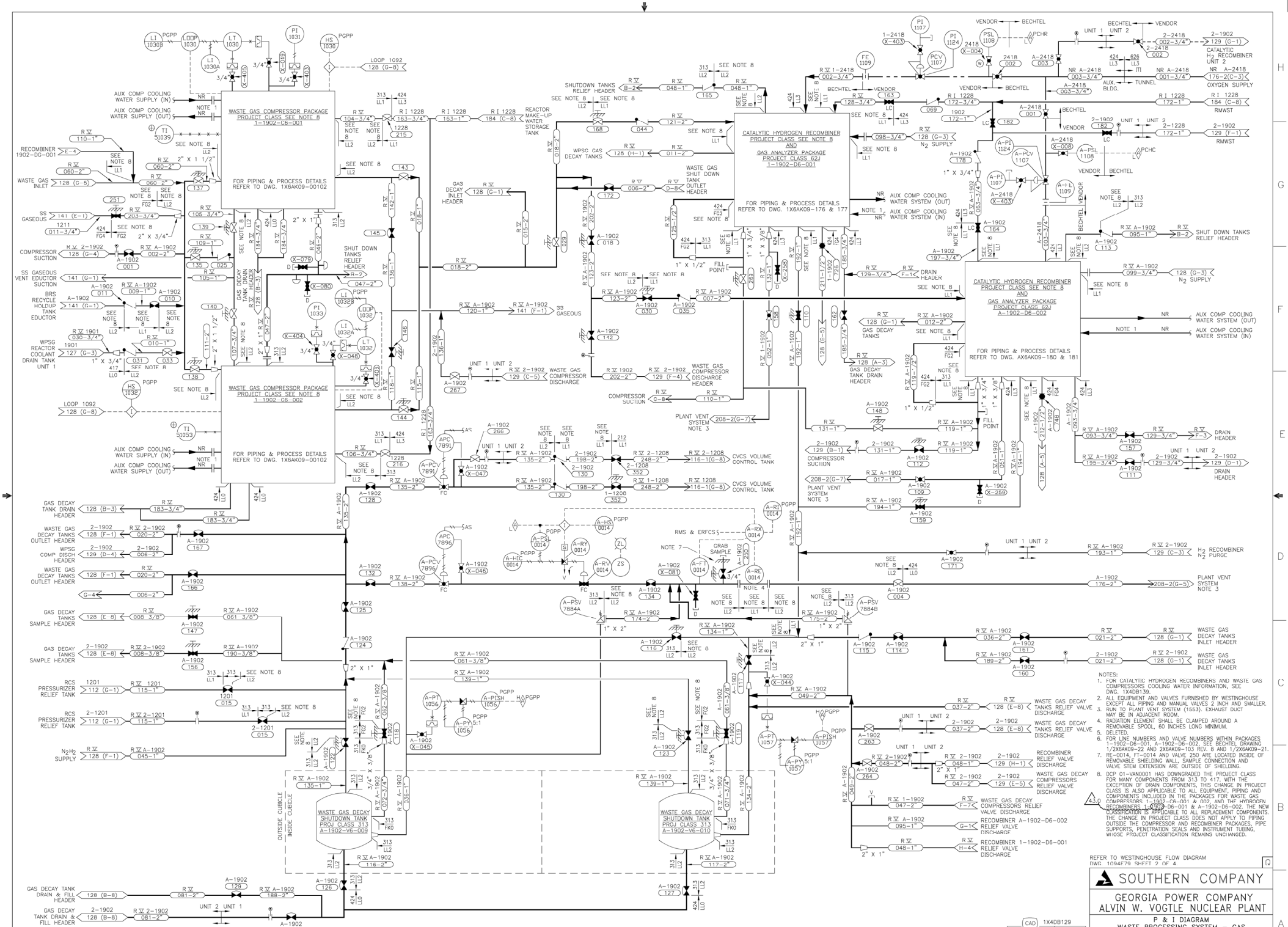
WASTE GAS COMPRESSOR PACKAGE
PROJECT CLASS SEE NOTE 8
1-1902-D6-001
FOR PIPING & PROCESS DETAILS
REFER TO DWG. 1X6AK09-00102
WASTE GAS COMPRESSOR PACKAGE
PROJECT CLASS SEE NOTE 8
1-1902-D6-002
FOR PIPING & PROCESS DETAILS
REFER TO DWG. 1X6AK09-00102
CATALYTIC HYDROGEN RECOMBINER
PROJECT CLASS SEE NOTE 8
AND
GAS ANALYZER PACKAGE
PROJECT CLASS 62J
1-1902-D6-001
FOR PIPING & PROCESS DETAILS
REFER TO DWG. 1X6AK09-176 & 177
CATALYTIC HYDROGEN RECOMBINER
PROJECT CLASS SEE NOTE 8
AND
GAS ANALYZER PACKAGE
PROJECT CLASS 62J
A-1902-D6-002
FOR PIPING & PROCESS DETAILS
REFER TO DWG. AX6AK09-180 & 181
WASTE GAS DECAY
SHUTDOWN TANK
PROJ CLASS 31.3
A-1902-V6-009
WASTE GAS DECAY
SHUTDOWN TANK
PROJ CLASS 31.3
A-1902-V6-010
OUTSIDE CUBICLE
INSIDE CUBICLE
SHUTDOWN TANKS RELIEF HEADER
WASTE GAS SHUT DOWN TANK OUTLET HEADER
GAS DECAY INLET HEADER
GAS DECAY TANK DRAIN HEADER
WASTE GAS COMPRESSOR DISCHARGE
WASTE GAS COMPRESSOR DISCHARGE HEADER
CVCS VOLUME CONTROL TANK
PLANT VENT SYSTEM NOTE 3
RMS & ERFCS
GRAB SAMPLE
AUX COMP COOLING WATER SYSTEM (OUT)
AUX COMP COOLING WATER SYSTEM (IN)
SS GASEOUS
SHUT DOWN TANKS RELIEF HEADER
WASTE GAS DECAY TANKS INLET HEADER
WASTE GAS DECAY TANKS RELIEF VALVE DISCHARGE
WASTE GAS DECAY COMPRESSORS RELIEF VALVE DISCHARGE
RECOMBINER RELIEF VALVE DISCHARGE
RECOMBINER A-1902-D6-002 RELIEF VALVE DISCHARGE
RECOMBINER 1-1902-D6-001 RELIEF VALVE DISCHARGE
H2 RECOMBINER N2 PURGE
CATALYTIC H2 RECOMBINER UNIT 2
OXYGEN SUPPLY
RMWST
N2 SUPPLY
COMPRESSOR SUCTION
WASTE GAS INLET
RECOMBINER 1902-DG-001
SS GASEOUS VENT EDUCTOR SUCTION
BRS RECYCLE HOLDUP TANK EDUCTOR
WPSG REACTOR COOLANT DRAIN TANK UNIT 1
WPSG COMP DISCH HEADER
WASTE GAS DECAY TANKS OUTLET HEADER
GAS DECAY TANKS SAMPLE HEADER
RCS PRESSURIZER RELIEF TANK
N2 H2 SUPPLY
GAS DECAY TANK DRAIN & FILL HEADER
REACTOR MAKE-UP WATER STORAGE TANK
WPSG GAS DECAY TANKS
GAS DECAY TANKS
COMPRESSOR SUCTION
PLANT VENT SYSTEM NOTE 3
AUX. BLDG.
TUNNEL
RMWST
SHUT DOWN TANKS RELIEF HEADER
DRAIN HEADER
N2 SUPPLY
AUX COMP COOLING WATER SYSTEM (OUT)
AUX COMP COOLING WATER SYSTEM (IN)
NOTES:
1. FOR CATALYTIC HYDROGEN RECOMBINERS AND WASTE GAS COMPRESSORS COOLING WATER INFORMATION, SEE DWG. 1X4DB139.
2. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND MANUAL VALVES 2 INCH AND SMALLER.
3. RUN TO PLANT VENT SYSTEM (1553). EXHAUST DUCT MAY BE IN ADJACENT ROOM.
4. RADIATION ELEMENT SHALL BE CLAMPED AROUND A REMOVABLE SPOOL, 60 INCHES LONG MINIMUM.
5. DELETED.
6. FOR LINE NUMBERS AND VALVE NUMBERS WITHIN PACKAGES 1-1902-D6-001, A-1902-D6-002, SEE BECHTEL DRAWING 1/2X6AK09-22 AND 2X6AK09-103 REV. 8 AND 1/2X6AK09-21.
7. RE-0014, FT-0014 AND VALVE 250 ARE LOCATED INSIDE OF REMOVABLE SHIELDING WALL. SAMPLE CONNECTION AND VALVE STEM EXTENSION ARE OUTSIDE OF SHIELDING.
8. DCP 01-VAN0001 HAS DOWNGRADED THE PROJECT CLASS FOR MANY COMPONENTS FROM 313 TO 417. WITH THE EXCEPTION OF DRAIN COMPONENTS, THIS CHANGE IN PROJECT CLASS IS ALSO APPLICABLE TO ALL EQUIPMENT, PIPING AND COMPONENTS INCLUDED IN THE PACKAGES FOR WASTE GAS COMPRESSORS 1-1902-D6-001 & 002, AND THE HYDROGEN RECOMBINERS 1-1902-D6-001 & A-1902-D6-002. THE NEW CLASSIFICATION IS APPLICABLE TO ALL REPLACEMENT COMPONENTS. THE CHANGE IN PROJECT CLASS DOES NOT APPLY TO PIPING OUTSIDE THE COMPRESSOR AND RECOMBINER PACKAGES, PIPE SUPPORTS, PENETRATION SEALS AND INSTRUMENT TUBING, WHOSE PROJECT CLASSIFICATION REMAINS UNCHANGED.
REFER TO WESTINGHOUSE FLOW DIAGRAM DWG. 1094E79 SHEET 2 OF 4.
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
WASTE PROCESSING SYSTEM — GAS
SYSTEM NO. 1902
CAD 1X4DB129
Acad2000 JJB - 01
SCALE: NONE
DRAWING NO. 1X4DB129
VER. 43.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL
43.0 REVISED PER ED SNC364416W001, VER.1.0 2-9-12 JJB LCF AAN X
NO. VERSIONS DATE DR CHK APPV DTL
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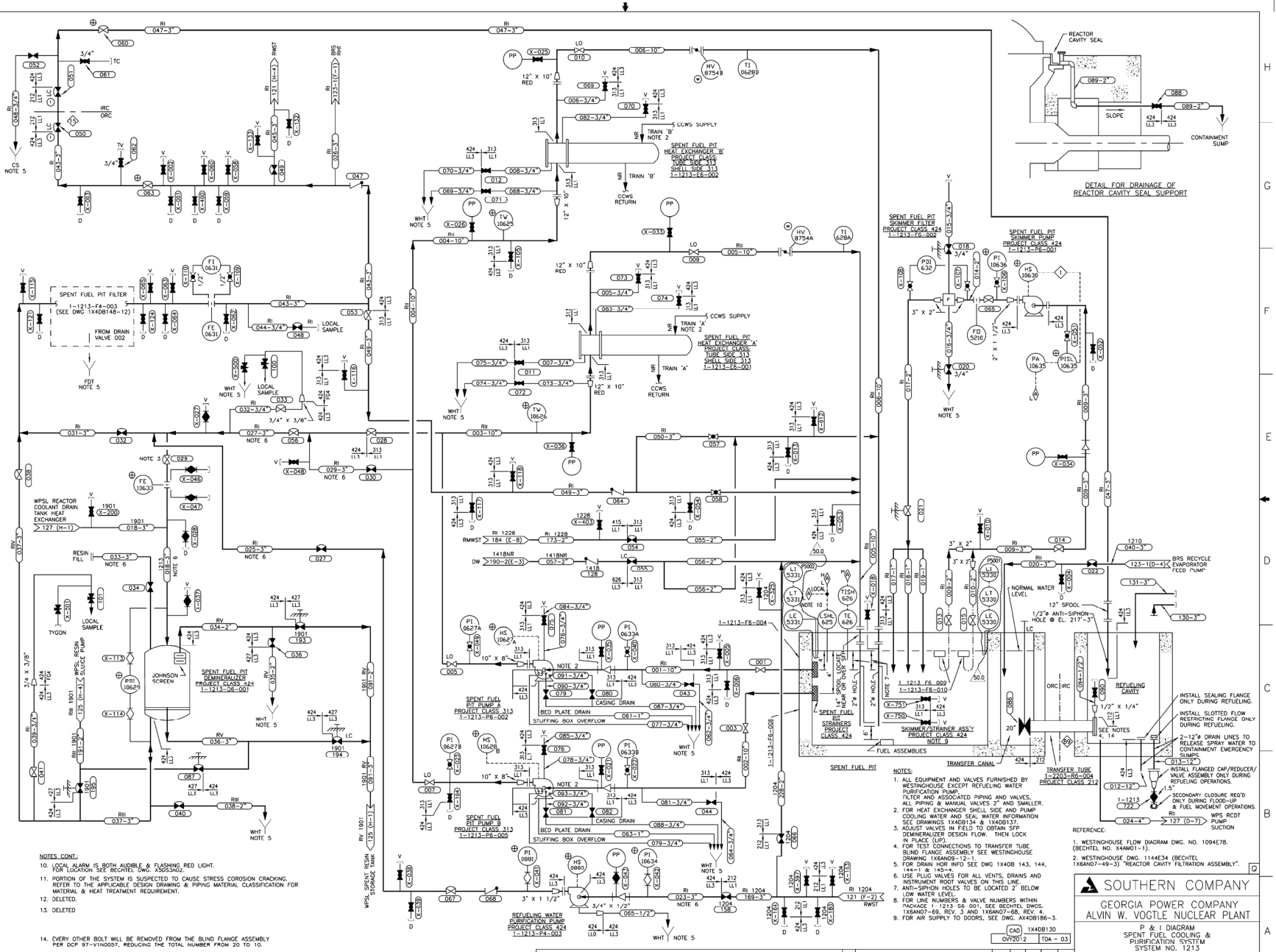


NOTES CONT.:

10. LOCAL ALARM IS BOTH AUDIBLE & FLASHING RED LIGHT. FOR LOCATION SEE BECHTEL DWG. X5D53A02.
11. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING & PIPING MATERIAL CLASSIFICATION FOR MATERIAL & HEAT TREATMENT REQUIREMENT.
12. DELETED.
13. DELETED
14. EVERY OTHER BOLT WILL BE REMOVED FROM THE BLIND FLANGE ASSEMBLY PER DCP 97-V1N0057, REDUCING THE TOTAL NUMBER FROM 20 TO 10.

NOTES:

1. ALL EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT REFUELING WATER PURIFICATION PUMP, FILTER AND ASSOCIATED PIPING AND VALVES, ALL PIPING & MANUAL VALVES 2" AND SMALLER.
2. FOR HEAT EXCHANGER SHELL SIDE AND PUMP COOLING WATER AND SEAL WATER INFORMATION SEE DRAWINGS 1X4DB134 & 1X4DB137.
3. ADJUST VALVES IN FIELD TO OBTAIN SFP DEMINERALIZER DESIGN FLOW. THEN LOCK IN PLACE (LIP).
4. FOR TEST CONNECTIONS TO TRANSFER TUBE BLIND FLANGE ASSEMBLY SEE WESTINGHOUSE DRAWING 1X6AN09-12-1.
5. FOR DRAIN HDR INFO SEE DWG 1X4DB 143, 144, 144-1 & 145-4.
6. USE PLUG VALVES FOR ALL VENTS, DRAINS AND INSTRUMENT ROOT VALVES ON THIS LINE.
7. ANTI-SIPHON HOLES TO BE LOCATED 2" BELOW LOW WATER LEVEL.
8. FOR LINE NUMBERS & VALVE NUMBERS WITHIN PACKAGE 1-1213-S6-001, SEE BECHTEL DWGS. 1X6AN07-69, REV. 3 AND 1X6AN07-68, REV. 4.
9. FOR AIR SUPPLY TO DOORS, SEE DWG. AX4DB186-3.

REFERENCE:

1. WESTINGHOUSE FLOW DIAGRAM DWG. NO. 1094E78. (BECHTEL NO. X4AM01-1).
2. WESTINGHOUSE DWG. 1144E34 (BECHTEL 1X6AN07-49-3) "REACTOR CAVITY FILTRATION ASSEMBLY".

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 50.0 | REVISED PER ABN SNC583177J002, VER. 1.0 | 02/24/16 | TDA | MWD | WLN | X |

CAD 1X4DB130 | OVY201-2 | TDA - 03

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

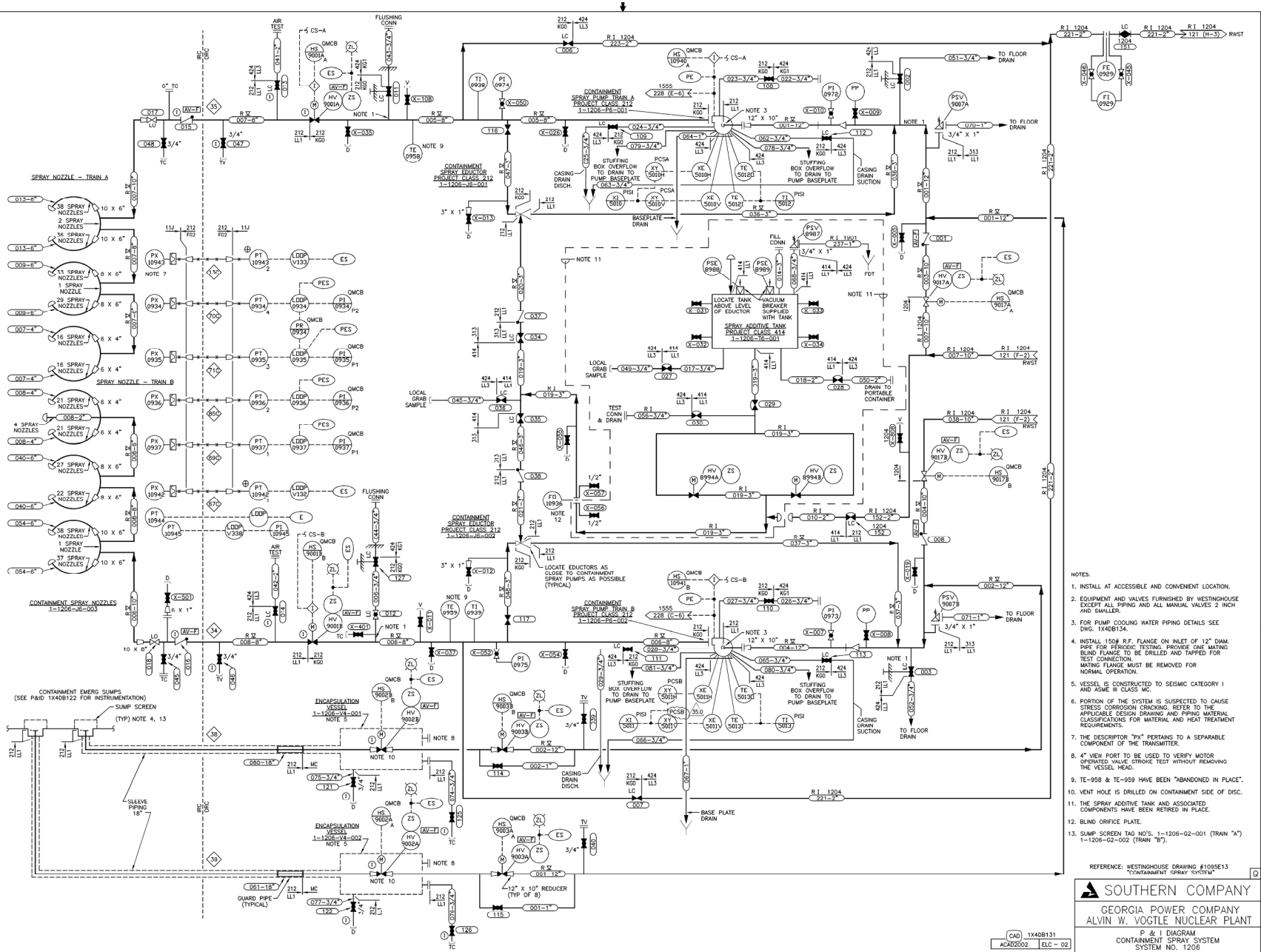
P & I DIAGRAM
SPENT FUEL COOLING & PURIFICATION SYSTEM
SYSTEM NO. 1213

SCALE: NONE | DRAWING NO. 1X4DB130 | VER. 50.0

JOB NO 10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. INSTALL AT ACCESSIBLE AND CONVENIENT LOCATION.
2. EQUIPMENT AND VALVES FURNISHED BY WESTINGHOUSE EXCEPT ALL PIPING AND ALL MANUAL VALVES 2 INCH AND SMALLER.
3. FOR PUMP COOLING WATER PIPING DETAILS SEE DWG. 1X4DB134.
4. INSTALL 150# R.F. FLANGE ON INLET OF 12" DIAM. PIPE FOR PERIODIC TESTING. PROVIDE ONE MATING BLIND FLANGE TO BE DRILLED AND TAPPED FOR TEST CONNECTION. MATING FLANGE MUST BE REMOVED FOR NORMAL OPERATION.
5. VESSEL IS CONSTRUCTED TO SEISMIC CATEGORY I AND ASME III CLASS MC.
6. PORTION OF THE SYSTEM IS SUSPECTED TO CAUSE STRESS CORROSION CRACKING. REFER TO THE APPLICABLE DESIGN DRAWING AND PIPING MATERIAL CLASSIFICATIONS FOR MATERIAL AND HEAT TREATMENT REQUIREMENTS.
7. THE DESCRIPTOR "PX" PERTAINS TO A SEPARABLE COMPONENT OF THE TRANSMITTER.
8. 4" VIEW PORT TO BE USED TO VERIFY MOTOR OPERATED VALVE STROKE TEST WITHOUT REMOVING THE VESSEL HEAD.
9. TE-958 & TE-959 HAVE BEEN "ABANDONED IN PLACE".
10. VENT HOLE IS DRILLED ON CONTAINMENT SIDE OF DISC.
11. THE SPRAY ADDITIVE TANK AND ASSOCIATED COMPONENTS HAVE BEEN RETIRED IN PLACE.
12. BLIND ORIFICE PLATE.
13. SUMP SCREEN TAG NO'S. 1-1206-G2-001 (TRAIN "A") 1-1206-G2-002 (TRAIN "B").

REFERENCE: WESTINGHOUSE DRAWING #1095E13 "CONTAINMENT SPRAY SYSTEM"

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTAINMENT SPRAY SYSTEM
SYSTEM NO. 1206

CAD 1X4DB131 | ACAD2002 | ELC – 02

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| NO. | REVISIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 35.0 | REVISED PER ABN ABN-52216, VER. 1.0 (DWG. CORRECTION) | 1-26-11 | ELC | TSL | JMR | X |

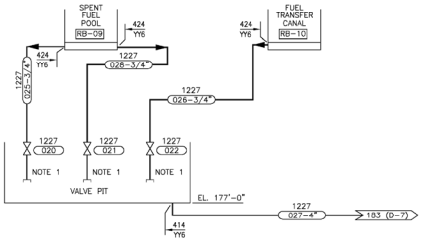
SIZE E 34044

SCALE: NONE

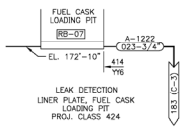
DRAWING NO. 1X4DB131 VER. 35.0

JOB NO.10604

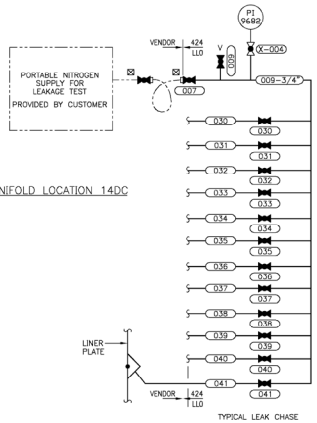
DRAWING CATEGORY: CRITICAL




LEAK DETECTION SYSTEM
LINER PLATE IN SPENT FUEL POOL, FUEL TRANSFER CANAL
PROJECT CLASS 424

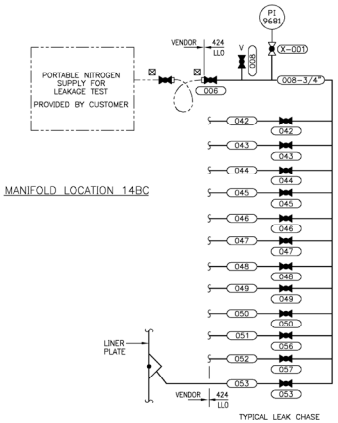



LEAK DETECTION
LINER PLATE, FUEL CASK LOADING PIT
PROJ. CLASS 424




MANIFOLD LOCATION 14DC

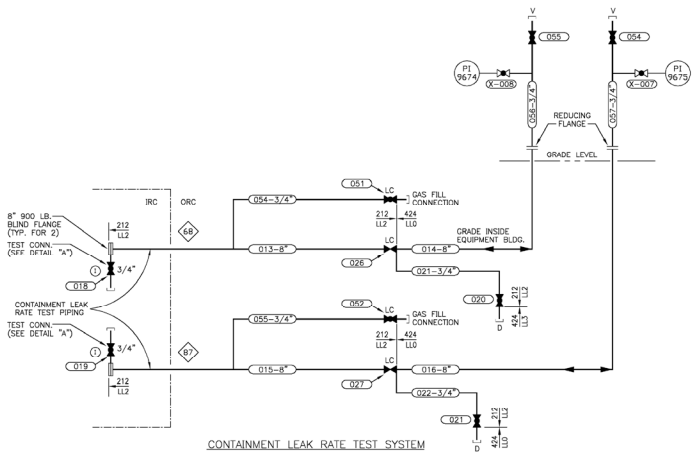
TYPICAL LEAK CHASE




MANIFOLD LOCATION 14BC

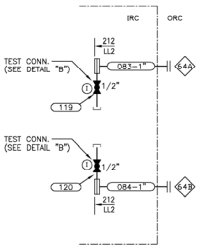
TYPICAL LEAK CHASE

CONTAINMENT LEAK CHASE SYSTEM

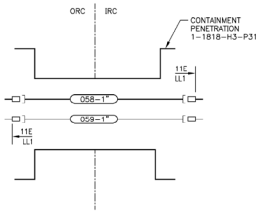



CONTAINMENT LEAK RATE TEST SYSTEM

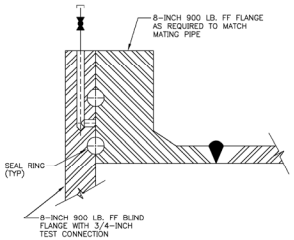
CONTAINMENT INTEGRATED LEAK RATE TEST PENETRATION



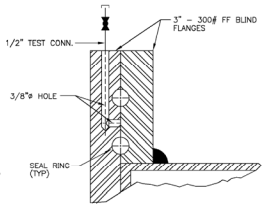

FLOW VERIFICATION AND PRESSURE SENSING PIPING




DETAIL OF FLOW THROUGH TUBE
FOR PENETRATION 1-1818-H3-P31 ONLY
(SEE DRAWING 1X34B03-00030)




DETAIL "A"




DETAIL "B"

NOTE:
1. CHICAGO FITTING.

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
MISCELLANEOUS LEAK DETECTION SYSTEM NO. 1222

CAD 1X4DB132
ACAD2002 CYN- 01

SCALE: NONE
DRAWING NO. 1X4DB132
VER. 18.0
JOB NO.10604
SIZE E 34X44

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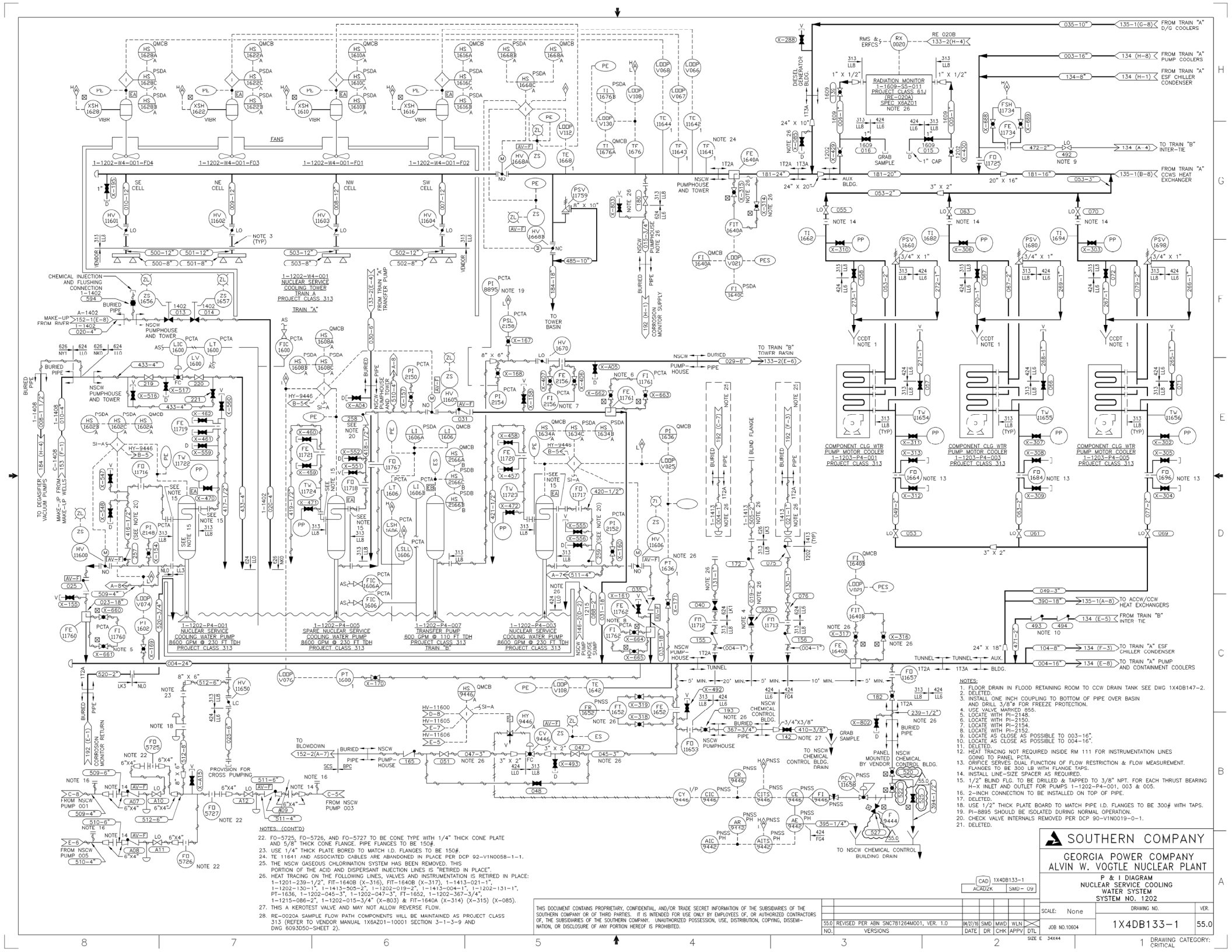
| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 18.0 | REDRAWN PER ABN-52271, VER. 1.0 | 1-30-06 | CYN | LCF | LKB | X |

DRAWING CATEGORY: CRITICAL




NOTES:
1. FLOOR DRAIN IN FLOOD RETAINING ROOM TO CCW DRAIN TANK SEE DWG 1X4DB147-2.
2. DELETED.
3. INSTALL ONE INCH COUPLING TO BOTTOM OF PIPE OVER BASIN AND DRILL 3/8"ø FOR FREEZE PROTECTION.
4. USE VALVE MARKED 855.
5. LOCATE WITH PI-2148.
6. LOCATE WITH PI-2150.
7. LOCATE WITH PI-2154.
8. LOCATE WITH PI-2152.
9. LOCATE AS CLOSE AS POSSIBLE TO 003-16".
10. LOCATE AS CLOSE AS POSSIBLE TO 004-16".
11. DELETED.
12. HEAT TRACING NOT REQUIRED INSIDE RM 111 FOR INSTRUMENTATION LINES GOING TO PANEL PCTA.
13. ORIFICE SERVES DUAL FUNCTION OF FLOW RESTRICTION & FLOW MEASUREMENT. FLANGES TO BE 300 LB WITH FLANGE TAPS.
14. INSTALL LINE-SIZE SPACER AS REQUIRED.
15. 1/2" BLIND FLG. TO BE DRILLED & TAPPED TO 3/8" NPT. FOR EACH THRUST BEARING H-X INLET AND OUTLET FOR PUMPS 1-1202-P4-001, 003 & 005.
16. 2-INCH CONNECTION TO BE INSTALLED ON TOP OF PIPE.
17. DELETED.
18. USE 1/2" THICK PLATE BOARD TO MATCH PIPE I.D. FLANGES TO BE 300# WITH TAPS.
19. PI-8895 SHOULD BE ISOLATED DURING NORMAL OPERATION.
20. CHECK VALVE INTERNALS REMOVED PER DCP 90-V1N0019-0-1.
21. DELETED.

NOTES: (CONT'D)
22. FO-5725, FO-5726, AND FO-5727 TO BE CONE TYPE WITH 1/4" THICK CONE PLATE AND 5/8" THICK CONE FLANGE. PIPE FLANGES TO BE 150#.
23. USE 1/4" THICK PLATE BORED TO MATCH I.D. FLANGES TO BE 150#.
24. TE 11641 AND ASSOCIATED CABLES ARE ABANDONED IN PLACE PER DCP 92-V1N0058-1-1.
25. THE NSCW GASEOUS CHLORINATION SYSTEM HAS BEEN REMOVED. THIS PORTION OF THE ACID AND DISPERSANT INJECTION LINES IS "RETIRED IN PLACE".
26. HEAT TRACING ON THE FOLLOWING LINES, VALVES AND INSTRUMENTATION IS RETIRED IN PLACE: 1-1201-239-1/2", FIT-1640B (X-316), FIT-1640B (X-317), 1-1413-021-1", 1-1202-130-1", 1-1413-505-2", 1-1202-019-2", 1-1413-004-1", 1-1202-131-1", PT-1636, 1-1202-045-3", 1-1202-047-3", FT-1652, 1-1202-367-3/4", 1-1215-086-2", 1-1202-015-3/4" (X-803) & FIT-1640A (X-314) (X-315) (X-085).
27. THIS A KEROTEST VALVE AND MAY NOT ALLOW REVERSE FLOW.
28. RE-0020A SAMPLE FLOW PATH COMPONENTS WILL BE MAINTAINED AS PROJECT CLASS 313 (REFER TO VENDOR MANUAL 1X6AZ01-10001 SECTION 3-1-3-9 AND DWG 6093D50-SHEET 2).

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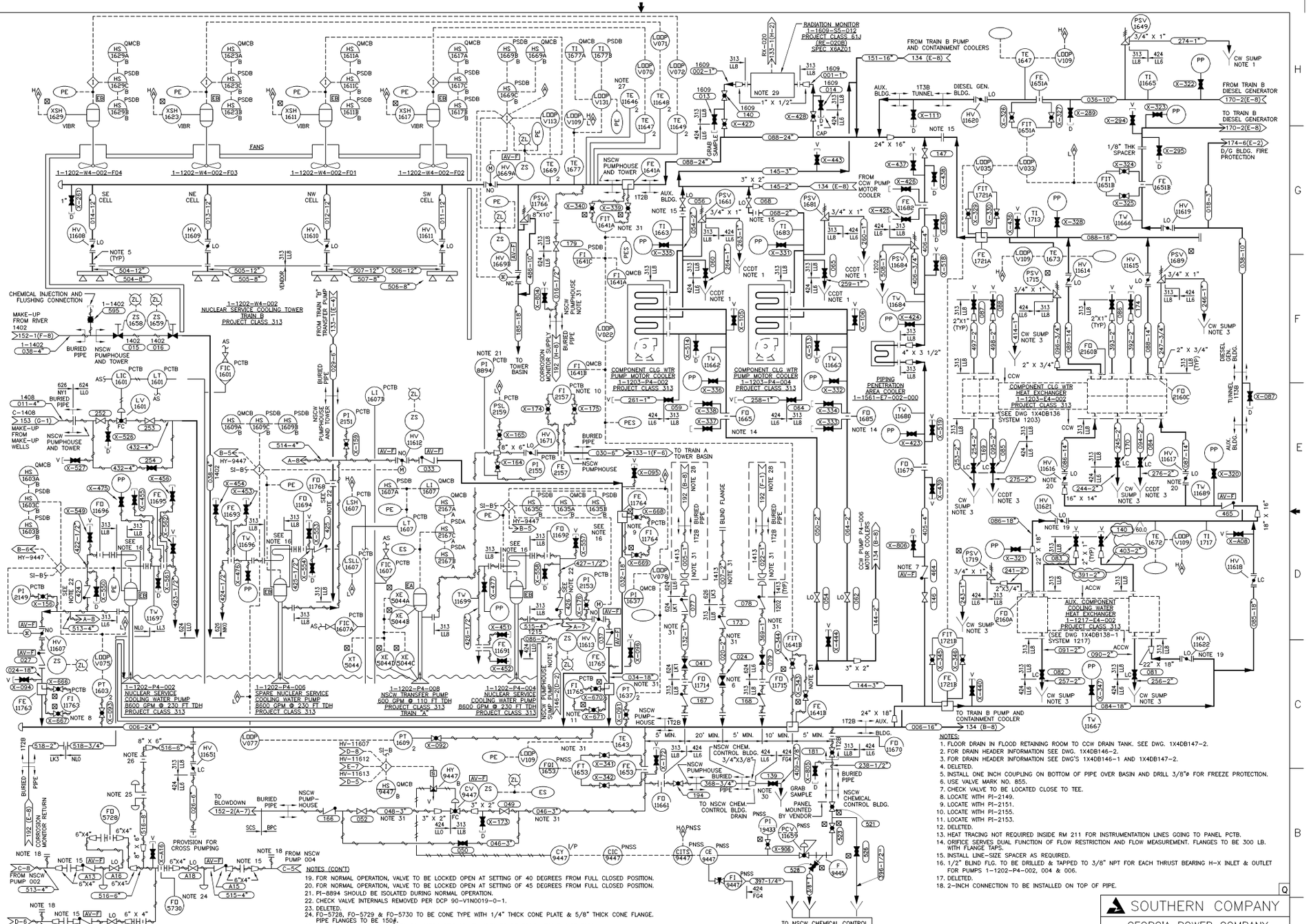
| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 55.0 | REVISED PER ABN SNC781264M001, VER. 1.0 | 04/27/16 | SMD | MWD | WLN | |

CAD 1X4DB133-1
ACADZK SMD- 09

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
NUCLEAR SERVICE COOLING WATER SYSTEM
SYSTEM NO. 1202

SCALE: None
JOB NO.10604
DRAWING NO. 1X4DB133-1
VER. 55.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL


FANS
1-1202-W4-002-F04
1-1202-W4-002-F03
1-1202-W4-002-F01
1-1202-W4-002-F02
SE CELL
NE CELL
NW CELL
SW CELL
NSCW PUMPHOUSE AND TOWER
1-1202-W4-002
NUCLEAR SERVICE COOLING TOWER TRAIN B
PROJECT CLASS 313
CHEMICAL INJECTION AND FLUSHING CONNECTION
MAKE-UP FROM RIVER
MAKE-UP FROM MAKE-UP WELLS
BURIED PIPE
TO TOWER BASIN
TO TRAIN A TOWER BASIN
COMPONENT CLG WTR PUMP MOTOR COOLER 1-1203-P4-002 PROJECT CLASS 313
COMPONENT CLG WTR PUMP MOTOR COOLER 1-1203-P4-004 PROJECT CLASS 313
PIPING PENETRATION AREA COOLER 1-1561-E7-002-000
COMPONENT CLG WTR HEAT EXCHANGER 1-1203-E4-002 PROJECT CLASS 313
AUX. COMPONENT COOLING WATER HEAT EXCHANGER 1-1217-E4-002 PROJECT CLASS 313
1-1202-P4-002 NUCLEAR SERVICE COOLING WATER PUMP 8600 GPM @ 230 FT TDH PROJECT CLASS 313
1-1202-P4-006 SPARE NUCLEAR SERVICE COOLING WATER PUMP 8600 GPM @ 230 FT TDH PROJECT CLASS 313
1-1202-P4-008 NSCW TRANSFER PUMP 600 GPM @ 110 FT TDH PROJECT CLASS 313 TRAIN "A"
1-1202-P4-004 NUCLEAR SERVICE COOLING WATER PUMP 8600 GPM @ 230 FT TDH PROJECT CLASS 313
RADIATION MONITOR 1-1609-S5-012 PROJECT CLASS 61J (RE-0020B) SPEC. X6AZ01
FROM TRAIN B PUMP AND CONTAINMENT COOLERS
TO TRAIN B PUMP AND CONTAINMENT COOLER
FROM TRAIN B DIESEL GENERATOR
TO TRAIN B DIESEL GENERATOR
D/G BLDG. FIRE PROTECTION
FROM CCW PUMP MOTOR COOLER
AUX. BLDG.
1T2B TUNNEL
DIESEL GEN. BLDG.
TUNNEL 1T3B
CW SUMP NOTE 3
CCDT NOTE 1
CCDT NOTE 3
CCW
ACCW
PROVISION FOR CROSS PUMPING
TO BLOWDOWN
TO NSCW CHEM. CONTROL BLDG. DRAIN
TO NSCW CHEMICAL CONTROL BUILDING DRAIN
PANEL MOUNTED BY VENDOR
GRAB SAMPLE
NSCW CHEMICAL CONTROL BLDG.
NSCW PUMP-HOUSE
NSCW PUMPHOUSE BURIED
NSCW PUMPHOUSE SUMP
FROM NSCW PUMP 002
FROM NSCW PUMP 004
FROM NSCW PUMP 006
NOTES (CONT')
32. RE-0020B SAMPLE FLOW PATH COMPONENTS WILL BE MAINTAINED AS PROJECT CLASS 313. (REFER TO VENDOR MANUAL 1X6AZ01-10001 SECTION 3-1-3-9 AND DRAWING 6093D50-SHEET 2).
NOTES:
1. FLOOR DRAIN IN FLOOD RETAINING ROOM TO CCW DRAIN TANK. SEE DWG. 1X4DB147-2.
2. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB146-2.
3. FOR DRAIN HEADER INFORMATION SEE DWG'S 1X4DB146-1 AND 1X4DB147-2.
4. DELETED.
5. INSTALL ONE INCH COUPLING ON BOTTOM OF PIPE OVER BASIN AND DRILL 3/8"ø FOR FREEZE PROTECTION.
6. USE VALVE MARK NO. 855.
7. CHECK VALVE TO BE LOCATED CLOSE TO TEE.
8. LOCATE WITH PI-2149.
9. LOCATE WITH PI-2151.
10. LOCATE WITH PI-2155.
11. LOCATE WITH PI-2153.
12. DELETED.
13. HEAT TRACING NOT REQUIRED INSIDE RM 211 FOR INSTRUMENTATION LINES GOING TO PANEL PCTB.
14. ORIFICE SERVES DUAL FUNCTION OF FLOW RESTRICTION AND FLOW MEASUREMENT. FLANGES TO BE 300 LB. WITH FLANGE TAPS.
15. INSTALL LINE-SIZE SPACER AS REQUIRED.
16. 1/2" BLIND FLG. TO BE DRILLED & TAPPED TO 3/8" NPT FOR EACH THRUST BEARING H-X INLET & OUTLET FOR PUMPS 1-1202-P4-002, 004 & 006.
17. DELETED.
18. 2-INCH CONNECTION TO BE INSTALLED ON TOP OF PIPE.
NOTES (CONT')
19. FOR NORMAL OPERATION, VALVE TO BE LOCKED OPEN AT SETTING OF 40 DEGREES FROM FULL CLOSED POSITION.
20. FOR NORMAL OPERATION, VALVE TO BE LOCKED OPEN AT SETTING OF 45 DEGREES FROM FULL CLOSED POSITION.
21. PI-8894 SHOULD BE ISOLATED DURING NORMAL OPERATION.
22. CHECK VALVE INTERNALS REMOVED PER DCP 90-V1N0019-0-1.
23. DELETED.
24. FO-5728, FO-5729 & FO-5730 TO BE CONE TYPE WITH 1/4" THICK CONE PLATE & 5/8" THICK CONE FLANGE. PIPE FLANGES TO BE 150#.
25. USE 1/2" THICK PLATE BORED TO MATCH PIPE I.D. FLANGES TO BE 300# WITH FLANGE TAPS.
26. USE 1/4" THICK PLATE BORED TO MATCH PIPE I.D. FLANGES TO BE 150#.
27. TE 11646 AND ASSOCIATED CABLES ARE ABANDONED IN PLACE PER DCP 92-V1N0058-1-1.
28. THE NSCW GASEOUS CHLORINATION SYSTEM HAS BEEN REMOVED. THIS PORTION OF THE ACID AND DISPERSANT INJECTION LINES IS "RETIRED IN PLACE".
29. RE0020B MECHANICAL COMPONENTS WILL BE MAINTAINED AS PROJECT CLASS 313.
30. THIS IS A KEROTEST VALVE AND MAY NOT ALLOW REVERSE FLOW.
31. HEAT TRACING ON THE FOLLOWING LINES, VALVES & INSTRUMENTATION IS RETIRED IN PLACE: 1-1202-238-1/2", 1-1413-007-2", 1-1202-048-3", 1-1202-368-3/4", 1-1202-020-2", 1-1413-005-1", FIT-1641B (X-344), FT-1641B (X-343), FT-1653, FIT-1637, 1-1213-088-2", 1-1202-016-3/4" (X-304), & FIT-1641A (X-339) (X-340).
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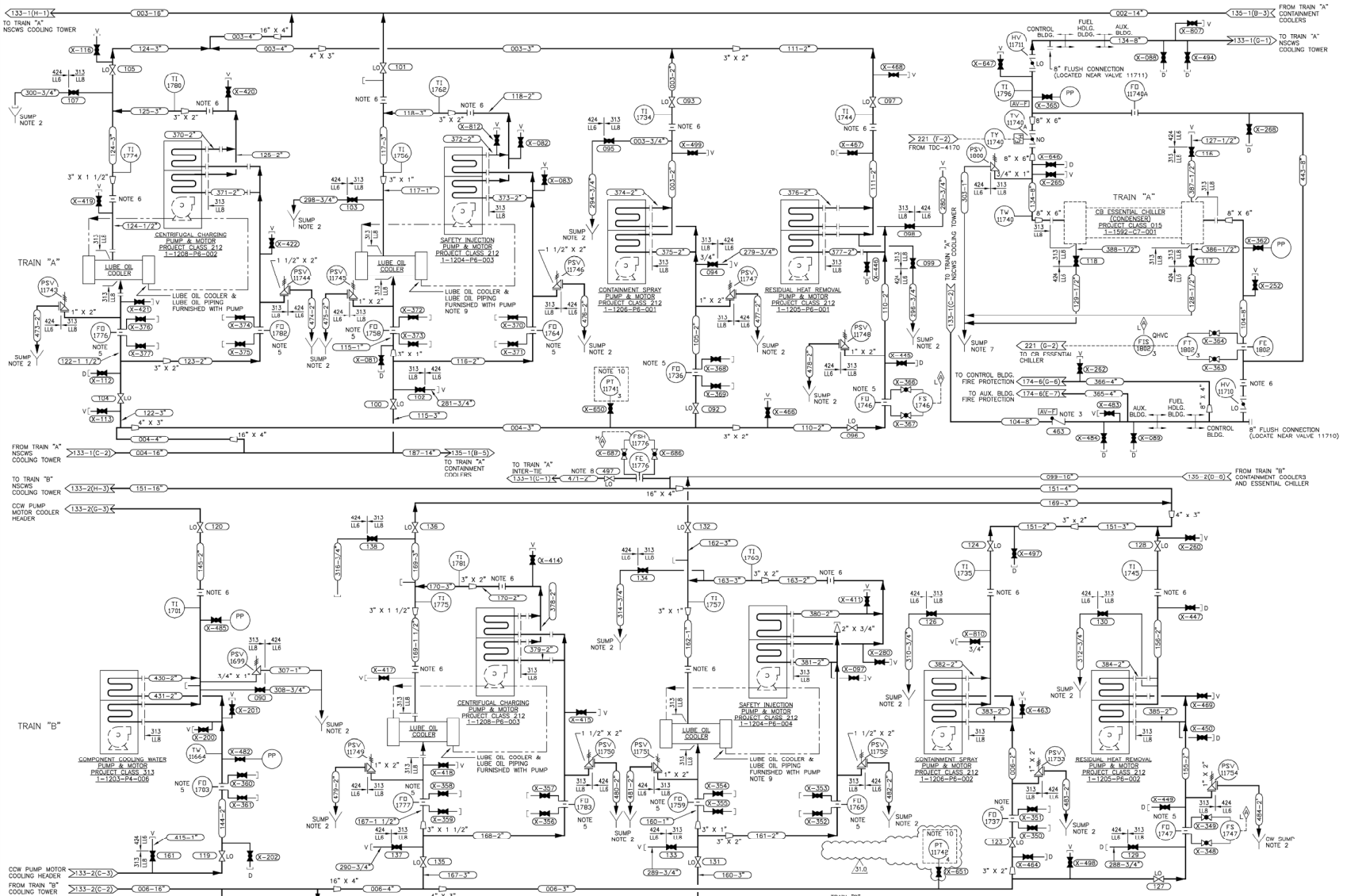
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
NUCLEAR SERVICE COOLING WATER SYSTEM
SYSTEM NO. 1202
CAD 1X4DB133-2
ACAD2007 PVK - 10
SCALE: NONE
DRAWING NO. 1X4DB133-2
VER. 60.0
JOB NO.10604
60.0 REVISED PER ABN-V02968, VERSION 1.0 4/16/12 PVK DHH AAN
NO. VERSIONS DATE DR CHK APPV DTL
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




| SAFETY INJECTION PUMP LUBE OIL SYS. INSTRUMENTATION VENDOR TAG NO. | PROJECT TAG NUMBERS | |
|---|---|---|
| | TRN. A PUMP 1-1204-P6-003 | TRN. B PUMP 1-1204-P6-004 |
| S003 INDICATOR | 1TI-920A | 1TI-921A |
| S003 - 1 WELL | 1TW-920A | 1TW-921A |
| S006 INDICATOR | 1TI-920B | 1TI-921B |
| S006 - 1 WELL | 1TW-920B | 1TW-921B |
| S013 INDICATOR | 1PI-920A | 1PI-921A |
| S011 INDICATOR | 1PI-920B | 1PI-921B |

NOTES:
1. SEE COMPONENT COOLING WATER SYSTEM (DWG. 1X4DB137) FOR COOLING WATER TO GLAND COOLER OF RESIDUAL HEAT REMOVAL PUMPS.
2. FOR DRAIN INFORMATION SEE DWG. 1X4DB146-1, DWG. 1X4DB147-1, AND 1X4DB147-2.
3. CHECK VALVE TO BE LOCATED AT LOWEST POINT BELOW ELEVATION 220'.
4. LOCATE AS CLOSE A POSSIBLE TO 098-16".
5. ORIFICE SERVES DUAL PURPOSE OF FLOW BALANCING AND MEASUREMENT OF FLOW. FLANGE SHOULD BE 300# FLANGE TAP.
6. FLANGES SHOULD NOT HAVE ORIFICE PLATE INSTALLED. FIELD TO REPLACE ORIFICE PLATE W/ 1/8" THICK SPACER. SPACER I.D. TO MATCH PIPE I.D. MAT'L THE SAME AS PIPING MAT'L CLASS.
7. FOR DRAIN INFORMATION SEE DWG. 1X4DB142-2.
8. LOCATE AS CLOSE AS POSSIBLE TO 099-16".
9. FOR INSTRUMENT TAG NUMBER, SEE TABLE AT A-8.
10. TRANSMITTER RETIRED IN PLACE.

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| NO. | VERSIONS | DATE | DN | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 31.0 | REVISED PER ABN-V01147, VER. 1.0 | 4-15-08 | Vep | RBH | JMR | X |

CAD 1X4DB134 | ACAD2002 | Vep – 03

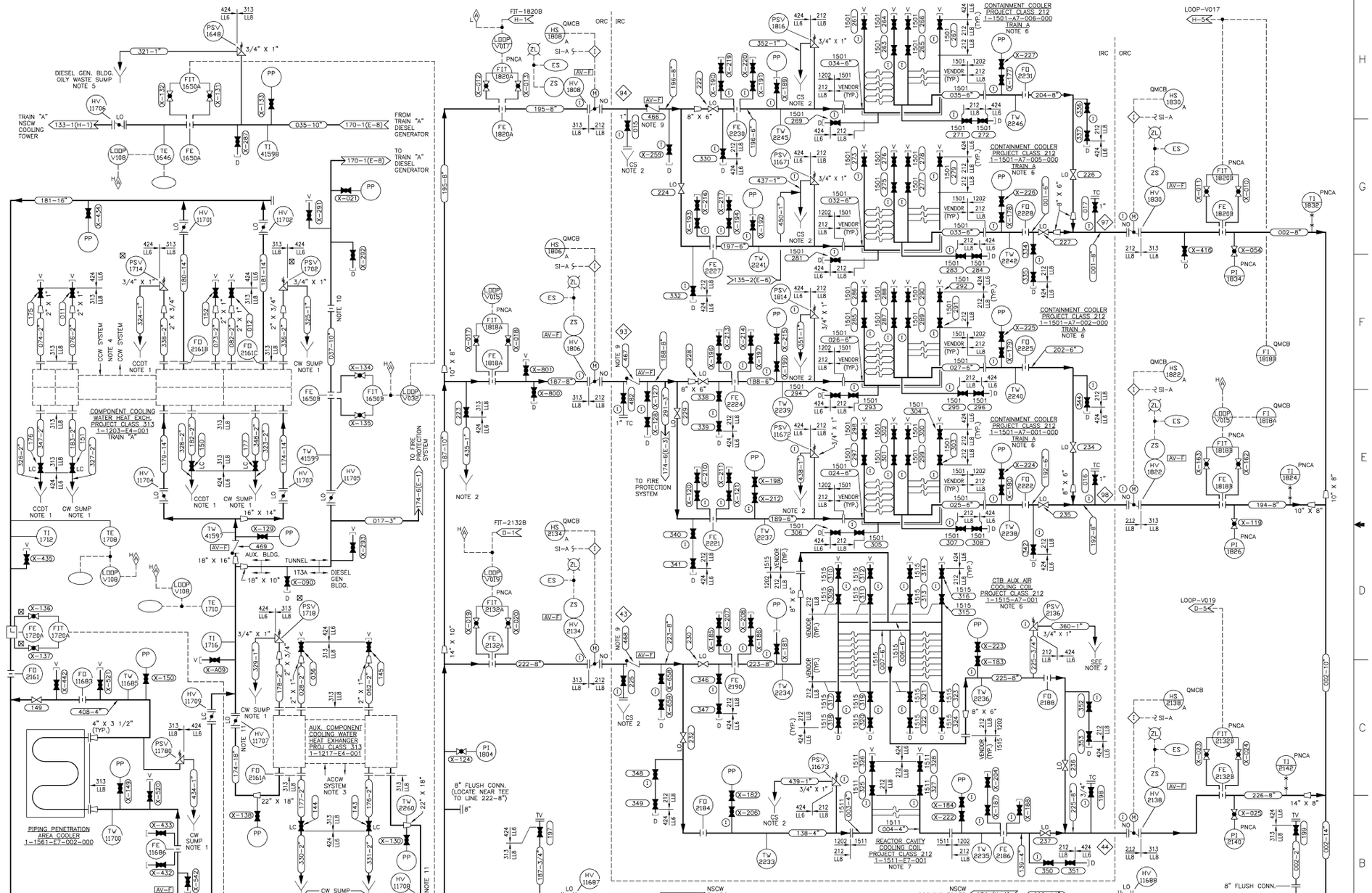
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SERVICE CLG. WTR. SYSTEM
SYSTEM NO. 1202

SCALE: NONE | DRAWING NO. 1X4DB134 | VER. 31.0 | JOB NO.10604 | SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG'S 1X4DB146-1 & 1X4DB147-2.
2. FOR DRAIN HEADER INFORMATION SEE DWG 1X4DB143, 1X4DB144-1 & 1X4DB144-2.
3. FOR AUX COMPONENT COOLING WATER SYSTEM AND SHELL SIDE CONN'S SEE DWG 1X4DB138-1.
4. FOR COMPONENT COOLING WATER SYSTEMS AND SHELL SIDE CONN'S SEE DWG 1X4DB136.
5. FOR DIESEL GEN. BLDG SUMP INFORMATION SEE DWG. 1X4DB146-2.
6. FOR CONTAINMENT COOLERS AND CTB AUX AIR COOLER AIR SIDE DRAINS SEE DWG. 1X4DB212.
7. FOR REACTOR CAVITY COOLER AIR SIDE DRAINS SEE DWG 1X4DB214-1.
8. CHECK VALVE TO BE LOCATED CLOSE TO TEE.
9. CHECK VALVE TO BE CLOSE TO PENETRATION.
10. LINE SPACER TO BE INSTALLED.
11. FOR NORMAL OPERATION, VALVE TO BE LOCKED OPEN AT SETTING OF 42.5 DEGREES FROM FULL CLOSED POSITION.

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| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 29.0 | REDRAWN PER ABN- 52215, VER. 1.0 | 9-7-05 | NJ | LCF | LKB | X |

CAD 1X4DB135-1
ACAD2002 NJ - 01

SOUTHERN COMPANY

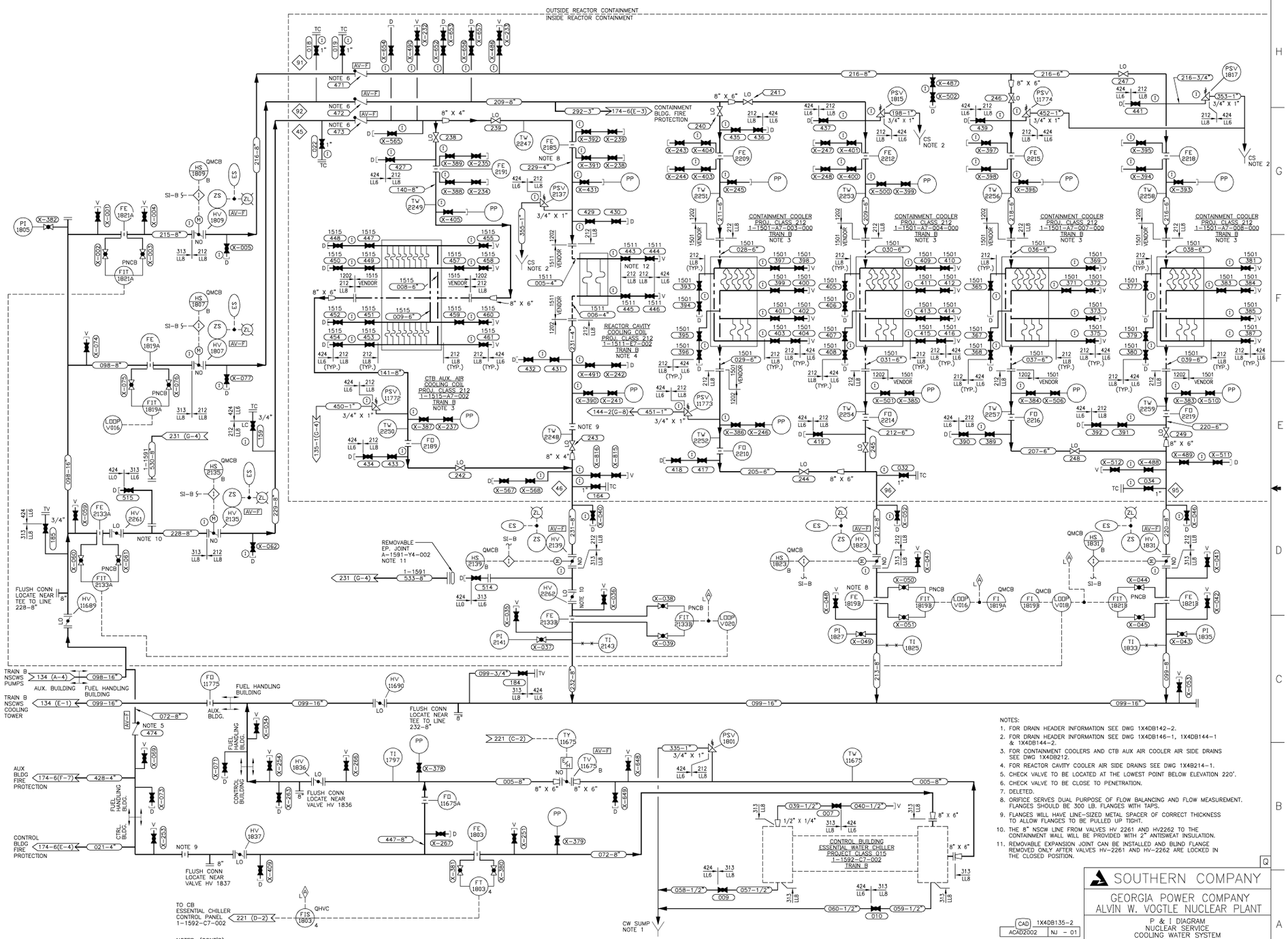
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SERVICE
COOLING WATER SYSTEM
SYSTEM NO. 1202

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB135-1
VER. 29.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG 1X4DB142-2.
2. FOR DRAIN HEADER INFORMATION SEE DWG 1X4DB146-1, 1X4DB144-1 & 1X4DB144-2.
3. FOR CONTAINMENT COOLERS AND CTB AUX AIR COOLER AIR SIDE DRAINS SEE DWG 1X4DB212.
4. FOR REACTOR CAVITY COOLER AIR SIDE DRAINS SEE DWG 1X4B214-1.
5. CHECK VALVE TO BE LOCATED AT THE LOWEST POINT BELOW ELEVATION 220'.
6. CHECK VALVE TO BE CLOSE TO PENETRATION.
7. DELETED.
8. ORIFICE SERVES DUAL PURPOSE OF FLOW BALANCING AND FLOW MEASUREMENT. FLANGES SHOULD BE 300 LB. FLANGES WITH TAPS.
9. FLANGES WILL HAVE LINE-SIZED METAL SPACER OF CORRECT THICKNESS TO ALLOW FLANGES TO BE PULLED UP TIGHT.
10. THE 8" NSCW LINE FROM VALVES HV 2261 AND HV2262 TO THE CONTAINMENT WALL WILL BE PROVIDED WITH 2" ANTISWEAT INSULATION.
11. REMOVABLE EXPANSION JOINT CAN BE INSTALLED AND BLIND FLANGE REMOVED ONLY AFTER VALVES HV-2261 AND HV-2262 ARE LOCKED IN THE CLOSED POSITION.

NOTES: (CONT'D)

12. VALVES 1-1511-U4-433 AND 444 ARE AGAINST THE WALL WITHOUT HANDWHEELS. REMOVE UPPER CLEANOUT PLUGS OF COOLING UNIT IF VENTING CAN NOT BE ACHIEVED USING VALVES 1-1511-U4-445 AND 446. NEW O-RINGS WILL BE REQUIRED IF PLUGS ARE REMOVED.

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| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 34.0 | REDRAWN PER ABN- 52218 VER 1.0 | 9-30-05 | NJ | LCF | MDS | X |

SIZE E 34X44

CAD 1X4DB135-2
ACADZ002 NJ - 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SERVICE
COOLING WATER SYSTEM
SYSTEM NO. 1202

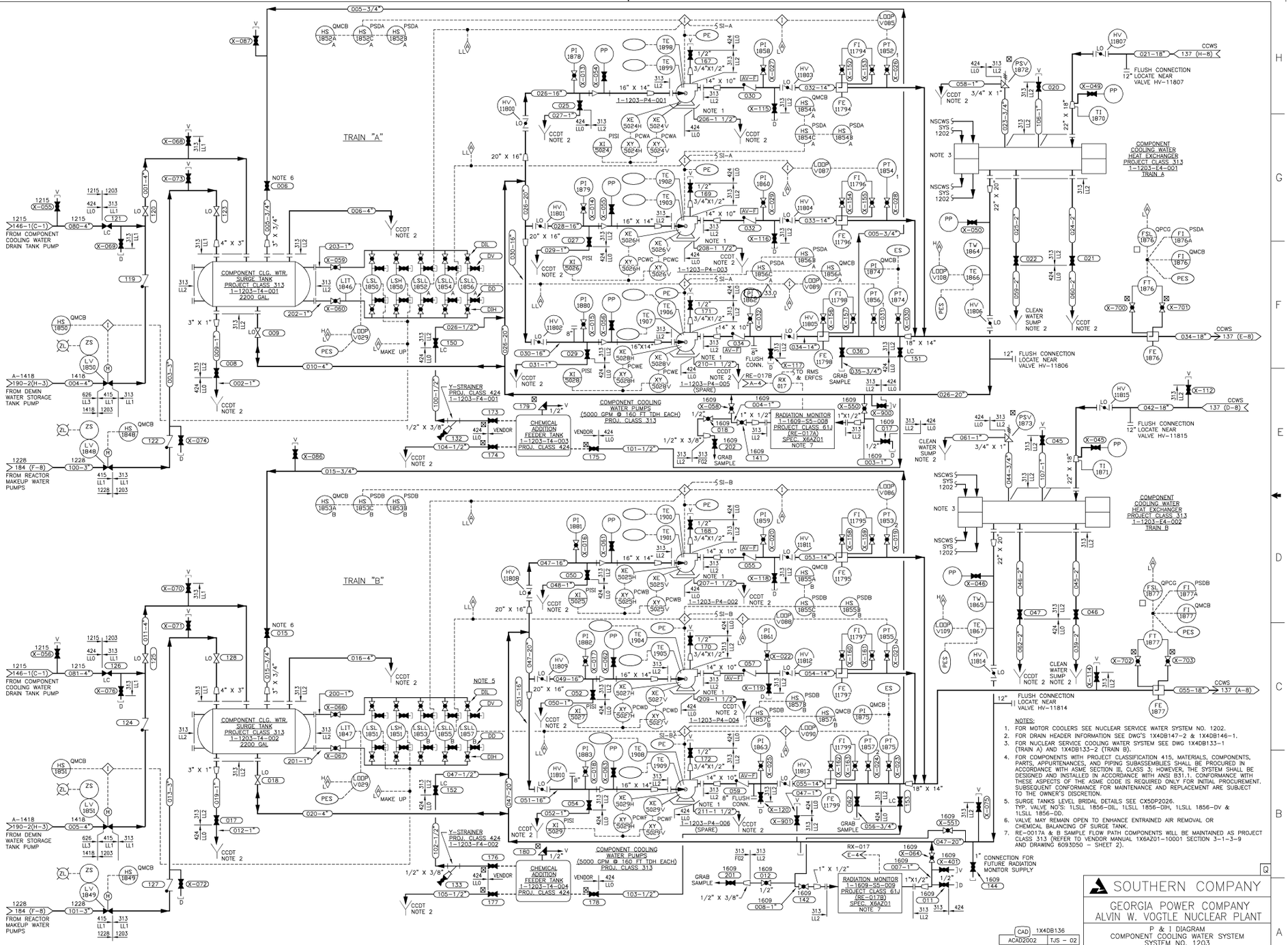
SCALE: NONE

DRAWING NO. 1X4DB135-2

VER. 34.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL

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| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 33.0 | REVISED PER ABN- V00871, VER. 1.0 | 7-23-07 | TJS | SRG | WLN | X |

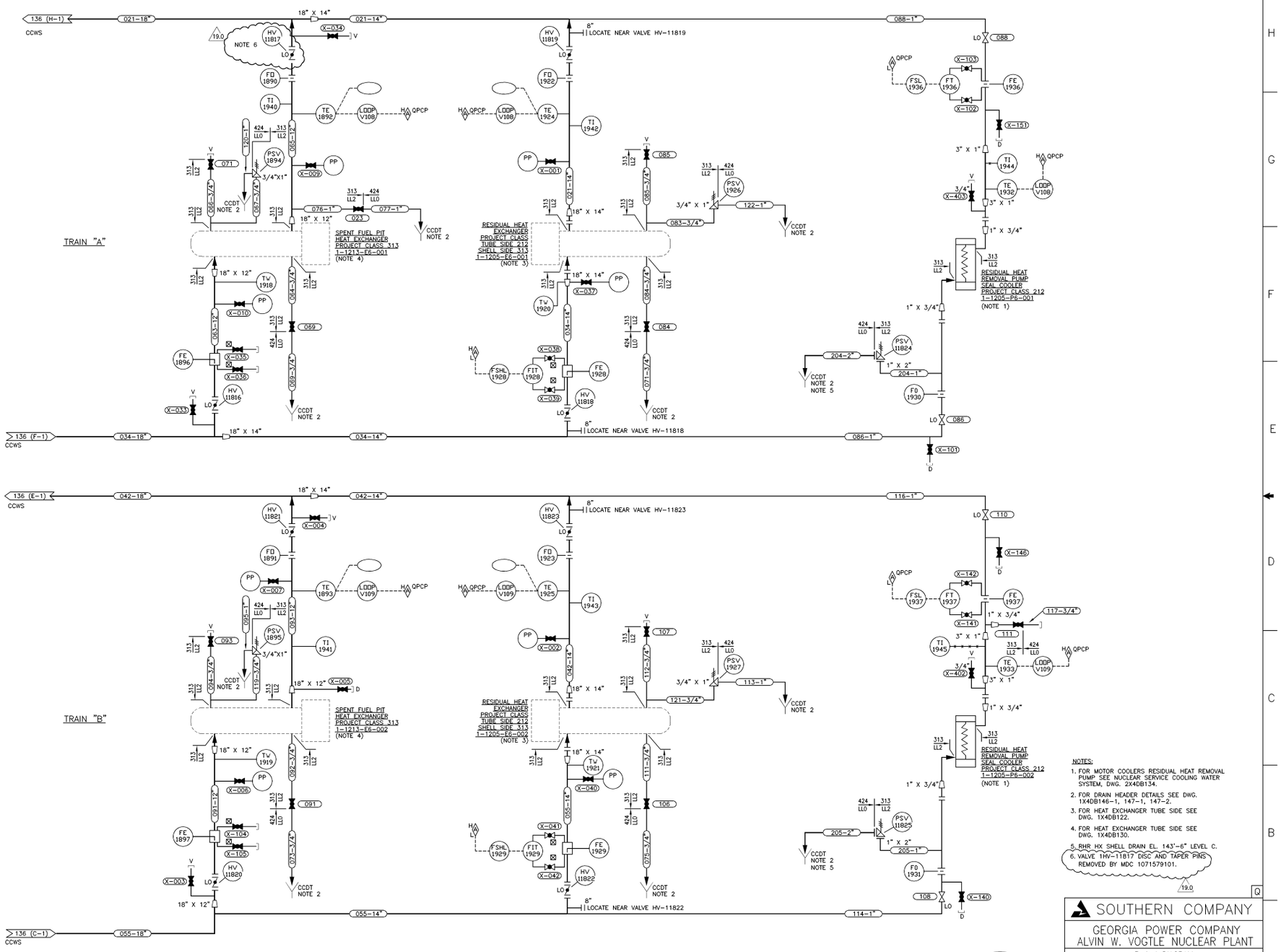
CAD 1X4DB136
ACAD2002 TJS – 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
COMPONENT COOLING WATER SYSTEM
SYSTEM NO. 1203

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB136
VER. 33.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL




CAD 1X4DB137
ACAD2002 TSL - 02

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| NO. | | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 19.0 | REDRAWN PER ABN- V00974, VER. 1.0 | 10/08/07 | TSL | RBH | JMR | X |

SIZE E 34X44

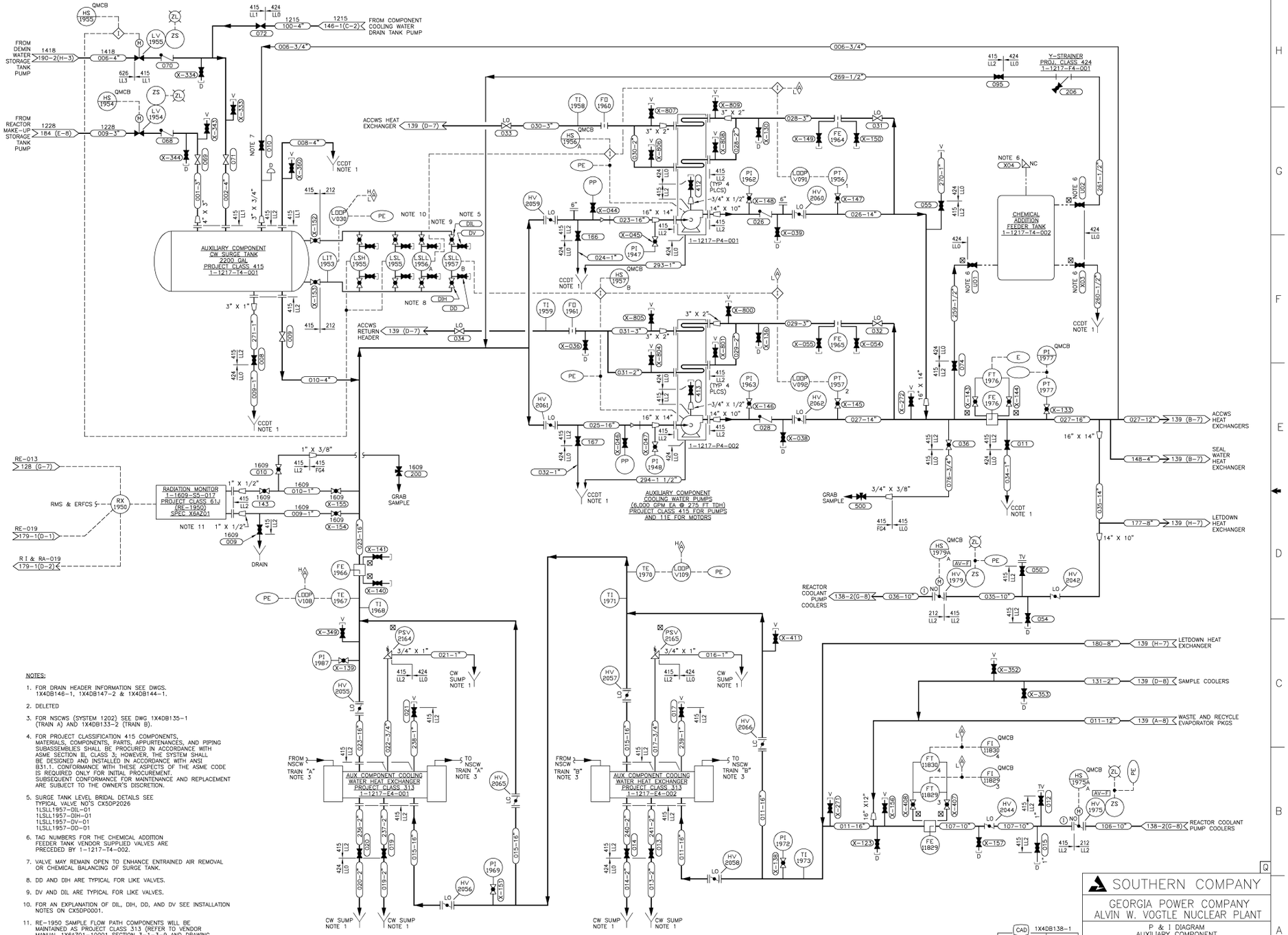
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
COMPONENT COOLING WATER SYSTEM
SYSTEM NO. 1203

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB137 | 19.0 |

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWGS. 1X4DB146-1, 1X4DB147-2 & 1X4DB144-1.
2. DELETED
3. FOR NSCWS (SYSTEM 1202) SEE DWG 1X4DB135-1 (TRAIN A) AND 1X4DB133-2 (TRAIN B).
4. FOR PROJECT CLASSIFICATION 415 COMPONENTS, MATERIALS, COMPONENTS, PARTS, APPURTENANCES, AND PIPING SUBASSEMBLIES SHALL BE PROCURED IN ACCORDANCE WITH ASME SECTION III, CLASS 3, HOWEVER, THE SYSTEM SHALL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH ANSI B31.1. CONFORMANCE WITH THESE ASPECTS OF THE ASME CODE IS REQUIRED ONLY FOR INITIAL PROCUREMENT. SUBSEQUENT CONFORMANCE FOR MAINTENANCE AND REPLACEMENT ARE SUBJECT TO THE OWNER'S DISCRETION.
5. SURGE TANK LEVEL BRIDAL DETAILS SEE TYPICAL VALVE NO'S CXSDP2026
   1LSLL1957-DIL-01
   1LSLL1957-DIH-01
   1LSLL1957-DV-01
   1LSLL1957-DD-01
6. TAG NUMBERS FOR THE CHEMICAL ADDITION FEEDER TANK VENDOR SUPPLIED VALVES ARE PRECEDED BY 1-1217-T4-002.
7. VALVE MAY REMAIN OPEN TO ENHANCE ENTRAINED AIR REMOVAL OR CHEMICAL BALANCING OF SURGE TANK.
8. DD AND DIH ARE TYPICAL FOR LIKE VALVES.
9. DV AND DIL ARE TYPICAL FOR LIKE VALVES.
10. FOR AN EXPLANATION OF DIL, DIH, DD, AND DV SEE INSTALLATION NOTES ON CXSDP0001.
11. RE-1950 SAMPLE FLOW PATH COMPONENTS WILL BE MAINTAINED AS PROJECT CLASS 313 (REFER TO VENDOR MANUAL 1X6AZ01-10001 SECTION 3-1-3-9 AND DRAWING 6093D50 - SHEET 2).

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 30.0 | REDRAWN PER ABN-V70113, VER. 1.0 | 08/21/06 | NJ | LCF | MDS | X |

CAD 1X4DB138-1
ACAD2004 NJ - 01

SOUTHERN COMPANY

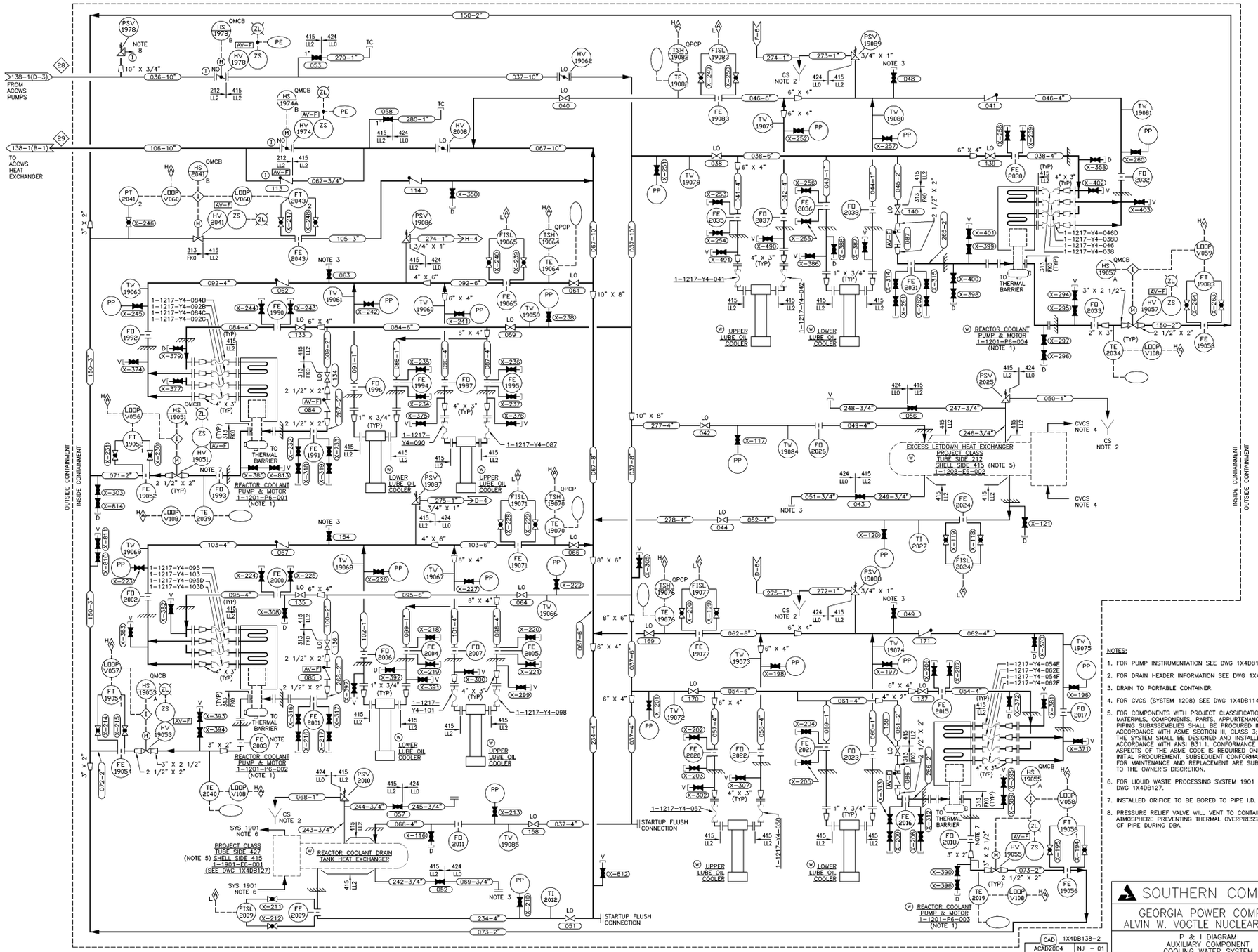
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY COMPONENT
COOLING WATER SYSTEM
SYSTEM NO. 1217

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB138-1 | 30.0 |

SIZE E 34X44

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR PUMP INSTRUMENTATION SEE DWG 1X4DB113.
2. FOR DRAIN HEADER INFORMATION SEE DWG 1X4DB144-2.
3. DRAIN TO PORTABLE CONTAINER.
4. FOR CVCS (SYSTEM 1208) SEE DWG 1X4DB114.
5. FOR COMPONENTS WITH PROJECT CLASSIFICATION 415, MATERIALS, COMPONENTS, PARTS, APPURTENANCES, AND PIPING SUBASSEMBLIES SHALL BE PROCURED IN ACCORDANCE WITH ASME SECTION III, CLASS 3; HOWEVER, THE SYSTEM SHALL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH ANSI B31.1. CONFORMANCE WITH THESE ASPECTS OF THE ASME CODE IS REQUIRED ONLY FOR INITIAL PROCUREMENT. SUBSEQUENT CONFORMANCE FOR MAINTENANCE AND REPLACEMENT ARE SUBJECT TO THE OWNER'S DISCRETION.
6. FOR LIQUID WASTE PROCESSING SYSTEM 1901 SEE DWG 1X4DB127.
7. INSTALLED ORIFICE TO BE BORED TO PIPE I.D.
8. PRESSURE RELIEF VALVE WILL VENT TO CONTAINMENT ATMOSPHERE PREVENTING THERMAL OVERPRESSURIZATION OF PIPE DURING DBA.

CAD 1X4DB138-2
ACAD2004 NJ - 01

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 19.0 | REDRAWN PER ABN-V70113, VER. 1.0 | 08/21/06 | NJ | LCF | MDS | X |

SIZE E 34X44

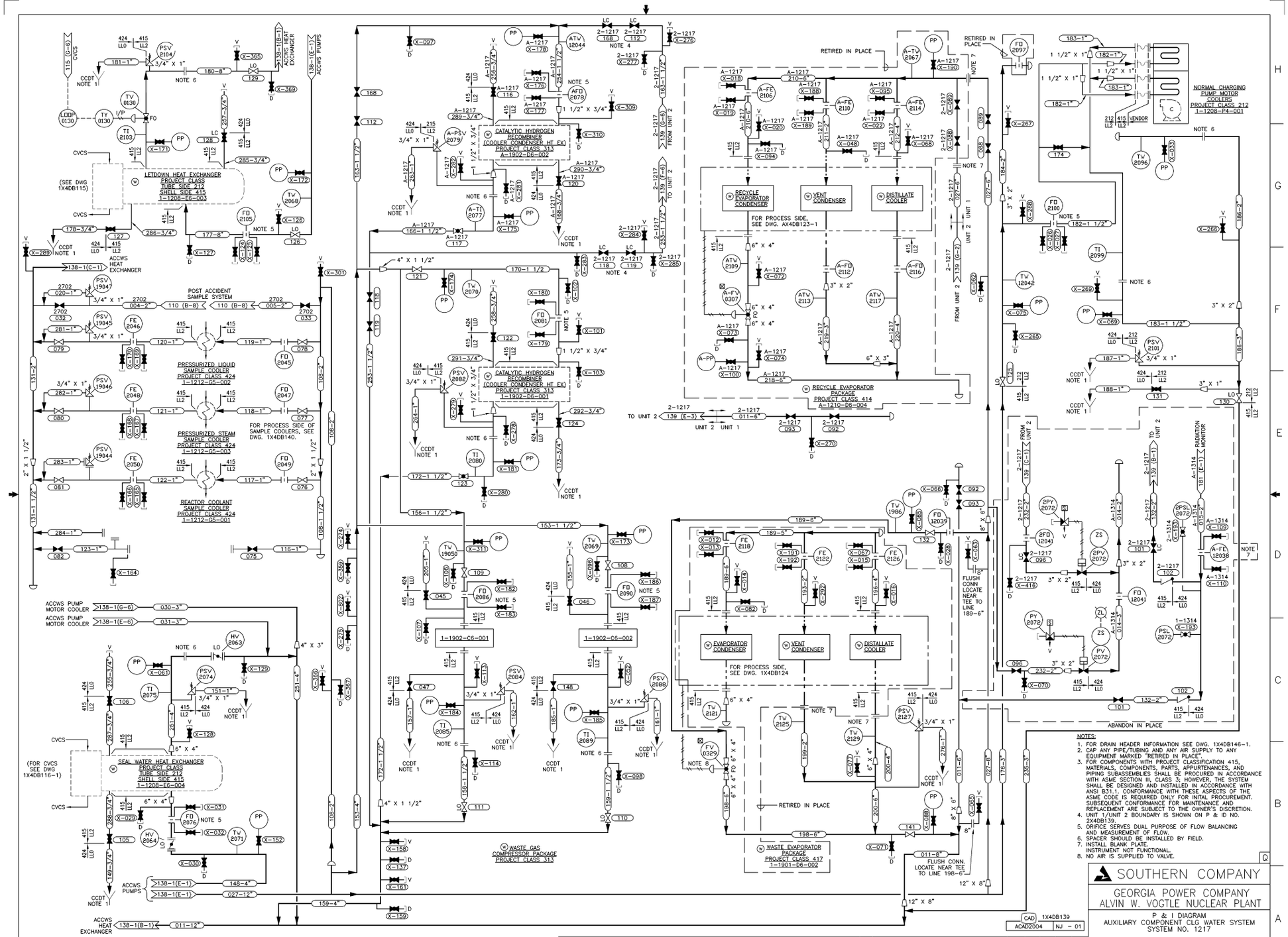
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY COMPONENT
COOLING WATER SYSTEM
SYSTEM NO. 1217

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB138-2 | 19.0 |

DRAWING CATEGORY: CRITICAL




NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB146-1.
2. CAP ANY PIPE/TUBING AND ANY AIR SUPPLY TO ANY EQUIPMENT MARKED "RETIRED IN PLACE".
3. FOR COMPONENTS WITH PROJECT CLASSIFICATION 415, MATERIALS, COMPONENTS, PARTS, APPURTENANCES, AND PIPING SUBASSEMBLIES SHALL BE PROCURED IN ACCORDANCE WITH ASME SECTION III, CLASS 3; HOWEVER, THE SYSTEM SHALL BE DESIGNED AND INSTALLED IN ACCORDANCE WITH ANSI B31.1; CONFORMANCE WITH THESE ASPECTS OF THE ASME CODE IS REQUIRED ONLY FOR INITAL PROCUREMENT. SUBSEQUENT CONFORMANCE FOR MAINTENANCE AND REPLACEMENT ARE SUBJECT TO THE OWNER'S DISCRETION.
4. UNIT 1/UNIT 2 BOUNDARY IS SHOWN ON P & ID NO. 2X4DB139.
5. ORIFICE SERVES DUAL PURPOSE OF FLOW BALANCING AND MEASUREMENT OF FLOW.
6. SPACER SHOULD BE INSTALLED BY FIELD.
7. INSTALL BLANK PLATE. INSTRUMENT NOT FUNCTIONAL.
8. NO AIR IS SUPPLIED TO VALVE.

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| NO. | VERSIONS | DATE | DR | CHK | APPV | DTL |
|---|---|---|---|---|---|---|
| 29.0 | REDRAWN PER ABN-V70117, VER. 1.0 | 08/22/06 | NJ | LCF | MDS | X |

SIZE E 34X44

CAD 1X4DB139
ACAD2004 NJ - 01

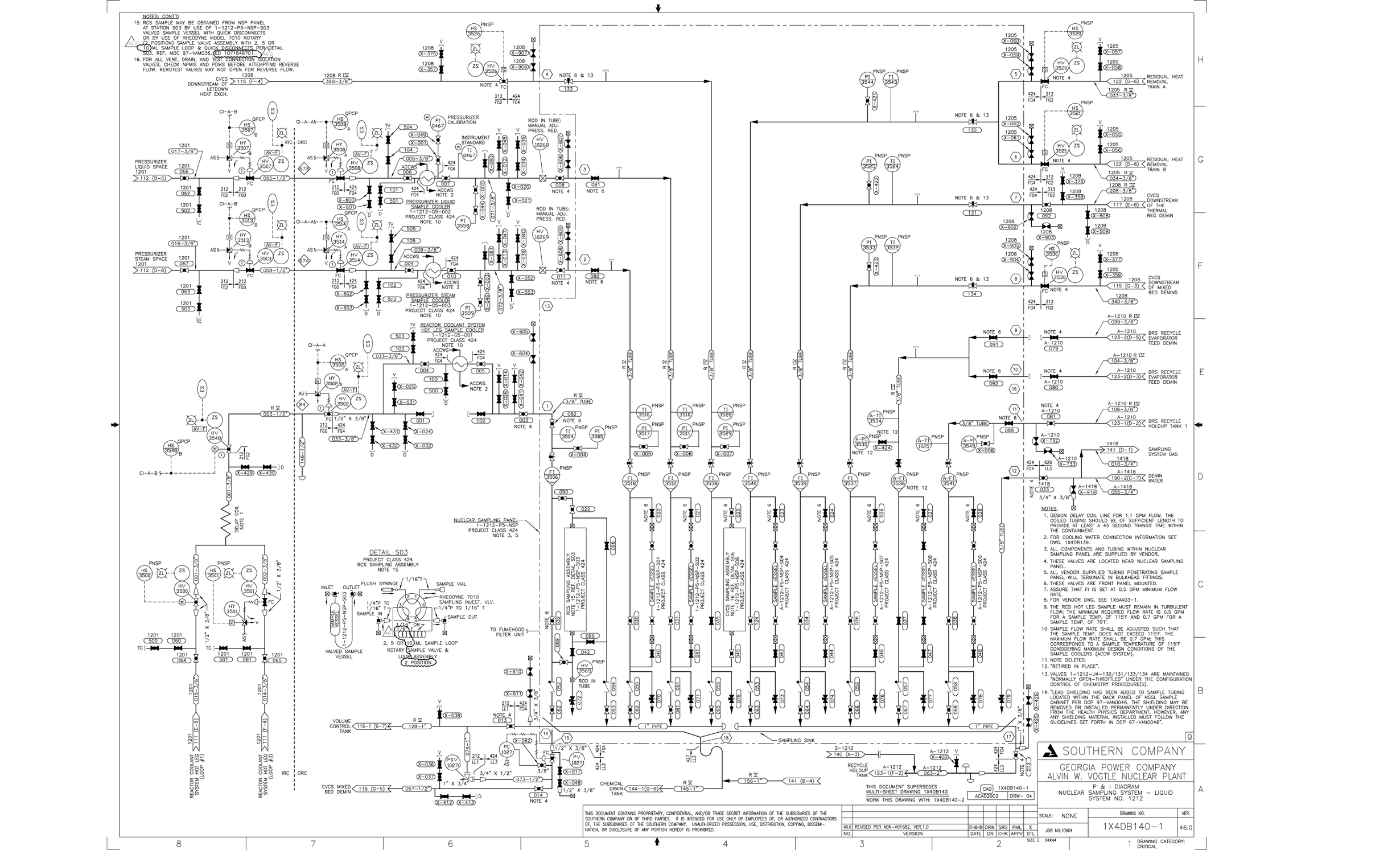
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY COMPONENT CLG WATER SYSTEM
SYSTEM NO. 1217

| SCALE: NONE | DRAWING NO. | VER. |
|---|---|---|
| JOB NO.10604 | 1X4DB139 | 29.0 |

DRAWING CATEGORY: CRITICAL


NOTES: CONT'D
15. RCS SAMPLE MAY BE OBTAINED FROM NSP PANEL AT STATION S03 BY USE OF 1-1212-P5-NSP-S03 VALVED SAMPLE VESSEL WITH QUICK DISCONNECTS OR BY USE OF RHEODYNE MODEL 7010 ROTARY (2 POSITION) SAMPLE VALVE ASSEMBLY WITH 2, 5 OR 10 ML SAMPLE LOOP & QUICK DISCONNECTS PER DETAIL S03, REF. MDC 97-VAN036, ED 10/71849/01
16. FOR ALL VENT, DRAIN, AND TEST CONNECTION ISOLATION VALVES, CHECK NPMIS AND PDMS BEFORE ATTEMPTING REVERSE FLOW. KEROTEST VALVES MAY NOT OPEN FOR REVERSE FLOW.
CVCS DOWNSTREAM OF LETDOWN HEAT EXCH.
PRESSURIZER LIQUID SPACE
PRESSURIZER STEAM SPACE
PRESSURIZER CALIBRATION
INSTRUMENT STANDARD
PRESSURIZER LIQUID SAMPLE COOLER 1-1212-G5-002 PROJECT CLASS 424 NOTE 10
PRESSURIZER STEAM SAMPLE COOLER 1-1212-G5-003 PROJECT CLASS 424 NOTE 10
REACTOR COOLANT SYSTEM HOT LEG SAMPLE COOLER 1-1212-G5-001 PROJECT CLASS 424 NOTE 10
ROD IN TUBE: MANUAL ADJ. PRESS. RED.
DELAY COIL NOTE 1
NUCLEAR SAMPLING PANEL 1-1212-P5-NSP PROJECT CLASS 424 NOTE 3, 5
DETAIL S03
PROJECT CLASS 424
RCS SAMPLING ASSEMBLY
NOTE 15
FLUSH SYRINGE
SAMPLE VIAL
RHEODYNE 7010 SAMPLING INJECT. VLV.
SAMPLE IN
SAMPLE OUT
2, 5 OR 10 ML SAMPLE LOOP
ROTARY SAMPLE VALVE & LOOP ASSEMBLY 2 POSITION
VALVED SAMPLE VESSEL
TO FUMEHOOD FILTER UNIT
VOLUME CONTROL TANK
CHEMICAL DRAIN TANK
RECYCLE HOLDUP TANK
SAMPLING SINK
RESIDUAL HEAT REMOVAL TRAIN A
RESIDUAL HEAT REMOVAL TRAIN B
CVCS DOWNSTREAM OF THE THERMAL REG DEMIN
CVCS DOWNSTREAM OF MIXED BED DEMINS
BRS RECYCLE EVAPORATOR FEED DEMIN
BRS RECYCLE HOLDUP TANK 1
SAMPLING SYSTEM GAS
DEMIN WATER
CVCS MIXED BED DEMIN
REACTOR COOLANT SYSTEM (LOOP #1)
REACTOR COOLANT SYSTEM (LOOP #3)
NOTES:
1. DESIGN DELAY COIL LINE FOR 1.1 GPM FLOW. THE COILED TUBING SHOULD BE OF SUFFICIENT LENGTH TO PROVIDE AT LEAST A 45 SECOND TRANSIT TIME WITHIN THE CONTAINMENT.
2. FOR COOLING WATER CONNECTION INFORMATION SEE DWG. 1X4DB139.
3. ALL COMPONENTS AND TUBING WITHIN NUCLEAR SAMPLING PANEL ARE SUPPLIED BY VENDOR.
4. THESE VALVES ARE LOCATED NEAR NUCLEAR SAMPLING PANEL.
5. ALL VENDOR SUPPLIED TUBING PENETRATING SAMPLE PANEL WILL TERMINATE IN BULKHEAD FITTINGS.
6. THESE VALVES ARE FRONT PANEL MOUNTED.
7. ASSURE THAT FI IS SET AT 0.5 GPM MINIMUM FLOW RATE.
8. FOR VENDOR DWG. SEE 1X5AA03-1.
9. THE RCS HOT LEG SAMPLE MUST REMAIN IN TURBULENT FLOW. THE MINIMUM REQUIRED FLOW RATE IS 0.5 GPM FOR A SAMPLE TEMP. OF 115°F AND 0.7 GPM FOR A SAMPLE TEMP. OF 70°F.
10. SAMPLE FLOW RATE SHALL BE ADJUSTED SUCH THAT THE SAMPLE TEMP. DOES NOT EXCEED 115°F. THE MAXIMUM FLOW RATE SHALL BE 0.7 GPM; THIS CORRESPONDS TO A SAMPLE TEMPERATURE OF 115°F CONSIDERING MAXIMUM DESIGN CONDITIONS OF THE SAMPLE COOLERS (ACCW SYSTEM).
11. NOTE DELETED.
12. "RETIRED IN PLACE".
13. VALVES 1-1212-U4-130/131/133/134 ARE MAINTAINED "NORMALLY OPEN-THROTTLED" UNDER THE CONFIGURATION CONTROL OF CHEMISTRY PROCEDURE(S).
14. "LEAD SHIELDING HAS BEEN ADDED TO SAMPLE TUBING LOCATED WITHIN THE BACK PANEL OF NSSL SAMPLE CABINET PER DCP 97-VAN0046. THE SHIELDING MAY BE REMOVED OR INSTALLED PERMANENTLY UNDER DIRECTION FROM THE HEALTH PHYSICS DEPARTMENT. HOWEVER, ANY ANY SHIELDING MATERIAL INSTALLED MUST FOLLOW THE GUIDELINES SET FORTH IN DCP 97-VAN0046".
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
NUCLEAR SAMPLING SYSTEM – LIQUID
SYSTEM NO. 1212
THIS DOCUMENT SUPERSEDES MULTI-SHEET DRAWING 1X4DB140
WORK THIS DRAWING WITH: 1X4DB140-2
CAD 1X4DB140-1
ACAD2002 DRW- 04
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NO. 46.0 VERSION REVISED PER ABN-V01682, VER.1.0 DATE 07-06-09 DRW SRG CHK PML APPV X DTL
SIZE E 34X44
SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB140-1
VER. 46.0
DRAWING CATEGORY: CRITICAL